

Time-Saver Standards for Interior Design and Space Planning

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Time-Saver Standards for Interior Design and Space Planning Joseph De Chiara Julius Panero Martin Zelnik McGraw-Hill, Inc. New York St. Loult San Francisco Auckland Bogota" Caracal Hamburg Uibon London Madrid Mexico Milan Montreal Naw Dalhl Parli San Juan Sio Paulo Singapore Sydnay Tokyo Toronto

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Foreword A resource of incredible range and detail, this volume was compiled by three remarkably inspired designers and educators. Because of their great knowledge of interior design and their sensitivity to the subject matter, they have created the most comprehensive source book for the field ever. The editors spent three years bringing this volume to fruition, culling the best project drawings by outstanding designers to illustrate much of the subject matter and tapping their own anthropometric expertise to address space planning and special function areas. They also address the importance of historic influence on present-day design with an impressive review of period furniture and interior details. All of these things have produced a reference work of such scope and inclusiveness that the reader will be relieved of many hours in the pursuit of details and information, time saved that can be used for more innovation and creativity in developing solutions for client needs. The authority and abundance of this book are a testimony to the maturation of this profession of ours and to the editors' appreciation and understanding of its importance. Jack Lowery, FASID, IDEC My pleasure in being invited to write part of the Foreword swiftly changed to respect and, in turn, awe at the scope and depth of this book. To say that it is an encyclopedic compilation and mass of information is obvious. But it is especially and uniquely user-friendly. It presents the written and illustrative data without a trace of pedantry; it meets a real need in our interior designer professional resources. The editors' effort, dedication, and patience, sustained during a period of over three years, are truly heroic. An astonishing number of hours of input have produced a reference of incalculable value. I offer the same cautionary advice mentioned in the Preface: If the book is a wonderfully comprehensive reference and support for interior design standards, historical material, suggested plan and design criteria, and regulatory limitations, it is not—it will never be—a substitute for the inspired, creative design act, for imaginative solutions are always driven by new cultural conditions, programs, and functional requirements.

So to all you designers: Continue to spin your dreams, but do not stray far from this great resource. Lawrence J. Israel, AIA, FISP xl

Preface Time-Saver Standards for Interior Design and Space Planning is a professional handbook dealing with the planning, design, and detailing of interior spaces. Its primary goal is to provide, within a single reference, information that typically is found dispersed throughout a multitude of sources, including manufacturers' catalogs, technical literature, books dealing with historic styles, and documents and drawings from various projects. This handbook can be used by the small and medium-size interior design or architectural firm to establish an instant reference library of design data and details by providing a broad selection of detail types and techniques. In addition, the large firm will be able to substantially augment and modify an existing library of details. Perhaps the most unique feature of this handbook is the vast array of construction and woodwork details reproduced directly from actual working drawings contributed by some of the nation's leading interior design and architectural firms. It is this that makes the handbook particularly useful to the interior designer, architect, and student alike. This book consists of five sections. The first, entitled Planning and Design of Interior Spaces, deals with residential, office, hospitality, and retail spaces in terms of the relevant planning, design, and detailing data specifically associated with each. The second section, entitled Construction Details and Finishes, deals with various basic interior construction components associated with most interior spaces. These components include partitions, wall openings, wall finishes, floors and floor finishes, doors, ceilings, stairs, fireplaces, and lighting. Details relevant to each component have been contributed by practicing interior designers and architects as well as manufacturers. The third section, entitled Architectural Woodwork, deals with standard joinery and casework details, customized woodwork details, cornices and mouldings, and furniture hardware. The fourth section, entitled Specialties, deals with various specialized areas of equipment, systems, furnishings, and decoration, including signage and graphics, audio-visual systems, window treatments, and accessories. Information for these subject areas is drawn from manufacturers, suppliers, and designers. The fifth section, entitled General Reference Data, provides the most comprehensive set of time-saving reference materials found in handbooks of this type, including tables, charts, formulas, and planning

guidelines. Of particular interest to the architect, interior designer, and facility manager are tables that can be easily used to determine carpet and wall covering yardage. Charts and drawings relative to human factors and planning standards are also provided. It should be noted that since the details and other information pre-

sented in this book have been compiled from so many different sources, it is difficult to ensure that all the data are entirely accurate or appropriate; for example, in some instances planning guidelines may reflect minimum acceptable standards and not necessarily ideal or preferred standards. In other instances the details indicated may have been perfectly adequate in the context of the total building design of which they were a part, but they may well require modification to reflect design conditions and the reader's intended use. It should also be noted that building codes, fire safety regulations, barrier-free standards, and many other laws governing the design and construction of buildings vary from state to state. Accordingly, the reader should consult all applicable local, state, and federal codes for conformance prior to applying any of the information contained in this book. Moreover, the reader is cautioned that the dimensional information provided in connection with furniture, equipment, appliances, accessories, etc., has been obtained from manufacturers and technical literature and thus varies from supplier to supplier and from source to source. Certain items may have been discontinued, others modified, and still others replaced. Although every effort has been made to ensure the reasonableness of the information, the reader is cautioned to consult the manufacturer of the item specified for current dimensional data. The reader is also advised that most drawings and other illustrative material have been enlarged or reduced for reasons of page layout and page size. The reader is cautioned, therefore, to disregard any scale designations and not to scale the drawings in order to determine any additional dimensional information. Finally, as mentioned before, the plans and details contained in this book were extracted from complete sets of actual working drawings prepared by many different contributors. They were selected both because they were representative of typical situations faced by the designer of interior spaces and because they were particularly informative. The authors would like to underscore the fact that these plans and details, as well as all the other material presented in this book, are intended to serve only as a helpful point of departure in

connection with the design process, and not as a substitute for original thinking and creativity. Although every effort has been made to present reasonably accurate information, the editors and publisher assume no liability or responsibility for damage to persons or property alleged to have occurred as a direct or indirect consequence of the use and application of any of the contents of this book. The reader is advised to view the subject matter primarily as guidelines for preliminary planning and detailing, and to properly review, modify, and process it to ensure conformance with local codes and practices and appropriateness of applicability. Joseph De Chlara Julius Panero Martin Zelnik xiv

Planning and Design of Interior Spaces

Residential Spaces Period furniture Furniture dimensions Living rooms
Dining rooms Bedrooms Bathrooms Kitchens Library/study
Family/recreation rooms Laundry/sewing rooms Closets/storage areas 5 44
61 80 87 100 149 190 196 200 206

Residential Spaces PERIOD FURNITURE 17th Century American:
Colonial THE EXTERIOR "Phe 17th Century immigrants brought to
America the building traditions of their native lands. The Parson Capen
house (1683) at Topsfield, Mass., for example, closely resembles English
houses of the same period. But the clapboards are typically American. In
the panels at right are close-up details of the Early Colonial background.
Moldings and trim Doorway surrounds FIREPLACE SURROUND
MOLDING JOHN QUIWf ADA.WS HOUSE IOHN WILLIAMS HOUSE.
QUINCY MASS. 1714 DEfRFIKO MASS. 1707 THE LIVING ROOM
rpms living room is typical of those in the more elaborate Early Colonial
homes. The crewel-embroidered curtains are blue-green with touches of
red. This is taken up by the upholstery—blue-green damask for the sofas,
red tapestry for the chairs. The Oriental rug and the portrait above the
fireplace are both in tones of red, brown and yellow, with red dominant. An
alternative color scheme would have blue and yellow upholstery
(needlework for the chairs, satin for the sofas). The walls would be pine-
paneled, adorned with silver sconces, the curtains a bright cotton print in
red, yellow, blue and white. Living-room fabrics NEEDLEWORK PANEL
Wing chairs, sofas, armchairs, stools DOUBLE CHAIR SETTEE Furniture

made in America during the Early Colonial period (the seventeenth century and the first quarter of the eighteenth century) was necessarily, and possibly also by choice, of the simplest type." The early colonists, particularly those in New England, had not time or equipment to spare for any but the essentials of life. Turning on the lathe was the simplest to achieve and thus the most common form of furniture decoration. It was also a process capable of infinite variations of design (some are shown in Fig. 1). Characteristic types of stretcher Turnings used for the corner BLAD TRUMPIT AND Bar Fig. 1 Motifs characteristic of Early Colonial furniture. 5

Residential Spaces PERIOD FURNITURE 17th Century American: Colonial THE DINING ROOM The color scheme in this dining room is keyed to the low tones of the pine paneling and walnut furniture, the soft gleam of the smooth polished brass chandelier. The banister back chairs have rush-bottom seats. Brilliant red and white printed cotton is used for the curtains. The hooked rug is in reds and greens. Alternatively the curtains might be of red and yellow crewel embroidery, the upholstery of red brocade. In the panels at right are furniture and fabrics suited to an Early Colonial dining room. Dining-room fabrics NEEDLEPOINT TAPESTRY Dining tables, table chairs PUNK-TOP KMG TOW TABLE enGUSH OATE. IEO TAKE C 1470 THE BEDROOM The little bedroom with its pine paneling and low ceiling is typical of the Early Colonial period. The bed, decorated with hangings of crewel work in an Oriental design, is the most important feature of the room. The chairs are upholstered in yellow damask. The green printed cotton used for the little draped window curtains is echoed by the greens in the hooked rug on the floor. Alternatively the walls might be painted a dark gray-blue, the curtain material being a red printed cotton on a gray ground. The furniture is of walnut and oak. Bedroom rubrics mm CREWEL [mBROIDE] W BASED ON TREE OF HFC DESIGN Beds, daybed, cradle WALNUT FOUHP05TEB C. 1710 DAVBEDOF 'CAIN STYLE C. Ifm Even the most costly furniture in this Early Colonial period was usually of solid wood unfinished except for stain or waxing. Veneering and shellacking, to gain carefully patterned graining and high finish, were still unexploited. The pine paneling on the walls might be left unfinished, waxed, or painted. Other woods near at hand in the forests and so commonly used were oak, birch,

maple, and walnut. Generally, American work is patterned upon English work of 10 or 20 years earlier. In Pennsylvania and Delaware, which were settled by colonists of Swedish and German descent (in addition to the English), much of the simple furniture was painted with its motifs transferred from European peasant art. In the later years of the Early Colonial period, when New Englanders were already beginning to trade with the Orient, much Chinese porcelain was imported. The Oriental influence was strong in textiles; the Tree of Life pattern was very popular at this period. Native textiles copied the patterns 6

Residential Spaces PERIOD FURNITURE 17th Century American:
 Colonial Armchair, side chairs Imrf ' CAiH CHAINS FLEMISH SCBOILS
 AND FEET, CANE BAC'.S AND SEATS CANE SEAT, PLAIN SPLAT
 LATE QUMN ANNE TIM, FLATTENED BALL BANNISTER BACK,
 AND CLAW FEET SPANISH FEET EAMY QUEEN ANNE TYPE WITH
 DUTCH SPLAT BACK UPHOISTEMD CHW*| WITH DUTCH FEET
 SHEU CM5TING, SHALLOW SEAT PIERCED BACK, BAHANDCIAW
 FEET Sideboards, dressers, chests Dining-room accessories HEAVY OAK
 CHEST OF DRAWER5 WITH APPLIED MOLDING5 OAK
 SIDEBBOARD, TWISTED LEGS (WILLIAM AND MARY TYPE
 PENNSYLVANIA WALNUT DRESSER, OLASS-fRONTED CABINET
 SPICE CUPBOARD C. 1700 PANELED DOORS SIIVES SCONCE
 WITH RAISED EMBOSSSED EDGE WALNUT-FRAMED MIHROIIS
 AMERICAN QUEEN ANNE STYLE, C. 1710 ETCHED MftfcOK
 SCONCE, CLASS DRIP DISHES Chests, kas, highboy, chests of drawers -
 TMJTCM KAS WITH BALL FEtl Desk, lowboys, night tables, stools
 UPHOLSTERED LOWBOY JACOBAN SIDE CHAIR WITH SHELL
 FRONT SLANT-TOP DESK ON A TURNED FRAME Bedroom
 accessories V HOG-SCRAPER CANDLESTICK WROUGHT-IRON
 CANDLESTAND HUSH IIGHT STAND CHINESE PORCEIAIN
 MIRROR SCONCE WITH SHELL TOP SMAL1 WAL1 MIRROR
 REFLECTOR CANDLESTICK DOUtlE CANOLESTAND WITH
 METAL SHADES WALNUT DRESSING MIRRORS 1700-1710 and
 colors of India, Persia, and China. The originals, or good copies of them,
 were usually imported from England, The colors in common use were of a
 piece with the solid, sturdy furniture. They seldom escaped from the
 conventional round of blue, red, gold, and natural gray. The only exceptions

were imported fabrics and the occasional hard brilliance of the Chinese porcelain found in the great houses of the day. Whatever luxury there was at this time expressed itself in textiles and silver rather than in furniture. Settlers in the South, many of them English aristocrats, maintained a higher standard of comfort than those in the North; they imported most of their furniture and fabrics from England and continued to do so for a long time. Early Colonial furniture taken as a whole is sturdy but not subtle. Furniture patterns in this country changed slowly. Paneling relieved the larger flat areas such as cupboard doors and drawer fronts. The latter were further decorated by quite elaborate fretted

Residential Spaces PERIOD FURNITURE 17th Century American:
 Colonial Fireplace designs A HORIEONUL MOIDINC ABOVE IH(FIREPLACE OFENPNG IS A RUDIMENTARY VERSION O* THE MANTEL SHEIE I PLAIN MOLDED SURROUNDS WERE SOMCTImEJ MARBtEIZED EOR VARIETY Eight typical paneled doors □E EH SQ Drapery treatments for various window types A-r^ffe*. iloKIP ill i/v\ v\J ON A PAIR OF LEADED CASEMENTS. SIMPLY DRAPED CUMAWS OF PRINTED COTTON fMBROIDERrAT THE BOTTOM OF THE CURTAINS REPEATS THAT ON THE VALANCE IOUR DIFFERENT TYPES OF VALANCE, USING FABRICCOVERED BOARDS, TAPE AND (RINGS r Corner cupboards, eabinets,secretaries, chest MAHOGANY CORNER CUP BOAS D HIGH CHEST, WIUIAM AND MAPY STYLE C 17-00 KD LACOUIR CHIWA CASINEI Occasional tables, desk, lowboys, pedestals Living-room accessories da! LACQUERED LOWBOY WITH CABRLOII LEGS HIE TOE TABLE WIFH IRUMPET TURNED IEGS AND CROSS STRETCHERS WALNUT DESK WITH BALI FEET SILVER SCONCE CANDLESTICK WITH Ft\ fj "II^S WSOUOMMRON GUARD EARLY DUTCH TYRE WAIL CLOCK ADJUSTABLE WOODEN STAND FOR BETTY LAMPS BRASS AND OECORATEO MIRROR SCONCE WALNUT-FRAMED MIRRORS TOP PANEL OF ETCHED GLASS brass and wrought-iron hardware (see Fig. 1). More carefully embellished than the earliest American furniture were the pieces imported by the colonists from their various homelands. These pieces, and the memories of others left behind, later served as models for American craftsmen. The dominant influence was Dutch, for the English had a Hollander, William of Orange, as king. He and

his queen, Mary, gave their names to a style of which elaborate stretchers (particularly on highboys, lowboys, and occasional tables) and scrolled legs are among the most obvious characteristics. Also from Dutch, Spanish, and Portuguese sources are derived most of the carved feet. Chair liicks VOLUTI CRtilihC (liuniniit I) Hi's ill' Icl> Fig. 1 (Continued) which distinguish this Early Colonial furniture and often give clues to its date and place of origin. Furiiliire liunlnri' Cnmimin lyiii's of I

Residential Spaces PERIOD FURNITURE 18th Century American: Colonial THE EXTERIOR The architectural details shown in the five panels at right are characteristic of the background for 18th Century Colonial decoration. As one (if the finest houses of the period we have pictured (at right) "Westover" the great mansion erected by William Byrd in Charles City Co., Virginia. Typical of this period are the brick walls and chimneys, the stone or white painted brick trim. In the North wood was in more common use than brick for the exterior, and the interior wooden trim was finely detailed.. Typical Colonial architecture Cornices ami trim Interior doorways TYPICAL McIntire DETAILS. CORNICE DADO RAIL ANOTHER CORNICE EARLY GEORGIAN CLASSIC MOTIFS'OF COLUMN AND SWAO DOORWAY FROM KINO'S CHAPEL. BOSTON THE LIVING ROOM The furniture, fabrics and accessories shown in these panels are all suitable to the living room, and they are all typical of the 18th Century Colonial style. The interior pictured at right is a fine Colonial living room carefully restored to its 18th Century state. The walls are Naples yellow, the columns and fireplace white. Red and green ure dominant in the Oriental rug, dark greens and browns in the portrait above the fireplace. So the sofa is upholstered in .striped satin, the armchair in yellow Venetian brocade, the wing chair in a printed linen. The urns are of Chinese porcelain. Another color scheme might be: pearly gray walls, oyster white columns and fireplace. Red would be dominant in the Oriental carpet, dark greens and red in the portrait. There would be red damask on the sofa, green rep on the wing chair, and gold damask for the armchair. Decorating a Colonial living room Fabrics for curtains ami upholstery Mi - Jri,7i'.»' \$k mi -L. .A BLOCK PRINT Wing chairs, armchairs, sofas CONNECTICUT WINC CHAIR PH!|AP£IPHIA VERSION MARTHA WASHINGTON CHAIR OF HEPPIEWHITF. PHILADELPHIA TYPE WING CHAIR CHIPPENDALE INFLUENCE MAHOGANY

CHIPPENDALE SOFA Whereas furniture of the Early Colonial period was often so primitive as to be referred to as "kitchen Colonial," in this succeeding era dignity and luxury prevail in the centers of taste. The furnishings reflect the fashionable contemporary styles of England and stately country homes, whether on New England farms or Virginian and Carolina plantations, followed these styles. This gave rise to a number of notable architects, craftsmen, and workers in metal and wood. The eighteenth century Colonial period was the first of the really great eras in American cabinetmaking. The manufacture of wallpaper in this country was begun by 1763. Before this it was from Europe. The "Pennsylvania fireplace" or "Franklin stove" was invented by Benjamin Franklin in 1742 and immediately became popular up and down the Atlantic seaboard. Philadelphia was a furniture style center, in fact the most active in the creation of taste, with Boston and Charleston following. A number of artists and craftsmen of this period bear mentioning. Among the architects were Samuel McIntire, Charles Bulfinch, John James, Richard Mundy, Peter Harrison, John Kirk, and Isaac Royall. These men were greatly influenced by the English architects Isaac Ware, James Gibbs, Robert Morris, Abraham Swan, William Halfpenny, Batty Langley and William Pain, who in turn were in debt to the Italian masters Palladio and Giacomo Leoni. Among the cabinetmakers were Moses Dodge, Stephen Dwight, Henry Hardcastle, Gilbert Ash, Robert Wallace, Charles Shipman, John Brincker, John Tremain, Charles Warham, John Brown, Bemsley Wells, Thomas and Benjamin Laskey, Jonathan Goodhue, and Job Trask. Among the upholsterers were Stephen Callow, Richard Wenman, Joseph Cox, and John Taylor; among the metalworkers were William Coffin, Wilkins, Joseph Liddell, William Bradford, John Bassett, and Peter Harby; and among the painters were John Singleton Copley, Joseph Blackburn, John 9

Residential Spaces PERIOD FURNITURE 18th Century American:
Colonial DINING ROOM The furniture and fabrics shown in the five panels at right would look well in any dining room; but for your guidance in the selection of materials and colors we illustrate at right a fine Colonial dining room as it might have appeared in the 18th Century, The pine-panelled walls are colored a light ocher, the niches Chinese red. Curtains are French blue. Blue, rust and beige predominate in the Oriental rug, dark green, blue and black in the portrait over the fireplace. Table and chairs are

of walnut, the sideboard of mahogany. An alternative color scheme would be light blue-gray walls with cream niches. Curtains would be oyster white silk, the Oriental rug having a greenish tan background. Decorating a Colonial dining room Fabric for curtains and upholstery STRIPED BROCADE BEDROOM In the bedroom at right, choice of color and textures was designed to achieve an impression of warmth and intimacy. The paneled walls are in two tones of gray- green, the ceiling ocher. Curtains are antique gray-green satin. Furniture is -walnut, except for the mahogany bed, which has a yellow taffeta spread. Fireside chairs are covered in crimson damask side chairs in turkey work. An alternative color scheme would be: warm gray walls with oyster white moldings. The ceiling would be cream, the carpet solid taupe, and the curtains of blue damask. The bed would have a white moire spread and blue valance. The side chairs would be upholstered in yellow dumiisk, the wing chair in turkey work. Decorating a Colonial bedroom Fabrics for curtains,upholstery, canopy PANTED tINEN PERIOD FURNITURE 18th Century American: Colonial DINING ROOM The furniture and fabrics shown in the five panels at right would look well in any dining room; but for your guidance in the selection of materials and colors we illustrate at right a fine Colonial dining room as it might have appeared in the 18th Century, The pine-panelled walls are colored a light ocher, the niches Chinese red. Curtains are French blue. Blue, rust and beige predominate in the Oriental rug, dark green, blue and black in the portrait over the fireplace. Table and chairs are of walnut, the sideboard of mahogany. An alternative color scheme would be light blue-gray walls with cream niches. Curtains would be oyster white silk, the Oriental rug having a greenish tan background. Decorating a Colonial dining room Fabric for curtains and upholstery Dining tables, consoles STRIPED BROCADE y i EXTENSIBLE DINING TABLE 1770 BEDROOM In the bedroom at right, choice of color and textures was designed to achieve an impression of lunrmlli JITlfl intimflpv Thn nnn. Decorating a Colonial bedroom Four-poster beds Ramage, James Peale, and Charles Wilson Peale. Important manufacturers were, of wallpaper, Jackson of Battersea (England) and, of window and bottle glass, Baron Stiegeland Caspar Wistar. Fabrics most commonly used during the Colonial period were damask, camblet, Indian gimp and binding, moreen (woolen drapery cloth), harrateen cloth, block-printed cotton and linen, cashmere, calico, dimity, durance, stout worsted cloth, turkey work (tufted "pilelike"),

paduasoy (strong silk), soy, shalloon, watchet, linsey-woolsey, fustian, silk muslin, chintz, Indian calico, tabby, sar- canet, taffeta, horsehair, camak. bancours, and brocade. Woods most commonly used were oak. ash, elm, red cedar, mahogany, walnut, maple, pine, and cherry. The Chippendale style merges at one end with Queen Anne, at the other with Hepplewhite, Sheraton, and Duncan Phyfe. The Rococo mounts to its zenith and starts to decline within these years. Walnut has a new rival in mahogany And American craftsmen produced pieces of a quality which compares favorably with English work.

Residential Spaces PERIOD FURNITURE 18th Century American:
 Colonial Armchairs, side chairs CHIPPENDALE SIDE CHAIR
 AMERICAN HEPPLEWHITE 1735 - 1795 PIERCED LADDER BACK
 CONNECTICUT 17th Sideboards, lowboys, chests Dining room
 accessories BENNINGTON WAKE CANDLESTICK BRASS AND IRON-
 17th LONG WHITE PORCELAIN FLUTED BOWL BRASS
 CANDLESICK Chests, kas, secretary or BLOCK FRONT CHES1 JOHN
 OODARD or PJ JOHN OODARD TYPE LOW CHIST OF DRAWERS.
 1770- 1780 Wing chairs, armchairs, tables S3 CAPE COD WING CHAIR
 Bedroom accessories UN CANDLESHM Marble was imported until after
 the Revolution when domestic marbles began to be used. Marble chimney
 pieces, window sash, lead roofing, and hardware were all imported from
 London. The size of glass window panes gradually increased as the century
 progressed. An order of small pilasters or columns supporting the mantel in
 a chimney piece was found only in imported work prior to the Revolution.
 Fireplace openings with neither cornice nor mantel shelf were long
 common. Ears on the architraves were almost universal, and a pediment
 (always broken) was very common. After 1760 the scroll pediment, or a
 similar treatment of the architrave, occurs. 11

Residential Spaces PERIOD FURNITURE 18th Century American:
 Colonial Fireplace designs CHIMNEY PIECE IN THE STYLE OF SAMUEL
 McINNESHE 3'6" WIDE, 4'-0" HIGH WOOD OPENING : -a- WIDE,
 3'-0" HIGH WIDTH MIDWAY 11" A- PROJECTIONS FINE COLONIAL
 CHIMNEY PIKE SHELF 11" WIDE, 1'-1" HIGH WOOD OPENING 5' 11"
 WIDE, 1' 11" HIGH WIDTH MIDWAY 6'-11" 3" PROJECTION Window
 details ft F It = ___ = X. 3 SMALLER DOUBLE HUNG WINDOW RACIO

ABOVE OADO 9 PANE SASH MORI ELABORATE RECESSED IVPE
 WITH WINDOW SEAT & E10W 17 PANE SASH Drapery treatments
 CLASSIC TYPE OF DAMASK DRAPING USUALLY ORNAMENTED
 WITH AN ELABORATE FRINGE TYPE OF DRAPERY USUALLY
 DONE IN TAFFETA AND LEFT PLAIN OR FRINGED Secretaries,
 highboy, clocks Desks, tables, firescreen MAHOGANY GRANOFMHERS
 CLOCK' MAHOOANIT BBEAKFRONT GRANDFATHER S CLOCK
 CHIPPENDALE STYLE SY). WILDE* 01 HINGHAM Living room
 accessories 6SASS AND IRONS, IV HIGH IS IOHG 12

Residential Spaces PERIOD FURNITURE 18th Century American: Federal
 THE EXTEHIOH <ymt0i%iMu&8m> AS a typical mansion of the Federal
 period we show Mappa ■*-Hall in Trenton, N.Y. It was started in the
 closing years of the 18th Century and completed in 1809. The portico and
 the simple pediment exemplify the prevailing Classic trend. In the panels to
 the right are some typical details from the Federal period background
 Cornices and trim Doorways for the interior 5 ^^ssLJ! I Q Q I DOORWAY
 OF FHHRAL TYPE GLASS FANLIGHT WITH LEAD CAMES DOOft sr
 SAMUEL McINTIRE THE LIVING ROOM rpilis is a fine Federal interior
 in its original condition. The ■*- walls and woodwork are painted pistachio
 green. The curtains are of beige damask, the sofa upholstered in red and
 gold damask. Gold damask is used for the armchairs, yellow damask for the
 side chairs. The Oriental rug is wine red in tone, the furniture, mahogany.
 The clock in of ox-blood marble. An alternate scheme would have light
 gray-blue walls and woodwork. The draperies would be yellow damask, the
 chairs upholstered in green damask. The furniture and fabrics shown in
 panels at right would also be suitable for the Federal living room Living
 room fabrics • "i ■*• ■"*.!■*■. SATIN DAMASK w » Armchairs, side
 chair, sofas SHERATON TYPERCHAIR MADE FOR GEORGE
 WASHINGTON NOW IN CITY HAlt. NEW YORK The Federal style is at
 its most suave and elegant in the furniture of Duncan Phyfe, a Scotch
 cabinetmaker who arrived in New York about 1795. He did not originate a
 style; he translated prevailing fashions into fine craftsmanship. Thomas
 Sheraton, then the current English favorite, and the French Di- rectoire
 cabinetmakers set the style. All these designers were profoundly influenced
 by a rediscovery of the classic splendors of Greece and Italy. Reeding of
 table, chair, and sofa legs and other framing members gives elegance to

Federal furniture. Contrasting color veneer is used to outline the edges of tables and desks and to lend interest to large plain surfaces. CARVED SUPPORTING POSTS IRONSHAPED TASSLE POST EYRE 'ROM CHAIR SACK STRINGS ARE STRASS OR WHALEBONE OVOID CHAIR SPLAT Fig. 2 Motifs characteristic of Federal furniture. 13

Residential Spaces PERIOD FURNITURE 18th Century American: Federal THE DINING ROOM IN THE dining room shown above the walls are mist gray, the chimneypiece ochre and white marble. The drapery and upholstery are both cherry silk damask. The Oriental rug is in tones of brown, blue and beige. The furniture is mahogany. An alternate scheme would include: soft gray-green wall, beige silk damask curtains, red damask upholstery. The sconces, clock and picture frames would be gilt. This original Federal period dining room will give you ideas for using the furniture and fabrics shown in the panels at right. Or reproductions of similar pieces are appropriate. JHiiiiii* room fabrics Dining tables, side tables, console FOUR SOFA TABLES JOINED TO FORM A DINING TABLE FOLDING TOP OCCASIONAL TABLE THE BEDROOM rooms bedroom shown above is typical of those found in fine J- houses during the Federal period. Walls, woodwork and chimneypiece are painted moss green. The upholstery is beige damask, except for yellow satin on the desk chair. The rug is in two tones of burgundy with a design of green, pink and white. On the walls are engravings in gilt frames, An alternate color scheme would have walls and woodwork painted peach color. The rug would then be olive green with a design in yellow and pink. The upholstery would be blue, except for red satin on the seat of the desk chair, Other furniture and fabrics suitable for this room are shown at right Bedroom fabrics 14 Another characteristic subtlety is the raised hairline of wood, known as a cock beading, which is used to finish off the edges of drawers. Phyfe used white wood linings for the drawers in his furniture, instead of the pine linings universally employed by other American cabinetmakers of this period. Brass ornaments (probably for the most part imported) are used extensively on Federal pieces. They have brass feet and casters, ring handles, and other types of applied ornament. Toward the end of the period, about 1825, china and glass knobs began to supplant brass rings as drawer pulls. The new United States was in its first throes of nationalism; consequently its emblem, the eagle, appears everywhere — on Four-poster and other types

of bed BED BT PHYFE FRENCH INFLUENCE transparencies in windows, painted on fans, inlaid in mirrors, desks, knife boxes, and brass work. The "Spread Eagle" became a favorite tavern sign. All kinds of historic scenes and patriotic emblems appear as decoration on clocks. And yet, the Classic influence was even stronger than the patriotic. Earthenware and porcelain such as Crown-Derby, Worcester,

Residential Spaces PERIOD FURNITURE 18th Century American: Federal Armchairs, side chairs Sideboards, chests of drawers Dining room accessories MAHOGANY SIDEBORD LION'S HEAD AND RING HANDLES TORCHERE, PAINTED BLACK AND SOLO Dressing tallies, wardrobe, chests CHEST OF DRAWERS I Q IY JOHN SEYMOUR MAHOGANY AND MAPLE VENEER DRESSING TABLE MADE BY PHYFE FOR MIS DAUGHTER ELIZA MAHOGANY BUREAU AND MIRROR BY WILLIAM HOOK Chairs, stools, tables, desks Bedroom accessories CLOCK. GILT BRONZE ROSEWOOD FOOTSTOOL ON A MARBLE BASE MOUNTED WITH ORMOLU and Wedgwood were molded in Classic forms and painted with delicate sepia figures in Classic robes. Silver and Sheffield plate (the latter replacing pewter) also followed Classic forms. Ireland sent Waterford glass. Fabrics most used were damask, brocade, satin, taffeta, haircloth, toile de Jouy, printed cotton, and silk. Woods most used were mahogany, cherry, and maple; and fruit woods in less splendid furniture. Curly maple often replaced the satinwood used in European models. After 1800 rosewood was used for the more costly furniture. The Federal motifs derive almost exclusively from classical sources, The acanthus leaf, the lyre, the saber leg, the lion's mask and paw, the bowknot, rosettes, thunderbolts, trumpets, and drapery swags are all to be found on the list of standard Federal furniture motifs. After the War of 1812, when the Federal era rose to its zenith of popularity, the laurel, cornucopia, and eagle motifs became especially popular (See Fig. 2.) Phyfe's treatment of the acanthus leaf is so typical that many of his pieces depend upon 15

Residential Spaces PERIOD FURNITURE 18th Century American: Federal Fireplace designs CHtMNW WEt WOM KaCQH HtU. BOWQN isiirsifir^Si CHIMNET PIECE DISHED BY CHAR LIS BULFtHCH Drapery treatments for five different types of window FEDESAr,

DRAPERIES SINGLE PANEL CURTAIN HEAVY MATERIAL FOR
 VALANCE. THROWN OVER TRACKS BALANCED WITH OTHER
 WITH NO OTHER PQR DUPLICATES RAISED CENTER VALANCE OF
 DIFFERENT COLOUR TYPICAL HEAVY FRINGES CONTRASTING
 COLOR LINING Secretaries, bookcases, cabinet, chest CHEST ON
 CHEST BR SAMUEL MONTGOMERY ROSEWOOD BOOKCASE MOUNTED)
 WITH ORMOND SATINWOOD SECRETARY, REGENCY TYPE Desks,
 tallies, piano, music desk PIANOFORTE WW MAHOGANY CASE
 MAHOGANY DESK SHERATON JURY Living room accessories . JH
 SCREEN TGFUNM CLOCK HEROIC CLASSIC MIRROR IN
 INFLUENCE MAHOGANY FOOTSTOOL TURNED AND SILT FRAME
 this for their identification. It is simplified into a series of rounded grooves
 and ridges with a raised tapering ridge up the center. The lyre was used to
 fill in the backs of chairs, to decorate the arms of sofas, and (split apart) to
 support mirrors on dressing tables. Two crossed lyres are used as support
 for a pedestal table. COGS FOOT LIONS FOOT £*j LION S MASK AND
 RING CHAIR LESS LIT MASS BRASS EAGLE FIAL BRASS
 HANDLE Fig. 2 (Continued) 16

Residential Spaces PERIOD FURNITURE 18th Century English: Georgian
 THE EXTERIOR Typical of the better country houses in the second half of
 the 18th Century, is this design from Abraham Swan's British Architect, one
 of the many handbooks of builders' designs, which at this period carried
 news of architectural fashions from England to America. At right are close-
 up details of the Georgian background Moldings and trim CORNER OF A
 DOORHEAD WITH PULVINATED FREEZE BELOW Interior doorway
 MOLDINGS AROUND NICHE cornice with acanthus leaf decoration and
 modillions CARVED LINE DECORATION COMBINES CLASSIC AND
 ORIENTAL MOTIFS THE LIVING ROOM The pine-paneled walls in this
 characteristic Georgian living room are left unstained. The silk curtains are
 richly embroidered in many colors on a yellow ground which echoes the
 gilt frames used for pictures and mirrors. The crimson upholstery of the
 mahogany furniture is given added quality by the olive green carpet. An
 alternative color scheme would be to have the walls painted dark gray-green
 with carving picked out in gold. The wall-to-wall carpet would be taupe, the
 upholstery of the wing chairs yellow Italian damask. In both color schemes
 needlepoint and natural leather would be used for upholstering other chairs

in the room Living-room fabrics Armchairs, sofas, settees
 TRIfI.rCHAIR.aACK SETTEE, GOTHIC FRET WATS " CHA'* BAI"K
 JETTEEl SWEPT WhORI TOR RAH. GEORGIAN TYPE CHIPPENDALE
 ARMCHAIR KWM CHAIR RACK SEITH CHIPPENDALE WITH
 CANTED ARMS WINS CHAIB MAHOGANY CHIPPENDALE SETTEE
 MAHOCANv CHIPPENDALE SOFA Chippendale was a dominating
 factor in the history of Georgian furniture design and his name serves as a
 convenient tag for the period centering in the reign of the second of the
 three Georges who provide the period title. Yet this English cabinetmaker
 achieved eminence not so much by his own work as by that of his copyists,
 They all used the designs in *The Gentleman and Cabinet-Makers' Director*,
 published by Chippendale in 1754. To fill this book Chippendale
 commandeered all the ideas he could lay his hands on and then embroidered
 them with his own fancy, adapted them to his own forms. He plundered the
 design manuals of China and the French rococo, of the ancient Gothic
 masters, and of his immediate predecessors in the English furniture trade.
 lii'iiDJiiln profiles ('muniiin l.v|jes ol' k'S mid Hurt IKtlltU BRACKET
 FOGT Fig. 3 Motifs characteristic of Georgian furniture. From the
 craftsmen of the early eighteenth century Chippendale borrowed such tested
 forms as the cabriole leg, the claw-and-ball foot, and the typical acanthus
 leaf ornament. But to each of them he added a grace and charm of which
 the earlier furniture makers had never been capable. Thomas Chippendale
 was a typical product of that brilliant English society which flourished
 during the mid-eighteenth century. He 17

Residential Spaces PERIOD FURNITURE 18th Century English: Georgian
 THE DINING ROOM Here the walls are pine-paneled, the wood being left
 its natural honey color. The consoles are also of pine. But brilliant against
 this pale background are the red damask curtains, and the mahogany
 furniture -with its red and yellow striped silk upholstery. Alternatively, the
 walls might be painted light blue as a background for yellow brocade
 curtains. The mahogany table and chairs stand on an Oriental rug which
 repeals colors found in the needlepoint upholstery. In the panels at right is
 furniture suitable for a room of this style Dining-room fabrics Armchairs,
 side chairs ELABORATED CHINESE INFLUENCE IN SHELL RACK
 DECORATION OF IACK SPLAT5 AND CABRIOLS LEGS AMERICAN
 CHIPPENDALE SIMPLIFIED CHIPPENDALE WITH CHINESE

ORNAMENT THE BEDROOM Bedroom fabrics Characteristic of the Georgian period are the richly embroidered Chinese silk draperies and the delicately fretted four-poster bed in this room. The dominant tone is yellow, against which is posed green upholstery, with a gun-metal carpet for base, putty walls for background. Alternatively the walls could be pale green, the carpet brown, the upholstery blue-green and yellow, the ceiling pale apricot. In the panels at right are other pieces suitable for a room of this type. Modern reproductions of such authentic pieces are available in good furniture stores

Four-poster and canopy beds

CHIPPENDALE IN ROCOCO MANNER COUCH BED DESIGNED FOR AN ALCOVE was a contemporary of Josiah Wedgwood, the potter, and of Edmund Burke, the orator, Boswell and Johnson, Benjamin Franklin, Garrick, Gibbon, and Goldsmith, all added their wit and intelligence to the creation of a sturdy culture, Thomas Chippendale served their changing taste and their fashionable whims. In his later years he was engaged in making furniture of classic, elegant simplicity for the brothers Adam. His earlier work to his own designs, his love of gilt and gaudy color, his fascination with the exotic - all typical of the age in which he lived - suggest that he might have made a brilliant stage designer, Chippendale is the first personality in the history of furniture style. This was due less to his fine craftsmanship than to his ability as a publicist. He was the first cabinetmaker to publish a book of furniture designs, The influence of his Director was particularly strong in Philadelphia, but the American cabinetmakers usually simplified his exuberant ornament to suit their clients' taste and 18

Residential Spaces

PERIOD FURNITURE 18th Century English: Georgian Dining tables, side tables, console Pedestals, wine coolers Dining-room accessories

FOLDING GATELEG TABLE CLASSIC INFLUENCE IN LATER GEORGIAN SERVING TABLE SIMPLE GATELEG TABLE OF EARLY PERIOD COMMODE WITH FOLDING TOP WHICH OPENS OUT TO FORM A TABLE MAHOGANY METALLINE D WINE COOLERS WITH BRASS ORNAMENT "CHINESE" FRETTE PEDESTALS BY EDWARDS 1 DARLEY CARVED AND GILT MIRROR CHANDELIER BY CHIPPENDALE IN THE '80s TASTE CRYSTAL CHANDELIER Chests of drawers, secretary, desk

CHIPPENDALE KNEE-HOLE CHESTS OF DRAWERS Dressing tables, night stands, stools

FOLDING JOP DRESSING TABLE STILL IN USE

Residential Spaces PERIOD FURNITURE 18th Century English: Late Georgian THE EXTERIOR HThe exterior of a later Georgian house, such as the one shown above, would have been finished in cream-painted stucco with stone trim. The Classic detail was in carved stone or molded stucco. At right are details of the architectural background at this period. Wall paneling and painted decoration FIREPLACE WALL IN THE LIBRARY OF SYON HOUSE. ISLEWORTH, ENGLAND DESIGNED BY THE BROTHERS ADAM FOR THE DUKE OF NORTHUMBERLAND THE LIVING ROOM /~im:EN brocade curtains, bound with gold, and green brocade upholstery on the sofa and adjacent chairs stand out brilliantly against the French white of these walls. A damask in tones of coffee and gold is used for the other chairs, a red moire for the other sofa. All these colors are repeated in the rug. The dark brown red of polished mahogany appears in the doors and furniture. Some of the smaller pieces are inlaid with satinwood. Alternatively the walls might be pale pink with white moldings. Upholstery would be blue green except for the chairs by the fire in lemon yellow brocade and the sofa in gold satin. Living-room fabrics mmmM SILK AND LINEN DAMASK BROCADED SATIN SILK AND VELVET STRIPE Armchairs, love-seats, sofas Chippendale went for inspiration to Chinese and Gothic decoration.'The great designers of the later Georgian period - the brothers Adam, George Hepplewhite, and Thomas Sheraton — were entranced by the recently discovered Classic glories of Pompeii and Herculaneum, and by the slim prettiness in vogue at the French court. The motifs most characteristic of the later Georgian period (see Fig. 4) are all of Classic origin: acanthus leaf and honeysuckle, ram's head, winged griffin and lion, laurel, and garland. Characteristic of this period is the perfect coordination between architects, painters, and furniture designers. The four Adam brothers - John, Robert, James, and William, who trademarked themselves the Adelphi (Greek for brothers) - were Scots by Furniture hardware ©I Chair and table legs CARVED AND TURNED DECORATION TYPICAL OF THE LATER GEORGIAN CABINET-MAKERS Matter capitals Fig. 4 Motifs characteristic of the later Georgian period. birth, architects by profession. They did not consider their job at an end when they had designed the shell of a house. Every detail of furnishing, decoration, and lighting was especially designed by the Adams to give a rounded effect. Nothing was too small or unimportant to deserve their attention. The best craftsmen would then be employed to carry out their

designs. Chippendale and Hepplewhite, perhaps Sheraton also, made furniture for the Adams. All these designers followed Chippen- 21

Residential Spaces PERIOD FURNITURE 18th Century English: Late Georgian THE DINING ROOM HPhese pale blue-green walla are relieved by grisaille paintings in delicate Classic taste, Gold appears in the leather chair seats, in the mirror above the consoles and in the binding of the white curtains. Green and beige enliven the carpet and painted ceiling design. Alternatively the wall paintings might be brighter and more varied in color, including Naples yellow, mauve and green. Curtains and chair seats would be cherry, the ceiling painting cinnamon brown and white. Dining-room fabrics ^6161 » :0'?Q? Dining tables, consoles HirttEWHM CONSOLS COMWttO TO FORM A DINING TABIE ADAM CONSOLS DECORATED WIIH RSEDINS TWO AQAM CONSOLES CQMBINID TO FORM A DININS TAtIE EXTENDIBLE DINING TABLE BY ADAM THE BEDROOM Dale colors are dominant here, The sofa-, painted oyster white, is upholstered in apple green satin. The mahogany bed is covered in white taffeta trimmed with apple green, and the armchair upholstery is cinnamon and gold-striped damask. Curtains are white silk, gold-trimmed. Alternatively the color scheme might be based on gold and white with blue green silk on the bed and yellow satin upholstery on the armchair for contrast. In the panels to the right are a number of authentic pieces which might be used in a Georgian bedroom such as this. Bedroom fabrics * (f A P SAIIN BROCADE T i ! i t 11 !! IP j } j T J | SATIN MMASK Four-poster beds and canopies SOFA tEO WITH CUPOLA IY SHERATON dale's lead by publishing design handbooks for the use of other less experienced and less imaginative craftsmen in this country and in the English provinces outside London. Here is seen the changing fashion: lowboys are being supplanted by dressing tables, highboys by wardrobes. Color and inlay become more popular than carving, with Sheraton as the champion of inlay against painting. Hepplewhite's work is usually characterized by his affection for curves, Sheraton's by a preference for straight lines. This was probably because Hepplewhite was more strongly influenced than Sheraton by contemporary French work, which was enlivened by a profusion of delicate curves. Of particular interest in Sheraton's work are his designs for ingenious folding and multipurpose furniture such as folding beds, combined bookcases and washstand, and

couches that folded up to become tables. These were designed for use in those bedrooms which were now doubling as parlors during the day. This later Georgian period has often been labeled the Age of Satinwood. All the designers eagerly exploited the possibilities of ve-

Residential Spaces PERIOD FURNITURE 18th Century English: Late Georgian Commodes, sideboards, cupboard IHniiv'-room accessories BRONZE ADAM SCONCES INLAID WITH WEDGWOOD CHINA CLAQUES Chests ol' drawer's, wardrobes, commodes ftr~-':—
HEPPLEWHITE INLAID MAHOGANY CHEST OF DRAWERS ADAM COMMOOI OF INLAID SATINWOOP WITH PANELING IN "ZUCCHI" STYLE INLAID MAHOGANY SWELLFRONT CHEST OF DRAWERS It HEPPLtWHITE SHERATON CABINET OF iATOWOOO WITH TULIPWOOD BASE INLAID SERPENTINEFRONT COMMUNE BY 1HERATON adam wardrobe with orawers below Dressing tables, secretaries, candlestands Bedroom accessories TAMBOUR SECRETARY BT SHERATON DRESSING TABLE R* SHERATON.ITTO ADAM MIRRORS WITH SESSO ORNAMENT MAHOGANY MANTEL CLOCK neering and inlay with woods such as satin- wood and amboyna, ebony, sycamore, holly, kingwood, and lime, Ivory and brass inlay were often used to mark key plates. Some of these motifs (the acanthus leaf, for example) had been in use by English designers for more than half a century, But now reintroduced from Italy by means of measured drawings, they take on a fresh elegance. Italian painters were brought in - Pergolesi, Zucchi, and Cipriani - to provide the background of decoration. Angelica Kauf- mann, a Swiss, filled their wreathed panels with neo-Classic figures. Yet the solid tradition of English craftsmanship remained intact beneath all these changing fashions, The basic proportions remain almost inviolate, Hepplewhite attempted (in his own words) "to unite elegance with utility, and to blend the useful with the agreeable." 23

Residential Spaces PERIOD FURNITURE 18th Century English: Late Georgian Fireplace designs Gkd ADAM MANTEL OF SIATUARY MAULE AND ANTIQUE SHEEN IEVWLY HALL. ENGLAND, IW A3AM MANTEL OF WHirE, SIENNA AND BROCATELIE MARBLE Interior doorways DESIGNED ir PESGOLESI DESIGNED Br THE 1ROTHEKS ADAM Drapery treatments for Late Georgian windows

ELABORATE CORNICE OF METAL OR OF CARVED, GILT WOOD
 \M7^k,WjljLjKLX1 OUTSIDE THESE TALL WINDOWS WERE
 SLENDER IRON BALCONIES Secretaries, bookcases, desks, cabinets
 Occasional tables, grandfather clocks MAHOGANY L«RARY CABINET
 IT SMEJIATON SHEMTON OCCASIONAL TABLES |N MAHOGANY
 AND SATINWOOO Living-room accessories GILT WALL SCONCE BY
 HEMIWHITE, \W DerorolJte panels and roundels THE WINGED SMHW
 K POfIJLA* ADAM MOTIF PWSTW CEILING ENRICHMENTS
 Uorclere and friezes URN AND SWAG GUUOCHE CUr LION WITH
 FOLIATED TAIL AND OX SKULL Fig. 4 (Continued) 24

Residential Spaces PERIOD FURNITURE Late 18th-Early 19th Century
 French: Directoire and Empire THE EXTERIOR Typical Directoire wall
 treatments rpHE typical Directoire chateau shows French Renaissance
 tradi- J- tion crossed with the newer Classic vogue. The center panel of this
 facade is of stone, the remainder in two shades of painted stucco, perhaps in
 such gay colors as salmon, tan and blue. VERY SIMM.! PANELING
 ELABORATELY PAINTED WITH MULTICOLOR DECORATION OF
 CLASSIC AND EGYPTIAN DERIVATION PLAIN COLOR DRAPERY
 PINNED UP WITH METAl ROSETTES PILASTERS AND CORNICE
 GILT THE LIVING ROOM A CHAHACKRIST1CALLY pale range of
 colors keeps this room in period. The walls are a pinkish gray, the doors
 gray and gold. The curtains are oyster white bound in gray and the rug
 predominantly white except for green and gold in the center. Green recurs
 in the upholstery of the armchair, side chairs and sofa, and gold (satin) in
 the sofa and meridienne by the fireplace. For added color the fireside pieces
 might be upholstered in red satin, the other furniture in gold and blue
 striped satin, In panels at right are other pieces suitable for such a room.
 Living-room fabrics Settees, meridienne, sofas &S1 SIMPLE
 DIRECTOIRE SETTEE UPHOLSTERED IN STRIPED SATIH GOLD
 AND WHITE PAINTED SOFA WITH GOLD SATIN UPHOLSTERY The
 Directoire was France's recovery period after the shock of a six-year
 revolution. The Directoire, established in 1795, lasted only a brief four
 years; but this was long enough for the designers to sketch in the outlines of
 a new style. Those outlines were to be filled in later as Directoire merged
 into Empire; these are but two stages in a single style. With the rise of
 Napoleon to absolute power, the delicate style of the Directoire was taken

over and developed "for the good of the State." It was to be made into a French national style thoroughly imbued with the political principles which were to guide the new state. Imperial Rome was found to provide the dignity and impressiveness required in the Frieze* mid limilernx SWANS AND IOWL BRASS KEY PLATES Sot'ii cutis Fig. 5 Motifs characteristic of Directoire and Empire. prototype, so all the Imperial symbols were converted to use. The symmetrical shapes of heavy proportion were taken over unchanged, copied in wood instead of being reproduced in stone or bronze. Most pieces displayed large surfaces of highly polished wood, usually mahogany. They were not, as a rule, decorated by molding or paneling, or even by carving. Ornamentation was almost always applied or inlaid. Most typically it took the form of gilded bas reliefs tacked to the smooth wood sur- 25

Residential Spaces PERIOD FURNITURE Late 18th-Early 19th Century French: Directoire and Empire THE DINING ROOM [[-!- "The rich brown of polished mahogany in this table is surrounded by chairs painted gold and white, upholstered in blue satin. The walls are painted oyster white picked out with yellow moldings. Above the doors are white Classic figure paintings with a blue background which is echoed in the blue taffeta curtains. Alternatively the walls might be painted green with the cornice picked out in white and gold. The chairs would then be upholstered in red. Other pieces suitable for a room of this type are shown in the panels at right. Dining-room fabrics Side chairs, armchair ARMCHAIR WITH LEATHER SEAT DESIGN INSPIRED I* FROM THE PALACE ENGLISH PRECEDENT OF FONTAINEBLEAU DESIGNED FOR NAPOLEON UPHOLSTERED WITH IEAUVAIS TAP3STRY A GROUP OF MAHOGANY CHAIRS DESIGNED BY JACOB MOTHERS MANY WERE PAINTED WHITE. UPHOLSTERY WAS SATIN OR TAPESTRY THE REDROOM **«£*'" * IS*** T)ink walls decorated in white and gold provide a good background for this mahogany and rosewood furniture relieved with brass mounts, Fabrics are gayly colored here: blue taffeta for curtains and bed canopy, striped yellow and red satin for the chairs, and yellow satin for the two stools (which have white-painted frames). An alternative color scheme would have dark beige walls, green taffeta for the curtains and bed canopy, Most of the furniture would be painted white and gold. At right are other pieces and fabrics suitable for this type of room.

Bedroom fabrics v a o © o c» o © n O 0 PRINTED SILK).<\$.<£ < SILK DAMASK Beds, chaises longues, meridiennes MAHOGANY COUCH UPHOLSTERED IN YELLOW SATIN MI-RIDIENNE WITH SOLO AHD WHITE FRAME, GREEN UPHOLSTERY MAHOGANY BEDS WITH UPHOLSTERED ENDS 26 faces. Painted decoration was more commonly used on walls and ceilings than for furniture. The general color scheme is rich, dark, and somewhat heavy. Rich deep mahogany, French polished and often stained red, was the favorite material, Rosewood and ebony were also in favor, Where other woods were used, their nature was concealed by staining to imitate the more popular species. Round tables were popular, They usually stood on a pedestal or tripod vase, The top was commonly of porphyry or marble, Beds developed into Classic ceremonial couches with scrolled ends, The popular craze for all things Roman extended to include women's dresses and Lucullan banquets. In the early (Directoire) part of the period fabrics were quite delicately colored, the decorative motifs still possessed some Grecian delicacy of form, and much of the furniture was painted and gilt. Later, under Napoleon's fist, fabrics were usually in deep primary colors, the motifs of Imperial Roman heaviness, the furniture of dark red polished mahogany. From each of his campaigns he brought home some new decorative motif which he would turn over to his craftsmen for use in the net batch of furniture made to his order.

Residential Spaces PERIOD FURNITURE Late 18th-Early 19th Century French: Dlireetolre and Empire Pedestals £=J* THESE STANDS, OF MAHOSANY AND BRON2E, WITH BRASS MOUNTS, WERE USED FOR DISPLAYING PLANTS AND STATUARY Wall tables, sideboards, consoles Dining tables PAINTED BLACK TOP AND 6L0BES, EGYPTIAN INFLUENCE MAH09ANV CONSOLES SUPPORTED BY BRONZt CARYATIDS WHICH ARE REFLECTED IN THE MIRROR PANEL AT BACK CONSOLE WITH LOWER SHELF OF MARBLE. TOP SHELF INLAID WITH BRASS MAHOGANY SIDE TABLES SATINWOOD TABLE SUPPORTED BY SPHINXES Cabinets, chests of drawers, secretaries, desk MAHOGANY CHEST OF DRAWERS WITH f RONZC MOUNTS AND GALLERY ffiUITWOOD CHEST OF DRAWERS WITH BRASS MOUNTS Dressing tables, stools, night tables, mirrors DRESSING TABLE MIRROR WITH SPHINX HEAD SUPPORTS NIGHT TABLE WITH BftASS BUSTS, FEET AND KEY PLATE £&>

DRESSING TABLE WITH BRONZE MOUNTS FOR THE LEGS
 CANDLESTAND BEDSIDE ADJUSTABLE PIER GLASS CHAIR IN
 MAHOGANY STAND BLACK PAINTED TABLE WITH GILT LINING
 NIGHT TABLES WITH TAMBOUR DOORS UPHOLSTERED PAINTED
 CHAIR DRESSING STOOLS IN PAINTED MAHOGANY Bedroom
 accessories MARBLE LAMP BRASS READING LAMP BRONZE
 HEAD WITH TOLE SHADE ON MARBLE STAND BRONZE SCONCE
 FOOTSTOOLS WITH WHITE-PAINTED FRAMES UPHOLSTERED IN
 STRIPED SATIN AND TAPESTRY The Egyptian campaign yielded an
 impressive collection of sphinxes, pyramids, obelisks, and lotus leaf
 capitals. From Italy came all the paraphernalia of Imperial Roman
 decoration, acanthus leaves, laurel wreaths, torches, winged victories,
 cornucopias, and the rest, including the famous wreath of bees Napoleon is
 usually accused of having appropriated from the arms of an old Italian
 family, the Barberini. The early Empire pieces (Directoire) are simplified
 versions of the styles current under Louis XVI. These pieces have grace,
 simplicity, and charm. The hampering restrictions on foreign trade led to the
 use of native fruitwoods instead of mahogany. 27

Residential Spaces PERIOD FURNITURE Late 18th—Early 19th Century
 French: Directoire and Empire French designs MARBLEIZEO
 SUMOUND SLIDING METAL SET IN FIREPLACE OPENING 6'0" x 6'0"
 6'0" x 6'0" METAL APPLIED. ORNAMENTAL RELIEVES THIS SIMPLE
 DESIGN Interior Doorways THE FRIEZE COMBINES CLASSIC AND
 EGYPTIAN MOTIFS. PAINTED DECORATION IN GREEN. SOLO AND
 MIS! paneled door with paintings of classic figures
 above decoration in gold white. blue and yellow
 for dining room windows simple light blue
 curtains edged with yellow binding tape yellow
 tieback green overdraft edged with solo and white
 embroidered tape 11' x 11' « 5' x white silk curtains
 trimmed with gold braid simple wooden cornice
 board Armchairs, stools SS mahogany desk chair white and
 gold chair from Hotel des Monnaies. Paris mahogany
 armchair painted green and gold white-painted
 mahogany armchair; by Jacob Brothers white-painted
 frame mahogany outer frame white-painted frame V

Residential Spaces PERIOD FURNITURE 17th and 18th Century

Residential Spaces PERIOD FURNITURE 17th and 18th Century

Residential Spaces PERIOD FURNITURE Late 18th-Early 19th Century

American; Federal (Duncan Phyfe) Duncan FWnc Stot. Oak. ffUCPLP
HOW.R.SHOE. SHAPED iEAT. 5>idc Chair. With Lyr.c Back.. Duncan
Phytic Arm Chair.. Two Back 5cttea, With Lyre. Ends. These. May &e
Made. With Twcec. Back.s £ Are. SoMtTints Caned . They JomCtimes
Have. Animal. FtfR 72iougA AwxmV C.ONSOL,C TaBLE.. VtWE-t-RCP
APRON. A—su'—J SEWING Ta6LC. THt 5iL*. 5a& Is Fastened lb The
Loiver. POTVCR, Which Has No Bottom. The. ■Jent-CIRCUAR.
ENPiARi.5CAE5 £ Th*. LIDS Have. lHviDifti.1. H-N&ES. Ll&RARY
TA&LE. WITH LYfU. E.NPS. . no" Lyre Base. Carp Taoi-e. PINI*. TaBLA.
ThRSEMAY BeM»E.In Tw>. THREE 0*. Five Sccrio«6. Wide Boards May
Also Be Ached Between Z»ch Section To UM6rHtt< The.Tables. Small
Venl(.ecd 5"0EflOApri3. Tnt Principle Pieces t/~ ru«nTu« CUoc By FHvrr.
Were Chairs. Tam-e&. iofAs 45ettees. Ml Also Made Jidcboardj. Bedsteads.
Mirrors. Washstahos. W<mn<, Desks. Ctc. SohC. j/"His Favoetc Morifs
Were. The Lyre.. The acanthus Leaf. Turned {ReejiedLe&s. Turnej
pcpestals supportep oh curvcp legs. f> Animal. FttT g/C*Ass. Tiit Principal.
WoOP Used Was Mahogany Often Vehecre.0. Duncan Phyfe style 31

Residential Spaces PERIOD FURNITURE 16th Century English: Early Jacobean FIG. 10 500~ 32 Early Jacobean period

Residential Spaces PERIOD FURNITURE 17th Century English: Jacobean I—I7'—«j f/6. /Z Jacobean 5idl Chair. Bannister. Back. Chair^ Carolan Chair. "J. F/G, /"t Wti.su DteisE*. FIG. 15 CromWeluan 5e.tte.e- FIG. 17 Mirror. Trame. Charles n 3e.tte£ Jacobean period 33

Residential Spaces PERIOD FURNITURE 17th Century English: William & Mary rte.r LACOOtFULD CAMNtT. Tall C loc 34 William and Mary period

Residential Spaces PERIOD FURNITURE 18th Century English: Queen Anne 5'DC Ch/mr. ThC f-haPCR. Wood (or Qwcln Ahm. fuRNITURC Is WaunuT. flAHCTjANT WA5 SOMCTltHSUJCO. Queen Anne period 35

Residential Spaces PERIOD FURNITURE 18th Century English: Georgian (Chippendale) STAKING Vj \$for t> OS jNilPt >IZ.f BtrWUH KfllLi 55'Wik j 78" I 50% 78' £ J6"<7fi'. Kail.s A«t Z'Thick £ Bo, Jptinoi Sl'til'i 51"j8?.'J. fco'itu'. Ini Upnonr«(.o bo» Scuxcij ARE tAMfTEL' fo ilHK BtLOW fill IWl. |. HAY &L &W4T1f Willi CU1- 0v' CoRMtfr. To fir ^eouno Ffor:. F^tukor' F/G.tf Chippendale style 36

Residential Spaces PERIOD FURNITURE 18th Century English: Late Georgian (Hepplewhite) f IATHK KS. Tut Bur tHAirj jO fl/*rHi> Tv«. Ant Tmr li- lowt*. Tn THL CtNTtK y^jrf* Tkont fc Back To fWt. Them Hott ConrottTArM-c. fClfUAH WfHvi °>Hir. i.D Rack Arm Chaw. 5hhi.p Back With CVau Dack. With Straight Tapj:r Lccs. Mahtha Washington Gai*. HIPPIIWHITI. SIRIBOARD Willi &ORM.R VtHIiPJO PRAWtP-i J CbOB.5 Pining Ta&ll In Thro. StCTioMb. These Sections Whin NotInUse M»T df. R,Acto Against The Wall. ■ — Inlay* TAMBOUR StCRMARV. Hepplewhite style 37

Residential Spaces PERIOD FURNITURE 18th Century English: Late Georgian (Sheraton) -14 f.AH o/"TOf> 5^55 3HEATH FIG./ Ann chair.- raaNT Uos £ Arm Supcoftrs /M!C RtUUD. fig. a .'not Chair. Wim Urn Motif In Pack. 5HER.ATOH ChLM j/'PbawI.RJ. 18- Urt 5»c.i!*ToN LiomSior. CA&ftr b Ch*jh of Tut. TVpx. That 10' To 13 BtCAMl VLH.V

Popular. In DeLP CUriLOt AMtRICA. I.AU 5HLf.ArOH CtlM jA f>A*j jo
With rliRRo^ J 'a 'i Plan o/'Ca&hz.j —1 t4-i ^ Chwa CaEiiii.1. rwjKHitu
SlPL&OAUd Sheraton style 38

Residential Spaces PERIOD FURNITURE 18th Century English: Late
Georgian (Brothers Adam) Tut Lip o/" This Uhn 1* F«rr.tu.0 To A Roo 6v
Means 3/ Which It Mav .De. Htm Sus- PENPED WHEN The. Contents Are
To De. REMOVtD. These fiiiorALS h*p shelves £ Drawers To Hold
Various Kiiids g/"PiNiN6 Room Aauttmo ACRELN -68- .POCK CASE.
r/G.s ■57"- oi'ril"»" "» The Srother.5 Awm 5t- gan The. Classical-Ira In
FURNITURt g/"THE. .Eighteenth Century. The Links asViwed From
Th^Front VAre itKAioHr. /4 s Mat fit. .Seen OrJ This Plate - The Plan
Views g/"TAtu.* \$ Commodes >3how .^aud fRXTS 4 0vAl- ^NTS. The
Variety ^Ornament Ltetp Was Very Great. Ir Was Rather Architectural. In
Character. Having &U.N iNFLUErtCtD JjY The. ^Tuoiti g/"Roman Runs
That Row.r.t APArI rVot. Window 3eat F/G.6 r/e.r Com mope., Brothers
Adam 39

Residential Spaces PERIOD FURNITURE 18th Century English: Late
Georgian (Brothers Adam) All The Furniture. CUsignid By Tut BkothlKs
Adam Wa* Built 5y o nuns. HtppuswHirc CHIPMNOAUt ^
OrneR»4X£CUKO Tha Commis&iows. floiT ^Th-CSe. Makers With Ihc
(tesiftLt, .Exception g/"CKIPP«.NDAL£. WtR«. INFLUENCED By THt
Work o/* Them. Aietim*. FIG. 9 ■>pc Chair rtLtuu While. The Furwturx Is
Very Formal In Character It Is Auao Very tStAynrut. \$ Tasteful.. No Detail
Was Too SmaluTo Receive Iheir Attention. Besides The ORPINARY
PlfCILS THEY DE5IONED LIOHIINS FIUTURJJ, UPHOLSTERY. \$>
NUMEROUS ACCESSORIES. FIG. 8 CAR.VEP Gilt ^FAINTtO
MtRIWA. 4 Console TA.DL*-. n' oilpth 5/* Seat Inside.. -a1- Arm Chair.
FIG. tO Tut BROTHERS AoAM WtRl. ARCHITECTS > The.
FuHiHIT^KtC They Desichip Was IntendeITor Definite Places In The.
Houses The-" Built. For Thij Reason Some, of The Pieccs Ws.rjl Uwt AS
IS THE CASE 0/ Recti SuchAj Tut. Ta&LE £ MKRpR. .Shown Aoovt, 5o«t
Booncases Were Mape Quite. Lon6. The. PrdWRTioMS HoWKVeOeem In
MoST CASUS To ^aVE LVeM .CxCELLENT. It Was Only Ojscause.
o/APtsiRe. To Have.fvery Detail Perfect In. The. Houses Thiy Jjuil.t That
They Designed THt FuRNiruREyty- Which They Are. Famous Today. . fct-

CAeifJtT The Poors of These. Casinms Wma Occasionally On The fñps.
 ThE'HaHP PAINTED PANEL Was THE. Work. p/"A Tahoos Artist . Tabus.
 /~/C?. 12 B_i Brothers Adam 40

Residential Spaces PERIOD FURNITURE 17th and 18th Century French:
 Louis XIV, XV, and XVI & Louis XIV Arm Chair. XJK XK / xv/. Louis
 XVI Chair. &ACK3 Sometime.* Tapercs Toward Tmc Fuior.- That Is They
 Webe. Wiper At The. Top Than At The. Tijook. Th Thio Case. Thb. Back
 Legs Wtic Square. Bttow The. Seat. Louis XVI Arm Chair. Louis XV Scat.
 L.OUIS XIV Siojl Chair. Lows XIV Furniture In Principle. c/"0E5Ign ■Is
 A Careful Comwnatioh g/"5rRAi&HT Lines \$ Wall Studied curves. Louis
 XV Furniturju Departs From The. Straight Line. Altogether., The.
 CuRVtbNtAR. Llement Is Supreme In This Style. , Tut Louis XVI &tylc
 Sucms More, Ivenly ftuANccD Pumls. Straight UNAS £ SIMPUL
 OUTLINE.! RtTURREO. Louis XVI Scat. Louis XV Chaise. Longue.
 Formed By Com&iNing 1hr.ce. Re.ce.s of Furniture. The. Fabrics Used To
 Ulster Louis XIV Fuwture. Were. TAftSTiao, Damasks. Cloth a/ 6ou>. £
 Satins. InTarestrics finoRAL^PjcH CoujRtDimns Wuit Usual, lows XV
 Pieces WeR£ Upholstered With The. 5AMt rLATE.RlAl.5 AsCMVEN
 ABOVE.. PECORATIVE MCTIFS PIFFERED Somewhat. The Pcwru.
 Element Gave. Way lb The. Highly Pecoraiive. Shejj^ Leaf Moti fs. In
 Lows XVI Piece.5 The. Fabrics Wiub. PtcoRATEji With foKrwk FLwEH
 60UOUEt5 . KIBBOHa.GAR.l^NPl g/"pAINTT FloWEM. CuPIPS. Zfc^
 &3FT CokoR.» PREPORllHATt. In Thij SiYlt.. French styles 41

Residential Spaces PERIOD FURNITURE 16th and 17th Century Spanish
 •J4Vo4o"- Three. Back Settle With Tooled Leather. E»ck £ 5eat Fastened
 With Large Headed Brass Nails Circa.Moo These, Cabinets Originated In
 Spain Durjno The It Th Century. They Were Often Used As A Desk. £ Are
 .easily The Most Important Contribution y"" Spain To Tut Furniture Ww,ld.
 W J II Th Century Chest With Chip Carved Panexs Laid Out In Geometric
 Designs Interupted Channel Grooves On Less. rVuLS. |> Stiles, Jpanish
 Chests Wry Grut- LY En CHAILAtTLC ioME. Am. DESICHEO V/ITH
 Round Lit». ioME Att Fastened To Tusncd Umed FRAMES. OTHERS
 ARC CovEREp WITH ToOL*-0 LEaTHC*. OrCacveo In Gothic Monfs.
 Etc. CuPOoARD g/"THClfcTH CENTURY . Fm-NAI35ANCE. FeRiOO.
 These. Cup&oards Were Used T0 Hold Food .5uch As Bread £ Chee.se' j,

Wine., fi"/Q.4 Front & Lno Views */" Turned Legged Tam.e, With Wrou&ht
Iron draeej ISth Century Spanish Side Chair. Spanish styles 42

Residential Spaces TABLE1 Period Style and Finishes Period style Early
English Tudor Jacobean Charles II Anglo-Dutch William & Mary Queen
Anne Early Georgian Chippendale Late Georgian Adam Hepplewhite
Sheraton Empire Federal Louis XIV, XV, and XVI Spanish Renaissance
Early Colonial Early American Modern French Provincial Victorian
Associated styles Italian Renaissance Spanish Renaissance William & Mary
Larger pieces of Queen Anne Chippendale Early Georgian Louis XVI
Smaller pieces of Jacobean, such as gate-leg table or Windsor chair
Chippendale Early Georgian Louis XVI Smaller pieces of Jacobean, such as
gate-leg table or Windsor chair Chinese Chippendale Louis XVI Duncan
Phyfe Di recto ire All late Georgian styles 1 or 2 pieces of Directoire Italian
Renaissance Early English Louis XIV All Early English styles William &
Mary Queen Anne wing chair Late Georgian Chippendale Queen Anne
Duncan Phyfe French Provincial Swedish Modern Chinese Chippendale
18th-century American Colonial Federal Colonial William & Mary Queen
Anne Walls and ceilings Oak panels Rough plaster with oak trim Parquetry
ceilings Papered Painted (in light tones) Hung with fabrics Paneled Painted
dado Painted Paneled Papered upper section Plain plaster Painted Papered
Large wood panels painted Gesso ceilings Large wood panels painted and
decorated Wallpaper in Chinese motifs Rough plaster painted Ceilings same
or beamed Oak panels Rough plaster with oak Parquetry ceilings Smooth
plaster, light trim Wallpaper, scenic and Chinese designs Paneling Ceiling
plaster Painted solid colors, striped, figured Plain papers Com binations of
above Smooth plaster Wallpaper in scenic or geometric designs Large-
patterned paper Floors Hardwood stained, dark strips and planks on flooring
Stone Tiles Hardwood flooring Parquetry Hardwood flooring Parquetry
Hardwood flooring Parquetry Hardwood flooring Parquetry Hardwood
flooring Tiles Vinyls in tile pattern Hardwood flooring or planks Vinyls in
jaspe pattern Dark hardwood flooring Vinyls in plain or jaspe patterns
Hardwood flooring Parquetry Vinyls in modern pattern Hardwood flooring
Parquetry Hardwood flooring PERIOD FURNITURE Period Styles and
Finishes Floor coverings Oriental and large- patterned domestic rugs Plain
rugs Oriental and large- patterned domestic rugs Plain rugs Plain or small-
patterned rugs or carpets Oriental rugs Plain or small-patterned rugs or

Residential Spaces FURNITURE DIMENSIONS Children's Furniture and
Tables CHILDREN'S FURNITURE ■J=1 •* ftrn tTi'r »"•"»"»" DIAPER
CHANGER POTTY CHAIR FEEDING UNIT H: 36"-42" H: 12"-13' H:
24"-26" W: 32'-42" W: 16'-17" W: 24"-28" D: 21"-24" D:- 14" - T6" D:
24'-25" HIGH CHAIR SIDE CHAIR H: 36"-40" H: 24" -26" W: 18"-22"
W: 14"-16' D: 18"-20" D: 15" -17" ARM CHAIR TABLE for 2 H: 23"-25"
H: 20"-22" W: 14" -18" W: 24"-30" D: 16" -17" D: 24"-25" TABLE for 4
H: 20" • 22" W: 36" • 42" D: 36" - 42" TABLES 1 1 - END/SIDE H: 20" W:
36" D: 20' END/SIDE H: 18" W: 20' D: 20' END/SIDE H: 18' Dlam. 18"
LOW/COFFEE H: 18: W: 36' D: 24' LOW/COFFEE H: 18" W: 36" 0: 36"
LOW/COFFEE H: 18" Dlam. 36" STOOLS H: 18" W: 18' D: 18" STOOLS
H: 38 W: 18 D: 20 ENTRY/HALLWAY H: 33' W: 48" D: 20' 44

Residential Spaces FURNITURE DIMENSIONS Bed/Mattress Types and Sizes Bassinet Infant crib Junior crib 14" . JSfr- -& Youth bed Twin bed * 10 r> 3 ? lilll ;;;,;;,;ii;!:x:;;,;;,;;,; mil 1111 Twin extra-long •» Queen eo" Queen extra-long <60" King California king 1 11 HI ■1 :•:«:■:•:■:■: 1 Si 1

W8M ill 11 • <0 lltilllil ::S::i:>>S::^:':::-::;::;::i::?:i:i min mill msxm mm®
 HB.1 Figure 1 provides the designer with an array of typical bed and mattress sizes with which rooms can be planned. Tables 1 and 2, however, suggest that within the bedding and mattress industries there exists a wide range of sizes from which to select. Many manufacturers use bed/mattress terminology that reflects different dimensional standards than that of other manufacturers. Ultimately, the designer, in consultation with the client, must verify exact measurements. Be sure to take your clients to see and test the bed or mattress selected. After all, they are the ones who will have to sleep on it. 7*" ■111 ■ III! ■111 liiii

TABLE 1 Juvenile, Youth, and Adult Mattress Types and Sizes

Mattress type	Bassinet	Portable crib	Junior crib	Youth bed	Bunk bed	Dorm bed	Hospital bed	Narrow twin	Twin bed	Full-size or double bed	Queen-size bed	King-size bed	Extra-long double	Super twin
Width (in)	Min 17	22	24	33	30	32	36	36	39	54	60	76	54	45
Max	23	26	32	44	36	33	36	36	36	39	54	60	78	54
Length (in)	Min 18	19	20	22	22	22	22	22	22	22	22	22	22	22
Max	27	30	36	36	36	36	36	36	36	36	36	36	36	36

Wore.- Many manufacturers also make and sell undersized pillows for cribs and youth beds as well as oversized pillows for the larger beds. 46

Residential Spaces WATERBEOS FURNITURE DIMENSIONS

Watsrbeds, Sofa Beds/Convertible Sofas, and Wall Beds

Twin Single 69" Double Queen King 5 A" Co" 3 ^ iO 78" 732EES££" LJ •:•???:*??,.. ~ ' SOFA BEOS/CONVERTIBLE SOFAS 96-in-diam round Oouble/puil Queen WALL AND SIDE BEOS Side 1 =i 1 34- A Width of Bed 39" 4B" 64" B Width of Clear Door, Opening 79" 79" 79" c 0f/>rt From Sac* Of Closet To Buck oj Doors 13" 13" 13" D Projection of Bed in Use From Back of Closet 43 W 52 W 57" HEIGHT: FLOOR TO TOP OF OPENING 44 1/2" for 39" lied. 53'A" for 48" lied. 59" for 54" Bed. Wall \L rv A B + Width of Bed 39" SI" 60"x80" 76"xB0" Width ajClear DtwrO/icnittn 42" 57" 63" 79" Depth From Bad Of Closet fa Qnek ofDtHirs 19" 19" 19" 19" Projection of Bed In Use From HotlivfChml ao'Stamlttrd 86" Extra Long SO" Stamford 86" Extra Long 86"Queen it" King HEIGHT: FLOOR TO TOP OF OPENING 80" ■ hut .IV mill .5-/" Ikih nj Shindtml 7,V /.<•(«" 86" • Ear Queen. King and lixirii Lang lleds HO" in Length 47

Residential Spaces FURNITURE DIMENSIONS Audio-Visual Equipment
 te-tctf A- - , "> 7,1.-1(3 *«0 Dfl •4- Airi'i-^ir I *i* -14* Irmji, T. U.l»" :in.mi
 '.•y'V.v.v.v.v.'.v.v 19-in television Rear-projection Front-projection
 television television -2. .i. Ill 1111 1111 m 3 i VCR 'A Laser disc player -> 1
 it-it TTtrr '£ + ->+■ <tP \€-lfi 00 1 00' tC-.i tU* a *—2* Tape recorder
 Turntable M + t _i Receiver CD player Storage rack 48

Residential Spaces FURNITURE DIMENSIONS Television Viewing Areas
 The shape of the viewing area is approximately as shown in Fig. 2. Its size
 is always based on the size of the image to be viewed. The human eye
 comprehends detail only within a limited cone angle (about 2V2 minutes of
 arc), and the length of chord subtending this arc, i.e., the image of width,
 varies with its distance from the observer. Thus an object 20 feet away and
 6 feet long appears the same as a similar object 10 feet away and 3 feet
 long. The size of the viewing area is determined by three dimensions: ■ the
 minimum distance (1), which is the distance from the nearest part of the
 image to the eye of the closest viewer ■ the maximum distance (2), which is
 the distance from the furthestmost part of the image to the most distant
 viewer ■ the maximum viewing angle (3), which is the angle between the
 projection axis and the line of sight of a person located as far from this axis
 as can be and still see all image detail in proper brilliance Practical
 minimum and maximum distances are both expressed as multiples of the
 image width (W). They vary both with the medium being used and with the
 type and quality of material being projected, and may be affected also, in
 some degree, by personal preferences. They have not yet been precisely
 determined by scientific methods, and it's doubtful that such data would
 have much practical value anyway. The generally accepted values, resulting
 from numerous studies, are these: Fig. 2 Minimum distance Maximum
 distance Size of M TV tube 17 in 19 in 21 in 23 in 24 in 27 in Film, slides,
 and projected TV 2W 6W to 10W inimum viewing distance 4W 4 ft 1 1 in 5
 ft 1 in 6 ft 4 in 6 ft 6 in 7 ft 5 in 9 ft 8 in TV receivers 4W 12W Maximum
 viewing distance 12W 14 ft 9 in 15 ft 2 in 19 ft 0 in 19 ft 4 in 21 ft 5 in 24 ft
 5 in a/iea 49

Residential Spaces FURNITURE DIMENSIONS I 9QT,mm I 1 3-°" I
 Small bullet I 1200 mm 1 Table lor lour I 1200 mm I Table lor six 1 800
 mm Table lor eight Fig. 3 Typical dining roam furniture. 6'-0- Table ior ten

Chesterfield I 900 mm I T 2-8' 1 9 ' 900 mm Television I 1050 mm I I TM I
 Arachau Coffee table I 750 mm I I 2''- J ■mall 1 1200 mm I I ^ I medium |
 1500 mm i large fiookcaeee SO 1.650 mm I I 2 2" I Occasional chair Fig. 4
 Typical living room lurnlture. J50 A End table I 1 000 mm I >. 4 a- | mm
 V.6- Desk Chair

Residential Spaces FURNITURE DIMENSIONS 20th Century Classic
 Chairs INGRAM HIGH CHAIR DESIGNER: Charles R. Macintosh
 YEAR: 1900 MANUFACTURER: Atelier International DIMENSIONS:
 18Vi"Wx17V2"Dx59WH WASSILY CHAIR DESIGNER: Marcel Breuer
 YEAR: 1925 MANUFACTURER: Knoll International DIMENSIONS:
 30%"W x 29"D X 28WV KUBUS CHAIR DESIGNER: Joseph Hoffman
 YEAR: 1910 DIMENSIONS: 36"W x 30Vi"D x 28V2"H MR. CHAIR
 DESIGNER: Mies Van Der Rohe YEAR: 1927 MANUFACTURER:
 Stendig DIMENSIONS: 21%"W x 32WD x 32WH HAUKOLLER CHAIR
 DESIGNER: Joseph Hoffman YEAR: 1911 DIMENSIONS:
 35/2"Wx.32"Dx37"H LCI SLING CHAIR DESIGNER: Le Corbusier
 YEAR: 1928 MANUFACTURER: Atelier International DIMENSIONS;
 23%"Wx25%"Dx25'/4"H MIDWAY CHAIR DESIGNER: Frank Lloyd
 Wright YEAR: 1914 MANUFACTURER: Atelier International
 DIMENSIONS: 16"Wx13"Dx35"H LC9 LOUNGE CHAIR DESIGNER:
 Le Corbusier YEAR: 1928 MANUFACTURER: Atelier International
 DIMENSIONS: 22"W x 63"D 51

Residential Spaces FURNITURE DIMENSIONS 20th Century Classic
 Chairs CESCO ARMCHAIR DESIGNER: Marcel Breuer YEAR: 1928
 MANUFACTURER: Knoll International DIMENSIONS: 22WW x 2 WD x
 3WH BARCELONA STOOL DESIGNER: Mies Van Der Rohe YEAR:
 1929 MANUFACTURER: Knoll International DIMENSIONS:
 23"Wx22"Dx14'/2"H BRNO ARMCHAIR DESIGNER: Mies Van Der
 Rohe YEAR: 1929 MANUFACTURER: Stendig DIMENSIONS:
 18"Wx23"Dx31'/2"H CHAISE LOUNGE DESIGNER: Mies Van Der Rohe
 YEAR: 1931 MANUFACTURER: Knoll International DIMENSIONS:
 23WWx47V2"Dx37'/2"H LC2 ARMCHAIR DESIGNER: Le Corbusier
 YEAR: 1929 MANUFACTURER: Atelier International DIMENSIONS:
 30"W x 27'/2"D x 26'/2"H ZIG-ZAG CHAIR DESIGNER: Gerrit Rietveld
 YEAR: 1934 MANUFACTURER: Atelier International DIMENSIONS:

14¹/₂"Wx17"Dx29"H BARCELONA CHAIR DESIGNER: Mies Van Der Rohe YEAR: 1929 MANUFACTURER: Knoll International
DIMENSIONS: 30"W x 30"D x 30"H PAIMO CHAIR DESIGNER: AlvarAalto YEAR: 1935 MANUFACTURER: Palazetti DIMENSIONS: 23¹/₂"Wx31¹/₂"Dx25"H 52

Residential Spaces FURNITURE DIMENSIONS 20th Century Classic Chairs BARREL CHAIR DESIGNER: Frank Lloyd Wright YEAR: 1937 MANUFACTURER: Atelier International DIMENSIONS: 21¹/₂"Wx22"Dx32"H MOLDED FIBERGLAS CHAIR DESIGNER: Charles Eames YEAR: 1949 MANUFACTURER: Herman Miller DIMENSIONS: 25"Wx25¹/₂"Dx31"H BUTTERFLY CHAIR DESIGNER: Harday, Boner & Kurchan YEAR: 1938 DIMENSIONS: 28"W x 27¹/₂"D x 35¹/₂"H DIAMOND CHAIR DESIGNER: Harry Bertolia YEAR: 1952 MANUFACTURER: Knoll International DIMENSIONS: 33¹/₂"W x 28"D x 30¹/₂"H MOLDED PLYWOOD CHAIR DESIGNER: Charles Eames YEAR: 1946 MANUFACTURER: Herman Miller DIMENSIONS: 21¹/₂"Wx19¹/₄"Dx29¹/₂"H LOUNGE CHAIR DESIGNER: Charles Eames YEAR: 1956 MANUFACTURER: Herman Miller DIMENSIONS: 32¹/₂"W x 32WD x 33¹/₂"H WOMB CHAIR DESIGNER: Eero Saarinen YEAR: 1948 MANUFACTURER: Knoll International DIMENSIONS: 40"W x 39"D x 35¹/₂"H OTTOMAN DESIGNER: Charles Eames YEAR: 1956 MANUFACTURER: Herman Miller DIMENSIONS: 26¹/₂"Wx21"Dx15"H 53

Residential Spaces FURNITURE DIMENSIONS 20th Century Classic Chairs ALUMINUM GROUP CHAIR DESIGNER: Charles Eames YEAR: 1958 MANUFACTURER: Herman Miller DIMENSIONS: 28¹/₂"Wx24¹/₂"Dx33¹/₂"H LOUNGE CHAIR DESIGNER: Richard Schultz YEAR: 1966 MANUFACTURER: Knoll International DIMENSIONS: 26"W x 28¹/₄"D x 26¹/₂"H SHERRIFF CHAIR DESIGNER: Sergio Rodriguez YEAR: 1958 MANUFACTURER: OCA TUBO CHAIR DESIGNER: John Mascheroni YEAR: 1968 MANUFACTURER: Vecta DIMENSIONS: 32"Wx32"Dx32"H HAND CHAIR DESIGNER: Pedro Freidberg YEAR: 1963 MANUFACTURER: Hand Crafted SAPPER COLLECTION DESIGNER: Richard Sapper YEAR: 1977 MANUFACTURER: Knoll International DIMENSIONS: 28¹/₂"W x

27Va"D x 3814-41 %"H PLATNER CHAIR DESIGNER: Warren Platner
 YEAR: 1966 MANUFACTURER: Knoll International DIMENSIONS:
 361/2"W x 25'2"D x 30'2"H BASIC OPERATIONAL DESIGNER: Niels
 Diffrient YEAR: 1979 MANUFACTURER: Knoll International
 DIMENSIONS: 25V2"W x 21"D x 321/2-36V2"H 54

Residential Spaces JEFFERSON CHAIR DESIGNER: Neils Diffrient
 YEAR: 1986 MANUFACTURER: Sunar/Hauserman DIMENSIONS:
 32%"W x 34"D x 43VS"H FURNITURE DIMENSIONS ZOth Century
 Classic Chairs NOTHING CONTINUES TO HAPPEN CHAIR
 DESIGNER: Horward Meisper YEAR: 1981 MANUFACTURER: Art et
 Industrie DIMENSIONS: 17"Wx16"Dx37"H OTTOMAN DESIGNER:
 Niels Diffrient YEAR: 1986 MANUFACTURER: Sunar/Hauserman
 DIMENSIONS: 25"W x 24"D x WaV LOUNGE CHAIR DESIGNER:
 Michael Graves YEAR: 1982 MANUFACTURER: Sunar/Hauserman
 DIMENSIONS: 32"W x 29"D x 29"H ED ARCHER CHAIR DESIGNER:
 Philippe Starck YEAR: 1987 MANUFACTURER: Driade Italy
 DIMENSIONS: 181/2"W x 21'2"D x 38'2"H QUEENE ANNE CHAIR
 DESIGNER: Robert Venturi YEAR: 1984 MANUFACTURER: Knoll
 International DIMENSIONS: 26V2"W x 23'2"D x 38'2"H STONE
 CHAIR DESIGNER: James Kutasi YEAR: 1988 MANUFACTURER:
 James Kutasi Australia DIMENSIONS: 19%"Wx19%"Dx351/2"H 55

Residential Spaces FURNITURE DIMENSIONS Traditional Bedroom and
 Dining Room Furniture BEDROOM FURNITURE ll,i'',"M'J-.g"ll|MJ -M
 B= " <o ■ Plan Elevation Bureau i-J nlui-4 Narrow widths usually used- PL
 Elevatic Chiffonier i""":i"ii"j,>_j0>""ir. -r i <o ■8 <0 ■o 1 ~1 — ."| 1-
 f-^ 'U\ .A v r Plan Elevation Chest of drawers \$1 ■S--5- —to- i - -4 ■;_ PL
 'apeA swing Elevations Dressing table Elevation Makeup or powder table -"
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 IH- Elevation Sideboard -*f- —S'-O" to C'-t,"- t 'I 1 •« i />/* ^ffeO=i —1_
 n io Buffet ■Trd _.* . O Elevahon V.-. ii t |i 4'-0~ho C'-O' 1 Plar, Dresser
 Elevation Elevation Cupboards Ele. vation I 4 }-2'-0' /o 3'-3"-<\ Plan i X
 El&varioni China cabinets El&^arjons Servers 56

Residential Spaces FURNITURE DIMENSIONS Traditional Sofas, Settees,
 and Benches £/&&<%/Zona ^J^^'■^^ >■>"'■' ■'■'•■'"L-l'"'-.-'^. 'f|'-

Residential Spaces FURNITURE DIMENSIONS Traditional Desks,
Bookcases, and Chests ■ ^ J t ., (---4,, ;■ W^ 3:6—t J*/k i-j-JL-U Z°rfl/u T
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Mens 'tlmb/*//as 2-'s"-J-b' Hymens' ■■ /■'«;"-2-' + (f/ei/ah'on 58 Chests
Umbrella stand

Residential Spaces FURNITURE DIMENSIONS Traditional Tables —S'-
o'fof'-o'- Pjorr JIJe E/et/oJ/on Library table Tea Tea and coffee tables t •<r3'-

O"t0-9'0~- P/at? £nd Serving table {••*.. {&*?. ■■•/ 9 •; ■■./fvare.; /ma// Larva Plans r *— //«-• ' - *fc > " N ' Night table •— 3'Wic -f-o"- Dressing table Plans Occasional table £levarier>. g-Gioa'-o:' Square Plans Fa/ding Card tables Thklrhesi foUed W" E./aual/on ,&a» ^ 53^ Uy ■/»/■<#>-• - - drop* ■ Plans <■£>u//erflt/ £pa/, weu/s Drop/eef ria/js spa ir/eu/s These are made /n auar/elyofj/jks jnopes, a/7a> be/aih, for ino/>if i Drop leaf and butterfly tables Plan -open*— closed—k-ofien- I1 || || t 5V ^Jmall Tilt table \Loro& 1 -vl w • >— -Closed—*- '•..(j8 o/e« ^-".<s'- ■ ► Closed-* /'■Ffiiam 1 _ . *" Loraæ fixe. . Small fixe. . *" 7#es«> ^re mode In a variety ofs/\$es 4 ^sed rat Dlmnf, Uiroryfclc. Plan £/evat/on Draw top tables Candle stand ...;>■ j-;^': tr-W:- to ■.". :: (V . . ' ••?-«'j -3-i»"- /**)// f£ i^W /arpe Gate leg tables Down Up Hutch table ^ /■rf-'^' ■/-/a- Console table Stands 60

The size of living rooms and the furniture arrangements contained within such spaces vary dramatically, depending on the size of the dwelling, the economic status and lifestyle of the user, and the relationship of the room to other areas of the dwelling. With regard to the luxury end of the scale, there are few limitations and no attempt has been made to identify the endless planning options possible. There are, however, minimum requirements and basic planning considerations that are applicable whatever the size of the space. Minimum Requirements A living room for a three- or four-bedroom dwelling unit requires more space for its occupants than one for a one- or two-bedroom dwelling unit. Luxury units will necessarily need more space to accommodate more furnishings. In any case, the minimum living room with no dining facilities should be approximately 180 ft² but preferably around 200 ft². Figures 1 and 2 show two living rooms With typical furniture groupings (no dining facilities). Figure 3 shows a living room with one end used for dining. This area often is arranged in an "L" shape to achieve greater definition or privacy from the living activities. Dwelling units with three or more bedrooms should have separate dining rooms or clearly defined dining areas. The minimum width of a living room should be 11-12 ft. This is extremely tight, however, and if at all possible the width should be at least 14 ft. Planning Considerations Planning considerations should include adequate floor and wall space for furniture groupings, separation of trafficways from centers of activity, and ease of access to furniture and windows. Circulation within the living room should be as

direct as possible and yet not interfere with furniture placement. Ideally, there should be no through traffic. If such traffic is necessary, it should be at one end, with the remaining portion of the room a "dead-end" space. During social activities, people tend to gather or congregate in relatively small groups. Desirable conversation distance is also relatively small, approximately 10 ft in diameter. When the living room is combined with the dining area, the dining area should be offset into an alcove or be clearly identified as an entity in itself.

Fig. 1 Typical furniture arrangement for a one- or two-bedroom apartment (12.5 ft x 16 ft, 200 ft²).

Fig. 2 Typical furniture arrangement for a three-bedroom apartment (12.5 ft x 20 ft, 250 ft²).

Fig. 3 Another typical furniture arrangement for a three-bedroom apartment (12.5 ft x 22 ft, 275 ft²).

Residential Spaces LIVING ROOMS Circulation (a) (b) (c) (d)

Fig. 4 When through traffic is unavoidable, pathways should skirt conversational or activity centers, as illustrated in (a), (b), and (c). (d) Illustrates a more ideal layout in which the entire room is bypassed.

Residential Spaces LIVING ROOMS Furniture Clearances Figures 5 to 10 show various groupings and related clearances. Figure 5 shows that a space 12'6" x 15'6" should be provided in order to accommodate seating for five around a 56-in-diameter cocktail table. The piano, sofa, and cocktail table arrangement shown in Fig. 6 requires a space at least 11'0" x 16'0". Figure 7 suggests that a space at least 12'9" x 13'3" is required to accommodate a grouping to seat 6 or 7 persons, while Fig. 8 indicates that a corner arrangement for two requires a space at least 6'3" x 6'6". When planning furniture arrangements, allowances for clearances should take into account the human dimension as well, as illustrated in Figs. 9 and 10. It should be noted that these diagrams are not intended as models for complete living room layouts. They are intended only as guidelines to illustrate minimum clearances for preliminary planning purposes.

Residential Spaces LIVING ROOMS Living Room Activities

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ACTIVITIES . 1 II. 1<i^:^wffi;;vv.-;;!K1:lJ'!-.^ ^.,:?MT...i1^ '><•'•• .va I>
LISTENING TO MUSIC WATCHING TELEVISION 64

Residential Spaces LIVING ROOMS Media Cabinet Details =5* = <* <#
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IiPPen HIW4«> At**/* (*>* PLAN 4_ . *-«' l'-i'-o" Fig. 11 Working
drawings of a media cabinet, Including plans, elevations, and sections of the
installation. The design of the cabinet should take into account the actual
electronic and other equipment to be housed and the clearances Involved for
operation. Power outlets should be coordinated and located so as to conceal
unsightly wires and cables. 65

Residential Spaces LIVING ROOMS Media Cabinet Delails ,*> I SEE 1
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Fig. 11 (Continued) - ■eopCIT- _ . -(Jo Burner 6> OF CABIMW fOf.
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Residential Spaces LIVING ROOMS Plans, Elevations, and Details 44^
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'J.LAi^:/Z-.I'D- .«««•♦ «■ .1 Fig. 12 Working drawings of a library/living
room, including a plan of the space, wall elevations, and some of the many
details involved. 67

Residential Spaces LIVING ROOMS Plans, Elevations, and Details ,HI-F|
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2s. SECTION G-G "SCAU . Vt' - I- 0" Fig. 12 (Continued) ± -W H l'-0" TO
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woe on PART, BD. TYR SECTION THRU KPT. SELF 3CALT . S'.r-O" 68

Residential Spaces LIVING ROOMS Fireplace Wall Elevation and Details
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DETAIL of- MANTLE /cale : IV - I' - O' IV/cale: DETAIL OP' PILA/TER
■&■ CORNICE 69

Residential Spaces LIVING ROOMS Fireplace Wall Sections and Details
*WIJ COLUMN COMPLETELY BURIED IN MASONARY FOR
STIFFENING 70

Residential Spaces LIVING ROOMS Fireplace Mantle Details Figure 13
shows a plan and elevations of modifications to an existing fireplace. Based
on these drawings and inspection and measurement of existing conditions,
the contractor prepares and submits shop drawings for the designer's
approval, Since at least two trades are involved, coordination of the trades
by the contractor and a thorough review of the shop drawings by both
contractor and designer are essential, It is important, also, that modifications
conform with all applicable codes. The extent of hearth extension, the
materials used, and the distance of combustible materials from the fire box
are among the numerous items governed by codes. www PLYHTU 1- A.
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«mwic«,ibovt A 1 . 14V SIDE ELEVATION A Fig. 13 PLAN IVtl'-o" 71

Residential Spaces LIVING ROOMS Bar Details snipsewtf will. I'■fe'rami
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Residential Spaces LIVING ROOMS Fig. 14 Floor plan, elevations, and details of paneled living room/library. 73

Residential Spaces LIVING ROOMS xfp~ NftffH Fig. 14 (Continued) 74

Residential Spaces LIVING ROOMS T.^MAX, (e'-d W. ko vjca <° Ku^ri was Q\>\JM'm\lM'\OH> 14 (Continued) wff: PaPR^JAMbl* fcfWMibdfp /.MeTAI/^P 75

Residential Spaces LIVING ROOMS Fig. 14 (Continued) ~U»tf g yiui. KU»J m Wflhtt 4uf f*.< frtaffyaE) ^ ® 6«hi«- p<afiL.fc fr«' u>^[hwihh-O rj\ }fy-A ■' fffw yoi^A t/.b st<' ■Haw*, 'f Ws. i;tr£. ■ ■ to p**et ?>?./ (* it >>>? *s =«'»■ "f —^ ft \ i -4"- TT^ it" £» >—^ I If? Kto oiftf li/ ^L K/<Wh;' W«* yiNfr< 7^- ' It ^r 76 ■fc ftggn^/»»» pfrffti»»»

Residential Spaces LIVING ROOMS Planning Data: Solas Traditional: roll arms, loose-cushion back, kidney shape, solid base Traditional: roll arms, fixed-cushion back, tailored skirt Contemporary: curved arms, fixed-cushion back, solid base Contemporary: dome arms, solid back, solid base Traditional: roll arms, fixed-cushion back, tailored skirt Traditional: roll arms, tufted Traditional: roll arms, one-piece back, skirted base Contemporary: slanted cushion arms, fixed-cushion back, solid base Traditional: roll arms, one-piece back, solid base Traditional; roll arms, loose-pillow back, shirred base Contemporary: curved arms, fixed-cushion back, solid base Contemporary: shaped sofa, shaped front view Traditional: roll arms Traditional: roll arms, loose-pillow back, tailored skirt Traditional: roll arms Traditional: roll arms, loose-cushion back, kidney shape, pleated skirt, Traditional; roll arms Contemporary: shaped sofa, shaped 'Y I ' ' I VI " ^ ± Traditional: roll arms, fixed-cushion back, solid base Contemporary: square arms, loose-cushion back, solid base Contemporary: miscellaneous slanted arms Traditional: roll arms, fixed-cushion back, shirred skirt Traditional: roll arms, tufted l V \ > \ 11 rl rr xrrrc S TTT Traditlonai: roll arms, fixed-cushion back, tailored skirt 77

Residential Spaces LIVING ROOMS Planning Data: Sofas Contemporary: straight arms, one-piece back, open base Traditional: roll arms, loose-pillow back, full skirt Contemporary: shaped sofa, shaped plan Contemporary: roll arms, fixed-cushion back, solid base Traditional: roll

arms, tufted Contemporary: miscellaneous slanted arms Contemporary: shaped sofa, partitioned back Contemporary: curved arms, fixed- cushion back, solid base Contemporary: square arms, loose- cushion back, solid base Contemporary: straight arms, one- piece back, open base Contemporary: miscellaneous slanted arms Contemporary: dome arms, one- piece tufted back, solid base Contemporary: roll arms, loose- cushion back, solid base Contemporary: square arms, fixed- cushion back, open base Contemporary: dome arms, loose- cushion back, solid base Traditional: roll arms, fixed-cushion back Contemporary: square arms, fixed- cushion back, open base Contemporary: curved arms, fixed- cushion back, solid base Contemporary: roll arms, fixed- cushion back, solid base Contemporary: square arms, fixed- cushion back, open base Contemporary: dome arms, solid back, solid base 78

Residential Spaces LIVING ROOMS Planning Data: Solas Contemporary: wood frame arms and legs, one-piece curved back, open base

Contemporary: curved arms, fixed- cushion back, solid base Traditional: roll arms, one-piece back, open base Contemporary: slanted cushion arms, fixed-cushion back, open base Contemporary: one-piece curved back and arms, wood legs, open base Contemporary: slanted cushion arms, fixed- cushion back, open base Contemporary: square arms, fixed- cushion back, wood trim, open base Traditional: roll arms, one-piece back, open base Contemporary: dome arms, channel quilted back, seat, and arms, solid base fr~^^ Contemporary; curved arms, fixed- cushion back, solid base Traditional: roll arms, one-piece back, open base Contemporary: curved arms, fixed- cushion back, solid base Contemporary: sofa/daybed Contemporary: slanted cushion arms, fixed-cushion back, solid base Contemporary: dome arms, one- piece tufted back and seat, solid base Contemporary; roll arms, fixed- cushion back, soft skirt Contemporary: shaped sofa, partitioned back Contemporary: curved arms, fixed- cushion back, solid base Contemporary: dome arms, solid back Contemporary; modular Contemporary: dome arms, solid back 79

Residential Spaces DINING ROOMS Furniture Clearances SPATIAL CHARACTERISTICS AND ARRANGEMENT Requirement Each living unit should contain space for the purpose of dining. This area may be combined with the living room or kitchen, or may be a separate room.

Criterion The amount of space allocated to dining should be based on the number of persons to be served and the proper circulation space.

Appropriate space should be provided for the storage of china and large dining articles either in the dining area itself or in the adjacent kitchen.

Space for accommodating the following sizes of tables and chairs in the dining area should be provided, according to the intended occupancy as shown: 1 or 2 persons: 2 ft 6 in by 2 ft 6 in 4 persons: 2 ft 6 in by 3 ft 2 in 6 persons: 3 ft 4 in by 4 ft 0 in or 4 ft 0 in round 8 persons: 3 ft 4 in by 6 ft 0 in or 4 ft 0 in by 4 ft 0 in 10 persons: 3 ft 4 in by 8 ft 0 in or 4 ft 0 in by 6 ft 0 in 12 persons: 4 ft 0 in by 8 ft 0 in Dining chairs: 1 ft 6 in by 1 ft 6 in Buffet or storage unit: 1 ft 6 in by 3 ft 6 in Figures 1 to 6 show the minimum requirements of the U.S. Department of Housing and Urban Development, Commentary Size of the individual eating space on the table should be based upon a frontage of 24 in and an area of approximately 2 ft'. In addition, table space should be large enough to accommodate serving dishes. Desirable room for seating is a clear 42 in all around the dining table. The following minimum clearances from the edge of the table should be provided: 32 in for chairs plus access thereto, 38 in for chairs plus access and passage, 42 in for serving from behind chair, 24 in for passage only 48 in from table to base cabinet (in kitchen). In sizing the separate dining room, provision should be made for circulation through the room in addition to space for dining. The location of the dining area in the kitchen is desirable for small houses and small apartments. This preference appears to stem from two needs: (1) housekeeping advantages; (2) the dining table in the kitchen provides a meeting place for the entire family. Where only one dining location is feasible, locating the dining table in the living room is not recommended. To living room Fig. 1 Dining room for 6-person, 3-bedroom living unit. 38" for chairs plus passage Fig. 2 Dining room for 8-person, 4-bedroom living unit. Fig. 3 Table for 2, 2'6" x 2'6". Fig. 4 Table for 4, 2'6" x 3'2". Fig. 5 Table for 8, 4'0" x 4'0". Fig. 6 Table for 8, 3'4" x 6'0". 80

Residential Spaces r-6" ■ a* i'4fj* i-8- DINING ROOMS Furniture Clearances (c) 4'0" MIN- FOUR PERSONS (1) ARMLESS CHAIR 20" x 21" 44" T 20" 32" 20" -b: -WALKING past seated person 36" ■ EDGING past seated person • ARISING AND DEPARTING • CHAIR PROJECTION (seated person) ■ MAXIMUM FOOT EXTENSION

Armless chair in place at table (0 (e) ARMCHAIR 22"x 23" WALKING past seated person EDGING past seated person ARISING AND DEPARTING CHAIR PROJECTION (seated person) MAXIMUM FOOT EXTENSION Armchair in place at table (9) Fig 7 (a) to (e) Illustrate, In plan and elevation, seating requirements and clearances for various dining table arrangements. (I) and (g) illustrate clearance guidelines for a typical armless dining chair and a dining chair with arms, respectively. It should be noted that the clearances Indicated relate to chairs with depth dimensions of 20" and 22"; clearances should be adjusted depending on the chair size finally selected. 81

Residential Spaces DINING ROOMS Furniture Clearances A dining room for 12. A hutch or buffet is typically about 18" deep, A 42" wide table is common. There is space behind the chairs to edge past one side and one end, and to walk past on the other side and end. Table space is 24" per person, the minimum place setting zone. With arm chairs at the ends, allow an extra 2" for each; add 4" to the room length. ^B 36"x 48" Table 32" 36" yu k Edge Past CO 6b ' Bench on one side. 7'-9" for a bench on one side and chairs on the other, Seating for four requires 31 ft². Bench 36" x 48" Table Edge Past i i Comer bench. Benches on one side and one end, and two chairs on the other side, seat five at a 3'x4' table in 44.5 ft². Bench and chair dining. Minimum width for table and chairs. 8'-8" for 36" wide table, 32" on one side to rise from the table and 36" on the other side to edge past. A 48" long table seats 4 and requires 34.6 ft². ;. ^S\ % ^F T ijj Table L G-> x ^S ^-Lj 5'-10" ~ Bench 36"x 48" Table 21" | 36" I 21 " <3 i 1 Dining space with benches. 6'-6" for benches on both sides of a 36" table. A 48" long table seats 4 and requires 26 ft². Round tables. A 36" round table with four swivel chairs fit in a 5'-10"x5'-10" or 34 ft² corner space. Figures 8 and 9 show clearances and room sizes for various dining arrangements. Since these data come from two sources, there may be slight disparities in suggested dimensions for similar conditions. Since these illustrations are intended only as guidelines for preliminary planning purposes, either set of any differing dimensions can be used. Fig. 8 82

Residential Spaces DINING ROOMS Furniture Clearances and Room Sizes KM D D W H-2-8"- : 9^ U 12-4" OOQOO" Q> iv-r —f— t ^m Fig. 9 83

Residential Spaces DINING ROOMS Furniture Clearances LMN«r ZOOM
 KrfOrteH LIVING aacw (b) Fig. 10 Minimum clearances for dining areas,
 (a) One end of table against wall, (b) Serving from one end and one side of
 table. 1C4.ftH^M To assure adequate space for convenient use of the dining
 area, not less than the following clearances from the edge of the dining
 table should be observed: 32 in for chair plus access thereto 38 in for chairs
 plus access and passage 42 in for serving from behind chair 24 in for
 passage only 48 in from table to base cabinet (in dining-kitchen)
 CNfTZAHCe. EN|C*M5fc Fig. 11 Minimum clearances and circulation lor
 combined living-dining areas. 84

Residential Spaces ROUND TABLES DINING ROOMS Dining Tables and
 Room Sizes Recommended Approx. Minimum Diam. Circum. Seating
 Room Size 10'0" 31'5" 12-15 20'0"X20'0" y'o" 2,X'2" ir-14 iy'o"x iy'o" 8'o"
 25'i" 10-12 1 ,S' 3 " x 18' 3" 7'o" 22'o" y-i r i6'o"x i6'o" 6'o" uS'io" 8~y
 14'rV'x H'fi* i'o" j 5' 9" 7-8 i3'o"xi3'o" 4'o" n'6" 5-fj ii'6"xii'6* 3'6" ti'o" 4-5
 io'6"xio'6" SQUARE TABLES w 5'o» 4'6" 4'o" 3'6" 3'o" L 5'o» 4'6» 4'o"
 3'fi" 3'0" Approx. Seating 8-12 4-8 4-8 4 4 Recommended Minimum Room
 Size i3'o"x 13'0" 12'0"X 12'0" n'6"x ii'd" io'6"x jo'6" 9'o"x p'o" Fig. 12
 Seating capacities for round, square, rectangular, and boat-shaped tables of
 various sizes and the recommended minimum room sizes to accommodate
 each. 85

Residential Spaces DINING ROOMS Dining Tables and Room Sizes
 RECTANGULAR TABLES Recommended Approx. Minimum W L
 Seoting Room Size — f>'o" iH'o" 28-30 iH'o"x4o'o" 6'o" 2(j'o" 26-28
 i«'o"X38'o" 6'o" 24'o" 24-26 i8'o"x36'o" 5'0" 22'0" 22-24 I5'0"X32'0" 5'o"
 2o'o" 20-22 i5'o"x30'o" 4'rt" i8'o" 18-20 i3'6"x27'o" 4'6" j6'o" 16-18 i3'6"x-
 25'o" 4'6" i4'o" 14-16 i3'6"x23'o" 4'o" 13' o" 12-14 !2'o"x2i'o" 4'o» 4'0" 4'o"
 4'o" 3' <>" 3'6" 3'6" 3'<i* 3'6" 3'0" 3'0" 2'6" 2'6" 12'0" I I'O" to'o" y'6" y' 0
 " 8' 6" 8'o" 7'6" 7'o" 6'6" 6'o" 5'fi- 5'o" 12-14 10-12 10-!2 8-JO 8-10 8-10 8-
 10 6-8 6-8 6-8 6-8 4-6 4-6 12'0"X 20'O" I2'0"X 12'0"X I2'0"X I0'6"X io'6"x
 I0'6"x io'6"x 10'6"x io'o"x io'o"x y'o"x y'o"x iy'0" 17'o" i6'6" 16'o" iS'6*
 i5'o" I4'6" 14'c 13' (■>" 13'o" 12'6" 12' o" BOAT SHAPED TABLES 3'S"
 8'o" 8-10 io'o"x i5'o" 3'8» y'o" 8-10 u'o"xi6'o" 3'n" io'o" 10-12 i2'o"xi7'o"
 4'3" n'o" 10-12 i3'o"xiy'o" 4'7" i2'o" 12-14 i4'o"x2i'o" 4' 11" 14'o" 14-16

i5'o"x23'o" }'}' ifi'o" 16-18 i6'o"x26'o" S'y i8'o" 20-22 i7'o"x2y'o" 6'o"
20'o" 20-24 i8'o"x32'o" 86 Fig. 12 (Continued)

Residential Spaces BEDROOMS Furniture Clearances and Arrangements

Most of the clearances and bedroom sizes shown here are minimum and intended primarily for preliminary planning purposes. Some building codes permit rooms of even smaller sizes, while rooms in many private homes and luxury apartments are much larger. Moreover, in the final analysis lifestyle, the size and scale of furniture, the activities to be accommodated, and barrier-free design are all factors that should be taken into account during the design process. Ideally, the recommended minimum bedroom size should be 10'0"x12'0" exclusive of closets, while the recommended minimum size for a larger bedroom or master bedroom should be 12'0"x16'0" exclusive of closets. A larger proportion of the bedroom floor area is occupied by furniture than is the case with any other room; windows and doors account for a large percentage of the wall and partition space. These two factors complicate the planning of bedrooms, especially when the rooms are small. Because of the room layout, some bedrooms with smaller areas better meet the needs than larger ones. The location of doors, windows, and closets must be properly planned to allow the best placement of the bed and other furniture. Privacy, both visual and sound, are desirable for the bedroom. Children's bedrooms should be located away from the living room, because conversation in the living room prevents the children from sleeping. Closets should be used between all bedrooms wherever possible. Each child needs a space that is his or her own to develop a sense of responsibility and a respect for the property rights of others. The ideal plan would provide a bedroom for each child, but since this is not always possible, there should be a bed for each. The minimum room width shall be determined by the space required for the bed, activity space, and any furniture facing the bed. Widths less than 9'0" will usually require extra area to accommodate comparable furniture. Aside from sleeping, the bedroom is the center of dressing and undressing activities. An interrelationship exists between dressing, storage of clothes, and the bedroom. Inevitably, in a small apartment, it is not only economical but necessary to plan the use of the bedroom for more than one activity. It is essential to incorporate in the bedroom other functions such as relaxation, work, or entertainment. A master bedroom should accommodate at least one double bed 4'6" x 6'6" or

two single beds 3'3" x 6'6" each, one crib 2'4" x 1'5" if necessary, one dresser 3'6" x 1'10", one chest of drawers 2'6" x 1'10", one or two chairs 1'6" x 1'6" each, two night tables, and possibly a small desk or table 1'6" x 3'0". Figures 1 to 3 illustrate three configurations and the furniture clearances and room sizes required. Ample storage is essential. Each bedroom requires at least one clothes closet. For master bedrooms, at least five linear feet of closet length is needed. For secondary bedrooms, at least three linear feet is needed. Clothes closets require a clear depth of two feet. Each bedroom shall have at least one closet that meets or exceeds the following standards: 1. Depth: 2 feet clear 2. Length (for primary bedroom): 5 linear feet clear 3. Height: a. At least 5'4" clear hanging space b. Lowest shelf shall not be over 6'2" above the floor of room 4. One shelf and rod with at least 12 inches clear space above shelf 5. At least one-half the closet floor shall be level and not more than 12 inches above floor of adjacent room

Q P 3'-.-^", I 4'.«" 6-6" - 2 — 1 4'-0 3LO.JZ □ 31.0" \ □^ I j££L . IV. 3. CLO. 3'-V _£=£! D \ / DOUBLE BED -ONE CLOS. lo!-3"xi2-o" Fig. 1 TWIN BED-SEPARATED Fig. 2 II-OXI6-0 3'-3" \ / 4-*2L I—■ □ Uii» J ii£> □ 1 OR, / n 3= £ TWIN BEDS -TWO CLOSETS l2-4"xi5-6" Fig. 3 87

Residential Spaces BEDROOMS Furniture Clearances and Arrangements 36' to use dresser *2" for dressing 22" on one side of bed (c) Fig. 4 (a), (b) Primary bedroom; (c) primary bedroom without crib. FURNITURE CLEARANCES To assure adequate space for convenient use of furniture in the bedroom, not less than the following clearances should be observed (Figs. 4 and 5): 42 in at one side or foot of bed for dressing 6 in between side of bed and side of dresser or chest FURNITURE ARRANGEMENTS The location of doors and windows should permit alternate furniture arrangements. 36 inches in front of dresser, closet, and chest of drawers 24 in for major circulation path (door to closet, etc.) 22 in on one side of bed for circulation 12 in on least used side of double bed. The least-used side of a single or twin bed can be placed against the wall except in bedrooms for the elderly 12" beside double bed ■ X. \ k2" for dressing 0>) Fig. 5 (a) Single-occupancy bedroom; (b) double-occupancy bedroom.

Residential Spaces BEDROOMS General Planning Data 9-6'x11-0' 104.5 t 9'-6'*U'-0* 104. »♦ Fig. 6 Although the recommended minimum size for a secondary bedroom is 10 0" x 120", these diagrams indicate how a double

bed, night table, chair, and dresser can be accommodated In a room only 9'6"x11'0". Fig. 8 Occupancy of a bedroom by more than two persons is not recommended. In cases where budgetary and/or space limitations offer no alternative, however, a dormitory arrangement may be necessary. The U.S. Department of Housing and Urban Development recommends the arrangement Illustrated in this diagram. I 3 300 mm T Fig. 7 Double occupancy bedroom. Net area: 14.7 m² (160 ft²). The most likely occupants of this type of bedroom are adults, school-age children of the same sex, children of different sexes who are less than 9 years old, and preschoolers.

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Residential Spaces BEDROOMS Built-in Storage Details w fr'-ll' Liwe » soUumn urn er o*fr - M*rr«M» ftJ-i.e-5 ■4" .I'l" *»■ W00D6WOGW6 rt*w UMiturt nxm> VVTWOO MM »U*T>oM t K**Tlt UMIUMfB MUWT«H wTV baui mat iex.1 MttfftMS WMIItmu. (Y|^) V flTWOOUOUrfJIA HI* 1¥7fS, l_i 4 j«!' ■■t r -QjfflMMm rU*nc UtMvi mM - LiOHT ppruM BASTION THH.ll *HIUf 4 STU>r.eew<ooM*4»*5 8C wwwo ***f«*ruu. rU*. UAM (S|M1»M MOT» ■■ CMWIMi OH IWUUT »»»T»mi<W tUMt xrf) **&T"J>j -THV> °9*** If^*)* 90

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Residential Spaces BEDROOMS Built-in Wardrobe Details 94 ••PART-
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Residential Spaces BEDROOMS Built-in Wardrobe Details a-o friction..
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DRAPERS 95

Residential Spaces BEDROOMS Plan and Elevation of Walk-In Closet
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Residential Spaces BEDROOMS Built-in Furniture and Closet Details FIN.
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r*»QAPU3 VALL4 ST (D 5SCTIQN <§> V. <'o" aw/ <?uw?rt-«s> to ee
tfoNTH? Fig. 9 (sj illustrates a typical hat shelf and coat rod, while (b)
shows relatively typical sections through a night table and a dresser, (c)
illustrates a typical closet. 97

Residential Spaces BEDROOMS Closets ;—Widrh of Closet 3 mm_
HooKs 4"ro 6 o.c ^Width I0"±_ 3)4" Hemisphere 1 Length I2"i Wood or
Meral Haf S"rand T Hat hung below shelf on slides Peg lack-may oe >-. ' a-
- sr^le bwall :f desired HATS Inferior or clear dimensions of" Drawer
determined by Dress Shirr, either 10"or 20" fide may be fronr. 4 Dress
Shirts — nyww ki'miuwrn DRAWER OR TRAY DIMENSIONS E^ ~M
TM iK Hardwood Tie Rack For back of Door Face of V Stile__ WOOD
TIE RACK -8 to 7" -12" AVERAGE MAN'S SHOES CLEARANCES-
VAR.IOUS ARTICLES OF CLOTHING -Additional Clearance necessary
for garments on hook srrcf4> : parallel to hanging pole HOOK STRIP Hook
slrip Clearance ..J,6" Mcfal Tte Rack- SECTION / tiki-. DOOR. JO' _i i.
Sack for vVomr > f, On Id-en.-:•■-; Shoes ->-, l\ZM-.M UTILIZING
BACK OF CLOSET DOOR Cabinet my extend up to jteif in. wonicns
Clasti ' ^ ' 1'? Clear Hanging; reighf (Mens Suifs.) * ' For Door Clearance ■
See T-SS. Serial "69 • Tray-open fronf "X. Trays-any type fronr ~\ -10"or
!-8" i j 4-<5' Gear Hanging _Hei^hf (Men). SVCWomen; 4 of Hanging Pole
5-10'above Finish Bedroom Floor (Ayerafc) Whanging wax above Cabinet
is not desired. Drawers may extend to under-side of slut/ above E -f-
Possible Storage Drawer. Shoe Racks, efc Finish Bedroom floor J > Rousjh
Floor CABINET FOR BEDROOM CLOSET ; 4of 3oie---- i"half - I or J /ps
. I ;J.U ^ iJ.U '•"-^g '—I"" Fe# —! I or 12—\ ."■jC- Haying Pole, i Moot

Residential Spaces BEDROOMS Closets ordinary treatment of space.
qPove door. Note poor access and 3m waste space j Space for I Rack for i
Shoes Hook Strip j space 6" !i extra ,i l-l -r~i ,, ,, '*- Appro* 12 Garments
on hangers MINIMUM CLOSET Hook strips for Ctuldreitr closets may be
provided on rear of door or side walls & may be adjustable, in, height. (See
Details) finished Ceilintf Lint 7 Floor Line- Individual compartment with,
separate, _ cabinet doors Shelf t Hanging Pole ifr* ^zy extend to shelf in
Wonwvs. Closets SHALLOW CLOSETS As much, as possible ot front
watts should be doors to make entire Length, of shallow closets accessible.
*M= Hi; l Short-/ Long/ Garmento i Garments 6 Suits i _ Dressesl -v- i I I /
i / i / I / i / i Sffece\ [cabinet,^ _ iShoc Drawers a Racksi ; \ il ____ . 12"-
-24^ J±_1_d ""4"v TT^ Hook strip space 6"exta z."^rf"zi:-t~' —r ^ '! 16"
desirable; SIWm§ ' Hansen available 10" to 24' and up I^Wi^V^^ Ujjj^
■'■■'"■●'■""■'V -4J6"— i—i,j ^vWm>>>>>i i_. Dead Horace Po\c —
btnef\i /\ __,-.._ W'UmnvmnmS.t H "1 2-2"Hintfed, folding or Sliding
Doors Sliding tum^er-rods accommodate maximum number of garments
without crush-in?. EXTREMELY SHALLOW —V» Approx. 18 Garments
on hangers FITTED CLOSET p\%.>>>>U "'s''''' Cupboard above is
impractical because depth is extreme. One or two shelves may be provided
Shelves over This dimension shwtdequal twice depth ot cabinet . _L 2;0'_2
s= '■2-or 2-0" w.vv^vv^ Possible cabinet 00"or ZO'frond Hook. *\$trlp"*1 i;
Extension I I p-4r3 C--I.lL.-_- crtihrd __E.Tt _H i-H-H---J --t—JUf;-3
^__L_ai_—i **■ ii-_ 2,"o wuu< ■. '_.'!<_« w mn : ■t; Possible Cabinet
;../(■ 7"Clearance for drawers at sr jamb unless depth, ot closet xrmuti
complete removal DEEP CLOSETS Closet Uf/tts desirable; controlled
preferably by door switch. Closet floors should be flush with top of the door
saddle.

lavatory, a water closet, and a bathtub or shower. Arrangement for fixtures should provide for comfortable use of each fixture and permit at least 90° door swing unless sliding doors are used. The bathroom should be convenient to the bedroom zone, and accessible from the living and work areas. Linen storage should be accessible from the bathroom, but not necessarily located within the bathroom. Each complete bathroom should be provided with the following: 1, Grab-bar and soap dish at bathtub 2, Toilet paper holder at water closet 3, Soap dish at lavatory (may be integral with lavatory) 4, Towel bar 5, Mirror and medicine cabinet or equivalent enclosed shelf space 6, In all cases where shower head is installed, provide a shower rod or shower door Each half-bath should be provided with items 2 to 6 listed above. Activity , Zone I Circulation Zone! G .-?!? H DOUBLE LAVATORY CLEARANCES fc in cm A B c) N : G H 15-18 28-30 37-43 32-36 26-32 14-16 30 18 21-26 38.1-45.7 71.1-76.2 94.0-109.2 81.3-91.4 66.0-81.3 35.6-40.6 76.2 45.7 53.3-66.0 Varies Vertical Grab Bar uj a Line of Wall Or Obstruction Activity Zone MINIMUM SHOWER CLEARANCES in cm A B C D E F G H I J K L 54 12 42 min. 18 36 min. 30 24 12 min. 15 40-48 40-50 72 min. 137.2 30.5 106.7 min. 45.7 91.4 min. 76.2 61.0 30.5 min. 38.1 101.6-121.9 101.6-127.0 182.9 min. WATER CLOSET BIDET BIDET AND WATER CLOSET 100 in cm A B C D E F G H I 12 min. 28 min. 24 min. 52 min. 12-18 12 40 18 30 30.5 min. 71.1 min. 61.0 min. 132.1 min. 30.5-45.7 30.5 101.6 45.7 76.2

Residential Spaces BATHROOMS Planning Data Circulation Zone Activity Zone =§> Mirror and Medicine Cabinet O W7^ Fig. 1 Lavatory: male anthropometric considerations. Figure 1 deals primarily with some of the more critical male anthropometric considerations. A lavatory height above the floor of 37 to 43 in, or 94 to 109.2 cm, is suggested to accommodate the majority of users. It should be noted, however, that common practice is to locate the lavatory in the neighborhood of 31 in above the floor. In order to establish the location of mirrors above the lavatory, eye height should be taken into consideration. Figure 2 explores, in much the same manner, the anthropometric considerations related to women and children. Given the great variability in body sizes to be accommodated within a single family, a strong case can be presented for the development of a height adjustment capability for the lavatory. Until that is developed, there is no reason, on custom installations, why the architect or interior designer cannot take

anthropometric measurements of the client to ensure proper interface between the user and the lavatory. in cm A B C D E F G H I J K L M 48 30 19-24 27 min. 18 37-43 72 max. 32-36 69 max. 16-18 26-32 32 20-24 121.9 76.2 48.3-61.0 68.6 min. 45.7 94.0-109.2 182.9 max. 81.3-91.4 175.3 max. 40.6-45.7 66.0-81.3 81.3 50.8-61.0 Fig. 2 Lavatory: female and child anthropometric considerations. 101

Residential Spaces BATHROOMS typical Plans and Fixture Arrangements Fig. 3 Two-fixture plans: water closet and washbasin. Fig. 4 Three-fixture plans: water closet, washbasin and tub. Fig. 5 Two- and three-fixture noncompartmented plans: water closet, washbasin, and shower. 102

Residential Spaces BATHROOMS Typical Plans and Fixture Arrangements BEDROOM, Fig. 6 Four-fixture compartmented plans: water closet, tub, and two washbasins. 103

Residential Spaces BATHROOMS Typical Plans and Fixture Arrangements 4' -0" 7' -6" 24" Door Minimum half-bath. 16 ft² is about minimum for standard fixtures; 4'-6"x4'-6" gives a more spacious feeling. 7'-6" 28" Pocket Door Generous half-bath. 22.5 ft² is a generous half-bath. A wall-hung lavatory instead of a vanity squeezes into 2'-6" width and 16.3 ft². n ■> -L I- f-6 -\ i^Q MWM 24" Pocket Door Corner toilet in a half-bath. A corner toilet and a small lavatory fit 13.5 ft². Consider this idea for installing a half-bath in a closet or under a stairway. 5'-?" 39" 15", 15' JO m 30" Door Corner shower. A corner shower, toilet, and lavatory fit in 33 ft². Very little storage space available. Towels And Supplies In 30" Vanity 32" Door Small, 3-fixture bathroom. A small 3-fixture bathroom with limited storage in a built-in vanity meets basic bathroom requirements in a space 37.5 ft². The door is 32" wide for a person with a cane or crutches. This bathroom is too small for a wheelchair. 8'-5" K 15V 30" 115" Storage jL itflHfflli cb cb £ 32" Door 18" J^k 60" Two-lavatory bathroom. A 2-lavatory bathroom with adequate room at the toilet and each lavatory. Note storage space under the lavatories and in a floor-to-ceiling unit. Area: 63 ft². 9'-5" 32" Door - 60" . 1 48" ■**— *■-. Separate tub and shower. This plan also includes a bidet. Storage is in the 48" long vanity. Area: 75.3 ft². Fig. 7 A wide array of two-, three-, four-, and five-fixture toilet plans. 104

counterlop lavatories. 31" -20 1VR' FINISHED FLOOR 12" 15'- — 20"- 4
-8" £±±3/ 17" 31" 20" FINISHED FLOOR 109

Residential Spaces BATHROOMS Lavatory Types and Dimensions o ^Q^
FINISHED FLOOD O II ■" -- A"~- -*^4- "i \ —,—c ? J a \ I size .18,15
IjO.! 1 1 " ! * I ' 11 IS I iV' 20 1 K 1 4 I 4 c l3Vi 19 0 8H 10 FINISHEO
FLOOH NOTE. 051-1644 and 051-1848 lavaiorles can be used with
concealed arm earner or wall danger -FINISHED WALL I 2W" FINISHED
FLOOR i. 1—i "19'17" 19", 17 iwfw it: JO .18; 20. IB 15'/»:10V«i
FINISHED FLOOR FINISHED WALL ' 20 -i <.{,l,u/..,l</l 18 1 - -: 4-K
*\$f) >— - | 15/i ^ I } FINISHED FLOOR FIWISKEO WALL JO .,
■•tv«vU<<!!<<<ii-t\r- ! 18 rT'^^P ♦ ♦ t Li FIHI5HE0 FLOOR Fig. 10 A
selection of wall-hung and pedestal-type lavatories,

Residential Spaces BATHROOMS Lavatory types and Dimensions \ 16 V«"
Fig. 11 Another selection of lavatories. 111

Residential Spaces BATHROOMS Lavatory Types and Dimensions 18
DIA. ———1- Fig. 11 (Continued) 112 w-A

Residential Spaces BATHROOMS Whirlpool Types and Dimensions JJVa"
^^ Fig. 12 A selection of whirlpool baths. 113

Residential Spaces BATHROOMS Bathtub types and Dimensions h -42" H
II -s\ \ n rr 37" 12" 1 -iiSSS 5--5V4- /c -^T ■ii"" 4W-5- M _ \ 30" c ■ ¥■'—
16 W :S> /c 32V." -LI ILL ISx- 16" /£ 2)\so- /£ 2^30" DIIt ID D 16/.' .1
Dimensions: 42 "x 37" x 12" Dimensions: 60" X3WX16" Dimensions: 60"
x3Wx 16" Fig. 13 A selection of standard baths. 114

Residential Spaces BATHROOMS Watercloset Types and Dimensions j»va
JJJ/4' MVT! Fig. 14 A selection of waterclosets. 115

Residential Spaces BATHROOMS Bidet Types and Dimensions Ml/J" Fig.
15 A selection of bidets. 116

Residential Spaces BATHROOMS Accessories PAPER. TOWEL.
DISPEN5ER 4" TO 10" DE-EP ELECTRIC HAHO DRYER TOWEL PIN

Residential Spaces BATHROOMS Plans, Elevations, and Details I ' i < 1 // " Ji. Y < * 4> ■ << i XX. * I ' l J \] — r M » II 1 i ——— HHH ' ' ^ K t ' i C0RWM TCP PLAN UV. ^ SPLASH [• [r l + ' ' ■ ■ ; ■ i ! _ ' 1 I ; ' , 1 1 1 ' i : ! 1 - L - n l " hrr p i . • I ' m ■ • 1 i ; 1 I i ■ \ L ■ i t > ■ ■ i . , ; i : - j . i : 1 : . , ' , 1 1 1 " i . : ' . _ * ~ 4 1 1 * ' 1 i ! , . . ■ ' ; i : I . 1 ! ' ' ' i . , , , , , 1 : ! 1 — i V _ J - r — ! K . ! 5 i 1 - JIMC TIIX ratrr scltction ly « WNCR ® ® Ucht IWIUNC I = - i i 7 - in tree * . < C - IDS t N . LOWCT4 CTWII ? RMTSKr puste u » * n : tli OX - Three - fixture bathroom with tub ' r — + - - nourmt. wwt fianrp vmc swwk m * L * 3232 - 1 PUN. LI I l ' IT | . — I — r - > no cow - | £ £ t < D ~ ! 5 ' ; ca. _ _ i cc , . . , _ li © ! ~ ar twiowto urn obnti iwxncc uow " ^ h - ios . md noae - Mi. CABINET * FUSTIC uniNATt. Three - fixture bathroom with shower 118

Residential Spaces WT £ G LI E 3 T bA. TIt BATHROOMS Plans, Elevations, and Details GUEST BAItM Guesr & wh W £ ST rv Jfe GUKST { SATH SOU TH Four - fixture bathroom with tub and shower guest b * m 119

Residential Spaces BATHROOMS Plans, Elevations, and Details W l 5 Ti. / Cou « rvj « Mv mm . -- 5 ! ~ ~ i . r ir , i . ® | l HON ! I r _ — . - . . " " r ^ i ^ _ _ 1 i : r ¥ 3 O I) T H E " A ■ 5 T © @ A f . A 5 T Four - fixture bathroom with bidet and tub NOt . rH 120

Residential Spaces PLAN l ' < V - f - ; 1 1 1 1 : . i I — i , v > - \ * ■ - + _ m _ _ ; . _ « . . ® - £ £ : JUJL m - FURRED - CEIL ' Q BATHROOMS Plans, Elevations, and Details 1 / i 1 . _ . ® Powder room U DD n I ! u UJ © - en (0 1 1 i ! 3 D ' : n , 8 u n I L J ■ ' H 1 = J 1 i © 121

Residential Spaces BATHROOMS Plans, Elevations, and Delalls UL. HSJ 1 ! + 4 > \ i ^ U - J & J ^ \ £ « lAH12f rim u . v . + * u * h P MHR ^ M « S * kMI « . 40AF " P » M ■ / . rn • . ^ U4HT e — ■ — ^ a - ^ f ' : j _ yz . Ju © AU < A * NtT < PfAWt * W ¥ H * HtVATIAJS. LI. 1 — ~ T | - J _ \ 1 V ' ■ — — ■ " ■ I l l] ®

Residential Spaces BATHROOMS Plans, Elevations, and Details « jti - sna * £ w . ia . ra . i l ta , i ta . r . * * ; r B - t ^ a ® . _ « ta . . • f l WWR ' / . / \ N - ANCEuqm - f 36 - 109 , — - 5 > .. - > - ' ! | , i i ' - imr . - t , , « " » . i Tiiu . . i ii i i , i , , , , iiii i I ! ' • ' ; ■ i j ' : : : . kj - fc i ■ * " — • ■ . . ii ^ — * " : . i ' l > ' 1 * A ■ ■ i ■ I 1 . • ' i ■ f w . r , , , , ... ® JWI - WfcH TiLt Her bathroom 123

Residential Spaces BATHROOMS Plans, Elevations, and Details r-1' l-»'
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CjJ- 5>t PEEP & MM!. LAV J TOP 4.SfUSH 3 -FWRBIBNE. swk
noiKp stnc ..+ 4854-1 FOLO - pew* 5CAT.DETAIL3 T0WUOW ■nuc
SH:i.r TILS 5WWH! OWNeR/SOTCTION PLKN MX CAHNC1T 4
DRKJTK DCOR5 AHTtPl>5lt LM1INATC NEVAM^R • —MARCHNND
VN-KNCC UQKT f 3(-l09 r d(i= LLa © 124

Residential Spaces BATHROOMS Vanities '•& I hAuiiuk CWLLHOUTf*
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LAVATORY FITTIUQ ARm.hlOgMi «MUR *kTH»5r ■- 125

Residential Spaces BATHROOMS Vanities W KM**rtM uwrninn
PWWflSP *«LP^ pet* soup oj« pane RX«tl5»3*7fT0N^ a: -weu-Hou**'
Plht* on u«rvn*n OOMT OH tWV» AS. 17^ TYPICAL 6U»gT/»H«JF
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BOTTOM NEW KfcDI&TO* WKW6TIC CATtH ntvv, unium coourtn,
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NUTWOOD DDOU mei. LAcauiM'imw (MNCIAL WINC.|> dlli U£4UtA
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Residential Spaces BATHROOMS Vanities fU.I*.|W- v*~|^ft(toPW(W F\
c£TAit'Wfr WApgg ■*KW.#* a-W. -wswr ~S\ CEWu»MaaasJ£ <sa&.
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B«m*S\$MUIQ 'jVl' Wood ffi i'o^, 0U ■UCNklt TOP (BOTTOM PA.IMT
PIUIW " iV'^o" 127

Residential Spaces BATHROOMS Accessory and Control Placeman! cm
AT LEAST 1/3 TUB LENGTH OR 18" MINIMUM L EQUAL ■4 ► - &-
Q) EQUAL -< p. 12"MIN. /f=3j^~\ +t (^ > r <D ELEVATION OF TUB
W/SHOWER CM +i io CM JI O) in (A h O h b lo FIN. BOTTOM OF TUB
ELEVATION OF TUB W/0 SHOWER SECTION AT SHOWER © 1a. © ©
© © SHOWER CURTAIN ROD: OF TUB OR SHOWER. KEEP WITHIN
INSIDE ENCLOSURE DOORS: IF SWINGING DOORS ARE USED,
PLACE HINGES ON THE SIDE OPPOSITE CONTROL VALVES. 2.)
SHOWER HEAD: SEE ELEVATION OF TUB AND SHOWER STALL
FOR RECOMMENDED HEIGHTS. PLAN GRAB BARS SHALL BE
MANUFACTURED OF SHATTER-RESISTANT MATERIAL, FREE
FROM BURRS, SHARP EDGES AND PINCH POINTS. KNURLING OR
SLIP-RESISTANT SURFACE IS DESIRABLE. RECESSED SOAP DISH
SHALL BE FREE FROM BURRS AND SHARP EDGES. WHERE GRAB
BAR IS AN INTEGRAL PART OF THE SOAP DISH, IT MAY HAVE A
MINIMUM LENGTH OF 6 INCHES. FAUCET SHALL BE
MANUFACTURED OF SHATTER-RESISTANT MATERIAL, FREE

FROM BURRS AND SHARP EDGES. ALL FAUCET SETS IN SHOWERS, TUBS AND LAVATORIES SHALL BE EQUIPPED WITH A WATER-MIXING VALVE DELIVERING A MAXIMUM WATER TEMPERATURE OF 110° ±5°F. 6.1 SHOWER STALL LIGHT: SHALL BE OF A VAPOR-PROOF FIXTURE WITH THE ELECTRICAL LIGHT SWITCH A MINIMUM OF 72 INCHES AWAY FROM SHOWER STALL. 128

Residential Spaces BATHROOMS Vanities; Lavatory Counters * ■ *■ + * 1 c»' L?* 1'Ce ^mas5-: o ^ TT II TT trJ * A^ ru-u* TOILET f^^M 0A& E^EV. ^—ft---.--- 4ff WALU f- V4 FW <5AI*- n,TWP- tV/r*ED OAF EPC/E'S P&TAIu2*W-c^ /T\ TcPILgT ROW CAD W riAfc ffUA&6 Miflfl^W^^UfSrAMe-. ■ft* UKiP&rvsApe. &<? <sl£;.. PUA€>nc: UAM. Plkl16H> by; giu«7MAf>r AUl. W^*?PVU?l»Vtf *t-KLL e>s firs* iseiArtf»ANT- fiu. i*ua?rt a 129

Residential Spaces BATHROOMS Lavatory Counters SHIM AS REQ'D BOWL Z-'/t PLYWOOD EPOXYTOBOTT. •wJw.s of marble top 1V<"x1"BRASSORSS PLATE BOLT TO WOOD J SECTION "A" (NO SCALE) Fig. 16 Typical details of a marble vanity-top installation. 130

Residential Spaces BATHROOMS Lavatory Counters NON STAINING CAULK LEAD EXPANSION SHIELD USE WITH 1'."THICK STOCK ONLY S* I I I I I I I I i iii r i i it mZZZZZZZZIZZZZL CER. BOWL ^Zfc^^A r NON STAINING CAULK ROUT OUT V* PLYWOOD TOP TO RECEIVE BOWL BOWL METAL CLIP & SCREW 4REQ'D NON STAINING CAULK WOOD BLOCKING AS REQ'D BOWL /4////A F V, PLYWOOD TOP Va PLYWOOD BRACE WOODSCREW TYPICAL SECTIONS BOWL CARRIER TRIM RING Fig. 16 (Continued) NOTE: Do not use oily putty or plumbing sealants with marble. CONCEALED MARBLE EDGE SECTION 131

Residential Spaces BATHROOMS Bathtub and Shower Details BATHTUB WALLS Wood or Metal Studs Cement Mortar -5/411 -(•«=> 11/4" CERAMIC TILE &£?udcoat moqtaq. e3ed SCriATCH COAT METAL LATH MEMWQAkje FLEXIBLE SEAUAUT TUR? HAHaEM^G EUO ft'ATrl TU& t=iEEf=iaaontJct whew qbsuiged [womea TGADE) W3PD "12 METALSTUDS Recommended uses ■ over dry, well-braced wood studs,

furring, or metal studs ■ preferred method of installation over wood studs for bathtubs TILE TUBS AND FOUNTAINS Glass Mesh Mortar Units Gypsum Board * p - c 1 • ; ^ fc \ , : < r \ I • — i = , r ; £t '■ E yS %< W-CEQAMICTILE PQESkUDEO DaY-5tT °n LATEX-t%?!2T LAUD CeMEUT t/DGTAO. GLASS MESH MOQTAO UWIT waoca metal studs FLEXIBLE SEALAkIT TU& HAtJaEr2°aELJD (SaAiw vv,s*?c> e>uas£ fATH TLJfc> T\mP\2<X>T\\ia VVHEIJ riE^UIQEQ t&frtKQ TQADEj Recommended use ■ in tub enclosures and tub showers over dry, well-braced wood studs, furring, or metal studs CERAMIC TILE LMCX-rePUTLAUD CEMEUT M^CTAQ ■=T2. ADHESIVE WATEI2 EESISTAkJT (GYPSUM KTAOD V*»D °a METAL STUDS r_EX:&LE SEALAkIT TU& HAKJ^EQ "E.EUD CrQAIW VY<2?D E>L<as-l£ fcATH TU£P pIGEPrcasFINCr WHEU peipuitarip (rV<5theq TQ.ADE) Recommended use ■ in tub enclosures and tub showers over water-resistant gypsum backing board on wood or metal studs SHOWER RECEPTOR RENOVATION Membrane Cement Mortar Cement Mortar exisTikior cer2AMc tile 4- MoaTArz mo COUT SEALAUT frEAD MEW FLASHIkICr MEWCEaAMIC- TILE 4- MOQTAD toED laePAIQ K4EME>|2AUE AUñ METAL LATH WE.W riEIMfWQCEp MoOTAC &ED UEW WATEQ.PP££?r MEMbPAUC SLOPED PILL STOUE* WOOD FORM CEQAMC- TILE NEAT CEMENT tOUO CO*X MOQ.TAQ £>ED SCQATCH COAT (|P DEEDED) metal lath ' (Preferred) \VATei2PI2Cor MEM&QAKJE C0NCQET5 TANK. CONCRETE TANK QEIMPDBCIMCr VEEP HOLES |CGuertED TILE "Q. STOMB Recommended use ■ overwood or concrete subfloors; where old shower pan has failed Requirements ■ waterproof membrane required except in slab-on-grade installations where membrane may be omitted ■ slope tank so that membrane will slope to the drain ■ flange drain with weep holes required ■ wood framing, if used, should be pressure treatedand designed to resist deflection and movement 132 Fig. 17 Typical Installation details for bathtub walls, tile tubs, and shower receptors,

Residential Spaces SHOWER RECEPTORS, WALLS BATHROOMS Bathtub and Shower Details Cement Mortar 1LJ i ?X2< T .^3AT , ieeim^s^eo 7 I M£»2.rA!2 &EP I" :! ^J ehtPWE..^. PAW '*-sloped pill. W5EP H^PL-ES CG-US^SC TIL! "3. S?i3*-E i»i:*j" Glass Mesh Mortar Units Wood or Metal Studs Gypsum Board Organic Adhesive CEI2AMIC

TILE PEE£>AWOEC> DGY-S6T °Q LATEV- fi&KTLAUO CE.MEUT
M£5QTAr2. <5rLASE> MESH Mc^tZr^a uuit weep °a metal studs ,
rl2tlMfe>ac:tP / lM<?GTAl2 ntO-l'T-iltf ISHOWEQ PAW «E
riMEMCHAUft ' f-QLCf^D PILL H — CE2AMIC TILE. — ACH23IV&
IWATEI2 r2ESISTAMT "Ii^YFSLIM E£>At2D — Vvteoy "2 METAL
STUBS — f LEXIfcLE SEALAUT JtjjALVAUIzeD i METAL LATrt |L5
LIKJEP Recommended use ■ over wood or concrete subfloors
Recommended use ■ in showers over dry, well-braced wood studs, furring,
or metal studs Recommended use ■ in showers over water-resistant gypsum
backing board on wood or metal studs COUNTERTOPS Cement Mortar
Thin-Bed Wood Base Glass Mesh Mortar Unit II 1 ? t~—i / T" / /r~
^IJiILJrZ f- Jt_ JL_ ^-::jL\ C'JT WALL MOQTAE MJD METAL LATH
HERE CEGAMIC TILE M<?t2TAQ DED METAL LATH MEMbOAUE-
W020 DASE PUU^HED METAL —i CEI2AMIC TI LE / EFfc>XV "12
*~" <PQGeM\G ADHESIVE CEI2A.MIG TILE LATEX-FOT LAUD
CEMEUT M^JETAR GLASS MESH M£>GTAn UWIT m--Av£fcs>D
vatebpocoi* meme3i2aue- - TILE A.DHE5IVE ;si7 Recommended uses ■
on countertops, drainboards, lavatory tops, etc. ■ preferred method where
sink or lavatory is to be recessed Recommended use ■ on countertops
where thin-set method is desired Recommended uses ■ preferred thin-set
mortar method on countertops, drainboards, lavatory tops, and similar uses
■ preferred method where self-rimming sinks and lavatories are desired Fig.
18 Typical Installation details for shower receptors, walls, and countertops.
133

Residential Spaces BATHROOMS Bathtub and Shower Details VC
*rRBP.WR,iVpe QK*MI^T1L*,BULUJ«*» * WO** T\lf*/z 4lt** BUU-
MG* irrtu. w/MjJt utMtu* *»Mt^t*V- sm?\ : mmmwjtawi)- £lsJ3 ISA
SE6TI<3N THKU SHOWER (TXfi. ¥~ SL 3RHC ^ --1UK inui*n<B»rna
SKW^S^^ £ W" in»W» -LBAPFN-J A9UE I '-VH<OUt'TILB-JI -I |-"«SS
—|g| -f- ^W WIHfrV^" :cnaj-n^aHAgLE.iiLE..Vi»u.<yBArH I Jj! M*«L6
WU-.V ftihi» -woe mkwnd W 'ema- fITUP *M»uJ all annul C*ULWWfr
yKUEM-f V PLYWCBP tnunum «.YW00P 6UTWrftY0 uu* <* ixiwufc
iva* W\ SECTION THIOL BWHTUgg E*TH*I (tVk). _____ sjAfia, 134

Residential Spaces BATHROOMS Whirlpool Details H'rWmo sespaj •
Kbftm engvisp- |<. (I.*" 135

Residential Spaces BATHROOMS Ceramic Tile Details TILE OVER TILE
 Interior Walls Interior Floors NEW CEQAMC TILE E3£5NOINCf
 M&reaiAL EKISTINOr TILE M CASE I CUT TQIM TO FIT NE\V CS-
 QkWC TILB fo£)NDING MATERIAL EyieTIMCr TILE CASE UE\V
 CEQAM\C TILE QOOOUT QOUOMCr MATERIAL EX|ST|M(3? TLCOO.
 1 ihiiiiill'7 '^ //// mew fl-ypsus', J50AQO °IZ PC MOT2.TAr2 NEV
 CEQAMIC TILE MATERIAL EXISTING TILE E CASE III NEW M£?
 r2TAr2 NEW CCQAMIC TILE fcONONC MATERIAL pmTIMCr TILE
 Recommended uses ■ for alteration of ceramic-tiled areas where
 modernization or a change of design is desired in residences, motels and
 hotels, restaurants, public rest rooms, etc. ■ also applicable to smooth floors
 of ter- razzo, stone, slate, etc. a case iv Recommended uses ■ for alteration
 of ceramic-tiled areas where modernization or a change of design is desired
 in residences, motels and hotels, restaurants, public rest rooms, etc, ■ also
 applicable to smooth walls of marble, stone, slate, etc. Requirements ■
 existing installation must be sound, well bonded, and without major
 structural cracks Materials, grouting, expansion joints, Installation
 specifications ■ for organic adhesive installation see Method W223 ■ for
 Dry-Set or latex-portland cement mortar installation see Method W202 ■
 for epoxy adhesive installation refer to manufacturer's literature Fig, 19
 Typical installation details tortile overtile. 136

Residential Spaces BATHROOMS Adaptable Bathrooms ACCESSIBILITY
 It is essential that the design of interior spaces, as well as exterior spaces, be
 responsive to the needs of those having physical disabilities. There is a
 proliferation of state and local legislation in this regard, and, more recently,
 federal legislation (Americans with Disabilities Act of 1990), that provides
 design guidelines and requirements. The designer should become familiar
 with those codes and other requirements in her or his area prior to initiation
 of design and, where possible, go beyond the very minimum standards. The
 design of the bathroom is perhaps one of those areas where the interface
 between the physically disabled and the interior space is the most critical.
 Accordingly, on this page and the following pages are design guidelines
 prepared by the Veterans Administration and the U.S. Department of
 Housing and Urban Development. removable vanity cabinet in knee space
 Small Adaptable Bathroom in Conventional Configuration recommended
 self- supporting shelf and countertop recommended additional connection

for hand-held shower head standard 5'-0" bathtub reinforced areas for possible future grab bar installation ANSI/UFAS clear floor space at each fixture exposed clear knee space under lavatory vanity cabinet removed and protection cover installed hand-held shower and grab bars added as needed Small Adaptable Bathroom in Adjusted Configuration clamp on tub seat added as needed 137

Residential Spaces BATHROOMS Adaptable Bathrooms offset controls clamp-on removable seat This sample bathroom meets the minimum space requirements of both ANSI and UFAS; note, however, that the space is very small and many wheelchair users will have difficulty using such a bathroom. More space should be allocated when possible. A Small Bathroom with Adaptable Features Plan 138 vanity base cabinet (removed)

Residential Spaces BATHROOMS Adaptable Bathrooms vanity cabinet installed in knee space reinforced areas for possible future grab bar installation offset controls standard 5'-0" bathtub with built-in seat Larger Adaptable Bathroom in Conventional Configuration vanity cabinet removed to expose knee space grab bars added as needed built-in seat at rear of tub offset controls Larger Adaptable Bathroom in Adjusted Configuration 139

Residential Spaces BATHROOMS Adaptable Bathrooms hand-held shower offset water valve standard tub Standard Bathtub with Removable Seat Standard Bathtub with Built-in Seat 1/ additional waterproof floor area for water control / proof floor area ' / for wntpr n ANSI Minimum Roll-in Shower ttc*itt* is-on >,* ■liMk Ms 1IIW ^^Hl a> ■ ■ ji ■i II ■ »*■■■■ >■■■■■ «■■■■■ «■■■■■ ■■■■■I §■■■»■■ Preferred Deeper Roll-in Shower 140

Residential Spaces BATHROOMS Adaptable Bathrooms REMOVABLE VANITY CABINET BELOW 20 OA. MTU PLATE • SECURED TO MTU . STUDS. 20 GA. MTU PLATE SECURED TO UTU STUDS. 20 OA. MTL PLATE SECURE TO MTU STUDS. (FINISH WALL BOARD DIMENSION) 4-6 1/2"] J, 2^il r-x V/r POWDER RM. PLAN WITH 27" REMOVABLE VANITY CABINET .20 OA. MTU PLATE SECURED TO MTU STUDS. BATHROOM PLAN W/20" LAVATORY 20 OA. MTU PLATE SECURED TD MTU STUDS. 4 yJ? BATHROOM TYPE WITH 24" REMOVABLE VANITY CABINET 141

Residential Spaces BATHROOMS Adaptable Bathrooms «v-*w j,y-y\W-
Pt,,/^r*^, T-i 1/1" (FIN. WAUBOARD (XU.) 4'-8 VV 20 OA, UTL. PLATE
■ SECURE TO UTL. STUOS, 20 CA MTL PLATE StCURE TO UTL.
STUOS. iLLJM is7 -20 <SA. UTL PLATE SECURE TO UTL STUDS.
BATHROOM PLAN WITH 24" REMOVABLE VANITY CABINET X-
THE OUTER EDOE Of THE WATER CLOSET WX NOT EXTENO WTO
THE CLEAR SPACE REOUired FROM THE DOOR OPENtNO. (ANSI
*,V,i J'-0"x V-0" AROUND SHO*ER (ANSI +.21.2) REMOVABLE
SHOW* DOOR (ANSI 4.21.8) SHOWS STALL WTH V MAX. CURS
HEIGHT SEAT (ANSI V2U) "CRAB BAR (ANSI ♦.J!,*) 36"x 36"
ADAPTABLE SHOWER STALL 20 GA. MTU PLATE SECURI TO MTL
STUO MUM BATHROOM REQUIRED W/ADDITIONAL SHOWER 142
20 GA. MTL PLATE SECURE To UTL. STUOS.

Residential Spaces BATHROOMS Wheelchair Accessible Design 1220
Possible wall locations 12 30S < L 36min 915 min i ;_ai 12 min '36S >
Clear Floor Space at Water Closets 54 toilet paper (a) Back Wall Sfe^ k
Grab Bars at Water Closets Side Wall c E O CO 17 min 430 : clear ; floor 2
; space '. • • O 48 min \ 1220 knee clearance 205 17min depth Clear Floor
Space at Lavatories Lavatory Clearances 143

Residential Spaces BATHROOMS Wheelchair Accessible Clearances
Floor-mounted waltr clotol Typical balhreom arrangomant Shower Mat
TrWififW W Pr**)J* ftlJPwKf -pUUIN* ^W>. C-fttlN* STB »A,r <f
STHtpUP fifiP IP pfengp Lavatory Bathtub "Roll-In" ihowor 144

Residential Spaces BATHROOMS Wheelchair Accessible Design TH«*(
*.»M.» W.£.T<>i/Mrl.YWl - THKB6HOUJ ■ =L#KIN fllpKAJM KV. H-
OfZOP—, ^FN SECTION THKU TH^e5ri0aP:TO6aMI°Ll/»j,AN6l4-i».fe
VA»J FULL 6.*Jt 145

Residential Spaces BATHROOMS Wheelchair Accessible Clearances 152S
SYMBOL KEY: • Shower controls <] Shower head ■^ Drain 48 min 1 % -
■ ■ =S II seat || J 1 i lavj clear floor space 60 min 1S25 (a) With Seat In Tub
c i 4t lav! HI MO clear door space <? 75 win 1905 (b) With Seat at Head of
Tub Clear Floor Space at Bathtubs 12 max 305 CO? 8 tafmuu 24max — o c
7 o XVVVVVVVVNNNSVV back (a) With Seat in Tub seat. head control area

-f= 24 min 610 LJ "O VSNVSVVVV foot back (b) With Seat at Head of
Tub L_7 wvwvvvs head 146 Grab Bars at Bathtubs

!1 :« 36 915 bick —4 m I- ^ .. Is - - . - - . ml _ J "T ir -I 11 ~i ., _L £ ;:: 27 n
I 1.1 I I ij 60 1220 I E • O ■0 e> (•) 36-ln by 36-ln (915-mm by 915-mm)
Stall (b) 30-ln by 60-ln (760-mm by 1525-mm) Stall Shower Size and
Clearances 23, 1-h, 16 max 330 Shower Seat Design W/////A side 18
WVSSNNV (a) 48, -^ NVVVvVVVV control area 1° 36-ln by 36-ln (915-
mm by 915-mm) Stall back eoniro (b) 30-ln by 60-ln (760-mm by 1525-
mm) Stall Grab Bars at Shower Stalls

Residential SpBces BATHROOMS Wheelchair Accessible Design ^Iw-ii-
o'1 \ fc>^^< r-ffwt \ \ -dm't-i/ni h \ \ I i—1< ' ^-^ ^^ ' I [T I i 1- w 4 i. i =0 •a
i must VAVfe,-1,-ok /4A»iy<(rigM»1 vriW-^»rrEP 6K1 WCPM U.3/ It"« H-
c" rr—i :ril 148

Residential Spaces KITCHENS Anthropometric Data Height 66*" Eye
level 62"— - Shoulder level 5411" I — Waist level 428" ■ Maximum
upward reach 78" (one handl "Comfortable upward reach 75" (two hands)
— \ Shelves visible from eve level 62" Outward reach, left hand 33" Knee
level 20" — — ..Bange 36" Ironing board 35" Mixing center 33" I ""-
Bottom of sink 32K" — " *- — Beach downward 28" | (one hand) I '
Outward reach, right hand 30" Fig. 1 The height of a kitchen workcounter,
the proper clearance between cabinets or appliances for circulation, the
accessibility to overhead or undercounter storage, and proper visibility are
among the primary considerations in the design of cooking spaces, All must
be responsive to human dimension and body size if the quality of interface
between the user and the components of the interior space are to be
adequate. In establishing clearances between counters, the maximum body
breadth and depth of the user of larger body size must be taken into account
as well as the projections of the appliances. Refrigerator doors, cabinet
drawers, dishwashing machine doors, and cabinet doors all project to some
degree in their open position into the space within which the user must
circulate and must be accommodated. Standard kitchen counter heights
manufactured are all about 36 in, or 91.4 cm. But such a height does not
necessarily accommodate the body dimension of all users for all tasks.
Certain cooking activities, for example, may be more efficiently performed

from a standing position, but with a counter height less than 36 in. In overhead cabinets the upper shelves are usually inaccessible to the smaller person, while the lower shelves are usually inaccessible to most without bending or kneeling. The logical answer is the development of kitchen cabinet systems capable of total adjustability to accommodate the human dimension of the individual user. Such a system could accommodate not only those of smaller and larger body size, but also elderly and disabled people. Figure 1 provides some general anthropometric data for establishing basic heights of cabinetry and appliances above the floor. Figures 2 and 3 show in more detail the interface of the human body and the kitchen environment, RANGE CENTER Fig. Z in cm A B C D E F G H I J K L M N O P Q 48 min, 40 15 21-30 1-3 15 min. 19.5-46 12 min. 17.5 max. 96-101.5 24-27.5 24-26 30 60 min. 35-36,25 24 min. 35 max. 121.9 min, 101,6 38.1 min. 53,3-76,2 2.5-7.6 38.1 min, 49.5-116.8 30.5 min. 44.5 max 243.8-257.8 61.0-69.9 61.0-66.0 76.2 152.4 min. 88.9-92.1 61.0 min. 88.9 max. 149

Residential Spaces KITCHENS Anthropometric Data Figures 2 and 3 illustrate the clearances related to range centers. Figure 2 indicates a minimum clearance between appliances of 48 in, or 121.9 cm. The anthropometric basis for the clearances are amplified in Fig. 3. The 40-in, or 101.6-cm, wall oven workzone clearance is adequate to accommodate the projected wall oven door, in addition to the maximum body depth dimension of the user. The standing figure shown in broken line, however, indicates both dimensionally and graphically that the 40-in clearance will not permit comfortable circulation when appliances on both sides are in operation at the same time. The range workzone clearance, also 40 in, is adequate to accommodate the open range door and the body size of the kneeling user. An extremely important, but frequently overlooked, anthropometric consideration in kitchen design is eye height, In this regard, the distance from the top of the range to the underside of the hood should allow the rear burners to be visible to the user. in cm A B C D E F G H I J K L M N O P Q 48 min, 40 15 21-30 1-3 15 min. 19.5-46 12 min. 17.5 max. 96-101.5 24-27.5 24-26 30 60 min. 35-36.25 24 min. 35 max. 121.9 min. 101.6 38.1 min, 53.3-76,2 2.5-7.6 38.1 min. 49.5-116.8 30.5 min. 44.5 max 243.8-257.8 61,0-69.9 61,0-66.0 76.2 152,4 min. 88.9-92.1 61.0 min. 88.9 max. RANGE CENTER Fig, 3 150

Residential Spaces KITCHENS Typical Layouts I Fig. 4 U-shaped plans. If dishwasher is desired, it should be located at sink center. The U-shaped plan is the most efficient. When not broken, it provides the opportunity and floor space for several simultaneous activities. The corridor or gallery kitchen is typically accessible from both ends, often converting it from a work space to a corridor. It sometimes is closed off on one end, thereby creating a variation of the U-plan, which although small can produce a fairly comfortable kitchen. The broken U-shaped plan often results from the necessity of locating a door along one or two of the three walls of a typical U-shaped scheme. The resulting through traffic reduces the compactness and efficiency of the plan. The typical L-shaped kitchen allows for the location of a small breakfast area in the opposite corner Fig. 7 L-shaped plan. If dishwasher is desired, it should be located at sink center. Fig. 5 Corridor plans. If dishwasher is desired, it should be located at sink center. REFRIGERATOR Fig. 6 Broken U-shaped plans. If dishwasher is desired, it should be located at sink center. 151

Residential Spaces KITCHENS Typical Layouts Service Entry U kitchen. L kitchen. E Service Entry D a irr,7,T ^H®®,^ Eating Bar 'UUUU c c c c About 13' < Service Entry Corridor kitchen. Broken U kitchen. Fig. 8 These diagrams illustrate further variations of the typical plans shown in Figs. 4 to 7. A triangle perimeter of 23'0" or less is usually indicative of a relatively efficient kitchen layout. Cooking/baking Oven I ov I I and/or i I MW I I 1 Minimum For combination counter f led work center 1 18"-24" -* ^ rontage. rs. Range oo RG OO Sink I 1 S J f > s J Mixing Center Refrigeration f 36" min 48"-54" R i_ 152

Residential Spaces KITCHENS Clearances Space Criterion The size of the kitchen should be determined by the number of bedrooms provided in the living unit. Work centers for the following equipment, cabinets, and space for their use should be provided: 1. Range space with base and wall cabinet at one side for serving and storage of utensils and staples. 2. Sink and base cabinet with counter space on each side for cleanup. Wall cabinets for storage of dinnerware. 3. Refrigerator space with counter space at latch side of the refrigerator door. 4. Mixing counter and base cabinet for electrical appliances and utensil storage. Wall cabinet for staple storage. Recommended minimum edge distance Equipment should be placed to

allow for efficient operating room between it and any adjacent corner cabinet. At least 9 in from the edge of the sink and range and 16 in at the side of the refrigerator is recommended. Circulation space A minimum of 40 in should be provided between base cabinets or appliances opposite each other. This same minimum clearance applies when a wall, storage wall, or work table is opposite a base cabinet. Traffic Traffic in the kitchen should be limited to kitchen work only. Serving circulation to the dining area should be without any cross traffic. Height of shelving and counter tops 1. Maximum height of wall shelving 74 in. Height of counter tops should be 36 in. 2. Minimum clearance height between sink and wall cabinet 24 in; between base and wall cabinets 15-in clearance. T . 0 oo y" 15" -r It If u' r "TC r-U m ;:i\:- ■;■■■!;■ o ■■.:*:;,■■ -a-- 3 -ESri,, ~^3 M y|f~" f* jdg.4 H:§: O~0 oo 0H£. 0£l?l2?eM r^MpBONT/«T6fe-(i? R.VAW>^Kflt'HeK& i ■*<>" t- *- o o o o te" o o o o r~3 4» tym v* 4^ *2" THe-fc£B6P12M.tJN<f ifto[iueex^£CH uwrf t-! I 00 oo 3- °r F~l — ■ ~"* w ~1 I I 1 1 I 1 MIM. EPtfG.p\<%~A.h4£&> 153

Residential Spaces KITCHENS Storage and Cabinets KITCHEN STORAGE Each kitchen or kitchenette should have (1) accessible storage space for food and utensils, (2) sufficient space for the average kitchen accessories, (3) sufficient storage space for those items of household equipment normally used and for which storage is not elsewhere provided. shelving that does not project past 60° may be Included as required shelving width (w) In (eet, times the depth (d) In (eet, times the number ot lull-depth drawers equals area of drawer space •Ho. 28" min.—sink 15" mln.—other ' range—see detail of previous figure area to be Included in base shelving where access is from one side = Vt depth of corner —where access is from both sides, allow full credit Area of Corner Base Shelving shelving—note #1 depth (inches) 4 to 6 6 to 10 10 to 15 15 to 24 Height, Depth, and Spacing of Shelving and Countertop min. spacing finches) wall shelving 2 s, ft. 2.5 3 4 total x2 = x2 = x3 = x3 = 4 s. ft. 5 9 J2 30 s. ft. 7 10 base shelving 4 s. ft. x4 = 16 s. ft. 5 x3 = 15 2 x2 = _4 total countertop 4 s. ft. x 3 - 12 s. ft. 2.5 x 1 = total 14.5 s. ft. 35 s. ft. Fig. g Example: measurement of shelf and countertop areas. ||j ^UftctuTTV^ Jiamau. if more than Va" increase clearance to cabinet by the additional amount elevation free-standing range e/e vat/on plan built-in cooking unit section CLEARANCES OVER COOKING RANGES In Fig. 10, dimension A: 2 ft 6 in minimum

clearance between the top of the range and the bottom of an unprotected wood or metal cabinet, or 2 ft 0 in minimum when the bottom of a wood or metal cabinet is protected. Dimension B; 2 ft 0 in minimum when hood projection X is 18 in or more, or 1 ft 10 in min. when hood projection X is less than 18 in. Dimension C: not less than width of range or cooking unit. Dimension D: 10 in minimum when vertical side surface extends above countertops. Dimension E: when range is not provided by builder, 40 in minimum. Dimension F; Minimum clearance should be not less than 3 in. Cabinet protection should be at least $\frac{1}{8}$ in asbestos millboard covered with not less than 28-gauge sheet metal (0.015 stainless steel, 0.024 aluminum, or 0.020 copper). Clearance for D, E, or F should be not less than listed UL or AGA clearances. Fig. 10 154

Residential Spaces 1 22" I 1 SINK 1 ABOVE SINK Above a sink, plan for a minimum of 22 in. to the bottom of a wall cabinet. Since the wall behind a sink often holds a window, measurement for a cabinet is academic. But if wall space is minimal, a cabinet over the sink makes good sense. RANGE 30" 36" ABOVE RANGE The use of large pans, pancake flips and similar cooking maneuvers dictate a distance of 30 in. between rangetop and wall cabinet bottom. A fan mounted in the wall is the means here to exhaust cooking fumes to the outside. KITCHENS Storage and Cabinets HIGHEST SHELF 15"-18" BASE CAB 72" 36" ABOVE BASE CAB A range of 15 in. to 18 in. is the proper span between standard base and wall cabinets. Opt for the 15 in. distance if you are 5 ft. 4 in. or less; a wider span if you're taller. The highest shelf: 6 ft. from the floor, is a reachable distance. Kitchen activities become tiresome in poor light. A single fixture, centered on the ceiling is insufficient. Your need for light is greatest over the work centers. A good light there reduces the danger of cutting yourself; eases the task of monitoring color changes during a mix, and so on. The best place to install fixtures for this purpose is beneath the wall cabinets (with a shield to prevent glare when you're seated in the kitchen). A workable alternative is found in fixtures installed in an extended soffit. Plan for light above a rangetop and over the sink, as well. Choose incandescent, deluxe warm white or deluxe cool white lamps for the fixtures to avoid poor color rendition. SOFFIT LIGHTING ' UNDER CABINET LIGHTING 18" 36" KIICHI N IIGHTING Utensil and General Storage Space for utensils includes storage for dishes, pots and pans, utensils, and appliances. With the

increased use of such electrical appliances, their storage becomes a significant problem. General storage requires space for linens, towels, and kitchen supplies. Included in this category are brooms, mops, and other cleaning equipment and supplies. TABLE 1 Minimum Kitchen Storage Required 40 to 60 ft² Area - Kitchenette Item Total shelving in wall and base cabinets Shelving in either wall or base cabinets Drawer area Countertop area Item Total shelving in wall and base cabinets Shelving in either wall or base cabinets Drawer area Countertop area 0-bedroom living unit,* ft² 60 ft² 1 24 10 4 5 Area and Over- and 2-bedroom units, ft² 48 18 8 10 -Kitchen living 1-bed 3- an room living unit,* ft² 30 12 5 6 d 4-bedroom living units, ft² 54 20 10 12 * Kitchen unit assemblies serving the kitchen function and occupying less than 40 ft² area in 0-BR living units shall not be less than 5 ft in length and shall provide at least 12 ft² of total shelving in wall and base cabinets. Drawer and countertop space shall also be provided. No room count is allowable for this type facility. 155

Residential Spaces KITCHENS Cabinet Dimensions *• «■ HGKEJT SMELT ADY!Silll If 1 30" >g-4r' ^ Lit' TT 34« ASOW SIK UOR MHI uon ouiatn cawa height rat sroou cawu men ra cru«s .*a. *- 1 . Example of the proper dimensional limits and relative placement of kitchen base cabinets and wall cabinets 156

Residential Spaces KITCHENS Cabinet Dimensions FRAMED CABINETRY -24" 18" 66" -12"- WALL CABINET -24" 34W ~T~3Vfe" 2W UTILITY CABINET W/WALL CABINET BASE CABINET |* 12W ~| DOUBLE-FACED WALL CABINET J- <zw 1 ,1 *.!■ 'n 1 U, ,11 DOUBLE-FACED BASE CABINET 31W VANITY BASES FRAMELESS CABINETRY WALL CABINET ■24W- -24"- 18" 83'>»" 65'A" L- W BUMPER PAD V— Vo" DOOR W'-^ J_4W 2V4" UTILITY CABINET 34W I 30" -12W'H ,, Vi° BUMPER PAD >-13W-4^ -Jlh-12"—ffiL. - 12" Hit— W DOOR- WALL CABINET ■ 24W I]> JH 2W BASE CABINET 30" k| VERTICAL DIVIDER DOUBLE-FACED WALL CABINET - 25W- P 34W T« 4W VERTICAL vtf-Hjf DMDER 2W DOUBLE-FACED BASE CABINET 2'/s" VANITY 157

WALL CABINETS Wall cabinets are available in heights of 42', 30? 24? 18" 15; and 12'. Most cabinets are available in widths ransins from 9" to

48;" in 3" increments. Framed wall cabinets are 12" deep, not including doors. Frameless wall cabinets are 12½" deep, including doors. WALL BLIND CORNER CABINETS Wall blind corner cabinets are available in heights of 42;" 30" and 24" Most wall blind corner cabinets are available in widths of 24;" 27;" 30;" 33;" 36;" 42;" and 48" DOUBLE-FACE WALL CABINETS Double-face wall cabinets are available in heights of 30;" 24" and 18" Most are available in widths of 18;" 24;" 30;" 36;" 42" and 48" Framed cabinets are 13,5/16" deep with doors. Frameless are 13 V deep with doors. BASE CABINETS All base cabinets are 34½" tall. Most are available in widths ranging from 9" to 48," in 3" increments, Framed base cabinets are 24" deep, not including doors. Frameless base cabinets are 24½" deep including doors. Four-drawer base cabinets are available in widths ranging from 12" to 24;" in 3" increments. Frameless base cabinets are also available in a three-drawer style in widths of 30" and 36" BASE BLIND CORNER CABINETS All base blind corner cabinets are 34½" high. Most are available in widths of 24;" 30;" 36;" 39;" 42;" and 48!" SPECIALTY CABINETS Lazy Susans: 36"-Wide Range Hoods: (framed only) 36" & 30"-Wide Wall What-Not Shelves; (framed only) 30"-High Base Open Shelves: (framed only) 34½"-High Pantries: (framed only) 36"x66" Utility Cabinets: (framed) 24"x66" 18"x66" In 12" and 24" Depths Utility Cabinets: (frameless) 24" x 65 V 18" x 65 V In 12W and 24¾" Depths Tilt-Out Range Hoods: (frameless only) 30"x24" Glass Door Wall Cabinets-, (frameless only) 30"&36"-Wide Microwave Cabinets: (framed only) 30"x21" Microwave Shelves: 30" x 22½" (framed) 30" x 18" (frameless) Oven Cabinets: (framed) 27"x66" 30"x66" 33"x66" Oven Cabinets: (frameless) 27" x 65½" 30" x 65 V 33"x65½" Up to six 6" drawers can be added to frameless oven cabinets.

Residential Spaces KITCHENS Cabinet Types and Dimensions I
 LAMINATE APRON PANEL • FINISHED SIDES CZZJ LIE -A , ■MU>)
 «~ L J J 1| LAMINATE N APRON PANEL ^~~- FINISHED SIDES =• «=■
 "^ — " \ ' 1 \ "i LAMINATE ^APRONPANEL ^'^- FINISHED SIDES ~—
 i 159

Residential Spaces KITCHENS Cabinet types and Dimensions ** \
 COUNTERTOP EDGED WITH CONTOUR MOLDING _ I LAMINATE
 ^APRONPANEL ^- FINISHED SIDES ' •=. — e==a — ^ \ 1 | 111 1

LAMINATE .APRON PANEL ^~- FINISHED SIDES " «» LAMINATE
APRON PANEL FINISHED SIDES 160

Residential Spaces KITCHENS Cabinet types and Dimensions Sink
Cabinets & Fronts SINK BASES SINK FRONTS CORNER SINK FRONT
SB24 L or R SF24LorR (Non-Trimable) SB27 SB30 SB33 SB36 SB42
SB48 ■ SF30 Lor R + SF30 SF42 SF36 SF48 + SF30 (Without L or R
Available in Nordic™- Not Shown) (Trimable 3" per side Except
Nordic™ Non-Trimable) : CSF36 L or R Oven Cabinets & Drop-In
Range Fronts DROP-IN RANGE FRONT 1— 1 L,- U ■, , : : : ■ n i 18 J
96" 84" -24"- OVEN CABINETS 60" Max. Cut Out an 37" Max, Cut Out
Nordic™ J 52" Max. Cut Out DRWR 1 16" J 27" 24" _ OV24DX96
OV27DX96 OV24D OV27D -27" 24" • OV24SX96 • OV27SX96 h OV24S
h OV27S * DORF30 (Trimable Height & Width) UNIVERSAL DROP-
IN RANGE FRONTS 24" •is—r6^ 22V2' 6"r] 34V6" * UDRF30 *
UDRF36 (Trimable Cut-Out) Utility Cabinets/Fronts & Pantry Cabinet
UTILITY CABINETS/FRONTS 24" PANTRY CABINET Desk Cabinets r
42" 30" ' _ 50' 4" 96" 84" UTILITY CABINET SHELVES (width)
UCS1812 (dePth) UCS1824 UCS2412 UCS2424 (Shipped 1 Shelf with 4
Shelf Supports) (Utility Fronts are Non Trimable) UC1812x96 LorR
UC1824x96 LorR UCF1896 LorR UC1812 LorR UC1824 LorR UCF18
LorR _ 24" I UC2412X96 UC2424X96 UCF24X96 UC2412 UC2424
UCF24 DESK UNIT I 28W I DRWR DRWR DRWR • KDB15
KNEEHOLE DRAWER I 30" 1 DRWR 16"1 16"" * PC2424 (Includes-Lazy
Susan Shelves & Racks.) Available in Oak Lines, Except Oakcrest,™
Euro.™ X-Line.™ & Spartan™ * KD30 (Trimable 3" per side) DESK
END PANEL 161

Residential Spaces KITCHENS Cabinet Types and Dimensions BASE
CABINETS DRAWER BASE CABINETS SPECIAL DRAWER BASE
CABINET COMBINATION SINK BASE CABINETS DRWR A«B9LorR
B12LorR (Concealed Drawer) B15 L or R B18LorR B21 L or R B24 L or R
DRWR DRWR B27 B36 B30 B42 B33 B48 | DRWR DRWR iiniir DRWR
jORWRj DRWR DRWR DRWR DRWR DB12 DB15 DB18 DB21 DB24 -
DB18CB (includes — Cutting Board, Cutlery Divider & Metal Bread Box.)
-' SB60 -' SB66 -* SB72 A 15" 15" 18" B 30" 36" 36" C 15" 15" 18" A No
Arched Door. ■ No Center Medallion. (Saxony) ROTATING CORNER

BASE CABINET 36" . Wall Space s Required 34 36" Wall Space Required
 r— h" t i V 1 H f \ > ft v. N * \ BLIND CORNER BASE CABINETS
 PENINSULA BASE CABINETS ■ CAR36 tqglt^J (L-Shown, 3" Filler
 included, must use to allow doors to clear.) BLB39 L or R (Pullable 6")
 BLB42 L or R (Pullable 6") • BLB48 L or R (Pullable 4") * PB24 ^d- ^
 -24^^^pr PB30 PB36 (Drawers open on one side only In Peninsula
 Cabinet.) BLIND PENINSULA BASE CABINETS q£=^ (L-Shown, 3"
 Filler Included, must use to allow doors to clear.) * BLPB42 L or R
 (Pullable 6") • BLPB48L or R (Pullable 6") 162

Residential Spaces KITCHENS Cabinet Types and Dimensions 1 42" 30" J
 I-12--I ' W1242LorR ▲■ * W930 L or R ' W1542LorR ' W1842LorR '
 W2142LorR ' W2442 L or R W1230LorR W1530LorR W1830LorR W2130
 L or R W2430 L or R WALL CABINETS * W2742 W2730 * W3042
 W3030 * W3342 W3330 * W3642 W3630 * W4242 W4230 * W4842
 W4830 L-12-J I 24" 18" J 1 "1 15" 12" > I 24" ' W1224LorR
 W1524LorR W1824LorR W2124LorR W2424 L or R W3024 W3624
 W4224 W4824 W2418 W3018 W3318 W3618 W4218 W3015 *
 W3015X24 ■ W3012 W3315 * W3315X24 ■ W3612 W3615 * W3615X24
 COMBINATION WALL CABINETS CORNER WALL CABINET II 18"
 130- I J _C_ J A B -* CWC60 15" 30" C 15" -* CWC66 18" 30" 18" -*
 CWC72 21" 30" 21" (Fixed Shelves) CW2442 L or R CW2430 L or R
 (Fixed Shelves) CWS2430 L or R (2 Adjustable Rotary Shelves) BLIND
 CORNER WALL CABINETS PENINSULA WALL CABINETS jrZ? ^2| 1
 30" J (L-Shown, 3" Filler included, must use to allow doors to clear/
 BLW2742 L or R BLW2730 L or R BLW3042 L or R BLW3030 L or R
 BLW3642 L or R BLW3630 L or R BLW4230 (2 Doors) L or R BLW4830
 (2 Doors) L or R * PW2430 BLW2724LorR BLW3624LorR BLW4242 (2
 Doors) L or R BLW4842 (2 Doors) L or R PW3030 * PW3024 * PW3018
 PW3630 * PW3624 * PW3618 • PW4218 BLIND PENINSULA CORNER
 WALL CABINET MICROWAVE CABINET 6V«"T fm=; 35" "-' 3Vs»
 MW3035 g1/S» A: 22" Minimum B: 13%" Minimum 28" Maximum 1W
 Maximum I MICROWAVE CABINET W/SHELF (30" , 15" u *s= ■12"
 6V4" -a; 35" 3W' t MWS3035X12 {^" 3V2" .23" ^fl~ (L-Shown, 3" Filler
 included, must use to allow doors to clear.) * BLPW2730 L or R 163

Residential Spaces KITCHENS Cabinet Types and Dimensions Partial
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Residential Spaces KITCHENS Cabinet Details ■■f-2" S0»,Oj», MEM can.
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Residential Spaces KITCHENS Cabinet Details STEEL- FWrTHM**
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Residential Spaces KITCHENS Cabinet Details b: X I RUM Cti taWT *l W
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Residential Spaces KITCHENS Sink Types and Dimensions rC5I 8- / ex- <l —(6—IHt *4 A M'J 4> 4 * 4"H • <>- f't 24" ■ H'/i-n J b o ■)' 18 W 0 18%" 13/. -*>■ 11-R h 32- T -(- 8"- i § 4" ><>- *■ ■<[] -■ - i 21'A' — V. i J H'H f- 24" - -.. £-, ' ¥ 4- I J 1' 2" 4" >~ i 13 W - ■ J - 170

Residential Spaces KITCHENS Appliances Ranges and Built-in Ovens Freestanding ranges and built-in ovens come in a variety of sizes and configurations. Some of the larger ranges consist of modular cooktops providing anywhere from two to seven heating elements as well as modular grills, griddles, and even downdraft built-in ventilators. Normally, a minimum clearance of 30" is required above any range or cooktop, but the designer is cautioned to carefully verify local code requirements. Manufacturers' specifications should be carefully reviewed for rough opening requirements and any venting requirements, particularly for self-cleaning ovens. STANDARD FREESTANDING RANGE W: 18--42- D: 24" • 27 1/2" H: 35" - 36" Dishwashers Built-in, freestanding, and undersink dishwashers are fairly well standardized in terms of overall dimensions. Access; to 'plumbing and waste lines is the major consideration, as is the method of securing the dishwasher in order to minimize vibration. Refrigerators Refrigerator door swings and clearances are of critical importance. While a 90° door swing may provide sufficient room for a person to observe storage within a refrigerator or freezer, a 180° door swing may be required to clean a refrigerator and remove storage bins. This is particularly true of the side-by-side door configuration. In addition, adequate clearance should be allowed between the sides and top of the refrigerator and any adjoining cabinetwork, especially if a built-in look is desired. The designer should check requirements with the manufacturer. While these drawings can be used for preliminary planning, final dimensions and clearance must be verified with the manufacturer. Often overlooked are clearances for refrigerator handles or pulls as well as coils mounted at the rear of the refrigerator BUILT-IN SINGLE OVEN W; 20-

24" D: 21"-24" H: 22"- 28" BUILT-IN OVEN/BROILER W: 18" -24" D:
 21" -24" H: 36"-41" BUILT-IN DOUBLE OVEN W: 18"-24" D: 21" - 24"
 H: 38" -51" STANDARD REFRIGERATOR W: 18" SINOLEDOOR D:
 24" H: 68 1/2" STANDARD REFRIGERATOR W: 23 1/2" - 39 1/2" (SIDE
 BY SIDE) D: 29" - 33" H: 64" - 69 1/2" STANDARD REFRIGERATOR W:
 24" - 33 1/2" (BOTTOM FREEZER) D: 29" - 33" H: 64" - 69 1/2"
 STANDARD REFRIGERATOR W. 24" - 33" (FREEZER AT TOP)) D:
 27 3/4" - 32" H: 63 1/2" - 68 1/2" UNDER COUNTER DISHWASHER W;
 23" - 24" D: 23 1/2" - 26 1/2" H: 33 1/2" - 34 1/2" FREESTANDING
 DISHWASHER W: 23" - 24" D: 23 1/2" - 26 1/2" H: 33 1/2" - 34 1/2" BUILT-IN
 /RECESSED REFRIGERATOR W: 30" - 36" D: 23 1/2" - 24" H: 84" 171

Residential Spaces KITCHENS Appliances 24" min. 24" max.
 CENTERLINE HOLE IN OVEN SUPPORT SURFACE FOR CABLE 3" MAX.
 ELECTRICAL SUPPLY JUNCTION BOX HOLE IN OVEN SUPPORT SURFACE FOR CABLE
 24" electric built-in double oven 24" electric built-in single oven Gas built-
 in oven MIN. 4" HOLE IN OVEN SUPPORT SURFACE FOR CABLE 3" MAX. ELECTRICAL SUPPLY
 JUNCTION BOX HOLE IN OVEN SUPPORT SURFACE FOR CABLE 3" MAX. Electric built-in single oven 27" electric built-in double
 oven 27" built-in microwave oven 14" MIN. HOLE IN OVEN SUPPORT
 SURFACE FOR CABLE 3" MAX. 46" ELECTRICAL SUPPLY
 JUNCTION BOX HOLE IN OVEN SUPPORT SURFACE FOR CABLE 3" MAX. ELECTRICAL SUPPLY JUNCTION BOX 1/2" INCH FLEX.
 CABLE 30" electric built-in single oven 30" electric built-in double oven
 30" built-in microwave oven Note: Dimensions shown are for planning
 purposes only. 172

Residential Spaces KITCHENS RECOMMENDED AREA MINIMUM CLEARANCES
 ELECTRICAL JUNCTION BOX (LOCATE ON REAR WALL AT
 COUNTER TOP) 4 FT. CABLE 3" DEEP SINK / 33" TO 36"
 MIN. DUX. SINK 3" MIN. Conventional electric cooktop 36" electric
 cooktop 4 FT. CABLE 24" MINIMUM DISTANCE OF COOKTOP
 AT FRONT TO FRONT OF COUNTERTOP. 4 FT. CABLE 18" TO
 24" MINIMUM DISTANCE OF COOKTOP AT FRONT TO
 FRONT OF COUNTERTOP, Glass cooktop 30" solid elements/glass
 cooktop 36" solid element/glass cooktop SIDE CLEARANCES TO

VERTICAL COMBUSTIBLE SURFACE: LEFTSIDE -114" FKGHTSIOE
 - W VK CIEAIUKf I TO wiitcm. cowutrau sunF«n linsn^r mghiide - r 30"
 gas cooktop 30" gas cooktop Note: Dimensions shown are for planning
 purposes only. 173

Residential Spaces KITCHENS Counterfoils DEFINITIONS & DETAILS
 FOR PLASTIC LAMINATE TOPS TOP MOUNT SIDE OR BACK
 MOUNT SQUARE BUTT JOINT. Where the junction of the counter top or
 back and end splash is made by two separate pieces. COVE BACK
 SPLASH SQUARE BUTT END SPLASH COVE (Integral Cove). A
 curved junction of the counter top and splash, formed by bending a
 continuous sheet of plastic. Jr 3 SELF EDGE CUSTOM GRADE 3 Jr TJ
 (Self- edged) SELF EDGE PREMIUM GRADE (TOP LAP OVER EDGE)
 NO DRIP BULLNOSE NO DRIP- TILT EDGE SELF EDGE. Application
 to the edge of plywood or particle- board core of a plastic laminate of the
 same pattern as the face surface. NO-DRIP EDGE. Where the edge is
 raised, formed by a continuous shee-t of plastic - may be either "A", a bull-
 nosed edge, or "B", a tilt or wedge edge. This is not a continuous sheet of
 plastic. 1 Tfc Ik ROLLED EDGE. Sometimes referred to as "Waterfall".
 Formed by a continuous sheet of plastic rolled over the edge, with no raised
 portion. WATERFALL. Curved upper portion of the back splash and the
 curved edge of the top, formed with a continuous sheet of plastic. ^ FULLY
 FORMED TOP N H S FULLY FORMED. A counter top made with a
 continuous sheet of plastic, combining a no-drip bull-nose edge, integral
 cove, and waterfall back splash. Radius may be from 1/4" to 3/4" TYPES
 OF SINK INSTALLATIONS Metal Sink Rim /m ^ lR' Self Edge ft ^dr
 TYPES OF EDGING TIGHT JOINT FASTENERS 174 L FLUSH METAL
 OR PLASTIC TEE TYPE EDGE SNAP ON STAINLESS STEEL EDGE r
 r i c 1 ft i Draw-Bolt . , 1 1 Fig. 11 Definitions and details for plastic
 laminate tops.

Residential Spaces KITCHENS Countertops TOP SECTION WITH
 SHELF V." WOOD MOULD Acid Resistant Coated If specified No. 10 - 2"
 F.H. Screws 24" o.c. Maximum Parkerlzed or equal. •SINK CUTOUT- y ^
 Vi" Standard Y*" Minimum A Sink Wall Attach to Cabinet with 1VS" P.H.
 Screws No. 8 or 10-Cadmium Plated- 30'-^ SHELF ■i- 1x3 Cleat cont. @>
 wall by Cab. Inst. Wall Line 231! TOP SECTION WITH SPLASH 1" Itf

-1'/' P. H. Screws No. 8 or 10- Cadmlum Plated ,,,. 23" 30" — Wall Line
FINISHED EDGE Jr ■Eased Edge ■ Relieved Corner - Drip Groove BUTT
JOINT Color Matched Epoxy Cement / -Ml'*- 1/8" Maximum 175 Fig. 12
Composition stone top and sink details.

Residential Spaces KITCHENS Plans and Elevations WALmJUs ((MNjULf
fiUNUt.)^ e- Pull out'pwwm.'. llvin an - ,..*- R17I X" \ / *>■' J: n L li= ,>fe
■ "MS E li^ . ■,,,iL....^j|: ^T^3^ n \1/ ,.-' 11 V _JI 1 M^ .^ ^^^ H v- 1 J J 1. ,
-1 r -1 x' i 'i" ■Yyiv. EAST MIM wcsr ^ST^Aut;)4' 176

Residential Spaces KITCHENS Plans and Elevations -iDRRtP riNBHSP
UAU. ID TPP Of CABINfTS PUST1C U"1- VOonA, COMT. I I ! TR
TOPS, MNP PRAWER CECT5 I | j -flff^HBLVK TO M- AMU5TAW.C.
OH, HCH-M PX.ILI.CP' l"O.C \MOj *cAi*-i jt'-11-^ ^-^ BRU5HCP
STWNLltt _ STtCL TO MUTCH KfcNpC F^NSE HOOD. ■TJAawM-frpJ
PUASTC LM1INAT • OE.INET P87KJ. / COINTCN TOPs.ANP /
PfiWJER p<<R5, UMN0TP COLOR ITtKTUK SEUCTD BY OJNCR
(Tff.; T\ HC^TrL KiTCl-ILM ELEVATION X2qJ -«six^p^-"F^, TILL
HLMHf fit BWK. >fLiH , "W W.IW ■4-n \ 12." CUM*. — wkh reHEw..
«ee mnt h»i- — HP. &LEAT T T5 LAMr RIBKP AIWORYUC SMT-IP
PIFFU6ER "AllCO AmVZ. CO" - UTTIE WCH-UliPES. «0. FIHTIUE —
FH1BHS0 KITCtfil WAU. =3 ZZ2 I , ■ I -rws i. wn.urr» eAftaw SB- IkTTS
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NOTE. ip. I SE.&TIOH PE-TAILfiUPfE-F- qg'itiE-TA AW 7 FULL %IZt
177

Residential Spaces KITCHENS Wheelchair Accessible Design *e n ^ (a)
mln V/M////////M imw* (b) Before Removal of Cabinets and Base Cabinets
and Base Removed and Height Alternatives Counter Work Surface WffZft.
(a) Before Removal of Cabinets and Base T^ V/A v 30 mln 760 '////////A •*
V/A l_28323G u * — ■-■——t=j,v 1 C- -B=©V' 5~'1 ,,'l o vmm (b) Cabinets
and Base Removed and Height Alternatives □ 30r 780 (a) Side-Hinged
Door 30min w wmmmm (b) Bottom-Hinged Door SYMBOL KEY: 1.
Countertop or wall-mounted oven. 2. Pull-out board preferred with side-
opening door. 3. Clear open space. 4. Bottom-hinged door. Kitchen Sink
Ovens without Self-Cleaning Feature 178

1 1 1 1 — -cr ~ w -111 oo Oo i i i i — -&- — — ft ! I oo Oo 1 1 1 oo Oo
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 ACCESSIBLE : BEFORE REMOVAL OF CABINETS AND BASE. i 1 ! * -
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 AND BASE REMOVED. COUNTER HEIGHT LOWERED. J L J L S S S r
 C D ! ^ P 3 B ACCESSIBLE; BEFORE REMOVAL OF CABINETS AND
 BASE. CABINETS AND BASE REMOVED. COUNTER HEIGHT
 LOWERED EXAMPLE OF ADAPTABLE KITCHEN -U- SHAPED
 PLAN EXAMPLE OF ADAPTABLE KITCHEN -L- SHAPED PLAN

Residential Spaces KITCHENS Wheelchair Accessible Design < 5'-4" 3
 FIN. WALLBOARD DIMENSION 2'-6" MIN. 3 . VARIES SECTION TO
 BE LOWERED ; 1 k • * cv 5 ^ 24 1/2" f l i < 2'-6" UIN, j SECTION TO BE
 LOWERED .. Q. (o (i VARIES 1 i in i i S 0 1 1 § * i ' , , / 2 t T - y * — * ■ 2'-6"
 2'-t 1/2" 2'-6" h i i — - - X & Z w t c / (TO REFRIG. 14 C, F. SINK I
 . DISHWASHER I 2'-6" 2'-6" S'-O" SECTION TO BE LOWERED 2'- y
 Minimum-sized adaptable kitchen or kitchenette 1 1/2' A 8 1 1/2" 1 1/2" 24"
 . 11, 3 < ' - MIN, t SECTION TO BE LOWERED , NIWS sws- i i h m n g i — OO -
 REFRIG. 1+ C.F. 'dishwasher' I UNDER _ [i i / r i h 1 < / * " T 1 " SECTION
 TO B « ? LOWERET. SINK BE I W E f t E B. . 95" (7' - M (FIN.
 WALLBOARD DIMENSION) Minimum-sized adaptable kitchen (galley
 type) j — t — i 1 W N I N O SPACE O A U £ A Y D I U. R O O U D M. Kitchen
 clearance dimensions (not to scale) 180

Residential Spaces adjustable height work surface set at 36" height with
 removable base cabinet in knee space removable base cabinet below sink in
 knee space adjustable height counter with sink mounted at 36" height meets
 ANSI/UFAS requirements for clear floor space, storage, controls,
 appliances, doors KITCHENS Wheelchair Accessible Design Requirements
 The ANSI and UFAS standards require accessible and adaptable features
 which make the kitchen usable by most people, The fixed accessible
 features specified in ANSI 4.32.5 and UFAS 4.34.6 include requirements
 for doors, clearances, clear floor space, appliances, storage, controls, and
 knee space. The adaptable features are removable base cabinets at knee
 spaces and counters that can be adjusted in height or fixed at a lower than
 standard height, The adaptable features for kitchens specified in the

standards are shown in Figs. 13 and 14. In Fig. 13, the kitchen is shown in a standard configuration with the counter height at 36 inches and the knee spaces covered with base cabinets. In Fig. 14, the kitchen has been adapted by exposing the knee spaces and lowering the work surface and sink counter segments. No other changes have been made to the kitchen. Since removable base cabinets and adjustable height counters are not now products that are readily available for purchase, they are usually custom-made items. Fig. 13 An adaptable kitchen In conventional configuration, lowered counter segment with exposed knee space lowered sink and counter segment with exposed knee space Fig. 14 An adaptable kitchen In the adjusted configuration. 181

Residential Spaces KITCHENS Wheelchair Accessible Design adjustable height sink counter segment with removable base cabinet, and pipe protection and appearance screen adjustable height work surface counter segment with removable base cabinet range with up-front controls recommendation: position refrigerator so door can swing back ANSI/UFAS complying refrigerator freezer with 50% storage within reach ranges ANSI 2.34 Fig. 15 A small kitchen with: i features: plan 182 removed base cabinet removed base cabinet Fig. 16 A small kitchen with adaptable features: perspective.

Residential Spaces pantry. -refrigerator removable base cabinets and adjustable height counter segment shelves microwave oven base cabinets carts stored under counter KITCHENS Wheelchair Accessible Design The kitchen shown in Figs. 17 and 18 is an example of a more elaborate kitchen having ANSI/UFAS accessible/adaptable features. This kitchen exceeds the ANSI/UFAS minimum requirements. overhead cabinets electronic media center rotating slide-out wire frame corner storage shelves Fig. 17 An elaborate kitchen with adaptable features: plan. refrigerator with large, low over head freezer - microwave within reach of a seated person recommended longer wall cabinets for additional reachable storage over work surface - knee space and lowered work surface on adjustable supports ■ recommended special sink with shallow disposal bowl lets disposal fit despite required knee space Fig. 18 An elaborate kitchen with adaptable features: perspective — tambour doors for easy access to storage shelving two rolling carts for - food preparation and serving; also 111 in knee space

in work surface electrical receptacles -recommended wide lowered countertop at sink standard range with up-front controls television and other electronic equipment mounted within reach of seated people 183

Residential Spaces KITCHENS Wheelchair Accessible Design Work Surfaces People who use wheelchairs and other people who must or wish to sit down while preparing food need at least one work surface lower than the usual 36-in-high counter (Fig. 19). The standards (ANSI 4.32.5.4 and UFAS 4.34.6.4) require that at least one 30-in-wide, adjustable-height work surface be provided in an adaptable kitchen, although a wider size is preferred. The wider work surface provides space for pots, dishes, and other utensils as well as small appliances, and makes it easier to work on several things at once or to cook using many ingredients. lowered counter segment for work surface Fig, 19 Seated person at lowered work surface, recommended wide lowered work surface and knee space min. clear knee, work, and floor space required Fig. 20 Use of a wider, lowered work surface. 184

Residential Spaces KITCHENS Wheelchair Accessible Design built-in oven adjustable counter with knee space located on the left or right of the oven Fig. 21 Work surface at non-self-cleaning oven with drop-front door. built-in oven adjustable height counter segment on latch side of oven; this could also meet the adjustable work surface requirement knee space and adjustable counter not required when oven's self-cleaning, but recommended Fig. 23 Pull-out shelf at non-self-cleaning oven with side-opening door. pull-out shelf 10" minimum in depth and full width of oven required with side opening door Work surfaces at ovens If a wall oven is installed, a lowered work surface with knee space should be installed next to the wall oven. The standards specify that when the wall oven is not self-cleaning, a knee space must be located next to the oven to permit a disabled person in a wheelchair to pull up close enough to clean the oven. Even if a self-cleaning oven is installed, locating the knee space next to the oven makes it easier and safer for a disabled person to remove hot items from the oven. When an oven with a side-opening door is used, a pull-out shelf located beneath the oven must be installed. The shelf is used as a transfer surface for dishes as they are placed into or taken out of the oven. When not needed, the shelf is pushed back into the oven cabinet (Fig. 23). When an

oven with a drop-front door is used (Fig. 21), the pullout shelf is not needed because the door serves as a transfer shelf. See ANSI 4.32.5.7 and UFAS 4.34.6.7 for dimensions and details of oven. knee space allows — someone to get close to oven knee spaces not required if ovens are self-cleaning Fig. 22 Use of knee space next to oven. 185

Residential Spaces KITCHENS Wheelchair Accessible Design Cooktops in Adjustable Height Counter Segments ANSI 4.32.5.6 and UFAS 4.34.6.6 permit use of a standard range if the controls comply with ANSI 4.25 or UFAS 4.27. The controls must be placed along the front or the side of the range so that a seated person need not reach across a hot burner to adjust the controls (Fig. 24). Some wheelchair users cannot use conventional ranges because the surface is too high and there is no knee space for maneuvering. Cooktops in lowered counter segments with knee space below allow some wheelchair users to get close enough to operate the controls and move heavy pots and pans (Fig. 25). Cooktops with smooth surfaces are preferred by people with limited hand and arm strength because they can slide pots of hot food on and off the cooktop rather than lifting them over raised burners and knobs. When a cooktop is installed in a lowered counter, the width of the counter segment and knee space should be at least 30 inches and should provide space to the side of the cooktop for utensils and maneuvering. An additional 30 inches to the side is recommended (Fig. 26). When the knee space is under a cooktop, the standards require that the bottom of the cooktop be insulated to protect against accidental burns. While this type of installation may be the only way that some people can cook, it does expose a person in a wheelchair to the hazard of spilling hot food in his/her lap. People who pull up beneath the cooktop must exercise extreme care and cool hot foods before moving them. front-mounted controls drop front conventional oven (self-cleaning preferred) optional lowered — cooktop and counter segment front- or side-mounted controls required additional preferred space Fig. 24 Standard range. surface and knee space wider than ANSI minimum Fig. 26 Lowered cooktop with knee space and wide counter. smooth ceramic surfaces allow easier sliding of pots and pans which can be safer than lifting bottom of unit must be insulated and covered to prevent people from being burned ■knee space provides essential maneuvering space for some people, but also creates a greater burn risk

from hot food spilled in the lap 30" ANSI minimum Fig. 25 Use of cooktop with knee space. 186

Residential Spaces 1* p* m>vsft, \ minimum ,iwm «hh.i» fgvu ffljf ■re
nttiiWfl KITCHENS Wheelchair Accessible Design Kitchan arrangement!
Counter-mounted cook top P ■MMMMM NIPTU f 3glWsI&»>T6P, Sink
KnM-ipace clearance Knee-recess work area Dlipoiol link Armrest
clearance 187

Residential Spaces KITCHENS Wheelchair Accessible Design sVm'M- /-&
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Residential Spaces KITCHENS Wheelchair Accessible Design mmfr-m\ t»
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*tfaTJ*ju»fte>. 189

Residential Spaces LIBRARY/STUDY Anthropometric Data It is difficult
to develop precise formulas by which to design residential library shelving
or to project the number of books that can be accommodated on a unit base
because of the many variables involved. The size of books, the types of
books and other reading materials, the reach limitations of the user, etc, all
have an impact upon the design requirements, It is possible, however, for
preliminary planning purposes, to apply the broad guidelines indicated in
Figs. 1 to 3. Seven volumes per foot of shelving can be used as a rule-of-

thumb to project capacity. The height of the highest shelf above the floor should be limited to between 78 and 81 in; 24 in is the minimum height above the floor to gain access to a shelf without squatting. Limitations for shelving to serve children will differ and are indicated in Fig. 3. TABLE 1 Library Shelving: Volumes per Linear Foot of Shelf Based on Subject (Standard stack section 3 ft wide x 7' ft high with 7 shelves) Subject Art (excluding oversize) Nonfiction Economics Fiction General literature History Law Medical Periodicals, bound Public documents Technical and scientific Average for overall estimating Volumes per foot of shelf 7 8 8 8 7 7 4 5 5 5 6 Volumes per single face section 147 168 168 168 147 147 84 105 105 105 126 125 " MAXIMUM REACH FOR WOMEN II" — " MAXIMUM HEIGHT OF SHELF 7.2" " " 0 NO STRETCHING : 64" 0 BROWSING SHELVES 5" " HORIZONTAL SCANNING AT AN " OPTIMUM «" • MINIMUM HEIGHT FOR 24" NO SQUATTING 0 SQUATTING REQUIRED 12" Fig. 1 Optimum shelving conditions for adults. MAN - 69" TALL WOMAN - 65" TALL HIGHEST SHELF BROWSING SHELVES HORIZONTAL SCANNING AT OPTIMUM ° MINIMUM HEIGHT FOR ' NO SQUATTING " SQUATTING SHELVES CHILD 14 YKARS OLD - 63" TALL Fig. 2 Optimum shelving conditions for teenagers. • BROWSING SHELVES ■ " yJIMUM HEIGHT FOB SQUATTING " SQUATTING SHELVES ; • • ' 2d (1 YEARS OLD 46" TALL 190 Fig. 3 Optimum shelving conditions for children.

•tt «a. lj» aq. tf **■ Hf {HAt+i* *// WIWWJ ofi*wrn) 5 V- l'-V ** a'HOC , i 1, E ■- s'-io" ^ «j. !H1», j PT«3 S«iKj*0 S*«6 I COWIK «WT. CiM«r-^ of weq-uou.) %- r ;: 'Ut <* vulwu ovea I W1W3 | WI LIBPAgy •- PLAN 'HP v j^ ^» ^ Onnvur PR ^ _ * ,> Eb ., *-°" | V-Hk." s'-io" omwfmtu " " gA LIBRARY = MOW.TH BL&VATIQM fc'.l'-d" ■■ "J h -1 ir- —i 3! i "?" r r —1 : tf-)»f ■>]])»' M]|tf W JIW K MT.ptfNUU** jkm u*« im B'-lfc" II" ±

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 Ovk_ PLAN SECTION 193

Residential Spaces LIBRARY/STUDY Built-In Bookshelves r~ w ."H-Sfil
 V* -J-: C-3" 71*1 &r VV pi 4 a i PLAKJ S/tf PLYWOOD WITH EDOE
 PAMPEP - - '«' PCiWOOD V+ MIRROR. APPLIED WITH ADHESIVE
 AMU R.PJWQ FLUSH WITH TRIM ECXje 1/4 CLEAR. PLOAT -)
 QLAS>£> SHELVES I oVi"-ioy+" 1/+ MiRKOR. APPLIED WITH
 ADHESIVE — AMD BR.IMG FLUSH TO &UTT JOINT ',* MIRR.CR
 APPLIED WITH ADHESIVE AMD BR1IUG FLUSH TO BUTT JOINT
 AWD TEIM 74 MIRROE. APPLIED "=■) WITH ADHESIVE V4
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 TEIMEOOE V» PLYWOOD WITH EDqE BAMPED ELEWATIOM
 SCALE 1/a =1-0 J SEC B-& THROUGH BAK. bcauevAi'-c)1 2-0 W' E 3A-
 vfl v*; =E== 1-IO'A 539'f v+- ^ib= 2.-0' -v/ E 5ECTIOM C-C THOUGH
 BAE_ ^i'.0" /■ V, , SECTIOM A-A 5CALE W» 1 -O TEIM 1/Z. PULL
 'brZ.E ~f- w ADHESIVE- M1RR.OR. V» LnJ bEC. D SEC. E, •bCAUS'M"
 =l'-o" 194 BUILT-IN BOOKSHELVES Here is a simple method of building
 in bookshelves, bar units, etc., for residences and other types of buildings
 by using an egg-crate system, The front of the shelf is supported by the
 vertical members and the back of the shelf is nailed to the plywood back,
 These built-in bookshelves and bar unit were developed for a residence on
 the Eastern Shore of Maryland, In this design Hugh Newell Jacobsen,
 FAIA, divided the built-in bookcases into units of three shelf widths and
 introduced a recessed vertical divider 3" deep by 7'/t" wide between
 bookcase units. The major trim piece is solid wood 1 1/2" X 1'1/2" with a 1/2"
 wide by 3/4" deep groove at the middle, This simple trim piece acts as
 framing for sides, top, and bottom of the bookshelves and also for the bar
 unit with glass shelves and mirrored back, sides, top, and bottom.

Residential Spaces LIBRARY/STUDY TOP FIXTURES FOR NO. 1
 ROLLING LADDER No. 210—Roll Type Top Fixture No. 230—Hook
 Slide Top Fixture "4J %. m D @. ■W No. 230 csnk 1*1 8=*" BOTTOM
 FIXTURES FOR NO. 1 ROLLING LADDER ,__,— No. 1 Bottom Fixture
 No. 270 Regular WHREL SECTION AT LINE A-A c I i No. 1 Bottom

Fixture No. 270 Old Style wi*h.i«ct»nata.a BRACKETS FOR TRACK No.
 1 Track Roll Top Hook Slide No. 210 , No. 230 Vertical Mount on Uprights
 No. 14 No. 14 H Horizontal Mount No. 27 No. 27H Top of Shelving Mount
 No. 29 No. 29H I-2-) NO.14 CSNK FOR NO. 12 FH WOOD SCREW 1—
 1V4—I Li NO. 14H i"Kh I—2y' H K—vl -2%. -v/r l VA- NO. 27 CSNK]
 FOR NO. til FH WOODl SCREW v NO. 27H \(@_ At -VA- -17.- o -IX-
 ■"■ '4T ORNO.il ' NO. 1 TRACK lit^sf) NO. 29GR CSNK FOR NO. H FH
 WOOD SCREW ffi— 4v._z_r I^^-ixJ NO. 29 PUTNAM ROU.IN0
 LADDERS ^ SHOWING APPLICATION (» SKcMShWTTK BUI D ® ®);
 195

Residential Spaces FAMILY/RECREATION ROOMS Arrangements and
 Clearances Recreational Activities Indoor recreational activities invariably
 require definite spaces for equipment and clearances for using it. Not all
 games occupy floor areas indicated as necessary for those diagramed on this
 page. But if interiors are planned to accommodate large units of equipment
 such as that required for table tennis, and provide necessary playing
 clearances, spaces will be adequate for many other uses as well.
 Dimensions of game equipment and floor areas required for its use are both
 subject to variation. Sizes noted here are comfortable averages, not absolute
 minima. U-5 ft 6 in to 6 ft-0 in Clearances for playing bridge OxS> 2 ft 9 in
 s^2ft9in-p v> vj/ Clearances for playing poker Folded S<* A'Krt-4ln tuftsin
 -JK," 1 L . 1 ft-6 in |-«-3to41n 4 to 6 in Folded bridge table Folded chair 4
 ft-0 in Folded poker table •6 to Bin LAVATORY ACTIVE GAMES
 CLOSET FOR RECREATION SUPPLIES Fig. 1 Play mom. 196

Residential Spaces FAMILY/RECREATION ROOMS Arrangements and
 Clearances HEIGHT2"-6" NET HEIGHT B-'a" k." © 3'-6" * LIGHTS 5'-0'
 ABOVE TABLE ■ 9-0" -V-6" H 5-0" TABLE HEIGHT 2'-r/8" 2 '-J 3/4"
 CHAIR FOLDED 3"xl"-IVt x3'-2" T^nr-r 3-0" ' S'-B'" -1-8 "-^ PING
 PONG CARD PLAYING 5-3" —7=A POOL AND BILLIARDS
 GYMNASTICS TABLE 1 Pool and Billiard Table Sizes (In feet) Size 3x6
 3'1/2 x 7 4x8 4'1/2 x 9 5 x 10 6 x 12 Where used Home Home Home
 Commercial standard in South America, Mexico, and Spain Popular U.S.
 commercial standard U.S. professional standard Commercial standard in
 Canada and England Standard ping pong table sizes are 3 ft x 6 ft; 3 ft 6 in

x 7 ft 0 in; 4 ft 0 in x 8 ft 0 in; 4ft 8 in x 8ft 6 in; 5 ft 0 in x 9ft 2 in; 5 ft 6 in x 10 ft 2 in; 6ft8in x12 ft8 in. 197

Residential Spaces FAMILY/RECREATION ROOMS Residential Bar i •
CDNTIVWOU VTi-a'iM". ! . "■;uriM',>NF VTi-a>iM". f •'-lUf-.M'ING 4
—H.A.' " ' ! i I I). _ :^K. r ^ i 1 =D = flak, ki\$. r=D SHELVES i ME1AL. I r-
n"! ,1 > I 11, '■K fW "s —1/4" PLYWOOD repwoop: TRINt.DIVIDBR
PACTIT.^S, ^ TOP, eOTTOM J r' t 4= DAKK. BKOWN LEATHeeETTE
If REDWOOD BUfcL. yENBSB. STAINLBSS. STE, I// PLYWOOD ■ ■ I
-^—TT- 3f ELEVATION scau* #-.-<* OAEKEKOWM LEATHERETTE
i/4 FOAM RU&BER. [reprid, =ETj: 11'-O." MATT WHITE PLASTIC
LAMiWATE FROSTED GL«&& LIGHT PPiNELS "t IN FOZREP
SOFFiT AeauE "|L ;;. i .. -.-. = ...-■,•- -. j.-.-.'r.-i .; PL AW SCALE i/W=1'-
o" -/i? FLYWCOD CR. F/A.KTICLE KV\p-0 /4 PLYWOOP . t/l"
PLYWOOP r-n/V WALL SECTION WOT TO ■uOALE =r"" >/1
PLVWCOO- xM •£- ^ -i/i, REPYVOO0 VEWEER.OIJ PLYWOOD
VINYL EASE ., AWDPLOOB- T 2-4!' ■ /4 PLVWOOD WITH
REDWOOD BURL V/EMEER. -I L WREDWOOD •• VELviESR. OM
PLYWOOD BAR SECTION hot to so-ve HARDWOOP '/IL PLYWOOD
RESIDENTIAL BAR This small residential bar with double lighting was
designed so that the back bar shelves would display all the types of bar
glasses and the liquor bottles as a decorative element, Note how the
recessed fluorescent tube fixtures indirectly light up all the glasses and the
2" open slot in the bottom shelf indirectly lights up the liquor bottles on the
back bar shelf, The entire front bar has recessed light fixtures in the ceiling
above; this allows for two different methods of lighting the bar area, Note
also that the bar front is slightly padded with foam rubber and the entire bar
top is finished with dark brown leatherette, 198

Residential Spaces FAMILY/RECREATION ROOMS Piano Sizes; Bar
Details UPRIGHT PIANO SIZES GRAND PIANO SIZES SECTION
THRU BAR CABINET 199

Residential Spaces LAUNDRY/SEWING ROOMS Laundry Room Layouts
Washer Dryer ^u.^ yx-.nP 62" Fig. 1 Angle arrangement. Fig. 2
Conventional arrangement. Ironing Board Hanging bar 13001 1400 mm I n
T ■C r, t Scale 1:50 Fig. 3 Arrangement of ironing equipment based on flow

of work. Fig. 4. Space around ironing board. HOME LAUNDRY ACTIVITIES Home laundry includes the processes from sorting through ironing of clothes and household linens, including pretreating, washing, drying, and sprinkling. General Planning Suggestions 1. It is desirable to plan space for specific laundry processes, 2. Moistureproof surfaces are needed for pretreating and sprinkling of clothes, 3. Drying areas should be accessible for use under all climatic conditions, 4. To control moisture in the room, dryers should be located to permit venting to the outside of the house, 5. Adequate storage for washing equipment and supplies should be located near the place of first use. 6. Facilities for hanging drip-dry garments after washing should be provided. 7. In locating the washing equipment consideration should be given to convenience of inter-related household activities, distances from the source of soiled clothes and the drying areas, and the isolation of clutter

TABLE 1 Space Requirements for Washer-Dryer Arrangements

| Type and size of equipment | Auxiliary equipment | Work area, in | Total floor area, in | Width | Depth |
|--|---------------------|---------------|----------------------|-------|-------|
| Stacked arrangement: washer, 31 x 26 in; dryer, 31 x 26 in | | | 43 | 62 | 63 |
| Angle arrangement: washer 26 x 26 in; dryer, 31 x 26 in | | | 43 | 62 | 63 |
| Straight-line arrangement: washer, 26 x 26 in; dryer, 31 x 26 in | | | 43 | 62 | 63 |
| Basket, 19-in diameter | | | 43 | 62 | 63 |
| Basket, 19-in diameter | | | 43 | 62 | 63 |
| Basket, 19-in diameter | | | 43 | 62 | 63 |

Figures 1 and 2 illustrate arrangements of laundry equipment. Space needed by a single worker in front of equipment or between equipment placed opposite is indicated. Overall dimensions of areas will vary with type and size of equipment selected. No allowance has been made between the back of equipment and the wall for electrical, plumbing, and dryer vent connections, The space required will depend on the type of installation used. Counter space is provided for sorting and folding three washer loads of clothes. The space under the counters has been used for bins, one for soiled clothing and the other for dry, clean articles that require further treatment before use or storage. Additional counter space can be provided by the tops of the dryer and washer, depending upon the type selected. A tall storage cabinet for laundry supplies would complement each arrangement. In this cabinet, an ironing board, iron, mops, and buckets (needed for cleaning the laundry area) may also be stored.

Residential Spaces Fig. 5 Automatic washer. A=24-30 In, B=26-30 In, C = 42in, D=36 In. Fig. 6 Automatic dryer. A=24-28 in, B = 24-26 In, C=42 in,

D=36 In. 23H h-23'4"-H 21%" K23J4" 32 Fig. 7 Compact washer. Fig. 8 Compact dryer. I 800 mm I 800 mm I V V "™*t 3 1 Corridor Scale 1:50 Fig. 9 When space is limited, it may be possible to locate the laundry space next to a corridor. LAUNDRY/SEWING ROOMS Laundry Room Layouts Fig. 10 Clearance in front of automatic washer and dryer. If the space in front of the automatic washer and dryer is a corridor, this dimension should be increased to at least 1200 mm (4 ft), This will permit a second person to pass through when someone is doing the laundry. If a washer and dryer are located opposite each other, this dimension should also be 1200 mm (4 ft). I 1200 mm I mp Scale 1:50 Fig. 11 Clearance in front of laundry tub. I 1000 mm I Scale 1:50 Fig. 12 Clearance in front of sorting counter or table. 201

Residential Spaces LAUNDRY/SEWING ROOMS Laundry Room Layouts
LAUNDRY LOCATION The ideal location of the laundry space is a matter of preference. The laundry area may be separate or combined with the bathroom, the kitchen, the utility space, or the corridor. The most frequently mentioned advantages and disadvantages of these various options are listed below.
Separate Laundry A separate space can be used for other activities such as sewing and hobbies, if it is large enough. Clothes may be hung for air drying without interfering with other household activities. Noise from laundry appliances can be shut off from the rest of the dwelling. Temporary holding or storage of clothing to be washed or ironed is made easier. Providing this extra room increases the cost of the dwelling.
Laundry in Combination with Bathroom When the bathroom is located near the bedrooms, the washer and dryer are close to where most laundry originates, This facilitates gathering soiled articles and putting away clean linen and clothing. Combining the laundry space with a half bathroom adjacent to the kitchen provides many of the advantages of a separate laundry room. The tops of the laundry appliances provide useful horizontal space on which to lay clothes. Floor and wall finishes in bathrooms are usually resistant to high humidities. Usually, additional plumbing costs are minimal. The bathroom sink may be used for hand washing. Mechanical ventilation can be provided economically for both functions.
Disadvantages A bathroom will usually accommodate only washing and drying facilities. Other laundry related activities such as ironing, will have to be carried out elsewhere in the dwelling. Occupants may wish to use the bathroom when laundry is being washed or dried. KEY TE ¥T TvENT 6 7 (OPEN} 8 POLE t SHELF

1, z 3. 4. 5. k 7 STORAGE CLOSET CLOTHES CHUTE SORTING SHELF LAUNDRY TRAY WASHING MACHINE DRYER IRONER 8. IRONING BOARD !5 i j. _ . POLE t SHELF .TO KITCHEN [6 VENT- si i SHELF-TABLE ON CASTORS TO DRYING YAED 7 . (OPEN) 202

Residential Spaces LAUNDRY SEWING ROOMS Laundry Room Layouts
 TE CAB I NET I WORK SHELF 'Ik 6 7 (open) 8 TO KITCHEN. IE JI
 Laundry in Combination with Kitchen Advantages Suitable in housing for young families because the person doing the laundry can keep an eye on the washing machine while doing other jobs and supervising the children. Direct access to the outside for clothes drying is likely to be easier than from laundries located in a basement or on a second storey. Kitchen sinks are usually sizeable and can be used for laundering. Additional plumbing costs are usually small, Disadvantages Danger of cross-contamination through the handling of dirty washing during food preparation. Grease and cooking smells can be passed on to clean clothes, Noise generated by running appliances cannot easily be shut off from the rest of the dwelling. Laundry in Combination with Utility Space In Basement Advantages Generally, as much space as needed can be provided. Noise generated by running appliances can be easily shut off from the rest of the dwelling. Disadvantages Laundry must be carried up and down stairs, although automatic dryers have eased the problem of carrying heavy baskets of damp clothes to outdoor clotheslines. Laundry In Combination with Corridor Advantages The space is used more economically (Fig. 9). The space above the appliances may be used as a linen closet. The appliances can be hidden from sight when they are not in use; they can be recessed into the wall and enclosed with doors. Disadvantages Noise generated by running appliances cannot be easily shut off from the rest of the dwelling. An alcove adjacent to a corridor will accommodate only a minimum-sized laundry area. Other laundry related activities, such as ironing, will have to be carried out elsewhere in the dwelling, 203

Residential Spaces LAUNDRY/SEWING ROOMS Laundry Room Layouts
 Planning for Efficiency The sequence of laundering operations dq- termines the planning of space and facilities and the placing of equipment. Convenience and time-and-step saving are easily achieved by placing the elements in their natural order of use: (1) clothes chute (with or without

bins or hampers), (2) sorting table or counter, (3) washing machine, (4) laundry trays, (5) dryer (6) ironer or mangle, (7) ironing board, (8) rack, "horse," or table for finished laundry, In addition, storage closet or cabinets will be necessary for soaps, powders, bluing, bleaches, starch, basket, clothespins, iron, etc. KEY •I. Z 3. 4. Jx 6. 7 8. STORAGE CLOSET CLOTHES CHUTE SORTING SHELF LAUNDRY TRAY WASHING MACHINE DRYER IRONER IRONING BOARD KITCHEN AND LAUNDRY LAYOUT ®r TO DRYING YARD _STOB6 _IN "A" IR. B'D SHELF- TABLE © SEWING MACHINE SEWING EQUIPMENT CABINET <D IR'N OPEN TRAY CUTTING TABLE SORT SH TO KITCHEN 3 I C w.u h- ^=itt LAUNDRY-SEWING-MENDING a SEC. ® ELEVATION WASH BOAED MAY BE HUNG ON THE INSIDE OF ONE DOOR f\\ \— / -J SEC, ® ELEVATION WALL AND BASE CABINETS ARE STOCK TYPES WITH SHADOW BOX ADDED, IF SPECIALLY BUILT, CAN BE PORTABLE. JL. T © UNIT MAY CARRY CLOTHES BASKET IF TOP SHELF IS REMOVED. KEY CLOS, STORAGE CLOSET CH. CLOTHES CHUTE SORT SH. SORTING SHELF TRAY LAUNDRY TRAY W.M. WASHING MACHINE DRY DRYER IR'N IRONER IR. B'D IRONING BOARD 204

Residential Spaces LAUNDRY/SEWING ROOMS Sewing Center Clearances General Planning Suggestions 1. An area especially planned for sewing, convenient to other activity areas, is desirable. 2. Most houses need storage space for sewing materials and equipment. The amount and kind of storage required varies according to the quality and frequency of sewing. 3. A minimum sewing area should include the machine, auxiliary work surfaces, a chair that permits freedom of motion, and storage arrangements. The work surface for layout and cutting may be outside the area for sewing machine operations and serve multiple purposes. 4. Consideration should be given to work surfaces at comfortable heights for the varying activities of sewing. 5. Light should be adequate for the activity. TABLE 2 Dimensions of Area for Layout and Cutting Garments Measurement Working surface Length Width Table, free-standing Table obstructed on one side Height Clearance for worker TABLE 3 Dimensions of Fitting Space Use of Space Viewing in mirror: Mirror dimensions, in Width Length Top to floor Clearance in front of mirror, ft Width Length Clearance while fitting self, ft Clearance while being fitted, ft Fitting garment on dress form, ft

Dimensions, in Minimum 56 Minimum 16 42 70 3 6-8 6x4 8V4 x 4 5x4
 Adequate 72 28 28 34-40 (range) 18 36 32 36 (median) 24 Adequate 18 60
 72 4 10 7x6 Fig. 13 Mean heights and clearances for sewing machine use.
 Work surface and/or storage Ironing board Fig. 14 Arrangement of sewing
 equipment based on flow of work. OROP LEAF 7'-01. r\ Fig. 15 Sewing
 room. 205

Residential Spaces CLOSETS/STORAGE AREAS Figures 1 and 2 show
 the vertical clearances related to male and female closet and storage
 facilities. Wherever possible or practical, the closet shelf should be located
 within human reach. The height shown for the high shelf has been
 established based on fifth percentile male and female data in order to place
 it within reach of individuals of smaller body size. Any shelf located at a
 greater distance should be used primarily for storage that requires only
 infrequent access. The location of the shelf just above the rod is essentially
 a function of rod height, The clearance between the bottom of the shelf and
 the top of the rod should allow for easy removal of the hanger, Figure 3
 illustrates two various types of walk-in storage facilities. Undoubtedly, it
 can be argued that the 36-in, or 91.4-cm, clearance shown between the
 hanging garment and the storage shelf or between opposite garments could
 be reduced about 50 percent. The authors contend, however that in order to
 achieve any degree of comfort in the selection and removal of the desired
 garment, a minimum of 36 in should be maintained, The degree to which
 this dimension can be reduced is a question of the level of comfort the user
 is prepared to tolerate in exchange for the floor space saved. The two
 drawings of the plan view of the human figure illustrate clearances required
 for donning a coat or putting on a pair of stockings. » D -. Shoe <|=~<=§)l
 Shell.<: D o 8 8 Fig. 1 Closet and storage facilities: male. Fig. 2 Closet and
 storage facilities: female. in cm A B C D E F G H I J K L M N O 64-68 72-
 76 12-18 8-10 20-28 34-36 10-12 60-70 69-72 76 68 42 46 30 18 162.6-
 172.7 182.9-193.0 30.5-45.7 20.3-25.4 50.8-71.1 86.4-91.4 25.4-30.5
 152.4-177.8 175.3-182.9 193.0 172.7 106.7 116.8 76,2 45.7 206 Fig, 3
 Walk-in closet and storage facilities,

Residential Spaces CLOSETS/STORAGE AREAS Types of Closets
 CLOTHES CLOSETS The capacity of a clothes closet depends upon the
 accessible length of rod. Three types of closets are common. Reach-in

closet The minimum front-to-back depth of space for hanging clothes is 24". The accessible rod length is equal to the width of the door opening plus 6" on each side. Edge-in closet By providing an edge-in space of at least 18", the accessible rod length can be much longer than the door width. This requires less wall space than a full front opening.

1 Walk-in closet This type provides rods on one or both sides of an access path at least 20" wide. A wider access space within the closet may be used as a dressing area.

! Rod Lengths and Heights The Minimum Property Standards of HUD (1973) require that each bedroom have a closet, with rod and shelf, with minimum dimensions of For double-occupancy bedrooms: 24" by 60" For single-occupancy bedrooms: 24" by 36" For closet at entrance to house: 24" by 24" A more desirable front-to-back depth would be 28" for bedroom closets and 30" for entrance closets to accommodate bulky outer garments. The average rod space per garment is about 2" for women's clothing, 2 1/4" for men's clothing, and 4" for heavy coats, Recommended heights of rods are 68" for long robes, 63" for adult clothing, and 32" for children's clothing, Shelf Space and Lighting The shelf is normally located 2" above the rod, and another shelf may be located 12" higher. Shelves higher than the rod may also be installed at the end of the closet. A fluorescent fixture over the door is recommended for lighting a closet. Deluxe cool white tubes match daylight for selecting clothes, Fluorescent light MINIMUM 4'-6" To 4'-0" - SHALLOW & WIDE -4'-0" To 6'-0" - NARROW & DEEP • 6' 10" & LARGER 207

Residential Spaces CLOSETS/STORAGE AREAS {Clothes Closet! Details Use when closet is 2'-2" or more. Vire coat hangers lit Floor SECTION SECTION 5'-0" , 1/2" inside 1/2" diam, pipe PLAN Floor- Clothes closet—Closets Na rabbef LTT Itl ELEVATION TABLE 1 Garment Dimensions Men's garments Allowance per garment, Women's garments Fig. 4 Bedroom closet designed for one person. This diagram shows dimensions for rods, shelves, and drawers to hold underwear, sweaters, shoes, hats, purses, and ties. Research shows that each person needs at least 48 inches of rod space for hanging clothing- Allowance per garment, in 208 Heavy jackets and coats Medium-weight jackets, coats, and raincoats Sweaters, light-weight jackets, and raincoats Work pants: Folded on hanger Hung full length Other garments: Top coats Robes Suits (trousers full length under jacket) Trousers Jackets Sweater jacket- Shirts {all kinds) Men's garments Suit jackets, other

jackets, shirts Trousers: Folded over hanger Full length Overcoats, robes 3 2
1 2Va 1% 214 2 3 114 2 1 114 Range of lengths, in 31-40 29-37 47-53 48-
54 Coats and jackets: Heavy Medium Light Sweaters Other garments:
Dress coats, winter Robes Suits, wool (skirt under jacket) Skirts Jackets
Blouses House dresses Other dresses: Average Full-skirted Straight-line
Womens garments Blouses, jackets Skirts, medium and short coats Dresses,
long coats, short robes Long robes, long evening dresses 3 2 1 V/a 314 2
21/2 1 2 1 1% 2 214 V/a Range of lengths, in 25-35 31-43 48-55 61-68

Residential Spaces CLOSETS/STORAGE AREAS Clothes Closet Details
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HANGING-SPACEr • SHELVING • • SLIDING -TFAYS • TYPICAL
DETAILS-OF-A- DRESSING- ^QOM. • WARDRObL. Action -mo- trays-
209

Residential Spaces CLOSETS/STORAGE AREAS Clothes Closet Details
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.1/11 523 3 1 Y1 210

Residential Spaces CLOSETS/STORAGE AREAS Clothes Closet Details
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ELEVATION SPACE * 4A ® sa*i.E. ^j* ■ |io / ac^e 2»"»i-tf> W.T*
OIRCH BMELVBS NATURAL LAOPVI FINISH aMCY a 3353 haj-johop
■ N /• B 330"! BOCWtT. 13 37AT&. HAT SHELVES, FAW ELEVaf ICN
SCAi.6 3« l-<5" 211

Residential Spaces CLOSETS/STORAGE AREAS Clothes Closet Details
VLK TMKv* 4c <°• ffl&-x**"&V'i&& ^m W/AB-6 ■|%" HP Vtot ItyfW,
[AH KINtfri Cfl-\0 PLAN SECTION AT COAT CLOSET 4+ fLHfcH
|"MU- fit H\H4tTof4&r. *i'plyWd k|/ 0*6 LAH. fiM. CPU-II) I' t?IAM£TE£
VERT. SECTION AT COAT CLOSET zx~& ttte&ne w&ca ficp ptowe.
nmaeHeRAte saucer t*c* fef^M^ COtfCU&gX ShfeUr 212

Residential Spaces CLOSETS/STORAGE AREAS Closet Wall Details
cttche,!,' ~ 3 a* -^ •* =tfm L aaiut /'■ u e. shell; :s 8'-lofe ELEVATION- OF'
CLOSET' POORS I DRAWERS; ■ SECTION-THRODRAWERS- A'. IO&'

J r; !>i/>^;>^Vi" T^vt :pLAtf SEGTrOJf TH££)'DODJL
SECTWTTTO'PRAWERS 213

Residential Spaces CLOSETS/STORAGE AREAS Closet Wail Details
ELEVATION-- OF • CLOSETS • (DOORS • REMOVED) • SECTION-■
THRO-TRAYS- rner -drawer-slides SECTION-THRQ //■ / - 1 SHOE •
SHELVING • J^L ^ PLAJf POTIOR THJLD TRAYS VERTICAL^CTrOK
THKO' TRAYS • 214

Residential Spaces CLOSETS/STORAGE AREAS General Storage
Shelving 1A.WgfH/W »cM«>Ul.)) FLrari mp* &LE.VATIOM OF
STOhAgg CLOS&1S 4'-0"OP.L£SS OJ LCUgTM 0if» ■"-«>»" \V|PTM(sM
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Residential Spaces CLOSETS/STORAGE AREAS Wire Basket and
Shelving Systems SPLIT CLOSET COMBINATION CORNER .•///
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lllllimUMWWWWVS «8 .!/// mmmmmmm wmmmsmm w,. j///
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m ADJUSTABLE SHELVING 3 TIER STORAGE and HANGING 216

Residential Spaces PANTRY CLOSETS/STORAGE AREAS Wire Basket
and Shelving Systems Front View 80" Door Opening Optional Storage
Rack Multiple-stacked, wrap-around storage shelving. Optional baskets and
door racks. (9', 12', 16" and 20" widths available) Top View All the
shelving you'll ever need for full-size family food storage. Sliding baskets
hold fruit, vegetables and other kitchen supplies. Optional door racks
maximize storage area by utilizing all available space. Front View Optional
Storage Rack Multiple-stacked storage shelving. Optional full-height door
storage rack. (9', 12', 16" and 20" widths available) Top View Standard
pantry design provides ample shelving and storage for canned goods and
other food items. Center pole gives extra support. Optional door racks
provide easy access to your most needed items.

HOUSEKEEPING/UTILITY ROOM > tU Upper Storage Linen Shelf
 jHiHUHMfIttlUIWifflflrat Imiim.'WishiimimiiiHiM TnUIC'fn''''1"
 litilliliakiiU. Liit Door Opening _ . Optional Storage Rack Front View
 Double, full-width upper storage shelving with stacked storage shelving.
 (12" 16" widths available) 80 Top View Makes housework easier to handle
 by storing household cleaning items just where you need them. Plenty of
 shelving space for cloths, detergents and brushes. Wide storage area holds
 vacuum cleaner, brooms, mops and small appliances. 13" 13' 13 13 15
 Upper Una tUIMIillMS ^IlfflililillS [umimiara lltlillllillltt Storage n Shell
 ummuiuimibt mnmunimnrf 1 £rmtn!innsj pHtlillillS.'lHl! . - Wall Dimension
 - Door Opening U Front View Double, full-width storage shelving with
 side-mounted shelving and basket unit and optional door/wall storage rack.
 (12;' 16" widths available) Top View Make a clean sweep of cleaning with
 full- length shelves that hold a variety of utensils. Storage baskets pack
 brushes, cloths and sundry items. Bottled detergents and cleaning products
 can be stored neatly and safely in optional door racks. 217

Residential Spaces CLOSETS/STORAGE AREAS Wire Basket and
 Shelving Systems BEDROOM APPLICATIONS ^E Upper Storage Shelf &
 Rod 3^ ^ ^ Shelf & Rod _ , T dLi Front View - Wall Dimension - Single and
 double hang with upper storage, center pole support and shoe racks. (12','
 16" widths available) Top View Combination convenience for single and
 double hanging clothes. The perfect his and hers closet. Extra wide shelf
 space for clothing, linen and blankets in your master bedroom. Plus lots of
 room for ■ her long dresses and coats —his shirts, suits and slacks. Shoe
 racks on both sides. st Upper Storage Shell & Rod Hod ,| JnnstBn Linen
 ffflpsnrria bsBIKU. Shoe Hacks , , Wall Dimension - Front View I Li= r—
 k 4 TTPfflTTT 1 1 3= Door Opening = _ = t~ = -.■ _ =: B Walk-In. Single
 and double hang with upper storage and central shelving unit with
 additional clearance and shoe racks. (12" 16" widths available) Top View
 Single hanging space for coats and other long garments. Double hanging
 convenience for shorter garments. Full shelves with central storage unit
 allow easy storage of sweaters, boots, sports equipment, tall and over-sized
 items. Tailor-made for couples with a 2nd bedroom. CHILDREN'S
 CLOSETS Upper Storage Shelf & Rod Sliding " Baskets llaUIIiiiililllUil:
 -.ja muid Front View 80 uUtbUltU&>Uiit'lilmillUk4iklJ(| Door Opening
 Full-width, double hanging with lower shelving height. Sliding basket

system and shoe racks. (12', 16" widths available) Top View Specially designed for the children's room. Extra low- hanging shelf makes it easy for kids to reach. Stores toys and sports equipment in easy-access sliding baskets. Shoe rack keeps sneakers and other footwear neatly organized. - Wall Dimension f+t(fl-HhHII(lir llllil Uiu illl Door Opening ilium Front View Standard. Double hang with shoe rack and off-center pole support. (12" 16" widths available) Top View Makes kids stuff out of chaos in any teenager's room. Plenty of storage space for footballs, beach equipment, basketballs, skates and other cumbersome items. Doubles as storage area for dresses and coats. Conveniently placed hanging rod for all your teenager's clothing. 218

Residential Spaces LINEN CLOSETS/STORAGE AREAS Wire Basket and Shelving Systems Linen Shelf)BnnBtS!fliff3E[]Btffif«fl ttitttti(HBIfIWIWffIS) bt«ttli!«4H4iS^ 1 BiHtHfl I TilillB^ Front View Multiple-stacked linen shelving with pole support and sliding basket system. (9; 12,' 16 "and 20" widths available) Top View Four extra-wide shelves for linen and blankets. Storage baskets slide out and hold dish cloths, pillowcases and smaller items. The perfect linen closet. Linen Shelf]wmuiuuuiHH[)twwwmtfffgB(ttfflimmuuuuaa(jBBJEfflBTHUffi^ *B 3B(^Hwwtwwmrm 80" Front View Multiple-stacked linen shelving, (9,' 12;' 16" and 20" widths available) Top View Bathroom linen closet stores towels, sheets and cleaning supplies in one easy-access area. Wall Dimension - Door Opening FOYER/FRONT ENTRY CLOSET UpporStorage Linen Shelf i!!i;,,,,,,,,,,,,; XST. Front View - Wall Dimension - Single hang with upper storage and off- center storage unit and shoe racks. (12,' 16" widths available) Top View A welcome addition to any home. Full- width, upper storage holds hats, gloves and sweaters. Off-center storage for umbrellas and winter items. Shelves, shoe racks and generous hanging space lets guests know they're welcome. nlll liiHlllHiliJl f f"HM|ff+- iiiiliiiiiii Door Opening wut4.mmun|iur liiiiriuiiiiiiutiu^ k Shelf & Rod t_3c ~^C ,__-?V JI- j Upper Storage Linen Shelf Shoe Rack -rff",. Front View Single hang with half- length shoe racks and upper storage. (12'; 16" widths available) Top View Holds coats, hats, shoes and guest so clothing with care. Upper storage area for visitor's bags and small cases. ■ Wall Dimension - n^umi' |i>» I'nii'im ^llililliKillullUui^rj Door Opening 219

Residential Spaces CLOSETS/STORAGE AREAS Wire Basket and Shelving Systems MASTER BEDROOMS N Front View Walk-In. Single hang with upper storage and central shelving/ basket unit and shoe racks. (12" 16" widths available) Top View Keeps shoes, shirts and clothing neatly organized. Sliding baskets for easy access to linen, underwear, etc. Full-length clothes storage for dresses, shirts and suits. Ideal for master bedroom. X^ Upper Storage Shelf & Rod Shoe Racks psi ^N I^M* SIBBB Front View -Wall Dimension - Walk-In. Single and double hang with upper storage, central shelving and shoe racks. (12', 16" widths available) 80 Tbp View Hang dresses and coats on one side, suits and shorter garments on the other. Central shelving actually replaces a piece of furniture in the master bedroom! Upper Storage K Linen Shelf]mimmimtuniiiimmiljiuifiHUMn3[Shelf & Rod JWuitiadr-jtetiiirrtttB Sliding ' Baakets — Wall Dimension - ^\ ■ttfl|t||H"ti(H|t<titHH|ij[HWfm^mf ■ti 11 HttfctAfctti tti ttbiJteH itttlunJj Door Opening Front View Single hang with upper storage and central shelving and basket unit, additional clearance and shoe racks. (12" 16" widths available) Top View So well designed it actually replaces a piece of furniture! Four sliding baskets provide multiple storage capacity for shirts, underwear, socks and sweaters. Full-length clothes hanging space, full-width shoe racks and lots of shelf space make this system a must for your 2nd bedroom. laufstmaaa mm Upper Storagn Linen Shelf HitmiiHmm I ^^ ^BMpq Front View Single hang with upper storage and full-width shoe racks. (12" 16" widths available) Top View Doubles shelf/storage space. Single hanging for clothes, coats, shirts and jackets. Expands easily to accommodate future needs. Two full-length shoe racks. -Wall Dimension-tiiitttiilWjjJiyiiiitiilli-t±Hiri±i| Door Opening 220

Office Spaces General offices and multiple workstations 223 Private offices 231 Electronic workstations 241 Conference rooms 249 Reception areas 260 Furniture, furnishings, and equipment 278

Office Spaces INTRODUCTION The amount of office space built during the past few decades can be measured in the hundreds of billions of square feet. Within these buildings, workers spend nearly half their waking hours and a third of their entire lives. Over the life span of a typical office building, the same spaces may be occupied by a succession of different tenants, each with their own programmatic requirements. Consequently

interior spaces may be recycled and redesigned many times, simply to accommodate the changing needs of new corporate users. In many instances redesign may be necessitated solely by the effect of technological change on the methodology of transacting business. Moreover, the escalating costs of land acquisition and construction and the increasing scarcity of urban building sites make it essential that the redesign reflects an efficient, cost-effective utilization of space, as well as one that is responsive to the human factors involved. It is necessary, therefore, for the designer to be familiar not only with the general planning criteria associated with office design, but with the architectural detailing of some of the typical interior elements contained within these spaces. Accordingly, this section includes general planning criteria and examples of actual working drawings of typical interior conditions, prepared by various design professionals. The details alluded to include such items as trading desks, elevated computer floors, library furniture, built-in storage cabinets, work counters, wall paneling, vanities, reception desks, and conference room elements. Also included are illustrations and dimensional data pertaining to typical office furniture, equipment, and electronic media storage. 222

Office Spaces GENERAL OFFICES AND MULTIPLE WORKSTATIONS

Planning Data: Basic Workstations The so-called general office takes on a variety of forms and configurations. In its simplest variation it may be nothing more complex than several standard desks with returns located within a room or space. In its more sophisticated and ergonomically designed form, the general office may be based on an open planning or office landscaping concept, involving a system of workstations. The workstations include desk surfaces, files, acoustic partitions, and a host of other optional components to suit the nature of the particular work tasks involved. The systems are extremely flexible, allowing the workstations to be configured in a variety of shapes. Provision for power and lighting is quite common. The design of the general office, like the design of the private office, requires a knowledge of the basic dimensional requirements and clearances of the workstation and, where applicable, of the visitor seating to be accommodated. In certain instances, where customized and/or built-in storage elements, work counters, credenzas, etc., are required, a knowledge of architectural woodworking, as may be related to the design of such elements, can be quite helpful. Accordingly, this part includes basic

planning criteria for general office design, in addition to examples of architectural woodwork details in connection with some of the more common customized components of general office spaces. The basic workstation, as illustrated in plan in Fig. 1, is the fundamental building block in understanding the anthropometric considerations for the planning and design of the general office. The worktask zone must be large enough to accommodate the paperwork, equipment, and other accessories that support the user's function. The work/activity zone dimension, shown in Fig. 1, is established by the space requirements needed for use of the typical return. In no case should this distance be less than the 30 in, or 76.2 cm, needed to provide adequate space for the chair clearance zone. The visitor seating zone, ranging in depth from 30 to 42 in, or 76.2 to 106.7 cm, requires the designer to accommodate both the buttock-knee and buttock-toe length body dimensions of the larger user. If an overhang is provided or the desk's modesty panel is recessed, the visitor seating zone can be reduced due to the additional knee and toe clearances provided. The specific type and size of the seating (i.e., if it swivels or if it has casters) also influence these dimensions. Figure 2 shows the typical workstation expanded into the basic U-shaped configuration. The work/activity zone dimension range is shown as 46 to 58 in, or 116.8 to 147.3 cm; additional space is needed to allow for drawer extension of the lateral file. Not only does it provide more storage, the lateral file unit is generally the same height as that of the worksurface and is often utilized as a supplementary worksurface. The distance between this unit and that of the primary worksurface must be sufficient to allow for movement and rotation of the chair.

Workstation
 Zone g ^ " Worktask=°^v= D |v? Zone Work/Activity Zone 3 <§> L =M
 Chair Clearance Zone K laaaaaoaaac d a ra a a » » in cm A B C D E F G H I
 J K L M N 0 90-126 30-36 30-48 6-12 60-72 30-42 14-18 16-20 18-22 18-
 24 6-24 60-84 24-30 29-30 15-18 228.6-320.0 76.2-91.4 76.2-121.9 15.2-
 30.5 152.4-182.9 76.2-106.7 35.6-45.7 40.6-50.8 45.7-55.9 45.7-61.0 15.2-
 61.0 152.4-213.4 61.0-76.2 73.7-76.2 38.1-45.7 Fig. 1 Basic workstation
 with visitor seating. 223

Office Spaces GENERAL OFFICES AND MULTIPLE WORKSTATIONS
 Planning Data: Basic Workstations « ♦ f* Worktask <^ Zone aaoaaaoc M
 C=IC3C3l=IE=ie3l=IC=3C]C3 Workstation Zone H Work/Activity Zone M
 94-118 238.8-299.7 Fig. 2 Basic U-shaped workstation.

Office Spaces GENERAL OFFICES AND MULTIPLE WORKSTATIONS
 Planning Data: Basic Workstations Minimum Square Footage Standards for the Open and Screened Workstation The Nonautomated Task. Square footage workstation standards for the nonautomated task are developed primarily according to task profile, equipment, conferencing, and privacy requirements. Open No requirement of equipment or task for privacy, concentration Screened Privacy required for reading, working, thinking, calculating, meetings, confidential phone calls, elimination of visual and acoustical distractions Task Profile: Processing paper on work surface with quick turnaround. d Continued flow of material is processed as it arrives at the workspace and is passed on to either another function or to group storage. d Storage for permanent files and reference materials minimal. d Reference material accessed infrequently. Telephone tasks may require concentration. No 30x60 (76x152 cm) No 3-4 0-2 Guest chair Primary work surface Secondary work surface File drawers Shelves 41 sq.ft. Task Profile: Typewriter the primary tool for processing paper. d Continued flow of material is processed as it arrives at the workspace and is passed on to either another function or to group storage. d Storage for permanent files and reference materials minimal. d Reference material access may be frequent. Tasks may require concentration. No 30x60 18x42 (46x107 cm) 3-4 0-2 Guest chair Primary work surface Secondary work surface File drawers Shelves 41 sq.ft. Task Profile: Typewriter the primary tool for processing paper. d Continued flow of material is processed as it arrives at the workspace and is passed on to either another function or to group storage. □ Storage for permanent files and reference materials minimal. d Reference material access may be frequent. Tasks may require concentration. Limited conferencing required at the workspace. d Need to see and hear co-workers or subordinates of secondary priority. 1 30x60 18x42 3-4 0-2 Guest chair Primary work surface Secondary work surface File drawers Shelves 56 sq. ft. 225

Office Spaces GENERAL OFFICES AND MULTIPLE WORKSTATIONS
 Planning Data: Basic Workstations The Nonautomated Task Open Screened No requirement of equipment or task for Privacy required for reading, working, thinking, privacy, concentration calculating, meetings, confidential phone calls, elimination of visual and acoustical distractions Task Profile: Same as 1 with addition of extended conferencing

requirements at individual workstation. 2 30x60 No 3-4 No Guest chair
 Primary work surface Secondary work surface File drawers Shelves 65
 sq.ft. Task Profile: Same as 3 with addition of extended conferencing
 requirements at individual workstation. 2 Guest chair 30 x 60 Primary work
 surface 18 x 42 Secondary work surface 3-4 File drawers No Shelves 65
 sq.ft. Task Profile: Data Entry. d Paper, material, or information processed
 and/or maintained. d Multiple reference sources may be used on a task. d
 Reference materials used frequently. d Limited volume of supplies and
 permanent records kept at the workspace. d Electronic equipment used for
 keeping records current, information inputting, and maintaining data and
 records. d Ability to see and hear co-workers may be desirable. d Tasks may
 also require screening for concentration. No 45x45 (114x H4cm) 30x30 (76
 x 76 cm) 1-2 0-2 Guest chair Primary work surface Secondary work surfa
 File drawers Shelves 48 sq.ft. 226

Office Spaces GENERAL OFFICES AND MULTIPLE WORKSTATIONS

Planning Data: Multiple Workstations +■ 4'-h" +^4 ■ > o "su+a-f^ 5»o*
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 £i(J J 9VP" I^H 6V0 6'-c'" TM"n *• I ** •X!'^vXl->I^Hv'!-;v!,l'!!v -+^H jr
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 /6C5C?.pT- i«i'-o" -vr -ir 1-71 sa. ft, Fig. 3 Depending upon function, the
 sizes of individual and multiple workstations vary dramatically.

Size of work surface, length and depth of return, chair size, and circulation
 patterns all influence the gross square footage requirements. 227

Office Spaces GENERAL OFFICES AND MULTIPLE WORKSTATIONS

Planning Data: Multiple Workstations * -v ^-^ jj'-6" 6'.o' \ 4*^ [i i i i i 11 Q
 jllll V Q ^.o^h^-Vt- In iD'O ZzT^Q-FT- Jc ^ V-e" 7'-<s>" + i >■'" +*^r D
 v.''.ViV.'!Kvi'.v" O. D 0 22o ^a-vr, ^r) A '-£,- ^ ^o &■■■&- ->c •J- *" \ [cie
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 flJL2!s^y»'.^H"^-aii d a *>oh +>o.?x 228 Fig. 4 Floating or free-standing
 workstations tend to utilize more floor area than workstations placed
 against a wall or sharing the same wall panel. Clustering of workstations
 will ultimately result in the use of less floor area, but at the expense of

major ergonomic considerations. Decisions relative to both acoustical privacy and personal space are often sacrificed in the name of economy.

Office Spaces 44* *|« FT GENERAL OFFICES AND MULTIPLE WORKSTATIONS Planning Data: Multiple Workstations Multiple workstations can result in efficient utilization of space and sharing of expensive computer terminals and equipment. If use of computer terminals is intensive, individual CRTs should be provided. Figures 5, 6, and 7 each show eight workstations, yet the setups range in area from 448 to 1012 ft². Furniture size, function, and ergonomic considerations all affect setup. Fig. 5 11' x 7' T*^t MV I gl-[^] i?t all a «.' sw +- D a! D D a D 1BV Fig. 6 10' x 6' ICO yf. 11' x 5' 5' > % - - ■ m t n'-*" 1,'Ai" "f[^]H 1-7'O 1 *W?" \W.iW1 ?'<>" 1 D ^TM* Q m o r II Q D Q :K/*: c 1 f[^]culf,-r\Jti 44>p" IOI-2 <6q. |PT Fig. 7 11' x 6' O Q ■t—y- 229

Office Spaces GENERAL OFFICES AND MULTIPLE WORKSTATIONS Planning Data: Office Layout Fig. 8 9 ft x 12 ft, 108 ft². Fig. 9 9 ft x 14 ft, 126 IP. Fig. 10 15 ft x 16 ft, 240 IP. Fig. 11 12 ft x 25 ft, 300 IP. Fig. 12 12 ft x 25 ft, 300 fP. Fig. 13 18 ft x 25 ft, 414 (P. It is not unusual to have two or more persons share an enclosed office space. In planning this type of office space, both circulation and clearance become critically important. Door swings, the extension of file drawers, and points of entry must all be carefully considered. 230 Fig. 14 24 ft X 23 ft, 552 (P.

Office Spaces PRIVATE OFFICES Executive Workstation The design of the private office requires a knowledge of the basic dimensional requirements and clearances of the executive workstation and, where applicable, of visitor seating accommodations. In certain instances where various aspects of the office interior are customized and/or built into the construction, a knowledge of architectural woodwork detailing is also desirable. This page and the following pages include the necessary planning criteria required, as well as details of certain customized components. Executive workstation and/or desk size and configuration can be customized depending on desired image, scale, and ambience. Desks are also available in generally accepted standard sizes. It is these standard desks that are most used in the design of the private office. Figure 1 illustrates the range of desk dimensions, chair dimensions, and clearances involved. Many

private executive offices are being designed with desks that do not conform in cm A B C D E F G H I J K L M N O P Q R S 30-39 66-84 21-28 24-28 23-29 42 min. 105-130 30-45 33-43 10-14 6-16 20-26 12-15 117-148 45-61 30-45 12-18 29-30 22-32 76.2-99.1 167.6-213.4 53.3-71.1 61.0-71.1 58.4-73.7 106.7 min. 266.7-330.2 76.2-114.3 83.8-109.2 25.4-35.6 15.2-40.6 50.8-66.0 30.5-38.1 297.2-375.9 114.3-154.9 76.2-114.3 30.5-45.7 73.7-76.2 55.9-81.3 with the basic rectangular shape. Such a situation is illustrated in Fig. 2, which shows a circular executive desk. Such a desk is often selected if the executive in question plans to hold conferences within the office and prefers the psychology of having either visitors or employees gather around the work surface in an egalitarian fashion. While a minimum desk size of 48 in, or 121.9 cm, is shown, this dimension is also influenced by the number of side chairs to be grouped around the desk. A circular executive desk must be supported by supplementary credenza or file storage within easy reach of the executive chair, Side arm reach relative to the work/ activity zone must always be studied carefully. Figure 3 illustrates a typical circular lounge grouping found within an executive office. Providing for the appropriate leg clearance of 12 to 18 in, or 30.5 to 45.7 cm, is also determined by the sitting zone requirements. Buttock-knee length must also be considered. Line of Wall Or Obstruction Fig. 1 Executive desk/visitor seating. 231

Office Spaces PRIVATE OFFICES Executive Workstation in 77-88 195.6-223.5 30 76.2 46-58 116.8-147.3 22-28 55.9-71.1 24-30 61.0-91.4 24-28 61.0-71.1 2-3 5.1-7.6 20-22 50.8-55.9 48-60 121.9-152.4 92-116 233.7-294.6 36-42 91.4-106.7 6-9 15.2-22.9 24 61.0 42-60 106.7-152.4 36-48 91.4-121.9 57-78 144.8-198.1 33-43 83.8-121.9 12-18 30.5-45.7 21-30 53.3-76.2 24-32 61.0-81.3 Fig. 3 Circular lounge grouping. Fig. 2 Circular executive desk.

Office Spaces PRIVATE OFFICES Planning Data: Typical Room Arrangements i4-.gr ->rH *.!lfl" MiNiiJ "SSS! "III:! i'ii/ Mi 4— ^ ■ n.i j.jtH- 4' l*lf * Fig. 4 12 ft x 15 ft, 180 ft². Fig. 5 14 ft x 12 ft, 168 ft². Fig. 6 13 fix 12 ft, 156 ft². 1-/1" \$-0' -rB \$ \$ Fig. 7 12 ft x 13 ft, 156 ft². Fig. 8 9 ft x 15 ft, 135 ft². Fig. 9 11 fix 14 ft, 154 ft². f-e>- lO'-O" VJI Fig. 10 10 fix 11 ft, 110 ft². Fig. 11 9 ft x 12 ft, 108 ft². 233

Office Spaces PRIVATE OFFICES Planning Data: Typical Room Arrangements #V • S'-o" Fig. 12 16 ft x 21 ft, 336 ft². Fig. 13 12 ft x 23 ft, 276 m. Fig. 14 15 ft x 18 ft, 270 ft². i < _«" iZ'-o 4- Fig. 15 12 ft x 24 ft, 288 ft'. JP--0" Fig. 16 12 ft x 16 ft, 192 JP. l'^-b" an Fig. 17 12 ft x 20 ft, 240 W. The private offices illustrated in Figs. 12 to 17 reflect middle to senior management functional, as well as status, requirements. Each office layout should be carefully reviewed with the client to ensure that all programmatic functions have been met. Offices of this size do not easily accommodate an independent conference function. 234

Office Spaces Fig. 18 14 ft x 22 ft, 308 ft². ZO'-g" PRIVATE OFFICES Planning Data: Typical Room Arrangements Fig. 19 14 ft x 28 ft, 392 m. f-eZ> cP La Fig. 20 19 ft x 33 ft, 627 ft². *a'-e>%% [__,',', '>' 1 D iii"i-"r" wt nnw,',?—n D [■WM.IM.i.i.'HI n ^H4+t A^-m .* . >HW -H.'WiWA'.v ■ ■ .■;!'!.'J J'.WJ: // 235

Office Spaces PRIVATE OFFICES Wall Unit Details *■» >/ ^1'^" ■ i'-fl' a? > I' MdHtxaAkW %oak W/MMMANV VUUtlt. 4 <<<PK>>. h EOkf*. PWUM U0LS* fafc. -*HIM 7a I4M. M> ¥*SO TOfor TVtIS W^llifoT^IH-t H91/TE0 WKDwD. ftw4I*ri/kl/j fiaivipe. t-ocje. om FX-e. orl iSr e*A4-je*.. 16 Bt U»o BO*yiJ<a. M*nos..v»wi«&«* 3SCCC te-rastf**^^ It' HNUO. *T (HUM 1' rwwa (M»4U*<r>#<-«-') W/HKDWP U&i 4 4T' #p.. &ECT10H 4HEWIW6? <>>vi flU» MW 1'--1'-o' toT» <vr ww* rvu.- ■WIM IB N*U. *) Rfc^P^ MAH04A** VEWfc« SN 2.0i*e ^T\ 6L0&BT ^6^TIQK Fig. 21 Custom architectural woodwork, or "built-ins; is often required for executive offices. These architectural working drawings reflect the custom design of a storage wall for a partner in a law office. Careful analysis shows the incorporation of file, book, and coat storage within a floor-to-ceiling mahogany wood unit. 236

Office Spaces PRIVATE OFFICES Wall Unit Details 01*01 AU4«L* Essftisa^- (Hid 4m*1* H/rf HWWO iOUt- flD mutt*.*. Lamt) AUU iflli IN6H USHT PWWat H/MB* CnM« t*HINp iWOT '-V1.MOU5 *iijq«t wxmu '*• Mv/ML, #c*i\j.v*a (TYP) avHWW 4 AcuUsTX&tJS- *H«I* H/HSDwD. W*«J ALL £*POT63 -mTSSOR 6>UX*:iwi A* **e 3 ■ »> ^g6TigM klALL UNiT -r-t *4>\ 5tCT\0H Of^Hij^.\, \IM^ 0H\T cevHttKTor N/Nr H' rwkio w/hwond. eaim I AAL-WMuf fUAH OXXtfl hytWOkAWV

VtNUK..HI>&l>*i| {3rM*TUJN£**U0 AW ivt \HOsAA hNUM <l
M.OCWN* MfMOO- "H ■"IS I -. a Fig. 22 These details represent typical
vertical sections taken through various storage components for the partner
wall unit shown In Fig. 21. Careful attention must be given to Integration of
electronic equipment, electrical wiring, and task lighting. ■ A. *IM TO
HALL *>. ■'ftSO exa^tju&ii^fiflce-. jZkll y •!>«<. wSK-W'5*MW-
cdr^c,C 2." ot4 p£hlTee. ■3-leL-f 11. *< _^H-l~V'Ttlf*-'— i. rtJL-e-'S»
«=■ c*3't> «-ra><je i z"»J Ce-HrejJ=- faupy"Ace-j»-r»Be: -» */tNMfl;
TN^EC'i^n - A JNiT ^PECtA t 7 'i' l. IB 237

Office Spaces PRIVATE OFFICES Waii Unit Details ■for FiuE wit H [5 ,|
fAStiUS^r , ! ft. •1/vcK ; 1 * i ! '? t l i *-6> J. i l J L£^AL Ab^fAUf
U^^KWA vUL e^neo. p LbU.n "V ^— ■v— 'jateo. ex&e - -2-a'tf V> ?-X
J^U-4- = v«>p 1. -v-y- ftSRTNeJS'e> H^EW^J-L. <s i<£Y6a^D "n^T
ta*£*kt ^sgiee-- wp. fi&Nfc). To MIL. p,L£ Mn- F.i^ , PT Hlu^Wp)cfp. TO
MAT^ti ft--? PP^TPIKI H0l*£ «p fApfNE.R'6 N^^NA-LL- Flg. 23 in many
instances, the utilization of standard wood moldings can enhance the overaii
appearance of an otherwise relatively simple workwaii unit. Other cost-
saving devices illustrated here are the application of a wood panei to a
standard metal fiie and the use of a fabric- wrapped tack board. The
incorporation of an undercabinet task light Is almost always required. 238

Office Spaces PRIVATE OFFICES Wall Unit Details r»t* . . n»j»tD wet o*
i M*r*y yv C-UMIT gT>MPA|gf=> fe^cgtfr/CPPTlQM »/ iiBoe»M*« ae»t
•fit*- «lJM*«V ^OHJKjIT STAHPAKP gjgaf |nae>M**^il Fig. 24 These
pian oblique drawings provide detailed design information to both client
and architectural woodwork contractor. These drawings are particularly
heipfui when the office project for which these waii units are intended
consists of many offices, and each office is to be customized within certain
constraints. 239

Office Spaces PRIVATE OFFICES Credenzas A'-db ib joint l »' l Of
MAKB* TOP [,a-L l!gl -:%■ Pt-TWCBP "T T f SRrueTj oak f*nr.#6ip&w
LIEU of PtoS**? (WHkTO MATCH EXI»TIKU. \ "lKIM n»(<PIN(>»e \^
T« MAT&M BVI*T4 SECTION 6ECTIQK Many private offices require
the detailing of custom credenzas and storage units. The sophistication and
complexity of such details can significantly influence the budget for the

space as well as the time of installation. Figure 25 represents a "high-end" approach, while Fig. 26 is more appropriate for offices with a moderate budget. a^teteigB it7"8' 3: -Ut'pfWOOP MCKIHS 3T ixzwo BiocXa JZ WO GWS* RUMNERS i5 2'-o" ac. SECTION! Fig. 25 S£*-nb& TO walu 240 Fig. 26 <•> (Oj-. r-o"

Office Spaces ELECTRONIC WORKSTATIONS Planning Data:

Anthropometrics New electronic technologies, together with the advent and proliferation of the microcomputer and the availability of inexpensive packaged software, have changed the complexion of the office workplace.

The ergonomic considerations related to th is new work environment have necessitated a re- evaluation of the traditional interface between the seated office worker and his or her workplace. It is essential that the design of this electronic workstation be responsive to human factors in order to avoid physical discomfort for the user. The location of the keyboard, angle of the visual display terminal, adjustability of the chair, field of vision, provisions for back support, and height of the seat above the floor are a few of the considerations in the design process. This page and the following pages provide a variety of anthropometric and ergonomic planning data and details for use as reference in the design of the electronic workstation.

Figure 1 illustrates guidelines for use in establishing preliminary design assumptions for a workstation display console. Since the types of displays and the nature of the tasks associated with those displays can vary considerably, Fig. 1 cannot be taken too literally. The configuration shown, however, is fairly representational. Certain basic factors should be noted anthropometrically. The use of an adjustable chair will permit the eye height of the seated viewer to be raised or lowered to view the display, as may be required depending on body size. An adjustment range between 15 and 18 in, or 38.1 and 45.7 cm, should be adequate to accommodate the eye height sitting requirements of about 90 percent of all viewers. Adjustability, however, will be of little value if the vertical distance between the underside of the desk and the floor is insufficient to accommodate the knee height and thigh clearance when the seat is adjusted to the appropriate position. If such distance is not less than 26.5 in, or 67.3 cm, the majority of viewers will be accommodated. The location of the top of the display should align with the standard sight line for optimum viewing conditions. Since the eye and the head can rotate within certain limitations and, in so doing, increase the area

that can be scanned, displays can be located above the standard sight line when absolutely necessary. It should also be noted that the more perpendicular the normal sight line is to the display plane, the greater the viewing comfort. Accordingly, consideration should be given to sloping the display plane since the normal sight line is about 15° below the horizontal. Stature is the vertical distance from the floor to the top of the head, measured while the subject stands erect, looking straight ahead. Elbow height is the distance measured vertically from the floor to the depression formed at the elbow where the forearm meets the upper arm. Eye height is the vertical distance from the floor to the inner corner of the eye, measured with the subject looking straight ahead and standing erect. Sitting height erect is the vertical distance from the sitting surface to the top of the head with the subject sitting erect. Sitting height normal is the vertical distance from the sitting surface to the top of the head, measured with the subject sitting relaxed. Eye height is the vertical distance from the inner corner of the eye to the sitting surface. Shoulder height is the distance taken vertically from the sitting surface to a point on the shoulder midway between the neck and acromion. Shoulder breadth is the maximum horizontal distance across the deltoid muscles. Elbow to elbow is the distance across the lateral surfaces of the elbows measured with elbows flexed and resting lightly against the body with the forearms extended horizontally. Hip breadth is the breadth of the body as measured across the widest portion of the hips. Note that a hip breadth measurement can also be taken with the subject in a standing position, in which case the definition would be the maximum breadth of the lower torso. Elbow rest height is the height from the top of the sitting surface to the bottom of the tip of the elbow. Thigh clearance is the distance taken vertically from a sitting surface to the top of the thigh at the point where the thigh and the abdomen intersect. Knee height is the vertical distance from the floor to the midpoint of the kneecap. Popliteal height is the distance, taken vertically, from the floor to the underside of the portion of the thigh just behind the knee while the subject is seated with body erect. The knees and ankles are usually perpendicular, with the bottom of the thigh and the back of the knees barely touching the sitting' surface. Buttock-popliteal length is the horizontal distance from the rearmost surface of the buttock to the back of the lower leg. Buttock-knee length is the horizontal distance from the rearmost surface of the buttocks to the front of the kneecaps. Buttock-toe length is the horizontal distance from the

rearmost surface of the buttocks to the tip of the toe. K- 4=c Display
 Controls - Possible - Keyboard Opt. Viewing | Zone Buttock-heel length is
 the horizontal distance from the base of the heel to a wall against which the
 subject sits erect with his leg maximally extended forward along the sitting
 surface. This is sometimes referred to as buttock-leg length. Vertical reach
 is the height above the sitting surface of the tip of the middle finger when
 the arm, hand, and fingers are extended vertically. Vertical grip reach is
 usually measured from the floor to the top of a bar grasped in the right hand
 while the subject stands erect and the hand within which the bar is grasped
 is raised as high as it can be conveniently without experiencing discomfort
 or strain. Side arm reach is the distance from the center line of the body to
 the outside surface of a bar grasped in the right hand while the subject
 stands erect and the arm is conveniently outstretched horizontally without
 experiencing discomfort or strain. Thumb tip reach is the distance from the
 wall to the tip of the thumb measured with the subject's shoulders against
 the wall, his arm extended forward, and his index finger touching the tip of
 his thumb. Maximum body depth is the horizontal distance between the
 most anterior point on the body to the most posterior. Anterior points are
 usually located on the chest or abdomen while the posterior points are
 usually found in the buttock or shoulder region. Maximum body breadth is
 the maximum distance, including arms, across the body. in cm A B C D E F
 16-18 16 min. 18 min. 15-18 adjust. 26.5 min. 30 40.6-45.7 40.6 min. 45.7
 min. 38.1-45.7 67.3 min. 76.2 Display and Set-Up Controls Fig. 1 241

Office Spaces ELECTRONIC WORKSTATIONS Planning Data:

Anthropometrics ■ Both the work surface and the display monitors must be
 lowered and raised as a unit with 31.8 cm of travel. ■ The work surface
 must be tilted anywhere between a horizontal position to 35° below
 horizontal. The work surface, at its lowest setting and with a 10° tilted
 angle, as is common in use, must be 63.5 cm in height at its front edge. ■
 The work surface must raise to a horizontal height of 104 cm,
 accommodating a majority of people in a standing position. ■ The monitor
 screens must be tiltable to any position between 15° forward of vertical and
 15° back. This lets the user adjust the screen to avoid reflective glare, and it
 accommodates various working positions of different lines of sight. ■
 Adjustment controls designed for hand operation must be located within the
 operator's extended reach envelope. ■ All surfaces must have matte or dull

finishes. This reduces the likelihood of reflective glare. ■ The workstation must be compact and relatively easy to move through a standard 81-in doorway. ■ No structural components shall exist which inhibit the workstations operation by users in wheelchairs, ensuring a barrier-free workstation. ■ Service personnel must have easy access to electrical components. ■ The digitizing surface must accommodate standard European and American D size drawings. ■ Screen depth of view must allow alphanumeric characters to be viewed at an angle between 20 and 28 arc minutes. 28 4 to 33 in (16710 83 8 cm) maximum reach 22 S10 27 m. 157.2 10 68.6 cm) extended reach 15 to 19 in (38 I 10 48 3 cm) normal reach Extra work surface and storage area Frequently used items within easy reach -Central field of view Extended field of view. maximum symbol recognition Binocular field of view maximum color discrimination Screen angle adjusts 15° from vertical. \ in either direction 242

Office Spaces ELECTRONIC WORKSTATIONS The Automated Task. Square footage workstation standards for the automated task are also developed primarily according to task profile, equipment, conferencing, and privacy requirements. Open No requirement of equipment or task for privacy, concentration Screened Privacy required for reading, working, thinking, calculating, meetings, confidential phone calls, elimination of visual and acoustical distractions Task Profile: Data Retrieval. □ Paper, material, or information processed, analyzed, and/or maintained. □ Multiple reference sources may be used on a task. d Reference materials used frequently, d Limited volume of supplies and permanent records kept at the workspace. d Electronic equipment may be used for easy reference, retrieval, keeping records current, and maintaining data and records. d Additional equipment such as microfilm viewers may be required, d Ability to see and hear co-workers may be desirable. d Tasks may also require screening for concentration. No 45x45 30x45 (76x114 cm) 3-4 0-2 Guest chair Primary work surface Secondary work surface File drawers Shelves 56 sq.ft. Task Profile: Shared Tasks. d Paper, material, or information processed, analyzed, and/or maintained, d More than one task may be performed concurrently. d More than one operator uses same equipment.' d Multiple reference sources may be used on a task, d Reference materials used may be used frequently. d Electronic equipment may be used for easy reference, inputting/maintaining data and records, retrieval, keeping records

current. d Storage requirements vary according to task. No Guest chair 30 x 45 Primary work surface 30 x 60 Secondary work surface (76 x 152 cm) 1-2 File drawers 0-4 Shelves 81 sq.ft. Task Profile: Administrative Specialist/Secretarial. □ Paper, material, or information processed, analyzed, and/or maintained, d More than one task may be performed concurrently, d Multiple reference sources may be used on a task, □ Reference materials used frequently. a Electronic equipment may be used for easy reference, retrieval, keeping records current, inputting/maintaining data and records. □ If supervising, ability to see subordinates may be desirable to direct activities, □ If monitoring, visual access may be desirable. d Moderate amount of storage required at the workspace, that is, casework, client accounts, supplies, No 45x45 30x60 3-A 1-2 Guest chair Primary work surface Secondary work surface File drawers Shelves 64 sq. ft.

Office Spaces ELECTRONIC WORKSTATIONS The Automated Task Open No requirement of equipment or task for privacy, concentration Screened Privacy required for reading, working, thinking, calculating, meetings, confidential phone calls, elimination of visual and acoustical distractions Task Profile: Administrative Specialist/Secretarial a Paper, material, or information processed, analyzed, and/or maintained, □ More than one task performed concurrently. □ Multiple reference sources used on a task. □ Reference materials used frequently, □ Limited volume of supplies and permanent records kept at the workspace, □ Electronic equipment may be used for easy reference, retrieval, keeping records current □ Tasks are complex enough to require concentration, □ Extensive use of telephone and additional equipment such as desk-top printer and microfilm viewer may be required, □ Need to see and hear co-workers is secondary priority. d Limited conferencing required at workspace. □ If supervising, ability to see subordinates may be desirable to direct activities. □ If monitoring, visual access may be desirable. + Guest) 1 45x45 30x60 3-4 1-2 Guest chair Primary work surface Secondary work surface File drawers Shelves 80 sq.ft. Task Profile: Word Processing. □ Time divided among administrative, processing paper, material, or information. □ More than one task may be performed concurrently. □ Multiple reference sources may be used on a task. □ Reference materials moderate but used frequently. □ Limited storage primarily for supplies. □ Ability to see and hear co-workers or subordinates

is desirable, d Typewriter and/or electronic equipment may be used to expedite processing and administrative tasks, for example, VDT, printer, transcriber, OCR, microfilm viewer, separate disk drives. No 45x45 (114x114 cm) 30x45 (76x 114cm] 3-4 1-2 Guest chair Primary work surface Secondary work surface File drawers Shelves 67 sq.ft. Task Profile: Word Processing (+ Guest), a Time divided among administrative, processing paper, material, or information, and limited conferencing at workspace, d More than one task may be performed concurrently. □ Multiple reference sources may be used on a task, □ Reference materials moderate but used frequently. □ Limited storage primarily for supplies, □ Typewriter and/or electronic equipment (VDT, printer, and so on) may be used to expedite processing and administrative tasks. p Tasks are complex enough to require concentration for analysis, or heavy equipment operations require acoustical screening. □ Work surface needed for organization of work. 1 45x45 30x45 3-4 1-2 Guest chair Primary work surface Secondary work surface File drawers Shelves 86 sq.ft.

Office ELECTRONIC WORKSTATIONS The Automated Task Open No requirement of equipment or task for privacy, concentration Screened Privacy required for reading, working, thinking, calculating, meetings, confidential phone calls, elimination of visual and acoustical distractions Task Profile: Technical/Systems Analyst/Programmer, d Time divided among administrative, processing paper, material, or information, and limited conferencing at workspace. d More than one task may be performed concurrently. d Multiple reference sources may be used on a task. d Reference materials may be extensive and used frequently. d Ability to see and hear co-workers or subordinates desirable. □ Typewriter and electronic equipment (VDT, printer, and so on) may be used to expedite processing and administrative tasks. d Moderate to extensive amount of storage required at the workspace for manuals, binders, computer printouts, coding sheets, supplies, permanent files, reference materials. 1 45x45 30x60 (76 x 152 cm) 6-8 3-5 Guest.chair Primary work surface Secondary work surface File drawers Shelves 128 sq.ft. Task Profile: Administrative/Managerial. d Extensive conferencing at individual workspace. d Analysis of reports, computerized materials, and soon. d Varied tasks or projects performed simultaneously on an ongoing basis. d Large amounts of storage extensively used. d Storage for client/project files, reference manuals, documentation,

correspondence. d Telephone used extensively. d Supervision of subordinates almost universal. d Electronic equipment accommodation is secondary priority, used primarily for communication/electronic mail, scheduling. 2 45x45 30x60 (76x152 cm) 5-6 3-5 Guest chair Primary work surface Secondary work surface File drawers Shelves 154 sq.ft. Task Profile: Administrative/Total Enclosure. d Extensive conferencing at individual workspace. a Analysis of reports, computerized materials, and soon. d Varied tasks on projects performed simultaneously on an ongoing basis. □ Large amounts of storage extensively used. a Storage for client/project files, reference manuals, documentation, correspondence. d Telephone used extensively. a Supervision of subordinates almost universal. a Electronic equipment accommodation is secondary priority, used primarily for communication/electronic mail, scheduling. □ Subject matter of job responsibilities requires confidentiality. 2 30x60 20 sq ft (15sqm) 5-6 3-5 Guest chair Primary work surface Secondary work surface File drawers Shelves 150 sq. ft.

Office Spaces ELECTRONIC WORKSTATIONS Trading Desk Details
 VENT SU=T: V_ ^fycH CIS 9" C^WNIX' •1 FUCH CI5 S" CtfTX^NIX
 PLASTIC L/VMNATE ^ <(7N5T. plan/sexti^ n Tf^At>INO CesK TYPE- '
 A' PLAN - Z^TH f S"HIGH " ^v- 0 F&PNT VIEW -1TAPINO i>ESrC
 TYPE 'A1 Fig. 2 Technologically and electronically complex trading desks must be economically correct in every respect. With little, if any, margin for error when *4b designing and detailing multiple workstations of this type, a full-size mockup Is always required.

Office Spaces ELECTRONIC WORKSTATIONS Trading Desk Details >-E
 (CNVEM^K. CdT- ^ r- —\f- "r \ "^^ ^[^jjm^ Stzjr^oc, ^ * - ©
 FF&OAOtAK FACTOR - PLAN © EL&VAn^N|/<5ECTT^N 0 TYPICAL
 PLAN LAYOUT ■FLASTK UAMII4ME, VIH£ «fL«N i=N
 LNDRSICC.^FOMc LBS AT QSWYCY^ 19 ->/»" INTEfU^ BNKn3>
 RAT BLACK CRAMMER) PLASTIC L/Mi^VfE- TSEa YKXXX.-
 WIREfcVU) BY ELECTXKIAW (AINTCb B«E ■GRAt£' P-Ar--
 LAMIN/NTC 2x4- rPAMiMQ Mk -fl^I? fAINTQ) 4-"tt\MeTER- IX RPC
 W Olf-^IUi FNI3H ."«"k i" stl's vet tC))1Jf tzSTpto HATE.H^LE VSP § 0
 SECTION THRU TYfKAL D55K. 247

Office Spaces ELECTRONIC WORKSTATIONS Trading Table Details ^
 *.!<?• TV QfiU^M 0? TKEASUPY T7WNO 1A£>U= QpAjUIAU
 ELCV^sTI^N Qscen&hi 05BCH^N 248

Office Spaces CONFERENCE ROOMS Planning Data: Table Sizes and Seating Capacities idooQac Circulation Zone Fig. 1 Square conference table. Consideration must be given to clearances and circulation around the larger conference table, as indicated in Figs. 1 and 2. A minimum of 48 in, or 121.9 cm, is suggested from the edge of the table to the wall or nearest obstruction. This dimension under ordinary circumstance allows for a circulation zone beyond the sitting zone of 30 to 36 in, or 76.2 to 91.4 cm, based upon a maximum body breadth measurement of the larger person. The greater dimension is recommended to allow for the chair in a pulled-out position. The actual dimensions of the conference table are a function of the number of people to be seated. The square table illustrated in Fig. 1 provides for eight people, with each side ranging from 54 to 60 in, or 137.2 to 152.4 cm. The larger dimension is more appropriate to accommodate people of larger body size and to allow for a more generous work zone for each person. This translates into 30 in, or 76.2 cm, per person, which constitutes a comfortable perimeter allocation. The circular table shown in Fig. 2 comfortably accommodates five people while allowing for a 30-in, or 76.2-cm, access zone between chairs. To accommodate both sitting zone and circulation zone, a space with a radius ranging from 72 to 81 in, or 182.9 to 205.7 cm, must be provided. in cm A B C D E F G H 1 J K L 48-60 4-6 20-24 6-10 18-24 30-36 54-60 30 72-81 42-51 24-27 48-54 121.9-152.4 10.2-15.2 50.8-61.0 15.2-25.4 45.7-61.0 76.2-91.4 137.2-152.4 76.2 182.9-205.7 106.7-129.5 61.0-68.6 121.9-137.2 Fig. 2 Circular conference table. 249

Office Spaces CONFERENCE ROOMS Planning Data: Table Sizes and Seating Capacities Round conference tables offer the advantages of intimacy, "equality," and compact' ness. On the other hand, if status is an issue, or if one wall within the space Is an audiovisual wall, this table shape can be less than satisfactory. The same problems can arise with a square conference table. In both instances, however, the total seating around each table shape must be viewed in the context of chair size, chair spacing, and tasks to be performed at the table. ROUND TABLES Recommended

Approx. Minimum Diam. Circum. Sealing Room Size 10'0" I r,n 3i'5" ra-15
 20'0"X 20'0" y'o" 28'2" n-14 iy'o"x ry'o" N'o" 25'i" io-r2 iS'3"x 18'3 " 7'o"
 22'o" y-n i6'o"xi6'o" 6'o" iS' 10" 8-y I4'6"xi4'6" \$'o» 15'j)" 7-8 i3'o"xi3'o"
 4'o" iz'6" 5-6 ii'6"xii'6" 3'C" n'o" 4-5 io'6"x io'6" SQUARE TABLES w 5'0"
 4'6" 4'o» 3'6" 3'o» L 5'o» 4'6" 4'o" 3'(3" 3'o* Approx, Sealing 8-12 4-8 4-8
 4 4 Recommended Minimum Room Size i3'o"x i3'o"]2'0"X I2'0" n'C'x 11'
 6 " io'6"x io'6" y'o"x y'o" 250

Office Spaces Rectangular and boat-shaped conference tables lend themselves toward formal settings where status and hierarchy are important. Both table shapes are also more suitable in a room where an audiovisual wall is placed at one end of the space, or where speakers are making presentations. The boat-shaped table also offers greater visibility of others seated at the table, as well as ease of circulation around its perimeter.

CONFERENCE ROOMS Planning Data: Table Sizes and Seating Capacities RECTANGULAR TABLES w Approx. Sealing Recommended Minimum Room Size fi'o" 28'o" 28-30 iX'o"x4o'o" fi'o" 26'o" 26-28 IX'0"X3X'0" fi'o" 24'o" 24-26 j8'o"X36'o" S'o" 22'0" 22-24 i5'o"X32'o" 5'0" 20'0" 20-22 T5'0"X30'0" 4'rt" fX'o" 1X-20 i3'r,"x27'o" 4'6" i6'o" 16-18 !3'r)"x25'o" 4'r," i4'o" 14-16 i3'r,"x23'o" 4'o" 13'0" 12-14 I2'0"X2I'0" 4'0" 4'o" 4'o" 4'o" 3'6* 3'r," 3'r," 3'r,« 3'r,- 3'o" 3'o» 2'r," l'6" I2'0" II'O" io'o" y'6" y'o" X'r," 8'o" 7'r," 7'o" <5'C>" r,o" S'6- i'o" 12-14 10-12 10-12 8-10 8-10 8-10 8-10 6-8 6-8 6-8 6-8 4-r, 4-r, I2'0 I2'0 I2'0 I2'0 10'r> toy, 10'r, 10'6 to'r, io'o io'o y'o y'o x 20'0" " x j y' 0" "x i7'o" "x n6'6" "x 1 r,' 0" "x t5'6" x 15' 0" "x r4'6" "X T4'0" x 13'r," " x 13' 0 " "X T2'r," "x r2'o" BOAT SHAPED TABLES 3'5" 8'o" 3'8" y'o" 3'it" io'o" 4'3" n'o" 4'7" i2'o" 4'11" i4'o" 8~io io'o"x i5'o" 8-10 n'o"xi6'o" 10-12 I2'0"X I7'0" 10-12 13 'o"x ty'o" 12-14 i4'o"x2i'o" 14-16 i5'o"x23'o" 5'3" i6'o" 16-18 i6'o"x26'o" 5'7" iS'o" 20-22 i7'o"x29'o" fi'O" 20'0" 20-24 I8'o"X32'0" 251

Office Spaces CONFERENCE ROOMS Planning Data: Table Sizes and Seating Capacities CONFERENCE/MEETING ROOMS Solid Conference For 20 people B»l 4-30"x72" tables 2-30"x 60" tables Race Track For 26 people \mmmJ 6-30"x72" tables 4-30" wide crescents Trapezold/Round For 12 people 6-30"x30"x30"x 60" tables V-Shape For 20 people .•2.S •••• •••? ••• I Boat Shape For 28 people ...^iii;... •••^T7TT.« BANQUET ROOMS 60" diameter tables Capacity: 180 people 30" x 96" tables

Capacity; 180 people Mill II III 18" x 72" classroom style tables Capacity: 162 people • • • • Table Centerline Size Capacity Spread' 48" dia. 6 persons 7'5" 54" dia. 6-8 persons 7' 10" 50" dia. 8-10 persons 8'3" 66" dia. 10 persons 8'8" 72" dia. 10-12 persons 9'1" Mill 1 1 1II Table Size Capacity 30"x48" 4-6 persons 30"x60" 6 persons 30"x72" B persons 30"x96" 10 persons Table Size 18"x60" 18"x72" 18"x96" Capacity 2 persons 3 persons 4 persons 252

Office Spaces 4E^pj»a CONFERENCE ROOMS Planning Data: Table Sizes and Seating Capacities f&*fr[Z. &eusfc,\o £&*& \%* &£&&> \Q 4x&sp,e> lit ii ii i,i «5g^=> & *&#&> e> &J*fr6 W^ 4&&ps> 4- «EAfS.;2. 1 <5€A-pb4- Fig. 3 These conference tables are useful in making initial space planning allocations. 253

Office Spaces CONFERENCE ROOMS Planning Data: Table Sizes and Seating Capacities Seats 8,323 ft2 \r-o" Seats 12, 324 IP H'-o XS'-O" Seats 12, 484 ft' Seats 12,532 ft2 0. M^^^m^m V- ' IAI P- ',0" Seats 50,1058 ft2 Fig. 4 These drawings provide the designer with a variety of conference room sizes, table shapes, floor areas, and seating capacities. They are useful in client discussions and In making preliminary area allocations. Chair size and circulation areas behind the chairs will, of course, cause overall dimensions to vary. 254

Office Spaces CONFERENCE ROOMS Planning Data: Room Layouts Seats 8,272 ft2 15'-0" TV PC£Z£ teq ¥1 ?b'-o* Seats 12, 338 ft2 /y'-0* Seats 10,294 ft2 f QQ fla^aa ^Kaa/v^ ^■-o" 23' ' _£>" Seats 18, 480 ft2 Seats 16, 690 ft2 255

Office Spaces CONFERENCE ROOMS Table Base and Edge Treatments It is important for the designer to understand and appreciate some of the important details that make up a conference table. The base treatments shown in Fig. 5 are but a few of the myriad possibilities. Perhaps even more important to consider are the finished edges of glass and wood conference tables, - - v&jWIT representative details of which are shown in IW-6^ n Figs. 6 and 7. Other edge details could be made of marble, granite, or even leather Fingers, hands, and arms make intimate clMMf*£*r'iSf contact with these edge details, something that should be carefully considered. IB" Diameter Cylinder 24" Diameter Cylinder Q ThrBe 12" Diameter Cylinders

Wood Slab (SI," thick) 4" Diameter Cylinder Lag » Cylinder Base 1^—"H
Fig, 5 Base treatment. QOMCKP <JOU. W&5E •people &©/&- Fig. 6
Glass edge treatment. pjitr voumfr pe^cit VcJ-isdex? p0j&-e. Tewtit-
e.&.&us cfr-eeiw- cowclev6?scierr r L ...// ^ -I I I Fig. 7 Wood edge
treatment. 256

Office Spaces CONFERENCE ROOMS Conference Table Details W*UP
TAPf-fftlif SECTION TABLE TtfP NoTpl VOT- VIV-P&V TAPeREC*
C.3SN- PREPARE *UP6TKA-TB A* RB^P 1*>K WBWIS4 . CLEAN f
pRg#S> WBH.t?£ A* KPO't? PLAN-CONFERENCE TABLE "-'B'V
fiKBWW VEWBSK^ fit\OP SPCUSH CENTER _ with *<snew rietfM top
o TASte 180 Pixv/oop SECURE L-E!4 U*INO 9CKBW* PfcJM
UNTSHftlPe (TT* 1*»Li! LC« liSB NO tSUJe 257 SECTION

Office Spaces CONFERENCE ROOMS Conference Table Details - l/i» 9*
Mm P iz'cbhteks EXCEPT tf HCLD JOIHT5 iwu&lc i/eV 3W <P FiCLP
JOIHT6 i/2* 6eJiT wi m W* iM* wioe * i'M> noh pixiw bore wa* hole
tnnausii «IO* CeNTCI»5ANotaUMTB« POKE A i/4*P«'*>* |'/4"0|AM
ON BACH SIDB. I/ft' P3U5HHP &R3MM ON POPLAR PRAMB
TYPICAL SECTION CONFERENCE TABLE TOP SECTION <3 FIElP
JOINT CONFERENCE TABLE /-1/zVaKTICLB BOARP WsGK / fltgawep
TO S/4"P'ARTIrt.B &P. „ - 3/4" PARTICLE B<V>BP- >
I'/S'^^'RIBSCMAPLe^ON 12" CBMTBBS^^- .I'/gNfi'/a" rI!<;tam«ulaR
tube w" thick 1/2* HONOUR** MAHOGANY ON PARTICLE Wi/
SECTION THROUGH SUPPORT FRAME .t*men i-lk aikch
I^WWHTULt MOAnp y Ctt- 7X ■ »?*' BOLT HUH ("UMB*, -■ t
WASHan y .*" I!^1SS¥~ FIElP JOINT SIPE SECTION CONFERENCE
TABLE SECTION! AT CONFERENCE TABLE IMIAK) T»fu IDT WW
MMCN MUMfIM* SOLID NUII LAWINITO frw incaiormn WH1TI OM
OH EITHC4 IK Of I0LO MOM UMTMUOUI MIL 258 SECTION AT
CONFERENCE TABLE

Office Spaces CONFERENCE ROOMS Credenzas and Heating Unit
Enclosures fVvSCIA.SOcRiT AMO — f# WALL A* n " ^"dTtTWALL
COW/ WsLLCovanihS yt'riSVBAI_ <?H aio« •VICVIITB JOh At G^a
flfcin 'p»t „i»r ^"inl ooone -«.th COWCSAuS UCNOJrtA'i A»At «J Al'h.
IMTBM. Custom credenza units are often designed to complement the

details of a conference table. They serve multiple functions, including storage, incorporation of electronic media equipment, display, and as a work surface. In addition, architectural woodwork is used to enclose existing convector covers and to frame window openings. It is Important for the designer to consider providing ease of access to the heating and air-handling elements behind the woodwork, as well as allowing the appropriate flow of air. A-mer eufcc* sxi&T window FHAMft AnO OLimO AU<>W covsn COUP CEUT^r CrEDEUZA BSWVAOI.B ^CUM., tO MA+CH AOJ /VAfH^-ll AL. TTP. IWI»H =LOOfI MEAT/MG UUIt cUCL06Jre 259

Office Spaces RECEPTION AREAS Planning Data: Receptionist's Workstation Proper design of the reception area is critical in communicating an organization's desired corporate image. Reception spaces are both the first and last areas with which the visitor interacts and, accordingly have considerable visual impact in communicating that image. Not only must the reception space look attractive, but it must function properly as well. The two most important planning elements in this regard are the visitor's seating area and the receptionist's workstation or desk. While most of the examples in this part are drawn from corporate interiors, the designer is urged to take into consideration the needs of special user groups who must interact with a receptionist. If small children are to communicate (or see or be seen), how high is the privacy wall? If a wheelchair-bound user is to approach the reception desk, is there room for the footrests to be accommodated? The designer must consider all user populations. This part deals primarily with basic planning data relative to the design of a receptionist's workstation and furniture arrangements of the seating areas. Also included are related details directly from the working drawings of design firms. For the purpose of privacy or security, the receptionist's workstation is often an area physically separated by built-in furniture and/ or partitions. Figure 1 shows a counter height receptionist's workstation. While the relationship of worksurface to seat height is key, other anthropometric considerations are eye height and sitting height normal. The minimum height of the opening above the floor has been established at 78 in, or 198.1 cm. Sitting height and eye height are significant in providing unobstructed vision. Figure 2 de- * C. . D ~ +B£B4| Fig. 1 Receptionist's workstation/counter height. picta a desk height receptionist's workstation. The depth of the worksurface ranges from 26 to

30 in, or 66 to 76,2 cm, allowing for thumb tip reach required for the exchange of papers and packages. Both Figs. 1 and 2 show in broken line an added counter top element often provided for security or as a visual screen of the work surface top. in cm A B C D E F G H I J K L M N O P Q R S T U V 40-48 24 min. 18 22-30 78 min. 21-27 36-39 8-9 2-4 4 44-48 34 min. 44-48 54 26-30 24 30 15-18 29-30 10-12 6-9 39-42 101.6-121.9 61.0 min. 45.7 55.9-76.2 198.1 min. 61.0-68.6 91.4-99.1 20.3-22.9 5.1-10.2 10.2 111.8-121.9 86.4 min. 111.8-121.9 137.2 66.0-76.2 61.0 76.2 38.1-45.7 73.7-76.2 25.4-30.5 15.2-22.9 99.1-106.7 D LU (1 0 0 Q 0 0 1) n n n 0 *F 0 CCO 0 in iiBiiii i in ihninn Fig. 2 Receptionist's workstation/desk height. 260

Office Spaces RECEPTION AREAS Planning Data: Seating Arrangements Fig. 3 The seating arrangements illustrated here provide some typical conditions that the designer must address. Individual seats are preferred over sofas. Corner seating arrangements must always consider leg clearance. Circulation between low tables and the edges of chairs must be adequate to allow far the legs of persons seated In the chairs. Convenient locations for side tables, so that magazines, ashtrays, artwork, or portable lighting can be placed on them, are important. 261

Office Spaces RECEPTION AREAS Reception Desk Details ,ftetSTMS »Y oeaK.y fia' v-*' f. ! . -BT—r- \ if i l . • *t (f)nAN view f= ~*rvi 4 ■rwe/V // *tv ii i i i i i i »^i i ! i / ■ g\^££ fl^£^_WW/_ ***<-* ■ ** "«'-'!o" ■ I ^ rteWPt fiLA&t rAtujeO **A***Tg *>K W PIYIMBOO fXIHISHeo OMPSX&. ruiiPt 2 tua uiuli men */ii*sw ,it*<jtr* fumui tsr**v* ■ MIT* 4f rt S"C 1*0 W'M-wtrH rt«v Tb *t+»1 HAvo or pck. t<j fu,7m. mtu it" »*«« cmo- rennet iw«m whits t-nnpimr CUPX/HUt AS MHVItEB r*"fT <?" BK'^W To MATCH UMiA/I/T rtiowp* itk" piam. fttitstep orat*** 4 o* to* 4 'A *^««m reor-r wau ■nwtnt IV/tS MAM&fMFNT »<« lie-eU iSHTCR-OH. WB TgOMcH. rw/oa &*je waujvt toouo o*s£ with AOjysi»BLE oruipcs i&"et.. Fig. 4 Depending upon the size of an office, a reception desk can be either relatively simple and small in scale, or relatively complex and large in scale, sometimes staffed by two or more persons. The reception desk Illustrated here shows a typical L-shaped unit with 44-in-hlgh privacy panel. Reception desks of this typB can either be custom designed or purchased from a manufacturer. 262

Office Spaces RECEPTION AREAS Reception Desk Details * t->£ " 1 U I
 II I i i © ne*- r r 1 L PLAN) - M V5 4° WO-1 ONLY- o'-cf Ei-EyATkgNi iv
 ^ ij/rn.) T\ I ,/K >*.2 / 7K~ k- VP.\ n*i I itvp^ QEL£VATLgSJ-Lgff -T*4fc-
 U<iHT EST- /jryegrvaJ- ,, — 1 ■ 1 Kl L i- <hEcr\£M- Fig. 5 More privacy
 can be achieved In the design of a reception desk when there is enclosure
 on three sides, as is shown here. When designing custom reception desks, It
 Is Important to fully understand the tasks that the person working there will
 be asked to perform, in order to provide for adequate storage, work surfaces
 at the appropriate height, the incorpratlon of electronic equipment, and task
 lighting. 263

Office Spaces RECEPTION AREAS Reception Desk Details Vt>-I N * r*r
 WP-I J" H , \PLAN K>2& © &L5/ATW-nz<?NT 1' '.>» i A G'-l' TfcSX. l-
 l>JT B= =| , i 1 t>u .. 1 1 i 1 II fr 1 * ; \ 1 'J — 6'-^ T-^-^—Tt < —|W>8
 fILt © ^E^-fkPN- © f&rw^- WP-I s. WP-I N bl.l.i. O " * i — " • t 7f\ i — '
 (7\U£XK\0^- LETT Fig. 6 The reception desk shown here Is designed in
 order to provide privacy on three sides with partial privacy on the fourth
 side. In this example, a right- hand typing return has been provided. Careful
 consideration should always be given to the height and placement of task
 lighting in order to ensure that the surface or task below is being lit
 properly. Many designers do not give this adequate thought, Overall costs
 of custom-designed reception desks can be reduced by integrating standard
 metal file compoments into the architectural woodwork, 264

Office Spaces RECEPTION AREAS Reception Desk Details PLAN
 RECEPTION DESK v>" - r-o" MARBLE THICKNESS VARIES SEE
 PLAN •A "SS DOWEL PIN EPOXVINTO WOOD BACKUP AND
 MARBLE DIE CAULKING BEAD SECURE MARBLE DIE BASE STOP
 TO WOOD FRAME WITH DOWEL PIN &ADHESIVE .!>..> ;■/ .*>
 MARBLE TOP /' WOOD BACKUP SECTION (NO SCALE) Fig. 7 A
 larger reception desk can accommodate work surfaces on three sides, as
 shown here. With this type of configuration, however, the designer must be
 concerned with the orientation of the open side. As with all custom
 reception desks, the designer must anticipate the integration of wiring and
 electronic equipment within the architectural woodwork. 265

Office Spaces RECEPTION AREAS Reception Desk Details K*** ^ z'-iVt
 f 2'-iV* ft g'-iVi f g'-iV* f a'-ifc^ III » i •fr^» >l*Z -ra,nm? ewuve caw*. -
 AHewfiW white a*. A e plan -!^ge?grTcM ceac C4 -KeaoiRssQ -z-?yi.
 rwuw&crmcrr.'WO -rewsHes? erwfi-w* <*u _^ aw-™ew harcw>p> tcj
 pp ue-av >u-f " "" " Au- intbkkp. rww. -p re 'Knou. wriuP5>' . , 'AWfWB'
 N'W 730/2 -CHAMm&NB 5f-"W a&tKX>WWq- 70% ■&IUK.,efE&IAU
 ASfTWC- 0AiWNe> uwrWf* to e* •• stacs locoz - n» ua py amspiu^
 uskthbp. AW5W5AH UBATHSI*. " CArRI" W 9KH. \W> -,VtArt3S»W ec
 I7CQZ -uo? pr ame<. i^wHee • 'c*tmtun\\& vroutob f* ^ i % 7^ s NT > ^L
 / ^ ^i>'*0%' w/wu- cmo ••■"s^y^ 3s,i—roua ~5rry t: WW- ^OgVATIW -
 K5>gfriQN CgS*. © scaw?; *f' * r-c" eieeuTKwi r?J9H*?JT3U9HK?
 3TAINLW* ~UW^SLW& ^ *; 3* J) 4sis * x* ^ - ■■ " ~> tB -butt jsnt& Tff
 - --» ALtir. -**iimim, (nr.~) « — ftX»HBO ©RAUre t ANP»(tAi< -rum
 rafrtra? staini»& 9twu U»«U>f» F15L^WlON - Pec^fTlCM ZttK e ftVt*: ^'
 ■ \'-0' Fig. 8 Total privacy of the receptionist's workstation can be achieved
 through enclosure on all four sides. In addition to enhancing visual privacy
 such a des gn can also provide added security and control by the addition of
 a door. Such a design might be particularly appropriate for a reception area
 where the designer might wish to control access by children. 266

Office Spaces RECEPTION AREAS Reception Desk Details ? 5* ^A-ifr*1
 *fc nr. — — '-T^- -Jsirrr. S^xpr^- g^38^ IBMBU V-tVU&H JP' *^fei T\gl-
 gVATI(?Nl - fZCZTVoti CPg5K e S<AW>: ** I'-C *« P 43* f-
 wrt.-^-.yJ^T? tT5 J 6> * ?rr^^PB^--,- 'v^... ^—* -rflhr-J £ T I! F^ n —
 renamm — b<xr**. ^gifi^TioM -FgcerrgN vee& (uw vwc ernKZ"? e
 •SCrMJf: %' ■o< -H3»\$5W' -fWWH 9T/9T. TNW -reue(«? «*aww -irese
 m*reo vcuoxw TASK ZWK>0 -Am.i«?nw£ -NHZPxM miT& ct*-V*B *•
 >» it- —f—= £ -fcwsnep e*AMnw < -UJ- ^-,1—{ | i m * -pSK?|UPW-.
 = -riuepwt.- «-. _.. -twp«& vfKAira? Ha»)7n5 7A#* e-uw»se -trrw?
 m*x> x~^ -> —MiencM wbt5 cut- &W /T\piEvwnoN • racemoi ere*;
 (wsw; surm^) INTO KVTZP 55B<5 ^.TYT7 -gWU'l-JoHNWJ W ■*\$*
 <&3\\ tax*.. 1 jV :fr Fig. 8 (Continued) 2-lfe's + * TsIOam-secret a ^%gmai
 r7ee<R33P. 267

PL-^N OF RgCEPTIOH DE,S^' . ■bP^CE 1S-Z I ~C?1»HNMtf 1"0!». /-
 =*uttu-oimt a tL v CJHWTHm. EjJUSnC*.- .V -fti-AXH, UtTHll*. c; «*.. ^

\$ D/<? V4-* »r-»a Fig. 9 A reception desk can often consist of two workstations.

Office Spaces RECEPTION AREAS Reception Desk Details I I da i'/a.-»i'-
3ae . m ^ -i-W. wjvnm* rnrneai-u worn. -jaKfc^lt-fctMINLflfcTM .
M*M=»L» ~»»rar»l,* - - -I-* T^T uAmxz 'fjfjpmMa fitrTrr JEujkfik.eKg
...■ ■ _., . t»LYW»«JO ■mmmmmatrs-.mmmmiRmtimimtam ■ / C4,P\p«T
ranfnwimifi immMiimmamnmie mm mm ie it Li »■,!'-"£.» ■ irfxJK.AUiJU-
yM1 jffl />- FULL. "UH.*.. [ti/l ti)m v ~ r-c Fig. 9 (Continued) 269

Office Spaces RECEPTION AREAS Reception Desk Details ia Vz*
SECTION) SECTION Fig. 10 A circular reception desk can make a bold
and sophisticated corporate statement. The designer is cautioned, however,
to carefully analyze the minimum radius required for chair movement.
Custom built-in files and drawers, if also curved, can become costly and
sometimes Impractical. 270

Office Spaces RECEPTION AREAS Reception Desk Details '**&' (*W9H
W./W.- - tew**, wrvrri?- gurr J01WT& <rrro TWM (CONTH" - -"HJ&x?
WTO© isky-r cp. HSUAi. trews-wftfn*? %t' fk?Moec7rp V**K 0UHf^« ^
•WJP&tJSU (TTP7 ewe.. euc*e> - puU. exna^cN •R^rrr- TYCV AUIWOP.
nu<& (TYIT) <SLIPT5>=> - /BO lb. <SAft\&lTY ruu- esjqrw&ioN /
•MKBAU wHrn5<a^c c*se (rrr? BNVPn^u eeCTicN » hish raagrjM. ^
9c+ve z>>+ ££t ,*rvep owe on ccmt. st./pt.—>2 7WM 2%' wax> >fc"
HCWeKW T7 #VCfSrt UXAJON Of *sm«-. »' -0" «S .vekttcaw ^esTW a
uav rscggrAL «ViO l'-C. e Fig. 11 A fully detailed reception desk will
require many large-scale vertical sections to explain the various storage,
drawer, work surface, lighting, and electrical requirements. Examples of
such details are shown here. 271

Office Spaces RECEPTION AREAS Reception Desk Details cnomi
6mcvai-pKH»i* SAftg, cm** **tneti~*oM tulip » yw«. »*** MARBLE-*.
~ - ~.J>^ CC JUBTOP-^* CORNER DETAIL ./.I-1 ;>v7. i 1 I ■Xj.
CURVED RIBS —V TYPICAL \ ^— SOLID is^S WHITE OAK WENQE
SOLIDS — - INLAID AS ACCENT STRIPS VERTICAL SECTION-
RECEPTION DESK 272 n o 8 i I '\$ tt B Z u 5 ■« V! »44. v««« >LP E ftAse
w= 4±/'« » j SECTION AT COUNTER FRONT V£ j?6 S'/H' /2" MEPITE
W/ PERMA-COAT PINI5H Flo. 11 f Continued) ^CHEMBTAL (* 8I»

BRONIB LAMINATE BASE^ SECTION OF COUNTER® STATIONARY FILE

Office Spaces RECEPTION AREAS Reception Window/Pass-Through
L_ ^..~Jl- 4 / nil to ^^Vwd. Li -1 JAMB DETAIL ATINTgeSECTTPM
.ZffgNgfg AT P><sfc3fe#gr VggTlcL^U- 'SISgTOiH THRU LP\$3.y
C0W3&DL. W1MQOV Fig. 12 A receptionist's workstation need not be
freestanding within a reception area, where security and privacy are of
critical Importance. A receptionist may be located on the opposite side of a
glass partition as shown here. Such a solution Is often suggested when the
receptionist performs multiple tasks such as typing and answering phones.
273

Office Spaces RECEPTION AREAS Reception Wlndow/Pass-Thuougli <2>
—i Z^: WHIT* F*L*ST'C LAMIWA.TW - - LINE OF SUSPENDED
CEIUNO.. -SYPLBM i«4ALL 60. -SLASTiCLAM, TEMDtlTe IS 14
MARIGOLD. - BUDING PANEL - 3l&' CLBAA UMN ■ OPEN -TOP CP
BAOA SPLASH . p"i_a»-rie LAj^iMirrm eouNTBI*-Tor» -FOBMICJk, ^n-
a* HONEYTOWS _TEAJ^ W/ WHiTS. PLA.&. LJW.. BOO*..
TBCTOUT* KftO -plywood snwNW ■ 2 3RAWEBs PUB 17\$S'w.30'D.
277s'H. W OTHERS -PLASTIC LAM. ■ 3<f PLY WC«?D BRACE. ■ »>,V
VMYL &A3E «■ «. .itfft'tlfr NORTH ELEVATION FROM SPACE MO. 15
S'A.* ^9 m Z'-flfc" o\v 5 J£ft _ I^LaJfc' PLASTIC LAM. / - PLASTC LAM
. BACK SPLASH . - PLASTIC LAM. COUNTED TOP, 7/6"thk •
f^CECFWALL r> 3CAif ill' > l-o" - 3fe"» VT' BTIFFNER -1ft FLY HOOD
BPtaONO Z DRAWER FILE. "7i»'X>'D' 27?A'H BY OTHERS. SCALE
lte'M'0. hHITS &»£M SHELVES, f 10 ^ S**TUI,AL- hASUM. FINISH 7 -
>J*5 ROD ' r-r « NORTH ELEVATION SPACE * 4A I ZZZ3 4231 -O % ®
SAS3Y S 3333 H*MCTRGO - |«l/ B 33«o) SOCKET ■i'.2 7F FILLER. M-
iTE BiHCH 3H6.VES NATURAL LA«*jeR FINISH - GARCY a S35S
HANOSOP w / B 3soi socurr. SEJ SCALE. Itfga \'-0 4CAi« 3"*l'0 CCK!
&. h-VKT SHELVES l«T \pAJg-T ELEVATION '* ' 'SCALE. 3's I-C" 274
Fig. 12 (Continued)

Office Spaces RECEPTION AREAS Reception Window/Pass-Through i I i
I 10 6 t: "V 2-0 -UNE OF SUSPENDED CL&. -GYPSUM WALL 6D.
"LABTIC LAM. TEWOLITE 1(814 M**ISOLD ■ PCVEALS AND

VERTICALS. Oaint to match WAUS - i,'a' CLEAR J»?<AN -OPEN .
 COUNTER TOP " TEXH3UTE 1400 WHITE • pLASTIC LAM
 TSX.TOUTE • |6|4 MARIOOLO. -V/a' vinyl »a.'ae SOUTH ELEVATION
 FROM SPACE NO. 14 SCALE }*."» |iO" auofO w»« pocwrr Wobar^wi
 TDPC^ B*£K9A^9H -v*.*D aiOOD "LLIB CASINO BSAD
 U.S.OYPSOM*200A OR EQUAL Wa'SYPauMSD.! MeTALSTUD- / .fe'j
 i*' ; *o' SCALE : HALF FULL SZE Fig. 12 (Continued) @ i£\ I.V jfe Hfl'
 5, PLABTIC LAM . COUNTED TOP -EXTOLITE H-floHHITE
 VERTICALS , qAiNT T3 MATCH k*ALLS. Jfi CLEA" LE*AN
 a"is'i*afc"swvv& =CB &LDINO DCC« !e'*y0' HASD WOOO FILLER
 jft'x 3?<|'VERTICALS)&>J# QUIRrt. ELASTIC LAM. 7/6 SLIDING
 DCOR PLASTIC LAM. TEXTOLITS 1814 MARIOOLD Ift"*^" QUIRK
 SCALE : HALF FULL 3IZB . 275

Office Spaces RECEPTION AREAS Reception Wlndow/Pass-Through
 WZTZ&r ', 4-»' ,»'-i».'kri.t, PLftVATION PEOrA -SfAC.1 2X>Z r^'xrc^Mt
 f latte £ui*s ^PAflB ZQ2. r«.*rrt«4 —, R SAIT SLEWXTloa- |-1 / a ',..
 i'-o' .UITI4 UK ^■2^ < tmtmjt J'M'-O" Rg. 12 (Continued) 7SOMBT»ilC
 OP 3TAIULJS) «TigL fc*«*lf 276

Office Spaces RECEPTION AREAS Wall Shelf; Coat Closet; Talaphone
 Shelf OCTAL. tf>/A&-\ FlfiN MCW 10 VvfcU- A .ft. UWIHATfi III lite,
 ^e^riow on u>v*#, ♦WITCH uaeATVN PETAIL - WALL, 6HEUr ff'r"Mtj
 DFTAI /■ fflONE 6HEO= PETAIL-TrP. 6£AT^ Fig. 13 While the
 reception desk Is typically the major element to be designed and detailed
 fora reception area, other custom-designed components must also be
 carefully considered. A phone shelf, a wall shelf, a coat hanging area, and a
 work surface are often items that must be carefully designed and detailed.
 277

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Desks
 and Sealing Furniture, furnishings, and equipment are the basic building
 blocks in the design of office spaces, The illustrations and dimensional data
 contained in this part are based on the product lines available from
 particular manufacturers, Although the data, to a great extent, are fairly
 standard throughout the industry, there will be so*me variations according
 to manufacturer, Accordingly, although the information presented is

adequate for preliminary planning purposes, the designer is cautioned to reconcile preliminary assumptions with the actual dimensional data of the manufacturer whose product is ultimately specified. Included in the data provided in this part are examples of filing cabinets, storage cabinets, conference tables, desks, and electronic media.

OS D 1 H DESK

DIMENSIONS DESKS DOUBLE PEDESTAL STANDARD 2'-6" 2'-5" 5'-0" RANGE 2'-0"-3'-3" 2'-4"-2'-6" 3'-9"-7'6"

SINGLE PEDESTAL STANDARD 2'-6" 2'-5" 5'-0" RANGE 2'-0"-3'-3" 2'-4"-2'-6" 3'-9"-7'6"

RETURNS FOR EXECUTIVE DESK RETURNS ARE AVAILABLE AT SAME HEIGHT AS DESK STANDARD 1'-6" 2'-2" 3'-0" RANGE 1'-3"-1'-8" 2'-1"-2'-3" 2'-0"-5'-0"

DESKS-SINGLE OR DOUBLE PEDESTAL

WORK TABLES ARE OF SIMILAR DIMENSIONS. FOR EXECUTIVE DESKS WITH RETURNS, RETURNS ARE AVAILABLE AT THE SAME HEIGHT AS THE DESK SURFACE. A MINIMUM CLEAR WIDTH OF 22" SHOULD BE PROVIDED FOR KNEE ROOM, 24" IS NORMAL. 5-18" RANGE 1'-0"-1'-6" 2'-0"-2'-6" 3'-0"-3'-6" 4'-0"-4'-6" 5'-0"-5'-6" 6'-0"-6'-6" 7'-0"-7'-6" 8'-0"-8'-6" 9'-0"-9'-6" 10'-0"-10'-6" 11'-0"-11'-6" 12'-0"-12'-6" 13'-0"-13'-6" 14'-0"-14'-6" 15'-0"-15'-6" 16'-0"-16'-6" 17'-0"-17'-6" 18'-0"-18'-6" 19'-0"-19'-6" 20'-0"-20'-6" 21'-0"-21'-6" 22'-0"-22'-6" 23'-0"-23'-6" 24'-0"-24'-6" 25'-0"-25'-6" 26'-0"-26'-6" 27'-0"-27'-6" 28'-0"-28'-6" 29'-0"-29'-6" 30'-0"-30'-6" 31'-0"-31'-6" 32'-0"-32'-6" 33'-0"-33'-6" 34'-0"-34'-6" 35'-0"-35'-6" 36'-0"-36'-6" 37'-0"-37'-6" 38'-0"-38'-6" 39'-0"-39'-6" 40'-0"-40'-6" 41'-0"-41'-6" 42'-0"-42'-6" 43'-0"-43'-6" 44'-0"-44'-6" 45'-0"-45'-6" 46'-0"-46'-6" 47'-0"-47'-6" 48'-0"-48'-6" 49'-0"-49'-6" 50'-0"-50'-6" 51'-0"-51'-6" 52'-0"-52'-6" 53'-0"-53'-6" 54'-0"-54'-6" 55'-0"-55'-6" 56'-0"-56'-6" 57'-0"-57'-6" 58'-0"-58'-6" 59'-0"-59'-6" 60'-0"-60'-6" 61'-0"-61'-6" 62'-0"-62'-6" 63'-0"-63'-6" 64'-0"-64'-6" 65'-0"-65'-6" 66'-0"-66'-6" 67'-0"-67'-6" 68'-0"-68'-6" 69'-0"-69'-6" 70'-0"-70'-6" 71'-0"-71'-6" 72'-0"-72'-6" 73'-0"-73'-6" 74'-0"-74'-6" 75'-0"-75'-6" 76'-0"-76'-6" 77'-0"-77'-6" 78'-0"-78'-6" 79'-0"-79'-6" 80'-0"-80'-6" 81'-0"-81'-6" 82'-0"-82'-6" 83'-0"-83'-6" 84'-0"-84'-6" 85'-0"-85'-6" 86'-0"-86'-6" 87'-0"-87'-6" 88'-0"-88'-6" 89'-0"-89'-6" 90'-0"-90'-6" 91'-0"-91'-6" 92'-0"-92'-6" 93'-0"-93'-6" 94'-0"-94'-6" 95'-0"-95'-6" 96'-0"-96'-6" 97'-0"-97'-6" 98'-0"-98'-6" 99'-0"-99'-6" 100'-0"-100'-6"

VARIOUS DRAWER ARRANGEMENTS FOR PEDESTALS 1 BOX 1 FILE ARTIST AND DRAFTING DESKS OR TABLES PEDESTALS FOR SECRETARIAL RETURNS WILL BE REDUCED IN HEIGHT THE EQUIVALENT OF ONE PENCIL DRAWER. STANDARD SIZE ENGINEERING OR ARCHITECTURAL DRAFTING TABLES ARE 37"V2"x 43"V2"Dx60"-72"-B4"Wx37"H. Fig. 1 Office planning: desks-sizes.

SECRETARIAL CHAIR 4'-w—1' i—d4 SWIVEL ARMCHAIR n i ■■ —D-, Hi ■ 1 —W —u H I* J I fr RIGID ARMCHAIR LOUNGE CHAIR 4'-w4 -J— d4 DRAFTING STOOL n -Mv- ■d4 SIDE CHAIR r -1 1 1 i w .

SOFA CHAIR DIMENSIONS w D H H| SECRETARIAL STD. 1'-5" 1'-7" 2' - 6" 1'-5" RANGE r-4"-r-8' 1'-6"-2'-0" 2'-5"-2'-10" 1'-4"-1'-8" SWIVEL ARMCHAIR STD. 2'-4" 2'-3" 2'-9" 1'-5" RANGE r-8"-2'-6" 1'-8"-2'-6" 2'-6" 3'-0" 1'-4"-1'-10" RIGID ARMCHAIR STD. 1'-10" 2'-6" 1'-6" RANGE 1'-6"-2'-3" r-7"-8" 2'-4"-2'-10" 1'-4"-1'-7" STACK CHAIR STD. 1'-9" 1'-9" 2'-6" 1'-5" RANGE 1'-6"-1'-10" 2'-4"-2'-9" 1'-5"-1'-6" RIGID AND ADJUSTABLE DRAFTING STOOL STD 1' -6" 1'-8" 3'-0" 2'-4" RANGE f-5"-?-Cf 1'-6"-2'-Cf 2'-ni-3,-6" f-5"-2'-10" SIDE CHAIR STD. 1'-8" r-io" 2'-6" 1'-6" RANGE r-4"-2'-0- 1'-6"-2'-B" 2'h4-MO" 1'-5"-1'-7" LOUNGE CHAIR AND SOFA DIMENSIONS LOUNGE CHAIR STD. 2'-6" 2'-7" 2'-6" 1'-3" RANGE 2'-6"-3'-4" 2'-2"-3- 4" 2'-1"-3'-4" 1'-0"-1'-6" SOFA D, H AND H,

SIMILAR 2SEATS-5'-0"-6'-7" 3SEATS-6'-0"-7'-6" 4 SEATS-7'-8"-9'-0" 278
 Fig. 2 Office planning: seating -sizes.

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Chain
 high back mid-manager grand class manager height ring! 33- arm height
 rungs: 251- M«t twlgbt rangti operator stool H mmhw operator mid-
 manager h_H^_j lEBS height renga: 391 M6t" pneumatic 391 M6l"
 vecmalle ■nn (Might renga: 31 "'381" pneumatic 31--381* vecmatic aeat
 (Might rang*: 221 "-291" pneumatic 221 "-29I * vecmatlc hilght ring*: 33"
 <37" pneumatic 33"-37" vecmatlc 34" fixed ■•at height range: 161"-201'
 pneumatic 161-•201" vecmatic 171* fixed height range: 33"-3 7" pneumatic
 33" .37" vecmalle 34" lixed arm height rang»g*t 251 '*29I" pneumatic 2S1
 "-29I * vecmatk) 26' lixed aaat height range: 161 "'201" pneumatic 161
 "'201" vecmatlc 17C fixed height range: 33) "-37I" pneumatic 331 "-371"
 vecmalle 341" lixed aaat height range: ir-21* pneumatic 17"-2r vocmatic
 171' lixed height: 33" arm height: 251" aeat height; height range: 391 "-43*
 * pneumatic 391 *-43J' vecmalle arm height range: 251 "-29J" pneumatic
 251"-2fl" vecmalle aeat height range; 17"21" pneumatic 17--21' vecmatic
 45 pounds Shipped Set-Up (Might: arm (Might: sa* arm height'. 251- mm
 aeat height: manager high back h——i h——i grand class manager sled
 bases I h A height range: 42M6' pneumatic 42".<(6" vecmatlc arm height
 range: 251 *<29I" pneumatic 251 "'291" vecmatlc seat height range: 171
 "-2H ' pneumatic 171 '-211 ■ vecmatlc arm height: 25' iMlghl range; Ml
 "Jfl - pneumatic 341' fixed arm height range; Z5l "-2W" preumatte 251-
 Fixed ' aaat height range: 171 **ll " pneumatic 171 " lixed (Might rang*:
 42-46" high back 341 "-39J * manager arm height range: 251 "-2M ' aeat
 height range: 171 ^211 ■ SO pounds Shipped Set-Up height range: 4ST-
 4XF m*-*** *nri height range: SKtMH 251'-»l' ■WMM aeat height range:
 171 "-211 * 60 pounds Shipped Set-Up Chair types are often associated
 with certain generic job titles, The designer, however, is cautioned not to
 make assumptions as to chair selection without a thorough understanding of
 the tasks the individual is to perform. Ergonomic considerations are to be
 carefully reviewed in order to select a chair with appropriate attributes, i.e.,
 seat height, adjustability back and arm support, firmness, etc. Overall chair
 size must be understood within the context of available clearances and
 workstation configuration. mid-manager height range: 331 *<371 *
 pneumatic 33C-371" vecmatic 341' IIKOd erm height ranges 251"-291"

pneumatic 251 "291' vecmalle 251' fixed seat height range: 17"-21"
 pneumatic 17"-21' vecmalle 171' fixed height range: 341 "-39) ' pneumatic
 341 '-39) ' vecmalle 35' fixed arm height range) 251 ".291' pneumatic 251
 '-291' vecmalle 26' fixed seat height range: 171 '-21" pneumatic 171 "-ail"
 vecmalle Z79

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Chairs
 height range*: 331 '.37J * pneumatic ■.«*™. 3'35" yMmaHc arm height
 range: i«mw?'H 241 "-2B1" pneumatic IWOTL* 22P-261" VMmatlc ■v".
 241" fixed seat height range: 1B*>22" pneumatic 16"20' vecma' IS" fixed
 height: 291" arm height: 251" MBthalght: 15 pounds Shipped Set'Up H h
 SWStafmil height range: 331' arm height range: »J»WI 25" height range:
 wmw* 43 pounds Shipped SaMJp 36 pounds Shipped Set-Up __2j2il_i I
 221 height range: 41 '-45" high back 331'-37" mid back seat height range:
 181 "Ztl" 67 pounds Shipped Sel- Up i I m I n height range: 331' arm height
 range: 251' seat height range: i/ helghl range: 411'-441' high back 371'-401'
 mid back 35">3Sr low back arm height range: 251" ■ 29" 19'dialr j I 24'
 chair " 221 stool ' ' 21" stool n 30' chair 331' stool arm height: 28' •eat
 height: 171" chair 30' SIool 21 pounds Shipped Sal-Up height range!
 411'-441'- highback 371'-401'- mid back arm helghl range! 26' • ZQ1" seat
 helghl: 171- 20 pounds Shipped Set'I helghl range: 41 '.45' highback 331
 '-37" mid back seat height range: IB1'.Z11' 67 pounds Shipped Set-Up
 height: 32' arm helghl: 25' seat height: 171' helghl range: 421'-451' high
 back 301"-4U- mid back arm height range; 241 --271' seat helghl range:
 IBC-S11- 64 pounds Shipped Sol-Up / H h-^H n Mi seat helghl: 161' 280

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Chairs
 Executive Chairs Tilt-Swivel Chair with Casters S4 Option: CA Non-
 Swivel Lounge Chair f^**" \ ^ Swivel Lounge Chair p; * Non-Swivel
 Lounge Arm Chair X H(y / ' 1 f" "1 Ottoman Lounge Non-Swivel
 Lounge Chair X Swivel Lounge Chair PC Non-Swivel Lounge Armchair
 pc; Swivel Lounge Armchair PC Tilt-Swivel Reclining Armchair / Swivel
 Lounge Arm Chair Ottoman 'X'X pc " Tilt-Swivel Reclining Arm Chair
 Straight Module Side/Pull-up Chairs Non-Swivel Side Chair fi Swivel Side
 Chair f* '■■■"■"> /" " \ Swivel Arm Chair 9-i £ Si Option: CA Executive
 Chairs Tilt-Swivel Chair with Glides 94 Lounge Chair Ottoman Lounge
 Chair and Ottoman Eames™ Chaise I Non-Swivel Arm Chair Sofa

Compact r^ Nelson Sling Sofa Tilt-Swivel Arm Chair with Glides Eames
Executive Lounge Chairs Tilt-Swivel Arm Chair with Casters Executive
Swivel Lounge Chair Adjustable Executive Tilt-Swivel Lounge Chair 94
Adjustable Tilt-Swivel 94 281

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT
Reception and Lounge Seating h-4' H — 5'8" — 8'6" 5'8" 23'6" ryjr 5'10"
361" T T T T A i A A i 10'6" 34' straight seating units %m All Mats: 2r height
16" seat halfht 34" depth All tables, 30" bench: 16" height 34- depth Full
arm: 27" height 34" depth wedge seating units arms Fig. 3 Reception and
lounge seating can assume various sizes, shapes, and configurations.
Modular seating units can offer a custom built-in look, and can often
Incorporate table and storage components. Overall sizes will vary from
manufacturer to manufacturer. 282

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT
Conference Tables ^=====^-=^=====5 Conference tables come in an infinite
variety of shapes and sizes. Figures 4 to 6 attempt to provide a
representative sampling of such tables, along with dimensional information
and seating capacities. The designer is cautioned to use such information as
a preliminary planning tool only, and to carefully lay out conference rooms
with actual furniture pieces that have been selected. Chair width and
spacing will ultimately dictate conference table seating capacity. 42"
diameter, seats 4 Veneer Tops 48" diameter, seats 5 54" diameter, seats 6
60" diameter, seats 7 "Reverse diamond" "Sunburst" 60" x 122", seats 10
60" x 180", seats 14 60" x 60", seats 8 53" x 151", seats 12 60" x 209",
seats 16 53" x 93", seats 8 60" x 151", seats 12 60" x 238", seats 18 Fig. 5
283

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT
Conference Tables 60" x 118", seats 6-8 60" x 176", seats 10-12 60" x 205",
seats 12-14 (hid 53" x 140", seats 8-10 53" x 198", seats 12-14 60" x 234",
seats 14-16 One piece top, 48" x 96", seats 8 One piece top, 48" x 117%" x
29", seats 6-8 «ii,i —■*■»■ V.- One piece top, 48" x 126V4", seats 10 Two
piece top, 53" x 146%" x 29", seats 8-10 Two piece top, 60" x 204%" x 29",
seats 12-14 Q Two piece top, 54" x 156", seats 12 Two piece top, 60" x
175%" x 29", seats 10-12 Two piece top, 60" x 233%" x 29", seats 14-16

Two piece top, 60" x 185%", seats 14 Two piece top, 60" x 216", seats 16
Two piece top, 60" x 245^', seats 18 Fig. 284

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Vertical File Cabinets TABLE 1 Vertical File Cabinets Outside dimensions

Description Cabinet height Depth Width Height 58%" Letter Legal 28"/i8" 289/is" 14%" 17%" 58%" 58%" 52%" Letter Legal Card 289/ia" 289/ib" 14%" 17%" 14%" 52%" 52%" 52%" 41W Letter Legal 28a/ia" 28^16" 14%" 17%" 41W' 41W 29%" 27%" Letter Legal Letter Legal 30" 30" 30" 30" 14%" 17%" 14%" 17%" 29%" 29%" 27%" 27%" Standard vertical file cabinets are usually designed to accommodate standard height drawers and half-height file insert drawers (optional). Cabinets are available in letter-size widths (147/s") and legal-size widths (171/s"). Vertical file drawers are usually 12" high and accommodate front-to-back filing. Standard cabinets are available in four heights: five-drawer (58%), four-drawer (52%), three drawer (41W), and two-drawer (29%" or 27%). The depths of the three-, four-, and five-drawer cabinets are 28W, while the depth of the 2-drawer cabinets is 30". Table 1 lists these dimensional data. It should be noted that although adequate for preliminary planning purposes, the data are based on Steelcase cabinets. Guidelines for Customizing Vertical Files Depending on the manufacturer, vertical file cabinets can be customized, Usually two half-height file insert drawers may be substituted for a 12-in-high drawer in any or all positions. Table 2 indicates the dimensions and linear capacities of such insert drawers, while Fig, 7 illustrates basic guidelines for customizing. SB" 52" Fig. 7 H>lf-h>lghl Drawers IPDrswm Legal il» 2ft" 285

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Vertical File Drawers; Dverllle Cabinets; Roll-Away Carts TABLE 2 Vertical File Insert Drawers _____ Inside dimensions Description Depth Width Height Linear capacity 53%"* (2 rows) 26%" Bills Storage Coin tray 26%" front-to-back 12V4" side-to-side 80%"* (3 compartments) 53%"* (2 compartments) 80%"* (3 compartments) 803/i"* (3 compartments) 53%" (2 compartments) See style no. C See style no, D •Deduct %" when compressor is used, TABLE 3 Drawers, Overfile Cabinets, and Roll-Away File Cart Inside dimensions Description Width Depth Width Height Linear capacity 12" Letter-size drawer - 27%" 12V4" 10%" 27%" front-to-back*

12" Legal-size drawer - 27" 16W 10V4" 27" front-to-back
 Outside/inside dimensions Overfile cabinet 29" 28 7/28" 29 7/27"
 Fits over two letter-size files Overfile cabinet Fits over two legal-size files
 Overfile cabinet Fits over three letter size files Overfile cabinet Fits over
 three legal-size files Roll-away file cart *Deduct Ye" when compressor is
 used. tDeduct V/f when ordered with sliding doors. Half-height (3x5 or 4x6
 cards) Half-height (checks) Half-height (cash) Half-height (box) Half-
 height (microfilm) Full height (letter size) Full height (letter size) Full
 height (box) Full height (3x5 cards) Full height (4x6 cards) Full height
 (legal size) Full height (legal size) 26" 26" * 8W 8 1/2" 4" 26"
 26V4" * 26" * 26" * 26" 26" * 26" * 26" * 26" * 6"(2) 10"
 3V4" 11V4" 2 1/4" 12V4" 4"(3) 6"(2) 4"(3) 15W' 5"(3) 6W'(2) 7 1/2"(2) 5"(3)
 4" 4" 1" 1" 1" 4" 4" 10 Ve" 10W 4" 4 1/2" 4" 10Va" 10W
 28'o" /25" 55" side-to-side on 2 shelves (1 adj., %" thick) 35"
 28 7/28" 2 44" 28 7/28, /z" 53" 28 7/28" 2 30W728" 35 7/33"
 28W' /25" 44 7/42" 28Vb725" 53 7/51" 28Ve725" 15V&712V6"
 22 7/13" 67" side-to-side on 2 shelves (1 adj., %" thick) 85V2" side-to-
 side on 2 shelves (1 adj., %" thick) 103 1/2" side-to-side on 2 shelves (1 adj.,
 %" thick) 25" Letter/legal

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Lateral
 File Cabinets TABLE 4 Lateral File Cabinets Description (cabinet inside
 height) 60" Cabinet width 30" 36" 42" 48" 30" 36" 42" 3" Moduleunll
 Outside/inside dimensions Depth 18 7/17 Ve" * 18 7/17 Ve" * 18 7/17 Vb" *
 18 7/17 Ve" * 18 7/17 Ve" * 18 7/17 Ve" * Width 30 7/28V/ 36 7/34V/ 42 7/40V/
 Height 64 7/60" 64 7/60" 64 7/60" 30 7/28V/ 52 7/48" 36 7/34Vz"
 52 7/48» 42 7/40V/ 52 7/48" Standard lateral file cabinets are usually
 available in three widths-30", 36", and 42"- and with 12"-high drawers or
 roll-out shelves. Some cabinets are designed on a 3" module to
 accommodate 3"-, 6"-, 9"-, 12"-, and 15"-high drawers and shelves. Ganging
 hardware is usually included with each file cabinet. Cabinets should be
 ganged with adjacent files or bolted to the floor. Counterbalance weights
 should be used for single-application files. Table 4 shows the outside and
 inside dimensions of four lateral file cabinets, and their loaded floor
 weights based on 12"-high drawers filled to capacity. It should be noted that
 the dimensional data and load factors are based on Steelcase cabinets.
 Although these data are adequate for preliminary planning purposes, it is

essential that the data of the equipment being specified are verified with its manufacturer. 36" 30" 36" 42" 18717 Ve"* 18717 M>"* 18717 Ve"* 30728V2" 36734V/ 42740V/ 41V4736" 411/4736" 41WV36" 24" 30" 36" 42" 18717 Vo"* 18717 Ve"* 18717 Ve"* 30728V/ 36734 V/ 42740V/ 28WV24" 28WV24" 28V4724" Loaded Weights Cabinet inside height Description 24" 36" 48" 60" Cabinet 30" 36" 42" width Loaded 285 336 391 weight in pounds 30" 36" 42" 401 475 553 30" 36" 42" 524 645 720 30" 36" 42" 610 725 843 •Deduct %" when compressor is used. 287

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Lateral File Cabinets Filing Arrangements Most lateral file drawers are designed for filing both letter-size and legal-size documents, in addition to EDP printouts. Lateral file drawers can usually accommodate materials in a front-to-back (F to B) arrangement or in a side-to-side (S to S) arrangement. In some instances a combination of the two is possible. The actual capacity in linear inches for each arrangement and for each particular drawer or shelf has been calculated and is shown in the "Linear capacity" column in Tables 2,3,6,7 and 8. It should be noted that the dimensional data in Table 5 are based on Steelcase drawers. TABLE 5 Filing Arrangements Description 30" width 36" width 42" width Letter/legal, 12" drawer FtoB FtoB StoS FtoB FtoB StoS EDP binders, 15" drawer FtoB StoS FtoB StoS EDP folders, 15" drawer FtoB FtoB StoS 288

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Lateral File Cabinets TABLE 6 Drawers, Shelves, Add-On Cabinets File cabinet Description width Style no. Depth Outside/Inside dimensions Width Height Linear capacity 12" legal fixed shelf with door 30" 36" 42" 161// 28V4" 10V6" 161/6" 34i/4" IO1/2" 161/4" 401/2" IO1/2" 281/4" side-to-side 341/2" S to S 401/2" S to S 15" legal fixed shelf with door 30" 36" 42" 161/6" 281/2" 131/2" 161/6" 34V2" 131/4" 161/6" 401/2" 131/2" 281/2" S to S 341/2" S to S 401/2" S to S Center hook filing hanger bar w 30" 36" 42" 17" 28i/6" 123/4" 17" 341/6" 12%" 17" 401/4" 12%" 12" high door) 15" high door) 12" high door) 15" high door) 12" high door) 15" high door) 281/4" S to S 281/4" S to S 34V6" S to S 341/2" S to S 40V6"StoS 40y2" S to S T-bar ?** -i 30" 36" 42" 17" 28V6" low 12%" 12" high door) 15" high door) 17" 341/4" 101/4" 12%" 12" high door) 15" high door) 17" 401// 10W 12%" 12" high door) 15" high door) 281/s" S to S 28i/2» S to S 34V2" S to

S 34V2" S to S 40i/4" S to S 40i/4" S to S Wire tape rack 30" 36" 42" 17"
 28V6" 71/a" 10i/a" 13 W 9" high door) 12" high door) 15" high door) 17"
 34i/6" 7/e" 10/e" 131/6" 9" high door) 12" high door) 15" high door) 17"
 4Qi/6" 7%" 101/a" 13/e" 9" high door) 12" high door) 15" high door)
 28i/4" S to S 28i/4" S to S 28V6" S to S 34i/4" S to S 341/2" S to S 341/2"
 S to S 40'2"StoS 40'2"StoS 401/2" S to S Add-on cabinet 30" 36" 42" 30"
 36" 42" 830-610 836-610 842-610 830-710 836-710 842-710 8716%"*
 8716%"* 8716%"* 8716%"* 8716%"* 8716%"* 30728'/a" 36734V6"
 42740V6" 30728'/8" 36734W 42740 Ms" 15W13V6" 15%713'/8"
 15%713'/e" 28V6»/25V6" 28V6725V6" 28i/6725i/6" 28Va" S to S 34i/e" S
 to S 401/s" S to S 28W S to S 341/a" S to S 401/6" S to S •Deduct %" when
 compressor is used. 289

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Storage
 Cabinets Standard cabinets often provide a fast, flexible, and economical
 solution to many storage problems. Table 7 provides dimensional data and
 capacities for four typical cabinet types. These cabinets are manufactured
 by Steelcase. The dimensions of cabinets of other manufacturers will differ
 somewhat. The data in Table 7, however, are adequate for preliminary
 planning purposes. TABLE 7 Storage Cabinets Description Outside/inside
 dimensions Depth Width Height Linear capacity Storage cabinet 18"/17"
 18"/17" 18717" 18717" 24723" 24723" 18717" 18717" 18717" 24723"
 367331/a" 36733W 36733 V4* 36733W 36733W 36733 Vtf 36733W
 36733 Ve" 36733W 36733W 411/4735%" 52%7463/e" 64%7583//
 801/2"/74'/2" 64B/a7583/4" 80V4774W 52%7463/8" 64%7583// 80W/74W
 64%7583/t" 99'/2" side-to-sid< 3 shelves (2 adj., 99 V4" S to S on 3 shelves
 (2 adj., 132%" S to Son 4 shelves (3 adj., 165%"S to Son 5 shelves (4 adj.,
 1323/t"StoSon 4 shelves (3 adj., 165%"S to Son 5 shelves (4 adj., Not for
 filing Not for filing Not for filing Not for filing 3 on 1" thick) 1" thick) 1"
 thick) 1" thick) 1" thick) 1" thick) Wardrobe cabinet S^ ^J 24723" 36733W
 80V4774V4" Not for filing Wardrobe/storage cabinet 18717" 36733W
 52%746%" 43'/i"StoSon 3 shelves (2 adj., 1" thick) 18717" 18717" 24723"
 24723" 18716%" 18716%" 36733W 36733W 36733W 36V33W 1871BW
 18715'/2" 64%758%" 80V4774V4" 64%7583/4" 80W74W 41Va735%"
 52V4747" 58" S to S on 4 shelves (3 adj., 58" S to S on 4 shelves (3 adj.,
 58" S to S on 4 shelves (3 adj., 58" S to S on 4 shelves (3 adj., 15'/e"onbar

15 1/2" on bar 1" thick) 1" thick) 1" thick) 1" thick) Wardrobe 187 16" 187 15" 64 VS 759 1/2" 15 1/2" on bar 290

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Interior Card Trays TABLE 8 Interior Card Trays (For use in vertical or lateral files)

| Description | Style No. | Inside dimensions | Depth | Width | Height | Linear capacity |
|---------------|-----------|---------------------------|---------|---------------|-----------|-----------------------------|
| 3x5 card | 4335 | 11 1/2" x 5 1/2" x 3 1/4" | 10 1/4" | 3x5 card | 4337 | M 14 1/2" x 5 1/2" x 3 1/4" |
| 5x8 card | 4355 | 11 1/2" x 8 1/2" x 4 1/2" | 11 1/4" | 5x8 card | 4357 | 14 1/2" x 8 1/2" x 4 1/2" |
| 4x6 card | 800-TN-46 | 12 1/2" x 6 1/2" x 4 3/8" | 11 1/2" | 4x6 card | 800-TW-46 | 15 1/2" x 4 1/2" x 14 1/2" |
| Tab card | 7201 | 11 1/2" x 7 1/2" x 3 1/2" | 10 1/2" | Tab card | 7204 | 14 1/2" x 7 1/2" x 3 1/2" |
| Coin and bill | 4388 | 8 1/4" x 5 1/8" x 1 1/2" | 1 1/2" | Coin and bill | 4389 | 7 1/2" x 3 1/2" x 1 1/2" |

• Dimensions of each of 6 bill compartments. t Dimensions of each of 5 coin compartments, t Card trays cannot be installed in 6"-high shelf located directly below a door Use 3"-high shelf and refer to guidelines. 291

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Storage Components Glossary Lower storage/Lateral flia Free-standing wall- or panel- mounted files with width dimension greater than depth dimension. Rails Mount In lateral file for drawer suspended filing, front-to-back or side-to-side. Media compartment kit Can be retrofit or factory assembled to 800/900 Series 6" roll-out shelf. Provides dividers and partitions adjustable for storing a variety of media- cassettes, mini-cassettes, cartridges, floppy disks, and more. Storage cabinet Storage for general supplies. Available In 2,3, or 4 adjustable shelves. Mobile padaatal Supports drawers In several combinations and has casters for mobility. T-bar filing for bound printouts Accepts EDP printouts In T-bar type binders. OverfMe cabinet For use above lateral or vertical files. Sliding door, lock, and shelf-modlfler options. Vertical file Letter- or legal-size filing cabinet with depth dimension greater than width dimension. For front-to-back filing only. Partition A double metal wall that mounts Into a lateral file drawer to divide drawer. Vertical Hi* drawer 6" high and 12" high drawers for letter or legal-size filing cabinets. Personal drawer For personal Items, An adjustable divider Is Included. 3" high. I5*, ••>^.

Wardrobe Provides full-width coat rod for

hanging clothes mounted beneath full-width shelf. Pull-out keyboard shelf Attaches beneath work- surface for computer keyboard support and storage, Wire tape racks Racks can be freestanding or built-in to lateral files for storage of magnetic tapes and disk cartridges on edge. Dividers can be positioned to accommodate media of different thicknesses. 292

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Storage Components Glossary Bookcase Units have adjustable shelves which can accommodate rows of standard ring binders and other bound materials. Cantar hook filing hanging bar Accepts printouts and magnetic tape reels with center hooks, Pile insert drawers For use In vertical files Instead of card trays. Hanging folder frame A metal rod mounted In lateral and vertical files for suspended file material, Can be mounted on partitions for front-to-back filing. Combination wardroba and atoraga cablnat Units are divided - space for hanging clothes and two or three vertically adjustable shelves. Compraaor A spring-loaded plate that supports file material. Can be moved and locked In position. Used In vertloal and lateral file drawers, pedestal file drawers and card trays. Dlvldar Metal plate used to separate and support file material. For lateral file and pedestal file drawers, fixed and roll-out shelves. Interior oardtraya Portable trays In various sizes for common card sizes: 3 x 5, 4 x 6, etc. Compressor Included. Latarfila drawer 3" tray drawer, 6" card drawer, and 9", 12" and 15" hlgh«le drawers. Letter or legal-size filing. Dividers, three sway blocks, compressor, hanging folder frames, rails, or partitions available. Lateral Ilia find shelf 12" or 15" high shelves with or without doors and with three dividers, Lateral file posting shelf Metal pull-out shelf option on 48" and 60" Interior height lateral files. When not specified, the space will be filled by a posting shelf filler. Double-door storage cablnat For miscellaneous storage below worksurface. Includes one adjustable shelf and two swing- arm doors. Lateral file roll-out shelf 3", 6", 12", and 15" high shelves extend for accessibility. File drawer For letter- or legal-size documents 12" and 15" high. 15" high drawers can also be used for computer printouts. For front-to- back or slde-to-slde or combination filing. Lateral flia workshelf 3" high roll-out workshelf with laminate surface. 293

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Electronic Media Storage Microfiche 4"x6" stm Description Microfiche is a

4" x 6" film transparency containing multiple rows of greatly reduced page images of any written, printed or graphic material. Image reductions range from 13 up to several hundred times smaller than the originals, A microfiche viewer enlarges the images so that they are readable. Labeling information is written or printed on a narrow strip along the long edge at the top of the microfiche. Microfiche may be stored in interior card trays, in lateral file 6"-high roll-out shelves, and in a lateral file media compartment, Microfilm 1"x 4" diameter reel (1V4"X4V4"X4V4"bOX) Description Microfilm is roll photographic film on a reel or in a square cartridge that contains images of pages of written, graphic or printed material reduced hundreds of times. A microfilm viewer enlarges the images so that they are readable. Microfilm on a reel is kept in a square plastic or cardboard box for protection and ease of handling. Microfilm is most conveniently stored on edge. Labeling is placed on one of the edges of the reel or box. Microfilm may be stored in the lateral file media compartment kit, in an interior card tray, and in lateral file 6"-high drawers or shelves. Print-out Paper Description Print-out paper, also known as continuous form data processing paper, is used in almost all computer printers and some word processing equipment. The most common types are recognizable by. ■ Small "pin-feed" holes along both edges which are used by the printer to grip and advance the paper. ■ Green or grey-shaded stripes across the paper which serve to organize the printed information. After printing, the print-out may exist in a "fan-folded" stack or it may be "burst," i.e., separated into individual sheets along the perforations that exist at the fold lines, If the print-out consists of a significant number of sheets, it may be "bound" for easier handling. Fan-folded printouts must be bound along the top or long edge. Print-outs that have been burst may be bound along either the long edge or the short edge. The binding may consist of only a narrow metal or plastic clamp or it may include a stiff plastic or fiberboard cover. Frequently the binding may include hooks at both ends so the print-out may be hung from two rails like a hanging file folder. Other hooks may be used to suspend it from special bars or rails. Identification information may be marked on one of the edges (depending on how the print-out is stored) or on the front sheet of the binding cover. Print-out paper may be stored in Steelcase lateral and vertical files depending on the paper size. Check the file which will accommodate your paper. Boxed paper can be stored in storage cabinets. Cartridges %"x 4" x 6" (cartridge) Va"x 4'A" x6'A"

(cartridge in case) Description Cartridges have 1/4" wide magnetic tape loaded into a reel-to-reel cartridge generally made of clear plastic with metal back plate. They look similar to an oversize recording cassette. Cartridges come in and are sometimes stored in a "flip-open" plastic or cardboard box. Labels or identification information are located on the long edge or on the side along the long edge of the cartridge or the box. Cartridges may be stored in interior card trays or in a media compartment kit in lateral file 6"-high drawers or shelves. 294

Office Spaces Cassettes Standard Cassette Case 1 1/2"x3 1/2"x4 1/2" Mini Cassette Case 1 1/2"x1 1/2"x2 1/2" Description Cassettes are available in "standard" and "mini" sizes and consist of magnetic tape loaded into a reel-to-reel configuration in a plastic case. Standard cassettes for electronic equipment are identical in size and appearance to those used for home recording. Cassettes may be used in microcomputers and in word processing or dictation equipment. They come in and are frequently stored in a flip-open plastic case. Labels or identification information may be located on the long edge or side of the cassette or its case. Cassettes may be stored in a lateral file media compartment kit. FURNITURE, FURNISHINGS, AND EQUIPMENT Electronic Media Storage Disk Cartridges 1" x 10" diameter 3" x 10" diameter 1" x 15" diameter 3" x 15" diameter Description Disk cartridges are round plastic cases which contain a series of rotating platters (or disks) on which data is magnetically recorded. The number of platters in a case varies with the height of the case. In use, the entire case is inserted in a computer disk drive unit where recording arms, which read/record information, enter the case through a slot with a spring door. Disk cartridges are flat in appearance with an elongated Y-shaped protrusion on the top. They can be stored flat or on edge. Labels for identification are usually located on the edge of the disk cartridge. Frequently disk cartridges have to be stored in a temperature/humidity controlled environment. Disk cartridges may be stored in lateral file 6"-high roll-out shelves, storage cabinets, and bookcases. Disk Packs 7" x 10" diameter 7" x 15" diameter Description Disk packs are round plastic cases which contain a series of rotating platters (or disks) on which data is magnetically recorded. The number of platters in a unit varies with the height of the plastic case. In use, the entire case is inserted in a computer disk drive unit where recording arms, to read/record information, enter the

case through a slot with a spring door. Disk packs are flat on the bottom and upright with the handle on top. Identification is generally located on the edge of the disk pack. These units should be stored in a temperature/humidity controlled environment. Disk packs must not be stored one on another. Disk packs may be stored in storage cabinets, bookcases, and on 3"- and 6"-high lateral file shelves. Floppy (Flexible) Disks 5 1/4" x 5 1/4" x 3/32" Description Floppy disks, also called diskettes or flexible disks, are small record-like disks each permanently enclosed in a square, stiff paper envelope. They are used to magnetically record information in all types of small computer and word processing equipment. Labels are placed on the paper envelope. Floppy disks may be stored in 12"-high lateral file drawers and shelves. 295

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT

Electronic Media Storage Laser Disks 12 1/8" x 12 1/8" jacket 12" diameter 4W diameter Description Laser disks look like long-play record albums complete with a paper protective sleeve. Data are stored and retrieved by laser beam. Laser disks may be stored in 15"-high lateral file drawers and shelves. Magnetic, Tab, and Aperture Cards 3 1/2" x 7 3/8" Description Magnetic cards, also known as mag cards, are a tab-size black plastic card with magnetic material coated on one or both sides. The cards are used to record or reproduce information in word processing equipment. In some cases one or more of the cards will be kept in a paper envelope. Identifying information will be marked on the face of the envelope. Tab cards, also known as keypunch cards, are 80 column cards or punch cards with small holes in them to represent bits of data. Although they may be stored in a workstation or central storage area, they are usually used and produced in a mainframe computer room or keypunch department, and are stored in specially separated decks. The decks are most frequently identified by markings across the edges of the cards. Aperture cards are tab cards with a piece of microfilm mounted over a hole in their center. They are most frequently used for microfilm images of engineering or architectural drawings. Aperture card reader/printers enlarge the image on a screen for reference and, if required, reproduce a full-sized copy of the drawing. Identifying information is printed in a narrow band along the top (long edge) of the card. These cards may be stored in interior card trays and in lateral file 6"-high drawers and roll-out shelves. Magnetic Tape Reels 1" x

Vh" diameter 1" x 8V2" diameter 1"x10'A"diameter 1" x 15" diameter
Description Magnetic tape is typically W wide and loaded on reels of
varying diameters. A flexible plastic strip locks around the outside of the
reel to protect the tape and prevent unraveling when it is not in use. This
media is used in large tape drive units that are generally found only in
computer rooms. The long-term storage of magnetic tapes is subject to strict
temperature and humidity requirements to prevent damage. Tapes are
labeled, both on the side and on the flexible strip that is placed around the
edge of the reel, They can be stored flat or on edge. For flat storage, handle
like disk cartridges. Magnetic tape reels may be stored in the lateral file
add-on cabinets, storage cabinets, or in free-standing wire racks on lateral
file shelves. 296

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT File
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Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT File
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Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT File
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Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Office
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Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Office
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S&CTIOM 3 vjAt-Hur vene«*- 301

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Office
Pantry SCALE '. Vt.V-0' Patt\$*h *iumi I'-o'/f 302

Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT
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Office Spaces FURNITURE, FURNISHINGS, AND EQUIPMENT Island
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Hospitality Spaces Restaurants 307 Bars 346 Hotels 374 305

Hospitality Spaces RESTAURANTS types and Sizes at Table Arrangements
The basic components of any restaurant interior are the chair and the table.
Depending upon restaurant type, menu, service, table setting, furniture
selection, and degree of intimacy required, table size and overall chair space
requirements can, and should, vary greatly. A restaurant that encourages
rapid turnover of customers will normally provide smaller table top and
chair sizes. On the other hand, those restaurants that encourage limited
turnover and emphasize the wining and dining experience will typically
provide largertable top sizes and larger, more comfortable chairs, with
greater distance between table groupings. There is no agreement among
even the most experienced restauranteurs and restaurant designers as to
what the optimal table and chair dimensions should be. In addition, many
other design factors will influence the final decision, including circulation
and egress, accessibility standards, methods of service, and the overall
dimensions of any given space. Figures 1 to 19 provide the designer with
restaurant planning standards that have been developed by many
experienced architects and interior designers. These drawings not only show
the various individual table and chairarrangements, but provide the designer
with groupings of these arrangements, as well as an indication of overall

size, floor area, and number of persons accommodated. These arrangements, however, should only be utilized for preliminary planning information. LUXURIOUS INTERMEDIATE ECONOMICAL 0 27." ir- zo" 17"-10" L 18" 16" 15" W 18"-20" It" '~T<t~~ A 18" 16" ~"16" PLAN SECTION CHAIR. DIMENSIONS LUXURIOUS A'-o" INTERMEDIATE 5'-4" io 5'-A" ? person? M"x 70"- i"" /I PfRSON? 4a" x w'-w ---L ECONOMICAL 5'-0" TO V-2" 1 PIRSOIK 4 PERSONS ■U'-IU'Al'iS' £ o" lo A'- A" /\jk- "-i1?" 1 \.-.niA«:i-' OK ROUND // ^■n,,,...^ y <;■:•:■ io y ts-r s'-a1 to 'y-io" / y yr- 34" n SQUARt OR ROUND 1__JL 1'-<?'■.!" TO /Ml" i5'-£ I io" \ SQUARE iOR ROUND. , CHAIR-AND-TABLE UNITS (OCCUPIED) AISLE WIDTHS: FIQ. 1 !<T> l'im' If CIRCULATION'. '-^" l.'IMIHUM CI I AK WIIUII \OR M^VICf ONLY ' 74" MIN. BT IIVI I N CHAIR BACKS FOR MAIN ENTRANCE : I AR.GE AS POSSI BLE 307

Hospitality Spaces RESTAURANTS Types and Sizes of Table

Arrangements 30"« 36" 2 PEOPLE r . ~ 24" JO" 60' I- " I 2 PEOPLE r _ IS" __ 2 A" "" 24" IB' 2 PEOPLE _«L_j 4 2"■ 42" rsj , 4 PEOPLE 4 PEOPLE 36" J . 36" "B" 4 PEOPLE 4 PEOPLE 1— - _ . 18" J J5_6" 30"* JO" -^A 4 PEOPLE / ^ LUXURIOUS SEATING SUCH AS OINING 14 SO. FT. /PERSON INTERMEDIATE SEATING CAFETERIA RESTAURANT I2S0 FT./ PERSON 4 PEOPLE ECONOMICAL SEATING SUCH AS BANQUET 10 SO,FT,/ PERSON r—1 1 { |—~| •-S , ■, .. B 13- n I., w I—-. V s tr s t:J 1 ; fl I ' a a "■ s 24" Is" f' 11 H"'-^ U J SL jblJi.Jij'I 36" 1 60" ' 36" , 60" " t SQUAR E SF ACING u 36" n 4 N * Dj SERVICE TOPS, 24' CHAIRS TABLES AISLES SQUARE SEATING 60" MIN. BETW. TA8LE AISLE PLUS TWO CHAIRS BACK TO BACK 36" 1" TO 18" FROM FLOOR TOSEAT, I7"PR£FERRED FOR WOMEN 29" TO 30" HIGH DIAGONAL SPACING SERVICE AISLES DIAGONAL SEATING 30" MIN BETW. CORNERS OF TABLE TOPS Fig. 2 308

Hospitality Spaces RESTAURANTS types and Sizes of Table

Arrangements a'-** ^'-fr* *i&" \ ■y.o- + ■*'-» // Fig. 3 Seating for 2. V^-2 q. Jr Fig. 4 Seating for 4. -V II t •* • 1 N -J +~^ ' ■T"" r- V.V.V..V. :;::;::;:-:~::: i J 1 I I ir TFr"" k .*.!.'i'i1!.1 1 I 1 '!.••i'i'i'i*. 1 J 11// :;::;::;:x;:: LLliii. -«r Jr *), ±JJ* U. HiT | WW MO £/ * c— 309

Hospitality Spaces RESTAURANTS types and Sizes of labia Arrangements
 1 jln Fig. 5 13 ft x 27 ft, 351 ft', seats 18. . i —■»,<—■ i HI.IMII. * v~m ii
 i t «■ - ' ■ —* « ■ ■> « * Fig. 6 8 ft x 27 It, 216 IP, seats 12. 71
 v&iAsc&'U&wse? oe>em&\oi-\ Fig. 7 33 It x 11 it, 363 IP, seats 12. ^ -V
 310

Hospitality Spaces RESTAURANTS Types and Sizes of Table
 Arrangements C e^ M A c u U A- r i M ■IHHiiiiiii v I
 mmHHW«B««twnttmHwmimwwmMfffrommH«mmmm Fig. 8 33 ft x 22 ft,
 726 ft2, seats 26. Fig. 9 11ft x 28 ft, 308 ft2, seats 12. 311

Hospitality Spaces RESTAURANTS Tables: Design Criteria • 5-10"
 AJ0"MIN- FOUR PERSONS 18'-20' -f ySJ// 24'«-30" r22,u-v 34"—4"—
 30"-+! Fig. 10 These drawings highlight several critical dimensions that the
 designer must consider. Aisie circulation must be adequate in width; other
 clearances to consider include chair depth from edge of tabie and clearance
 between chairs. While laying out chair and tabie arrangements, a designer
 must anticipate the potential conflict between a patron ieaving a seat and a
 tray-carrying waiter. -v-io"->j«-r-8,-> o lli—f ran, 6" 5'-3" 2'-8' k '-6' Access
 Access 312

Hospitality Spaces RESTAURANTS Types and Sizas of Banquette
 Arrangements t WAU- OfL WEt£&^ CC^fEUCfl Ot^y Fig. 13 21ft x 9 ft,
 189 fP, seats 12. Fig. 14 27 ft x 10 ft, 270 ft2, seats 12. 313

Hospitality Spaces RESTAURANTS IVpes and Sizes of Banquette
 Arrangements 314 |V«' v* l'«'Tt>f * ..y-f Atf >N» It Y *B. 75*1 he? li C. C
 O t- A. T \ cH ICM—^ Fig. 15 19 ft x 9 ft, 171 ft2, seats 12. mm t H -1 V 1
 hi r* '*. '< X •* * r | r ■ iHfiir . .v.v!.v 1 i^S^!;: 1 1 ;:::~*~!~*~;: j 1 I 8\$\$! if
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 1 \ \ I 1 1 1 III/ ,,,i,i,,i,. :#\$\$S: :###: :\$#:\$ Fig. 16 19 ft x 15 ft, 285 ft2,
 seats 24. mlfol Kw&l elffli teH i ■iWA 3G1 <t-i p-c-U l> a. T I ° M ^ i f-p-
 jfj f-g- j ■■,«' j t'-q" jri *i»» j ■■-<* j »'o" >-■ *w j ;» «MU^lili^iwi^^

lllvMi H ' ■-*■*■*■*.*' ■'■' I ■'■'.'■■ I.I.I'■'!'!.'■■ <., Fig. 17 21 ft x 19 ft. 400 ft², seats 24.

Hospitality Spaces RESTAURANTS Types and Sizes of Banquette

Arrangements C-6.' *'">'. <' .>. <', "., ". _ '.! '<:' ■*.!. !. •!,•!* ,;* '!**■** :>>*.; •,'."!'A •.* I*I! ■ mmmmmm wmtimm wn ^y^^A^ay'^S^S^r^Wrtii'i
mmmmmmmm l I II I II I It IMtlM' n ii*!,, *"" ,/>)'«■■ "t"" J »H*', '^ -J ***,
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T1.'!..I.L.I'TI.,T,T'I,I,!IX"! «:::.!:!>■::.:■> JT !«WWW 1,1,1,1,1,1,1,1,1,1!
i'i'i'i'i'i'il'S Fig. 18 24 ft x 20 ft, 480 ft², seats 36. A = main circulation, B = activity zone.
4—— Fig. 19 13 ft x 13 ft, 169 ft², seats 16.

315

Hospitality Spaces RESTAURANTS Dining Room Seating Restaurant and Dining Room Seating Dispersed seating suitable for guests with restricted mobility should be available in restaurants, coffee shops, and dining facilities. As a guide, the Uniform Federal Accessibility Standards (UFAS) require a minimum of 5 percent of restaurant seating to be accessible. Accessible aisles should connect the entrance to these seating locations, public restrooms, and self-service areas such as salad bars, condiment stands, or buffet tables. Comfortable seating for waiting should be available to customers near the entrance. A variety of accessible seating should be available, suitable for large and small dining groups. Small tables may not be accessible to guests in wheelchairs because of the restricted kneespace. Therefore, a party of one or two may require a table usually set up for four. Restaurants or coffee shops with built-in seating, such as booths or banquettes, should also provide some chairs for guests who have difficulty getting into and out of bench seating. These chairs can be removed to seat guests in wheelchairs. Where seating areas are raised on platforms, accessible seating and similar services should be available on the main-floor level or a ramp to the upper level should be provided. Aisles serving accessible seating should be at least 3'0" wide, which typically requires a 6'0" clearance between parallel tables, or 4'6" between rotated tables. (See Fig. 21.) Aisle widths should also provide room for customers to be seated at tables. At least a 2'6" clear space should be available behind each seating location. This space allows chairs to be withdrawn from the table and staff to assist guests reposition chairs close to the table. For wheelchair seating, a

3'0" to 3'6" aisle is necessary, depending on the width of the kneespace. (See Fig. 25.) Wheelchairs positioned at tables project approximately 5" further into aisles than most chairs. To allow guests with restricted mobility to turn around, seating arrangements should also include a 5'0" diameter circle or T-shaped clear area at dead-end aisles. Dining Tables and Chairs Accessible seating locations should allow guests With restricted mobility to dine with ambulatory customers. Tables should provide kneespace for customers in wheelchairs, and dining chairs should be coordinated to provide comfortable seating at the same table height. Dining room chairs should be stable to maintain balance as guests seat themselves, and comfortable to sit in during dinner. Chairs should be light and easy to reposition. The seat should have a slight slant to the rear to transfer body weight to the back of the chair. However, an exaggerated incline makes it difficult to rise. The seat should be approximately 16" deep and at least 16" wide to allow space for customers to reposition themselves during the meal. Padding or cushions on the chair seat should be firm, and the chair back should also be slightly inclined to the rear. To help guests sit and rise, dining chairs should have armrests 7" to 8" above the front edge of the seat. (See Fig. 22.) Supports or cross-bracing should not 5" (13). Fig. 20 Restaurants or coffee shops with fixed seating should include some movable seating for guests in wheelchairs and guests who have difficulty getting into and out of the bench seating. Number in parentheses is dimension In centimeters. Fig. 21 Aisles serving accessible seating should provide a path at least 3'0" wide for passage and clear space for guests to seat themselves at tables. Numbers In parentheses are dimensions in centimeters. 316

Hospitality Spaces RESTAURANTS Dining Tables and Chairs interfere with kickspace below the seat, so the feet can be positioned to rise. The front edge of the chair seat should be low enough to allow the feet to rest on the floor, but not so low that it is difficult to rise. This is determined by the lower leg length (popliteal height) which varies between 15" and 20" for most adults. The height of the chair seat should be 10 1/2" to 11 VS" below the top of the table. Common seat heights vary between 14" and 18". Because the height of wheelchair seats is typically 19", a relatively high chair seat is necessary to coordinate with the table height. A chair with an 18" high seat is comfortable for most ambulatory guests and closely approximates the height of a wheelchair seat. Dining room tables should

have a stable surface at a convenient height and knee- space and legroom below the tabletop for customers in wheelchairs. Narrow table configurations allow face-to-face seating, which reduces the distance between diners, making conversation easier and table lighting more effective. For safety, the corners and edges of the top should be rounded. Full-height wheelchair kneespace is 2'6", which requires tabletops to be at least 2'7" above the floor, too high for most seating. Many wheelchairs now provide adjustable or two-tier armrests, which allow customers to sit close to tables in a kneespace only 2'3" high. To provide this kneespace, the tabletop (without an apron) should be 2'4 1/2" to 2'5" above the floor. This is 11" to 11 1/2" above the chair seat, 10 1/2" to 11" above the seat of wheelchairs, and convenient for both. This kneespace also permits the armrests of chairs to pass below the tabletop so seated customers can draw close. This combination of tables and chairs is suitable for the majority of wheelchair users and most ambulatory guests. Footroom is important for customers with wheelchairs or leg braces. The footrests of wheelchairs are 2 1/2" to 3" above the ground and angled slightly forward, which requires 17" of footroom, measured from the edge of the tabletop. The outside width of footrests is only 1'6", but 2'6" of side-to-side clearance is necessary to maneuver into position beneath the table. To provide kneespace, table legs should be at least 2'6" apart, and the tabletop, for face-to-face seating, should be 3'6" wide. Pedestalbase tables should have low, tapered bases and a minimum diameter of 3'6", although 4'0" is preferred. A portable raised leaf should be available to modify tables for customers in wheelchairs with high armrests. The leaf should be approximately the size of a place setting, 1'4" by 2'0", and secured to the underside of an accessible table with clamps. The raised leaf should project 6" beyond the edge of the table and provide 2'6" clearance above the floor (See Fig. 24.) 10 1/2" to 11 1/2" (27-29)

■ f 3 So* «5 La Kickspace I Fig. 22 Dining room chairs should have a seat 10 1/2" to 11 1/4" below the top of the table and armrests 7" to 8" above the seat. To coordinate with an accessible table, the seat height should be 18". Numbers in parentheses are dimensions in centimeters. Fig. 23 Accessible tables should provide kneespace at least a 2'3" high by 2'6" wide with 17" of footroom. To increase the kneespace height, a raised portable leaf can be provided. (See Fig. 24.) Numbers in parentheses are dimensions in centimeters.

• W Kneespace Width 2'-6" (minimum) 3'-0" or greater Clearance 3'-6" 3'-0" Fig. 24 A portable raised leaf can be provided

for accessible tables to accommodate customers in wheelchairs with high armrests. Numbers in parentheses are dimensions in centimeters. Fig. 25 The necessary maneuvering room required to access a kneespace depends on its width. 317

Hospitality Spaces RESTAURANTS Banquettes: Design Criteria Figure 27 shows the basic dimensions for the design of banquet seating. The lack of armrests makes it difficult to define seat boundaries. The user, therefore, tends to establish a territory by assuming a desired sitting posture and placing personal articles next to him or her, such as a briefcase, purse, or package. Since the nature of this type of seating can permit some form of body contact, hidden dimensions and personal space also play an important part in how close the users sharing the banquet will sit. Because of the many hidden psychological factors involved, the actual efficiency of this seating type in terms of capacity is questionable. Figure 27 indicates two possible seating situations, each dictated by the anthropometrics involved. One arrangement is based on the premise that the user's elbows will be extended, possibly in conjunction with some activity, such as reading, or simply as an attempt to stake out additional territory, as would be the case in the strategic positioning of some personal article on the seat. In this situation it would be reasonable to assume that each user would take up about 30 in, or 76.2 cm, of space. The other diagram shows a more compact seating arrangement. Figure 26 shows a section through a typical banquet, in cm A B C D E 18-24 15.5-16 16-17 30 24 45.7-61,0 39.4-40.6 40.6-43.2 76.2 61,0 SECTION Fig. 26 Banquette seating LOW DENSITY Fig. 27 Banquette seating. HIGH DENSITY 318

Hospitality Spaces RESTAURANTS Banquette/Bench Seating Details aV 'H' rt&rU tOtOO V—' L Aif.V t'-fj. . '^Sfjj i 1TT»'<5' f ^ 7 1 /g\ ifctwWrf cWU a ft** fthWuffte-' canity tg* Banquette seating provides the designer with one of the few opportunities to custom design restaurant seating. While there can be a great variety of aesthetic solutions achieved through use of various materials, ergonomic considerations must be analyzed carefully. Specific attention should be given to depth of seat, slope of seat and back, height of back, and relationship of seat height to table height. /-\ flH»M4< ■I »c*tMH«TY|0*M ton*rrypow-i «h- «««; Mept-*f«st*(* |u»<wi«<fYfia»M >-»* .V 61, *s. »«- M*W7 e^ifC -pit* <*<»>(. TVPl^L

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Hospitality Spaces RESTAURANTS Banquette/Bench Seating Details
Banquette seating can be detailed relatively simply, as Figs, 28 to 30
suggest. The simplest form of banquette seating may take the form of a
plywood seating platform with a removable seat cushion, or a box cushion
seat and back support. Such seating is appropriate in fast food or quick
turnover restaurant operations. -kw*s**l Finish o*- 0 ©I7E.TAIU of
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6.'rtM»ni,'>«J&-ul. vPtr. kA££ ES*A_ tV '4"- acAU-iit'-i'-o" Fig, 30 321

Hospitality Spaces RESTAURANTS Lunch Counters: Design Criteria
Figure 31 shows some of the basic clearances required for a typical counter:
36 in, or 91.4 cm, for workspace behind the counter; 18 to 24 in, or 45.7 to
61 cm, for the counter top; and 60 to 66 in, or 152.4 to 167.6 cm, between
the front face of the counter and the nearest obstruction. Figure 32 shows a
section through the counter and back counter. Most counters are about 42
in. or 106.7 cm, in height. The clearance from the top of the seat to the
underside of the counter top and the depth of the counter top overhang are
extremely important. Buttock-knee length and thigh clearance are the key
anthropometric measurements to consider for proper body fit. Footrest
heights should take into consideration popliteal height. In most cases this is
ignored, and 42-in counters are provided with 7-in, or 17.8-cm, footrests
that are 23 in, or 58.4 cm, below the seat surface, which cannot work. The

popliteal height of the larger user, based on 99th percentile data, is only about 20 in, or 50.8 cm. Therefore, the feet dangle unsupported several inches above the footrest and the body is deprived of any stability. The footrest shown in Fig. 32, although high, only serves a portion of the seated users and is intended primarily for standing patrons. The most logical solution is a separate footrest, integral with the stool.

| | in | cm | A | B | C | D | E | F | G |
|---------|-------|-----------|-----------|-------|-----------|-------|------|-----------|-----------|
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| | 10 | 12-13 | 152.4- | 167.6 | 45.7-61.0 | 91.4 | 61.0 | 30.5-45.7 | 88.9-91.4 |
| | 106.7 | 76.2-78.7 | 27.9-30.5 | 25.4 | 30.5-33.0 | | | | |

Fig. 31 Lunch counter. ^ Work/Activity Zone ^ Circulation Zone 322 Fig. 32 Lunch counter.

Hospitality Spaces RESTAURANTS Miscellaneous Counter Details
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Hospitality Spaces RESTAURANTS Waiter Station/Host Counter Details 1
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 N a LI Fig. 33 Walter stations and host/hostess counters can be designed as
 freestanding elements or Integrated Into the Interior architecture, as shown
 by these details. Special attention must be given to specific drawer and
 storage requirements. 328

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Hospitality Spaces RESTAURANTS Wheelchair Accessibility to Self-
 Service Areas Self-Servlce Areas Salad bars, buffet lines, condiment stands,
 and other self-service areas should be accessible, Cafeteria or food-service
 lines should have a minimum width of 3'0", but a width of 3'6" is

recommended to permit ambulatory customers to pass customers in wheelchairs. The tray slide should be 2'10" above the floor, the maximum height for customers in wheelchairs and convenient for ambulatory guests. The tray slide should be continuous, if possible, from the entrance to the cashier. Tray slides restrict access to the counters and therefore should not be wider than necessary (1'0" recommended). In this instance, the reach of a customer in a wheelchair is extended if the wheelchair can be angled or positioned perpendicular to the tray slide. This is possible if the lower face of the counter is recessed to provide low knee-space. (See Fig. 34.) For guests with a limited range-of-motion, food, beverages, utensils, or other items should be displayed near the edge of the counter where they are easier to see and reach. When duplicate items are displayed, a vertical rather than horizontal arrangement allows customers to select items at the most convenient height. Self-service systems, such as beverage or ice-dispensers, should be easy to operate without fine hand function, instructions and price information should be prominently displayed in large clear lettering. Salad bars and buffets should provide a 3'0" -wide clear space for access on all sides and plate slides, or areas to temporarily set plates, at a maximum height of 2'10". This permits customers to serve themselves with one hand, without simultaneously balancing the plate or bowl. Kneespace 2'3" high below the counter or table allows front wheelchair approach, to increase customers' forward reach. Condiments should be located as low and close to the edge of the counter or table as practical. A tilted mirror above the food display at salad bars also aids customers in wheelchairs and children. (See Fig. 35.) For some customers with restricted mobility, poor balance, or limited hand function, it is more difficult to carry a plate. Therefore trays should be available at both salad bars and buffets. Fig. 34 Cafeteria lines should be wide enough to accommodate guests in wheelchairs. Food and beverages should be within a convenient vertical and horizontal reach. Numbers in parentheses are dimensions in centimeters. Fig. 35 A plate slide is recommended at salad bars and a kneespace at the counter. A mirrored surface above the bar is a further aid to guests in wheelchairs. Numbers in parentheses are dimensions in centimeters. 330

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Hospitality Spaces RESTAURANTS Lunch Counters: Cashier Station \ w
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Hospitality Spaces RESTAURANTS Service Counter; Host Cabinet;
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Hospitality Spaces RESTAURANTS Condiment Cabinet; Dining Room
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Hospitality Spaces RESTAURANTS Salad Bar/Serving Counter HOOP
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Hospitality Spaces RESTAURANTS Tray Pick-Up Details; Condiment
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Hospitality Spaces RESTAURANTS Condiment Counter Details attr.ep.
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<\Q*\\d*. .\$.<?.' DETAIL SECTION AT CONDIMENT COUNTER 342

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AT SALAD BAR /GRANITE COUNTER 343

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Hospitality Spaces BARS Bar Section Details The distance between bar and
backbar should allow adequate workspace. A minimum of 36 in, or 90 cm,
should provide space for one bartender to serve and another to circulate
behind. Maximum body depth and maximum body breadth are the primary
anthropometric considerations in establishing clearance. A one-bartender
operation would require a 30-in, or 75-cm, clearance. In regard to bar stools,
clearance between the stool seats is more critical than center line spacing,
and it should allow patrons of larger body size a comfortable side approach
and departure from the stool without body contact with the next person. A
12-in, or 30- cm wide stool on 24-in, or 61-cm, centers, which is quite

common, will allow only less than 5 percent of male users access to the stool without disturbing the next patron, while a 30-in, or 75-cm, spacing will accommodate 95 percent of the users. The tradeoff however, would be the loss of two seats for every 120 in, or 300 cm, of bar length. A spacing of 12-in stools on 28-in, or 70-cm, centers is suggested as a compromise. The ultimate decision is an individual one and must reconcile human factors with economic viability.

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
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| 54 | 18-24 | 24 | 30 | 16-18 | 24-30 | 30-36 | 28-38 | 100-128 | 42-45 | 11-12 | 6-7 | 7-9 | 6-9 | 22-26 | 60-69 | 36-42 | 137.2 |
| 45.7 | 61.0 | 61.0 | 76.2 | 40.6 | 45.7 | 61.0 | 76.2 | 91.4 | 71.1 | 96.5 | 254.0 | 325.1 | 106.7 | 114.3 | 27.9 | 30.5 | 15.2 |
| 17.8 | 17.8 | 22.9 | 15.2 | 22.9 | 55.9 | 66.0 | 152.4 | 175.3 | 91.4 | 106.7 | 000 | O | 000 | O | BAR AND BACK-BAR | I* | K=ae»K=ara^)(=aDi=ii=iai=icacai=ii=ii=ii=ii=iddQqO ' Customer Activity Zone #> JiooQaDaoam «=l «=l 1=11=1 tS> CM ^wlkT°^ Activity Zone F Daaoc n BAR / SECTION Fig. 1 346 |

Hospitality Spaces BARS Counter Types and Sizes r4' 3Daaaaaaao ® High Divider / (No Visibility) X_r.j paQaaQaaaaaaaaoaaq] * » Activity Zone Fig. 2 Bar/clearances public side. To ensure proper circulation and interface, adequate clearances in front of the bar are illustrated in Fig. 2. A customer activity zone of 18 to 24 in, or 45.7 to 61.0 cm, should be provided to allow for seating, standing, and access, in addition to a general circulation zone of at least 30 in, or 76.2 cm. If a supplementary drinking surface or shelf is provided, a smaller activity zone of 18 in is suggested in front of the shelf. The shelf can be 10 to 12 in, or 25.4 to 30.5 cm, deep. Figure 3 shows suggested clearances for 18 or 24 in cocktail tables, in cm

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
|-------|-------|-------|------|------|-------|------|--------|----------------------------------|-------|-------|------|-------|-------|-------------|
| 76-84 | 66-72 | 10-12 | 18 | 30 | 18-24 | 76 | 54-56 | 6-9 | 7-9 | 42-45 | 24 | 29-33 | 32-36 | 193.0-213.4 |
| 167.6 | 182.9 | 25.4 | 30.5 | 45.7 | 76.2 | 45.7 | 61.0 | 193.0 | 137.2 | 142.2 | 15.2 | 22.9 | 17.8 | 22.9 |
| 106.7 | 114.3 | 61.0 | 73.7 | 83.8 | 81.3 | 91.4 | Fig. 3 | Cocktail tables/seating for two. | 347 | | | | | |

Hospitality Spaces BARS Bar Shapes: Planning Criteria Fig. 4 U shape: 16 ft x 11 ft, 176 ft², seats 9. Bar shapes, seating capacities, overall dimensions, and "footprints" of bar areas vary greatly. Figures 4 to 22 show examples of bar designs drawn at a scale of W = 1'0". Careful study of these designs would suggest that seating width, spacing, and circulation areas must be given special attention. Fig. 5 Straight/enclosed: 20 ft x 10 ft, 200 ft², seats 9. Fig. 6 Straight bar: 25 ft x 10 ft, 250 ft², seats 8. 348

Hospitality Spaces BARS Bar Shapes: Planning Criteria Fig. 7 Angular: 30 ft x 12 ft, 320 ft², seats 10. Fig. 8 Enclosed/rounded end: 22 ft x 9 ft, 198 ft², seats 10. 349

Hospitality Spaces BARS Bar Shapes: Planning Criteria Jr. 1. Fig. 9 L shape: 30 ft x 13 ft, 390 ft², seats 15. Fig. 10 Angular: 16 ft x 16 ft, 256 ft², seats 11. 350

Hospitality Spaces BARS Bar Shapes: Planning Criteria 2. Fig. 11 Octagon/partial: 26 ft x 18 ft, 468 ft², seats 16. Fig. 12 L shape: 27 ft x 20 ft, 510 ft². Bar seating, 17; additional seating, 10. 351

Hospitality Spaces BARS Bar Shapes: Planning Criteria Fig. 13 Polygon: 36 ft x 11 ft, 396 ft², seats 18. Fig. 14 U shape: 21 ft x 20 ft, 420 ft², seats 22. 352

Hospitality Spaces BARS Bar Shapes: Planning Criteria Fig. 15 Curvil Fig. 16 Straight bar: 40 ft x 10 ft, 400 ft², seats 24. 353

Hospitality Spaces BARS Bar Shapes: Planning Criteria Fig. 17 Octagon/freestanding: 28 ft x 21 ft, 558 ft². Fig. 18 Circular/freestanding: 22 ft x 22 ft, 334 ft², seats 26. 354

Hospitality Spaces BARS Bar Shapes; Planning Criteria Fig. 19 Polygon Irregular: 20 ft x 25 ft, 360 ft², seats 27. Fig. 20 Octagon: 25 ft x 25 ft, 429 ft², seats 28. 355

Hospitality Spaces BARS Bar Shapes: Planning Criteria Fig. 21 Horseshoe/oval end: 34 ft x 19 ft, 546 ft², seats 34. Fig. 22 Racetrack: 50 ft x 19 ft, 750 ft², seats 48. 356

Hospitality Spaces BARS Bar Plans, Elevations, and Sections The detailing of bars and backbars can vary from the very simple and basic to the complicated and intricate. Figures 23 to 40 provide the designer with selected examples of working drawings from some of the most experienced

addition, local building codes and health codes must be consulted. <KIUN3
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Hospitality Spaces BARS Bar Plans, Elevations, and Sections &W> —I-,
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(Continued) 358

Hospitality Spaces BARS Bar Plans, Elevations, and Secllons .^"'^ArT. "D
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Hospitality Spaces BARS Bar Plans, Elevations, and Sections 0 0 0 0 0
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1/2" SCALE Fig. 25 S2"GRANITE FLOOR 360

Hospitality Spaces BARS Bar Pians, Elevations, and Sections SECTION
THRU BAR CABINET 1'-4'/B' *o ; : 1 > ^ V» POPLAR-\ PLEXI <" LENS

POPLAR-^' fOUKINT Fig. 26 VERTICAL SECTION AT METAL POST
AND LIGHT BOX 361

Hospitality Spaces BARS Bar Plans, Elevations, and Secllans ©PLAN Q
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Hospitality Spaces BARS Bar Plans, Elevations, and Sections -suqni»n«* *
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Fig. 30 Hospitality Spaces BARS Bar/Serving Counter Details ■ 1,T.
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Hospitality Spaces BARS Bar Plans, Elevations, and Sections U'o' tiunfc
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Hospitality Spaces BARS Bar Plans, Elevations, and Sections »■-*■ 1 I 't
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STORAGE UNIT Fig, 37 CHERRY VioVcherry VERTICAL SECTION-
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Hospitality Spaces BARS Bar Section Details H ■^JACKt BACK BAR
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 VERTICAL SECTION AT BAR Fig. 40 r-aVy «' mE" SECTION AT BAR
 373

Hospitality Spaces HOTELS Guestroom Plans It is interesting to note how trends in hotel design have headed off in two directions, especially in regard to the design of rooms. On one hand, an effort is being made to provide more luxurious multipurpose rooms and suites. The hotel room as office away from work or as fantasy sleeping/relaxation environment often results in rooms with work areas, living rooms, and hot tubs, just to name a few of the more popular amenities. On the other hand, there is a trend toward economy accommodations. Hotel rooms are being designed as a place to rest and sleep, a place to feel comfortable and safe at a reasonable cost. Accordingly, these rooms use less floor area and provide less secondary or frill items. With both of these approaches, however, designers must ensure that the room or suite layouts are accessible to the physically challenged. In that regard, various room layouts and bathroom plans are provided in this section that address this issue, g-S Id) Fig. 1 (a) Uris Brothers Hotel, New York, (b) Americana Hotel, New York, typical tower room, (c) Loews N.Y. Motel, typical room (d) Causeway Inn, Tampa, Florida. 374

Hospitality Spaces HOTELS Guestroom Plans with interior (e) (d) Fig. 2 Guestroom plans, (a) Typical double-double finishes plan: vinyl wallcovering (WC), paint (P), carpet (C), ceramic tile (CT) identified and keyed to legend, (b) King-studio (Holiday Inn): standard layout with armoire unit and large lounge area including a convertible sofa, (c) Reversed layout (Sheraton, Washington, D.C.): unusual room with bed placed in front of window and lounge area near bathroom, (d) Luxury king room (Sheraton Grande, Los Angeles): oversized room with shelf/ledge in place of headboard, large desk surface, and lounge area; four-fixture bathroom. 375

Hospitality Spaces HOTELS Guestroom Plans Fig. 3 Motel rooms - exterior entrance. Fig. 4 Motel rooms - interior corridor. 376

Hospitality Spaces HOTELS Accessible Guestroom Plans

Accessible guestrooms have design features and floor plans that provide the maneuvering clearances for guests with limited mobility. Figures 5 to 9 show sample plans of guestrooms and bathrooms with the required:

- Widths and clearances at the entry, connecting, closet, and bathroom doors
- Maneuvering space in front of the closet, in the sleeping area, and within the bathroom
- Clearances to use and transfer to fixtures in the bathroom
- Clearances to open dresser drawers, to maneuver into kneespace at the desk, and to access the bed, bedside table, windows, blinds, and thermostat

Clearances may depend on the design of specific furnishings. The width of the access aisle at the bed is determined by the design of the bedside table. Access to dressers is determined by the width of the drawer. The maneuvering space to turn into the desk is determined by the width of the kneespace.

15' Bay-spacing 14' Bay-spacing Fig. 5 Bay-spacings of (a) 14', (b) 15', and (c) 16' can easily accommodate guests with restricted mobility.

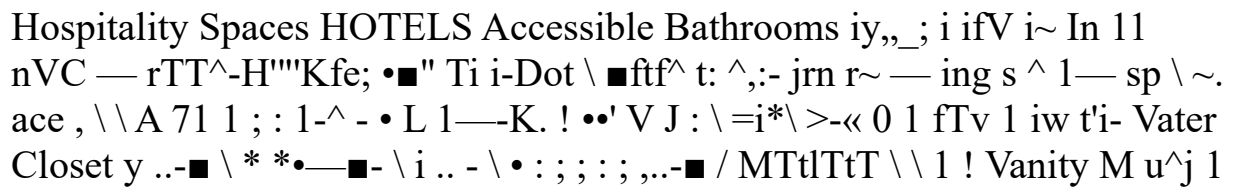
1'-0" (30)--1 16' Bay-spacing 377

Hospitality Spaces HOTELS Accessible Guestroom Plans 12' Bay-spacing Fig. 6 This alternative 12'0" bay-spacing design requires the dresser to be offset from the foot of the bed. The bathroom wall is stepped back to provide clearances for the bathroom door and connecting door. The heating/cooling unit projects into the room to allow access to the thermostat. If balconies are provided, a minimum depth of 5'0" is recommended to allow guests with wheelchairs to turn around.

13' Bay-spacing Fig. 7 A 13'0" bay-spacing provides room for wheelchair clearances, including a turning space in front of the closet and at the foot of the beds, an access aisle between the beds, a T-turnaround at the window aisle for access to temperature controls and blinds and drapes, door clearances, and a bathroom that meets ANSI standards. 378

Hospitality Spaces HOTELS Accessible Guestroom Plans Suite with 14' Bay-spacing Fig. 8 Accessible suites should meet the same requirements for accessible guestrooms and guest baths. Because suites are usually more generous in terms of space, providing accessibility is less difficult. If a small kitchenette is included, a kneespace 2'3" high should be provided below the sink. A countertop height of 2'10" (2" lower than standard) is suitable for both ambulatory guests and guests in wheelchairs. A pull-out

lapboard at a height of 2'6" provides a workspace for guests in wheelchairs. The kitchenette should include a 5'0" turning space. 379

Hospitality Spaces HOTELS Accessible Bathrooms  Fig. 9 These two diagrams illustrate the same bathroom plan with the required clearances for door operation and turning space and access to each fixture, including the tub/shower, vanity, and water closet. Clearances for maneuvering space, door operation, and individual fixtures can "overlap" because of the vertical characteristics of wheelchairs, clearances can include toespace (9" high) below water closet and kneespace (2'3" high) below vanities. 380

Hospitality Spaces HOTELS Registration Desk The hotel registration desk serves as both a symbol of hospitality for the arriving guest and the operational nerve center for the hotel. With the advent of electronic check-in procedures, credit cards, and computer-aided management, the registration desk has become a sophisticated electronic workstation not unlike a trading table or an airline reservations desk. At the same time, this electronic data processing capability is meant to be maintained at low visibility for reasons of hotel image and confidentiality. Accordingly, the designer must be able to project the appropriate hospitality image while at the same time integrating all of the required technologies. Figures 10 to 13 show examples of architectural working drawings and details that meet many of these requirements. The design of a front desk or registration desk can take many forms and be constructed with a variety of materials.

Regardless of the design vocabulary used or architectural style, certain important design considerations must be observed. 1, The number of persons actively staffing the counterlike facility will dictate both the width and overall depth of the front desk. It is suggested that between 5-7 ft be allocated per staff workstation and that one workstation be allocated for every 125-150 rooms. For every additional 125-150 rooms, an additional workstation should be provided. Peak check-in/check-out loads could require even more staff workstations. 2, The front desk should be easily accessible from and to the main hotel entrance. "Easily accessible" strongly

implies clear visibility. «■ i^tW^tMf^V.1 ENLARGED PARTIAL PLAN
 REGISTRATION COUNTER GRAND FLORIDIAN HOTEL SECTION
 AT REGISTRATION DESK GRAND FLORIDIAN HOTEL PARTIAL
 PLAN REGISTRATION COUNTER GRAND FLORIDIAN HOTEL
 SECTION AT COUNTER GRAND FLORIDIAN HOTEL Fig. 10 381

Hospitality Spaces HOTELS Registration Desk 3. Elevators servicing the hotel guest rooms should be readily visible from the front desk. This is not always feasible in extremely large hotels, 4. The front desk should be designed in such a way as to take into consideration the various users it will accommodate, Special attention should be given to the fact that hotel guests may be physically challenged or chairbound. The overall height, writing surfaces, and overhangs should be designed to accommodate a hotel guest seated in a wheelchair. 5. The basic front desk design should avoid, wherever possible, visual obstructions that block sightlines or create blind spots. Accordingly, columns and high walls should be avoided. 6. Equipment and custom elements that are typically incorporated within the front desk include computer monitors/CRTs with keyboards and printer, room racks, reservation racks, information racks, room status displays, mail drawers, key drawers, alpha guest listings, message-waiting display, credit card imprinters, fax and telex, guest/ employee paging system, automatic wake- up system, electric receptacles, cable chases, alarm systems, and file and cash drawers. □ □a □a □ --_ -000R FRAME fLEYONO [SHOE 1 I SLIOE 1 * FOfcwitoO * ' F0RWAR0 * ELEVATION AT SELF CHECK MONITOR SELF CHECH MONITOR (PAINT) SECTION DETAIL AT SELF CHECK MONITOR MARBLE MASTIC SECTION DETAIL- SETTEE ELEVATION/SECTION- REGISTRATION COUNTER MARBLE FACING-BACK FLAT SEGMENTS-FACE CURVEOTORADIUS- CUT FROM 2" 1 THJCH SLAB, BACKING FOR BASE- CUT FROM lifi-SAWCUT SACK a BENO TO RADIUS SECTION-CIRCULAR SETTEE BACK +5ifo Bfr* „j, _/fCi« Wr RECESS T j WXIOIVIOERS 3* T t--;v 3 THIS 1EI£W0HE , f;- tVf MOUHTINQ PiHELLol iiL r 8E RECESSED 4 1 ft ELEVATION- REGISTRATION COUNTER ELEVATION/SECTION- REGISTRATION COUNTER Fig,11 382

Hospitality Spaces HOTELS Registration Desk — CAWIE CHA&E
 Ml«H4EP IH- &'■£>* L.V. FvIBISONLITE. j 2.*Z S[l AN6UE /]
 CONTINUOUS FIN. 6RAPE HAKPWoop VEMEER, PLYIWJOP 2-Vz»
 iyp, blUPb. Ifc" CJ.tt. /[^]LBPSE. 5ECTIQH K£CBrx\o\4 pi&6\< eroK/[^]&
 jMiJ ftv-i-e: lli'-[^]c" Fig. 12 383

Hospitality Spsces HOTELS Regltrallon Desk PARTIAL ELEVATION-
 FRONT DESK TROPICANA HOTEL VERTICAL SECTION
 REGISTRATION DESK TROPICANA HOTEL Fig. 13 384 VERTICAL
 SECTION AT RECEPTION DESK

Retail Spaces Shops 387 Banks 396 Department stores 401 385

Retail Spaces SHOPS Planning Data Public Circulation Zone Clerk's Aisle
 Fig. 1 Seated customer/high counter height. Circulation Zone ooooooo [^]3 a
[^]3 a a ooooca csaa est en a a a est ea □ a a c Customer Activity Zone The
 essential function of retail spaces is to display and sell merchandise. The
 design of these spaces involves the manipulation and coordination of
 architectural, interior design, and merchandising elements as necessary to
 meet the programmatic needs of the client. It is critical that the space in
 which the customer and store personnel function is of the highest quality.
 Ensuring this quality requires a knowledge of the planning and design of
 the various interior components that constitute the building blocks of retail
 spaces. Figure 1 shows the clearances involved for a 42-in, or 106.7-cm,
 high counter to service a seated user By filling the recess with an additional
 display, however, the counter can also be used exclusively as a typical sales
 counter. It should be noted, however, that although sometimes used for
 special display situations, such a counter height is not recommended. Both
 the customer and the sales clerk of smaller body size would find coping
 with such a height uncomfortable anthropometrically, particularly when
 one considers that the counter would be higher than the elbow height of
 slightly over 5 percent of the population. From a merchandising viewpoint,
 where customer convenience is of paramount importance, it would be
 unwise to exceed 39 to 40 in, or 99 to 101.6 cm, as a counter height. In
 addition, the smaller sales clerk forced to tend such a counter for extended
 periods of time could be subjected to severe backaches and pains. Getting
 on and off a high stool for elderly and disabled people or those of smaller

body size can be not only difficult, but hazardous. Figure 2 illustrates the clearances for a typical sales counter. in cm A B C D E F S~" H 1 J K 26-30 18-24 42 28 84-112 18 18-24 30-48 18-22 35-38 72 66.0-76.2 45.7-61.0 106.7 71.1 213.4-284.5 45.7 45.7-61.0 76.2-121.9 45.7-55.9 88.9-96.5 182.9 Fig. 2 Typical sales area/standing customer. 387

Retail Spaces SHOPS Planning Data Figure 3 shows the clearances required for a medium height display counter, the suggested seat height of 21 to 22 in, or 53.3 to 55.8 cm, requires a footrest for the seated customer. The counter height shown will allow the display to be viewed by both the seated customer and the standing sales clerk. The customer activity zone allows adequate space for the chair. Knee height, buttock-knee length, popliteal height, and eye height sitting are all significant human dimensions to consider in the design of counters to be used by a seated customer. Figure 4 shows a low 30-in, or 76.2-cm, display counter also for use by a seated customer. The anthropometric considerations are the same. Although the counter height is responsive to the anthropometric requirements of the seated customer, it is less than ideal for the standing clerk. For the standing user's optimum comfort, the counter height should be-about 2 or 3 in, or 5 to 7.6 cm, below elbow height. This will allow a person to handle objects comfortably on the counter surface or use the counter as support for his or her arms. The 30-in height is too low to permit such use. in cm A B C D E F G H I J K L 36 26-30 18-24 30 mill. 10 21-22 5 23-25 4-6 34-36 30 16-17 91.4 66.0-76.2 45.7-61.0 76.2 mill. 25.4 53.3-55.9 12.7 58.4-63.5 10.2-15.2 86.4-91.4 76.2 40.6-43.2 Fig. 3 Seated customer/desirable counter height. Public Circulation Zone Clerk's Aisle Seated customer/low counter height, 388

Retail Spaces SHOPS Planning Data Shelving is probably used more than any other single interior component for the storage and/or display of merchandise. Not only must the merchandise be within reach anthropometrically, but it must be fairly visible as well. The heights established must therefore be responsive to vertical grip reach dimensions as well as to eye height. In establishing height limits, the body size data of the smaller person should be used. Since in retail spaces, departments may cater exclusively to members of one sex or the other, two sets of data are presented. One is based on the body size of the smaller female and the other

Retail Spaces SHOPS Storefront 3 Aluminum 3 MA. 'ytdian. 5/14 SCALE
CULcULuls a£. 3~scal& 7tcfcrn.m. \ r^\ n .-WALL \ CLA«-' / SLIDING
CL, DOCKS JU^ 1 -M-r- W ^sp 5 -II-6- BULKHEADS DOTTED- £iW. 2-
1- 5/14 SCALE . CEKAMICTIUS FLOWi WIOB CEMSNT JOINTS
>tctidYL.9 (!.^e'r^m^ V ll»IIIBlllllll|lllll|l|ll|l|l|llltlllll»llllllllllllllllllly
METAL SADDLE 390

REFRIGERATED CASE REFRIGERATED CASE BBf—Pr CEILING
MOULDING "IUOYIII — iw Htm* UK it 10 II —trim-him wic- wmsiai,
kiu TYPICAL TRIM 1X"-"^ ,;r"^^' "■" •■■"ffl iFfHiziS m CROSS
JECTION^IS PLAY-CABIN ET ISLAND DISPLAY

Retail Spaces SHOPS Display Cases SIGH HOLDER TAN ELS
C£MEMTEO TO ^TH.ceNTB^. PANEL w/I/&,xV,z." P6XI BAHp <?N
THRSS 6IP&9 (ToP<?P£HJ I^TH-rUTWP.ceMTEi DIVIDER |»t.LAM.
FIN. ALL, ©ipes r-T-M.n-.LAM. PLEX! CS-LASS oic^H HOLPER5
PANEL, PL.UH. N RM,-AU-°IDg6 - 3/»*CL.rLCXI C1L. SHELVES W/
CLIPS ^"cLCAR /"LEX I f^MEuS M fLYwo.&^e TL.LAM. FIH.AU sipes,
wood "ART FRONT ELEV' SCALE J BvV= l-o" VERTICAL eSdloN F-
CTO rpw^IPE FINGER CUT-' OVTWEteU 9|Chl HOLDER ^THK.fLEXI
PANEL CEME- HTEPTOJ/8" PANEL W/ — i/g,«xt<»"pl-£XI ' &AHP
ONl*i&IHs> SC:s"»|i<>' 3/fl,OL,pLE'X| ©HELVES ON OL ALLIff^TOp,

CLIPS- SCALS ! -3/^" * 1'-o" U-CLEAR. PLEX I ALLIGATOR CDC
"3/bTth-cleap; PLEXI PANELS ALLIGATOR CUP •b/bO qL. SH&LF ,5/8'
SVa'^1 PETAIL sc.-s'^r-o11 DETAIL 0 393

Retail Spaces SHOPS Sales Counter \11 SCALE FLUORESCENT
LIGHTING FKOSTED GLASS u -1-2- -SLIDINiS DOORS HINGfiD -
DOORS £m &cc£um. Vf=C F=RO*T6D CLASS FK0ST6D GLASS 1,
LIGHTING _ SPACF LIGHTING — SPACE TllaM. th/ut CotontvL.
Tlicon. <x6cnrt CcrUAvbvt. Section. \ i 394

Retail Spaces SHOPS Counter Details COUNTER, WITH VERTICAL
FRONT 34»» 1'-O" COtTNTER WITH SLOPING FRONT 34'= 11 O"
MABBLE DEAL PLATE, AND PUGSE SHELF/ COUNTERWITH
RECESSED FtXRSE SHELF 3 4' -1'-O* COUNTER ON- -WOOD
FRAMING 3 4"-1' O" 395

h *• n ■W f-»j/7~ r-1 riA>T!C UAMINATq ftrttCrftlV/eK*-*
3/*t"m.YW«W«LrAT — «V 2 BOTTOM PUATB SECTION AT CHECK
PESK (DEPOSIT SUP AREA-) SECTION AT TELLEP, COUNT6P.
VERTICAL SECTION AT TELLERS COUNTER

Retail Spaces BANKS teller's Counter VERTICAL SECTION-TELLERS
UNITS PLAN OF TELLERS COUNTER 397

Retail Spaces BANKS Teller's Counter a'-ita' 3M" 8IRCK PLYWOOD,
1*2' BIRCH PLYWOOD W TACK BOARD HPDL ON 3/4'
PARTICLEBOARO ■: j"1.-1.. ■.■:...■.■...■. ' .^3 f.ifa' HPOL ON V
FWtTICLEBMRO [REMOVABLE PANEL) 3/4'MAHOOANY
PLYWOOD 2 x BLOCKING (\$ 2<0A. VERTICAL SECTION-TELLERS
COUNTER VERTICAL SECTION OF TELLERS COUNTER
3M"R*OHJS^ * - ■-- W* MNEL WITH l«0» BRONZE CL*D*I«C? ^—
CARPET AT HARDWOOD- BACKEO WITH BALANCING SHEET ALL
AROUND- AT KNEE SPACE EIWHI] TYP TO BE KKEWED TO
SM'auEl AND ATMOVail MOVABLE CARTS SQLIO -J MhHOOAHY
8«E . i'/z'TH «loVIG* W BLOCKING ■ LEAVE LOOSE 3/4'FIR
PLYWOOD W/COLOR-TH VERTICAL SECTION AT TELLERS
COUNTER SECTION AT TELLERS COUNTER

Q BSSSm /.V it'I ttv m 3/4 MOF PS, MMOMNY VENEER 4, HEL~7 ^
SCREW INTO METAL STUD WALL RECESSED HCDL BASE STOP
FLUSH W/COLUMN -COVER BTEEL MOUNTINQ- PLATE - ATTACH
TO FLOOR VERTICAL SECTION AT CHECK STAND VERTICAL
SECTION AT CHECK DESK HALF PLAN THRU CHECK DESK

Rstaii Spaces BANKS Counter LAMINATED PLAiTIC.^ ^tcVLiryL^. "5/4
•• <cale hi in— —ilr"! open "JHH—HHR IP ini a. f -■j'-e'/V- .r-2"OPEN_
I7--51— | a £1 K3 1 C3 | METAL CABINET* J L _2'-2"OPEN- 400

Retail Spaces DEPARTMENT STORES Fashion and Fine Jewelry Back
Island Details 94 PLYWP.ee>NSTJN. PL. LAM-FINISH F-c.TO HUWIPB
POWER TRAC#t.«50, L-flSI OR L-C52 WITH LAMP* L atll SIMEKL
Rlr4tf| LAMP HffIDER W/ WTftSRAU S3UP «TATB 1ZVTRANSK?
RM«lW/ aSWIZV MR 16 LAMPS ,/4" CU. <3|LASS POOH ON PIV<7T
HINGES FABRIC CDVeR'p. REMOVA&LE UfM MASONITE PAD BY
FIXT- CDNT'R- -iVxiV'ftANGLg EXTRUDED AtUM» kio substitute 3/t"
PLTWD.DKAVVERS 7 PLY TYPE ON FULL EXTENSION r-PROVIDE
VENT HOLEe UPPER SECTION/X\ SCAU5J 3»=*'\-d>". \TV GRANT
DRAWER .e.UPES»B5S 7l PL* LAM. -J "PIVOT HINQE - - ^GLBAR
SLABS DOOR VARIES SEE PLAN PLAN SECTION ©GALE: 3" = 1'-0'.
B |^TYP. aECTION SCALE: 3>/4«*|i0!f PUPLE* OUTLET ACCESS
FttNEL 6EB DWfij.ecM F.C.TOPROVIDB ACCESS PAKIBL TO
TRANepOrtMBR TYP. TRANSPARENCY 0B& DWGj.SD-4
'/4"ct,.fiL.8HELF W/MB.TAL 9rVCH.BT» FAawc covbk'd. '41'
MASONITE PANBL TYr. PRAWHR BASE UNIT OF ^4*
PLYWD.CONSTR'N. PULAM.PN.Z* <3AN<5 U>CKED NOTE: BACK
FIXT. BUILT IN SECTION* *• BOLTED TOOBTHER ON SITE. PL.
LAM-FINI6H rxceeesEo oAas ~ <D J4.»" SECTION / ^ 401

Retail Spaces DEPARTMENT STORES Showcase lor Men's Cosmetics
and Fragrances VARIES. ®1 ■ VMies PLAN GLASS jc?f ANP FRONT
PL.LAM.FIN. M.F-W. AT MEN 4-"H.RBCC6SBD FRONT ELEV. r#/-
(N.F.W.ATMBN*) FRONT ELEV' cut-our pop, ttLBCTRIC
RECCFTACLE ty CU5AR MIRROR SUPINE ttoopd " W/PUIXAUX* PL.
LAM-FIN. N-F.W.ATMBM9- REAR ELEV, 4'h. nectao»p BAsg n-LAM-
HN. TTHp W, AT HPN*T 1/4" CLEAR <5LAS5 ffWNT «f T^r

fOL.CHR.FIN. METAL PKAMH I -3/31|CLR-CjL.0HELYE6 _JP(>ps pl;
 wm. bn- .PULL t, UXM N.p.W.AT MENS ML euT-ouT ro* LECTNC
 RICEPTAOU REAR ELEV I'-O" N'dL.MIRRoft SLIDING poops, PL.
 UAM. pr W/PULL * U>CK W.f.W. AT MENS VABLS PISPIMt&P. . ^ rfp-
 pf?AWER6Ase ■^ W/PL.UM-FIN-A 4'L. PULLS W/U»K fN.F.W.)AT
 HEN* SECTION BAOE MBNS ?| / J5C i-3/4 SIDE ELEV- SIDE ELEV.
 402

Retail Spaces LEVITON *1374-W ■Euscrmc receptacle W/ IS'-O" LONG
 CORD I TWIOT LOCK PLUO 2"0VBRLAr DEPARTMENT STORES
 Showcase for Men's Cosmetics and Fragrances PLAN SECTION SCALS S
 HALF rULL 6IZB,

Retail Spaces DEPARTMENT STORES Showcase lor Men's Cosmetics
 and Fragrances #RM4AKM3 PU. LAM'OS- SECURITY CUPS—J
 CtARCY ♦'rt-lotftffWU POM CAM- <—IOCK*3I0-O!6 Ho •UBSTITUTrn
 P^TI^HT JE^ L# .\O'-O*\jMGO0fIDHt TWIST UXK PUIO. SOLID
 WOOD •BASE PL.LAM.RN.—> eECTIQN SCALE: 3/4* »l'-o' ®
 LQMQ.SECT>/a\ eoALtT:a/4»-ll-o" _ / pl.l*m REJ-V>/A6Lf. BtlILP-UP
 V MA50NITE REMOVABLE DISPLAY PAD. R*©SIC COVER 'P eW
 BU«DNB& -4; A6 . REQUIRED DISPLAY PAD DETAIL SCALE : |/4* =
 ONE INCH 404

Retail Spaces DEPARTMENT STORES Television Cabinet Details
 ATEHPfr 6HT. R|4» CURTAIN WALL W PAINT. FINY CEILING L.|K/e
 AT 11 '•O" At P. P. SEE ELEVATIONS FOR SPECIFIC TRIM DESIGN
 FASCIA 3/4"PLYU/0 W/ PL. LAM. FIN. i^'pl-mooo oooa W/ PL, LAM.
 PIN A MAQH&TIC CAIGHee AT TOR 4~ BOTTOM OH CONCEALED
 PIAHO HINC3B& WALL CABINET #f PLYWOOD W/TH PL. LAM.
 FIN- CUTOUTS IN OORfc TO ACCOMMODATE' J.V.C.TELEVISION
 46 M0OEL N0.C-2626 V4" FLUSH PLYWOOD ACC£<*e> PANELS 14/
 MAGNET/O CATCHER REMOVABLE BAi>S 5T ^VPLYWa PL. LAM.
 FIN- ^^CABiWE HINGEP DOOg , V4" PLYWOOD W/ PL. LAM. FIN'
 RECBS9E«P BASE , S/4" ©OLIO WOOD W/ PL* LAM PIN/
 ELEVATION scale; ^'si'-d" 405

Retail Spaces DEPARTMENT STORES Television Cabinet Details
 CEILING LIKE AT 11'-O" A.F.F. 9KT. BK.. CUPTAINWALL 6BB

ELEVATION 6 FOR 6 SPECIFICATIONS * — SOFFIT 4 FACE CIA
 W/PLYWOOD. W/ PL. LAM. PIN- 1/2" DIA. VENT HOLE 9 COMBINATION
 AM/PM 4-4000, ANTENNA OUTLET 9 1 VOLT Q.C. HOSIZ. BT p.RC. 1/2"
 DIA. VENT HOLE* PAKIEL COOJ»/2'-4" 6, W.* 41-71 1/2". 1/4" PLYWOOD. W/
 PL. UAM. PIN- ON CONCEALED HINGES/4 NO OUTLETS FOR TV's.
 a" o/A. wife pin holes CABINET 4" PLYWOOD. W/ PL. LAM. PIN, flush
 Access panels, W/PLYWOOD W/ PL. LAM. PIN. ON CONCEALED
 HINGES AND MAGNETIC CATCHES REMOVABLE & AS CABINET
 2/4" PLYWOOD W/ PL. LAM. PIN. COMBINATION AM/ PM & -A OOO ,
 ANTENNA CUTLETS 2'-0" O.C. HORIZONTALLY» PURNISHED #
 INSTALLED BY P.F.C 4" PLYWOOD ©HELP W/ PL. LAM. ON
 PILASTER 7 AMD AKC CONCEALS HINGED DOOR VA" PLYWOOD.
 W/ PL. LAM. F/N. W/TH LOCK AND 4» MORE PULLS SECTION SCALE
 no* ro 406

Retail Spaces DEPARTMENT STORES Shelving Details -tfr DUPLSX
 REC-ePT. 0 Y E.C, rj. 1". ■ PERIMETER BACKWALL -FABRIC
 WRAPPED RAHCU . . 1'-4" WEPTH. MILJFC WHITE* PL3XI *
 <*LA<S*9 PLUOA. LIGHT 5 TQIP' C.3". LIGHTING 1" IG02 SHELF
 DETAIL MET. ANGLE AT ENDS W/ HK.MIU WHITE PLFXI -GLA*6 -
 a" DIA. HARDWOOD TITIM COL. LAQ. PIN. 6 HELF BOX W/PLYWOOD. +
 PL. LAM. P/M* ALL EKP'O- MST.AW«LE^ PAINT WHITE BAIT
 INTERIOR WHITE OUPBK BECBPTACLE PER SHELF *Y e.G,
 PSRIMSTE 3 ACXWALL REMOVABLE BAD© W/ FABRIC cov'fi BY
 P.P.C BULLNOSE TRIM 2"D)A« HARWOOD FLUOR. LIGHTSTWP
 &Y PBaiM. FIKT.coNTR* MODIFIED MBTAL %*ACU AT5NP6
 W/SHBLP&LJPS BY RP.C. SHELF EtX, 2^t" PLYWOOD PL. LAM. PINi
 PLAN SECTION @ 5 HELF/on [<3> *tt- ,ft_-,t -I ^1 ■ tmt mi) iiiiii
 iiiiiiiiii'iiii nil i:i,'i'iii':ri"!iini mi irr A j n mil REMOVABLE,PAD6V\
 FABRIC COVO MI Lid WHITE PLEKI AT 6 HELF SHELF &0< MILL
 NHITS PLEXI AT BASE CA6IWET BASE CABINET PLAN NHEgE
 SHELVES ASUTT 407

Retail Spaces DEPARTMENT STORES Shelving Details LUOR. LISHT-
 STRIP BY ELEC. CONTff. \0 MET>AN0»l.e9 I e'-a'A.RP. T-"70/
 DUPLT (ZEC6P' ^— * AT EAOi -&HBLF \ * by e.c. N < SLASS
 5SHELVE5 6ECUF?ED TO MET. 3R/tfji6T£ w/-clips &r SINIVirVG

COHTfZ PANELS BY P-R6. FABRIC WRAPPED "5Y P. SEE PVVfi, y P-3 ^ 2/8"THH.MIUit WHITE PLKXI 4 +- CONTINUOUS MET. ANGLES FLUOR. LIGHT 5TRIP BY P.P.C, G,J.*I00O OR EG*. - ^J.BOX PROVIPEP * *> N STALLED 5Y E.C» s<s>lid wood BA&a- REMOVA&LE BASE CABIN&TJ 2^" PART. 5D. PL. LAM. PIN-Q^A.KF. -SHELF ASSEMBLY, LIOHT FIKTUBB '*' DUPLEX OUTLET BY PESIM.P.C. FABRIC WRAPPBP PANELS BY RRC SEE DW6< 1/ P-3 ^S'THK. MILK. . WHITE PLEXI it CONTINUOUS MET. ANGLES 1 y-2.1 dia. r/ HARDWOOP TftlM 17 COL. LAQ. FIN. FLUC?)?, LI0HT STRIP &YRF.C ■J StfX' BY E.C, Rec&eeeo BA5-E -4"H. scuio wcoo SECTION scale; 3/411* i'-o'l O SECTION SCALE? 3/41 r?^r0 408

Retail Spaces DEPARTMENT STORES Shelving Details /•removasle. pap^ j/ W/ FABRIC COV© ♦H&L-F0N Mar azT^tLfIT "Fuuok. u-le Bv>rFf> B* 1°-^ FLUCK", L<&HT STRIP DETAIL ^Tn SCALE'.)^»s|/-C7' VCV 5EOJ(2eP TO MEJ' STUD mn- . IMMOVABLE PAPS >\ ev p.ko. _ O. x |/'4"*V' w &Y B.C. ,«i NPIVIPEJZ PANEL. DETAIL FLUOP, LI<SHTIM6 FIXT BY g.C, •see elevat/^ns for ecAustt p/^M'-o1 -SPECIFIC PSSI6N'DETAILS .ME-pANGLES Bf>>.C, BUILSIOSE TRIM PL.LtAM.FI^ . A^0 fluor. stjzip lar arc PLVIPBH PAhleL 3/4"PAfaT.6D ^/ PL. LAM, FIK/J RECE&e«0 ,4-6"L. PLUGMOLO »Y g.C. W/OUTLET'S' 0'toC REM0VA&UE PAP5 »V/ FA&KI^ cWo |2>Y FiF.c, ■cEe'Dwa, i/p-3 SECTION scale t cv-^-r-o" CONTIMU0U5 HARWOOp &ULLNOSE THIM 2."DIA. COL. LAO.FIM' REMOVABLE SHAM BA66 4/411 PABTICLE &OASD \MI PL, LAM.PIM.tfN ALL- EX P'C* &0&&O&9 RB6B&4&0 &A&B 4"H< 3/4" SOUP WOOD W/ Pi-/ LAM.F/N 0 409

Retail Spaces DEPARTMENT STORES Triple Mirror Details "LIQHT FIXTURES SUFpLIB BY aVNER.; ls4srALLep £ COWECTED'B.S.C. 410 SEcriow - . .SCALE_34l. - I'-O1 "bT/UMAKK11 AIA>M. . HARD Wp £D<5E" PL LAM, PINltM.

Retail Spaces DEPARTMENT STORES Back-Lighted Transparency Display 3M*x. aM r&u <jhr. metal TUBIKlii FRAM& i— j/&* MiutwHiTt n.w PLAN rrz SS V \\ WfcUOf? MBfAU •star in vl ■' Vt,* X t'A-'xt'/f'W. OHWMfr W^WUMOep SMOOTH WeUPEP -JJIUfi, - Vfe." m i l-k-niHiTt rLexi SECT TYPICAL TRANSPARENCY DETAIL 411

Retail Spaces DEPARTMENT STORES Feature Wall Lighting

■bpeF*"TD~AI30K/£. t\$X*jgreafteifuifo-tt;. ■ 3im^ma~f^eftcmfiG\$s&-.
Jo I H<^^7U'OVm^~" .ui-rvt op^ri vauahc^ -Smf■«»- cp" ov6*u».p
sec-fic?N ^>/Noi-e/oouesL& HAHGi-fSoo "SINOL& LAMP -: ..•.:
STeppED •etft=|_ Vhlfe_ Mei<<sHAWc»i*B txepuef WITW -+"
oVEffUtp.!. "S&S-p &YA "H gtf&HAND I SBr;p.l*»pUY 412

Retail Spaces DEPARTMENT STORES Feature Wall Lighting ^BC-riO.W

w IOl2)--5lM5PsrAMr WMW *U|03«SB.Ti©IWEp... •BHIBLC>IH<k
MAT6P1AU . : :. : ; f-tep<iHAHOH&e «-psi<r/i<{>» J5BCTIOIS xtr&e
fiwrw unite-. JSOR5IT fIOH-KBfwEcf IVE. t5M->i*Mot«S(e with
Ace©4r_ .w<*Mr6 •trtiO-<lH<ftLe LAMP BuucafiB6ceHT..aw,Hfciaje.....
l>«~<SKlBUp-WnH" FONT ■Base ©£. -f>A!seoiJT*" TBATUK& UAUU
DISPLAY IWITU IUl UffITS AT£D~■" SOFFIT ano.ACCEnt..U6MT 4'-
o' 's&c-rizs]^:: UAMp _ . . AT ttJOft iBiKa3raffi=KR?ns«aATe=_ pp..6t-
H55Ms U|rt4 fla*W p& Afu.se iiasciziasBG&r
K&oe6!aBia.jMCAHn6«agHT Mtf*4T: Miivl-.Mic3erESBi«BV.
-5IC<SlA»e..OS..AI«=fWO&K^ _ ..betcttoit:"":. :. _ 413

Retail Spaces DEPARTMENT STORES Feature Wall Lighting O

■Mt**U£~M;MP'~. egue^fen. -oppofite. / !■'
Bmf^sr<3i*&A...-<«et«]W5> wi-ru a*-&" ovaewp BeKCj4AN016B
SfoftiAB : ?^^^E^zi^Lsiazxa6PLAY Ui-ru _rt-.uur?iiMATi6o.
WaiflftftewOTrMStl^ocBWfc;... C* COHpMU6U6:-IRii(i:..W>flt." or
:6HER?-'TaaBrja5ffiHirr gc>i>e of "pACBouT";"" :uitm pecssaeD h&ht.
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Retail Spaces DEPARTMENT STORES Valence and Cove Lighting Details

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Retail Spaces DEPARTMENT STORES Valence and Cove Lighting Details
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Retail Spaces DEPARTMENT STORES Valence and Cove Lighting Details
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417

Retail Spaces DEPARTMENT STORES Wall Display Systems (- 2'-0" -j
2'-0"—j-j 2'-0" -i 2'-0" -j WATERFALL FACE OUT & WATERFALL
Average Quantity of Garments Per Post 46 I I POST FACE OUT WITH
STRAIGHT HANGING & FACE OUT STRAIGHT HANGING Average
Quantity of Garments Per 4'-0" Section: 94 -ft^H K K -.Hrfl '.K; ■ft—B -
H H~B THr.-.Hz 418 FACE OUT WITH DISPLAY 'Compatible with
Average Quantity of Garments universal W slotting Per 4'-0" Section: 49 •
Unique new oval hangrail FACE OUT Average Quantity of Garments Per
Post: 46 1 STRAIGHT WITH HANGRAIL FACE OUT Average Quantity
of Garments Per4'-0" Section: 96 L 21" - 23" ROD DISPLAYS 6 or 7 Rods
Per 4'-0" Section

Retail Spaces • Design continuity from wall to floor carries theme
throughout the department or the store • Designed for high volume
merchandising • Flexible merchandising > Designed to be compatible with
other Pam International Systems 1H-H ll i FT EHBf k&w MULTI-NET
SYSTEM DEPARTMENT STORES Wall Display Systems h~ 1 L 4'- 0" m
- ie " _» ^ TK BINNING SYSTEM ■ 35y," 4'-0" BASKETS & SHELVES
-25Vi" WOOD OR GUSS SHELVES - 25W ,. MULTI-NET PANELS

MESH PANEL SYSTEMS •All Multiples/Systems 2™ upright posts are engineered with easily changeable post covers (Pat. Pend.) •All metal components are coated with a durable, long lasting, baked on epoxy powder finish. w— GRID WALL PANELS 419

Retail Spaces DEPARTMENT STORES Rack Display Systems 4b 16" Straight Arm y", Ob 16" Slant Arm with t billl db 16" Strilght Arm ind 18"Slint Ann with I B.lli 16" Slant Arm! with 5 "J" Hooki Two-arm costumers FOUR WAV RACK 18" Slint AtmiwIttiJ Bills 11" Slant Arm with B "J"Hoo«l Four-way racks Rictingulir Kmgrlil REVDLVINO IK" Oil, lliugtill Circular racks 420

Retail Spaces DEPARTMENT STORES Rack Display Systems OUTRIGGER All Chtomt Will Fin Snglt Vu Rtvolveinfl Bait Rick £t) Gt> Two Tlir Rivolvlng Bilt Rack 1«"dMO 16V dMO whh id|uiubli cornlet mount IWditp NoR-odJutubla cmlco mount Specialty racks Outriggers, wall tins BO" Long Doubli Hingrail Tondom Hangrill Rectangular racks FoMlng-60" long w/16" Pullouti r'Squiro Tubing 0=5 60" Long All !<" Squin Tubing 54" High 66" High 74" High 60" Long All Rictingulif Tubing IK" Hingnll 1 1/18" tubing with 12"Pullout Upright! rimovabti lor itoftgi, 60" Long Rolling racks 421

Retail Spaces DEPARTMENT STORES Display Systems Standards MEDIUM DUTY-!*" SLOTS 1" O.C. 0 1/t6"-«-jJ»- lj: 13/32" -n[11/16"|"*-1 For Oantnl Um Tho Sundard of Wall Sundardi 24" 30" 3«" nH 46" 0@B ■l 1/16' i «" L- vr -J 1 For Midlum Load Sngla or DouMa Bracksts. M" 72" •4" HE ^ 0 0 0 0 0 3/32 :t/i6- It/IB ConcMltd wall standard far 6/1" or lass wall pawls. 72" M" II" 13/32" (UL ~<j 11/16-'*J-a—' For additional atrangth. 10" 72" M" n ii ■ i ii 3/32' "'1h JL Ntad far haaviar loads. M" 72" M" HEAVY DUTY-1" SLOTS 2" D.C. 1/8"* LJ 11/16" *7/(" ■fT For your heaviest load requirements. 60" 7Z" 84" 76" 1/6" 11/16" Concealed wall standard for 5/8" or less wall panels. 72" 84" 96" Ragular Duty Brackets for uh with 1" O.C. Slotting. V, hard itnl .013 thlcknail .Polishad chroma. Fait llflad to pravant scratching of hangrall 12" For 1" and I 1/16" Tuba 12" 14" For IK" and) 5/16" Tuba 12" 14" .Fallihad Chroma. Fait llnad to pravant scratching hangrall .Satin Zinc. For 1" and 1 1/16" Tuna 3" For IX" and 1 5/16" Tuba 3" Haavy Duty Brackati for utt with 2" O.C. Slotlin. X hard itaal

.126 Thlehnan 12" .Pollthsd Chroma. Fait llnai to pravant scratching ol
hang rail. 12" 14" .Satin Zinc.

Public Restrooms, Toilets, and Coatrooms Restrooms and toilets 425
Coatrooms 460 423

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plumbing Data TABLE 1 Minimum Number of Plumbing Fixtures
Required by Building Occupancy Type* Type of building occupancy
Assembly-places of worshipt Assembly—other than places of worship
(including but not limited to auditoriums, theaters, convention hells) and all
spaces classified as F-4 Dormitories - school or labor, also institutional
Single room occupancies for sleeping accommodations only Dwellings -
one- and two-family Public buildings, offices, business mercantile, storage;
warehouses, factories and institutional employees* Public bathing Schools:
Elementary Sacondery Workers' portable facilities Industrial - foundries
only Type of fixture Weter closets 1 for ea. sax for ee. 150 parsons No. of
No of fixtures persons for each sex 1-100 1 101-200 2 201-300 3 301-400 4
Over 400, add 1 fixture for ea. sex for ea. additional 200 persons 1 for ea.
sex for ea. 8 persons 1 for ea. 6 persons i 1 for each dwelling unit No. of
persons No. of each sex fixtures 1-15 1 16-35 2 36-55 3 56-80 4 81-110 5
111-150 6 1 fixture for ea. additional 40 parsons 1 fixture for ea. sex for ee.
30 persons 1 fixture for ea. sax for ea. 35 students 1 /30 workers No. of No.
of persons fixtures 1-10 1 11-25 2 26-50 3 51-80 4 81-125 5 1 additional
fixture for each additional 45 persons Urinals Urinals may be provided in
toilet rooms in lieu of water closets but for not more than Vi of the required
number of water closets Urinals may be provided in toilet rooms in lieu of
water closets but for not more than Vi of the required number of weter
closets Urinals may be provided in toilet rooms In lieu of water closets but
for not more than Vi of the required number of water closets Urinals may be
provided in toilet rooms in lieu of water closets but for not more than Vi of
the required number of water closets when more than 35 parsons Urinals
may be provided in toilet rooms In lieu of water closets but for not more
than Vi of the required number of water closets Urinals may be provided in
toilet rooms in lieu of water closets but for not more then Vi of the required
number of water closets 1 /30 workers Where more than 10 men are
employed: No. of No. of men urinals 11-29 1 30-79 2 1 additional fixture

for each additional 60 males Urinals may be provided in toilet rooms in lieu of water closets but for not more than $\frac{1}{6}$ of the required number of water closets

Lavatories 1 No. of No. of persons fixtures 1-200 1 201-400 2 401-750 3 Over 750, add 1 fixture for ea. 500 persons 1 for ea. 12 persons 1 for ea. 6 persons 1 for each dwelling unit

No. of No. of persons fixtures 1-20 1 21-40 2 41-60 3 61-90 4 91-125 5 1 fixture for ea. additional 45 persons $\frac{1}{60}$ $\frac{1}{50}$ pupils $\frac{1}{50}$ pupils Over 300 pupils: $\frac{1}{100}$ pupils

No. of No. of persons fixtures 1-8 1 9-16 2 17-30 3 31-45 4 46-65 5 1 additional fixture for each additional 25 persons

Bathtubs or showers 1 for ea. 8 persons; for women's dormitories, 1 bathtub shall be substituted for 1 shower at the ratio of 1 for ea. 30 women 1 for ea. 6 persons 1 for each dwelling unit $\frac{1}{40}$

In gym or pool shower rooms, $\frac{1}{6}$ pupils of a largest class using pool at any one time 1 shower for each 15 persons exposed to excessive heat or occupational hazard from poisonous, infectious, or Irritating material

Kitchens for public or employees dining Dwellings - multiple or apartment 1 for each dwelling unit or apartment 1 lavatory for the personal use of kitchen employees 1 for each dwelling unit or apartment 1 for each dwelling unit or apartment

Drinking fountains 1 for ea. 1,000 persons except that there shall be at least 1 fixture at each assembly floor level or tier 1 for ea. 75 persons $\frac{1}{50}$ persons but at least 1 per floor At least 1 per floor equivalent area, 100 workman 1 for ea. 75 persons

Other fixtures Where motion picture projection booths contain more than 2 projectors, at least 1 water closet and 1 lavatory shall be provided on the same level and within 20 ft. of the booth

Laundry trays -1 for ea. 50 persons Kitchen sink — 1 for each dwelling unit

Other fixtures One machine or a 3-compartment sink for the effective washing and sanitizing of all cutlery, dishes and glasses before re-use Kitchen sink-1 for each dwelling unit or apartment. Within each dwelling unit, not designed for use by transients, one laundry tray or automatic laundry washing machine; or in a readily accessible location within a general laundry room. 1 two-compartment tray for each 10 dwelling units or 1 automatic laundry washing machine for each 20 dwelling units.

The population used in determining the number of fixtures required shall be based on the number of people to occupy the space but in no case shall the population be less than that determined by allowing 125 sq. ft. of net floor area per person. Such facilities may be in adjacent buildings under the same ownership or control, and shall be accessible during periods when the assembly space is occupied. Facilities for

employees in a storage building or warehouse may be located in an adjacent building, under the same ownership, where the maximum distance of travel from the working space to the toilet facilities does not exceed 500 ft. horizontally. 425

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plumbing Fixture and Accessory Heights While Fig. 1 provides specific vertical dimensions of both plumbing fixtures and accessories, the designer is cautioned that every plumbing fixture and accessory must be carefully analyzed in light of the users to be served. Plumbing contractors will follow the manufacturer's recommendations or their own standards unless the designer provides this Information on the working drawings. In large-scale projects, it is suggested that the designer carefully provide all fixture mounting heights on all interior elevations or on a separate diagrammatic drawing, such as is shown in Fig. 1. TOILET UE1NAL LAVATORY MIEEOE/TOWEL MIRROR/TOWEL 4 'PI5PEHSEE SOAP PISFENSER v?Mk <s4v flo I y1—*- \$ £5,6.54" LE.W5TH OOI2.PM-WALL "TOWEL "SLOT com + 5L0T HANDICAP TOILET STALL TOWEL \SPLi5ie/WASTE. EECE.PTACLE 5AMITAE.Y WAPI^IW VCND OG. Fig. 1
Fixture heights. Partition Mounted Units & Handicapped Washrooms Fig. 2
Suggested mounting heights for various bathroom accessories. Fig. 3
typical back to back male/female washroom stalls using partition mounted units to accommodate a handicapped stall and one standard stall. If room permits, grab bars should be placed on alt three sides, resulting In a "U"-shaped configuration. Most codes require toilet stall doors to open outward. 426

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plumbing Fixture and Accessory Heights Provide i/lmirnor for handicapped .PART EL E V AT I <7 N ^— Pin. f L i M E N'S PART PLAN T O I LET K-00 M Fig. 4 This drawing of a part plan and part elevation of a men's toilet room demonstrates how mounting heights of plumbing fixtures and accessories are indicated. In addition, spacing of plumbing fixtures is Indicated by use of a horizontal dimension from centerllne to centerllne of the lavatories. Many designers prefer to show horizontal dimensions on the plan, a = recessed waste receptacle, b = recessed towei dispenser and soap dispenser with shelf. 4Z7

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
 Plumbing Fixture and Accessory Heights <r ■ .-' i + ! 1 -\ - i-j—] i =p 1 M l
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 i..i 4 V-^3^=« _ ip^v -- i;?-S -f ». L _ -j- | _ii so ^-Fin.fl- ine .PART
 ELEVATION PART E L E VAT l<?t>l Hook * 7- '—HooK lligtrr
 M2"x5'9helf ^ Fin. PART PL AM T H ® I—p .Luall line ,im □a PAR.T P L
 A hi WPMEN 5 TOILET R.00M, Fig. 5 Mounting heights or vertical
 dimensions are always taken from the finished floor. When Installing
 accessories on tile walls, the tile module and dimensions should be taken
 Into consideration, a = full length mirror, b=recessed feminine napkin
 dispenser, c = recessed towel cabinet and waste receptacle, d= recessed
 soap dispenser with shelf. 428

Public Restrooms, Toilets, and Coatrooms RESTROO MS AND TOILETS
 Plumbing Fixture and Accessory Heights Tile a fi-Fin.clq, me i>- 1 i _ —i—
 • i t- & i r _ ^ tr * T= 1 V rr1 l "7 h (J -H b 1 -4V V> PA.eT ELEVATION]
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 Incandescem or fluorescerrr liejrr 'LA LE Kl T F .£ ->c >N, i b t %) _C (J r
 '- \ Fig. 6 The mounting heights of plumbing fixtures and accessories for a
 private toilet are, In many instances, determined by the physical
 characteristics of the primary user. A person 6'6" tall might require the
 mounting height of a lavatory, mirror, or shower head to be higher than
 usual. Note that any electrical outlets near a lavatory or shower must be
 specified with a ground fault interrupter, c = first aid cabinet and medicine
 cabinet. 429

Public Restrooms, Toilets, and Coafrooms RESTROOMS AND TOILETS
 Plans 4-8"min. clear. 5'-0"prefer red r 1 8-0*mln.clear. 9-0""max. clear.
 Wheelchair Enclosure Recessed towel dispenser, soap dispenser, shelf and
 mirror Recessed waste receptacle. 3'-0] ^H MEN Feminine napkin
 dispenser- Recessed soap dispenser and shelf. S XD \ SIDE ENTRANCE
 WHEEL CHAIR TOILET ENCLOSURE (This Is not preferred) 4 - 8"mln.
 clear. 51- 0"preferred Full length mirror Rtcessed towel cabinet and waste
 receptacle. WO MEN 430 Fig. 7 These drawings show minimum
 dimensions both for toilet enclosures and between partitions and walls.

These layouts are recommendations provided by the General Services Administration, but they may not be in conformity with other codes or desired bathroom layouts, especially in regard to accessibility. Remember, too, that codes provide minimum, not optimal, standards. "

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plans and Elevations « ^jfl r-tM-T rz^: ufi^yA^rr^r^v pfcs >>a \ *ia in tk
^8-- UA^>frerv£?F*-*v M At.OV«- <^M - 4 4'-0"t •*-o \ ,M j | I i a Hi . .J
TJ H <g> Fig. 8 These working drawings provide both vertical and horizontal dimensions for placement of plumbing fixtures and accessories. Note that accessories are identified or "called out" through the use of letters, which would be coordinated with either a legend or a schedule. 431

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plans In multiple-fixture public toilets, at least one watercloset and lavatory must be designed to conform to barrier-free or accessibility standards. Si jUdOl f^3" is) +f\ L*fi*w/±?T4>m>/ UA>s/>6cr<y f*."v im i-
-?tt a'-o« 1*.* IF SL'-0» ---V-t71 H' «ovc 3 S^ —II i Ml aw* €> ^IQ^..11 JJ
<*£ ,'. s" LA-g ^3~^f-J is r 4-*" k 5U? 1 £m i'-<s" .*-« u/^v^>crg>T^gy q-
^M. ^u<hwt 432

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plans and Elevations •S-*" Al.«r .alef . ai-'rf i^ a'-*" a'-V x'-«" |w«»li^ i^J-
4*' M^Jbi^ Tg?lL-S / 57 2 5Z sz CM □» -^ n» Ck =3S ' V" ! i d> a- 1 ■ ' \ \
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Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plans and Elevations The women's room shown in Fig. 9 requires approximately 250 ft² for the toilet area and about the same for the vanity area. Wall elevations for the two areas are shown in Fig. 10. The designer should carefully analyze the number of lavatories and water-closets specified for a given facility. Research suggests that most "fixture counts" provided by city or state codes are too low and do not adequately reflect the amount of time that women require. As a result, it is not unusual to see long lines in front of women's rooms, particularly those that service places of public assembly. Note that the plan in Fig. 9 provides supplemental vanity or counter surfaces. t^mtamssas=—a=ai ammmm L to ui -K A,fkOv* • (.»»

»'.*• -.'-»« -.< ■»• «►*► >i»r\t*iy,|,*'■>'],)»«'•*',l~*' -v-1- *-»w fe-rr K**
j^'£^^L^f.u^ Fig. 9 434

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plans and Elevations ■ 4 \ \ \ \ .J2 _ , s \ \ \ > / / w =c= s • s • ■■■■ . ,!.: \ / \ \
 \ • \ \ \ \ / M j \ \ * .) / \ \ { - i ; ©■ (£K2> r Ml «■. !<■. OI»- »UTT-JO|MT«|3
 ■VA.N |TV J t ; MOTBE.' W.L.1«.v"B> I* 5iMIL.dk.pt. TO «L.msl'r» Fig.
10 Wall elevations for the women's room plan in Fig. 9. 435

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plans and Elevations mm un»m a.m«l« [Hjilllfl Fig. 11 This men's room
and women's room complex, Including a janitor's closet, requires slightly
more than 400 ft² of floor area. Corresponding wall elevations are shown In
Fig. 12, 436 Fig. 12 Wall elevations for the men's room and women's room
complex shown In Fig. 11.

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plans and Elevations V \ \ \ \ \ \ \ Y|x-Jrm Al tzt. \ \ \ \ \) / \ \ \ \ \ / It \ \
 \ \ \ \ \ / M di> ,tt> ~r~r" _r ~1 \ Fig. 12 (Continued) 437

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Miscellaneous Elevations L " i ,,1 Ill I'- Ml IT 4 MIAMI.© pa. * 1 —.p. —
p —p i—t LT! 0 _ _i rr ,. '* H 'i r' - - 4 j N S A. L ■ \ ^ i *B,LSa. ~i 3 In / Li
rs# \ 1 < V / ,M i | I i i 3 (D : TJ f \ LJ Fig. 13 These drawings show how
same designers Indicate the heights of certain fixtures and bathroom
accessories. 438

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Miscellaneous Elevations KEN WOMEN CBfl..T!LE- •w»msa eusser, I
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Clf I ^OTITIC ft. £ S? t , " eb < ajoserr s * K 4-2M- 439

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Plans and Elevations men's toilet 440 L' i . -V^ TIL* v-»- S&^jl >aj -a-g-j

a'.g-j *1 *ji-«r«rw-i :01 showers (jlsjCJVA^pLja. . crst*w>5 liu: *-»■
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Public Restrooms, Toilets, and Coatrooms ,. 6n*th t<>>' -im) nu.i^Mj.-
nu.UALi- r*rei* rifl-wn* RESTROOMS AND TOILETS Plans and
Elevations <j?ir«WM (AOkiteP y 1 YWKttcm. shj»bwsbp lam. Ja"-|I.om
j&?\\\$x.KMm &l&\\jKr\0K& Fig. 14 Detailed large-scale wall elevations
such as this are required to show materials, accessory mounting heights, the
coordination and placement of plumbing fixtures, and even manufacturer's
model numbers. Fig. 15 The large-scale counter detail shown here provides
all the Information needed to construct this essential bathroom element. Not
only are the construction details carefully defined and described, but all the
other design relationships are clearly shown. Note the relationships of the
mirror, soap dispenser, and lavatory to the plastic laminate counter. Other
lavatory counter details are shown In Figs. 16 to 19. 441

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Lavatory Counter Details Fig. 16 y'.. I'. n~/i»' FU1. l*M. •J»Vl COMT.- '
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if—* * %'•«' , f.ft* ^ lav-typ: sperA«wL SLO LAWOWE5 442

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Lavatory Counter Details im e>rzic?ie* mom * 105 i MCTA.U. '-T*
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rz" 3£7 MIRROR FiBBKSUASS ON WW^ GVMS'P. "~N, 4 CER.T11_e

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J.. ,,,_&!__ 1. rU.r. ^H ^' w'p. -J— ■ v^ —^- 33 UAVATORT Fig. 17 443

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Lavatory Counter Details PLWJie LAMIUVJf uee of iA.v/kTo«i »<
viwepx*kRe hob of euo nete fcovesBD w Detail ©LAVATogr 444 Fig. 18 "
("AMfc |V<2 ' CAN?*- tfO&&/>&C\7SO AWMihUM -* - MARBL.& \ it*
^ei>!L c3 lavatory t*f\ i i '^ia' A- V »'■ -*, V -\ - tl

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Lavatory Counter Details c>u a/4' evt+ia. in ^ ® cabinet \4 '«—' *
eWfTOPT PK9\$! ft- 1 9-VIHMrt ll») * It * »'--*• (b.6oh nil —■—■> » ,
PK.ywti.i- wriT ^ -f -U'4"fM.r I •! _ , <k -£*" Krt rt) «V* =^L , Ut.l A2V
Uri.nO ss^Hy it KM** MM _£/¥.#//<? et*.r itl tievmm <p mm- m. //###/£ A
TPL 1 -•"/R&tJb over 3TUP VVALL -"CCKtAMIC TIL.B- LAMkjATPD
fLAOTC rcwv«b couMTerf- TO f» w/ 4-McavCC SfUASrl (CMC fcioe i
back 4 W m<=>n DRirEC*ae) Vxm, CLASTIC fK* TO MATCH TO r 2X
ourroKT ri<AMB SBCURCP TO WALL STBEi- »«A.CKi.eT- Stanley
nantc => fcR^CKST. PAiMT T<=> MATCH _A,M. PL^BTIC -SEB INT'C-
SV, B 7 K> Pu^sticv ua.m. tor, &?6&\$ ^ruA5H + ■^JW— VAMILY.
P&TMU Fig. 19 445

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Lavatory Cabinet Details 1+ rtcotwt- • * I * 1 'Lo'. °"?t * " ■ - 1 H+CK
fe*r\&\ ftM^l'—■ it « »Le36*»4ei /* R*»fe = J < ■ • 4 vc/" \ 17\
CAtotNET PCXNL »T FOUPEr P«5M 37*o**! i|»"i'-e' .HftwrfHt4.*"!!
■tto rM+toot**^, oit- fttiri vchpi aktwl* rt**ace i«e*r >i 6*&MET
CfcTA|L,->T UgHEKi *y soh*: ir-i^o" wo r-wws.AWf^i. piHiw fLK^L.
u*UXlfK f-H, FiAtJ, euv. < seer »f gnaj= «.i i-tsHeu Fig. 20 Elegantly
detailed lavatory cabinets are shown here. Note the use of an exposed oll-
flnlnshed red mahogany frame or edge surrounding a verdl antique marble
top. Complementary telephone shelf details In plan, elevation, and large-
scale detail are also shown. 446

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Toilet Stall Details WW\||||||| \ \ |||| 32" . * 34" 36" 32" 34 36" 32" 34" 36"
32" 34" 36" \ * 56" \ \ \ . -60" \ ||||| ^ \ \ \ \ \ J— »•• 4—24' PLAN PLAN ^
W W W\W \W \ts"rV\W \ *W WWY \ \ ||||| ^" FRONT ELEVATION SIDE

if a user loses her or his grip. front transfer scale: 3/8"=1'-0" scale: 3/8"= t-0" grabLbars. section scale: half scale: 3/8"=T-0" 449

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Wheelchair Accessible Design Lavatories need not be specialized designs to be accessible. Utilization of clearances shown will do much to make lavatories accessible. Because persons with loss of sensation in their legs cannot feel pain (and because they heal at a slower rate), hot water lines and drains must be insulated. Also, under several state codes, faucets are required to be lever, blade, or multi-arm handle operated. Single lever controls are preferable. Spring-operated faucets must have time delay devices. At least one mirror must be located with the reflecting surface mounted at 3'4" a.f.f. (3'2" or lower preferable). Where possible, full-length mirrors are preferable. At least one of each type of toilet accessory must also be located at 3'4" a.f.f. or less. Note that this dimension is measured to the highest control required for operation. Controls that require twisting and grasping motions should be avoided. Because people that use wheelchairs require increased fluid intake, drinking fountains become more than convenience items. While there is not space here to address all configurations, the following concerns are typical to all: controls should be operable without the need for precise grasping; the faucet should not direct spray away from the user and must be located as near the front edge as practical; the units must be free of sharp edges and corners and overhead obstructions. lavatory scale: 3/8"=1'-0" lavatory rg scale: 3/8" = 1'-0" \J3 mirrors rr accessory scale: 3/8"= 1'-0" to malch 450 scale: 1/4"= 1'-0' alcove scale: 1/4"= 1'-0" scale: 1/4"=1'-0"

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
Wheelchair Accessible Design Urinals, if provided, should have elongated bowls with the opening of the basin located at 19" a.f.f. or less, or mounted level with the main floor. Many state standards specify maximum mounting heights of 15"—16" a.f.f. These lower dimensions are preferable. The toilet room itself should provide a clear floor area with minimum dimensions of 60" x 60" to facilitate maneuvering wheelchairs. Additionally, provide a minimum of 3'6" clearance in front of accessible toilet stalls to facilitate entry. Similarly, adequate clearances must be provided at entrances. The spaces shown in details I/3 too/3 represent typical dimensions specified in

state codes. Note, however, that federal and many state standards require 12" clear jamb areas adjoining both sides of all doors. A clearance of 18" or more on the strike side of a door is more effective. In vestibules having doors in series, there must be space for a wheelchair to clear one door prior to opening another clearances scale: 1/4"=1'-0" clearances scale: 1/4"= 1'-0" 451

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
 Wheelchair Accessible Design 42 min latch approach only, other approaches 48 min (a) Standard Stall 1 42, 1065 12 max CM CO CO <h 42 1085 o -r-e; 12 max 305 54 min 1370 66r IS/T 1745 w. wall mounted w.c. 69mln w. fir, mounted w.c. CM CO T~0: \ 42 min latch approach only, other approaches 48 min Fig. 23 Toilet stalls. 12 max 305 54 mil 1370 (b) Alternate Stalls 36 min 915 o clear floor space o 00 o CO 56min 1420 59mln 1500 w. wall mounted w.c. w. fir, mounted w.c. (al) Standard StaU (end of row) 36 min j 915 AW' M — ? CO CO (c) Rear Wall of Standard StaII 12 max sos - <0 CO /— 4Q|42lmln -alternate 1015110651 136 max i ! 915 I =a> toilet paper* O) If c 1 O) IWWWLWA.' (d) Side Walls 452

Public Restrooms, Toilets, and Coatrooms In Figs. 24 to 30, various generic toilet installation vary greatly. Placement of ac- accessories and grab bar configurations are cessories in relationship to plumbing fix- illustrated. While most manufacturers have tures, door swings, and interior circulation is similar accessories and grab bars within their to be carefully studied by the designer, catalogs, overall dimensions and methods of RESTROOMS AND TOILETS Toilet Accessories .y^wii Facial tissue dispenser \|L>17W Dual feminine napkin/ tampon vendor ,.--x Sanitary napkin disposal ■>- Dual feminine napkin/ tampon vendor S a -10'/i" Sanitary napkin disposal 8^ 4" Recessed dual napkin/tampon dispenser and disposal 414"" Recessed seat cover and toilet tissue dispenser U" \ p Wall urn ash tray HV,"-^, T 12V," Toilet seat cover dispenser Liquid soap dispenser Tr k ^ 4'/i"Ny *"■ ■ Recessed powdered soap dispenser 1514" -^- t 4" " I Recessed horizontal soap dispenser and shelf 4«" k All-purpose unit with concealed towel cabinet Fig. 24 4"h Multipurpose unit with mirror, shelf, towel, and liquid soap dispensers £\ m «"^\f 7>/«" Disposal valve soap gun 453

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
 Toilet Accessories ,..K &>\\ T r Paper towel dispenser a ^ Paper towel
 disposal i\\ r''' > SI Paper towel dispenser and disposal 454 Fig. 25 11V \\ T
 17%', J- r-' 12%' ■ X 141/, " \\ 12%', " -X >r \\ l -1114" ' \\ \\ r 4"Kp .^w-n Paper
 towel dispenser Paper towel dispenser and disposal Paper towel dispenser
 Paper towel dispenser 12%', " iis W\\ .S r" T Ml*" J- Towel dispenser and
 Dispos-Valve soap gun \\ -Rj P irJ>-, 0U-1W Paper towel disposal Paper
 towel disposal \\ a \\ rRb^ T Paper towel dispenser and covered disposal
 Paper towel dispenser and disposal * Roll paper towel dispenser and
 disposal \$ rf T Paper towei dispenser and disposal Roll paper towei
 dispenser and disposal Paper towel dispenser and disposal Semirecessed
 waste receptacle

Public Restrooms, toilets, and Coatrooms RESTROOMS AND TOILETS
 Toilet Accessories fcd :x_ M Waste receptacles Shelves Medicine cabinets
 Fig. 26 455

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
 Toilet Accessories 305MM ,. \\ \\ *r-jr-*~ OUBMM 1 18" MMM A M ^
 Sanitary napkin dispensers Sanitary napkin disposals iW-MM V o • C T \\ \\
 ~\\ 7 — * Combined sanitary napkin dispenser and disposal Hand and hair
 dryers \\ cai "M Paper cup dispenser IQ5MM ', mlum \\ \\ Paper cup disposal
 * * "^ 17/. " Multipurpose cabinet Fig. 27 456

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
 Toilet Accessories V M«MM N \\ M 29;i6" * ' |V " i" "V "» \\ \\ B V S DM /
 ■^rV v £,N-»-).ffi- Toilet tissue dispensers \\ 1S2MM * \\ • r 10SMM * \\ 16 V
 409MM J . 308MM . , f 6 V OSMM / k 3 t -r~ 20" 151/. " tOOMM Paper
 towel dispensers io';.- 27BMM ^T J. 3\\ u 1U1MMV « " 15^ 389MM 541/s"
 1375MM 8V|6" 205MM ~~V V<" return r-_ x r 56" 142SMM 351/s"
 092MM _173/ie" 437MM H " " Soap dispensers Fig. 28 457

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
 Grab Bar Configurations 32" JM. & ~4\$ii bar swings full 180° I W -30':
 33" 30" /■ left hand shown right hand opp. right hand shown left hand opp.
 ^ right hand shown left hand opp. toft hand shown right hand opp. "This
 dimension 11" when IVi" O.D. tubing specified. left hand shown right hand
 opp. ■©» 32' .97° \$w» •This dimension 11" when 1V4" O.D. tubing

specified. M1 hand shown right hand opp. Projects 3" below finished floor, has slip flange at floor. Fig. 29 458

Public Restrooms, Toilets, and Coatrooms RESTROOMS AND TOILETS
 Grab Bar Configurations right hand shown left hand opp, right hand shown left hand opp. right hand shown left hand opp, right hand shown left hand opp. right hand shown left hand opp. r12'K right hand shown left hand opp. right hand shown left hand opp. 36" from side wall to side wall Fig. 30
 L>Rii 24" 33" right hand shown left hand opp. &>^ 459

Public Restrooms, Toilets, and Coatrooms COATROOMS Floor Area Requirements Coatrooms typically fall into two categories: those that are self-service and those that are controlled by one or more attendants. The latter category of coatroom can be more compact because only one, or perhaps two, attendants have access to the coats. A self-service coatroom must have more space between rows of coats so that several persons can enter and get their coats. Self-service coatrooms are susceptible to theft of property, particularly expensive outerwear. Therefore, it is desirable that these coatrooms be visible to someone at all times, such as a maitre d' in a restaurant or a receptionist in an office. In those situations where a supervised self-service coatroom is inappropriate or cannot be provided, self-service keyed locks offer a viable alternative. In addition to being able to provide secure coat storage, lockers can also store briefcases, packages, o; other encumbrances. TABLE 1 Floor Area Requirements for Public Coatrooms

| Number of coats (capacity) | 50 | 100 | 150 | 200 | 250 | 300 | 350 | 400 |
|----------------------------|------|------|--------------------------------|------|------|------|---------------------------------|--------------------------------|
| 450 | 500 | 750 | 1000 | 1500 | 2000 | 3000 | Floor area (with attendant) Mln | |
| 35 | 65 | 90 | 125 | 150 | 180 | 200 | 225 | 260 |
| 300 | 450 | 600 | 800 | 1000 | 1600 | Max | | 50 85 120 |
| 150 | 190 | 225 | 260 | 275 | 320 | 365 | 500 | 650 |
| 900 | 1250 | 1850 | Floor area (without attendant) | | 40 | 80 | 110 | 150 |
| 180 | 215 | 240 | 270 | 310 | 425 | 540 | 720 | 950 |
| 1200 | 1900 | 60 | 100 | 145 | 180 | 230 | 270 | 300 |
| 330 | 370 | 430 | 600 | 780 | 1080 | 1500 | 2200 | Floor area (electric conveyor) |
| N/A | N/A | 80 | 100 | 110 | 125 | 140 | 150 | 100 |
| 110 | 125 | 140 | 150 | 160 | 175 | 175 | 190 | 250 |
| 275 | 325 | 350 | 450 | 480 | 600 | 675 | 1000 | 1150 |

Note: The above floor areas are approximate and should only be used for preliminary space planning requirements. Since the number of coats per linear foot of hanging can vary from 4 to 8, the floor area can vary dramatically. A lightweight overcoat, for example, can measure 1-1.5 inches in width. A medium weight to heavyweight coat might measure from 2 to 4

inches. A fur coat might require a minimum of 4-6 in. The designer must consider the overall size of coatroom based upon the following critical factors: (1) geographic location/climate; (2) attendants required or not required; (3) aisle clearance; (4) peak entry/exit loads for coat retrieval; (5) assumed garment thickness or garments per linear foot; (6) linearfeet of countersurface and overhead shelving; and (7) other storage components, i.e., hats, umbrellas, briefcases, packages, etc. 460

Public Restrooms, Toilets, and Coatrooms Three basic types of manufactured or prefabricated coat storage units are shown in Figs. 1 to 3, Exact coat storage capacities are provided by the manufacturer. All units can be customized to suit various room configurations. Note the adjoining counter space to speed operations. Coat capacities relative to length are listed in Table 2, COATROOMS Types of Checkroom Systems 15'-0" 10'-0" 5'-0" Fig. 1 Electric carousel coat storage. •17'-0" • 11'-6". 6'-0". Fig. 2 Rotating reels coat storage. ■18'-0"- •12'-0" ■6'-0"- = b» 5* Fig. 3 Stationary coat storage. 461

Public Restrooms, Toilets, and Coatrooms COATROOMS Electric Checkroom Systems 0«MU MNOM Fig. 4 nrm 111111 TOP VIEW RAILEX SYSTEM 1102 TABLE 2 Overall length with garments* 7' 5" 7'1W 8' 6" 9' 1" 9' 7" 10' 1½" 10' 8" 11' 2½" 11' 9" 12' 3½" 12'10" 13' 4½" 13' 11" 14' 5½" 15' 0" 15' 6V2" 16' 1" 16' V/2" 17' 2" 17' 8V2" 18' 3" 18' 9½" 19' 4" 19'10½" 20' 5" 20'1W 21' 6" 22' OW 22' 7" 23' 1½" 23' 8" 24' 2½" 24' 9" 25' 3½" 25'10" 26' 4// 26'11" 27' 5Vf 28' 0" Length of hanging capacity 13' 0" 14' 1" 15' 2" 16' 3" 17' 4" 18' 5" 19' 6" 20' 7" 21' 8" 22' 9" 23' 10" 24'11" 26' 0" 27' 1" 28' 2" 29' 3" 30' 4" 31' 5" 32' 6" 33' 7" 34' 8" 35' 9" 36'10" 37'11" 39' 0" 40' 1" 41' 2" 42' 3" 43' 4" 44' 5" 45' 6" 46' 7" 47' 8" 48' 9" 49' 10" 50'11" 52' 0" 53' 1" 54' 2" Coat capacity; no, of slotst 144 156 168 180 192 204 216 228 240 252 264 276 288 300 312 324 336 348 360 372 384 396 408 420 432 444 456 468 480 492 504 516 528 540 552 564 576 588 600 "Add 4" minimum clearance to each end and each side when adjacent to walls, columns, obstructions, or other machines, tThis provides 1.1" per coat. In areas or facilities where bulky coats are customary, the actual capacity may be reduced one-third. ONE SHELF RAILEX SYSTEM 1112 TWO SHELVES RAILEX SYSTEM 1122 THREE SHELVES RAILEX SYSTEM 1132 FOUR SHELVES RAILEX SYSTEM 1142 " 1 n

fc in 10a R nmann ._. A Al M k. ^J M i'V 1 s : "8 y\ 1- R , * oc L ^ * With '
462 Fig. 5

Public Restrooms, Toilets, and Coatrooms COATROOMS Stationary
Checkroom Systems 122-1-5 w Wall Mounted ■■■SiE.-,l 5 SS fe\ H^-
M"—^ , H— 24"—*) noituH ' ~T- |.i.,'iii.iii il J \$ a ^Ss, Length Number of
Coats Number of Shelves Wall Mounted 3'0" 4'0" 5'0" 36 48 60 0 1 2 0 1 2
0 1 2 k :■:" '*; Single Face 3'-4" 4'-4" 5'-4" 36 48 60 0 7 2 0 1 2 0 1 2
Double Face ii i - .I "" i 3'-4" 4'-4" 5'-4" 72 96 120 0 1 2 0 1 2 0 1 2 463

Public Restrooms, Toilets, and Coatrooms COATROOMS Coatroom Plans,
Elevations, and Sections Bpi. acVT»«f1fW 4 "\ H>js|i coftiteoM aaJhJTBP,
""|L«" r -tttahki pqusJ que pea etfV«s. (MNTIM. CUS«r. VK ar\ scCno<
ccATW.coCKl.s^bgiEee gfo A<sJ liM'-o" 4 -! IT 9 0 1 h / < V«V IW* t- r
—to."**_-*** -Y w n^, -""H 7 AegcTTQM'CcAmM.gca-ff®fi>e6--ffltooH
-r x: ^. T Ictvmr W**f-o»f |IKH enJt^t F\sU\fcTTav|< co*n*>eM m; v-m
Fig. 6 A typical coatroom configuration will often consist of a counter with
an access door plus the required shelves and hang rods, The overall size of
the coatroom will vary with the number and types of coats to be stored, In
high-volume coatroom situations, the design should provide appropriate
counter space for those persons working behind the counter. 464

Public Restrooms, Toilets, and Coatrooms COATROOMS Coalraom
Shelves Details -HAKDIVOOD <oH*LVB<i ■>\ETAL HAM4ROD see -
E.cu60ui.ife ■plM- f=L. 3, :? ■^7^ a'-o" mx. -// =E SM SCHBDUUE
NOTE.: C7.A- OI=B.MIM© TO RMS' OIVIDttX* INTO EQUAL
6ECTIOMS UMUttS. OTWR.F*.\V><»»». SHOWH UHDB.R S'-Q" '1/4"»
|'-Q" grUfe-VATIOH Of= LIMITS oveR e'-o" 1/4" = t'-oM * M wee of=
WAL.L 60CK.ET Moire ; \H sypsu/A. BOARD pARTmONS. PROMISE
WOOD K.UOcUIH<» OR.6HeET AEXAi. PUATq. TO RE.CEIVJB
4CR&SVS -- 1', --"-C *- % 1 Vic" <£ «%a.tinI caRo/At HA.HGHOD
PA.R.T feL&VATION SECTION FOT3t fi,CHBJ3LU-SL; _ <® V=*i'-o
TYPICAL OPEN COAT ARfeA, 4.CM.E-" A.«=> NOTSO 46S

Construction Details and Finishes Partitions and wall finishes Floors and
floor finishes Doors Ceilings Stairs Fireplaces Lighting 469 516 566 641
660 724 743

Selecting the appropriate partition or wall type is both a science and an art. In fact, there are so many options available to the designer that it is not unusual to refer to the partition or wall as a system, a combination of framing, sheathing, and finish elements, all working together to meet aesthetic, functional, code, and economic requirements. In that regard, this section explores the great variety of wall and partition types, examining all of their characteristics with the exception of load bearing capacity and cost of labor and materials. With respect to load bearing or structural capacities, while many of the wall and partition types are able to carry superimposed loads, it is not the intent of this book to discuss structural issues. With respect to cost, too many factors and variables make this a topic that is difficult to analyze with any precision. Information on both traditional and contemporary partitions and wall types is provided. Many traditional materials and methods of construction, such as solid gypsum plaster and plaster on clay tile, are cited, thus providing information to the designer who is redesigning or altering older structures. A large portion of this section is devoted to the detailing of contemporary partition systems. In addition to providing examples of partition types, these pages place great emphasis on the detailing of unusual interface conditions that many designers often leave to the contractor to work out in the field. It should be noted that most, if not all, of these details have been selected from the working drawings of outstanding architectural and interior design firms. While general information has also been provided about acoustics, sound transmission, and fire ratings of various walls and partitions, the designer is cautioned to verify all such information with manufacturers' certified test results, as well as with those building and fire codes having jurisdiction. It also should be noted that while test results may demonstrate a certain fire rating or sound transmission classification, it is important to determine if the results have been accepted by the local building or fire department. It is often necessary to apply a finish to a wall or partition. Again, both traditional and contemporary methods to apply wood paneling, ceramic tile, and stone are clearly illustrated through the use of architectural details. Finally, walls and partitions must ultimately meet floors and ceilings, and, of course, have doors and openings penetrate them. While some examples are provided in this section, the designer will also find important information in the sections Floors and Floor Finishes; Doors; and Ceilings, which follow.

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Characteristics of Interior Partitions DRAWING AND DESCRIPTION
 SOIL AND DAMAGE RESISTANCE ACOUSTICS REMARKS COST

COMPARISON 4 Inch face brick, tooled joints: Actual thickness, 3 3/4 Inches; Weight, 40 lbs. per square foot of wall surface Installation cost Incombustible, with one hour fire-rating Good Very good; transmission loss, 45 decibels Low maintenance, but limited flexibility: a good-looking wall, but poor light reflection maintenance and Insurance cost for 20 years 4 Inch concrete block, tooled joints, two coats of paint on each side; Actual thickness, 3 3/4 inches; Weight, 30 lbs. per square foot Incombustible, with one hour fire-rating Good transmission loss, 40 decibels Inexpensive; attractive If constructed neatly; frequently used for corridors, gyms, assembly rooms, etc. no flexibility 4 Inch cinder block, 1/2 Inch layer of plaster on each side, 2 coats of paint on Incombustible, each side; with two hour Actual thickness, 5 1/2 Inches; fire-rating Weight, 30 lbs. per square foot Poor Good; transmission loss, 43 decibels A smooth, dense finish; a good light reflector If painted a light color no flexibility 3 Inch cinder block, 1/2 Inch layer of plaster on each side, 2 coats of paint on each side; Thickness, 4 1/2 Inches; Weight, 21 lbs. per square foot 4 Inch structural facing tile, glazed on each side; Actual thickness, 3 1/2 Inches; Weight, 40 lbs. per square foot Incombustible, with two hour fire-rating Incombustible, with a fire-rating of less than one hour Poor Good; transmission loss, 39 decibels Very good Good; transmission loss, 35 decibels A smooth, dense finish; a good light reflector If painted a light color; no flexibility Used well In classrooms, corridors, also In toilets and showers; care must be taken with the design to avoid bright reflectivity; no flexibility 4 Inch concrete block, 2 coats of vinyl plastic spray over entire surface of each side; Actual thickness, 3 3/4 Inches; Weight, 38 lbs. per square foot 2 by 4 inch wood studs, spaced 16 Inches apart; metal lath and plaster, 2 coats of paint on each side; Thickness, 4 1/2 Inches; Weight, 20 lbs. per square foot incombustible, with one hour fire-rating Good Good; transmission loss, 40 decibels Combustible Poor Good; transmission loss, 39 decibels Sleek finish, but no flexibility Good light reflector, not much flexibility 470

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Partition and Wall Types B I W-2 *? 1 W-1 W ' 1/2 WOOD STUDS-1/2" O.C. Vt
 3-PLY PLYWOOD NAILED BOTH SIDES WOOD STUDS -1/2" O.C.,

METAL LATH, GYPSUM SCRATCH I BROWN, WHITE FINISH BOTH SIDES WOOD STUDS-16"QC. Vt FIBERBOAED, JOINTS PILLED, BOTH SIDES SAME AS W-3, WITH W SCRATCH, BROWN £ WHITE, GYPSUM BOTH SIDES W-4 BSSSSSS ES=jS=saa= 7", 4" WOOD STUDS, N ,, STAOOERED, 6"O.C. V-i M 2"«t" STUD AT EDGES "-" [M W EIBERBOARD W-5 W-7 1 333H3S333333333 S^B W-8 TS5 W-9 W-lo ^Trr" NAILED BOTH SIDES SAME AS W-5, WITH W SCRATCH, BROWN, t WHITE OYPSUM, BOTH SIDES WOOD STUDS-14" O.C., OYPSUM LATH, ATTACHED WITH STIFF CLIPS WSCRATCH, BROWN, WHITE OYPSUM PLASTER BOTH SIDES SAME AS W-7, EXCEPT ATTACHED WITH SPRING CLIPS ft" PLASTER BOTH SIDES. ?"SOUD OYPSUM PLASTER ON PERFORATED OYP LATH't ^"CHANNEL STUDS, SMOOTH WHITE BOTH SIDES V SOLID OYPSUM PLASTER SAME AS W-9 EXCEPT EXPANDED METAL LATH 2>V SOLID OYPSUM PLASTER SAME AS W-IO W-l <2&x3zfz5&3&&G>*- W-12 toflpctfris aqf W-l? W-14 }" M ETAL STUDS - It" O.C., METAL LATH, Vf SCRATCH BROWN, WHITE OYPSUM PLASTER BOTH SIDES TWO PANELS. NOT JOINED; VA" CHANNEL STUDS. EXPANDED METAL LATH, SCRATCH, BROWN t WHITE OYPSUM PLASTER BOTH SIDES FACE TO FACE -.10" SAME AS W-l 3. EXCEPT PACE TO FACE-4V SS, ,, •*" BRICK PARTITION, // Yv brown; white finish *C4 GYPSUM PLASTER BOTH SIDES SAME AS IV-17, EXCEPT 8" BRICK PANEL ' W-18 vmg. , SAME AS W-17, EXCEPT ONE LAYER &. 'OF BRICK LAID ON EDGE W-19 W-to BRICIC LAID ON EDGE. I"« 2" FURRING, WIRED.J OYPSUM LATH PLUS Y1" BROWN t WHITE OYPSUM PLASTER BOTH SIDES. iii^IL^j^KS 3 , I2\ 12"- J CELL CLAY TILE, L _—II ll _J L _J: W BROWN t WHITE H.i.-.:■...■.'....■...<.rr* -#—~~ OYPSUM PLASTER BOTH SIDES. W-21 wm, rtssi^si! W-22 ANOTHER PANEL BUILT AS NEARLY LIKE W-21 AS POSSIBLE SAME AS W-21 EXCEPT A\ IV, VI' J-CELL TILE _SAME AS'W-21 EXCEPT &M2M2" J-CELL TILE SAME AS W-21 EXCEPT flVK'iir J-CELL TILE DOUBLE CLAY TILE; ity*Wi\ V 8\ I2\ 12" JV BROWN AND WHITE OY"PSUM PLASTER BOTH SIDES W-2& niaianBiia, nrTKiiHiinmnMiniffIWirmimiTiiimiiiMiitiijiinni t W-27 DOUBLE PARTITION WITH AIR SPACE. TWO WALLS OF 3"k I2"« 12" 9" CELL _ CLAY TILE r'FLAXLINUM BUTTED ihf TIOHT BETWEEN TILE. NO T PLASTER. I\4" FLAXUNUM STRIP AT

BOTTOM, SIDES & TOP OF ONE PARTITION PUMICE & PORTLAND CEMENT 1-CELL TILE 4"x8"x1/2" NO PLASTER (VERY POROUS) SAME AS W-2S, BUT 5/8" GYPSUM PLASTER ON ONE SIDE ONLY W-2? W-1? 3/2" W-U 3"x12"x30" GYPSUM TILE 1/2" BROWN, WHITE GYPSUM PLASTER BOTH SIDES : 3"x12"x30" GYPSUM TILE. RESILIENT CLIP. METAL LATH, 3 COATS GYPSUM PLASTER, 2 COATS GYPSUM PLASTER ON TILE, , OTHER SIDE (WHITE FIN. BOTH SIDES} W-30 W-31 SAME AS W-2B. BUT 1/2" GYPSUM PLASTER ON BOTH SIDES GLASS BRICK 3"x4"x4" 471

Construction Details and Finishes PARTITIONS AND WALL FINISHES Partition and Wall TYPES 1. LATH, PLASTER BOARD. OR WALL BOARD INSULATION, SOUND CONTROL BY STAGGERED STUDS 1. 1/2" LATH, plaster board, * plywood, etc. 2. 1" BLANKET TYPE. • INSULATION BETWEEN STUDS. - * -" S SOUND CONTROL- SIDEWISE STUDS AND BLANKET CENTER LATH; PLASTER BOARD, OR PLYWOOD, ETC, joint placed Loose between studs 2. 1/4" SILL WALL BOARD INSULATION SOUND CONTROL-STAGGERED STUDS AND WALL BOARD SOUND CONTROL-BLANKET OVER GYPSUM BLOCK PLASTER ON LATH, PLASTER BOARD, PLYWOOD, ETC. 3. 1" BLANKET TYPE INSULATION WOVEN BETWEEN STUDS 4. 1/2" LATH SOUND CONTROL- STAGGERED STUDS AND BLANKET CENTER GYPSUM PLASTER 2" OR 3" SOLID GYPSUM BLOCK WALLS- 1" BETWEEN SOUND CONTROL-BLANKET BETWEEN GYPSUM BLOCK MEETING STUDS - • CORNER PROOF: PAPER ROUGH PLASTER METHOD FOR WALL AND FLOOR FOR EXTENSION OVER PORCH GYPSUM SOLID OR HOLLOW PARTITION TILE-1 TERRA COTTA BLOCK PARTITION TO HEIGHT OF GLAZED TILE GYPSUM SOLID OR HOLLOW PARTITION TILE-2 FRAMING FOR • ABOVE SUB WIDE OPENING • STRUCTURE TERRA COTTA HOLLOW PARTITION TILE 472

Construction Details and Finishes HOLLOW MASONRY UNITS bonding unit HOLLOW WALLS CAVITY WALLS ended facing. PARTITIONS AND WALL FINISHES Types of Masonry Walls and Piers COLUMN OF MASONRY INTEGRAL ISOLATED COLUMN WITH WALL OF

MASONRY PIERS MASONRY TIES; METAL TIES; MASONRY BACKING MASONRY BACKING METAL TIES; MASONRY BACKING METAL TIES; FRAME BACKING VENEERED WALLS FACED WALLS 473

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Modular Engineer Economy Double Roman Norman Norwegian King
Norman Triple SCR Fig. 1 Typical clay brick. SCR SCR < WALL
THICKNESS 12 X 12 8X8or12 10 2/3X12 5 1/3 X 12 5 1/3 X 12 5 1/3 X 12
S'WALL THICKNESS 12 X 12 12X 12 8 X12 8 X 12 6" WALL
THICKNESS 12 X 12 12X12 8X12 5 1/3X12 8X8 5 1/3 X12 8 X 12 5 1/3X12
10 2/3X12 6 2/3X12 8X12or16' 8X12 10" WALL THICKNESS ^ 5
1/3or8X12or16 12 X 12 12X12 IS- WALL THICKNESS 12 X 12 12 X 12 8
X 12 8 X 12 8 X 12 Fig. 2 Structural clay tile. STRETCHER SCORED OR
UNSCORED SOAP i>, SOAP 6" STRETCHER 8" STRETCHER
SCORED OR UNSCORED I bfi *S\ STRETCHER *^ ,;k£<H W^ MS ^ wr
^ STRETCHER SOAP SOAP Fig. 3 Structural facing tile. 474

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Types of Masonry Z B I B m\ 4m aa\ m zM 6" Fig. 4 Solid brick: bearing or
nonbearing (sections). A=brick, B = nominal wall thickness, C=finish. B 8"
A v ' /A B 1 _ 12" Fig. 5 Hollow brick units: bearing or nonbearing
(sections). A=brick, B = nominal wall thickness, C=fnlsh. C . V M- 6" Fig.
6 Structural clay tile: bearing (sections). A=structural clay tile, B = nominal
wall thickness, C=finish. 475

Construction Details and Finishes PARTITIONS AND WALL FINISHES
types of Masonry A B C- Flg. 7 Faced or veneered construction: bearing
(sections). A=brick; B = sheathing; C=corrosion-resistant metal ties spaced
24 In on centers, vertically and horizontally; D=wood or steel studs;
E=plaster or gypsum wallboard; F=masonry bond; G = masonry backing
unit. Tg&Tfrfe s 5? £ S 55 s & s SS S7 s 10" 2jon inn i n ■• r. ,;" F -■' mrt
*z I- 1 S sri_da IT10 If I. t Fig. 8 Cavity type: bearing (sections). A=clay
brick, B = corrosion-resistant metal ties spaced to provide one tie to each 3
ft2 of wall surface, C=gypsum plaster, D=structural clay load-bearing tile,
E=concrete masonry units of load-bearing grade, F=exterior face ot wall. ,,
A / q i n u _ —yv-. 1 ft tw L * s is > a ! 8", 10" 3", 4", 6" (a) Fig. 9 Hollow

concrete masonry units (sections), (a) Bearing, (d) Non-bearing.

A=concrete masonry units conforming to ASTM, Standard Specifications for Hollow Load-Bearing Concrete Masonry Units; B = nominal wall thickness, C=nominal shell thickness, D = gypsum plaster. 476

Construction Details and Finishes A C- ssa 3", 4", 6" A C -^S ss=i' 4", 6", 8" A : c < B| 4* pSPQ B «:■ = : >.*; 5 #* - E*V i:1^1 n - »:• : " '»•'■ - '. I ■' T : - ■>' = m : ; . (i 6" PARTITIONS AND WALL FINISHES Types of Masonry 4", 6" Fig. 10 Structural clay tile: nonbearing (sections).

A=structural clay tile, B = nominal wall thickness, G=fnlsh, D=f 1 A C \ U ' B « V ■ft .'fiV 2" 3:4" Fig. 11 Gypsum tile or block: nonbearing (sections). A=gypsum block, Binominal wall thickness, C=fnlsh. A n i c B ".*.<! 6" \ Fig. 12 Structural clay facing tile: nonbearing (sections). A=clay tile, B = nominal wall thickness, C= piaster, D= glazed or smooth- surfaced side of tile. 477

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Brick Types and Bonds HEADER BULL HEADER QUOINS

STRETCHER BULL STRETCHER PLACEMENT OF BRICK '— BED
SIX FACES OF BRICK LOSER 0- i\$> CLOSER KING CLOSER QUEEN
CLOSER HALF OR BAT SPLIT THREE QUARTER METHODS OF
CUTTING BRICK i 1 II II _L_ P II | I | i JjLJULJZIDLLO ■ .JLJLi _Jl II 1
II II « ni ii ii ii ii 1! II II II II II I t II II II "Till II II 1 .imm—i. n:-: .ir' tn
iivjllxji: ~l ll il i 'I ii II II II 1 COMMON HEADER BOND HEADER
COURSE EVERY 6TM COURSE 7,88 BRICK PER SO FT ~l BASKET
PATTERN 6 75 BRICK PER SO FT TT STRUCK xzzc nrr lt^ ir—ir ll f
RAKED STRIPPED FLUSH OR PLAIN CUT V-SHAPED CONCAVE OR
ROUNDED WEATHERED FLUSH AND RODDED BRICK JOINTS
3CZC 3CPC jn~r XZZ3C rrr aoociE 3 C izzx znnrx I X I xzc
aSHHEfflEZaEaEEIQ COMMON FLEMISH BOND ENGLISH BOND
ALTERNATE FULL HEADERS EVERY 6T" COURSE HEADER EVERY
S1" COURSE - HAL F BRICK 7 15 BRICK PER SO. FT. USED FOR
HEADER COURSE EXCEPT EVERY 6TM 7,88 BRICK PER SOFT. .,l,
ll:'1.----' II II II •■ < —rw~ i-w ZE3= J _zoz ,ii.; "i.:""-zjui ii::n Zl _ir ii . ii n
Liii ■ ii-ii ~Tf :■ r,~ -: —pj rr — imr II ■ -,l 1! .' ft--.=: ■•■-■-« —FaT-
t«4 flI II II ■ ii «f ■■■■! 'i ■■ ir- .■■ _ij_ -ii i M l',-;I II 'II li T.- .■ JI 1 ll
ll ll ll I _JUUI_J_ILJU_J 1 1 II ll ll ll ll ll fs-^M^M^mi-hJl MMim-K ,, i i|

ii II II l| II 1 II II 1 l| ll ll ll II II II II 1 II II I 1 ll II 1 II II II II , > ', ■ *"T--
i'.l Mt.-fi' ii ii n II 1 ■■•f.-r-M-'-i-ir: ii FLEMISH CROSS BOND
ENGLISH CROSS BOND ALTERNATE FULL HEADER EVERY
6'«COURSE CONTINUOUS FULL HEADERS EVERY 6T"COURSE 7
15 BRICK PER SQ FT 7 88 BRICK PER SO FT 478

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Masonry Walls i%ms%* ■'.V ""'/!.'"/■.V'i ?d WW- Sti-xiynt-i ..'!■-,% i
■...*;; zmmm i:!"':j:.V ■.■'•'••« ■m p^I ~""!^/".'./ :«:;i^'fl fo^V, *!,'!*:!'
■■*,,: * *, *, *.■'■*i[wm »m II H ^..■iti ^!T^1 iu_J •| i.." ■■.■■;■'• ' '■ 1
;".•,•:,, -1 1 '■ ■ ■"- '■'!,! ■ii ■■ ■■—i-. . .iL-wi—w/ , ,i Ha [?
■■"•.-. i^r^ k"" ..:-Jv 7^ F "•":'• , T^ i, ,,,;•..."•■ , "4f <:>l ■■,■■,■-, i
::■'■-■.■'!■. ■-■.■.■.-!'■•.J ^..■.■.■"■.■.■'■=-J ••">>l':!."-i.:.-.<J
:■*■;■ ■.-J ^■.■.■' Full and Half-Height Units L !■, ■ ^it^-*J ' n_M'j_- i- i
■ ^• lap^^^ffijii - > * ■'■ ■ ■ J-*r ■ * ■ * ifii- • i"- ' *' ■■v.Wrt'.v.iy-iiij Full,
Hulf and Fractional Size Units Fig. 13 Designs of standard-size hollow
concrete-masonry units. (a) II I I I II ll I 1 1 /Vy I ..■,.,.—. ,,,.,.,. , , 1 rL 1
——.—«,. RANGE ASHLAR .V I E3 (fc; RANDOM RANGE ASHLAR
RANDOM RUBBLE 1 II I rz>; i i i 5/\ lOKEh H 1 1 i > RANL 3E Ash
iL> * AR _tU^— 3~? m: . Afir. T. .. " , , tl 53 (d) COURSED RUBBLE (f)
ROUGH OR ORDINARY RUBBLE Fig. 14 Stone ashlar and rubble
masonry. 479

Construction Delaiis and Finishes PARTITIONS AND WALL FINISHES
Masonry Partitions Stkuctur* a»ov» - -tore: uusuxu *u>ut. ■=*E=£| J
<7T^ \ ;bxt*mp puwranTTo ■ UUDUWIO* 0* ■JTFUJttTUn.B a.-- ■*;
<:■« iro»c»o (A&TUAU.) T^ . ,&g,n: (uqm,\ 2*.s,4'^%,iiV8i;<to>'i E stf-
j ■ A* Q'ta" frlOMIMAO 3*4,5V»;!%'(ACTUAL)" ;•%/ 4^, Cl', a'
ftBMINA) sV-'W%"(Acw-i 5% 73tf <>V - -fCRATCU .COAT fuWfcl
©SLAZEE OK. HWPINMHBP cone, ok. cmoag. bus. COUC. OB.
CmPH.lt. RUE, plA*TBE BP BOTH ■itSe* © FLA6T6g./glittA»iAte.
"HUB. 6GT IH MOB.TAH. MASOSiR-V P^ETITIONM5 TYPICAL SIZES
AND SHAPES OF CONCRETE BLOCK Coping Screen (decorative)
Stretcher (3 core) Half - Corner Double - Comer Bullnoao Return (or 'L)
Corner 480

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Metal Stud and Gypsum Board m Of\' # \'; minm.jiniMi rll'IIMIII 60«P
flN.OA. 'ft' -CQtJTislUipLte s/Aft**. ie>o<aaie« <? I I A / Weyr- <*)*j?.
AHA*. worth, pf.. cant. caMWUtP Ho\m£H- J AcooetvcM- P/Wit*io*J 481

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Metal Stud and Gypsum Board t>%' "*" I—] ■LM»*f flMH»H*P + —*/>■
4S-U.P., wrwasw WPItfA-Ttf 0N <WN0T: FUN6 16' ^>.<f. -
W+*pUL.^PflHtt»H J 4>0., pPTTOrt |«UWNB^ - I (tOMD^a-M-**- * w
Metal stud and gypsum board: braced to slab *■»««,• IT r——p-^ fc-r rs*
i—UNW*il AU> V«9ir>» ■fsr cc tux* rt-i-tne.«w»hip mar ■wr>usrui»»ei r
L.Necriwi»H«r» 6#IWN£> u: -ZW.- MUfAL ^fUP 16* <*£ «x*Ouu*P
fiHIWt Metal stud and gypsum board partition: floor to slab uMPc-fs^iPp.
or <?K McTal 6TU»,A* iwo'c OUPfeTrXTK- &CHK?U4.p fV»f- *H' i^ I
MM»-<?f nNHsHBWS -#«5P>iJU3? FINISH Metal channels and gypsum
board: wall furring JfcS& iu»«w)e> * & &u.& a-V KflfclAi- <&fUP
;6*5i iT,} {H»5JJW'l*» Metal stud and gypsum board partition: floor to
slab 482

CDttDflinii' I jvjiinnuoa WAtff. flN, &*■*& *5Hep.piN,rj-p. Construction
Details and Finishes PARTITIONS AND WALL FINISHES Metal Stud and
Gypsum Board JNPenfotPB- or *>TMJ<yrrur>e I" asilP INOIUIVM UMff-
Cf r"IMI4»HH? C-ISIUIMi-i ILATCn **1s> Ci.uJ.p. l(&»^ . W eM «* to
TOP Cf HIH-WK HiutwK/ t*&e <s. f-i only «rru^Tul** t>*>pex?U Metal
stud and gypsum board: underside of ceiling *£WW.FN.FLf. a^epfiiJ.^iu. *
«HE* flN &***• i /—^SHeP.FlrJ.FMZ. y*P*JT Metal channels and
gypsum board: wall furring 483

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Metal Stud and Gypsum Board H£AP m It: a ZDL cz PWAN -V*fBft
f*BW> il il if w ft tf v,; «!»' 3" pNonwwBUWH funurtutt wrcM ■timtHr
FUB-OUT GYPSUM B'D QM METAL FUHRINQ CHANNEL «gPW t
WW PftflTiTIQN ^mw^ re itoeR&oe. or* rf. -1 una» vtf *t" *) «6CM fltPi
es^uw^T t<=t ^ e«?rwt»i 1 ' I'M I I I I IN i ' I I I I I IT ^�t*tHO. -
MMMftllU, ill k. • 9CMBO «uo«l*n Metal stud and gypsum board:
shaftwall Metal stud and gypsum board: movable 484

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Metal Stud and Gypsum Board mum T ,1311158 V -zJi" nTi. rwiW«p- -
Hurtle, i\lt c\+. 'f %" *F esc' ^ MH */PfeC W U(JfISf6|D6 *|><WftAfcnfē,
csvirtiixxt,. es c-rma^ H^wiThgM) ♦ffdlcyJp^L «U» zfe'ria Naffer *><«*..
<a>)JphJibii* aJL>^>. '£• 2 (4-txiw, Aan»nflM> r»*TMv>H) A^^-fli* ei*.
Wt.C. WITH TJtflWtlK. few <b\o&t>io ***«*«- eotiy&tax, cm**k> © ffft
% ££-ltei^Fy&-^lTOftCTTk?H) U « T •v- IV" J IUH'H'.Ht— , -. »■* *■<,
• (r -l#|W,r"l%- -As*a*fiOTHeel*. |*V.<5. |Jif r\$t Uf«fe ""'^«Tr!*i'l1 W»ff -
ll'tUufdrU* - tf.flfllltloJ*. 0h1U*J«- O.I. /TvTy^ - H-\ ftosf^ ^/^ .fi'^rrp <gur
48S

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Metal Stud and Gypsum Board PARTITION TYPES AND
DIMENSIONING SYSTEM A riwsH a» . SCHE.D. %'. PLAN OIMEN. «-
*► w s% 1, CEIUN<3 GYP- BOARD _MET/M. STUQS 0\«ac. >6A' w
PLAN OMEN. A PUAM DIMEW. FINISH gypsum e©*.«D FIRECOOe
"C" U-S.S.OR AWOVeO B3UAL shaft wau- plan dmen Plan omen THICK.
LINER PHLS. EXTEND fLCOR To STRUCTURE - CAULK.
PERIMETER. PLAN OIMEN. A CAULK / PERIMETER PARTITION »
U.L.O(E*1(5M_j HO. U 411 K PINI&H A* £cneo. ^ W 2&1 »-7«<» tt-
UNDERSIDE or structure OR FICOR METAL STUDS * IS 'IOC. PLAN
OMEH. :i:w 1 CEILINS ■ gyp. eosro riBECOOE "c" WML TO
jCMITWuf - To box, or. •STRUCT. A I'm'/ Va' — MUM.ANSIE Wood
(SIOCKIN© PLAN DIM'S. n CIS. AS 9CHED. PARTITION TYPE a op. 4
uHDeRetoe cr STRUCTURE. OH FLCCR 74 THICK AGOU6. eyP bo,
wall H/ i,by*1 MAW VEMEEft- ^ ' A 5'/t* ^, PA.RTITIOH TYPE 'Z OR 4
EkAISl OYPSUM bO. METAL &TUD& M.AR&LE VENfEK MAP-6LE
BEINBapciMG PLASTER SRANITE &ASE 486

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Metal Stud and Gypsum Board Details EXP066P -SPLINE SRID
CHAUtIEL ACOUSTICAL , BAN EL CZIUHG, PAINT JO MATCH
PARTITION TOP RUNNBR SHT. RIC PANBL ON METAL STUD
NOTE! SBCURB TOP RUNNER TO •WOCIA W/ TOS&LE BOLTS
/^OUSTICAL PANEL CBILINS METAL ANGLE PAIWT ■"" CBILIN
PAINT TO MATCH «HT.Rfc» PANEL ON METAL STUD PSTAIL (!)
DU6TPROOF MEMBRANE PURBIKG CHANNEL «<SUP FURRING

CHANNEL V5"6HT.RK.CL<3<. PAINT TO MATCH PAIQTITION TOP
 RUNNER 5HX RK. PANEL ON METAL GTUD I'/a'd?, * CHANNl &. ^d"
 SMT. RK. VEILING FURRING CHANNEL BUILDINfi WALL DETAIL
 /fg\ SCALE.'a»*lkPlIv3' DETAIL SCALEJ S"/-©1 <■) TT 4--V8" 3V»* -
 v- □ t ■5/6" %?% I1/*" y^-ft™ i / v L — SHEET ROCK PANEL ~TTr/ -
 METAL STUD TIM. FLOOR Li SHEET ROCK. PANEL — FURRING
 CMAHH&\~ — BUILDING WALL Li -4J'M. VINYL BASE BY FL'R
 COV'G CTONTR. FIN. FLOOR BOTTOM RUNNBR BASS PSTAIL
 scale; 3"~r-o' <2> BASE DETAIL SCALE : 311* l'-o" 487

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Metal Stud and Gypsum Board Details q\ fLA^i- <z>tc\xt?^ - fcn^ ^gf®
 pgpifig|igfg, Ua-lx- (z^flpg^ ^^ Atei***^ 3^7 I*'f0«*-H(14<»)* '^ eAtfl
 wHiVriTui** l^USr i-.s-'^— 3**s 2 I -wrr: »o r^iit1 "«f «*iM^e V «*-- I *
 —yg&rf**^^ W FC M Sfl XIZZ Sat -><i^vl«r>Kf«f«fiil<t¥ pps -
 i«i«Nip't.*»jyp*fii \ -TfrteM. tffvt p^TW ' ^H*d«Sfr?Tiy|J-IWI11*J
 AUfrfJg!7 iJ/flgE uttyc" LL fcMW rk T p«n*tTMS'wupi* — nevwwwAi
 -•nwnvpiWTi*') ©TJMi- 'tSai^M- IfrKTtTW £ H/ PICK l^T«ri>k»ijat»4 tj>
 Man ?—7f "WH"*** a/Wmh. ^iisr, l|J*t|l>IWU: 488 V-<rr <«*#•; im'1..!'-
 *'

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Metal Stud and Gypsum Board Corner and End Conditions DRYWALL
 CORNER BCAd ^ -TAPE AND JOINT COMPOUND © 0\Q° CORNER
 (p)l,T" CORNER ««5fr TAPS AND JOINT" COMPOUND MODIPIED
 CORNER BEAD-i TAPE AND JOINT COMPOUNDj ■SO SA.'METAL.
 auJe^ET plat 59 itQi'LONa , ,, . SPACED 3'-0'b.&. METAL ~Tur ~ £>) 45°
 CORNER "CROSd" CcSRNeR W A-W j. , aW j, j W -v 'J* SHEET l?OCK
 ■ METAL , .. -DRYWAJ-U coRNes g»EAD ® ^) PARTITION END
 OBTAIU TAPE* AND JOINT COMPOUND 'Y CORNER 489

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Metal Stud and Gypsum Board: Partition Conditions WUM9
 .TT«kiUA.TIOM HOTg.' u».a us. aoo. MfiTA.U TR.IWI U^(J.
 ACOU'STIC*.- «XI*TINCi WINDOW MUUJOM *T~ wihPOW
 MUUUIOM Metal stud and gypsum board: partition to mullion detail
 K.IMDMIS * »-WOM .."■ an. «quo.l - cuLQumjtca. ccsmtinu lau« A* -

ACOUSTIC CmiUHa P*kHM-^» MOUNTIHO CUIF»» SuRPMft MouN-
nisat ci.1T* W^aP/Vim MOUMTltJ UYT»»P/^M Metal stud and gypsum
board: partition to column detail ■pU<VAHt>O^I HUUioU augur,
utaer&iE Eg***/1?0** »=l£WJDlMT Metal stud and gypsum board:
partition to mullion detail 490

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Metal Stud and Gypsum Board: Column Enclosures and Fireproofing A
&H3<f ch r\)z&* &G>n& _ ^>" c^r, tec? on 'M Max. tc oHte^aeice VC^£
\\ NWH <35U)MH lb -see M<£>re 44|t>/.vz7 ATTpi IK: f-A^& ME^L- La?
Tn -Tie UAT^ TZ? <2?L, W/ + r0> Orf/iS* <a*D/. -er'C. VSrIIZB. FINi.
£25L.WIPT-H U.^.nJ 491

Construction Details and Finishes PARTITIONS AND WALL FINISHES
492 ll£ar«-pt<w ■ff> aw *v ■/)*iw*«lht* "FHi»*p^JrrngHl ^jWSU**^!*&
*t^F-*i*<Kr«F» [•tzianxnu?ii*t _ . _ ' l;*M!ilMllllil ■<* \ *| (T^iT^'gj
pirrtt*J'pu><frwrr Tempered glass "Wr^ ^19*0? ^n-roe MJ/cf«Ci w
^,k»wp«, j*jOu^T|J P!PW*TVM<W, 6*^£4M. ^,^£^^ upk^V^f V>
tJ<**U», (i> fl|"HV) * WHEPHj: |U^ . fun i)HP*««*YHM" TS£
Demountable

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Miscellaneous Metal Stud and Gypsum Board Details ••■VA' \ ' t
TMiriirml 0 i: h ii in v*&" DETAtj for FIRE RATEP PAHTN. & A.C7
DUCT ABOVE iw^mwi*.—, rSLffl?Ara nee jam I fltttfW «W> - I'4HNT
W»«< ^ 9ttTIQH f METAL PUCT ENCLOSURE ami rt M jr] |" ! **■'-'
■•ii.T!"". » I -T* ON& IAYWC ^IBIX?, 1W* 1**B6W(7B--
llt'TWUKMNHw- &MXK oijertm] WW 5\ COLUMN FURRING © sir**
at 8M>nY aw **** ENp^Qs8.e S^ILl P H \\ rnv rnaniM ./ / BI ■ _ II ■ ■.
M|*. Z X Z s> -® coMcmuoot n» (T) rum nwoiuii 2"x 2" audli to «ui. rum
tor unon or nmncm V 10 TWTOHJUW AMU AM (MCT ITUM.
(KAIITTAIK 1" AU If AO. UMBO nana tm iu*.) rtm TOaitu amm ioajb
laiu(i) to ow iim or ntM. __ (TS una 2"i 2" commnoi awli <win V" retr sun
to igi) to «uu 'O' OUT <M JBQX IIMI STUM 10 THJ AMU). "■* CD
XMtAU. UM SA1 MUUAL KM. UMM (TOM. '©• mr (1) wm tppvoui
ansu am uruus mm »"x >" m»uwwi awu (win V rtu aum to iim). rAim va\$
awu to «u« oau. Q"|QFrTERMINATTQN OF PARTITIOffi

"■"■»■«—"l|j »4 TUKwC • VjAflV 7f->/ +- *# ri j%±\timtr*6PW&
M*4< WIWXT rfarnaU (&iu«») 493

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Sound Insulation and Transmission 494 The resistance of a building element, such as a wall, to the passage of airborne sound is rated by its sound transmission class (STC). Thus, the higher the number, the better the sound barrier. The approximate effectiveness of walls with varying STC numbers is shown in the following tabulation:

| STC No. | Effectiveness |
|---------|--|
| 25 | Normal speech can be understood quite easily |
| 35 | Loud speech audible but not intelligible |
| 45 | Must strain to hear loud speech |
| 48 | Some loud speech barely audible |
| 50 | Loud speech not audible |

Sound travels readily through the air and also through some materials. When airborne sound strikes a conventional wall, the studs act as sound conductors unless they are separated in some way from the covering material. Wall Construction As the preceding STC tabulation shows, a wall providing sufficient resistance to airborne sound transfer likely has an STC rating of 45 or greater. Thus, in construction of such a wall between the rooms of a house, its cost as related to the STC rating should be considered. As shown in Fig. 5, details A with gypsum wallboard, and 8, with plastered wall, are those commonly used for partition walls. However, the hypothetical rating of 45 cannot be obtained in this construction. Good STC ratings can be obtained in a wood-frame wall by using the combination of materials shown in Fig. 5D and £ One-half-inch sound-deadening board nailed to the studs, followed by a lamination of Vi-in gypsum wallboard, will provide an STC value of 46 at a relatively low cost. A slightly better rating can be obtained by using B/s-in gypsum wallboard rather than 1/2-in. A very satisfactory STC rating of 52 can be obtained by using resilient clips to fasten gypsum backer boards to the studs, followed by adhesive-laminated 1/2-in fiberboard (Fig. BE). This method further isolates the wall covering from the framing. A similar isolation system consists of resilient channels nailed horizontally to 2- by 4-in studs spaced 16 in on center. Channels are spaced 24 in apart vertically and 3/4-in gypsum wallboard is screwed to the channels. An STC rating of 47 is thus obtained at a moderately low cost. Thus use of a double wall, which may consist of a 2 by 6 or wider plate and staggered 2- by 4-in studs, is sometimes desirable. One-half-inch gypsum wallboard on each side of this wall (Fig. QA) results in an STC value of 45. However, two layers of

5/8-in gypsum wallboard add little, if any, additional sound-transfer resistance {Fig. 68J. When 1/2-in blanket insulation is added to this construction (Fig. 6C) the STC rating increases to 49. This insulation may be installed as shown or placed between studs on one wall. A single wall with 3/2 in of insulation will show a marked improvement over an open stud space and is low in cost. The use of 1/2-in sound-deadening board and a lamination of gypsum wallboard in the double wall will result in an STC rating of 50 (Fig. 6D). The addition of blanket insulation to this combination will likely provide an even higher value, perhaps 53 or 54.

WALL DETAIL n -2x4 I ■■■■'■'■■■. ■■■■■■■■■■mi. ■ ■■ ■
 ■"■■■*■'■■■■■■' BE A I BE B ■■i'-T ■'? M I s. D 'I ' 'T ■ ' ' ' ■ " 'I IT "■■
 I" Fig. 5 Sound Insulation of single walls. DESCRIPTION STC RATING
 1/2" GYPSUM WALLBOARD Qfe" GYPSUM WALLBOARD 3»fe"
 GYPSUM LATH (NAILED I PLUS W GYPSUM PLASTER WITH
 WHITECOAT FINISH (EACH SIDE) B" CONCRETE BLOCK W
 SOUND DEADENING BOARD (NAILED I 1/2" GYPSUM
 WALLBOARD (LAMINATED I (EACH SIDE I RESILIENT CLIPS TO
 ^8 GYPSUM BACKER BOARD 1/2" FIBERBOARD (LAMINATED)
 (EACH SIDE I WALL DETAIL A ~W-z" sa B ^BETWEEN OR "WOVEN"
 IF- D DESCRIPTION STC RATING 1/2" GYPSUM WALLBOARD *fe
 GYPSUM WALLBOARD (DOUBLE LAYER EACH SIDE) 1/2"
 GYPSUM WALLBOARD I '<4" FIBROUS INSULATION 1/a SOUND
 DEADENING BOARD (NAILED) 1/2" GYPSUM WALLBOARD
 (LAMINATED) Fig. 6 Sound Insulation of double walls.

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Acoustical and Fire-Rated Metal Stud and Gypsum Board WU1 i^Jqi FC#£
 4CK& Cf.'jgtW(.VpJ> * IjHE HOUR Fine HATEP MHTITICM^ ^ 1d
 SAME AS SOUNO A" TloNBUN*ftT 2* ""ii 1, GO m twq Hoyp Fine
 RATEP PARTrTKJN I^WAAT^uiTI«li^UNKET ~B A^ i-*Nt,f. ^O *^ b
 ""& |W i9 FirriniHw ;<- "5C*Wif-' ^•(fiSSK^ 40*17* T" v J-CJ :S2
 "fiWgJS^1 N*-"*^ 3* SOUND PARTITION /CLASS A' / SOUND
 PARTITION ITA ASS &L 495

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Acoustical and Fire-Rated Metal Stud and Gypsum Board ^V».«uJo fy>^i
 ww'^H DRYWALL PARTITION. USE TO ALIGN WITH partition III @

non* fire rated area SAME AS fgl BUT WITH y1 Sound attenuation blanket
 — — ^""^ USE TO ALIGN WITH PARTITION §3*9 (i .NON-FIRE RATED
 AREA M ra «hj SfeSWa, DRYWALL PARTITION TO 6* ABOVE
 CEILING 13 I^AtTINuifloffSL, ANKET /'CLAtt CV ##** PARTITION /
 USE WITH PARTITA LUMN ENCLOSURE SAME AS f71 BUT WITH ■.
 ,...- SOUND ATTENUATION BLANKET 1&&0Z®^ N—**!*« 8.
 S^ND^fttINUAM^LANKET 496

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Acoustical Wood Veneer Plywood Wall Paneling PA£JL »P> WALL 1+u
 AIP-. trACl. MOLD 2." TWIC)*, 4 Lb, «KUftlTV FlfeR.*«LA«ft • *
 MIW»ILAL VV»«L AC*1J»T)«AL ti.AMKBT-A« MAWur, fey
 ftALfrWU- «UfrST MIU. 1»IV, Of HIIMI feO^,r*MTI*W' V
 UAfrDWOoD VIHElfc fbVWMB FAMtL* PIT A-1 PE^FOISATID (VIB
 M«T») OUT A-a ULlI*ll*F*»»SAT*D Ut Cfl£l<« <* W*a»,£LUfc
 PA»BL« Ttt ■tfONJTIWUdU* '4" * 1" Ply W##» tTfclfc* MAILED T« »*»
 piUmMNI^ ■Z"xZ" >,vooc> FUfMMM** 24"e».c N *<S>TM
 DIMiSTioMS SWIMS A* IH*«»'0> T* <IT *U**-OM4 FLUMft 4 TM4IU
 TAMEL« SMALL WAV*. 1)4" ^lA.WOUl* 1^" o.tf. MOiuM* SMALL hi
 &*ILL*C> W»T *UN«H«» OP»W A*KAT« »&> AT UU«T 11%
 A£OUST)£ WALL Th&ATMEUT DETAILS ■a" » ^—=»* *<—»-
 MUU-» ' TO CD i CD i CD i O-^—i: I I (p ; Cp ; (p i Q-r- (J)-.■ 0- 6 -0- ©
 • 6—A~

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 wood frame partition systems for sound control PARTITION SYSTEM
 single stud walls Basic construction is 2" x 4" Bgi studs 16" o.c. with
 double top p plate and single or double bottom plate. Faces are V thick fire
 resistive type gypsum board applied, taped and fin- g ished in accordance
 with man- F ufacturer's recommendations. Resilient channels are applied to
 studs 24" o.c, as shown with k a Vx3" gypsum nailing strip f at the bottom.
 Absorptive material is paper-backed glass fiber or mineral wool batts .g
 stapled in the stud space as il- f lustrated, Sound deadening / board is
 sound-rated organic jf fiber board with a 15-18 pcf 1 I density, L# no. 3
 double stud [fl walls with a common plate basic construction is a double
 row of 2"x3" or 2"x4" studs, each row 16" o.c. and each row aligned with
 an opposite edge j of the 2"x6" top and bottom plates. The rows of studs are

offset 2" to 8" to prevent any chance contact. Other details and materials are as described for single stud walls. no. 11 | I / P./ double stud] walls on separate plates Basic construction is a double wall of 2"x3" studs on separate plates about 1" apart. Studs of j each frame are 16" o.c. with H the studs in one frame offset 2" | to 8" from those of the other. fl Other details and materials are H , as described for single stud 3 walls. | n0" 1 3 IJJ It WALL NUMBER 1 2 3 4 5 6 7 8' 94 10 11 12 13 14 15 16 17 18 WALL FACE Single gypsum board each side, applied with screws; no resilient channels Single gypsum board laminated and nailed1 over sound board each side; no channels Single gypsum board applied with screws 1 side; opposite side on resilient channels Single gypsum board laminated and nailed1 over sound board, opposite side on resilient channels Single gypsum board on resilient channels each side Double W gypsum board, base sheet vertical; face sheet horizontal; applied on resilient channels one side Single gypsum board each side, applied with screws (2x3 studs—16" o.c.); no resilient channels Single gypsum board laminated and nailed1 over sound deadening board each side (2x4 studs—16" o.c.); no resilient channels Single gypsum board nailed one side, Single gypsum on resilient channels opposite Single gypsum board laminated and nailed1 over sound deadening board 1 side. Single gypsum board on resilient channels opposite (2x3 studs—16" o.c.) Double gypsum board [W over %") nailed one side; single gypsum board on resilient channels opposite (2x4 studs—24" o.c.) Single gypsum board each side applied with screws Single gypsum board laminated and nailed1 over sound board each side Same as wall 13 Single gypsum board laminated and nailed1 over sound board 1 side; single gypsum board on resilient channels opposite Double gypsum board; nailed each side Double gypsum board each side; outer layer laminated and nailed1; base layer nailed Double gypsum board laminated and nailed2 one side. Single gypsum board on resilient channels opposite STC 34 45 50 52 53 59 49 4'j 50 53 56 51 53 CO bU 51 5'J 57 1 Design No. 5—1 Hr. combustible (bearing wall) Underwriters' Lab, Inc. (10) 'Face laminated vertically with three 6-inch wide strips of construction adhesive and nailed with about half the usual number of nails.

Construction Details and Finishes PARTITIONS AND WALL FINISHES
W &rf»WM KSM9 (T> • 'z'ruP.KjN& channels - EXISTiw©
COMCWITfc W.O&C N^ULS CXTVMQ ^ iNTWV^D CO^NEK. 3gTA|L

(T^CAI) . -T // ^ / Ay " / . ' ' ' ' / A A ■ ' A ',//, - ■ *2 ffly-SJrv- MA^O - 2.'
 PUKR.INJ& CMANN£15 -1' IMdUI.M'iON (RiOlP) A^AAM J? ■//■///
 ,yy^y-. V////////A,;. scaCE i/s"» r-c ©* UL,U SCALE McTAL CORNIER
 MAP rtEiNnDRCEMBNT 4 TuASTftK ■/i" ©T^SUM &OAJ*£>
 CXJ5"ING CONCRETE &I.OC* VLL I1 IM3ULATION ('KICip) f
 rUWJNS CHNNNficJ US(J N^CTAL rU^^NO CHANNEL. p-i vimvl tkiw
 sr js -GysuM EX.5T<N© »ViNpc>V „AMft bx-.st.ms Glass - >OTei_ The
 ADOVC PCTAiLS A« SCX-UTiOnS IWTMt 5AWB COwpiTION \£/
 SCALE 1/2*»r-0' Metal "Z" furring and gypsum board: exterior walls /\
 "(JE.S.iNS AT V.NPOW Vsjv;e> ' SCALE VH?1 'T-O' WCTAt, COKJNft*
 MAP WiNirCWa=1 hJU-n- 4 TLASTBlT r-l vIM/L- TWM fc^ US-OMWJM
 e?uarTir-ia p00* ^vsm* ru^fslNe AT POOPn
 JAM&tT>ricALtiNTewon,tAiL» ruu. scale, ~ ' Metai "Z" furring and
 gypsum board: door jamb details 3 KjQNrOUCCWeNT -a.M9TIN» BOOR
 JAM6)riAMN& AT POCK, JAMB fwoMacxrcmonAAuQ rUU-SCALC
 499

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Wood Vendor Plywood Wall Paneling v-dx(/ Y|£UH*. fkVMMP PAMU. 7
 4«IMT - '4' UAlS»W**» VIMVIA M.IV»D NN»L 'tfoHTWWtfWS '4.'"*
 5" fi.VvV«<» STMCS MAILW 7oi»WM»wu6 ffcVIV0#D PAkjat wool*
 c^fttfUMD* J.IW* dp F<Lat^ IMSTALL'ATloW MfcTMOP ^>»A<tbm
 PuhfeiuA T» ^iU- WH»P|I tj«ilMT« IW V»M«Mt VVIM.
 <IT.MAM|MMM'I'<o' •« t>BT«fcMIM»t» »V «**«*» C/&II* STPVIPS
 L_i S6ALK i 3"»l'-o"~" ' ' "~~ 500

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Wood Veneer Plywood Wall Paneling SOI

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Wood Wall Paneling Details and Conditions MA&mqrI ffr ft <iJVIbtt lwv.f:
 walls.. en -==—^ »■ « 1—> **£ea.-L(P ■tfw.V.R (&**) •tH&BC-
 %Twi%WM?item nffinc y x / ^Types') . gANTVftRV/^=PVER^ . ~ OFfi\£tIA
 (With no syppir) , y (TVPE I) / WVP W4LL / ACTIU5 c^JUNcs/ PA^C/A
 INTERSECTIjiSNS 502

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Wood Wall Paneling and Wainscoting Details and Conditions -H? Tilling

ledge is left on, cap piece to cover slight variations due, to the plaster
work to be planed off to suit condition*. Wainscot & K Ml woodwork to be.
best grade, well matched for grain, and thoroughly kiln-dried, ■\$.
~P^ 75T In. some cases the use of the cover mould is specified to avoid the
Jill-lef-ton- cap (to cover joint). Mouldings run, solid on, Si I. will, rail
routed-out to receive panel. The 1 neH10d.wk.ile used in the tie-aper class of
work does not allow for any adjustment of panel variations. \ •Panel rail
R-ised moulding Li J-fly ve-ered panels are like room, tended for interior
world*/in. preference- to solid panels. Exterior panel line should always be
solid, & ?\u .panels (the ply is 1/2 in, thickless.): Cabinet work no! be
installed until the wood is thoroughly dry. Dae,- ,om. position floor, i V
•".CeiiCcofoi' • ~\ Straight panel All paneling fits snug and field se-
cured in manner shown, allow > 1/8 in. between panels or contract for
joints are never to be nailed fast. All wainscot, 'lil' cladding, base
and kick, particular the panels to be top-painted as like nill, to
prevent the wall dampness from entering or in m™<! *ff«f% Ike
wainscot after erection*. This dimension, should never be 'eS5 7kan|: WA f
H 5C0TIH0 W(TH-PAK&L' HAVJRQ-£AI5£D«M0ULDI)lGo
WAIDCGHVITH-fANLL" HAVIH«FLU5H°M0ULD- WAIDCOT-
WITH-PARLb HAVIJkMUI3ED-M0LJLD- WAIDCOT-WITH-fAHLb
HAVI KG-FLUSH-MOULD" 503

Construction Details and Finishes PARTITIONS AND WALL FINISHES Wall Paneling Elevation yz,pMs*#H& 504

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Ceramic Tile Wall Finishes Wood Studs or Furring Cement Mortar Metal
Studs Cement Mortar .■: 1 i 1 - 1 ; 1 CSQAMIC TILE to OUO COM
work. to SCQATCtt O OAT METAL LATH MEMDGHJE VVO0O
STUDS -D 1 ^-1- F if id— 5 . " \ *4' : CEQAMIC TILE DI2r-9ET "Q
LATEX- ROT2TLAUD CCMCUT MSaTAG to O^0 COO- CUQED
MOOTAJi WD S5CT2ATCH COiCT METAL LATH METAL STUD-
MIM,^" T° Recommended uses ■ over dry, well-braced wood studs or
furring ■ preferred method of installation over wood studs in showers and
tub enclosures Recommended use ■ over metal studs Gypsum Board
Organic Adhesive !c -CEQAMIC TILE - ADHESIVE ISIUGLE op
MULTIPLE 1 LAYER OYPSUM to telO -MffTAL STUDS Recommended

uses ■ over gypsum board screwed to metal studs, single or double layer
 Installed in accordance with GA-216 ■ where a gypsum board, non-load-
 bearing partition is desired with durable, low- maintenance finish ■ for fire-
 resistant, sound-insulated, ceramic-tiled walls (fire-resistance and sound-
 insulation ratings calculated on partitions before tiling) ■ for dry areas in
 schools, institutions, and commercial buildings Wood or Metal Studs
 Gypsum Board Dry-Set Mortar or Latex Portland Cement Mortar =
 CEQAMIC TIL6 DOT-SET "0. LATEX- FOETLAWD CEMEWT
 MOQ.TAQ &OUVOOK OYPSUM tOt-0.0 I WOOO "12 METAU I
 STUDS Glass Mesh Mortar Unit Dry-Set Mortar or Latex Portland Cement
 Mortar >;... 'd ■» 0- •B 'g. P' j. ,f •'0 t' 0. sL "V f t» _ CCQMIC TILE
 tOQY-SET °Q LATOf- RSQTLAUD CEMEUT UoQikatouocopx (3-LA33
 MCSM MOQTAQ. UUIT VOOO °0. MCTAL STUDS Recommended uses
 ■ dry interiors over gypsum wall board ■ for dry areas in schools,
 institutions, and commercial buildings Recommended uses ■ in wet areas ■
 over dry, well-braced wood studs or furring ■ over well-braced metal studs
 505

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Ceramic Tile Wall Finishes Masonry Cement Mortar CEQAMIC TILE
 &OKJD GOAT MOOTiKQ. DED SCQATCH <5^AT METAL LATH
 MEMEPQAkJE *>/"T°l|! Recommended use ■ over masonry or concrete
 on exteriors Dry-Set Mortar or Latex-Portland Cement Mortar CEQAMIC
 TILE DOY-SET <=tt LATEX- POOXLkUD CEM6UT MOI2TA.a CCUD
 COAT W/a /■ ■//// % \ Recommended use ■ over clean, sound,
 dimensionally stable masonry or concrete Masonry or Concrete Cement
 Mortar Bonded p H W. % mwT 2>/&» " m7\ CERAMIC TILE e?ONO
 COM S/1C>aTat2 teEO MAtONQY Recommended use ■ over clean
 sound, dimensionally stable masonry or concrete Cement Mortar One Coat
 Method Solid Backing Organic Adhesive CEQAMIC TILE boUD C<=AT
 MOQ.TM bED SCI2ATOH COAT METAL LATH MEMEsQAUE solid
 iSACwua; worn PLASTER, MASOWQy "Q GtYPSUM fcOAab iVe"
 GEttA-MIC TILE ADHE5IVE solid epAc^iNa: PLASTEI2. MASONQY,
 CrYPSUM t^&QD, °r2 CrLASS MESH MOQTAI2 umits ovea \v<aac) aa
 METAL OTUD3 Recommended uses ■ over masonry, plaster, or other
 solid backing that provides 'firm anchorage for metal lath ■ ideal for
 remodeling or on surfaces that present bonding problems Recommended

uses ■ over masonry, plaster, or other solid backing that provides firm anchorage for metal lath ■ ideal for remodeling or on surfaces that present bonding problems ■ ideal for remodeling where space limitations exist ■ preferred method of applying tile over gypsum plaster or gypsum board in showers and tub enclosures Recommended use ■ interiors over gypsum board, plaster, dimensionally stable masonry, or other smooth surfaces 506

Construction Details and Finishes PARTITIONS AND WALL FINISHES

Ceramic Tile Wall Finishes - ^ ^ ft HORIZONTAL SETTING Fig. 15

Standard trim shapes and designations (6 by 4Vi In wall tile set in conventional mortar bed), V*i«™ 1" >EAD 6" ■ »" ^ \ « cove 6"i J»" COVE 4W» 13 SUKACE IUUNOSE 4" . 1" SURFACE (UUNOSE IV » 4 V," CUIt CURt 4" « 4" BUUNOSE •UUNOSE 4".3 V." SINK TRIM 4" « 2" Fig. 16 Standard trim shapes and sizes. S07

Construction Details and Finishes PARTITIONS AND WALL FINISHES

Marble Veneer Wall Finishes WA , ... _L SECTION - ■^MAHbLE

WAINSCOT MAHbLE WALL tSgALL- 5" M'o' -3<£ALL 3"«1'C walCsect]on~" ^LSOa^ME ASH LAH WALL* MARBLE CEIL. Fig. 17 Marble treatment for walls and wainscots. 508

Construction Details and Finishes PARTITIONS AND WALL FINISHES

Marble Veneer Wall and Ceiling Finishes CORNER, BLOCK BBVULKD BLOCK R 3ST .A. Xj CORNERS BUTT JOINT CORNER COVE TYPES OF INTE3 -, .^ —. _ w -v ~- — -v Not©: TYPICAL JOINT THICKNESS FOR INTERIOR MARBLE SHALL NOT EXCEED '(6* UNLESS OTHERWISE SHOWN OR SPECIFIED WIRE ANCHOR (T YHCALJ •*> —■ >x ^ Utf Mi 1 I tf £S£ B U T T JOINT T Y JP E S QUIRK JOINT O F U CRAMP (TYUCAL) £3 3C ilOLDBD JOINT TERN CORNER, BLOCK J-, CO R. BABBBTBD JOINT N" EG R S AnoKorage note : it is the responsibility of the warble contract ob to show complete anchorage ON THE MARBLE SHOP DRAWINGS. SEVERAL TYPICAL METHODS OFANCHORING MARBLE TO VARIOUS SACK-UP AND STRUCTURAL MATERIALS ARE SHOWN ON THIS AND OTHER DETAIL PLATES TO ACHUAINTTHE ARCHITECT WITH SOME OFTHE METHODS AND MATERIALS U 5 U A LI Y E M P I O Y E D Hsi^gg^ CONCRETE BACKING -WIRE HOOK SHELF CLIP ANGLE

STRIP LINER r=D0WELS n FASCIA SUPPORTED EYE BOLT
SUPPORT WITH STRIP LINER AT SOFFIT JOINT METHODS OF H-A-
STRIP LINER CROSS SECTION n a i n a s STRIP LINER. LONO-'T,
SECTION O F F I T S Fig, 18 Anchorage details. 509

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Marble and Travertine Veneer Wall Finishes TLAVE|tfl ML"
WALLOUHFACL- 510

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Slats Panel Veneer Setting Spots of Epoxy or Latex Mortar SECTION
Anchors »8 Dead Soft Bronze Wire or Soft Copper Wire Natural Cleft
Face-Gauge Back Surfaces PARTIAL ELEVATION *>S5>yWWy
4*>666£<S6^ Sealant To Be Slate Color Polyethylene Back-up Rod 1~Ji"
Min. 2 X 3E^^ SECTION "Y-Y" 3"- I'-O" Fig. 19 Slate panel veneer. J^ . ■
Cement Plaster SECTION "X-X" 3" - I'-O" • 6 dead soft ^ bronze wire ". -:l
turned into slate t" ELEV. Fig. 20 Slate panels applied to concrete wall
SECTION 511

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Reception and Pass-Through Windows ?. , g2?MTRC?L W(NP<3W
V£en<^u _ siscton thru L^rey ^KTreot. wino<=>w 512

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Bl-Fold Door Details viN*tfu m*mt*Ac fi>h Foi-D H AsW.O - ^* &A&~
P1.VNWO0 Bat-f*OUP DQOI*. *HAUL i fg>KsrVnu,-y AsLCQVE 513

Construction Details and Finishes PARTITIONS AND WALL FINISHES
Column Fireproofing Details COKHMt BttAb Doui,t_ * <7HAwo lb a*, ti*
wiam '/a* viRMicuuiTB ah. avp«ur>\ Pftiurt- pgutmn.' HttxAao«*,U MMU
57C* L COUUMti \$Bftj£rXBUICTUItM. bV46f 4 HK. EATING FLUTEC
THtCKMK** JTYF^ tyttW 3 HP ■JWR 1MB IUR. Tumi;. W 2' 'Vb "i
TVP« VHftMICUttTl ©«. . flvr*ij* 1 *AMD • C^MuM LATH PUA.4TKPI
A«R T*BVB FOR. -■ i/TEEL CoL, 4St -CrfuJCrUllM. ftWC* 4te TA-M-t*
xaa-ufthfirkAQ 1*4" V»M-MICUUIT» Alt' £WB«U1A efcGLMTft
GYPSUM t-ATH 4 PLASTER. RK.£pK,OOftHq 4 Hit. RATW6 METAL
LATH j PLASTE- RGLEPfcOORMG F - pLi*.STI(l.(i*n. FIULiH
4CH«bUL.») ^7^ 2^Z iju^i X ■THICK M«4 il»T»»L« POAWIMG4

hAATEMA.U afraid* (.ftuftHto cu^v oft -shm*) ItRICVi (64UC. MM*1)
 coijcbm« ftL.ocx,ftilei£,on.TiLi BkctPT ciudbs .COMCmt* UMIT4
 HoLLOVI CmDfett. oft C0MQB.1TH. BLOCK #TIL* Hk/lklS A
 COWPHK4IV* *nV»IJCTH OF AT LSMT TOo'Viam Of 6VJ>1\ AliAA
 ■iOLIG CVP4UM ftUOfitd PttOv/mUD TU»t 1*0 OdTAlM 4-MO-
 RATING , ftLOOii M4AL.L He PUAtTftn«0 WITH VT LCAvT 1/1"
 CWP*UWi r*4 _A«-T»JC UoLLOVJ oil -«,<3l.i[> bL)|tN»D CLAV TILI2
 QH Cawli- \WfcTIOw aF TILB: ♦ COMillfrTfc Hollow c^muka bloc*.,
 paovioeo that to 0[%TA.ig A 4-Htt. EATIMG , IftLOCU* 6MAU- fcB
 PLA*TSfcSO With At LfiA4T '/l." <ivp*UKA PLAVTfffL THve^MH4i IU
 \W 1 MR 2V^' ru 2/4 'V* 1" iV ■a" tut »V.»" 1'4" 2U«.' x" I"! 1." 3" 3. He
 *V iW *V4" 1' V a" V H¥* + MB. ** lift" »V" 1*1 1* *V »" kAA6QNK.V
 FI REPROOF-IMG 514

Construction Details and Finishes PARTITIONS AND WALL FINISHES
 Column Flreproofing Details Uili DUKAtuB OK USUAL Cot., DIM. + tV-l
 ftaAiu a*, iquau «»»*U COV.UWU «■» Mot* •■ Ol>A»46loa ^B0»A PACt
 OP Column .tAAY »»« IUCH.lA.-t80 |p, M«D TO ALIaU .wnrt pac« op A-
 iomiria pMctnrioki PAttTmoH WitBTt COLUIAM GVP*UM fob. a mil,
 BOARD FIR.6PB.00F*W« — 1 E. AT IU C ADO 1 i'/2 MB. BATIUC I
 tk.V*tt> OF | HE. KATIk|£ 1. " " ApblTtaMAU 6WP4UM »0 LAVISH. Op
 ^TAHftABP) Me. «/a" EATIWfi Fimcooi CSVC'tUM, AoAflf ■4*t*-*1
 Vm'-«' IMI OF LATH M«T*k LATW pot Tvr» an. Prruti PLMT*t. *•■
 Tmcvcum RAT1IU 1 HE. 2MB.. 3HK. AHE. ' pLAHU. . TUICtt. Vfl" i"
 iHT «r UATM ».+1* PIAWV1P IAWH (A«TAL LATH . J?t A»AO*
 MPTA.L lp puuuue ID tAfrtH LATH U<T»i M.V.c. coot -*na. urriuci (aav
 »ft O»TAIH»0 VUITH I* VCKMIOIUT* pl _A*T«0- OU «.(>, LATH 4
 6ACK.KH-L. OP LOO*«. YWlAICUkl-ra Vfcg.M\CUI-IT& Og.
 GtVPSUKA Pfeg-LITlfc PHZ.E.PR,OOflMQ i!£t »Ma-li.*3iMa oimu «i
 out- vasv CHKCI4. LOCAL »U>G tDBB 4 VAAUUFACTUPJH*
 •*r*eir|ffA-(ioW ■6PK.AV&P OKf FHZE-PIZOOFIMa 515

Construction Details and Finishes FLOORS AND FLOOR FINISHES The
 designer must be familiar with the great variety of floor types, finishes, and
 patterns in order to specify and detail architectural flooring properly While
 some examples of "soft finishes" such as carpeting and resilient flooring are
 shown, this section explores in depth the installation and detailing of "hard"

or architectural finishes. It is important for the designer to research the various characteristics of the floor finish being specified. While aesthetics and color are obviously important considerations, the designer must also analyze other factors. Among these factors are wear resistance and durability, soil resistance, maintenance, resiliency, flammability, costs of installation, and life cycle cost. Once these factors have been analyzed, the final specification and detailing of the architectural finish must be developed. Examples of standard patterns are provided, but the designer must become familiar with the infinite number of pattern possibilities. The inherent limitations of materials control their sizes and thicknesses. The patterns of certain materials are dictated by both the thickness of the material and the weight or "dead load" of the material superimposed upon the structure. For example, a large pattern of marble or granite will necessitate a slab of material that will weigh much more per square foot than that of a smaller pattern. This greater weight might have structural consequences, as well as make floor transitions more significant. Transitions between flooring materials, particularly under doors or at entrances, and transitions between flooring and walls are some of the key material interfaces that have to be detailed. Again, this section provides such information using both traditional and contemporary approaches. Finally, a portion of this section is devoted to the detailing of raised computer room floors. While not traditionally a floor finish, raised computer room floors seem appropriate for this section. While generic architectural details are provided, the designer should always develop final details in conjunction with the manufacturer(s) being specified. 516

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Typical Characteristics of Floor Finishes DRAWING AND DESCRIPTION
 WEAR RESISTANCE SOIL RESISTANCE, CLEANING AND
 MAINTENANCE RESILIENCY REMARKS COST COMPARISON
 3/4 Inch hardened cement finish on concrete slab Cement base costs little, Poor;
 frequent Is too hard a floor to be cleaning needed; comfortable; Infrequently
 must be refinished Very hard used in classrooms, some- every ten years
 times used in corridors, shops and Inexpensive toilet rooms Installation cost
 maintenance and insurance cost for 20 years 3/4 Inch terrazzo finish, with 1/2
 inch cement underbed on a concrete slab Very good Very good; needs
 cleaning once a week with detergent and water Terrazzo base Is easy to

clean and sanitary, but not resilient and some- Very hard times noisy;
seldom used in classrooms, often used In corridors, vestibules, toilets and
shower rooms Ceramic mosaic tile, 1/2 inch setting bed on concrete slab i/s
Inch asphalt tile finish Installed In mastic on concrete slab Very good Poor,
usually needs replacing every ten years Very good Fair; must be cleaned
and waxed once a week> Very hard Fair Used in toilet rooms, showers,
food service areas, but seldom used In classrooms Low first cost; finish
requires careful maintenance i/b Inch linoleum finish installed in mastic on
concrete slab Good Fair; must be cleaned and waxed once a week
Serviceable; a sanitary floor for classrooms, corridors, assembly and
administration rooms Vs Inch cork tile floor installed In mastic on concrete
slab Fair; needs frequent cleaning and waxing Used primarily In libraries
and kindergartens; floor Very good Is subject to Indentations by chair, legs;
acoustically good Ve Inch rubber tile finish Installed In mastic on concrete
slab Good Fair; needs cleaning and waxing once a week Very good Subject
to slight Indentation by chair legs 1/4 Inch vinyl tile finish Installed In
mastic on concrete slab 25/32 inch maple strip flooring set In Va Inch hot
asphalt mastic on concrete slab 25/32 Inch maple finish; 1 by 4 Inch cypress
sub- floor laid diagonally; 2 by 6 Inch cypress sleepers, 12 Inches apart, set
In two Ve Inch layers of hot asphalt mastic Very good Very good Fair; needs
a weekly cleaning and waxing Good; requires monthly cleaning with
steelwool and a wax finish Good; requires a monthly cleaning with
steelwool and a wax finish; sand and re-finish every 2 years Very good
Subject to Indentation Steel angles necessary to cover expansion Joint; Fair
used In gymnasiums and playrooms; not suitable for damp areas or climates
Excellent A deluxe gymnasium floor 517

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Floor Construction Details F-1 2-«a'JO|STS-l6"O.C. »AV OAK
FLOORINO, SUB-FLOOR, METAL LATH. W OYPSUM PLASTER I W
HARD WOOD FLOOR, SUB-FLOOR, VtV WOOD JOISTS, WOOD
LATH, VI', PLASTER SAME AS F-2, EXCEPT W CIB6RBOARD
BETWEEN ROUGHK FINISH FLOOR W HARDWOOD FLOOR, SUB-
FLOOR, V4"<rNAIUN0 STRIPS (SEE NOTE) &-FIBERBOARD, SU8-
FLOOG, ■Z'<4' WOOD JOISTS, WOOD LATH, tV PLASTER.
NOTE.ROUOH (. FINISH ELOOt? NAILED TO NAILERS; NAILERS
NOT NAILED THROUOH FI9ERBOARD: MERELY RESTING

THEREON. SAME AS F-4, EXCEPT HZ' FIBERBOARD BETWEEN
 ROUGH & FINISH FLOOR F-5 W HARD WOOD FLOOR. SUB-FLOOR,
 V&V WDDD JOISTS-14"O.C. SUSPENDED CEILINGS'. 4" JOISTS
 -14"O.C. FIBERBOARD, H" PLASTER NOTE - CDMOM-END
 SUPPORT FOR BOTH JOIST SYSTEMS; NO INTERMEDIATE
 CONNECTIONS BETWEEN LATH (. "ZLTS. P-6
 IH>I>I<!WWW,fl.)Mi('OHIW|t F-12 T" . "f"rrL"ulw;>,i'l^i;By,.lTm
 fflSBJswfai '■l',Vt' " 'I P-IJ P-14 h%iii|nii?.i)*fiinl|i|r-'*.' '■.●.■':jitt F-15 F-
 16 MJrnr.iwiimi. SAME AS C-11, EXCEPT W%1' SLEEPERS PLUS
 BOUOH & FINISH FLOOR ADDED, TDP ."< SAME AS C-IJ, EXCEPT
 ^"FIBERBOARD PLACED UNDER SLEEPERS 8" COMBINATION
 FLOOR, k'i\Vt\ 5-CELL TILE, W TWO-COAT GYPSUM PLASTER
 CEILING i.;,'K':|j'i'i;^iJ.^nj;;.ii^V:iiV:!.,'i!)i;' ^A'^!?!^i'^:!'tf^')>W'i!^
 SAME AS F-14, EXCEPT 2'CINDER CONCRETE PILL PLUS r
 CEMENT TOPPING ADDED FOR FLOOR 4"COMBINATION FLOOR,
 i'i'r>n" 3-CELL TILE, *i",,2" CURRINO-14'QC. W FIBERBOARD LATH,
 ff GYPSUM PLASTER SAME AS P-14, EXCEPT 4V< 7"SLEEPERS
 PLUS ROUGH & FINISH FLOOR ADDED, TOP F-17 SAME AS P-S
 EXCEPT W FIBERBOARD DN PRIMARY SUB-FLOOR. . &"« V
 NAILERS <!EeT NOTE WITH F'4) PLUS RDUOH & FINISH FLOOR. F-
 IB SAME AS C-17. EXCEPT HZ' EISCRBDARO PLACED UNDER
 SLEEPERS F-6 V CONCRETE FILL. STEEL FLOOR. SECTION,
 SUSPENDED METAL LATH, W PLASTER m F-19 FINISH & ROUGH
 FLOOR ON *4'<2" SLEEPERS'14'O.C RESTING ON (V FIBERBOARD,
 4' COMBINATION FLOOR, 4".12'«12" 3-CELL TILE, SUSPENDED
 r«4'JOISTS, PLUS W FIBERBOARD LATH, W GYPSUM PLASTER !wr-
 l,s->i:i.-';.^WT' Mm ■ Mill ill mi in |li>i<w samb w p £xcept \\\I\I
 Hi'EMULSIFIED ASPHALT \ U U APPLIED BEFORE T V CONCRETE
 FILL. F-9 V;t||piMw8iiiMtr""*'-'^ ,vt.Xlll|ll|l|lill)|n|ji|ffllWlll P-20
 Wj.OAK FLOORING NAILED TO 2"«TSLEEPEHS-14"O.C, SLEEPERS
 GROUTED ON f COMBINATION FLOOR (SIMILAR TO F'19), H"i
 TWO' COAT GYPSUM PLASTER CEILING wmmmmTOWMffiMKMu
 F-lo BATTLESHIP LINOLEUM, 1W CONCRETE FILL ON HIGH-RIB
 METAL LATH, 8'MAC-MAR JOISTS, HIGH-RIB METAL LATH, 3
 COATS GYPSUM PLASTER 4'REINB CONCRETE SLAB, Mf« T
 FURRING, YT FIBERBOARD LATH, Yi- GYPSUM PLASTER 35 F-21
 SAME AS F-7o, EXCEPT 2'»2" SLEEPERS CARRIED ON RESILIENT

STEEL CLIPS, 1/2" HYDROCAL ON 5/8" GYPSUM PLASTERBOARD
 1/4" SLEEPERS-14" O.C. ON RESILIENT STEEL CLIPS, 4"
 COMBINATION FLOOR SIMILAR TO FIG. 1 TWO-COAT GYPSUM
 PLASTER CEILING P-22 518

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Floor Tile Patterns UDD D<X By groups of four squares as a unit
 separated by wider joints, the scale is increased. A diagonal pattern of
 square tiles is emphasized by a border. rTJbHriu npM M-ir-r Q dc; WD
 WtfB When the small squares are less than one-quarter of the area of the
 large squares, the pattern runs off at the side. When the small squares are
 one quarter of the area of the large squares, the pattern has more repose.
 Another way to increase the scale with small tiles. i^QQSQQE hbjhl aDDB
 A decorative pattern that can be made on the job. 01 main 1 1 ... By a few
 rows of broken joints, an effect produced in a field of square tile of border is
 es. E [J DDDCCO JZL LED By breaking joints in one course, the border is
 made wide. An arrangement adapted to large rooms. DDZG The simplest
 floor of square tiles is interesting if the joints are in scale. iHHri When
 square tiles are laid with broken joints, long lines in one direction are the
 result, When double squares are laid "basket pattern," the necessary
 allowance for joints adds interest. A good pattern for corridors. GHBH
 □QE□ Q^OQQQt mrED Varieties of "herringbone." [ffIEBS Two
 combinations suggesting plaids. H IS ii i ZIP □a m DD- inflmi i ED ED A
 simple device for a panel or a floor for a large room. 519

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Marble Floor Patterns and Details NOTE1 Size of marble tiles vary with
 design, if several varieties are used, the abrasive hardness (Ha) of each
 should be similar. (ASTM C241). MARBLE BORDERS MARBLE TILES
 MARBLE BORDERS 4s -»■ FEATURE STRIPS OR JOINTS MARBLE
 BORDERS TYPICAL MARBLE FLOORING DESIGNS 3/a"=r-o"
 MARBLE TILE ^■■:/Z?#L ■'4-. '* . i i —N * [1] MORTAR BED
 MARBLE TILE MORTAR BED. REINFORCING. MEMBRANE- ■SUB
 SLAB H [3] THIN SET MORTAR [5] MORTAR BED *-+ ■■WTZS&. H
 MARBLE TILE MORTAR BED -REINFORCING 4—MEMBRANE
 MARBLE TILE — ADHESIVE — STRUCTURALLY SOUND WOOD
 SUBFLOOR SUB SLAB 520 [2] MORTAR BED Fig. 1 Marble flooring

details. [4] ADHESIVE [6] ADHESIVE METHODS OF INSTALLATION
half size

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Marble Floor Patterns and Details ' ' MARBLE FLOOR TILE •J? Vt 21/2
MIN. ' ROUGH CONCRETE ' ,OR CINDER FILL OPTIONAL- TAR
PAPER, W.P. FELT, POLYETHYLENE FILM, OR THIN BED OF
CLEAN, WELL SCREENED SAND. •/!■ MARBLE FLOOR TIL.E 71
»>;• T /•/: ■/./■ ,--y..:/ ■•A/" .8,,1 f.h &-THIN-SET CEMENT '(T £f:.OR
MASTIC BED/.f ,>■ ■■!■:■. ■ ■ ■ ■ ■:■! . ■.■./■ ■/.. ' ■' • ■,'... i'.'4' :...: '■ ■
' ,>'! •*«'•■ SMOOTH CONC. OR >'!•' ' .■'!•. WOOD SUBFLOOR '...!',
, '? L' ••••! . '■ • :!• v- * •• I MIN. a. Standard floor tile, preferred method b.
Standard floor tile, thin-set method Fig. 2 Marble floor setting methods. \
f'jfiw Garble \iiy[<pii" ' CEMENT SETTING BED!' 1 'A /:.\V.:/ •' •' ■'•■•*
•': 6 I ^ .-: ■ ■ >subfloor ■ ■ :.j [Wf ■J: v .*"•" e. Thin marble tile,
preferred method \#"TrM MARBLE TILE .^Y'/Z ■■-i; ••.:;■ ■';v-
■ ■:;■ ■'. ■.:.■-< •V'-VWN-SET CEMENT,'.4: ■/./■:• or mastic bed •,•:,•.,
■/:' ■ '!.'su'bf'lo'or'?-'."-^ 5/» MIN. :-C-vr- d. Thin marble tile, thin-set
method a. Random rectangular Fig. 3 Flagging patterns. a. On sand bed b.
Coursed ipMimHlllWippipipilWllWBB1 * 1 ■i; ?fpr ■'•*/ '■'
hMiUfflnipiiijii^i; ^^^^^^^^^ b. On concrete slab on grade SIMS c. On
wood joist construction Fig. 4 Flagstone setting methods. c, Irregular or
mosaic Flagging 1 to 1'/j in. Sand bed 4 In, Flagging 3A to 1 in. Setting bed
1 to 1 Mi in. Reinforced concrete slab 4 in. Gravel or cinders 4 in. Flagging
3A to 1 in. Setting bed 1 to IVa in, Reinforced concrete slab 4 in. Wood
subfloor 3A in. 521

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Saddles/Floor Transitions t*f>kt T» MAT«M P**wa wi>th p*4*.tiMrr tit*
«••* an.- — •#M«, *-t tfHfcf\2"Z/ ^| ■»"• !>-* iDMI 1 snsgs ->».*ti K •' "
- & ti & " *■ ■'" •'. •>,' '»' ■ <> . *!' " *' D •':*■' dl»*"«l'.'* «*fcr«T. ft*/
M*1*?ft ~TM>>^ Ml tU.* smnprnmoaee: 522 ^p:r-y" •>';>- 14) JI'.M'1

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Saddles/Floor Translllons ZSfe**67t 1.1 Ml*, MAT .eAm>>>rf tew s#.flf**T
1*2*4/ 4 V«l'-#M THI* tM>><f 5 •• fc> 0 . I> Ai« , -rita fr-o, «*•' l'-«" 523

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Saddles/Floor Transitions Mt.-raa >"7/ a'V* •!-'* AkUMIHMM *A>>>M
m>>um. MWrltt «u*mhy tit.* Zlfc* A>JU«TA>L» ff*||>p|M« If/ £ *».'rL^
hi*. Tlft a 11M11 rri i fi 1111J i m i Mifu i il n ni ii t>nrm=tr r* a* •**•>•*
T7>;. •>.M.T V:'*\ *", a ,

Construction Details and Finishes FLOORS AND FLOOR FINISHES Door
Saddles ventOw2Jt : u (o- is a & &. Y'L) *« i>',J- ii" „**' ^*3*>***?
Pivot*, ciosem. S-2 / pux?rt W^ ■ i rf AM <J «••*», u (!>'.^i jasaatotooit -
leasnufMT nusc W"V/ yt?lo» S4 floofe. <:cosb*-PZ. rr 525

Construction Details and Finishes FLOORS AND FLOOR FINISHES Door
Saddles SADDt-C TO MATCH HZkWE WIDTH pbsturner.tub 2V2" A-
tMIHUM^- S-7 &#*>* _(?E5iypMT: 1ME. I Am VO0&
mii;i«>>uiiiiiiilmiiniiiiili'iiiL,i>iitfM.iiinlim tmttth -It 6 o- • £>' _ - -6 J. '
. " ". f "' ./i>' -', k '• S*m.\iO,[VTIH4 SrtW /J^ «5*f*PCT-
nuiMiiiiiiiuintmiiiiiiinuiiM'l'iilnifi unfit >;. ootnz ij 1] iMBaic««&e^?
rs^n^. gewt saddle iitmtitr;[ffitnMirtwirrrmti

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Saddles/Floor Transitions I t<nwt FtooR Isy w. f. ccrTf: 6/WF.«HH>> I 6
HIL K*V£T>llUht law -;V miH. rAo:6i>;i> w§ e^^fei 5rv?. H.e.1. t>'
IfWHIMUHI ,Sfe! \Mnrs\,B -C. u At?M>^>tviS — ,|Btntl.Ahllda.B K>T
ALiaU -i r- M/ii>-irsUa ,iswir>T«Mi««5>er.-B.iww«jaL m>>>
«<wntJUemwf:/stwJilt rilV•2^^(>'«l*rr.*MIJau« |^v-i«<:*i««->'rH;2)o"«.
(;u" ■AwlBatMia I—review m>i*N" -SuUstr-j Ji>UTIfW<*J* WW!*
M«WU AM4L81 rr/ie^-rwi* c>/u>->r«.T 527

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Saddlas/Flaot Transitions #• kW r wmw.* »f h>iiu> i*>ht«. vIMfi.
fU>>>>im, »H n**rto I — 3/t" WiwoiUw^w* —I tiiBnr'B.wtfc, WapM^vc- -
n.i. 4AOHM>lvft. IfS-Mf. *'HIT>- H«fAt- AH4UL '</>>> ■s-iiu-HW-TH-l.
|-CIMMIC TIL* MAiM /M^La JoTji'eWMMiiPTli.K, H> f «ATUI»M . -
THHH.il • TftAH*moi-l iivr;>vuiT«M>wu fv.e.T. tfs' caHTWI/txM
WHIT* MMVIL _A0HirerJ0 -'m' V.o.T. ccurejuaun <u>nv win. t-KSus or
HARtli couwoom 528

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Floor Finish Transition Details ctnpst oJ ftcantn - • 0»TPBf til* ^MXe&lv
^affife I ' • • • ©CAP-yffSf "HUO tc CAM3&T ^ cap-tp&t tiLeTtp s»jtffe.
its o&perr iM) pii, are se^ J fl ■" Cafcl-t:) rum* MMt >MO(-)t), MEOW ^
—' ?\ cBp-i»Mio nte to oj^f^r pOU S*Zf SOU*- HIUt 9ZS. SOJ.& *&&. 4-
\ G<»rTfU»'l TO SB&- ce^U/£\ JMbcVET. tr - -' ■'" U jiLs to ecsJc^e^ «!
■!>■-■ F rii!:iin.iii' 'i- '!.It rOPrTiirffTi Si > 'S^CBJhfiMie'tile* tv c*>nf&r
-pie -T^CEPTAMIOflL& ft3 \Z)nIyU -flUS/ 9tPkJg 1p cSthfrMlg' fig fuU.
tKE 3CW* -4- 529

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Floor Finish IVansitlon Details wuimihiitewmi}-*^ L .0*fW ■*t i -VW<I>
111* Lfm — 1/2-FvS. e «FT Tue e J^mhub |C>ftM6 Til*
A>f«R<*pi(po«fW + Hcwep (or. swau^ MpoJfcs) OPF*T tW«ww-W**ih
aw*, <jObhit 'gN -Ster^ fftg* TBMg.|T.M^, cPt/c*^C. 530

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Edgings; Tile/Carpel Joiners; Reducer Strips . . Vinyl Snap Down Divider |
—y*"—) my,"wide removable section ~T^W for joining carpets || Vinyl
Snap Down Divider r—1* *7~H ■ Wwlde removable section jJTf—." ""."
»). for joining carpets HH Vinyl Tile Carpet Joiner ■ Provides smooth
carpet/tile transition Vinyl 3/16"Glue Down Reducer ■ Undercut, flange,
and transition combined ■ Undercut 3/16,"smooth transition ,-\ ^ >A >s\.*>
—85 JX Vi." {T II Vinyl Corner Guard ■ Protects corners and columns
from bumps, abrasion, wear Vinyl Fillet Strip ■ Superb backing for flashing
up the wall ■ Black only |»-r->| L—!%»—J 5-TV U—w—A *!* U I—
1%"—J L—1V(«—J Vinyl Snap Down Edging ■ V'wide removable
section for joining carpet to tile Vinyl Snap Down Edging ■ Y/," wide
removable section for joining carpet to tile Vinyl Single Flange Track Vinyl
5/16" Glue Down Reducer ■ V." butting gauge undercut ■ Long flange,
smooth taper Vinyl Custom Edge ■ For sponge backs ■ General purpose
edge and cap T 1 P ir U—1"—»J Vinyl Carpet Cove Cap ■ Trim for coved
carpet ■ Undercut 3/16"(flexible) Vinyl 9/32"Cap ■ For capping coved
carpets, ceramics, and paneling Vinyl Tackless Carpet Bar ■ V<" butting
edge to accept- tackless strip ■ Deep undercut Vinyl Tackless Carpet Bar
Reducer ■ 1/4" butting edge to accept tackless strip I—'—I r—i'—\ h'—I
Vinyl Underslung Reducer ■ Binder-bar type edging for resilient flooring ■

Undercut .105" for use with 1/16" to 1/8" material Vinyl 1/4" Glue Down Reducer ■ 1/4" butting gauge undercut ■ Long flange, smooth taper Vinyl 1/16" Tile Reducer ■ Beveled edge for resilient flooring material ■ 1/16" (063") butting gauge Vinyl 3/32" Tile Reducer ■ Beveled edge for resilient flooring material ■ 3/32" (094") butting gauge Vinyl 1/8" Tile Reducer ■ Beveled edge for resilient flooring material ■ 1/8" (125") butting gauge P Vinyl 1/4" Square Cove Cap ■ Undercut 1/4" ■ Cap for coved carpets and wall paneling in Vinyl 1/8" Square Cove Cap ■ Undercut 1/8" ■ Cap for coved sheet vinyl 531

Construction Details and Finishes FLOORS AND FLOOR FINISHES Base Details MIUH) ifllllliir'^! .Siu-^l cavm.Jt»*a- •*B*.-rti« f-uood 1? -*i - .VBKTIOM. Ml. L.fkOAfcp* JfTWJSWT VIHVI.I>A4« y+MAiioVHtoo \tatmc. pjrwtno. -fjuuu-s .• »':WU:>»HI«tHI|H BAS£ ..TYPES- 532

Construction Details and Finishes FLOORS AND FLOOR FINISHES Baseboards ELISABETH COIT Plaster. woodcap moulding ~ Wood ease. (Cowtes Lewis Xouse, Jarmington, Conn) lineat'masonry support Plasterline , Metal ground—» iM'Plaslic Magnesia base arid ftoormg- Cement. 1/4 Sase coal^ . Concrete \ sub-floor PlasUr wall— Wood shims. Mrdal strip * ' C/xumtid) JAMES EPPENSTEIN ftud line mad rraU covering Plaster-Wood base. C £e Moyne Jiousc.) Itiishuiglon. County Perwst/lvanta. PtaJterr (Doneyjiause.) patented metal ~mzshai\$ton County ^ c/,M»f Pennsylvania • ■ - Piaster Wnll Wood skims— ± Aielal } dividing—i- i Cpiueiiltid) \ Line of \ masonry § support 1 Terrazzo 1 baseand 1 floor- jm) 4 AYMAR EMEiURY U Wood wainscot - and Capmotilii. \ Marble- base ? uush wood floor <i%M& LAWRENCE J. KLEIN n Line or roug. wall Wood paneling-. Wood base- /- 7uush wood floor■ j£ \$*£&"->-**!. N.CHESTER SORENSON Paujh wall — Plaster 2ogauge metal dipt Wood base ~ Jtrush modftoor^ (Xmgslcu Mouse) piaster or -fewteven,&nn. oftiemaU Plaster- PAUL OERHARDTJR. Wood, base (Aiuseum of fine v Arts, Poston) j Wood wainscot. cmdCap moulding. Marble base finish wood flooring 533

Construction Details and Finishes FLOORS AND FLOOR FINISHES Stone Floor Patterns and Details cmtna> CSSrcit r\gtOie FLOOniUG PATTEftU PETAL pr. Picon »r twu 4»"fi. ■yikft-dt COLOm *at a*ec. ou

out A-^p A ■ VSHD« AVBH WU6HBD C ■ AMOM OL*On OVHJITE
m.lSM«0 loppt elev. shaft BXISt. BLBV -■Mist extnjDED ALUM. 9ILU
Ft**CIA TN StOUe PET. <3 ELEV POOD SILL -W-^j-, 1 UODDY H'-
StONB PLOOMINO OW MO«T*M SBtTlh* WO ' FULL HilSWT V*300
FKWBL9 OM LOBBnr iMfcu. Aide sm IOO0T »-HVAt10N» 3fl«ie#
T14t*r TYFTStOUe FLOOniMG PET Z\ ® LOPPY PAU=L WALL 2»»»«i'-
o« StUN6 FL00MIM5 ou nmr*r> SEfT" (WO ME STO "IN.CLOOM
C*WI°HT pflwtwc e**#* 03Nt WV.M AtJGUS OIVIOSM 6TII» .'»'1*>
SBUhED T" SLA» A6 nfiffl. ^STOWE. f=LCOr?IUG DgT.e EkrtTJY
DCO^S ifitta».';ioi =—' ' 5TOWB ("LCWniUS OM MbtAn 3atriMO o*o-\
6eA eit^Mt erac. ,«r- . -r-fp Durt JOIUf b\1~YP 5tOUE fr'Jft JOIMT
PEtALL rfwo^a^i-o* 534

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Terrazzo Floor Construction Details MASONRY OR STEEL. 'spinc-funx.
CSuog. ou col.) LOCATION A* EETCHMINEP »•* AUCH/E-Uq.
AT>PROKIMATt.1-Y OH CKJJ.TE.'R— VATilE-S ^FmE-R. EXPANSION
MATER.IAU* fULL -De_pyn expansion joint COMPACTE.13
qnA«uL»ariLL 1,Y OfHt-TiS SAW CUT I I 4 SEAL. A NT
.pUASTICTJIVIPe-B. 3TB-IP wim*. Mesh 2x2« I4> qAuce. TYTICAL.
G-0*\-V-0\ IOINf "E.U5XIC- T/E.'K.'e.AZ.Z^O ~ STTaXJOTURAL.
SySJE^M, O VE^2_ q-E.A N LI LAW. "f I l_ L_ MAiONR-V O-H.
STEE.L. 'otr.ucxu'o.e. feux;. or pou) LOCATION AS BE.TE.'KtMniEJD
T>¥ ■ATyzU/E-HfZ, APPU4PXIMATfc.L.V IB |=T. ON CH-WTE.*>
VAP.IES vriB't-'B-'ElKpAN&ION MAT>TI.IAI_ fuI_1_ XIElpTH
E^XPAMSIOW JO|MT 5TKJUCTUtt.AU eOKJC by othetas PLASTIC
XMVIDETi STTiIp TYPICAL, CONTROL JOINT £jJSJ|C JE.-R-
R.AZIT.O-'SOM'DE-D SYSJE-M WITH 5E.TTIN^ BSD ~ OVER ST12-
UCfU12-Ai- CONCE£.J£ ^ MA3QNRS OTL SfJEEL. STU-UCfU-RX.
<TjLT>q, OR COL.) LOCATIOM AS ■DE.TfB.MINED T>Y
rA1LCH./E.WQ, AT-pH-OXIMATE-LY \ IB fT' OH CfCnTCR. ■'A1ME.O
.COWC. t>Y TtUSMIO INSTALL. fR. act spe.cs>. •PTLOTE.CTION
tiOARO BY OTUES.S TXVSS- SLAH t>Y OTHMIS - FIDE.TL.
EXPANSION MATE.TJ. AL. - fuI_1_ TjETpJH OpANSIOH JOIMT s*w
CH1 IJL^^lant -puAOTJC divide stp.it> WI-RJ2. ME.SH Z«2<IS QAUCE.
3 Mil-. ■Polye.THE.i.e.ne. THICKNE-5S VAPJE.S WATK.T4.PR.OOF
mLUBPAHC fcY OTHtXS TVPICAL. COMT,U_Ol_ JOIMT ■Rustic TE.-

ra_Ti-Az.z.o Ove-tj. Sla-bs With We-Ateb 5?ac! ■BE.L.OV <i. OVEL-p.
IMSULATE-TJ ^W'ATE.Tc'P'ROOrE-'D 5l_A£>3 535

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Terrazzo Floor Construction Details V WTRE MESH 2» x i" I 16 ga or
equal ISOLATION MEMBRANE 1/16" or DUSTINC PF SANE, "^
Divider Strip ^ 3 ' . ' !, '... ' " . ..", ' . /■ i •'CONCRETE SLAB " ■ •.*.
SAND CUSHION TERRAZZO Standard Divider 3trln Control Joint Two
"L" Strips Back to Back N !«— Neoprene Filler Between Two Standard 1&
Gauge Metal Strips .(Mfeced Baok To Back. e TERRAZZO TOPPINCi
IJNDEBBEE m&m % OF BONE, Divider Strip 7 @mwm % CONCRETE
SLAB i" 1-3/4" Touwaa col.t> jo Monolithic Terrazzo OlfWHON
MKTUIIM. Divider Strip RUSTIC TERRAZZO TOPPINGS
UNDBRBED^-i POINT OF BOND / . 'ConcVete, Slijb. ByiOthers • i ^ _H
1-3/ v RUSTIC TERRAZZO Bonded to Concrete e- lifvtdti- Strln bJSIKF
"tTTl ' i ' • : =7 H= •ittia rill ft otmn ?mg^ BONDED TO CONCRETE
w^mM mQfo R^\$ l> nmr« b Kll Pdlyithylint ^ijwr lirrlir By ' ■' " =*
iliilSj';' "«»' VS "L" Type Strip yfe^i TERRAZZO BONDING ^^S^K^
AGENT . «Concrete"Slab By Others : ^ : BONDED TO CONCRETE i"
xmviPe-p. ft-phUT— i^S^1 tSe^^ 31 ISOLATION MLMliUM<t, 1
^=SS==^J^^^^. e RUSTIC TERRAZZO Q POLISHED TERRAZZO FUU
J»E ^S FULL SIZE STRUCTURAL TERRAZZO SYSTEMS Terrazzo
Over Wood 536

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Terrazzo Floor Construction Details 4 Divider Strip . ' ■ •.,""■;.. " ' . "
Underb-ed Minimum Thickness 2dt , 2x2 X 16 ga. \V.Wlre Mesh
4.\180latlon Membrane r 1/16" or Dusting of Sand SAND CUSHION
TERRA220 OVER PRECAST CONCRETE TYPE DECK PIVIOCTt.
fflT«.H> XSMMMtt <fe Jol»*r m^wmv&im trffiegeS |+j_ . i V/ V EPOXY,
POLYESTER, TERRA220 FLOOR & BASE TERRA220 OVER
CORRUGATED METAL TYPE FLOOR PIVIPE.13. aTP-IP TWUA.WO
^^i^^^^ij^l^ag 3-r aisj-wats. Fig.5 TERRA220 OVER CELL TYPE
FLOOR %0¥?mmmm j.,. . POLYACRYLATE TERRA220 FLOOR &
BASE u Slab Control joint VF1 *- . ■;■ i:4 GF Isolation Joint FT Tsra
Construction Joint V •■: Pig-5 Angle or "L" strips Single Angle Strip Two
Strips positioned Back to Back Directly over saw cuts 537

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Terrazzo Base Details SLOCI WA>UL- vTVPt, H, SclltLP PRECAST
 TERRAZZO BASE POURED TERRAZZO BASE Sheet Rock
 IK1STAUV.E.X5 'BY OTHERS —" T-E-vaB.A.xz;o Flush Tvp^ U_
 FINISH ■ Vail finish Metal channel The Terrazzo base hetp.ht should be 6
 Inches or more to use this detail. SHADOW TYPE H TERRAZZO BASE u
 —^ 2 layers ► pp*--^ . Sheet Rock Metal Base Bead Flush Type 3/8"
 Terrazzo Underbed Galv, Metal Lath anchored to steel studs R-1" or IV'
 ;w^!j« S t ., •■■,, > -. XtA5 _'t£S= _ FLUSH OTa. T=«.OJtlCTIHG iTYW
 H.V-jy==. , E.Tt_ T=L A.X?=-Q RE-VEAL TYPE Note: Provide Dimension
 in Space Indicated "Varies" Sheet Rock ■ S-puAsY Type. Sheet Rock
 INSTALUfOBV OTHERS ■.•.Vj'.:.;^ "pTa-OJccTiNca Type. VAPOR,
 BAB.B.IE.*?. VERTICAL TERRAZZO UM-DLVK.tbV.O 7C SCOWCH
 COfcT V O T Ht V-.S 3CA»LC 3* - I- O" ■;uTJ:o,ir<a C!!/SWKL15 - i 7."
 o c, £-.7 UR, DIAMOND ML.SH \- /VT M •a.fc.-s'c 538 SOLID
 PARTITIONS 5c A.ut a* - i' o"

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Terrazzo Stair Tread Details PRE-CAST TERRAZZO STAIRS ASR-
 ftSIVE. IMS CO, NOTE; Abrasive inserts should be positioned 1/16 inch
 higher than Terrazzo surface ■ S" STE.tl _-PU.KTE-/ CTtlUUeD 1
 TWPC.O, WCLHE.TJ -TO "atlMFOECINQ I>A.T-S Self Supported Tread
 ■ g-MAT<a.>x 4 rmi5H -PUASXCW. ■ £> _ ,/>>■-> TEXTURED MOSAIC
 Epoxy, Polyester or Polyacrylate 539

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Ceramic Tile, Terrazzo, and Brick ttty^r.'?'.!;" tikv^irVj 5-lEZS il Xr~T .4 .
 ∴ &OMD£D TEFIRMZO V/t'T*Hn*-Z-ZQ TorPIHO --ois/iDimi •»-rRii=
 (viTt(pfteren ■ CfIR*,/Mc TIIJft CK4\VN*r /AOhTAP. SfcTTIHCr IWH.D
 c&HAi^lC,.TIUfe TftT IN .MQRT^R - OIVIPIH0 ^Tnjp(viTH. PHQPHR.
 ■ ^a'TERRAIXO TOPPING'" - AOftTWt ISrftt). : LIGHT
 tUUMJSOfFcCIMfi #v*JM4 ■ TAP* PAP PA K UAPpBP t* ■ */4* R.SD
 .OF BJ.HO ■ v. 1 ■ ' t it n i n rpiTifninTf rrrnmrrj « .0 '■ ' ■'(C*LR.*v>MC
 TIUMI .CERAMIC . TILS. IN PR.y-SEi.T MQFVTAR, 5.A.ND-
 CUSHION TEHBA7ZO PIVIOIMQ, ftTO-IP ■ ITFIUCTUR^L *»).Aft
 DIVIDER. «TR-|PB Tp »«. DRAVH 4 trHCIfItfD OH *|H±-L WMKIHck
 DR>WWtt» £0VJC «U.A.fa CtPA/MC T't-*. *TT«-UCTUF*.*KU «LM<b

C&RA.MIC TiLfc "5E.T IN APMfe.'StlVe *—— 6AHI"*T fcHIHIftiJ \ r—
<SAC-.P«T .H.kMmmV"l I' lllllllllll I A' 7*3, -A I CAP-.PK-T b^i^m
pxvams

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Marble, Resilient Tile, Slate, Wood, and Quarry Tile / /-

w/MmMmM/'M+w r . <• ■ n . ■ . ■ « ■'..*; < " . . . ■ , '■.' '- .'. *■ ■ ■ - -A
.'.. eTPHfe ROOR.IH& R\$*il'*nt 1 * ...I^ FLOORING WTOHU 1M*L -
T*.R. f*F»»- , Vi *■ ***VT POWKTHVL1HM riU*. OB^ «■«*! k«p OP
Ct»**\ VH4.U 4C*K«.U*D a*Mo • 9TBuTuR*,L *L>-* TU«- CI*«T A*
PBW. , ■ — ■atRUCTUf^A-L «L**R> ■.;- "<^H 7/ WOOD FLOORING r-
fa' rtyvaoD «u»'*rt.ooaUHii liPACVD Ifc* O.C« C»fv«MT ftLL
•ftTNUC'rimM. *1_*vffa iiiiiMiiuiiihu.tiiiiuyiJiiiiuii.ii' / *- aum .: ;'.*•". ./ J,
;,-.---;'■ .4 .. -I- ~* QUA.RRV .TIWfc —iVORTKR IfemO
-♦TRvJCTUraAu mv-A-b QUA.R.P.T TIU.& MORTAR BED METHOD
SLATE FLOORING, NATURAL CLEFT FACE AND BOTTOM -,
GROUTED JOINTS, 2 PARTS FINE SILICA SAND T01 PART CEMENT
1" MORTAR BED, 3 PARTS SAND TO 1 PART CEMENT SLATE
FLOORING, NATURAL CLEFT r-FACE AND BOTTOM WATERPROOF
MEMBRANE OR CONCRETE BOARD ON PLYWOOD SUBFLOOR
THINSET METHOD SLATE FLOORING, NATURAL CLEFT FACE
AND GAUGED BOTTOM -i ACRYLIC LATEX GROUT. CAN BE
TIGHT JOINT WHEN SLATE IS 1/4" ACRYLIC LATEX MORTAR BED
APPLIED WITH NOTCHED TROWEL, DEPTH NOT LESS THAN
HALF THE THICKNESS OF SLATE (MIN, Va"), PLYWOOD OR
CONCRETE BOARD-TO-PLYWOOD SUBFLOOR ■ CONCRETE SUB
SLATE FLOORING, NATURAL CLEFT FACE AND GAUGED
BOTTOM 541

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Wood Floor Construction Details WOOD FLOOR OVER PRE-CAST
OYPSUM CEMENT FLOOR OVER PRE-CAST GYPSUM BASE AND
CEILINO - WOOD JOISTS BASE AND CEILING — WOOD JOISTS
TILE FLOOR ON PLAT-TOP SOUND-CONTROL WITU STAGGERED
AND BEVELLED WOOD JOISTS JOISTS AND SUE-FLOORING
SOUND-CONTROL WITH INSULATION BLANKET SOUND-
CONTROL WITU STAOOEKED BETWEEN ROUGH AND FINISHED

FLOORING JOISTS AND INSULATION BLANKET PARTITION OVER PARTITION PARTITION NOT OVER PART- PARTITION OVER PARTITION PARALLEL WITH JOISTS ITION-ON DOUBLE JOISTS AT RIGHT ANGLES TO JOISTS 542

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Wood Strip Floor Construction Details Perhaps the most widely used pattern is a $\frac{1}{2}$ - by $2\frac{1}{4}$ -in strip flooring. These strips are laid lengthwise in a room and normally at right angles to the floor joists. Some type of a subfloor of diagonal boards or plywood is normally used under the finish floor. Strip flooring of this type is tongued-and-grooved and end-matched (Fig. 6). Strips are random length and may vary from 2 to 16 ft or more. End-matched strip flooring in $2\frac{1}{2}$ -in thickness is generally hollow backed (Fig. &4). The face is slightly wider than the bottom so that tight joints result when flooring is laid. The tongue fits tightly into the groove to prevent movement and floor "squeaks." All of these details are designed to provide beautiful finished floors that require a minimum of maintenance. Another matched pattern may be obtained in $\frac{1}{2}$ - by 2-in size (Fig. 66). This is commonly used for remodeling work or when subfloor is edge-blocked or thick enough to provide very little deflection under loads. Square-edged strip flooring (Fig. 60) might also be used occasionally. It is usually $\frac{1}{2}$ by 2 inches in size and is laid up over a substantial subfloor. Facenailing is required for this type. Fig. 7 Application of strip flooring: A, general application; B, starting strip. S43 Fig. 6 Types of strip flooring: A, side- and end-matched - $\frac{1}{2}$ -in; B, thin flooring strips - matched; C, thin flooring strips - square-edged.

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Wood Parquet Floor Patterns IP! 11 ill m MM II 544 PRODUCT
DESCRIPTION AND PATTERN STANDARD Pallorn Unfinished—paper-
faced STANDARD Pallorn Unfinished—Web Beck or Mesh-Back
STANDARD Pallorn Unfinished-WebBeck {For Industrial Use}
STANDARD Pallorn Unfinished WebBeck (For Industrial Use)
STANDARD Pattern Factory-Finished and Unfinished (Available In various
colors) STANDARD Pallorn Factory-Finished Foam-Back Tila ANTIQUE
TEXTURED (Factory-Finished and Unfinished)—Kartsawn Various colors
available ANTIQUE TEXTURED [Factory-Finished and Unfinished)—

Wire brushed Various colors available MONTICELLO Pattern Unfinished
—Paper-Faced— Pre-Finished. Mesh-Back HADDON HALL Pattern
Unfinished—Paper-Faced- Pre-Finished. Mesh-Back HERRINGBONE
Pattern Unfinished—Paper-Faced SAXONY Pattern Unfinished—Paper-
Faced CANTERBURY Pattern Unfinished—Paper-Faced Pro-Finished,
Mesh-Back RHOMBS Pattern Unfinished—Paper-Faced BASKET
WEAVE Pattern Unfinished—Paper-Faced ITALIAN S DOMINO Pattern
Unfinished - Paper-Faced 'PANEL SIZE 5/16" x 19" X 19" 16 equal
alternating squares 5/16"x12"x12" 4 equal alternating squares
5/16"x19"x19" 16 equal alternating squares 5/16"x11"x11" 4 equal
alternating squares 5/16" x 19" x 19" 16 equal alternating squares 9/16" x
19"x19" 16 equal alternating squares 11/16"x11"x11" 4 equal alternating
squares 11/16"x12"x12" 4 equal alternating squares 5/16"x6-
11/32"x6-11/32" 5/16" x 2" x 12" (slats) 5/16"x 6" X 6" 5/16" x 6ft" x 6ft"
5/16"x6"/, "x6" Individual unit 5/16"x6"x6" Individual units .., %" foam,
2 lb. density 5/16"x6"x6" Individual squares 5/16"x6"x6" 5J" Individual
squares 5/16"x6"x6" Individual squares 5/16"x 11" x 11" 4 equal
alternating squares 5/16"x6"x6" S/16x 6"x6" individual squares 5/16
x 6" x 6 Individual squares used with 5/16" x Vx 8" pickets 5/16"x 13"
Cx 13U" 4 equal alternating squares 5/16x 13!Tx 13J4" [Factory Finished)
5/16"x 14!<"x WA" 5/16"x 13!Cx 13K" 4 equal squares ' 5/16" x 13!<"x13X"
[Factory Finished) 5/16"x2"x12" individual slats 5/16" x 14X"x tail"
[Approximate overall) 2 - "V" shape courses wide end 11 slats long 5/16"x
19"x 19" 4 equal squares on diagonal and B equal half squares 5/16' x 13!Cx
13K" 4 equal alternating squares with diagonal center slats 5/16" x 13J!<x
13!4" Hexagonal Shape 5/16 x 15J!<"x 1514" 12 equal Rhomboids 5/16 x
15-1/5X 19" 4 runs of 3 slats and 5 slats alternating 5/16"x 19" x 19 400
equal size pieces butt-jointed Select » Better Rustic Select * Better Rustic
Select 4 Better (Par 4 Better) Rustic Select 4 Better (Par J Better) Select
Rustic A Better Rustic Select S Better (Par 4 Better) Select & Better (Rustic
S Better) Select & Better Natural & Better Freeform Rustic Cabin & Better
Cabin Natural & Better Cabin & Better Cabin Select Natural & Better
Select 4 Better (Par 5 Better) Rustic Fireside Natural & Better Cabin Select
4 Better (Per 4 Better) Rustic Natural & Better Select & Better [Par 4
Better) Rustic Natural 4 Better Select 4 Better [Par 4 Better) Select 4 Better
[Par 4 Better) Select 4 Better [Par 4 Better) Natural 4 Better Select 4 Better
(Par 4 Better) Select 4 Better (Par 4 Better) Select & Better (Par 4 Better)

SPECIES Cherry, Maple, Red Oak, White Oak, Cedar, Pecan, Walnut, Rhodean Teak, Angellque (Gulane Teak) Red Oak, White Oak, Pecan Pecan Red end White Oak Maple, Red Oak White Oak Pecan Red Oak. Maple White Oak Pecan, Maple White Oak, Ash Oak, Pecan Maple Red Oak S White Oak Red Oak S White Oak Oak Angellque (Guiana Teak) Red Oak. White Oak Black Walnut Ash, Maple Angelique (Guiana Teak) Red Oak. Willie Oak Black walnut Angellque (Guiana Teak) Red Oak. White Oak Black Walnut Angellque [Guiana Teak) Red Oak. White Oak Angelique (Guiana Teak) Red Oak. White Oak Black Walnut Red Oak S White Oak Angellque (Gulene Teak) Black walnut Angelique (Guiana Teak) Red Oak. White Oak Black walnut Black Walnut Angellque (Guiana Teak) Maple. Red Oak White Oak

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Wood Floor Patterns Pattern #1 Same length of any size I l l' l l i l' i i i l i
Pattern #6 Same length of any size llllll Mi'l'i Pattern #11
36"and18"lengths Pattern #16 18" lengths Pattern #21 36" and 27" lengths I
I I I :I | T I I ■ - I I I I I- Pattern #2 Same length of any size I I Pattern #7
Same length of any size Pattern #12 Same length of any size H I H I I H I H
H I H I 11 111 H I H I H I H I h -II H I H I h - H Pattern #17 36" and 9"
lengths ~ — — — I I T — = — — I II — . _ I I Pattern #22 27" and 9"
lengths I I I I I I I I I I I I I I I I I I I Pattern #3 Same length of any size 1 1
1 1 1 1 1 1 1 1 1'1 1 1 I I ■ • 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 Pattern #8 Same
length of any size llllll llllll llllll MINN llllll llllll Pattern #13 36"and 9"
lengths H H H f—I H h efee: e E 33 HHE I—II—II—II—I \— Pattern #18
9" lengths Pattern #23 Same length of any size J1!1] Tll± tltl tJ-tJ- jiji TiTi
jiji tlIt Pattern #4 Same length of any size Pattern #9 Same length of any
size 1-1^1-r- in Pattern #14 9" lengths HH^\HH HHHHH HHHHH
HHHHH HHHHH Pattern #19 36" and 9" lengths A. ^\ ...-E:. I Pattern #24
36" and 9" lengths I.' 1' 1' 1' 1' 1' 1 ---!;i;iil|--- i - !l'I'Jv 'T'i'iVi'il ■ Pattern
#5 Same length of any size Pattern #10 36"and 18" lengths HI HI HI m in
mi in in in in in in mi Pattern #15 18" and 9" lengths L — ~:i — T" Nf I I
Pattern #20 18"and 9" lengths he n n 31 Pattern #26 36" and 9" lengths
Pattern #27 18" and 9" lengths Pattern #28 36" and 9" lengths Pattern #29
9" lengths Pattern #31 27" and 9" lengths Pattern #25 36" and 9" length ' s
— S4S

Construction Details and Finishes FLOORS AND FLOOR FINISHES.
 Wood Parquet and Plank Floor Patterns S/16"X 13-1 M*X 13-1/4" S/ie-
 x16-1/4*x18-1/8* SlatLength5-1/2" 5Wx19"x19" Wxn-xir 3/4"x3",5",47"
 3/4"x3",5",47" 3/4'thk* design tamed with6"x6"blocksand 2-V4'x14-
 1M"pfckets 3/4'x 2-1/4"single slat. Lengihs: 6-3/4", 9*. 11-1/4", 13-1/2*,
 15-3/4", 18" 3/4"x3*singtos>at. Lengths:6",9",12",15",1B" 3/4"x15"x15"
 3/4"x16"x16" Units 3/4* x 3", 5". & 7" 546

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Wood on Concrete Slab Floor Construction Details CONCRETE
 ASK4T04-CEMENT HAIINO CONCRETE WOOD FLOOR OVER
 CONCRETE WITH UNDERLAYER OF NAILINO CONCRETE WOOD
 FLOOR OVER CONCRETE FINISHED FLOOR DIRECTLY ON
 SLEEPERS SET IN MASTIC CEMENT i NAILED TO CONCRETE . i • 1
 \i ?* '•■* •■'*• G> CINDER, ' CINDER-CONCRi OR OTHER DAMP
 POSITION CONCRETE 2Ai£ WOOD FLOOR OVER CONCRETE
 WITH SUB-BASE OF SLEEPERS t SLEEPER FILL WOOD BLOCK
 FLOOR OVER CONCRETE WOOD FLOOR OVER CONCRETE IN
 MASTIC CORIWIL6 FLOOR\ CEMENTED AND BLIND* NAILED
 •XINOER-CONCRETE PILE DEEP ENOUGH TO COVER CONDUITS
 MIXTURE OC ONE PART TO FOUR "S. RT4 5ANO- >< 4URCACE
 MU4T ^ BE LEVEL. CONCR CORR TILE FLOOR OVER CONCRETE
 WOOD FLOOR APPLIED OVEJ?; EARTW 547

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Floor Construction Sound Insulation Sound insulation between an upper
 floor and the ceiling of a lower floor not only involves resistance of
 airborne sounds but also that of impact noises. Thus, impact noise control
 must be considered as well as the STC value. Impact noise is caused by an
 object striking or sliding along a wall or floor surface, such as by dropped
 objects, footsteps, or moving furniture. It may also be caused by the
 vibration of a dishwasher, bathtub, food-disposal apparatus, or other
 equipment. In all instances, the floor is set into vibration by the impact or
 contact and sou nd is radiated from both sides of the floor. A method of
 measuring impact noise has been developed and is commonly expressed as
 the impact noise ratings fINR). The greater the positive value of the INR,
 the more resistant is the floor to impact noise transfer. For example, an INR

of -2 is better than one of -17, and one of +5 INR is a further improvement in resistance to impact noise transfer. Figure 8 shows STC and approximate INR(db) values for several types of floor constructions. Figure 8A perhaps a minimum floor assembly with tongued-and-grooved floor and 5/8-in gypsum board ceiling, has an STC value of 30 and an approximate INR value of -18. This is improved somewhat by the construction shown in Fig. 8B, and still further by the combination of materials in Fig. 8C. The value of isolating the ceiling joists from a gypsum lath and plaster ceiling by means of spring clips is illustrated in Fig. 8D. An STC value of 52 and an approximate INR value of -2 result. Foam-rubber padding and carpeting improve both the STC and the INR values. The STC value increases from 31 to 45 and the approximate INR from -17 to +5 (Fig. 8E and F). This can likely be further improved by using an isolated ceiling finish with spring clips. The use of sound-deadening board and a lamination of gypsum board for the ceiling would also improve resistance to sound transfer. An economical construction similar to (but an improvement over) Fig. 8F, with a STC value of 48 and an approximate INR of +18, consists of the following: (a) a pad and carpet over 5/8-in tongued-and-grooved plywood underlayment, (b) 3-in fiberglass insulating batts between joists, (c) resilient channels spaced 24 in apart, across the bottom of the joists, and (d) 5/8-in gypsum board screwed to the bottom of the channels and finished with taped joints. The use of separate floor joists with staggered ceiling joists below provides reasonable values but adds a good deal to construction costs. Separate joists with insulation between and a soundboard between sub-floor and finish provide an STC rating of 53 and an approximate INR value of -3. Sound absorption Design of the "quiet" house can incorporate another system of sound insulation, namely, sound absorption. Sound-

| DETAIL DESCRIPTION | STC RATING | APPROX. INR |
|-------------------------------|------------|-------------|
| 16" 2x8- FLOOR | 30 | -18 |
| 7/8" T. a G. FLOORING CEILING | 31 | -17 |
| 5/8" GYPSUM BOARD | 45 | +5 |
| 3" SUB FLOOR | 48 | +18 |
| 3/4" FINISH FLOOR CEILING | 52 | -2 |
| 3" V FIBERBOARD | 53 | -3 |
| 2" 2x8 FLOOR | 54 | -4 |
| 3" 2x8 FLOOR | 55 | -5 |
| 3" 2x8 FLOOR | 56 | -6 |
| 3" 2x8 FLOOR | 57 | -7 |
| 3" 2x8 FLOOR | 58 | -8 |
| 3" 2x8 FLOOR | 59 | -9 |
| 3" 2x8 FLOOR | 60 | -10 |
| 3" 2x8 FLOOR | 61 | -11 |
| 3" 2x8 FLOOR | 62 | -12 |
| 3" 2x8 FLOOR | 63 | -13 |
| 3" 2x8 FLOOR | 64 | -14 |
| 3" 2x8 FLOOR | 65 | -15 |
| 3" 2x8 FLOOR | 66 | -16 |
| 3" 2x8 FLOOR | 67 | -17 |
| 3" 2x8 FLOOR | 68 | -18 |
| 3" 2x8 FLOOR | 69 | -19 |
| 3" 2x8 FLOOR | 70 | -20 |
| 3" 2x8 FLOOR | 71 | -21 |
| 3" 2x8 FLOOR | 72 | -22 |
| 3" 2x8 FLOOR | 73 | -23 |
| 3" 2x8 FLOOR | 74 | -24 |
| 3" 2x8 FLOOR | 75 | -25 |
| 3" 2x8 FLOOR | 76 | -26 |
| 3" 2x8 FLOOR | 77 | -27 |
| 3" 2x8 FLOOR | 78 | -28 |
| 3" 2x8 FLOOR | 79 | -29 |
| 3" 2x8 FLOOR | 80 | -30 |
| 3" 2x8 FLOOR | 81 | -31 |
| 3" 2x8 FLOOR | 82 | -32 |
| 3" 2x8 FLOOR | 83 | -33 |
| 3" 2x8 FLOOR | 84 | -34 |
| 3" 2x8 FLOOR | 85 | -35 |
| 3" 2x8 FLOOR | 86 | -36 |
| 3" 2x8 FLOOR | 87 | -37 |
| 3" 2x8 FLOOR | 88 | -38 |
| 3" 2x8 FLOOR | 89 | -39 |
| 3" 2x8 FLOOR | 90 | -40 |
| 3" 2x8 FLOOR | 91 | -41 |
| 3" 2x8 FLOOR | 92 | -42 |
| 3" 2x8 FLOOR | 93 | -43 |
| 3" 2x8 FLOOR | 94 | -44 |
| 3" 2x8 FLOOR | 95 | -45 |
| 3" 2x8 FLOOR | 96 | -46 |
| 3" 2x8 FLOOR | 97 | -47 |
| 3" 2x8 FLOOR | 98 | -48 |
| 3" 2x8 FLOOR | 99 | -49 |
| 3" 2x8 FLOOR | 100 | -50 |

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Floor Construction Sound Insulation batts between joists, (c) resilient channels spaced 24 in apart, across the bottom of the joists, and (d) 5/8-in gypsum board screwed to the bottom of the channels and finished with taped joints. The use of separate floor joists with staggered ceiling joists below provides reasonable values but adds a good deal to construction costs. Separate joists with insulation between and a soundboard between sub-floor and finish provide an STC rating of 53 and an approximate INR value of -3. Sound absorption Design of the "quiet" house can incorporate another system of sound insulation, namely, sound absorption. Sound-

absorbing materials can minimize the amount of noise by stopping the reflection of sound back into a room. Sound-absorbing materials do not necessarily have resistance to airborne sounds. Perhaps the most commonly used sound-absorbing materials is acoustic tile. Wood fiber or similar materials are used in the manufacture of the tile, which is usually processed to provide some fire resistance and designed with numerous tiny sound traps on the tile surfaces. These may consist of tiny drilled or punched holes, fissured surfaces, or a combination of both. Acoustic tile is most often used in the ceiling and areas where it is not subjected to excessive mechanical damage, such as above a wall wainscoting. It is normally manufactured in sizes from 12 by 12 to 12 by 48 in. Thicknesses vary from $\frac{1}{4}$ in. to $\frac{1}{2}$ in., and the tile is usually factory finished ready for application. Paint or other finishes which fill or cover the tiny holes or fissures for trapping sound will greatly reduce its efficiency. Acoustic tile may be applied by a number of methods — to existing ceilings or any smooth surface with a mastic adhesive designed specifically for this purpose, or to furring strips nailed to the underside of the ceiling joists. Nailing or stapling tile is the normal application method in this system. It is also used with a mechanical suspension system involving small "H," "Z," or "T" members. Manufacturers' recommendations should be followed in application and finishing.

ESTIMATED VALUES DETAIL DESCRIPTION STC RATING APPROX. INR 2x10 FLOOR * $\frac{1}{4}$ " SUBFLOOR (BUILDING PAPER) VV FINISH FLOOR CEILING GYPSUM LATH AND SPRING CLIPS WGYPGUM PLASTER 82 2x10- FLOOR $\frac{1}{2}$ " PLYWOOD SUBFLOOR $\frac{1}{2}$ " PLYWOOD UNDERLAYMENT $\frac{1}{4}$ " VINYL-ASBESTOS TILE CEILING $\frac{1}{2}$ " GYPSUM WALLBOARD -17 \ iiliniV ' ' 1 . ' T i i ' ' I i I I ^MMit I ' i'i' ' I B 2x10 FLOOR $\frac{1}{4}$ " PLYWOOD SUBFLOOR $\frac{1}{2}$ " PLYWOOD UNDERLAYMENT FOAM RUBBER PAD $\frac{1}{4}$ " NYLON CARPET CEILING $\frac{1}{2}$ " GYPSUM WALLBOARD 40

Fig. 9 Relative Impact and sound transfer In floor-ceiling combinations (2- by 10-in joists).

549

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Floor Construction Sound Insulation conventional wood floor joist systems for sound control FLOOR SYSTEM FLOOR NUMBER FLOOR COVERING conventional $\frac{1}{4}$ " vinyl asbestos tile on $\frac{1}{4}$ " plywood underlayment CARPET & PAD ' PLYWOOD SUBFLOOR 0>WffMt=p<^^

1/8" vinyl sheet on 1/2" plywood underlayment -H4 8 JOISTS 16' OC y CLASS FIBER FLOOR NO. 3 The basic construction is illustrated by floor No. 3 although floors 4 and 5 have 2"x10" joists and 1/2" subfloor. Except in floor No. 1, the ceiling is fire-resistive type gypsum board applied with screws to resilient channels 24" o.c. Standard carpet is 44-ounce (sq. yd.) drop point over 40-ounce hair pad. Carpet and pad directly over subfloor 1/2" oak strip floor over subfloor Carpet and pad added to No. 4 conventional With Floated Floor Over Wood block (s-h") laminated to underlayment 7 Carpet and pad 1/2" UNDERLAYMENT, 1/2" SOUND BOARD v V SUBFLOOR v 8 Vinyl flooring laminated to underlayment applied over sound board with 4-inch circular globs of glue Vinyl covering like 8 with sleepers glued between sound board and underlayment FLOOR NO. 8 The basic construction is illustrated. Sound deadening board (15-18 p.c.f.) is laid over a 1/2" plywood subfloor, with or without stapling, and W T&G underlayment grade plywood glued over the sound board. The ceiling is 1/2" fire-resistive type gypsum board on resilient channels; absorptive material is 3-inch thick glass fiber batts 10 Oak strip flooring (V) nailed to 2x3 sleepers glued over sound board strips 1/2" glass fiber between sleepers 11 Vinyl flooring (0.07") on W T&G plywood underlayment glued to 2x2 sleepers glued to subfloor 16" o.c. Sand fill over subfloor to depth of sleepers (1/2"). Balance as in basic construction conventional With lightweight Concrete or Gypsum Cement Added 12 FLOOR NO. 11. r si'JJjJi'i. ■ 1 VLT WT CONCRETE, 1/2" SUBFLOOR, 3" MINERAL WOOL 13 14 \, \ 2« 10 JOISTS 16" 0 C ■ *** GYPSUM BOARD ON R C The basic construction is illustrated by floor No. 14, The floor topping is 1/4" thick cellular (foamed) concrete (100 p.c.f.). Ceilings are fire-resistive type gypsum board on resilient channels, 24 inches o.c. Absorptive material is 3" thick mineral wool batts. Floor coverings for impact tests are 44-ounce carpet over 40-ounce hair pad or vinyl floor covering, approximately 0.07 inches thick. Note variations from basic construction drawn in plans 12-16. 15 16 Ceiling nailed to joists; no absorptive material; with carpet and pad Ceiling nailed to joists; 3" glass fiber with carpet and pad Basic construction—(no floor covering) with carpet and pad Add W sound board between concrete and subfloor with vinyl tile with carpet and pad Basic construction—but with V thick gypsum concrete in place of 1/4" thick cellular concrete; W gypsum ceiling without floor covering 550 The improved resistance to airborne sound transmission gained by isolating the

ceiling with resilient channels and adding absorptive material is evident by comparing floors 2 to 5 with No. 1. A 10-point increase in STC reduces the loudness of transmitted noise by one-half. Improved resistance to impact noise transmission is gained by adding carpet and pad as is evident by comparing floor No. 3 with No. 2 or floor No. 5 with 4. An IIC of 51 is often recommended as an acceptable level of impact insulation.

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Resilient Base Details Standard toe base No-toe base: Adds a decorative touch to carpeted interiors. Butt toe base: Engineered to butt precisely to W floor coverings. Long toe base: For special applications. Features a long toe extending 1" to cover wide irregularities between floor and wall. TABLE 1 Cove Base Specifications Type Sizes available Standard toe base No-toe base Butt toe base Long toe base 2 1/2" 4" 6" 2 1/2" 4" 6" 4" 6" 4" Length: 48" ^ ^ Dimensionally stable. Won't shrink. Sealing lip ensures light fit. - * » > \ 5/8" Standard Toe 8 asc Ribbed back for long-lasting adhesion. No-Toe Base Butt Long Toe Base 5/8" Toe 8 asa 1" TABLE 2 Corner Specifications Type Inside/outside Underlap outside No-toe outside Long toe outside Length of return 2W 3" (with underlap) 2V 2V Sizes available 2W 4" 6" 4" 2y2" 4» 4" oc Oulsids ^ NTOC No Toe LTOC Outside Long Toe Outside ULOC Underlap Outside 551

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Ceramic Tile Patterns Unglazed quarry tile Ceramic glazed tile 6" X 6" 6" hexagon 4" X 4" □ W X 2W □ 1" X 1" 2%" X 6" 3"X8" 8"X6" 8"X8" 8" elongated 8" X10" hexagon hexagon Valencia Thickness from Vi\ aYH' Fig. 10 Ceramic tile shapes. 2" hexagon 2" X 2" 2"X1" □ 1%" X 1%" 4" X 4" 4V4" octagon 6" X 4W' Hexagon Valencia 6" X 6" Thickness from V4" tn ' // I II J XT^7 A/yA & X X y X X £^ Fig. 11 Ceramic tile patterns. 552

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Ceramic Tile Patterns SMALL CIRCLE fO" DIAMETER 8arp«ntlna Clroln {8 Ft. Module) Ouanlity lor 100 sq. leol: 3,2 pes circle, 70 pes small wedge; 256 pes medium wedge. 600 pes large" wedge Meandering Serpentine Quantity lor 100 sq (eel 3 pes circle; 58 pes smell wedge, 220 pes medium wedge; 610 pes large wedge 8arpantlne Clrolas {8 Ft. Module) Ouanlity lor 100 sq. feet 5.7 pes. circle; 126 pes small wedge; AGO pes medium wedge:

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Construction Details and Finishes FLOORS AND FLOOR FINISHES

Ceramic Tile Patterns Due to the complexity of mounting 1" Hexagon border pattern corners which require a number of special If a Hexagon border is required, you must provide a plan of sheets on smaller jobs, a

premium charge Is made. To the area with dimensions because the Hexagon avoid this, It Is suggested that on smaller jobs the configuration precludes interchanging sheets. We will border be formed using 1" squares with a hexagon field. provide specific sheets for those areas and setting plans. Note that on three of these border patterns a full sheet Is used for the corner. Some designs, however, will require a half sheet for the corner as shown in SB-1404. In this case a right and left corner will be on one sheet and the sheet is cut in half before placement. Fig. 16 1" hex border patterns. All patterns master-set 12" x 24" sheets. 557

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Ceramic Tile Patterns Single sheet repeat pattern, Repeat for overall pattern. Three sheet repeat pattern. Three different sheets complete the pattern, then repeat throughout. Fig. 17 1" hex overall patterns. All patterns master-set 12" x 24" sheets. 558

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Construction Details and Finishes FLOORS AND FLOOR FINISHES Bailc
Quarry Tile Patterns Qr-62It SPIRAL Shown: VAT x 8" |«W, 8" K 8"
(33'/i%) Qr-6222 1NTP.RLOCKING SriRAL Shown: 3Wx 8" (100%)
AlsoUsc:o"x|2'/'»" Qr-6223 BLOCK RANDOM Shown: 3'/'«" x 3'/'«" (15%).
YV y R" (41%) B'mi'Hn) ...:. -I- "-.-.' ..- . _ . — — .":... _™ — I- — ■ — -
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L. ii": . QM22-) VERTICAL IIKRRINGDONI: Shown: J%"" 8" (100%)
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(i225IMNW|IHt:i. Shown: l'/'»"v 3W(II.I%), 3*' ■ fV'IHHAM.) Also Ust: 6"
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Ql'-6227 1)01'111 l: IILRRINGIIONF. Shown: 3!,i""^ 8" (1001.) AI-.ii Llsr:
fi" - I2"«" QP-622R Hl.OCK RANDOM VARIAI ION Shown: 3W'x 37."
(11,2%), 3™" * H" (•)•.)-%). B"x8"(-H..|'*.> Ql'-(>22'l RANDOM
SQUARE Shown: ViT • l'."(|0' ,N). 8" H1 (HIV .•%> Qrr>230 DIAMOND
IIUKUINCDONF; Shown: 3/'«" « 3v4- (33%'W. V.V * 8" (GMVM
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. _ " — QP-6231 RAMBLING SQUARES Shown; 3'/'«"x 37»" H0%). 3%"
■* 8" (60%) Al.soUsc:6"*6'.'6"x12H" Ql'-6212 DIAGONAL MRil'l Shown:
3:/«"> B" CilVX.). ff • 8"(Sin.i 560

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Floor Conilrucllon Detalli: Ceramic Tile on Wood and Concrete Subfloort
FLOORS Cement Mortar Wood Subfloor Organic Adhesive STAIRS
Cement Mortar CEQAMIO TILE <frt>UD C&tX MtfUTAA &ED,,
M0MIUAL 11/4." |2.EIMPK>nciMcr CLE/WASB MEM&I2AUE

euorL<=>a -CBQAMIC-TILE -ADHESIVE - COUDLE V<5CD riCOQ. '
 MI I 111 111 llll.lin1, .1 ■ >->>>>>>y>>:>5?i5^ ^ mmmmmmmiam
 Recommended use Recommended use ■ over all wood floors that are
 structurally ■ over wood floors exposed to residential sound traffic only
 Wood Subfloor Epoxy Mortar and Grout -CEQ.AMIG TILE -EFfcPXY
 OrtZCJUT ICRS^Y MC3QTAQ 1p£JUD CC3AT JCtAP IO&TVVEEW
 "IPLVW^^O SHEETS -OOUOL.Z W030 flooQ Recommended uses ■
 over wood floors where resistance to foot traffic in better residential,
 normal commercial, and light institutional use is desired with thin-set
 construction ■ where water, chemical, and stain resistance is desired ■ for
 tilework exposed to prolonged high temperatures, use high temperature,
 chemical resistant epoxy mortar, and grout Glass Mesh Mortar Units Dry-
 Set Mortar or Latex-Portland Cement -CERAMIC TILE DQY- SET "O.
 LATEX- >5tad eopo .toCTAD VOUD CCAT OLAO& MESH MORTAD
 UNIT Recommended uses ■ over structurally sound plywood where
 lightweight construction is a factor ■ where water resistance is desired ■
 eliminates necessity of recessing sub- floor to accommodate portland
 cement mortar bed C6I2AMICTILE — VOUV COXT CONCRETE
 STAIRS CEI2AMIC TILE BOUD CCW MOI2TAC2 |>?&D
 iaEIUR3I2clM.it- METAL STAIRS IIIIIIIIIIEfrK CE.I2AMIC- TIL&
 ■scbatch cottr ■ METAL LATH WOOD STAIRS CEILINGS, SOFFITS
 »4"-1V MOQTAO DKD OVi ©uePEUDCP M&TAL LATH 4 SCRATCH
 COAT, COMCaETE. °0 (3LASS MEetl M^CTAQ. tUUIT Recommended
 uses ■ over a mortar bed ■ over glass mesh mortar units ■ over clean,
 sound, dimensionally stable concrete ■ over metal lath attached directly to
 the bottom of wood joists ortrusses; spacing not to exceed 16" on center
 561

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Ceramic Tile on Concrete Slab Floor Construction Details CONCRETE
 SUBFLOOR Cement Mortar Cleavage Membrane Cement Mortar, Bonded
 Dry-Set Mortar or Latex-Portland Cement Mortar CEDKWICTIUE
 R>£>MP aoa MOTzrAQ eep,, U9MIWAL P/411 aaufOQGMGr
 CLBAVAOrE MEMeQAME iiiiiiuiiiiiiiiiiiiiiii'irn "a o'* Recommended
 use . ■ over structural floors subject to bending and deflection
 Requirements ■ reinforcing mesh mandatory ■ motor bed thickness to be
 uniform, nominal VA" thick Cement Mortar Epoxy or Furan Grout

CeiiAMIC TILE ER3KY °0. FUI2MJ aQoW E*?kID CCW MCOTAO
tJED Ut?MIUAL 11/4" U£.\UfOOCAUOt PltkVAO,! MCMEGAkie
iiitnnnnnnnnnnnnTtwr .5/ Recommended uses ■ with tile set by Method Fill
requiring good stain resistance and resistance to erosion caused by
occasional contact with mild chemicals such as found in commercial dining
areas, photographic dark rooms, public toilets, public foyers, etc. ■ for use
with quarry tile and paver tile CSQAMIC TILE - fsauo coat JMOI2.TAI2
&ED IUOMIM^L I'4" Recommended uses ■ on slab-otvgr.ade construction
where no bending stresses occur ■ on properly cured structural slabs where
deflection does not exceed 1/360 of span ■ on properly cured structural
slabs of limited area Dry-Set Mortar, Epoxy or Furan Grout CEQAMIC
TILE PR3VY *& pJBAOJ OTOU7 PGY-SCT "12 WTC*.- rK>t2.TLM-
4DCEMEMT M02.TAI2 tCKXO COKC TTtnT 11 iiiim|iiiiimiiiiiiiiiriiiimm
<3 Recommended uses ■ with tile set by Method F112 or Method F113
requiring good stain resistance and resistance to erosion caused by
occasional contact with mild chemicals such as found in commercial dining
areas, photographic dark rooms, public toilets, public foyers, etc. ■ for use
with quarry tile and paver trie obo.ha\c tile oar- set °a LATEY-ftSGTLAUD
CEMEUT MOQTAQ R*2UD CCW Iiiiiiiiiiiiiiiiiii IIMIIIIIII'I'1 A* <3
Recommended uses ■ on plane, clean concrete ■ on slab-on-grade
construction where no bending stresses occur Organic Adhesive or Epoxy
Adhesive - GEI2AMIC TILE ■ADHESIVE Recommended use ■ for use
over concrete floors in residential construction only; for heavier service
select Method F113 562 WATERPROOF MEMBRANE Cement Mortar
Bed ceEAMIC TILE 6CIJO caxr MOI2TMJ &ED WCMIUAL I'//
GEIUfc>t2C|U<5r vATEBpiaoor MEMIbCAUE COUGQtlt "O VOOO
©U^FLCOQ iitiiiiiiiiiii iiniiiiiiiiiiiiiBmi Recommended use ■ wherever a
waterproof interior floor is required in conjunction with ceramic tile
installed on a portland cement mortar bed Thin-Set CECA.MIC TILE DQV-
SET "Q LATEX.-FtoBTLMJ D CEMEkJT M£>Q.TAQ. WS>UD C<SAT
\VATBQFQi23F MCMbEAUE: SHEET "E LIQUID APPLIED
MEWbCAME te?UOe<*,T &£tJC&ETE "G CUI2ED M£>QTM2 OEP
IIIIIIIIIIIIIIHIMilHIIIIIIIIIIir ' <5 4' Recommended use ■ wherever a
waterproof interior floor is required in conjunction with ceramic tile
installed in a thin-set method

Construction Details and Finishes FLOORS AND FLOOR FINISHES
 Ceramic Tile on Concrete Slab Floor Construction Details CERAMIC TILE
 erpxy aaoui ERPXY MOQTAQ Ct^UD CCAT LLM1II lillililHIIMlllllll.1 £
 Recommended uses ■ for setting and grouting ceramic mosaics, quarry tile,
 and paver tile ■ where moderate chemical exposure and severe cleaning
 methods are used, such as in commercial kitchens, dairies, breweries, food
 processing plants, etc. ■ for tilework exposed to prolonged high
 temperatures, use high-temperature, chemical-resistant epoxy mortar and
 grout EXPANSION JOINTS Vertical and Horizontal CEQAMIC TILE
 EF^yy adorn MOI2TA.O VZO HCMIUA.L 114" niMiiiiiiiiiiiiiiwiiiiiii
 '^'^a'O.'A Recommended uses ■ where leveling of subfloor is required ■
 for setting and grouting ceramic mosaics, quarry tile, and paver tile ■ where
 moderate chemical exposure and severe cleaning methods are used, such as
 in commerical kitchens, dairies, breweries, food processing plants, etc. ■
 for tilework exposed to prolonged high temperatures, use high-temperature
 chemical resistant epoxy mortar and grout J W THICK. fi&GI£IW«-
 IH<cuee Tiue J ACID QESISTAUT "laaollT and rA?QTAI3; W THICli
 ACID GESISTAWr MEMDCAUE M^t2TAD K>£D °Q c&ucntTZ fill IMP
 AiLniiiiiiiiimiiiiiiiiiii immioi ■••,'■■:;; f^f. . . ■ , ;■;• ■•.■.<■. V. ■'•'" ■ . *
 a t .»'•• A ' .■' "■." ■.. ■ '<3 Recommended use ■ for setting W thick packing
 house tile in areas of continuous or severe chemical exposure where special
 protection against leakage or damage to concrete subfloor is required
 CQAMIGTILe PUQAKl £*r20UT QESIU MOG.TAQ
 iiii finifiiiuiiiiiiriirii.i,iJT i,.iiiiinw.iirmT <\$■ Recommended uses ■ for
 setting and grouting quarry tile and paver tile ■ in kitchens, chemical
 plants, etc. CEQAMIC TILE OEiuFoacep MOETAI3 foED CLEAVAGE =a
 WAT ED PHCO r MEMfc>|2AWE COUCI2ETE °I2 WOOD ©EALAUT
 C£MYQS.<XA*»LX. tACIC-UF '////////Amtem. SEALAMT DEPTH 1/2
 WIDTH -CEQAMIC TILC ■ M0I2.TAI2 foED ■ CiPMCrIETEi ■
 eEALAUT 4 KACli- U P CERAMIC THE fcCUD C£>AT G2U2t2ETE "E
 MASOUI2Y 6EALAWT TTTfrlTTTTTI b£>UD DQEAUEI2 TAPE
 B*CI£-UP 5T0UCTP0AL JSIUT D" ■'. \V|TM HO fSOUD T" SEALANT
 ICiPMpaESSI&LE l&ACIMJP ividth 4 times 1 expected Imovement ••■"
 Use these details for control, contraction, and isolation joints 563

Construction Details and Finishes.' FLOORS AND FLOOR FINISHES
 Raited Computer Room Floors pKrVHUfflH. .^^Ti^J-'pgj: pHp-f>H^

w*»«? ^re;^f»*^ fte*<w* , , /T\ TVH-»P n~«K>i» guw? pp=pti,,«?
gami/ftsijutfr ^\ rA**pp r^'^^ f*^*" •^g'g'fi'^ ^ftgn^J-pq-.^ffV^iUfrTg^
i>*Ht »Tv« ■vWfO s^?6VHkTHr-j rvrgcAMU "fi \$y*;Ly rVn*=»
&fu0*&%, p,*?* £>'!*& ?g>tU *-|L* 564

Construction Details and Finishes FLOORS AND FLOOR FINISHES
Raised Computer Room Floors CLOSURE PLATE ~ I.T55?, I— LOCK
NUT r '4-20 SCREW (SELF-TAPPIN6) j>eoestal "coupling c^ORID
CHANNELS 12-24 F.H. SCREW NUT 8 LOCK WASHER ^-PEDESTAL
BASE NCHOR r ELEVATION FLOOR PROJECTION WITH CLOSURE
PLATE AND NOSING PANEL AND GRIOWORK CUT TO FIT IN
FIELD -COVE BASE as %'y.>'\\:.T.;V'''.'06Nc.'FLboR''', *'i:v:'■'■. ►,':!';
'■>'"." ELEVATED FLOOR" AGAINST MASONRY WALL PANELS OF
STEEL HONEYCOMB AND STEEL SANDWICH CONSTRUCTION
TILE FINISH — STUD PANEL HONEYCOMB TYPICAL ELEVATED
FLOOR i;' ;■. •.'.'■'• t' ' ' ,>'! ' PANEL AND QRIDWORK CUT TO FIT
IN FIELD A 4 rm r^% rfh ipr^\\ fffl ELEVATED FLOOR-A
CONTINUATION OF EXISTING •FLOOR ANGLE CLIP PARTITION
EDGE OF ELEVATED FLOOR AIR GRILLE 565

Construction Details and Finishes DOORS Hollow Metal Door

Construction The design, specification, and detailing of a door can have serious consequences for functional considerations such as accessibility and sound transmission. The door is also one of the most important architectural elements with respect to design image and aesthetics. A door can be a major part of design expression: a monumental door to a church or synagogue, the main entrance to a residence, the doors to a corporate board room - all of these doors have symbolic importance. Doors come in a variety of standard heights, widths, and thicknesses, yet they may also be custom designed, assume a variety of shapes and forms, and be constructed with a variety of materials. The design, specification, and detailing of a door is, in fact, a rather complex task. A door is typically set within a frame or jamb, but may also be installed within a wall without a frame or jamb. The frame/jamb interface between door and wall partition is another area requiring special attention by the designer The design of a door is never complete without the specification of hardware. Hinges, locksets, closers, stops, and thresholds are but a few of the hardware elements that a designer must consider. This

section on doors provides the designer with extensive information on door types, materials, doorframes, and materials and methods of door construction and installation. Details show doors and frames installed in all types of walls and partitions, including wood and metal stud, masonry, concrete, and glass. Of special interest to the designer are examples of less standard door types such as elevator doors, sliding pocket doors, and fabric-covered doors. The majority of the details in this section are taken from the actual working drawings of successfully executed projects.

PANEL CONSTRUCTION There are two basic types of panel construction: Steel stiffened: Face sheets supported by steel stiffeners, which are channels, Z-shaped sections, hat-shaped sections, or similar members, positioned vertically. Sheets are attached to these members by spot welding. Laminated core: Sandwich construction employing a core of impregnated kraft paper honeycomb, plastic foam, or structural mineral blocking, to which the steel face sheets are laminated, using a structural adhesive.

stiffeners Insulation face sheet **IMPREGNATED HONEYCOMB STEEL STIFFENERS** **FOAM LAMINATED CONSTRUCTIONS** **MINERAL BLOCKING CHANNEL**, ' *W-i - spot weld ■ **ZEE OFFSET CHANNEL** m> ■" | - arc weld at ends **I HAT SECTIONS** **HAT SECTIONS 566** **REPRESENTATIVE STIFFENED. SECTIONS** Other sections used by some manufacturers

Types of Construction The four basic types of construction for hollow metal swing doors are illustrated and identified in Fig. 1. The type usually specified in commercial work is the continuously welded edge seam construction, Type A, and it is this type which is the basis of NAAMM Standard HM MA 861. Most custom hollow metal doors are of the full flush type with continuously welded edges (Type A). When glazed openings, recessed panels, or louvers are to be provided, they are built into the door during fabrication, rather than being cut out of a flush panel door by field modification. Fire-rated doors may differ in certain details of construction; see NAAMM Standard HMMA850, Fire-Rated Hollow Metal Doors and Frames.

Construction Details and Finishes **DOORS** Hollow Metal Door Construction **D FULL FLUSH WITH CONTINUOUSLY WELDED EDGE SEAMS (NAAMM STANDARD HMMA 861)** **FULL FLUSH WITH UNFILLED EDGE SEAMS (NAAMM STANDARD HMMA 860)** **FLUSH STILE AND RAIL RECESSED PANEL(S) (MANY**

VARIATIONS) Fig. 1 The top edge of Types A and B doors may have only an Inverted channel (standard construction) or may have an additional closing channel. Types C and D have tubular rails and stiles, with no edge seams. S = stile (hinge stile Is stile at edge where hinges or pivots are located; lock stile Is stile In which a lock or latch Is Installed; and meeting stile Is stile adjacent to another door, In a pair of doors). TR = top rail. CR = center rail. BR = bottom rail. P = panel. P/G = panel or glass. 567

Construction Details and Finishes DOORS Hollow Metal Door Types
 nominal door width (frame opening) actual door width I * \ S hinge ' cutouts
 varies as specified (Vt" maximum for labeled doors) 4- - floor surface
 DIMENSIONS AND HINGE LOCATIONS Hinge locations shown
 represent the Industry standard, but may be altered to suit requirements.
 MOST COMMON SIZES FOR 1V4-INCH THICK DOORS' Width of
 Opening 2'0" 2'4" .2'6ff 2'8" 3'0" 3'4" 3'6ff 3'8" 4'0" 6'8" 6'8ff 6'8" 6'8ff 6'8"
 6'8ff 6'8l" 6'8l" 6'q" TO" TO" TO" TO" TO" TO" TO" TO" TO" Height of
 Opening 7'2" 7'2" 7'2" 7'2" 7'2" 7'2" 7'2" 7'2" 7'2" 7'10" 7'10" T10" 7'10"
 7'1<r 7'10" 7'10" 7'10" 7'10" 8'0" 8'0" 8'0" 8'0" 8'0" 8'0" 8'0" 8'0" 8'0" 'Sizes
 shown are for single doors only; for pairs of doors, use twice the width
 Indicated. OTHER DOOR SIZES: The sizes listed are those most
 commonly used, but custom hollow metal doors are available In any width,
 height and thlpkness desired. It Is not uncommon to supply them In widths
 of 5'or more and/or heights of 10'or more. Standard doors, on the other
 hand, are generally available from Inventory only In the most commonly
 used sizes. LISTING DESIGNATION: Always preface the door listing with
 "SGL" or "PR," followed by the designation of the opening size. For
 example, a single flush door for a A'O" x 8'0ff frame opening Is listed SGL
 4080F, and a pair of flush doors for an 8'0" x 8'0ffframe opening Is listed as
 PR 8080F. REPRESENTATIVE DOOR DESIGNS 1. i 03 FV4 FNLV
 FNV2 FMV FRV FDV 3 FOV FCV I ■ I »i FQ2L FQ3L 1Q 1G3 1Q6 L1 _
 1 — u u L2 FDG1 FL FV FVL FNV FNV L FQ FQL NOTE: Some
 manufacturers may use differing designations for some designs if 1 FQ4
 568

TYPICAL HARDWARE PREPARATION Construction Details and
 Finishes DOORS Hollow Metal Door Hardware NOTE: CUTOUTS
 AND/OR REINFORCEMENTS OF SIMILAR NATURE ARE

PROVIDED FOR ALL OTHER HARDWARE ITEMS SUCH AS FLUSH BOLTS, SURFACE-MOUNTED CLOSERS, FIRE EXIT HARDWARE, PULLS, ETC. 569

DOOR SCHEDULE 10 11 12 13 Column numbers are for 14-«— reference here only Opening Number IO1 102 103 104 105 106 107 ioa IO<9 I \0 III •Us DOOR Type F 1G RSL FG FV F FL F - F<&L F e metric i Mat'l. HM AL WD HM HM HM HM WD* — HM HM Jrirts if dt Nominal Size* No. i 2 1 1 3 2 1 1 - 1 1 aired; Width 3-0 &-o 3-0 3-0 4-0 7-0 3-0 2-iO - 3-0 3-0 1 inch = Height v-o s-o 7-0 7-0 7-0 7-0 7-0 7-0 — 7-0 7-0 25.4 mn Thkns. 1*A 13/4 114 134 134 !34 l« l% — l^ I* i, 1 foot Silt Detail -^7 ^7 25"l7 ^7 *Xl TM/n 7Vn *y* — ^,7 *y* = 0.305 Louver W 23 23 Tl. H 12. 20 Glass - TEMP y+TEMP 4*v«R£ ^"TtMP - — — 'A-'temp — Spec"! . Detail — **At — — — — *Vr7 — — 3Ym FRAME Type I 2 i I 5 2 3 1 1 4 1 Mat'l. HM AL HM HM HM HM HM HM HM HM HM HM Sections Jamb Va V17 Va '/.7 M7 /■7 '/I7 V17 5/,7 VX7 8/i7 Head V.7 6/I7 v.? % 2/7 '/I7 '/I7 Vi7 ^i7 '/.7 S/I7 Sill — — — — — - — - — 3*1^7 — Fire Rating — — — c — A — — — — — Hard- Ware Set i a 4 & I 5 7 4 — 1 2 Remarks Conhn .d\om"tnovn-ttnn?sVtfld Mullions ^7 Transom bar '% Plastic fcard door 1 Cased opening Side ligntmullion "717 Sound re+ardant" Opening Number Number all openings individually, with the numbering system reflecting floor numbers if practicable. Door Type Use alphabetical designation for types, as shown on elevation views on facing page. Elevations should show door configurations and all features such as louvers, vision lights, etc. Do not use one elevation with dash lines to indicate variations. Door Material Designate material from which door is made: HM = hollow metal; AL = aluminum; WD = wood. * indicates special facing as noted in Remarks column. Type of core construction should be stated in the specifications. Nominal Size List number of doors per framed opening, plus width, height and thickness of door. State head and jamb clearances in specifications, using Hollow Metal Manufacturers Association recommended standards unless special conditions require otherwise. a Sill Detail Reference siii detail, which shows siii clearance, threshold if any, and any special condition. Reference number shows detail number first, followed by sheet number. 6. Louver Note width and height (in inches) of louver panel. Louver types may be either specified or shown in detail drawings. 7. Glass Note thickness and type of glass to be used in glazed opening. 8. Special

Detail Reference detail(s) showing special features such as astragal (on pair), dutch door shelf, flush transom panel or other. 9. Frame Type Use numerical designation for type, as shown on elevation views on facing page. 10. Frame Material Designate material from which frame is made, using same symbols as for door materials. 11. Frame Sections Reference details, showing frame sections at head and jamb, and details of such members as transom bars, mullions and other special features. 12. Fire Rating State fire rating, if any, required for opening. 13. Hardware Set State applicable hardware set number as described in specifications. 14. Remarks Note here any special characteristics or required features of the opening, to insure that the contractor or supplier will be properly informed.

Construction Details and Finishes DOORS Hollow Metal Door Schedules
 REPRESENTATIVE DETAILS ACCOMPANYING DOOR SCHEDULE
 DOOR TYPES: \—i © © H i=tfct FRAME TYPES: © © © © jit I2_ 1 16 Jt
 16 © DETAILS: bsEEr 0© ©*© w © © ,. + u—u s^sS oii»ss ^n 1—r—rh
 —T^T fcZJl t 1 -L J- 571

Construction Details and Finishes DOORS Hollow Metal Door Schedules
 Bass,, scfecas BfldfroL M '?'£ © !jssSe! © TO1" f^4 ® j^a © 1^0" TOFSg
 1CT7 © 8M 4^ ?, '-&■. uwtioh CELL** 6XITB caLM'SaiaEliM
 ou.,wr,/aaWTP, WWW BHiKi- Ttr» WISH WMH m»n fUK« NtMWW.
 tWlcW HBW. (Husvl WTH. itou~«r,/m«ui,a** «l=W kWTAL. «ipa* greet.
 fl HH *f,9,0, W*. r.rro,o, *T=ti- (*w*) l»»».wse.,«ic-w MoTftg' >■ mm ') ©
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 'fltfu.e". Qg) esDPMS«cfi« ft. ni>a< T£3ccwn.rwA,L.» WW t*KO iwfeurp
 Jtii. © ■wy wAwmmut iu** W4WW (&M6 m«ia 2'-ol i" © BAtWtooM*
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 IWH HoU-.KGT «J*L fegito. ■Fo^ > © »»PffaaMt hW* o 572

Construction Details and Finishes DOORS Hollow Metal Door Frames (a)
 Masonry with plaster, one or both sides -GET (c) Wood stud and plaster "i~
 it o o o a JJ k o - 2 or 2| 2 or 24- |^_1ior2^|tL-ilS lb) Two-inch plaster wall
 (e) Steel stud and plaster Fig. 2 Typical jamb installations. (f) Masonry
 block. Wrap-around or butt joint 573

Construction Details and Finishes DOORS Hollow Metal Door Frames
 BACKBEND OR PLASTER STOP - See optional profiles below 1 1/4" door
 varies for other door thicknesses FACE varies; Vmln.tor 16 ga. width and
 profile vary 3 variations below DOUBLE RABBIT TYPE i varies '
 SINGLE RABBIT TYPE any face and backbond profile '*' :■■.'■■. VI •L
 CASED OPENING BLANK JAMB BASIC DOOR FRAME PROFILES
 AND THEIR PARTS outer surface of face n iu7 1 ~l minimum flange width
 — %,*(for 1B ga.) BACKBENDS NEED NOT BE THE SAME ON
 OPPOSITE TRIM FACES - ANY COMBINATION MAY BE USED
 TYPICAL BACKBEND or PUSTER STOP PROFILES finish floor rough
 slab N 7Tf- below floor dimension (If applicable) VERTICAL FRAME
 DIMENSIONS 4 i i PREPARED WALL OPENING WIDTH |1/4,,(overall
 frame width + '/*")y4;J ♦I 1 J »• *•» DOOR OPENING WIDTH
 HORIZONTAL FRAME DIMENSIONS JZJZ-J Frame depth to fit any wall
 thickness or finish Any face profile may be combined with any backbond
 profile splaye.d lamb drywall REPRESENTATIVE FRAME PROFILES
 574

Construction Details and Finishes DOORS Hollow Metal Door Frames
 HEAD (perforated or corrugated strap) STRAP & STIRRUP (MASONRY)
 SOLID PLASTER METAL CHANNEL STUD WOOD STUD COMMON
 TYPES OF JAMB ANCHOR FOR PRESET FRAMES WOOD STUD ■
 Removable stop CUTOFF (SANITARY) STOPS ADJUSTABLE FLOOR
 ANCHOR CABINET JAMB SPLIT JAMB FIELD ASSEMBLED PRE-
 ASSEMBLED FRAMES WITH ROUGH BUCK ANCHORAGE OF
 FRAMES IN PREPARED OPENINGS 575

Construction Details and Finishes DOORS Hollow Metal Door Frames ■
 terrazzo or other ba9e mtl. FLOOR STILT. FIXED MULLION ANCHOR
 SPAT SPAT May be used with either cutoff or full length stops Light gage
 stainless steel wrap-around covering Stainless steel same thickness as frame
 and flush with all jamb surfaces SPATS head 12 or 14 ga. channel Used on
 wide openings to prevent deflection and possible Interference with door
 operation SHOULD NEVER BE USED IN PLACE OF STRUCTURAL
 LINTEL HEAD REINFORCEMENT ^&& ma. i-fa^t. frame head' HEAD
 ADAPTER For frames extending from slab to slab 4"± adjustable gasket —
 i door CEILING STRUTS jamb & head section SOUND BARRIER

FRAME Lead lining in frame provides barrier to x-rays, which travel in straight line, in gap between lead-lined wall and door LEAD-LINED FRAME r> A KEY ELEVATION at center of head ALTERNATE HEAD SECTIONS "A" DETAILS OF DOUBLE EGRESS FRAME ! J I open ends capped -\J . I "•>,, / i i at center of head I I J " JAMB SECTIONS 576

Construction Details and Finishes DOORS Hollow Metal Door Frames — MK:*OM(a>V ttWS8k.4s-.tf.tf © : ccmc Lox. -xmosft;. Rug Awewon.—, jk-, n 4' * »• «..«ffjBtna. *J3bJ 4i' 6" 8" hi. 7T 3* dbUMMWt^AM;: S1 * ««© 2)■ COMCL Og."CI.WDKCL.: BUtC iffi X-WJAft oPe. ;.*aA4jomk.v -AUCHOB. ;aMbKM« . 'o«». :WAAona.V *MtHO»A - 4-" 3< 6" ,J% Yr..CQMc. occistDEg. aum. 7 - ^>WA,yrgg. H.OTH *>>PR» 4-' 6" 8' ' PUfr>»TBB. OUtt «IPB. • i94 i-j * 1—» .4— r- i < V) ©' j: _ **':i _eeai*% opa ,^AV<3k»«.V A**B1*OK- ^ * £ in 0 Irconcega'c or cimpi&p. fn->c. (0 —*-

Construction Details and Finishes DOORS Hollow Metal Door Frames IMLtoHftV ftHeuof*. BBAMfc OPd, ®. ' ' F»LA*Ts«.a.P qua ■Sipe a A £INPEtt. OSU COHC. W.K. -TPUKTKE^" 7 CERANMC TILfc. IM CBM, hftOJLTATf- j,, CBJUKMlC TIU6 M^40Mje^j^UfiM(S' imi o*-rmu4*T tu« MIC TIC*. 4' cT 8" " 7- CfrRAhMCTIIB-TUIM 4gT KfrORTAft 0 ®. j CgK-AvqC TIUBr -*-.! . >j4cttofi:- kJ3SL -*?• FEAWO*»> rt^ . «* .** 1 - TJ»^»m. tal*T»n> prion rd*u>rruut ->i,a,*-t*u. / 578 EO.Q CUT-PUASTEK. / ^MA^ONttV ftLQCW rf, PLAJTtiV.

Construction Details and Finishes DOORS Hollow Metal Door Frames Jams Aucuae. Clip* see vj^a. haudaooi^ . pea. <pAciM<a WfcUCJEP TO »UCK i TX\ JSTEE^^/STOb^ WITH S/fe," SYPlIM l°.,aoAg.t> both «>pg« CMM WMAW To IWeic BOTH <doaj. ■* TVPICAL. JAM* . ANCHOR..CUVP4 © |4) 1%■STgEL. <Tot>« WITH 5fc>" (SVP^UKV C6R&S/UC TIL* -. *«T WITH H=y?4ai^W»ftHC«oa CUM — .—..— ■vj), —g—n ii nm | 1/ ut» —TM i^1^ K *# BOTH \$10** < to V3 2%," 3%"" - STML -JTOD ©6T6&1. *TUP* W\TH ^a"GVP«JKA ftp -fefu libs* -catt-AkMcTiue, dUf ^PB -Z.fi' Sty" 4TWL 4TV0' aTfeeL 4.TOO* .!■..CLIIM_ © • a \ £-tt=el 4tod\$ - <svp^uka lath 9Mifl da iff 4" tf* 5TUB. DI>A. 579

Construction Details and Finishes DOORS Hollow Metal Door Frames (Type I WALL- *>/**err. go, oN T/1"HU. STUOS , «<=t. Ma. DooR AS

scHeo, HEAD/JAMB "i'/q1 T/PF+ WALL- 9HAPT WALL —rill IIf r~"—u
—' JR \wyii* 1 i'V,. 4V&1 HEAD/JAMB % tt- «,<,/" CERAMIC t|L6 oh
trrs i wau n CtooP TPAM6 - * 1%' iV1 I'W. 6'A" jS HEAD/JAMB M i'Ai
sVV i-£/4ii. E i"Vw' !»/ &' p)7g' J l'*/n' JAMB -MAP-lbLE ON TyPE I
WALL HOLLOW METAL rP-AME — DOOp A-3 wseMia ON rWiNS
WD. fcLK1© flier. rv«. WAIL-J I '///YA \l n NEW CESAfflc TUB
fMft|l8D To <!X|9TIH<2 JAMB V4i nV^'-Exier. kJAii M thickness- vtMrr
_____ IcoHHtH MAO / / I! PIASTE* --••", cWcrtl1 / CSW9IHK9 6f^P '
«oL. MIL, TFAMt / x O^our rULL ^ JAMB/HEAD V*yp. t>o. en %' rlwrite
on e«i"=T, MA^oHur K^LL I JAMB PLASTEI"- P/vRTIHON flol. rttL.
FRAME W/Q^OUT PILL <— Door* A3 SCHEDULED HEAD/JAMB 580

Construction Details and Finishes DOORS Hollow Metal Door Frames
*U*U*lh (TtriCAI All -N ^-«'» M.M.PfcAMl«> • ClfeAMI* Tit* H.M.
«U-»- E^AM* V»I»TM (f^lfflffl, P-V f FIAHU TM Ml till, O.M. ^ . /'
fUVlh-, 'J»i rM.M ft 2—* C43W tU«Tl*> M.M. KAIT»h 581

Construction Details and Finishes DOORS Hollow Metal Door Frames
HWt. T*>IM (f T^'2H*HM Wi-WiAMWtia ♦ fi.A*Ttw HWp, TPtM flA*
WI»TM 8 |>kA.**TilS 6M.M. r»,AMi - Nw». «. ^l . * h *vimm4 in < % " it
I MiAM* @B^# M>W—» MUIUUh4 -101 MA*»WM» PfeAMt m«am*
/C |<5\Tyfl*AU H.M, MULI.I** 582

Construction Details and Finishes DOORS Hollow Metal and Wood Door
Frames <9 v>/'-«>" avr*«M »». ^V11-*" HHf^Mt "101 (4 » "• i'-»" 583

Construction Details and Finishes DOORS 'Hollow Metal and Wood Door
Frames WW8, thlM *r«#0 fAU*L ®- ma,*pw^p pmAui <» i"i r-/" -
M.4*Tft* M.M.I»*A,M» -IW* QVM.1M« Hi.m»W**» FhAM) A »•• ■'-#" 584

Construction Details and Finishes DOORS Hollow Metal and Wood Door
Frames HIV».TMM em ami* Tin , ^ > wBATMHm»T^(^ AT
U»A»*.MMfr- <@> «••■ ^ •»>• i'-* LAMII4A>T»» w#H-#vv MI7AL
»u**T»fc 585

Construction Details and Finishes DOORS Hollow Metal and Wood Door
Types IT) d*"""" IIIAAMI* TIH-, "HW»,1»,M <£ I «!'-•• :-2 « J'. l'-#" 586

<WI>TH »mi4fc lM *. *T«IM#H H*UL»<W MUttl PHA.M* ,41 i*<. ■'-
'—' 586

Construction Details and Finishes DOORS Door Types :«£W IV 'OOP
veHEEie. '!'■", -iUil WJ 1 llllll ■I i 1 a&^i WIA "ft .~g*"J""UiJ © 587

Construction Details and Finishes DOORS Door types WW vrmih
MIHftlMfcf m M VIMHH tAMfe. A lkj**i.ki e y ^v-v-- Keuuu H ML ^
.!««->» SWHN- i mi :vpmm !-U f* I'i "III.)w Fn i %St* 41 pLT i' tMMvri
M* — nf 'i—p L 4 « ■L w- m.u«n m*,*v**» y«H»«M Hh ^^ § •Uft
FAM1L 588

Construction Details and Finishes DOORS Hollow Metal Door Hardware:
Hinges width © © © / bearing © © bearing © height backset FULL
MORTISE BUTT HINGE Two equal square-edged leaves, one mortised
Into door edge, the other Into frame rabbet. Two bearings, as shown, on
regular weight hinges, four on heavy weight. Typical Uses: By far the most
common type of hinge for both Interior and exterior hollow metal and wood
swing doors, in all types of buildings. Usual Sizes (see NOTE below):
heights — 4 1/2"; 5" for doors over 36" w. widths — 4 1/2" for 1 1/2" door
and 1 1/2" trim clearance (dimension A); 5" (or more) for thicker doors or
larger clearances. HALF MORTISE BUTT HINGE One square-edged leaf
mortised Into door edge; the other leaf, bevel- edged, mounted on face of
frame. Typical Uses: Used with hollow metal or kalamein doors In
structural channel frames, usually in industrial type buildings. Usual Sizes:
4 1/2", 5" and 6" heights. door edge leaf - 1 1/2" o © « t) 0 2" ■—* ^r; U
PLAN HALF SURFACE BUTT HINGE One leaf, bevel-edged, mounted
on face of door; the other leaf, square- edged, mortised into frame rabbet.
Typical Uses: Used with hollow metal or kalamein doors in hollow metal
frames, usually n industrial buildings. Heavy weight type also used on lead-
lined doors. Usual Sizes: 4 1/2", 5" and 6" heights. FULL SURFACE BUTT
HINGE Two bevel-edged leaves of differing widths, one surface-mounted
on door face, the other on frame face. Typical Uses: Used with hollow
metal or kalamein doors In structural channel frames, In Industrial
buildings. Heavy weight type may be used on lead-lined doors. Usual
Sizes: 4 1/2", 5" and 6" heights. © door u NOTE: Anchor hinges and pivot
hinges should be specified for heavy doors and doors with high frequency

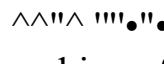
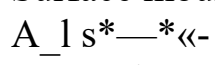
use, such as entrances to large department stores, office buildings, theaters, banks and schools, or to toilet rooms in schools and airport buildings. Regular weight hinges may be specified for doors with average and low frequency uses such as corridor doors in public buildings and doors in residential buildings. **PLAN ANCHOR HINGE** Heavy weight hinge with each leaf extended at its top edge and bent to form a flange that fastens to top edge of door and to frame head rabbet. May be used as top hinge on heavy doors and doors having high frequency usage. **THRUST PIVOT UNIT AND HINGE SET** Pivot unit for top of door, with both jamb and top plates for both door and frame. Used, with conventional butt hinges, on wide doors that may be subjected to abnormal abuse. The hinge is almost invisible when door is closed. **PIVOT REINFORCED HINGE** Heavy weight hinge with added pivot on the same pin. Leaves of pivot are interlocked with hinge leaves. Used with conventional butt hinges on doors subject to abnormal abuse, particularly with overhead closers. 589

Construction Details and Finishes **DOORS Hollow Metal Door Hardware:** Hinges **FULL MORTISE** Both leaves bent; one mortised into frame rabbet, the other into door edge. **HOSPITAL "SWING CLEAR" TYPES** These hinges have their pins located approximately 2" beyond the door edge, providing an unobstructed clear frame opening width when the door is open 90°. They are used on hospital corridor doors to patients' rooms, operating rooms, emergency rooms, or wherever a completely clear opening is required in hospitals, institutional or public buildings. **Swing clear hinges** are available only in heavy weight *| 1 W door -g^* **PLAN HALF SURFACE** Offset bevel-edged leaf surface- mounted on door face, bent leaf mortised into frame rabbet. **HALF MORTISE** One bent leaf mortised into edge of door, one flat, bevel-edged leaf surface-mounted on frame face. **FULL SURFACE** Offset bevel-edged leaf surface- mounted on door face, other bevel- edged leaf surface-mounted on frame face. Full mortised, centered on door thickness. Hinge is completely concealed when door is closed. Full mortised; door leaf usually centered on door thickness. When door is closed, only the knuckle is visible. door **PLAN PLAN © O) n <0) ■& JO) ■Oi INVISIBLE HINGE OLIVE KNUCKLE HINGE (PAUMELLE HINGE SIMILAR) 590**

Construction Details and Finishes DOORS Hollow Metal Door Hardware: Pivots alternative floor-mounted bottom pivots PLAN PLAN OFFSET TYPE Used on single-acting doors only. Need for Intermediate pivot depends upon the size, weight and usage of door; recommendation of hardware manufacturer should be followed. Pivot knuckles visible when door is closed. CENTER TYPE Used at top and bottom of double-acting doors only. Pivots are completely invisible when door is closed. Pivots are stronger and more durable than hinges and are better able to withstand the racking stresses to which doors are subjected. Their use is generally recommended on oversize doors, on heavy doors such as lead-lined doors, and on entrance doors to public buildings such as schools, theaters, banks, store and office buildings. NOTE: Because of adjustments that must be made during the installation of doors with bottom pivots, it is recommended that reinforcements be furnished in blank and that drilling and tapping be done in the field by the contractor. 591

Construction Details and Finishes DOORS Hollow Metal Door Hardware LOCKS, LATCHES, AND DEADLOCKS The selection of the proper lock type is very important. The types shown here are those most commonly used, but are by no means the only types available. Their names serve to identify either the type of lock construction or the type of installation. Mortise locks provide the greatest variety of lock functions, the best security, and excellent durability. Another popular type, with rugged construction and easily operated, is the preassembled lock, which is completely assembled at the factory. It does not have as many lock functions as the mortise lock, but can have a separate deadbolt. The bored lock is the least secured type and is not available with a separate deadbolt in the lock. BORED (CYLINDRICAL) LOCK This type of lock uses the key-in-the-knob principle. It is installed in a door having one hole bored through the thickness of the door and another bored in from the edge. The assembly must be tight on the door, without excessive play, to avoid binding. MORTISE LOCK The mortise lock is so named because it is installed in a prepared recess (mortise) in the door. Working parts are contained in a rectangular case with holes for cylinder and knob spindle. Anti-friction split bolts are available for smooth retraction of the lock bolt. Lock front may be armored to protect against burglars getting at cylinder screws and lock fasteners. Lever handles may be used if desired, and trim

may be either sectional or full plate. **UNIT LOCK** This lock is preassembled in the factory and consists of a one-piece extruded or cast brass frame within which all parts are contained. It is installed in a rectangular reinforced notch cut in the door edge. Lever handles may be used in place of knobs. **MORTISE DEADLOCK** This is a mortise lock with a deadlock only, (A deadlock is a lock bolt which has no bevel or spring action, and is operated by a key or thumb turn.) It is often used for locking a door having push or pull plates or for providing added security on doors with cylindrical locks. **BORED (CYLINDRICAL) DEADLOCK** This is a cylindrical type of lock having a deadbolt only. It fits into the same type of cylindrical cutout as that required for the bored lock. 592

Construction Details and Finishes **DOORS Hollow Metal Door Hardware:**
Overhead Closers  Fig. 3 Surface mounted, on hinge face of door. Also available with concealed arm  Fig. 4 Concealed in door, with exposed arm.  Fig. 5 Surface mounted, on stop face of door.  Fig. 6 Concealed in head, with concealed arm.  Fig. 7 Concealed in head, with exposed arm.  Fig. 8 Concealed in transom bar. ELEVATION 593

Construction Details and Finishes **DOORS Hollow Metal Door Hardware:**
Floor Closers OVERHEAD AND FLOOR CLOSERS Overhead closers (Figs. 3 to 8) are hydraulic devices, containing a piston, fluid chambers, and a spring. When the door is opened the piston is pulled back, the spring is compressed, and the fluid is moved from one side of the piston to the other. With release of the door a reverse action takes place, closing the door. Closing speed is controlled by an adjustable valve or valves. Overhead closers may be installed on either single- or double-acting doors. Floor closers, generally more durable than overhead closers, provide concealed closing mechanisms often appropriate for doors having a high frequency of use. As shown, the type of closer used depends on whether the door is hung on hinges, offset pivots, or center pivots. Both overhead and floor closers are available in a range of sizes for various door sizes, locations, and job conditions. The manufacturer's recommendations should always be

followed in determining which size and type should be used. Where surface-mounted closers are specified, internal reinforcement plates shall be provided in the door and frame by the manufacturer. Drilling and tapping for the closer shall be done in the field by the installer. Only after the door is installed and adjusted can the closer be mounted for proper operation. If drilling and tapping have been done at the factory, the necessary field adjustments become difficult if not impossible.

1 1 1 r . * T < ** 1 , 1 ^ m
 ELEVATION r floor -i SECTION FOR HINGED DOORS =p i-ly--r £"°r
 ~£ ELEVATION SECTION FOR OFFSET PIVOTED DOORS single
 acting & 'tr"--r' £ "oor X ELEVATION 4-- _LI -rp SECTION I I i 1 FOR
 CENTER PIVOTED DOORS either single or double-acting S94

Construction Details and Finishes DOORS Hollow Metal Door Hardware
 PANIC AND FIRE EXIT HARDWARE Types of Installation Panic
 hardware is tested and labeled for casualty only, fire exit hardware for both casualty and fire resistance. Only the latter may be used where fire rated doors are required. Both types are always releasable from the inside by depressing the crash bar. The mortise type (Fig. 9) and the concealed vertical rod type (Fig. 10) are the least conspicuous, and either of these types is readily applicable to custom hollow metal doors. Rim and mortise types are used on Single door Active door of pair Both doors of pair with mullion Vertical rod types are used on Single door Active door of pair Both doors of pair Where rim type (Fig. 11) or exposed vertical rod (Fig. 12) exit devices are specified, internal reinforcement plates shall be provided in the door and frame by the manufacturer. Drilling and tapping for trim and mounting plates shall be done in the field by the installer The hardware can then be more readily adjusted for best operation. In preparing the door for a lock, the drilling of three bolt holes (Vfe" dia. or less) and/or the drilling and tapping for sectional or full trim plates shall be done in the field by the installer and not at the factory. After the lock is installed and adjusted, the trim plate can be applied to suit the final position of the latching device. If thru bolt holes or tapped holes are provided at the factory, this adjustment becomes difficult if not impossible. The manufacturer shall drill for all function holes, i.e., cylinder turn piece, and knob. Door Coordinators
 Coordinators (Figs. 13 and 14) are used on pairs of doors having overlapping astragals and closers. When both leaves are open, the coordinator holds the active leaf open until the inactive leaf is closed,

preventing interferences of the astragal. Fig. 9 Mortise type exit device. Fig. 11 Rim type exit device. Fig. 10 Concealed vertical rod type exit. LBS3 —'—" Fig. 12 Eicposed vertical rod type exit device. frame head Inactive leaf active leaf Inactive leaf ■ Fig. 13 Surface-mounted type door and coordinator. Fig. 14 Mortised type door coordinator. 595

Construction Details and Finishes / DOORS Hollow Metal Door Hardware
FLUSH BOLTS These bolts are installed on the inactive leaf of a pair of doors to secure it in the closed position to serve as a latching point for the active leaf. They may also be used as auxiliary locking devices for added security. Bolts may be either surface-mounted or flush (concealed rod); only the latter type is illustrated in Fig. 15. There are many variations of these flush bolts; only the more common types being shown in Fig. 15. Due to the variety of frame construction encountered, the selection of the most appropriate type of strike is particularly important, and clearance at the floor must be very carefully controlled to insure proper engagement. The manual type (Fig. 15A) requires hand operation of the operating lever for both latching and unlatching. The variable length of the extension rod, however permits convenient location of the operating mechanism in the door edge. The self-latching types (Fig. 15B and C) latch automatically when the inactive leaf is closed, but must be unlatched manually. The automatic type (Fig. 15D) both latches and unlatches automatically when the inactive leaf is closed or opened. None of these types of flush bolt should be used on doors that are intended to serve as emergency exits. NFPA pamphlet 80 should be consulted for the selection of bolts for fire-rated pairs of doors. strike 0 0 "<fe_ of plate door edge * wm C3 0 © © ^Ttr "^ „J- concealed rod I I |»-- door edge I Interior [III surface- mounted operators -. Q exterior door edge - MANUAL TYPE SELF-LATCHING TYPES AUTOMATIC TYPE U.L.-APPROVED BOUS ARE REQUIRED AT BOTH TOP AND BOTTOM OF INACTIVE LEAF OR FIRE-RATED PAIRS OF DOORS Fig. 15 Except for Type C, only top bolts are shown; bottom bolts are similar In all cases. 596

Construction Details and Finishes DOORS Door Hardware shock absorber
OVERHEAD DOOR HOLDERS top of door hold-open mechanism pivot block set flush with head rabbet **CONCEALED TYPE** For single-or double-acting doors These are devices used to limit and control the swing of the

door or hold it in the open position. By controlling the door action they serve to protect against damage to the door and/or hinges caused by abusive usage, and damage to the holder caused by violent opening of the door.

EXPOSED TYPES For single-acting doors only **PUSH LEVER** rubber bumper © 0 direction of normal swing in door, -i stop **PLAN VIEWS**

ILLUSTRATING OPERATION stop depressed **PULL LEVER ON**

OPPOSITE FACE direction of emergency operation / 4- **HOSPITAL**

DOOR LATCH Designed primarily for use in hospitals, on corridor doors leading to patient rooms. May also be used on any door requiring push-pull operation, particularly by forearm or elbow, when hands are engaged in carrying objects. **EMERGENCY DOOR STOP** Intended primarily for use in hospitals, on doors between patient rooms and toilets. This stop permits door to be opened from the stop side in the event that an

incapacitated/patient should block the normal swing by falling. Door must be hung on center (double-acting) pivots. 597

Construction Details and Finishes DOORS Hardware Locations 10 < o UJ frame head rabbet hinge < O hinge < £_ of hospital arm pull —«. (vertical type) \. (L of roller latch and —y of hospital push-pull latch (L of door pull grip —' and of push-pull bar t of knob on lock or latch —' and of cross bar on fire exit device hinge finish floor level 1: (^_ of deadlock r-§_ of push plate 60" * 48" 47" 45" 42" 38" 1 'Except when used with push-pull plates cut for cylinders. 598

V-BEVEL BULLNOSE Used on double-acting center-pivoted doors

RABBETED PARALLEL BEVEL These two types may be used on double egress doors **Construction Details and Finishes DOORS Hollow Metal Door Edge Treatments V-BEVEL** with flat surface astragal **RECESSED**

ADJUSTABLE ASTRAGAL Surface-mounted type also used - ^e^"

PARALLEL BEVEL with molded surface astragal **RECESSED**

WEATHERSTRIPPING COMMON MEETING STILE EDGE PROFILES

I ^1 1 est ,. spot welds-. I _J no channel or angle may be used in place] of plate All joint seams continuously welded and ground smooth **STILE**

EDGE DETAILS - TYPE A DOORS 1 1 —V r— sealed If desired - I 1

L _ ^ _ \ \ —*— **STANDARD Inverted channel FILLER CHANNEL TOP**

EDGE DETAILS HARDWARE REINFORCEMENTS are provided on doors wherever hardware is to be attached, to insure that it is firmly and

securely fastened. 1—* """"l 1 —* *- _ 1 1 ■&—*— STANDARD FLUSH (closing channel) Other designs available as required BOTTOM EDOE DETAILS AUTOMATIC WEATHERSTRIP rn transom panel filler on ext. doors, optional on others door TOP EDOE DETAILS WITH FLUSH TRANSOM PANEL ABOVE 599

Construction Details and Finishes DOORS Fire-Protected Wood Doors 13,4" min solid wood core crottbanding veneer SOLID WOOD CORE FLUSH DOORS 1%" min. l» -5-inch min. width battent and brace -2-inch nominal T & G or iplined stock not over 6 inches wide l'/j" min. ^3^^ ^ BATTENED DOORS two layers of I-inch nominal T & G stock not over 6 inches wide,with one layer vertical and the other homontfil, and one layer of atbeitoi paper between iy," min. ■ ?*- m .-o-jyv.* ^ 13/," min. spring hinge or door closer on doors less than I 1/2 inches in thickness, cover latch stile with minimum 28-gage metal minimum throw of latch 3/," doori more than 5 ft high: minimum 3 butts not less than 4" by 4" ■ wood panel -sheet iron, copper not less than 28 U. S. gage, fastened on each side to stiles, rails and edges with nails or screws wood panel Vl-inch asbestos millboard or %-inch gypsum wallboard sheet iron, copper not less than 28 U. S. gage, fastened to side on which protection is desired, to stiles, rails and edges with nails or screws PROTECTED PANEL DOORS fire-retardant treated wood fire-retardant treated wood stop, glued and nailed to frame on not more than 6-inch centers f/g-inch minimum actual dimension TREATED UNTREATED METAL PRESSED WOOD COVERED ROLLED OR STEEL HARDWARE 600

Construction Details and Finishes DOORS Sliding Glass and Aluminum Doors AV/M.S.A. -WAP . 6#M*. w/a,f.A. flMI«H UShlUhM A*6WIT»CTU*AL AfrfiAPIA Ai.UMIUWM /-jr ceMcs t\i vh' imuLaMWj m~~ VV/HIA* CMANU1L • W/l FIK«t» JAM* AM* I 1.ATCHIWC JAM* i \V/ INTEfcl'6J<|«M J AW^WttP AUUM.PIUI5H ,JU IWSULATIM4 HUA»* \ r-U..*ftV-#" nrUpiT7rffT7^grrr— A- t> & 3" « I'-* 601

Construction Details and Finishes DOORS Thresholds Thresholds are essential for nearly every type of door. Usually a standard section is satisfactory. Where conditions require, special sections may be- designed. ^VWMit^ ^mVMW/.'M.W.WM2?fo)li ^HL mMM-m-A'/ML ^ Thresholds

of plain surface, extruded or rolled. Thresholds with fluted surface, extruded or rolled. WIDTH Thresholds cast with plain or abrasive surface. ^a. SZZ2SSZE2Z2 Thresholds for weather strips. j Holes for fastening not to exceed 12 " o.c. for threshold less than 3" wide. For thresholds over 3" fastenings should not exceed 15" o.c. On wide sills holes may be staggered. ^_ Threshold fastened to wood with wood screws. Threshold fastened with screw in fibre plug or expansive metal anchor. Floor may be cement, terrazzo or similar construction. ^rTf^/WW} C///^*^ A .6 h ■ i . i 0 If mm % %^ 602 Threshold fastened clips set in cement. with screws tapped to Threshold fastened with screws tapped to steel angle set in floor construction. THRESHOLD SIZCS AND ME.TAL rits. 14 IS 16 17 18 19 20 Zl 22_ "23 24 25 26 WIDTH zV 3" 4" 3" 4" ■5" ~1 6" iXt 3" 4" S" 4%" 4" 5" & HEIGHT Va* W VI' *fc" &• Vi" /*." J4" y%° %* *tv CAST a 0 0 0 5TEE.L 0 0 0 0 0 0 BRASS 0 0 0 0 0 0 0 BRONZC 0 0 0 0 0 0 0 ALUMINUM 0 O o 0 0 0 o o 0 HICKtL 5IU VH 0 0 CasI metal may be iron, aluminum, bronze, or nickel silver, with or wltoul abrasive surface. These thresholds are representative of a great many sections produced in various metals, widths, heights, and types of surface. For other sections refer to manufacturers' catalogs.

Construction Details and Finishes DOORS Thresholds and Joint Strips 7~JW WIDTH ANGLE. STIFFENEIRi (wELDIId)' Steel loading door threshold anchored to concrete WIDTH eo&^s^ =-'—4--J * '- ' ANGLE. STIFFC:NE.R5 (WELDtD) Steel shipping door threshold anchored to concrete Steel shipping door threshold screwed to floor WIDTH WIDTH Elevator door threshold for double doors cast with grooves. Surface may be abrasive or plain of cast iron, aluminum or bronze. Elevator door threshold for double doors of rolled sections, steel, bronze or aluminum. \r~ < • n— tf / *% J Ik "~* PIVOT § \\ T • <i " 3—i -yy.*. a 4 • WIDTH ' * ' . * ■ u]..3 4rfJ 4 4 Threshold for single acting floor check. HtlG^T 1 3^_ 5TE1EL FACIA Elevator door threshold for single doors cast with grooves. Surface may be abrasive or plain of cast iron, aluminum or bronze. Thresholds with concealed steel anchors are usually fastened to anchors with flat head machine screws, the anchors independently fastened to the floor construction. t 1 t5 ? "Si U"^PIVOT / S> 1 Threshold for single acting floor check. m pa Threshold for double acting floor check. < -■+ _ 6> ® ® © s ® PIVOT ® ■ IQ SCALE. l£" I-O" Threshold for double acting floor check.

HEK3HT 3 ^ ««WM Joint strips, also called dividing strips or division bars, used for separation of floors of different materials, may be of steel or non-ferrous metals. They may be of angles or other sections with anchors attached, or of a patented design. SCALE. Thresholds for floor checks may be obtained in the same metals and sections as standard thresholds or may be designed to fit special conditions. All thresholds fitted to floor checks must be designed with removable cover plate. Screw spacing must fit floor check. Dimension "A" is determined by type of floor check, usually 5'A", 6'A" or 7Va". Dimension "B" may be same as "A" or less as specified. SPECIFY! Type, location Width, length Metal and finish Show detail of special requirement! Terrazzo strips for design or pattern work In terrazzo floors are not considered architectural metal. 603

Construction Details and Finishes DOORS Thresholds and Edging Strips
 life" Ceramic tile adhesive applied or cork. Hie flooring —
 llllllllllllllllllM^ 11.' Screws with 1 T expansion »heid6,i_ ' oneateacnerxi
 and intermediate one« a"o.o- Normal floor lin* y«&" resilient flooring—1
 Extruded qlunn. +h re* hold T-1 Vd" resilient flooring — Normal floor I in*'
 jiUijuu'ffUfcmmui"" -Vinyl plastic edging strip T-2 Finished floor -^Ala IT
 Material a* required »ee v, othVdwgS' » T-3 K Zinc dividing strip ^ %'
 resilient flooring Norma line — floor Screcu« lul+b expansion ahieIda one
 at each end and Intermediate one& »"o.c. Solid metal edging strip a&
 negulfed T-4 Marble threshold- i-normal floor line /e"reslll«nt flooring —
 Carpet - T-5 Aluminum threshold - -Normal floor lin*» /©"reolient
 flooring — fcarpet- — <>- T-6 Provided* fujo roais of<BcreuJs uiBb
 expansion sn'Kalds,tuJo at each end and innsrmeeJi - ate ones staggered and
 spaced Wo.c. in each roai. HeoprenCi weatherstrip jfe'reallient flooring —
 Carpet -Aluminum Hires hold UsL Screws with expansion shialdsj one at
 each end and intermediate ones &"o,c. NOTES'- 1. For door swing, see
 other drauXnq=j. 2. For ^schedule of floor Pinisbes, see other drawings. 3.
 For gauges of metal see specif icatfonfc. 604

Construction Details and Finishes DOORS Bank Vault Doors
 AWM.IWEAU IsWINUSS srttL^M.) I— >■ TJIH. tff VAULT COOrl.
 MP4..fWSW.Uirt. L»MIN*.1IC TO 'A.' PLYUOSO >E>ftCl^1**S BY
 VWJUT D03H. ■ Mfts, ■(imsi alCkz. ■ 8-A52 ELEVATION A39 1/2"
 1'-0" £S»WI £S»K.V«JCT ."MTUPUIViJNii 'J-KV hJgyre: au. E*P=>66D

AUM- To B* WoJ26 F«Jr«(4et». '9-A52 SECTION A52 1/2" V-0" ^■3"
EXCESS 12-A52-BEfAiJL A39 1/2" 1'-0" 605

Construction Details and Finishes DOORS Light- and Soundproofing of
Wood and Hollow Metal Door Frames *~V6'X.Vi' foam, tt ^~ZfcRd*3<>U
Auto* ^TKiui ft Jamb HiHGJfe Q>f*S POOR. .B>OTTOt>A XIGHT t
SOUKiDPiZQOFiMG FORL WOOD DOOBL ,Fg.Ar vUfe .. -T^OH-OW
KABrTAL. PIUAfcMI M. • _ . jan\is» POL-VUB-tTHAUt <5AL
HikiGfe'JAvMB A.LTERNATEL XTG-MT. t 60UNDP12QOFIMG FOE '
HOLLOSff^ET<FRftlftE Jjtmn&l DOOR. BOTTOM -SIMILAR. TO
WOOD DOOSL AB.QVB. __ <fcU*.FACft.. fAOUUTiLD A UTO NsAT *
C I5QOO. ftStTTCJA 05* ZBl<3*44 6 606

Construction Details and Finishes The prime functions of the door frame
are to hold the door and its controls in the opening, and to trim the opening.
But frames often serve other esthetic or functional purposes also, such as
trimming a wall opening having no door, or enclosing glazed areas that
provide through-wall visibility or admitting light and/or air. Hollow metal
frames, which are strong, sturdy, and durable, serve all such functions
economically. The variety of configurations available in custom hollow
metal frames is virtually unlimited. Illustrated in Fig. 16 are some of the
more common and representative types. DOORS Hollow Metal Door
Frames no door single door pair of doors CASED OPENING THREE-
SIDED FRAMES may be removable DOUBLE EGRESS MULLION
FRAMES shapes, sizes and locations optional BORROWED LIGHTS
muntlns optional In all types mulllon optional SIDELIGHT FRAMES
WITHOUT TRANSOMS t I glass or p] anel . , .v- optlo ^ !: I -■ may remo
be vable I I I ll flush panels -+ WITH TRANSOM BAR WITHOUT
TRANSOM BAR TRANSOM FRAMES consult mfr. for limitations
SPECIAL FRAMES :.\ ,.--X- muntlns optional, all types MULTIPLE
OPENING FRAMES Fig. 16 607

Construction Details and Finishes DOORS Hollow Metal Door Frames 608

Construction Details and Finishes DOORS Hollow Metal Door Frames A <:
opofe. U3L t* ft w,w 5",li<9" ' «SE PS-OMi_4/A-2'» ft** door**/I KPCVy-
CCWINS- (4.M- f"TC*Mfl! ?"-|J«5" ft-,o\ rt>M, m Vn'/r-IV K>M,
MUXtoM H,M> PIMMBE '609

thickness MAX. SIZE CORE STILES RAILS FACES CROSSBANDS
 VENEER MATCHING PREMATCHING OPENINGS SPECIAL
 DETAILS FINISHING WARRANTY STANDARDS PARTICLE v/r. 13A"
 4'-Q-xi2-(r Mai-iormed particle board conforming to ANSI A2061-1L1 iVj"
 lace matching or compatible to face veneer, rruil option inner-pty Glued to
 core Maximum 5" STAVE 1W. VA". 2'<" tf-0-x12'(r [21-27 pcf) Low
 density wood clocks bonded together One species per core Vs" maichmg or
 compaiiflie lo lace veneer. Glued lo core 1V«" mill option hardwood glued
 !0 core standard. Nominal Ph", 5". 8" and 12" optional. HOLLOW ivi: i^A"
 4'0-*12'<T Resin impregnated honeycomb — '/'?" cell 1 'h" 2 ply Matching
 or compatible to face veneer, mill option inner-ply. 2'A" mill option
 hardwood- ACOUSTICAL- * STC 31', 36', 38', & 40' 13&" 4' 0- * 10' 0"
 Special materials and assembly to meet ratings shown. STC 3i — Slave
 Ve". panicle 1V«" matching or compatible to lace veneer. Glued to core.
 STC 36,38, 40 — 2"- Face compatible outer-pry. Glued to core. STc 31 — 1
 Vs" m'dl option hardwood glued to core. STC 35. 38.40-3". LEAD 1%" —
 lead lh'ickness to Vt". 4'(Tx1Q'tT Divided core. Mal-lormed particle board
 conforming to ANSI A205.1-tM3. Lead over '/'*" reinforced with lead plugs.
 1 7?" matching or compatible to face veneer. Glued to core. Z*A" mill
 option hardwood glued lo core. All available domestic and toreign veneers
 Medium density overlays. High pressure laminates Min in" Hardwood
 Virtually unlimited in standard veneers, end matching in door and transoms
 with wood grain plastics STILE AND RAIL 13A". 2>A" 4'Q"x12'0" Stiles
 and rails: Low density wood blocks bonded logeiher & to edgebands.
 Panels: Mill option, mat-formed particle board or edge-glued lumber core.
 Vb~ Maiching or compatible id lace veneer. Glued to stile. '/';" edgeband
 compatible to lace veneer. Glued to rail. Alt available W* sliced hardwood
 veneers on stiles and rails. Standard veneers on panel faces. '/'»" mill option
 hardwood. Limited veneer matching. Prefitiinq. mortise lot appropriate
 hardware. No preparation (or surface mounted Mm 5" margins edge of door
 and adjacenl hardware cut-outs. Max opening 40% of door area or 50% of
 height Cui light openings: install metal and wood louvers, wood beads, and
 safety glazing. Cut light openings!"Install metat and wood Standard beads
 or architect's detail al hghi and louver openings. Applied mouldings. Dulch
 louvers, wood beads, and safety glazing doors and shelves Standard beads
 or architect's detail al fignr and louver openings Cut tight opemngsr Sately
 glazing. Standard wood, acoustical and lead lined beads. No applied

mouldings. No dulch doors. Min. 5" safes, minimum 10" bottom rail, min. 5" top rail- Solid slicking or moulding per architect's detail. Wood louvers and safety glazing factory installed. Gardall ll. primed, painted, sealed, oiled or waxed as specilied. Interior — Lile ot original installation Exterior — 2 years NWWDA1.S-1 AWl Section 1300 PC Federal LLL.D581 Type I B, II. Class 1 NWWDA I.S.-1 AWl Section 1300 SLC Federal LLLDSai Type 1 & 11. Ctasst tnlerior — 1 year. Exterior — Not recommended- NWWDA t S -1 Interior — U'e ol original installalion. Exterior — Not recommended. NWWDA I.S.-1 AWl Section 1300 SR - ASTM E90-70 Federal LLLD581 Type IV. Class A NWWDA 1.S.-1 AWM30Q-U and £413-73 Federal LLUJ5B1, Type IV. Class 5 Interior —2 years. Exterior — Nol recommended. AWl Section 1400 Federal LLLD581. Type III THICKNESS MAX. SIZE CORE STILES RAILS FACES CROSSBANDS VENEER MATCHING PREMACHMNG OPENINGS SPECIAL DETAILS FINISHING WARRANTY STANDARDS •Footnote on openings — Minimum margins per AWl Section 1300. "No rating guaranteed on doors with lites or pairs. PARTICLE STAVE HOLLOW ACOUSTICAL STC 38 LEAD STILE AND RAIL

Construction Details and Finishes DOORS Sliding Doors and Handles natural linen ijlrjt^gr^s^i^umns^ colored jUnw bandilB^aBliaH UiTAlION Mtunl linan ELEVATION J PLAN natural linen IB fifth •! II ill m A V \ \ V fl ilsvat:ok f!!!:, nil1 ;!l!ll Wi 1 li A !■ ■' 1 W* ' life flfel 1 I i iil i!i iiii! -li |! 'n 1' "il :.,. Hi!..' !l,!■', 'I ill' ! i1. I!1 1 i! eer r 1 ll s ill lii' ■'Ml .oor a 111 !; Hi f if Il PS!« iHi; 'i. € •D linen or a «lophone r Wk l\ -\ ii ILIVATIOK A- Al nti >> V \l 0° O »o«l. 1' 5' 3' 4' 5' 611

Construction Details and Finishes DOORS Door Types -« 1=™«.jp-- - 4 J . 6Um V 1. r 4 ■t-jjo _i— ^ r^n «► Mirror one side -41- -'i-- 5JP. -* -JP -T — ?" -*• i-- 4 -4 ..4 -Jtr Vf> —r- ^ '4 M« ■•a» ■ t -i_ 4 Fig. 17 Typical interior doors showing ihe dimensions of stiles and rails. Dotted lines indicate that "doors fre assembled withh~><.5" wood dowels Lock rail heights, width of stiles, and width of rails as noted on all elevations are minimum and maximum dimensions as used by the various manufacturers. STANDARD SIZES OF ONE, TWO, SIX PANEL DOORS AND TWO PANEL tR 4toS{ 3t?lu Panel 2'- 2'- £'- 2'- 2'- 2'- 2'- 2'- 2'- 2'- 0"x6'- 0"x6'- 0"x6'- 0"x7'- 4"x6'- 4"x6'- 6"x6'- 6"x6'- 6"x7'- 8"x6'- 8"x7'- 0" 6" 8"

.KAA60MKLV BU*. ^ PLA^TEB. FHZ.6fgOQF WOOD PBAWB-
Arnj^W 614

Construction Details and Finishes DOORS Wood Door Frames FT
MASTER BATHROOM MASTER I&60BOOM PLAN 2'-4' ^=SF «* d
IZM 4>- r- -tyrtpim A6 PtU StMBDOLE ^-TYP. BASE A* /^ PER
SCHEDULE DETAIL e JAMB 615

Construction Details and Finishes DOORS Secret Door " ^Floor
■ELEVATION.SHOWIRS'SICMT'DOOfc
■LOCATE.D°)N'J)Al(ELL))^Oc3)&i"PcX button, releases latcL, 4 permits
pu.rk.outspnitij> to tliroW door open,. 3ECTIOIUL-M.A(l» THEU-
5ECBET-DOOI LOCKi- Scale. 5'-HO"» Tk,ere are Several .secret door
open.; era procurable/, AllparetTed, _y L. fl H d 0 700.^3 :i -Pivof iamb
liinoe. teoular jamb deHa'il. uedd as lunge is so design.- ed tk.arclearji.ee is
pnpvictec 'or by k-iioe. /or projectl'io iiouiaii»?ljebli'b'o/llj lurujes wiE
slow. Alternate.; TKe Wist bis- Uitoe.. Use 2 -3" butts. HI n-Qc dcstqued
for li clearance.. ■Projectiip base requires groove At sito-ilar to de tail
sLowiu. fiorm. panel oil uuide o/door. /ace . Cur c/roon, base 4 door jamb
depends on, clearan.ee/ required wten. door opeits -Dotted liruo indicates
tolow crowe. iil jamb lecexranj to allow projectiio base/ to clear __ jamb.
SroovcJobejusf li'oL '*! eiouoL to allow base to cleat/:
5ECTJON»TH£U»5ECR.ET-DGOL<{5HOWINq"DEVEL«
CUT/'IH&ASLJAMB^DOOILjOX-'C-C"Seats 3-f-OS •5SECTION-THE.U
•OKA-A"- •HUJ>4-TOp.|c.All- Ocale3"-1L0"- _ builr-up door wifL ven-
eered raised pan, ;l tkorouokJij secured. ply veneered 31 I j—r—i
arcoriueiyi norouoWi oil ed pan«l. ,, .i Hai/ed mouldiiq.rn.it;:rec ood i -Tins
dislaiLce lo be as little, as >rach'c- able.. Dotted li'nei kdica^ location/ of
floor .pivt >r. 1* "k. Floor •5tCTIOI^THE.U' DOTTOM •TACHLD-
DA3L- OK«"b-&"- ■E-AIL-i-AT- 5ca|c3".IL0' 616

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
617

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
LIGHTS ISOMETRIC CUTAWAY Fig. 22 Door for an architect's office.
618

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
TCTfULTOM acvwfPfiBva. t*w.waVTt> un. MJC THUMB POCH TD-
ftCMNkl WW UO-HTO tWKH RT6 FU DT NWU BOB WV&TDltfc ,
WW WV1 TO WKH nu. WHwaa grrSCttKRJo wyt TP«fHtflOVE*'f F-JDF
HU-E-. TAMPE UEWKWi -*ME*1WB0TP ..'xTCH BTO .STILE CTJ /
PKTCHI cat. tor..Hca«uc. TDtfct-TOW 1U7T. W-tnitVEi--^
KTTTVJLTMM 4SJ5J ■ 1ft-ro ' i . * , ■MMurm ifon Hia-t. ru to mam
E6tim& -r-tov "M & fl***"1* 4auo WOOD , !OTffSft**nnphi : THIS IWr
to ** -Ui*-JBo T WtTH, Tor »r. HlfMn

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
SECTION "A-A" S & &.!«,! lfe" » i'-O" ■ rq&. 7 iff • T^ypupq ,-
ELEVATION S c » I c : %' - I'—0 620

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
E L E (VAT1 ON • S ft. ». I e. '. VT - I'—O" SECTIONED id.lt,: \V - I' O"
621

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
(W? • E L E V A T 1 O N i c. i i e. : 'h' - I'— O- 622

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
FUiu K SECTION CC £ c ilt : V4* - 1'— o" • "C- "W3 • E LEVATION —
,, •IMTERIOR.'ELEVATION' S ECTI ON " ft- E>" •UF-DOOIL'ScJc^-j'-o-
■ * S ciLi_i 3/V ■ ■!'— °- 623

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
ife"Sfiefct!jinCf HEAD ■ s'one. fl&< TYPICAL SECTION ■ KALLi lit' B
!'— O" Door 6 Trim a.rc 7Fii77 is h^nj Jietvn^ Lc»cU<J dL«.P*.ne
SECTION "E>-E>" scale : i!4" - i'—o* f &. , Screen Door 1 SE CTION
"A-A" ■ Ha.IT XfiDcr Scale: \%h" « t—o -&' -SheJfima ■ Wtferproof Pa.
per ELEVATION 5 wLt- : 'h' - I'— o" 624

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
E L E VAT I On S cl A L I '■ &• - 1'—o 625

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
•ifc'-SCALE- DETAIL- OF- f-f- •STRAP •&• DCDL-OOCKfiJL- £

LE.VAT10N 5 c \] c : fe' - 1—0' 626

Construction Details and Finishes DOORS Extorter Wood Entranco Doors
Lath £■ Pla/ter 2'x7/g" Rirnnq -In/ulahon DETAIL-OP- DOOR-PANEL
/tale : Half • Pull -/i^e ■ 627

Construction Details and Finishes DOORS Exterior Wood Entrance Doors
FLASHING BLOCKING HEADER CASING HEAD JAMB EXTERIOR
DOOR COMBINATION DOOR INSERT GLASS -M J- A RABBETED
STOP AND WEATHERSTRIP CASING SIDE JAMB EXTERIOR DOOR
THRESHOLD i e\ -i—-- COMBINATION DOOR •EXTERIOR DOOR -
THRESHOLD AND WEATHERSTRIP Fig. 23 Exterior door and frame.
Exterior-door and combination-door (screen and storm) cross sections: A,
head jamb; fl, side jamb; C, sill. 628

Construction Details and Finishes DOORS tf&lepi-jaw paPR- p^jit'is.'j
Wood clad metal door frames xrr. rt>" eevsn. .^_ |_- & "■" l^//r h ,,,
~w°V!e»v p»p Fjan& 55H6.P cz«7e. UO?P fctt?e, M£AP tf frffcira £ 4' \o
Urt^f t>h&t.pt\Jb-\ -m ,r ,! , 1 4 — *VfcM& (la*.fttt BELOW - ^he, t\\A*\\m
JAMS? £ £L£VA7<?I2. Marble jamb and head details at elevator 3TB& £
DHfcl\telf>te '1 " *4 & £ HEAP <2 ELEV/AT£>r2-. ■^ 629

Construction Details and Finishes DOORS Metal Door Frames e
HWD.TRiH MTfavsM. Pate, oj&s B> one-Sloe. .-rtcin"f<cm«KTt> ess
aw*«c> —«■ ■/.» • „. i 630 ©3 H.M.f=TC*M»- HAROVMoOP>/i?NE
SIPC ■U<9* ' «3£E PST^i. ^/^ -SO FW* IVOfc^/l

Construction Details and Finishes rf.FTB. Eftaey-owiHs- DOORS Metal
Door Frames <*.KS.

Construction Detalts and Finishes DOORS Bl-Fold Doors VIM-?l_
F5<MfcW.W£ V4? H^fftOWCOO V«HE.« SWA*. «*«««. / n —I ■ptvweoo
r>>i-i=oi-ir> ooowl_ fia_ U«S.H 6BtT. aim. *«?U/*u huu Z pprrr— 1-4
G&Y*. rout. »h ***<*>* \ UV /KUCOVK 4ILL 632 I*" Pl.A4TI*
bitMtlkMTk | |U» f SWtl'l #l*«4UAK Folding shelf for dutch doors

Construction Details and Finishes DOORS Sliding French Door 633

Construction Details and Finishes DOORS Hardware 36"-38" 'HINGE Hardware for doors may be obtained in a number of finishes, with brass, bronze, and nickel perhaps the most common. Door sets are usually classed as (a) entry lock for exterior doors, (b) bathroom set (inside lock control with safety slot for opening from the outside), (c) bedroom lock (keyed lock), and (d) passage set (without lock). Hinges Using three hinges for hanging 1 $\frac{1}{2}$ -in exterior doors and two hinges for the lighter interior doors is common practice. There is some tendency for exterior doors to warp during the winter because of the difference in exposure on the opposite sides. The three hinges reduce this tendency. Three hinges are also useful on doors that lead to unheated attics and for wider and heavier doors that may be used within the house. Loose-pin butt hinges should be used and must be of the proper size for the door they support. For 1 $\frac{1}{2}$ -in-thick doors, use 4-by-4-in butts; for 1 $\frac{1}{4}$ -in doors, 3K2- by 3 $\frac{1}{2}$ -in butts. After the door is fitted to the framed opening, with the proper clearances, hinge halves are fitted to the door. They are routed into the door edge with about a 3-16"-in back distance (Fig. 26A). One hinge half should be set flush with the surface and must be fastened square with the edge of the door. Screws are included with each pair of hinges. V $\frac{1}{2}$ " OR MORE Fig. 24 Door clearances. FRAMING STUDS 3/16 HINGE ■ DOOR EDGE HOLE FOR SPINDLE v.-., A ^ HOLE FOR' KEY -FACEPLATE ROUTING ■ DRILL FOR LOCK STRIKE PLATE • (ROUT) // fl 1/33" CLEARANCE mjxju DOOR HI MCE SIDE B PLAN VIEW Fig. 25 Door details: A, Installation of strike plate; B, location of stops. LOCK SIDE -FACEPLATE ROUTING 634 Fig. 26 Installation of door hardware: A, hinge; B, mortise lock; C, bored lock set.

Construction Details and Finishes DOORS Hand of Locks; Lock Functions MORTIS 6 LOCK OUTSIDE LEFT MAND RtOMT MAND OUTBIDS LEFT HAND REVERSE BGVEL RIGHT HAND REVEfUE BEVEL UAWD DOOR LOCK CUPBOAROS, CABINETS, BOOKCASES Determination of the hand of mortise or rim locks STANDARD BEVEL Locks not designated as reversible are made right-hand, left-hand, right-hand reverse bevel, or left-hand reverse bevel. The hand of a lock is invariably determined from the outside of an entrance door or from the corridor or hall side of a room door. An easy method of determining the hand of a lock is to imagine oneself on that side of the opening from which

the lock is controlled or operated by the key. Viewing the opening in this position, note which one of the following is true: (1) If the door swings in and is hinged at your right hand, the lock is right-hand; (2) if hinged at your left hand, the lock is left-hand; (3) if the door swings toward you and is hinged at your right, the lock is right-hand reverse bevel; (4) if hinged at your left hand, the lock is left-hand reverse bevel. You may find that many locks are marked "reversible," meaning that they are interchangeably right- or left-hand, and in these instances no reference to hand or bevel of lock is necessary. These are locks which operate alike from both sides or locks which can be inverted in order to reverse the locking functions. Features

Deadbolt " Latchbolt, Anti-friction——. g Z ■ ^Auxiliary (guarded) Latch
 —Stop Button Symbols *l t A — — Cylinder Outside knob or lever Inside
 knob or lever Solid spindle Two piece spindle » I- <- ■ Turn piece Cylinder
 turn Emergency turn piece Occupancy indicator Features of and symbols
 for door hardware 635

Construction Details and Finishes DOORS Lock Functions ll ¥ s s ♦ I I
 ♦ iiniu OFFICE For Inner Office and Area Entry Doors, • Latchbolt retracted
 by lever or knob from either side unless outside is locked by stop button. •
 When outside is locked, latchbolt is retracted by key outside and lever or
 knob Inside. • Auxiliary latch deadlocks latchbolt when door is closed. •
 Latch holdback available. APARTMENT ENTRANCE For Apartment
 House or Office Building Entrance Doors • Latchbolt retracted by lever or
 knob either side unless outside is locked by key from inside. • When locked,
 latchbolt retracted by tenant key outside, lever or knob inside. • Auxiliary
 latch deadlocks latchbolt when door is closed. INSTITUTION For
 Permanently Locked Passage Doors. • Latchbolt retracted by key from
 either side. • Both levers or knobs always inoperative. • Auxiliary latch
 deadlocks latchbolt when door is closed. • Latch holdback available.
 STORE DOOR For Store Entrance Doors, • Latchbolt retracted by lever or
 knob from either side unless outside is locked by stop button. • When
 locked latchbolt retracted by key outside and lever or knob inside. •
 Deadbolt operated by key from either side. • Auxiliary latch deadlocks
 latchbolt when door is closed. STORE DOOR For Store door, Storeroom or
 Utility Room Doors. • Latchbolt retracted by lever or knob from either side.
 • Deadbolt operated by key from either side. CYLINDER X TURN PIECE
 Deadlock • Deadbolt operated by key outside and turn inside. w I DOUBLE

CYLINDER Double Cylinder Deadlock • Deadbolt operated by key from either side. • Bolt automatically deadlocks when fully thrown.

CLASSROOM Classroom Deadlock • Deadbolt operated by key outside. •

Cylinder turn inside will retract deadbolt but will not project it. • Bolt automatically deadlocks when fully thrown. CYLINDER X BLANK

Deadlock • Deadbolt operated by key from one side. • No trim on opposite side. • Bolt automatically deadlocks when fully thrown. *¥ 636

CLASSROOM • Latchbolt retracted by lever or knob from either side unless outside is locked by key. • Inside lever or knob always free for immediate exit. • Auxiliary latch deadlocks latchbolt when door is closed.

STOREROOM • Latchbolt retracted by lever or knob inside, by key outside. • Outside lever or knob always inoperative. Knob is free spinning. •

Auxiliary latch deadlocks latchbolt when door is closed. • Latch holdback available. «l «l 4 ¥ ENTRANCE For Commercial and Residential Entry

Doors • Deadbolt and latchbolt retracted by lever or knob trim either side unless outside is locked by stop button. • When locked, key outside and lever or knob inside retracts deadbolt and latchbolt simultaneously. Outside remains locked until stop button is reset to unlocked position. • Deadbolt operated by key and turn piece. Throwing deadbolt automatically locks stop button. • Auxiliary latch deadlocks latchbolt when door is closed and locked.

CONVALESCENT For Convalescent or Bedroom Doors •

Latchbolt retracted by lever or knob from either side when unlocked, •

Deadbolt operated by key outside and turn piece inside. • Throwing deadbolt disengages outside lever or knob. • Turning inside lever or knob retracts deadbolt and latchbolt simultaneously for Immediate exit and unlocks outside.

CLOSET For Closet, Storeroom, or Utility Room Doors •

Latchbolt retracted by lever or knob from either side at all times. »

Deadbolt operated by key outside. DORMITORY For Dormitory or

Bedroom Doors. • Latchbolt retracted by lever or knob from either side at all times. • Deadbolt is operated by key outside and by turn piece inside.

HOTEL For Corridor Doors to Guest Rooms. • Outside lever or knob always inoperative. Knob is free spinning. • Latchbolt retracted by guest key outside except when deadbolt is thrown by turn piece inside. • When

thrown occupancy indicator is engaged, all keys inoperative except emergency or display keys. • Turning inside lever or knob retracts deadbolt and latchbolt simultane- eously. Aux. latch deadlocks latchbolt when door is closed.

HOTEL For Corridor Doors to Guest Rooms • Same as 85P function

except that visual "DO NOT DISTURB" plate replaces occupancy indicator button. Available for 1 1/2" doors and escutcheon trim only.

Construction Details and Finishes DOORS Lock Functions A, N, S, I, No, Grade A10S C10S D10S F10N F75 WH D12D F89 HD» A20S *0| A25D D25D m A43D F79 *B A44S D44S «D» A30D F77 2 D30D 1 F30N 2 A40S F76 2 C40S 1 D40S 1 F40N 2 A170 D170 F170N « A. No N S, I, Grade a# u ^, , ANS/A156.2 Non-Keyed Locks Series 4000 Passage Latch: Both knobs always unlocked Exit Lock: Unlocked by knob inside only. Outside knob always fixed. Closet Latch: Outside knob and inside thumbturn are always unlocked. Exit Lock: Blank plate outside. Inside knob always unlocked, (Specify door thickness, 1 1/2" or 1 3/4".) Patio Lock: Push-button locking, Turning inside knob releases button. Closing door on A & D series also releases button. Bath/Bedroom Privacy Lock: Pushbutton locking. Can be opened from outside with small screwdriver or flat narrow tool. Turning inside knob releases pushbutton. Closing door on A, C and D series also releases button, preventing lock-out. Communicating Lock: Turn button in outer knob locks and unlocks knob and inside thumbturn. Hospital Privacy Lock: Push-button locking. Unlocked from outside by turning emergency turn-button. Rotating inside knob or closing door releases inside button. Dummy Trim Single Dummy Trim: Single dummy trim for one side of door. Used for door pull or as matching inactive trim. A, N, S, I, No, Grade F51N F81 «D» D50PD F82 A53PD F81 2 C53PD F82 1 D53PD F82 1 Mm- A55PD F92 2 D55PD 1 »«D» D60PD F88 1 D66PD F91 1 A70PD F84 2 C70PD 1 D70PD 1 D72PD F80 • 1 A73PD F90 D73PD D76PD F85 1 ♦MB* ANSI A156.2 Keyed Locks Series 4000 Entrance Lock: Unlocked by key from outside when outer knob is locked by turn-button in inside knob. Inside knob always unlocked, Entrance/Office Lock: Push button locking. Pushing button locks outside lever until unlocked with key or by turning inside lever. Entrance Lock: Turn/Push button locking: Pushing and turning button locks outside knob requiring use of key until button is manually unlocked. Push button locking: Pushing button locks outside knob until unlocked by key or by turning inside knob. Service Station Lock: Unlocked by key from outside when outer knob is locked by universal button in inside knob. Closing door releases button. Outside knob may be fixed by rotating universal button. Vestibule Lock: Unlocked by key from outside when outside knob is locked by key in inside knob. Inside knob is

always unlocked. Store Lock: Key in either knob locks or unlocks both knobs. Classroom Lock: Outside knob locked and unlocked by key. Inside knob always unlocked. Communicating Lock: Key in either knob locks or unlocks each knob independently. Dormitory Lock: Locked or unlocked by key from outside. Push-button locking from inside. Turning inside knob or closing door releases button. Classroom Hold-Back Lock; Outside knob locked or unlocked by key. Inside knob always unlocked. Latch may be locked in retracted position by key. 637

Construction Details and Finishes DOORS Lock Functions A, N. S. I. No. Grade A79PD «D| A80PD F86 2 C80PD 1 DBOPD 1 F80N 2 ♦HO* D82PD F87 1 A85PD F93 D85PD *HD» ■• ... ANSI A1S6.2 Keyed Locks Series 4000 Communicating Lock: Locked or unlocked by key from outside. Blank plate inside. Storeroom Lock: Outside knob fixed. Entrance by key only. Inside Knob always unlocked. Institution Lock: Both knobs fixed. Entrance by key in either knob. Hotel-Motel Lock: Outside knob fixed. Entrance by key only. Push-button in inside knob activates visual occupancy indicator, allowing only emergency masterkey to operate. Rotation of inside spanner-button provides lockout feature by keeping indicator thrown. A, N. S. I, No, Grade B250PD E2121 1 ♦ICII B252PDtE2111 1 ♦ICU B270D E2181 1 Oil Schlage A. N. S, I, Number No. Grade No. Grade Deadbolt Locks ANSI A 1S6.S B160N E2151 2 B460P 1 B560 1 ♦IQII B162Nt E2141 2 B462Pt 1 B562I 1 4DI» B461P E2161 1 B463P E2171 1 ♦I0II B464P ♦ID B180 B480 E2191 DII Single Cylinder Deadbolt Lock: Deadbolt thrown or retracted by key from outside or by inside turn unit. Bolt automatically deadlocks when fully thrown. Double Cylinder Deadbolt Lock: Deadbolt thrown or retracted by key from either side. One-Way Deadbolt Lock: Deadbolt thrown or retracted by key only. Blank plate inside. Classroom Deadbolt Lock: Deadbolt thrown or retracted by key outside. Inside turn unit will retract bolt only, Cylinder Lock: Deadbolt thrown or retracted by key from one side. No inside trim. Door Bolt: Deadbolt thrown or retracted by turn unit only. No outside trim, A. N. S. I. No. Grade Deadlatch Locks ansiais6.s Night Latch: Deadlocking latchbolt retracted by key from outside or by inside turn unit. Rotating turn unit and activating hold-back feature keeps latch retracted. Double Cylinder Deadlatch: Deadlocking latchbolt retracted by key from either side. No hold-back feature. Exit Latch: Deadlocking latchbolt retracted by inside

turn unit only. No outside trim. Rotating turn unit and activating hold-back feature keeps latch retracted. Lever Functions Lever Passage Latch: For use on passage, closet and doors that do not require locking. Rotating either lever retracts latchbolt. (Specify door hand.) Single Dummy Trim-Double Dummy Trim: For use on single or pairs of doors when fixed turn is required. (Specify door hand.) Grip Handle Sets Entrance Lock: Unlocked by key from outside when thumb-piece is locked by inside turn-button. Entrance Lock: Deadbolt thrown or retracted by key from outside or by inside turn unit. Latch retracted by thumbpiece from outside or by inside knob. Double Cylinder Entrance Lock: Deadbolt thrown or retracted by key from either side, Latch retracted by thumbpiece from outside or by inside knob. tCAUTION: Double cylinder locks on residences and any door in any structure which is used for egress are a safety hazard in times of emergency and their use is not recommended, Installation should be in accordance with existing codes only, 638

Construction Details and Finishes DOORS Lock Functions A. N. S. I. No. Grade A. N. S. I. No. Grade Dummy Trim Outside and Inside Dummy Trim: For use as door pull or as dummy trim on an inactive of pair of doors. Fixed thumb- piece and inside knob. Thru bolted dummy cylinder. Outside and Inside Dymmy Trim: For use as door pull or as dummy trim on inactive leaf of pair of doors. Fixed thumbpiece and inside knob. Dummy cylinder with inside plate. Interconnected Locks ANSI A156.12 Entrance—Single Locking: Deadbolt thrown or retracted by key in upper lock from outside or by inside turn unit. Latch- bolt retracted by knob from either side. Turning inside knob retracts deadbolt and latchbolt simultaneously for immediate exit. Entrance—Double Locking: Deadbolt thrown or retracted by key in upper lock from outside or by inside turn unit. Dead- latch retracted by key in outer knob when locked by pushing turn-button in inner knob. Outer knob may be fixed in locked position by rotating turn-button. Inside knob retracts deadbolt and deadlatch simultaneously for immediate exit. Storeroom Lock: Bolt may be operated by key from outside or by turn unit from inside. Bolt automatically deadlocks when fully thrown. Lock may be opened by key from outside. Inside knob will retract both latch and deadbolt. Latch automatically deadlocks when door is closed, inside knob always free for immediate exit. Outer knob always fixed. Hotel-Motel Lock: Deadbolt thrown or retracted by key in upper lock from outside or by

inside turn unit. Deadlatch retracted by key in outer fixed knob. Push-button in inner knob activates visual occupancy indicator, allowing only emergency masterkey to operate. Rotation of inside spanner-button provides lockout feature by keeping indicator thrown. Turning inside knob retracts deadbolt simultaneously for immediate exit. No. Grade Dummy Trim Single Dummy Inside Trim: Snap-on rose and knob. Concealed mounting screws. Dummy Trim Inside and Outside: Snap-on rose and knobs thru-bolted. A. N. S. I. No. Grade H110 F95 4 «ID *D i * H153 F97 4 ♦ID ♦He i *■ H180 ♦ID ♦Hfl i * H185 F100 4 ♦ID ♦WC i * H170 * H172 1 4 4 L9010* F01 I L9040 F22 t L0170 L0172* » «• L9175** J L9176** 1 A. N. S. I. No. Grade Mortise Locks ANSI A156J3 Non-Keyed Series 1000 Passage Latch: Latch bolt retracted by lever or knob from either side at all times. Bath/Bedroom Privacy Lock: Latchbolt retracted by lever or knob from either side unless outside is locked by inside turn piece. Operating inside lever or knob or closing door unlocks outside lever or knob. To unlock from outside, remove emergency button, insert turn piece (furnished) in access hole and rotate. Single Dummy Trim: Lever or knob on both sides fixed by mounting bar. Pair Dummy Trim: Lever or knob on both sides fixed by mounting bar. Single Dummy Trim: Lever or knob on one side fixed. Includes lock chassis and armor front. Pair Dummy Trim: Lever or knob both sides fixed. Includes lock chassis and armor front. Keyed Locks Office and Inner Entry Lock: Latchbolt retracted by lever or knob from either side unless outside is made inoperative by key outside or by rotating inside turn piece. When outside is locked, latchbolt is retracted by key outside or by lever or knob inside. Outside lever or knob remains locked until thumbturn is returned to vertical or by counter clockwise rotation of key. Auxiliary latch deadlocks latchbolt when door is closed. Apartment Entrance Lock: Latchbolt retracted by lever or knob from either side unless outside is locked by key from inside. When locked, latchbolt retracted by key outside or lever or knob inside. Auxiliary latch deadlocks when door is closed. Classroom Lock: Latchbolt retracted by lever or knob from either side unless outside is locked by key. Unlocked from outside by key. Inside lever or knob always free for immediate exit'. Auxiliary latch deadlocks latchbolt when door is closed. **When armored front is required as strike for inactive door, specify L9177 for single or L9178 for pair of dummy trim. Specify door hand. 639

Construction Details and Finishes DOORS Lock Functions A. N. S. I. No. Grade L9080 F07 .1 L9080EL Y L9080EU L9082 X L9453 F20 L9456 F13 1 L9465 .1 ! L9466 F14 ♦iini. L9473 F21 «l Keyed Locks Storeroom Lock: Latchbolt retracted by key outside or by lever or knob inside. Outside lever or knob always inoperative. Auxiliary latch deadlocks latchbolt when door is closed. Storeroom Lock: Electrically locked. Outside lever or knob continuously locked by 24V AC or DC. Latchbolt retracted by key outside or by lever or knob inside. Switch or power failure allows outside lever or knob to retract latchbolt. Auxiliary latch deadlocks latchbolt when door is closed. Inside lever or knob always free for immediate exit. Storeroom Lock: Electrically unlocked. Outside lever or knob unlocked by 24V AC or DC. Latchbolt retracted by key outside or lever or knob inside. Auxiliary latch deadlocks latchbolt when door is closed. Inside lever or knob always free for immediate exit. Institution Lock: Latchbolt retracted by key from either side. Lever or knob on both sides always inoperative. Auxiliary latch deadlocks latchbolt when door is closed. Entrance Lock: Latchbolt retracted by lever or knob from either side unless outside is locked by 20° rotation of thumb- turn. Deadbolt thrown or retracted by 90° rotation of thumbturn. When locked, key outside or lever or knob inside retracts deadbolt and latchbolt simultaneously. Outside lever or knob remains locked until thumbturn is restored to vertical position. Throwing deadbolt automatically locks outside lever or knob. Auxiliary latch deadlocks latchbolt when door is closed. Dormitory/Exit Lock: Latchbolt retracted by lever or knob from either side. Dead- bolt thrown or retracted by key outside or inside thumbturn. Throwing deadbolt locks outside lever or knob. Rotating inside lever or knob simultaneously retracts deadbolt and latchbolt, and unlocks outside lever or knob. Closet/Storeroom Lock: Latchbolt retracted by lever or knob from either side except when deadbolt is extended. Deadbolt extended or retracted by key outside. Store/Utility Room Lock: Latchbolt retracted by knob or lever from either side except when deadbolt extended. Deadbolt extended or retracted by key from either side. Dormitory/Bedroom Lock: Latchbolt retracted by knob or lever from either side except when deadbolt is extended. Deadbolt extended or retracted by key outside or thumbturn inside. A. N. S. I. No. Grade L9486 F15 | MWIBIIUWp A. N. S. I. No. Grade L9460 F17 ♦ill 1 L9462 F16 «i!Dli» mil L9463 'Ir L9464 F18 4 1 Keyed Locks Hotel Lock: Latchbolt by key outside or by lever or knob inside. Outside lever or knob always fixed.

Deadbolt thrown or retracted by inside thumbturn. When deadbolt is thrown, all keys become inoperative except emergency or display keys. Turning inside lever or knob retracts both deadbolt and latchbolt simultaneously. Auxiliary latch deadlocks latchbolt when door is closed. Hotel Lock: Latchbolt retracted by key outside or by lever or knob inside. Outside lever or knob always fixed. Deadbolt thrown or retracted by inside thumbturn. When deadbolt is thrown, "DO NOT DISTURB" plate is displayed—all keys become inoperative except emergency or display keys. Turning inside lever or knob retracts both deadbolt and latchbolt simultaneously. Auxiliary latch deadlocks latchbolt when door is closed. Deadlocks Cylinder X Thumbturn: Deadbolt thrown or retracted by key outside or thumbturn inside. Double Cylinder: Deadbolt operated by key from either side. Classroom Lock: Deadbolt thrown or retracted by key from outside. Inside cylinder turn retracts deadbolt but cannot project it. Cylinder Lock: Deadbolt thrown or retracted by key from one side. No trim on opposite side. 640

Construction Details and Finishes CEILINGS Suspension System Types

This section provides the designer with information on both suspended ceilings and ceilings directly attached to the structure above. It starts with a review of generic suspension systems and then provides details and discussion of the various suspended ceiling types. Large-scale details show how, in addition to standard acoustical tiles, other ceiling materials such as plaster, metal panels, baffles, gypsum board, and wood can be attached to suspension systems. A variety of unusual conditions are also detailed, including curved and vaulted ceilings, wall conditions, light coves, and lighting fixture framing. The designer is cautioned that in many jurisdictions, suspension systems must attain a higher level of structural integrity than most other architectural elements. For example, wire hangers may not be an acceptable method of suspending channels from the structure above. Rather, steel rods of a minimum diameter or flat bar hangers of a minimum width and thickness may be required. Local or state codes should always be consulted prior to finalizing such details. In many situations, the ceiling "skin" takes on further importance beyond aesthetic, acoustical, or visual requirements. It can also be used to complete an envelope that provides a fire-resistive rating to the structural members above. Again, it is necessary to thoroughly investigate the building and fire codes that might

govern ceiling design. Acoustical Tile and Lay-in Panel Ceiling Suspension Systems WAUSfcQ VJCrt-L MOULOIWG' ACOUSTICAL LAY iu paMtu-s CARRYING CHANNEL M&ISJ QUMU&U M&IU QUWU6B') UfcUGfcE CL.QQYIU& CMAuUtL' fa) MUU6 ■±>U6PfeU610U dyATfrH ACOUSTICAL "TILfe •WALL MOULDIWO MCIU OUUUBC UAU&fctZ. &&CU.IW1& E>D Rjneiuo UCMLIU& E!akC3 e>6.ii* p*,p nunc Ava-rfcH -xjppoaT cup FOB PUOtIUG CARRYING CHANNEL AOOUIbTIOAL. TILE:

Construction Details and Finishes CEILINGS Suspension System Types OAT ollf. 4BL.SI ik.T.*li.-M-4M (00 OTHIR WfOUinii) HilKUXllitU, tROtt TCE OR UKKlljG »kl MVJ1T 61 iMSC 2. TMt tOTM. yjEUilr OF k UjlfT f l<WR£ MID OTrfER EOJIPMUT (MR 60><£4, ETi.) MJD ttllill^ MkTtttlM. WPPOBTID »V Tilt Mkivl eO*A MlllT UOT EWttD THE ►UOWJvBLE , MflectIOvI Of Km or rf(4PwJ. ■DEftUTIOll BMi. MiiVT Bi rORk)\(l)EP wjo aeTirnc »vrnt mmJofwtuher. 3. iiVM OR PELLpEljT mTIIRE* MOM «£ IHDEPEwlOtkTtIY <IW>PORTEt> «OM it tlAttf leaJ OB FROM FIOOK OR ROOF IOUiTtUtVlOkI. tfi.B. IM*. SlEtI *OSR iMwl. ,475 M>PRO!<. •V-i, 0.1. (UOTE' IWJ4ER. OR Uem/IES £iW. M»,Y BE l»£D WMEM REOR ey 9t,R, ioPSJ EXPOSED T GRID SYSTEM 642

Construction Ostafis and Finishes CEILINGS Suspension System types SIDE MOOUTIlki lWEUll<i Ul« B/FIXTURE Mfi,, iilrM-iAnoil »y EiEi-raitimJ m p»at of fixture ■ jper ooe wall Mom-onltvOnHM corl. wall spruM clip Z BAR «XlpLIUIJ!-!>.-<1 >JH«C PdRUtl OUUB4 KT LWVT, TOP LIP »tt»U M 8U4T OP TO PRtVtvJT 4LIP FROM SUOI^A orF 6a*.Tiou "A.-A' I- rirfTilRi \A.R« WtTH MM TILE QPEUIU4 IV"<:.R. 'id*.. STEEL SUSP. tttMJlJEL .«/ fcPPROX. *-4' 0,4. IkloTE I LARM.R OH ~ HEWltR MmJ.MA,V£>EU(ED WdEkI SEQ'P BV 6LDH. COPE 1, HjTHt E^iUT k L0^lll4 BKO dullltfr BE IXWETEP BUM)li OF iOitTIUllloUS M T<ldE>«, m«T5, TH£ 1 ttJt <H<M1. U-iFrilgfPTO-TIIE-4VMIIII4 MEMBER IV MtkHS «J, MUX, «<Tldt>llj VL RIVET, SHUT MM. «M»/,OR OTHeR MEMIS. 2, he tom. WeiW or*, lkiit ruruiE mio onln EOillPME»JTf*IR IctXEiUU A*IP£lltld< MMERIM. .. ' difWRTEO trm s«rwt> i B*a kust ilpt EWUPTHE ALLOWABLE OCfLUT'O*! of ^Of IT* i»J. OtFIE4T«»l DAT*, MUST B< FU«fc)liMB WlpaRTIFW BYTHI

MMMFtXTURIP,, 3, 5llW*iE OR PEUPUJT FIXTURE? MOST at
 IUDtPtviDEMtiy SUPPORTED FROM l(>Wt IROil trtA*1 OR FROW^
 FLOOR OR RoPF ep>l^tRue.TioVJ xaos/E, ♦ELF'LEl/ElIlli, TotJfiilt f
 £R£GYc JO\ilr «,k.T SLIP *6l.SL,a,T*1&.M\$M (OR OTHER »),PPKO/ED
 ME«4* EXPOSED Z BAR SYSTEM 643

Construction Details and Finishes CEILINGS Gypsum Board Suspended
 Ceilings ASSEMBLY REMARKS GYPSUM BOARD, ATTACHED
 DIRECT TO FRAMING 16" 10 24" O.C. SPACING OF SUPPORTS
 secured directly to framing members or to solid furring, most widely used
 in residential and light commercial construction. two layers may be required
 for an improved fire resistance rating or for better resistance to sound
 transmission. directly affected by deflection and/or expansion/contraction in
 supporting framing. FURRED-DOWN SPACING OF SUPPORTS METAL
 OR WOOD FURRING CHANNELS ONE OR TWO LAYERS METAL
 RESILIENT CHANNEL ONE OR TWO LAYERS hat-shaped or resilient
 channels may be used. furring will minimize effects of deflection and
 expansion/ contraction in framlng>upon membrane. resilient channels also
 used to Improve resistance to sound transmission. furring will also
 minimize effects of streaking due to temperature differential which may
 occur with direct attachment. GYPSUM,BOARD, SUSPENDED
 PRIMARY SUPPORTS ONLY 60"to98" + o,c, SPACING OF SUPPORTS
 24"lo46"o,c. l=r-1 —i— OPTIDNAL MANCER I 4—in—i-
 NAILINGCHANNELSOR METAL JOIST FURRING PREFABRICATED
 METAL SUSPENSION GRID PRIMARY AND SECONDARY
 SUPPORTS 60" 10 98"+ o*. 60" 10 96" +o.C | 5| *| SPACING OF
 SUPPORTS tZZZ^T"^ FUHRINGCHANNELS MAIN CHANNEL
 FURRING CHANNELS MAIN CHANNEL SECONDARY SUPPORTS
 when framing is spaced more than 24 inches on centers, or when a plenum
 space for mechanical/electrical service lines is required, a
 suspension/support system consisting of wood or metal sections or special
 nailing channels is generally provided. prefabricated metal suspension
 systems are available. primary suspension system may also include a
 secondary system of furring channels used to align the primary system
 and/or to provide resilient mounting of the membrane. it is a high cost
 assembly and not widely used. resilient furring channels generally used
 with wood framing. 644

Construction Details and Finishes CEILINGS Plaster Suspended Ceilings
 ASSEMBLY REMARKS PLASTER, ATTACHED DIRECT TO
 FRAMING 24" o.c. SPACING OF SUPPORTS 24" o.c. METAL
 LATH GYPSUM LATH metal or gypsum lath secured directly to framing,
 membrane will be directly affected by deflection and/or
 expansion/contraction in supporting framing. metal lath may be backed for
 machine application of plaster, fire resistance ratings for different
 assemblies have been established. FURRED-DOWN 24" to 60" o.c.
 SPACING OF SUPPORTS 24" to 60" o.c. LATH supported by furring, furring
 will minimize effects of deflection and expansion/ contraction
 upon membrane. large areas of membrane should have expansion joints and
 should not be restrained at the perimeter, corners of openings In gypsum
 lath membranes should have metal lath reinforcing. FURRING
 CHANNELS 16" to 24" on CENTER FURRING CHANNELS 16" to 24"
 on CENTER PLASTER, SUSPENDED PRIMARY SUPPORTS ONLY 16"
 to 24" o.c. 16" to 24" o.c. SPACING OF SUPPORTS METAL LATH
 FURRINGCHANNELS 16" to 24" on CENTER GYPSUM LATH
 FURRINGCHANNELS 16" to 24" on CENTER suspended membrane with
 furring channels only Is similar to furred membrane except that furring
 channels are suspended from, rather than directly attached to, framing
 members, suspension of membrane may be a requirement In some fire
 resistance rated floor or roof/ceiling assemblies, spacing of hangers Is quite
 close and limits the size and/or extent of mechanical/electrical service lines
 in plenum space. PRIMARY AND SECONDARY SUPPORTS 24" to 60"
 o.c. 24" to 60" o.c. SPACING OF SUPPORTS 24" to 60" o.c. METAL LATH
 FURRING CHANNELS MAIN CHANNELS when
 spacing of framing is wide and/or the number of hangers must be reduced, a
 primary support system consisting of main carrying channels may be used;
 the furring channels are then a secondary system, secured to such primary
 supports, for wide hanger spacing, metal joists Instead of carrying channels
 may be used, GYPSUM LATH FURRINGCHANNELS MAIN
 CHANNELS 645

Construction Details and Finishes CEILINGS Exposed Grid Suspended
 Ceilings ASSEMBLY REMARKS EXPOSED GRID: FLAT UNITS
 SQUARE EDGE DUCTAND NON-MODULAR DIFFUSER -HANGER

OPTIONAL CARRYING MAIN [CHANNEL RUNNER / T\ OPTIONAL
 BOX & FIXTURES •-HANGER •—HANGER OPTIONAL
 HOLD-DOWN * CLIPS SECONDARY SUPPORT J- MAIN CHANNEL
 f>1 IPC StUUNUAMr RUNNER A OPTIONAL BOX ,__« LIGHT
 FIXTURE C^S SUPPORT I MJ FIXTURES J0 A| rjr i ■!(___ " * H I . _.. I
 ..! _ UAIUDIIIUUCn CICin^lITTIIC WALL ANGLE CROSSTEE MAIN
 RUNNER MAIN RUNNER FIELD CUT TILE lay-In panels should be
 secured In place by clips when assembly requires a fire resistance rating;
 also against uplift due to pressure differential. fixtures generally have to be
 boxed-In for fire resistance rating; fire dampers must be provided at all
 openings, such as dlffusers. hangers secured to framing members, structural
 deck, or to secondary framing system. RECESSED EDGE
 FIELD CUT TILE AS REQUIRED CARRYING CHANNEL i MAIN *[]
 RUNNER , S l4= • HANGER OPTIONAL HOLD-DOWN CUPS A \t.
 HANGER-, ' LIGHT FIXTURE WALL ANGLE CROSSTEE gg^Si
 clearance required for ail lay-in panels for tilting them into place,
 suspension system used Is the same as for square edge tile, but tile only
 available in 2x2 foot size, may be used in fire resistance- rated floor or
 roof/ceiling assemblies; clips to secure tiles in place and opening protection
 generally required. MAIN RUNNER EXPOSED GRID, SHAPED UNITS
 INLAY PANELS, CORRUGATED, RIBBED •-HANGER LIGHT
 FIXTURE i OPTIONALS3UND ABSORBING BLANKET OPTIONAL
 SECONDARY SUPPORTS •-HANGER pi-T . II ,X MAIN RUNNERS I
 MAIN RUNNERS '- MAIN RUNNER ' ,, ,,, ' CROSS Tec -HANGER
 OPTIONAL SOUND ABSORBING BLANKET MAIN RUNNERS '-
 MAIN RUNNER INDIVIDUALLY SUSPENDED metal panels generally
 perforated, with sound absorbing blankets, plastic panels generally solid;
 used in luminous ceiling installations. corrugated/ribbed metal or plastic
 panels generally used with main runners only, flat plastic panels generally
 either 2x4 or 2x2 feet in size, used with main runners and cross tees. PRE-
 ASSEMBLED MODULES LIGHT FIXTURE LIGHT FIXTURE RE'
 PTIONAL AIR ■>■ VENTILATING TILE TV -\ CROSS RUNNER/TEE
 MAIN RUNNER MAIN RUNNER AND LINEAR DIFFUSER flat pre-
 assembled modules are also available. when pressurized plenum and
 ventilating tile are used, air return must be ducted through plenum, with
 ventilating plenum, dirt streaking may result unless the membrane is made

completely air tight. may be used in fire resistance- rated floor or roof/ceiling assemblies. 646

Construction Details and Finishes CEILINGS Concealed Spline Suspended Ceilings ASSEMBLY REMARKS CONCEALED QRIO, SHAPED UNITS METAL PAN TILE SOUND ABSORBING BLANKET VENTILATING TILE LIGHT FIXTURE tile may be repeatedly repainted without loss in sound absorbing characteristics. heating/cooling piping may be incorporated into the system. combination lighting/infra-red heating fixtures may be integrated into membrane. secondary suspension system generally required. tile may be used for supply/return air. LINEAR PANELS HANGER MAIN RUNNER AIR SUPPLY OPTIONAL SOUND ABSORBING BLANKET MAIN PANEL- J MAIN RUNNER INTEGRATED LIGHT FIXTURE formed prefinished metal panels in long lengths, air supply/return and lighting fixtures may be integrated into the system. may be used outdoors In protected locations, such as large soffits, canopies. some assemblies may be used as required components in fire resistance rated floor or roof/ceiling assemblies, membranes may be curved perpendicular to direction of panels. BAFFLES MECHANICAL/ELECTRICAL SERVICES ABOVE MAIN RUNNER baffles available in shaped metal, with or without sound absorbent material cores, or in faced sound absorbent material, various arrangements available, such as linear, radial, hexagonal, used to: provide additional sound absorption In selected locations; for visual interest, or to conceal mechanical/electrical services. METAL SHAPES SOUND ABSORBING MATERIAL CONCEALED OF WFT, *LAT UN»Tt ***** ig""jw*yr'w*T', "^'w *****-! "-""w^"1 KERFED EDGE SPLINE CROSS TEE ACCESS TILE ^^^ (3-MAIN RUNNERS-ft ABB MOLDING tile, generally 12x12 Inches In size with kerfed edges secured in place by main runners In one direction, and cross tees or splines In the other, secondary supports, such as carrying channels may be used to reduce spacing of hangers to framing system. may be used as component In fire resistance rated floor or roof/ceiling assemblies, special panels available to provide access to plenum. 647

Construction Details and Finishes CEILINGS Suspended Ceiling types ? 3 5 l&"w*!m»-&-" (biz fvwr ttt ■ SUSPENSION SYSTEM. tile TtStt .

I2VI7V5*' nee-asrAePANT I-He CWtl«N — OHl-MSUAL-. 1z1tW*1N*.I*
]'z"Rmiti*s & «"PWEIH» & * '<fr m — MflrfIKL LAtU .I44J2P WHITE
 ■^y/PSUM PLdStEK Vf=Vt*eJl. JOISTS, Erz. J 2§"<CVR *c
 ifFyiTRlNoS- fit ? sp~2_ s ■P.w.<;.Fi/ReiMc» C » 1"*5 ■Sp/Wn*- #" wv.r.
 (amt) JH '«. •IW !* AN ^ Concealed 2-spline system with acoustical tile
 Concealed 2-spiine system with aluminum-clad acoustical tile Suspended
 piaster ceiling Suspended ceiing with gypsum board Suspended ceiling
 with plywood finish 648

Construction Details and Finishes CEILINGS Suspended Ceiling Types -
 iVWU. fW#4 -UBS. ft*ik\ PIAM •fW HOW\Hif\ t «tf FUcRimsi I6*W,
 ^HANNBU £UP| VSfIW! WkN(WER€> - ¥*>" OC, MAX, (- #,T,0c*puC
 (MtlUN^ Vf- TYPICAL HUNG GYP.BOARD CEILING DETAIL <ZK0(&
 TVS- H \ MAIN TEC (. \h PtSE-lMeTET, WAU. C-HAUWEL- L)HYUK-
 fivii«7 U*r-IN CGUA& liUs5 Z'-Z-' T- EilMNesfM wipe HAN«Pfc - 44-o"
 <9^' I L |WAU- pNg* - <=«* pMlfeft ftAN TYPICAL LAY-IN TILE
 CEILING DETAIL Hz- yjtrfieN <y\AHNeu IV | —1 wze H*H&tC - O-'-ef
 OC, max, —\VOVG\H& - 1 (>C*V?U\&> WALU TYPICAL CEILING
 DETAIL 7 VPICAL CONCEALED SPLINE CEILING DETAIL 649

Construction Details and Finishes CEILINGS Suspended Ceiling types
 emeTNO 9T«JCTL«*a. mwm. H/SPRAYON FWBI?ROOP»<*>.
 >aruE4uiRMP. EMOTMS- tMEN WALL MNTCPeXNO. P.W.C. rURBING
 CHANNELS. yi5' GYPSUM BO. FASCIA. PfTV OUTWW. ACOUSTIC
 MOUUDlf-ta SUSPENDE ACOUSTICAL, TILC CLO. ^ - ijfc'
 FLHAINO CHANNEL MACS. AS RESUIRED. - SUSPENDE
 ACOUSTIC TILE CLQ, AT el-g,' TTP. • 'W' MVEAL MOLOINO , W»M
 75 (OH tflUALj So' OlWuM BO. RA4C1A. - D.W.C. FURRING
 CHANNELS. FTTT DRY WALL ACOUSTIC MOLOINO. =PA ■ BIS CK
 ESWAL. - BUSPtNOED ACOJ6T1CAA. T1LX CLQ. CEJLINC hEIOHT
 S'-O' A-[DST. NO. 2 CEILING HEIOHT f- S' AT D*T- NO. 2 A ijfc'
 BLACK IRON o.w.c. ^LftflING Channel FLUORESCENT LIOMTINQ, 2
 RONS, BTAGOeMO. •a' Channel hangers. D.W.C. FURRINO
 CHANNELS. 5/3' STPSUM SO. il'uVi' lcuvbrs. we plastic (i't^'V *—
 MNUM L3.iA.0N WITH ELEV. CO* HEAQ J ^6' GTPSLM BO. FASCIA.
 3SACINO AB SXau\HMO. suepei-oep acoustc til& clg . ="*Y P^TT"
 ACOUSTIC NIOLDING FDA - S2S OK. E4UAL. 650

Construction Details and Finishes CEILINGS Curved Ceiling with
Recessed Lighting fi«ein« % "(uf At h*t»i. tuartHMOtt *mc* w*. «* ae. 5
<9F OWCL CBUMcS V*ULT5 » f- 651

Construction Details and Finishes CEILINGS Vaulted Ceiling HfiAP^eS-
^M^E-l RPKfe-L JAH^©f?l?TAL £<_au&. i&"« I'-i g£^>%J§ffL eouJMFJ
4*4 ^s, •eeneoMduaKfc* 1 !'■<£ tWe/h-H-*? 6S2

Construction Details and Finishes CEILINGS Miscellaneous Details of
Suspended Ceilings Dznnr © t?ETAiL-e<?r-Hr u cafbteeia at kmc*>h wall
© P'B.fAIL- 4,ap-PIT IM C*f-e""E.|£|A <-|L0' ifrutKM& - GtlWNfr e
pfeTMI-LIM5A|t t?iP"-U»eR, e iw. MITHEP WIW1* ^ PtTAIL- Cg^E AT
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0*tfift 4V*-eM-*ll LitriTIM* «aH»DUUE. e Mi C*. 653

Construction Details and Finishes CEILINGS Miscellaneous Details of
Suspended Ceilings ;Jj\ If ^'Ml5faU"3ftJ05 MWkIphkIkIB^C- >WfeU/J
«Sf? U^IM AOWf: tL£'CBIW<3 W" >bft?Tl<?N THKU twr, &(?. WWW
^ &W teW) WNPWW A^i 4 NKTIW^H 1 »>»*iaoJ *wokn^5 n~v b tvi'i'i'!
1 «■— r — t UkOHyVCGGP aesfrir^-- , t&rm- Ia" h£D,l atoA&^.ffcfal
•susp Accusf, jrUrdV At, sn kJINRPW MgAP - AW. r^we fl(/crr 3*1%, 654

Construction Details and Finishes CEILINGS Miscellaneous Details of
Suspended Ceilings wi.esBfi*s£ afi aw vpwwu«e - v<p rc*Jcu— ■*£? —
he**—c—r —on AT d*-ir unaj* be: — *fc *U W-W&l * t>'Cb ■—iftrm?
6-£.k>*H tfff-EP. CEjl^kj&i p£.TijL ^5/ iOLA ' V MeTaj- T.kJ CEiUMfti
PE-TAIL 'pCT ^ - *yjr ~tx*uo o-m caul ^up-ocr ^ y\ e>iHrig< me.ial ieu-
m^ octail. V«i'-e* /y\ rV^HO>i. f»\ME.^ <^e-iuikjai P&>^i- _ *—H'*fIK" -
H£/W- *T-JPi»- 3F—' 3=1 V£ -v-^vrw K^vstc rAwe-J L / 03 z. ■k nsanc
52=3 3H3E —XMJtT" Tile ■(&*+ *ay-iM.) ^*s, ?£->"u- gf RA.1&&P rV-
HOuy^T p-j-jfew CEJi^imC; 655

Construction Details and Finishes CEILINGS Suspended Ceilings:
Perimeters and Drops %.VIAaK«MKM.,%»S3MMMkSU 0 SOT -i
>>rsj/>>>////. iJIKfc V 'V6 sfzzzp FTrnfc 1 <JU,.HT. tf-l* \ hLM* i W^T
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I (Ur. «JjJ wm^mmfi 3Z: AUtffJ T^T 5* y /»* ivr 2\ ^N|*. AlfK, ^Lfr. - TTf,

TO 4 O •nu> a*. Qt-lSA.f!f, t& ± _ _ TIW «Upp/ ir'-n. rmwN iki&,N6,
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klIMCPld MS^P C? Q*"fla>K 656 1- All«iw -eqr-tiv t&< WmsmssEE mm
■me #<\$t, w w (^ am w.A6& -m

Construction Details and Finishes CEILINGS Gypsum Board and Plaster
Suspended Ceilings © VAttft FVOH, tUariHA Vf CA.T4L0A. POt
«»"6C1F«CA!TIOI*4 1 iXPLAw€>TE.12: QR METAL LATU J, 6TEEL
FURRIHA.: / — Hwsxsee*-sam t>6rrA»i_*i a»ov» — :tya" *uaa.in6
chankicl« -V-ofo.©. © ^.twaD;- r. PLA/STSE. OK "GYPSUM L&TM' j
6TEEL FUP.g.l^g ■.yAmelia* - 4t» PKT*iu -i- A&ovtt. .. • 3&'fiiYP^Y^A
»OA>«JS MITAL FUaClkKl C* CUPPED Od. W1IM6 cw»etc w-o<sv
coo* fii.A^Jvm^ .CATAi-oa, FOJt.*si«.» RATIMft*

Construction Details and Finishes CEILINGS Suspended Ceilings >i.ih» op
Wall TW BAB. CLVfi *■ * Haugshis -«*■ deta»l*i pv06*ci Ji. I)T2
I"U1*1W HOI I L^-: > 5 J I ^aaaffii:_i l'-d' o.c. ■ £&E puMM»Bi WALL
CMA.UUEL OB. AU6L9 1 -ftoftoeo. tilr clip 4)_ Acoustic tile concealed
"Z£& £u»pfc>aa\ou ^tbm HAUGfe«.4-Sft* OWTAlU#l t>VJfi*C-t l—
ra*o«*:T»*. • - ■■'H'RUMMaw. © ^600«T»C TIL* \X X-1.4 »Vt". WALL
CMAMUtl 00. AUGLH 10W>*B.TIL6 CUP ACOUSTIC TILE y *T
SU6PE.NSIQM £Y4TEM Vefctt* Of WAUL. -aft- cto<s *LLi ta*o.e.
^lj^mW ll/11LFUB.K.»U6 CHAUnWU* i-d' ?Acou*ric tile \ix24" -.'Wall
cUAuet. a a. AuauL fEM.feAJL a-0*O;cu I®. 658 ACOOSTIC TILE ZiL
'SUSPEN'SION SYSTEKN

Construction Details and Finishes CEILINGS Reflected Ceiling Plan TO -
n* I « lAYIH :• UHIT=> ■■^o *& ^f ■r%v' Completed design for an office
space CEILING LEGEND I i I 14 DESCRIPTION iff tss&Mo monten
rMIM I" SBPSHPfD METAl BBoam rfW oshcemsd swif AeansriaM. nmeia
io'x4i'Kcsmso ction, n*ruee ih s'*s' susWDeD mep-i opto wirn ncfxGct /
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El«*»S1 MJIIWST «HHI OP HtlW Al* Sftll (see MBsHi RECC5SEO
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O'X-fRKPSseP fUaR. r*r«se m sosp nri. swo 659

Construction Details and Finishes STAIRS Planning Data Purpose The six
diagrams in Fig. 3 represent unit plans for types of nonwinder stairways

which are most frequently encountered in the average residential planning problem. Tabular information with each was developed from data contained in Table 1, Unit plans are drawn to V_s" scale and therefore can be supplied directly as a check of stair layouts to sketch plans and elevations, Each represents an average condition with a stair pitch well within the comfort zone. The basis is a 9'6" floor-to-floor height with 16 risers each 7.13" in height. Width is 3'0" from wall to wall. Tabular data with each unit plan indicate dimensional variations which occur when stairways of substantially similar pitches are planned for floor-to-floor heights from 8 to 11 ft. Width is the only critical dimension missing from this unit plan information. This varies with requirements of design and stair use and should be selected from data in Table 2. Width is a dimension controlling critical clearances on all stairs that contain a turn. Winders have not been included in these unit plans because they represent a stair condition generally regarded as undesirable. However, use of winders is sometimes necessary due to cramped space. In such instances, winders should be adjusted to replace landings so that the narrow portions of treads at the inside of the turn are at least equal to $\frac{1}{2}T$. When this is done, dimensions of L₁ and L₂ are decreased by approximately $\frac{1}{2}T$ the exact figure depending upon the width selected, The practice of adding a winder-riser to bisect the landing diagonally from the corner of a newel is to be avoided in all cases for it produces a dangerously narrow step in a particularly undesirable place. Application of Unit Plans Diagrammatic data can be used on sketches as a graphic check as noted. Tabular data can be applied to either sketches or working drawings to eliminate the necessity of developing experimental stairway sections to determine run, proportional rise, horizontal and vertical areas, and location of under-rake minimum headroom. Dimensional data have been confined to a single pitch for all floor-to-floor heights. The pitch indicated is that most generally desirable for human comfort. Data for other pitches listed as tread and riser proportions in Table 1 can be substituted for values of L₁, L₂, and M.

STRAIGHT RUN L₁ 11' J₁ 11' J₂ 11' J₃ 11' J₄ 11' J₅ 11' J₆ 11' J₇ 11' J₈ 11' J₉ 11' J₁₀ 11' J₁₁ 11' J₁₂ 11' J₁₃ 11' J₁₄ 11' J₁₅ 11' J₁₆ 11' J₁₇ 11' J₁₈ 11' J₁₉ 11' J₂₀ 11' J₂₁ 11' J₂₂ 11' J₂₃ 11' J₂₄ 11' J₂₅ 11' J₂₆ 11' J₂₇ 11' J₂₈ 11' J₂₉ 11' J₃₀ 11' J₃₁ 11' J₃₂ 11' J₃₃ 11' J₃₄ 11' J₃₅ 11' J₃₆ 11' J₃₇ 11' J₃₈ 11' J₃₉ 11' J₄₀ 11' J₄₁ 11' J₄₂ 11' J₄₃ 11' J₄₄ 11' J₄₅ 11' J₄₆ 11' J₄₇ 11' J₄₈ 11' J₄₉ 11' J₅₀ 11' J₅₁ 11' J₅₂ 11' J₅₃ 11' J₅₄ 11' J₅₅ 11' J₅₆ 11' J₅₇ 11' J₅₈ 11' J₅₉ 11' J₆₀ 11' J₆₁ 11' J₆₂ 11' J₆₃ 11' J₆₄ 11' J₆₅ 11' J₆₆ 11' J₆₇ 11' J₆₈ 11' J₆₉ 11' J₇₀ 11' J₇₁ 11' J₇₂ 11' J₇₃ 11' J₇₄ 11' J₇₅ 11' J₇₆ 11' J₇₇ 11' J₇₈ 11' J₇₉ 11' J₈₀ 11' J₈₁ 11' J₈₂ 11' J₈₃ 11' J₈₄ 11' J₈₅ 11' J₈₆ 11' J₈₇ 11' J₈₈ 11' J₈₉ 11' J₉₀ 11' J₉₁ 11' J₉₂ 11' J₉₃ 11' J₉₄ 11' J₉₅ 11' J₉₆ 11' J₉₇ 11' J₉₈ 11' J₉₉ 11' J₁₀₀ 11' J₁₀₁ 11' J₁₀₂ 11' J₁₀₃ 11' J₁₀₄ 11' J₁₀₅ 11' J₁₀₆ 11' J₁₀₇ 11' J₁₀₈ 11' J₁₀₉ 11' J₁₁₀ 11' J₁₁₁ 11' J₁₁₂ 11' J₁₁₃ 11' J₁₁₄ 11' J₁₁₅ 11' J₁₁₆ 11' J₁₁₇ 11' J₁₁₈ 11' J₁₁₉ 11' J₁₂₀ 11' J₁₂₁ 11' 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10/1' 10 tf IOW If w II' s 13 13 13 13 13 13 13 10-3% W IO'-b»W 10-b'tW
IO'-9"tW ii'-o'.w 1 r-o\ w 11'-0" W 2 " L f 2 z 0 1 2 3 4 ~iT b w w 10W ♦ w
l'-9rf tW 5f-T»w 3-6" »w 4-7". W M 10'-3' IO'-b" 10- b" 11'-O' 11-4' It'-?
Il-b" Fig. 3 662

Construction Details and Finishes STAIRS Planning Data ' L, - 1_ -W- p^ I
w 1 L M WIDE L ■ 1 w 1 I, ul r—1 H- t w 1 1 NARROW u HEIGHT
FLOOR TO FLOOR. 8'-0" 8-b' 9-0" 9-b' 10- 0" 10- fe" ll'-O" W RISERS
13 14 15 lb 17 16 19 2 7.38 7.29 7.20 7.13 7.06 7.00 b.95 1 iow' IO'/j" 10
ft" IOV4" II" If II" a 2 Ll z 7 7 8 B 9 9 10 5'-l'A\W 5'-3\W b'-l/i\W
t'-3W".W 7-4\W 7-4\W 8'-3\W a b 7 7 8 8 9 9 4'-3'/4. W 5-3" 4 W 5-3' .W
6'-3tftW fe'-5'. W 7-4" ♦ W 7:4" 4 W M - A'ti I'-IVi' l'-lim' 2-W ' 3-7' 4-5"
HEIGHT FLOOR TO FLOOR s'-o' B'-b" 9'-0" 9-6' 10-0' IO'-f ll'-O" N»
RISERS 13 14 15 lb 17 18 19 5 7.38 7.29 7.20 7,13 7.0b 7.00 b.95 TREAD
IO'/V iO'fi IO'/a" IOW II' II" II" a UJ 3 7 7 B 8 9 9 10 L, 5-W+W 5'-3%W
i'-l/j\W b-3'/4", W 7-4+W 7-4\W &'-3\W ! L2 4-3WVW 5-3'4 W 5'-3'.W
fe'-W.W b-5".W 7:4'.w 7:4\W M - 4ft" I'-ltf I'-IISTf 1-3« 3-7" 4'-5"
DOUBLE "L" •Li- :—w-i L2 I t_ W M- ■ TT j ; w , fa dJ L2 fcd_1
HEIGHT FLOOR TO FLOOR 80' 8b' 9'0" 9'b' 10-0' 10'b' 11-0" N° RISERS
13 14 v\ Riser 7.38 7 59 7 70 lb 7 13 17 7 0b ,8 Lilj 7 00 r*b.95 TREAD
10/T IO'V IO'/j" IO>/» I L, * 13 12 II 10 IO'-V i2W 9-7/i,2W 8-9' , 2 w 1 8-
OW. 2W II" j_ 9 7-4" , 2W ii | a II' 7 b-5' .2W 5-b' ,2W z 0 1 2 3 A 5 b L2 w
w iO'/i'-iw r-9'A'i w 2'-9" * W 3'-8%W 4'-7\W M id-3".w 9'-7Vi.W 8'-9%W
8-!V<\w 7-8'4W b'-IO' + W b-o\w rf 1 , —w- L rw~i pE t w t I —1 M * ■3
WIDE u HEIGHT FLOOR TO FLOOR B'-O' 8-6' 9'-0" 9'-6" 10-0" IO'-b"
ll'-O" N" RISERS | 13 14 15 lb 17 18 19 RISER 7.36 7.29 7.20 7.13 7.0b
7.00 b.95 IOfe' 10 ft" 10ft" IOW 11' II" 11" 8 a L | X A A A A A TI 4 2-
feWt2W 2-7/i't 2W 2-7li* 2W 2-6»i%2w 2-9' * 2W 2-9' »2W 2-9" +2W N°
RISERSJ 4 5 5 <o b 7 7 1-2 2-bVf-iW 3-b' 4W 3-b" .w 4-5'A\ W 4-7' 4W 5-
b' tW 5-b" tW K 1 5 V-3 z 5 5 b b 7 7 8 3-5" +W 3-b" +W 4-4VI4W 4-
5V44W 5-b" tW 5-b" tW b-5" .W M - 4'.V l-l'ft" MM' 2-9'// 3'-t" 4'-5' Fig. 3
(Continued) 663

Construction Details and Finishes STAIRS Wood Stairs K10-P.5-
DETAIL/OP.MAiKi./TAIR.IVAY..TVVQ./TOa.Y COLON 1AL-MOU/L
4^." cA-o c. PLAN.AT tih/t. £LOOR.-= JOuA ELEVATION-AT-B-
-/ECTIOM-AA- 666

Construction Details and Finishes STAIRS Miscellaneous Stair Details IA»
l/a. 'i: ■ ■■ ■*/. ■'// , 't- ■+• £Zil ' '7 :i>4" ,£>A>V*- f-l/v.HC>t3LA^|i_
"Z." <^ OAvK It-R-AvCK-F.T^ •>> RbraLA^cKTST-^ P>E ra-pf/b^K.- cJP
VVA.UL. ,<^>Avfc^- 4—lA^MTopa./^ ■GLUE FOR THICKNESS AS
REQD SOLID POPUW PRIMED 1/2"BIRCH PLYWOOD 1/2"
PLYWOOD VENEER CORE W SHIM SPtCE FURRING &
SXEETROCK PARTIAL ELEVATION STAIR TYPICAL SECTION AT
STAIRS SECTION AT STAIRS

Construction Details and Finishes STAIRS Wood and Steel Stairs t,«lWV
"~£—mfitspr***" 'i II, I I II II II llli-1 II..I II IT 2N S6CTIOSI THRU
UPPER pllftff LCCXiN^MqglH ' \b) V «I'-o" ^H.ti^-fl' + *Ua!V 1 I II II I!
II III uiy^w1 L**'^ 4 llfai-a* T\ WEST BLBWIOl Aisy M'-i'-o" ■2«»4(LC
2'^«TUV»oMt- 1BikX(»«tl.C = z^oifjtc lAiy |i"»i'-o" eaaag-ftkift Of fill.
MOUNTING.! ec*rw-io-pt*»<o t*6EeTWSM3 SiTWW— 670

Construction Details and Finishes STAIRS Wood and Steel Stairs SiU.
tm.v*lt Ai« 0»iVN cjuw.aioiwfa-', W|,il|'MJ»«MK\ owyna !>■> UicJ I""*."-
i'-o" (tffrwiuiiw* -ajeicMwa? two Y fiCOEW imlJMtl^Wlft; J -
stsasnassa?) ^flMtas* ID -ixz*it4-'w«i' iwiLCM ^<4«lii.elir tiMacn»
FIN.ftDofc ^ ^ -4x(isrt..ccur=«i^' ft1) aula* laxizFUmjaera. 671

Construction Details and Finishes STAIRS Spiral and Circular Stairs
SPIRAL STAIRS I—^—I 5' diameter -spiral CIRCULAR STAIRS 5'
diameter -spiral 6' diameter -spiral 15'diameter-circular 674 Note: All
drawings left-hand turn (looking down). UP Z'-9fe" TREADWIDTH
IMV'I"* 11'diameter-circular w-w 13'diameter-circular - FINISH
OPENING / UP 3'-3V TREADWIDTH LANDING 15'diameter-circular

Construction Details and Finishes STAIRS Circular/Oval Stairs Elevation
675

Construction Details and Finishes STAIRS Spiral Stairs WOOD TREAD
SCREWED TO STEEL PLATE a"WOOO TREAD PLA V "t SECTION .
■*■ b in 0 |i^ *■ ■ o • ^ ■ ■ W /. ,!;* i -iL ri i* ■■;■■'.■ u S'-O" . , 0 ■'■#;
<a'R, .>■'." » tf .o •; o> 6 *** FIN. GRAOE l2"Xi2"XS/e"ST, PLATE
SECURED ^WITH FOUR S/B" BOLTS V-v ' "#";■ £■'■ M>? . ■\$.*■ 676

Construction Details and Finishes Types of Stairs Four types of stairs are defined: straight stairs, circular stairs, curved stairs, and spiral stairs.

Straight stairs are by far the most common type, representing the bulk of the stair market. Though the term "straight" is self-explanatory, for purposes of classification a straight stair is defined as one in which the stringers are straight members. Straight stairs, unlike stairs of the other three types, may be arranged in several different ways: Straight run: Either a single flight extending between floors, as shown in Fig. 5A, or a series of two or more flights in the same line, with intermediate platforms between them, as shown in Fig. 5S. Parallel: Successive flights which parallel each other and are separated only by one or more intermediate platforms, as shown in Fig. 5C. Angled: Successive flights placed at an angle of other than 180° to each other (often 90°), with an intermediate platform between them, as shown in Fig. 5D or £ The type shown in Fig. 5D is often referred to as a "trussed" stair. Scissor: A pair of straight run flights paralleling each other in plan and running in opposite directions on opposite sides of a dividing wall, as shown in Fig. 5/f Circular stairs are stairs which, in plan view, have an open circular form, with a single center of curvature. They may or may not have intermediate platforms between floors. Curved stairs are stairs which, in plan view, have two or more centers of curvature, being oval, elliptical, or some other compound curved form. They also may or may not have one or more intermediate platforms between floors. Spiral stairs are stairs with a closed circular form, having uniform sector shaped treads and a supporting center column. Classes of Stairs The class designation of stairs, as already noted, is a key to the type of construction, the quality of materials, details and finish and, in most cases, the relative cost. As stairs of all classes are built to meet the same standards of performance in respect to load carrying capacity and safety, these class distinctions do not represent differences in functional value, but in character and appearance. It is important to recognize that where function is the prime concern, and esthetics are of minor importance, significant economies can be achieved by specifying one of the less expensive classes. The following four classes of stairs are listed in order of increasing cost (as a general rule); the general construction characteristics of each class are described. Industrial class. Stairs of this class are purely functional in character and consequently they are generally the most economical. They are designed for either interior or exterior use, in industrial buildings such as factories and warehouses, or as fire escapes or

emergency exitways. They do not include stairs which are integral parts of industrial equipment. Industrial class stairs are similar in nature to any light steel construction. Hex head bolts are used for most connections, and welds, where used, are not ground. Stringers may be either flat plate or open channels; treads and platforms are usually made of grating or formed of floor plate, and risers are usually open, though in some cases filled pan type treads and steel risers may be used. Railings are usually of either pipe, tubing, or light steel angle construction. Service class. This class of stairs serves chiefly functional purposes, but is not unattractive in appearance. Service stairs are usually located in enclosed stairwells and provide a secondary or emergency means of travel between floors. In multistoried buildings they are commonly used as egress stairs. They may serve employees, tenants, or the public, and are generally used where economy is a consideration. Stringers of service stairs are generally the same types as those used on stairs of the industrial class. Treads may be one of several standard types, either filled or formed of floor or tread plate, and risers are either exposed steel or open construction. Railings are typically of pipe construction or a simple bar type with tubular newels, and soffits are usually left exposed. Connections on the underside of the stairs are made with hex head bolts, and only those welds in the travel area are smooth. Commercial class. Stairs of this class are usually for public use and are of more attractive design than those of the service class. They may be placed in open locations or may be located in closed stairwells or in public, institutional, or commercial buildings. Stringers for this class of stairs are usually exposed open channel or plate sections. Treads may be any of a number of standard types, and risers are usually exposed steel. Railings vary from ornamental bar or tube construction with metal handrails to simple pipe construction, and soffits may or may not be covered. Exposed bolted connections in areas where appearance is critical are made with countersunk flat or oval head bolts; otherwise, hex head bolts are used. Welds in conspicuous locations are smooth, and all joints are closely fitted. Architectural class. This classification applies to any of the more elaborate and usually more expensive stairs, those which are designed to be architectural features in a building. They may be wholly custom designed or may represent a combination of standard parts with specially designed elements such as stringers, railings, treads, or platforms. Usually this class of stair has a comparatively low pitch, with relatively low risers and correspondingly

wider treads. Architectural metal stairs may be located either in the open or in enclosed stairwells in public, institutional, commercial, or monumental buildings. The materials, fabrication details, and finishes used in architectural class stairs vary widely, as dictated by the architect's design and specifications. As a general rule, construction joints are made as inconspicuous as possible, exposed welds are smooth, and soffits are covered with some surfacing material. Stringers may be special sections exposed, or may be structural members enclosed in other materials. Railings are of an ornamental type and, like the treads and risers, may be of any construction desired. STAIRS Classification of Steel Stairs General Requirements, All Classes of Stairs All fixed metal stairs, regardless of class, are of fire-resistant construction and are designed and constructed to carry a minimum live load of 100 pounds per square foot of projected plan area or an alternative concentrated load of 300 pounds applied at the center of any tread span. Railings and handrails are designed and constructed to withstand a minimum force of 200 pounds applied in any direction at any point on the rail. I / Uf VP_ Fig. 5 677

Construction Details and Finishes STAIRS General Purpose Steel Stairs STRINICS TAP-5CRE.WE.D TO P05TS PLATrORM L PLAN B-B START-AT POST TRE.ADS PLAT. . I WALL 5STRING SCALE. ■START-AT WALL I -O PLAN A-A LAYOUT FOR: MULTIPLE STORY INTERMEDIATE PLATFORM TWO RUNS PER STORY SPECIFICATION FOR STAIR AS DETAIL5D, Furnish and erect steel stairs and railings complete as detailed, Strings 10"kl Vi "x8,4 lb. channels with 1 Va "x 1 Va "x/i " angle brackets, facias same section bolted to newels and floor construction. Headers of channels baited or welded ta newels ond strings, Angle struts ploed in wall, bolted to strings and to floor construction. Risers and sub-treads of 14 gauge steel, sub-platforms of 12 gauge steel reinforced with angle ar tee stiffeners. Fill, 2" for treads, 3" for platforms, by others, Newels 4" square pipe, railing balusters Vi" square spaced 4 Vi" and welded into l'V/2" channels top and bottom with hdndralt section as shown. All surfaces to be cleaned and painted one shop coat. Shop drowings, to show construction methods and fastenings, are to be approved before fabrication, ALTERNATE SPECIFICATIONS! Stringt may be channels, flat plates, ar farmed plates. Tread Brackets may be other size angles, or bars. Riser Brackets may be omitted. Hanger Rods may be

used In place of struts. Sub-Treads, Risers and Sub-Platforms may be heavier gauge. Newels and Railing* may be of other construction as designed by architect. Wall Rails, where required, may have same handrail section as railing. Prime Coat may be red lead, black graphite, line chromate, or other approved paint. 678

Construction Details and Finishes STAIRS Steel Stairs NUMBER Or B.15ER3 VARIABLE. -sncTioN c-c „PLAT TREADS sgzzz ■SECTION F-F SECTION K-K ROUGH WE.L.L, PLAN J-J RQUgH WE.LL r—K PLAN H-H DN CALE. LAYOUT FOR: MULTIPLE STORY INTERMEDIATE PLATFORMS PLATFORM AT FLOOR TWO RUNS PER STORY Strings and rails finishing against face of rectangular newels, allowing minimum hand clearance between strings and rails. For wider center well, two square newels replace one rectangular newel. <£- RAILS PLAN P-D LAYOUT FOR. MULTIPLE STORY INTERMEDIATE PLATFORMS THREE RUNS PER STORY Open center well allowing Intermediate stair runs at 90 degrees. One square newel at each platform. l'-o" PLAN g-G LAYOUT FOR. MULTIPLE STORY INTERMEDIATE PLATFORMS ONE AND TWO RUNS PER STORY Arrangement for stairs in corridors, or other restricted spaces, either closed or open well. f H IT If IT 679

Construction Details and Finishes STAIRS Steel Stairs A LY£ RNATE, T RE.AD CON5TRUC TION 5TART- AT POST 'I I] I WALL STRING il I ANGLE STRUT 5CALC. %" - I START-AT WALL 680

Construction Details and Finishes STAIRS Steel Stairs V , PLATFORM ^- —^—ij'* p. WALL RAIL ENWRE.TURNE.D 3TART-AT POST WALL 5TRINS SCALE. 3 * |-o" uL ii'4 PIPE. WALL RAIL CLOSED END WITH BRACKET 681

Construction Details and Finishes STAIRS Steel Stairs ALTERNATE ALTERNATE ■STRING CONSTRUCTION TREAD CONSTRUCTION SUPPORTING BRACKET5 ARt OMITTED WHEN TREADS ARE WELPED TOSTRIN«3 ?*--TIRE IRON XT" HALF OVAL ALTERNATE. RAIL CONSTRUCTION 682

Construction Details and Finishes STAIRS Stair Platform Construction The platform (Fig. 6) is shown constructed with a steel channel, A, of adequate strength to span the well on line X, through Sees. 1, 2, and 3, and supported at both ends by the wall strings. Newel posts rest on this channel through angle clips, around which the platform plate is cut (Sec. 3). Face strings have welded end plates with flathead screws tapped into the newels (Sec. 2). The two platforms with two intermediate risers (Fig. 7) are shown constructed with the load carried on line Y by string B, post C, and channel D, which are shown bolted together (Sees. 8, 9, and 10) with through bolts. The load is also carried from post C on line Z in the same manner. The members at post C may be brought together and welded and the post fitted over the connection, or the entire unit welded.

FJg. 6 Plan—One Platform. Load carried on Line X, Wall strings supported on masonry wall, SCALE. '•'-° Fipl. 7 Plan—Two Platforms with Two Intermediate Risers. Load carried on Line Y and supplemented on Line Z. Wall strings supported by struts. Stairs are supported by one or more of the following methods, (a) String at floor rests directly on floor construction; (b) String at landing or platform extends into adjacent load-bearing wall; or (c) String at landing or platform is supported by struts extending to the floor below, these being of angles, I-beams or pipes either set in the wall or exposed; or (d) String at landing or platform is supported by rods hung from the floor above, either set in walls or exposed; (e) String paralleling load-bearing wall may have shelf brackets on the back of the strings and set in wall; similar brackets may be used with struts or hanger rods. Angle, strut (omit on fig. q) 7 CONDITIONS ILLUSTRATED; Concrete or terrazo fill. Open unplastered soffit. Square steel newel posts. Steel Channel Strings, 683

Construction Details and Finishes STAIRS Stairwell PLATE[^] USUALLY ONE-HALF / * Or 3/4 IN. WIDTH "x: Width of stair is usually considered center line of rail to finished wall. When wall rail is required allowance should be made for clearance to comply with any legal requirements as to net width. 41: HAND CLEARANCE -NOT LESS THAN 44 IN. VARIES \ ONT-MALr Of HANDRAIL WIDTH ROUGH WELL METHOD Or ESTABLISHING, THE. WIDTH Of A STAIR WELL - Pt/UTER Height of riser and width of tread vary to fit the type of stair and its use. Legal requirements often limit the minimum tread and maximum rise. A tread of 10" and a rise of 7" to 7 1/2" are considered average, Stairs of

easy runs are often 10' x 10" treads with risers under 7". Stairs used exclusively by maintenance and operating men are often constructed with a rise and tread to equal a pitch greater than 40°. Tread width is always face to face of riser. BEAM ■ HALF or CLEARANCE METHOD OF ESTABLISHING THE LENGTH OF A STAIR WELL 684

Construction Details and Finishes STAIRS Stair Width Dimensions Width of stair is usually measured from inside face of balusters or newel to finished wall. However, governing codes should be consulted for points of measurement. Q A minimum clearance of 3" should be allowed between edge of stringer and wall. REFER TO GOVERNING CODES TO ESTABLISH DIMENSIONS 3" maximum projection into required egress width. If more than 3/4", egress width should be increased by the excess of projection over 3/4". "x: Minimum 4" hand clearance between rails, and between handrail and wall, or other obstructions. — STAIR WELL WIDTH — DISTANCE BETWEEN WALLS Minimum code requirements are usually measured from finished wall to finished wall. When establishing rough stair well dimensions, allowance should be made for thicknesses of any finish materials to be applied to the rough walls. Platform width not less than width of stair, usually measured from inside face of balusters or newel to finished wall. However, governing codes should be consulted for points of measurement. 685

Construction Details and Finishes STAIRS Stair Length Dimensions REFER TO GOVERNING CODES TO ESTABLISH DIMENSIONS Height of riser and tread run vary according to governing codes. A tread of 10" and a rise of 7" to 8" are considered average. Stair treads for more comfortable runs are often 10 1/2" to 11" with risers less than 7". Treads and risers should be so proportioned that the sum of two risers and one tread run is not less than 24" or more than 26". In establishing stair well dimensions, tread run is always face to face of riser. 686

Construction Details and Finishes STAIRS Tread and Riser Construction Steel riser and sub tread with formed nosing, angle supporting brackets. Steel riser and sub tread with abrasive safety nosing, sanitary cove, flat bar supporting brackets. -V- Steel riser and sub-tread with ^grooved safety nosing and tile or linoleum fill, angle supporting bracket. Other types of

safety nosing or tread covering may be used. -<, v~ I1 " 12. I-O Grating Tread. Steel riser and sub tread with formed nosing, angle supporting brackets. "Steel riser and sub tread with "rolled steel or extruded nosing and tile or linoleum tread, angle supporting brackets. Other types of safety nosing or tread covering may be used. -u- Steel floor plate tread, angle tread bracket, with or without steel riser. Steel riser and sub tread with lead filled safety nosing, sanitary cove, angle supporting brackets. Steel riser and sub tread with marble, or pre-cast tread, angle supporting brackets. Cast metal abrasive tread, with or without steel riser. Steel riser and sub tread with precast or wood treads, angle supporting brackets. Stairs with concrete or terrazzo fill may be constructed with the top of supporting bracket 2" below the tread surface and 3" below the platform surface. These thicknesses may be less for narrow stairs or where use is limited. Fill is always considered the distance from string bracket to finish tread. PLASTER SOFFIT Steel sub-riser and sub tread with marble tread and riser, angle supporting brackets. Treads and riser brackets may be 1 1/2" x 1/2" or 1/2" angles, welded or riveted to string, or 1/2" x 1/2" bar welded. Treads and risers are usually bolted to brackets with round head bolts. Cost or groting treads are usually bolted to strings with two A balls at each end. Brackets back of risers may be omitted when more economical construction is desired. 687

Construction Details and Finishes STAIRS Tread Sections Steel sub-tread and riser with formed nosing at 45 degrees, with or without sanitary cove. Concrete filled tread. Steel sub-tread and riser with or without sanitary cove, with square formed nosing. Concrete filled tread. Steel sub-tread and riser with riser sloped to meet formed nosing. Concrete filled tread with resilient tile covering. Steel sub-tread without riser, concrete filled and reinforced. Steel sub-tread and riser formed to receive pre-fabricated tread such as pre-cast concrete. Steel floor plate or aluminum tread plate formed tread and riser. Steel floor plate or aluminum tread plate formed tread with sheet steel or aluminum riser optional. fTaTTl fT] Steel or aluminum grating with nosing. End plates welded to grating and bolted to stringer. Extruded aluminum tread. Steel channel sub-tread concrete filled. 688

Construction Details and Finishes STAIRS String Sections r<3\ PIPE RAIL r> .Jl NON-SLIP 5UP-FACT ^ Flat plate strings with floor plate tread and pipe railing. JttL — RAIL tf nJiii "iiM>m ni fcsagssaawam 11 "

'i^"TT7L,ing I m, r ^ Face string at post. Moldings not mitered unless shown or specified mitered. Steel channel strings with filled pan type tread. The stair should be installed before the face brick or tile walls are built to Insure close joints. Bent plate strings with cast tread, with or without abrasive. f^- ■^1 .SEE NOTE MS,8 31 Structural steel channel strings with grating type treads. Face strngat post. Moldings mitered only when shown or specified mitered. JZ5L Steel channel strings with ornamental molding on face string and channel wall string; plastered soffit. L_ r 3tt note. r\ e,, '8 it Box type face string of plates and two channels. These string sections Include a majority of the various types of strings employed for steel itdirs. Other types also are used, and other methods of combining with railings are sometimes desired. The various types of trim moldings shown are only illustrative of the possibilities of design. The various forms of box lype strings shown illustrate several methods of accomplishing this type of construction. Because of the greot number of extrusion ond rolling dies now In use for the manufacture of moldings af steel, aluminum, bronze and other metols, the architectural plans should give the manufacturers molding numbers selected. If the mold- fngs shown are designed specially for the protect the plans should so state. two B°x type face string of channel, plate and moldings; channel wall string* t m£ TTiMflfTTTFOT "1 CLIPS FOR METAL LATH /IS* TO IS" C.C, =d-t=™ Box type face strings. Fig. 8 Box type face string of steel channel and ferrous or non-ferrous exposed face; steel plate wall string. 6B9

Construction Details and Finishes STAIRS Stringer Sections ' yfaii'tt i?- a ft t?~t7*i foEy/V ft iff/ "ttj£ ^ <"&& ■/■/■/ // it rr ft ~rr iL..j./L./.,lz:s.:./j:::, "^i ^ & Steel plate stringers, carrier angles, floor plate treads, pipe railing on side of face stringer. Aluminum tread plate may be used when specified. Wall not plastered. Steel plate stringers, carrier angles, steel sub- tread and riser, concrete filled tread. Pipe railing on side of face stringer, wall not plastered. pa }£ yr-n Optional Closure Piece iiLhi" "" "" >' "" "" >"=\ \ii " "" "" "" "" ,,,,) Standard steel channel stringers, grating tread bolted or welded to stringer, pipe railing bolted or welded to top flange of face stringer, Wall not plastered, —\-L- Steel junior channel stringers, carrier bars, steel sub-tread and riser. Concrete filled tread. Railing with bottom channel fastened to top fiange of face stringer. Optional closure

piece fastened to top flange of wall stringer In the field. Wall not plastered.
690

Construction Details and Finishes STAIRS Abrasive Nosings and Treads
mmwmi -zr f z^irz-,£l^£-*~\ Cast abrasive nosing with short lip, available
In Iron, bronze or aluminum as specified. Standard drilling with wing
anchors, bolts and nuts or drilled as required, Cast abrasive nosing with
deep lip, available In Iron, bronze or aluminum as specified. Standard
drilling with wing anchors, bolts and nuts or drilled as required. Extruded
aluminum base with epoxy top, containing abrasive. Available In colors.
Integral anchors for fresh concrete. Also available drilled to specifications
without the anchors. Extruded aluminum or brass with abrasive filled ribs.
Concealed Integral anchor runs full length of tread. Also available drilled to
specifications, without the Integral anchor. Extruded aluminum with
abrasive ribs. Special design for pan stairs with sloped risers. Drilled to
specification or furnished with strap anchors or wing anchors. Barrier free
design to meet standards for the physically handicapped. Aluminum base
with epoxy containing abrasive top. Integral anchors for fresh concrete.
W/S7TJ!* Cast abrasive structural tread, available In Iron or aluminum.
Integrally cast end lugs for bolting directly to stringers. ^^f^^j? Cast
abrasive double nosed tread, available in Iron or aluminum and Is
reversible. Supported by carrier angles bolted to tread and either bolted or
welded to stringer. Cast abrasive tread, available In Iron or aluminum.
Nosing and toe plate can be drilled for attaching flat plate risers. Supported
by carrier angles. 691

Construction Details and Finishes STAIRS Handrails Stair platform or
landing with pipe railings, railings not connected, for stairwell having
minimum clearance. Short newels, supported on header. Stair platform or
landing with pipe railing, one post at return. Lower rail returned into post,
two or more posts used at wide wells. Short newels, supported on header.
Stair platform or landing with rectangular or square newel, pipe railing
members capped and welded to newel post. «STie£U...W£sMpi*A\u ' a."
•*■*•" ->7v*.inu*s* *-TTEWl_ MA>HDBA|U 1 <=.4k.ui_K-ir-<es
b<3TTgM m.l«S»g.F». W/iOHTIHUCJa* fil;u>x S(V«(-.tt • 692

Construction Details and Finishes STAIRS Newels and Railings s\\\\\\\\\\\\^
s\\\\\\\\\\\\ \\\\W s\\\\\\\\\\\\\\\\V Stair start with square newel, baluster type railing with
channel top and bottom, pipe handrail. Stair start with short newel, parallel
bar type railing with end and intermediate posts of square, rectangular or
round section, extruded handrail with mitered, forged or cast terminal. Stair
start with square newel, parallel bar type railing with intermediate posts of
square, rectangular or round section; extruded or rolled handrail section
mitered to form cap over newel. Z3P3 3 H3- Stringer PLAN ELEVATION
Railing Railing 3 Welded Stringer- uncoped PLAN 3- ht i\\ iB .if' L-' ^ F
f>*\ u PLAN Bolted Stringer- coped on U ELEVATION PLAN Railing
Stringer Square or rectangular newel, pipe rail fitted with offset lug to
center on stringer. Square or rectangular newel, pipe rail fitted with offset
lug for positioning Inside of stringer. Section showing fastening for
intermediate posts to stringers. Rectangular newel, pipe rail and stringer
welded or bolted to face. 693

Construction Details and Finishes STAIRS Newels and Railings PLAN-
RAILING -jA- i D Stair landing, with stringer and fascia at right angle,
Landing extended on down flight to set-forward riser, producing easement
in handrail. Stair landing with stringers and fascia framed square, Square
railing return, end balusters centered on newels and landing extended on up
flight to set-back riser, PLAN-RAILING 694 Stair landing with stringers
and fascia framed square. Radius railing return, parallel bar type railing
with end balusters centered on newels. Landing extended on down flight to
set-forward riser. Stair landing with stringers and fascia framed into full
height newel, baluster railing with channel top and bottom. Continuous pipe
handrail offset from balusters and newels by brackets.

Construction Details and Finishes STAIRS Ornamental Railings Figures 9
and 10 indicate typical railings for decks, platforms, balconies, roofs, and
similar locations, adapted for residential, apartment, or hotel construction.
These railings may be fastened with wood screws or lag bolts to wood, or
with expansion bolts to masonry. On roofs or decks the setting of the post
bases should be waterproofed. LENGTH Fig. 9 ft—J 4 © © I .SPACE.
FOR ORNAMENT 'PE.SIGN BY ARCHITECT OPENING ./ | Fig. 10 'ii—
ji' Drh DHa Railing with balusters and bottom longitudinal member
supporting balusters. Posts .extending into masonry. SUCTION H Methods

of constructing railing top members. Specify kind of metal. Specify finish. Specify size of members. Give height. Provide details of ornamental, flange and bases. Specify method of fastening or have fabricator provide fastening detail adapted to each condition. SPACE. FOR ORNAMENTAL Railing without bottom longitudinal member, each baluster set in masonry and fitted with slip flange or base. Masonry specifications should specify holes. 695

Construction Details and Finishes STAIRS Ornamental Railings d)
 Ornamental -Z- Railing panels set between columns or jambs. Posts extended to floor construction for support. Railing for balcony or mezzanine with double posts and panels. Posts extended to support fascia and fastened to floor construction. Railing for balcony or mezzanine with curved section. Fascia fastened to floor construction. Railing fastened to fascia. Fig. 11 * 4 .
 ■4 ■d (D- B -S- -6- 696

Construction Details and Finishes STAIRS Center Railings Center railings are recommended for wide stairs. They may be a single pipe or tubing railing or they may be designed with double rails and panels of interesting design. Note: A number of codes require that railings have a level extension beyond the nosings at the floors as indicated in Fig. 11 by dashed lines, This applies to both wall and center railings. * i- Center railing of double pipe, posts extended to subthead. HRr* Fig. 11 Center railing of single pipe or tubing, round, square or rectangular steel, bronze or aluminum. Posts set Into floor as at ends, or extended to subthead and bolted as at center. Flanges loose or fixed. { n U.JJ % Center railing of double pipe, posts set into floor. Railing post set over dowel which is anchored to pan tread and reinforced with structural tee member supported between stringers prior to placing concrete fill. ALTERNATE SECTIONS PLAN A Handrail with square return mitered. PLAN B Handrail with round return. Center railing of two members, with brackets to square or rectangular, posts, steel, bronze or aluminum. Posts set Into floor as at ends, or extended to subthead and bolted as at center. Flanges loose or fixed. HINGED BRACKET Center railing of single member centered on top of rectangular post by means of hinged bracket. Non-ferrous or stainless steel sleeves may be used on exterior rails to prevent staining masonry or concrete, 697

Construction Details and Finishes STAIRS Railings .cl.au aft. fc«LT To
 ■wall w/ |KUMtl«MIKLTi fit TAR.Lt fom Spaai'U4 fth#UMP »M*«TH 1
 ri.o«m KM OF — E-LELVATIOkl o. WELDED PIPE. MILIM6 9 V'. i*-©'
 c^& I'/^PIPE POIT *1T M Flf'. o*i 1HM.iT M»TAL *1_l»V«, J>M«U»fe
 VW/COHC. a* SULPHUR. FLAW*!. W1A.V M, L.00SB, Oh FAtTIMI*
 TO P*»T. M«T*V MOLTIU LIAt> • * U»*t» VM«««U MAV tft Ut«» f«M
 AU«M«*|M4, WMUNt THB TBWdiUtfVT* FL«W /(N*Tf AUCT«m
 IUUOLUIMii VThlWATH MITAU IUftlVC •"J"kv »*TTOM rt..
 VVILMSBTO <3 »»» i1-©" 698 4 i'aVri't I'l'uV fLATB. «7kb»ED TO
 -9TmiM<s»i»i r V7gLLC>i.& PlfE PtXILIM6 PBTAIL FtfH1^ STftt.1,
 STAINS 3 V» »'-«H tilt Of ftfl V" V" iy £" L" 2iL 3" MAXIMUM •
 PACIU6 8' IO'

Construction Details and Finishes STAIRS Railing Posts TYPICAL
 EXTRUDED ALUMINUM AND BRONZE POST SECTIONS 1 V 2%"
 2%« Aluminum only n £21 1%" 2%" Aluminum only Aluminum only
 Various \ f Lfcfj Flanges for box and solid stringers. -+—r y j+t 1'// Flanges
 for channel stringers. uno '(1 c uxr[^_ Post mounted on box Post mounted
 on stringer. channel stringer. ,, , L . t t Post mounted on concrete step using
 post anchor, Sections — Railing posts with brackets, ** Elevation;
 Intermediate post set on face of box stringer. Fig. 12 Section; Intermediate
 post set on face of box stringer. Stair landing with box stringers attached to
 sides of newel, parallel type bar railing supported by brackets at newels and
 Intermediate posts. Risers offset to allow metal soffits of stair to meet at
 intersection with soffit of landing. Bottom and top rails must be the same
 and have symetrleai cross section to obtain proper mitered connection. Fig.
 13 Elevation of Fig. 12: rectangular newel post. 699

Construction Details and Finishes STAIRS Handrail Sections TYPICAL
 EXTRUDED ALUMINUM AND BRONZE HANDRAIL SECTIONS * A
 «-| A = 1-5/8", 1-15/16", 2-1/4" Aluminum and Bronze Km A = 1-3/4", 2-
 1/8" A = 1.66" and 1.90" typical. Aluminum and Bronze Aluminum and
 Bronze Fig. 14 A = 2-3/8" Aluminum A = 1-1/2", 2", 2-1/2" Aluminum A =
 3-1/4. 3-3/4" Aluminum A = 2-1/4", 2-3/4 "Bronze A = 2-3/4", 3"
 Aluminum and Bronze A = 2-7/8" Aluminum and Bronze 2-3/4" Aluminum
 Most of these sections can be mounted on channels or flats, secured by
 screws from below. Some are designed for mounting on handrail brackets.

The use of channels instead of solid bars often simplifies the attachment of baluster and ornaments. The channels may be of the same or a different metal. TYPICAL ROLLED STEEL HANDRAIL SECTIONS A = 1-3/4" Steel A = 2" Steel A = 1-15/16", 2-1/4" Steel A = 1-3/4" Steel Fig. 15 WA 1-3/4" Steel Various Steel Various, 1.66" typical Steel Most of these sections can be mounted on channels or flats, secured by screws or welding from below. Sometimes they are welded directly to the baluster (see Fig. 15) or attached to handrail brackets (see Fig. 14). The use of channels often simplifies the attachment of balusters and ornaments. 700

Construction Details and Finishes STAIRS Handrail Sections

REPRESENTATIVE EXTRUDED AND TUBULAR STAINLESS STEEL HANDRAIL SECTIONS L mzz^j, h-3/4"-H 2-1/4" and 2-3/4" Stainless Steel Extrusion Stainless Steel (other sizes available) Stainless Steel A = Various 1.66" and 1.90" typical Stainless Steel Stainless Steel (other sizes available) Stainless tubular handrail sections usually have a wall thickness of .065". Stainless Steel PLASTIC HANDRAIL COVERINGS v=f. // /> ti -77-77 h o1 Fig. 16 Fig. 17 ^/////////[^ f h—A—1 Fig. 19 Fig. 18 TABLE 4 Table of Dimensions for Plastic Handrail Coverings Fig. 16 17 18 19 A Inside width 1" 1W 1W 1 </i" 1W 2" 1W 2" VA" 1 Vi" B Inside height 'a" %" Va" '// C Outside width 1W 19/16" 1W 1 "/16" 2W 2%" 3W 1%" 1%" D Outside height ' 13/ie" 9/16" 9/ie" "/ib" 9/16" 1*ib" 1b/ib" Caution: Consult manufacturers for fabrication limitations. General Information Functional and decorative plastic handrail mouldings of polyvinyl chloride plastics are available in a variety of sizes and profiles, several of which are illustrated in Figs. 16 to 19. Consult suppliers' current literature for variations in details and features. Plastic handrail mouldings are not structural and require bar, tube, or channel members to support vertical and horizontal loads. Plastic handrail mouldings are produced in a range of colors from subdued to bright, to suit either formal or informal design situations. The color is integral with the plastic, which is highly resistant to wear, weathering, and corrosion. The thermoplastic material becomes pliable when heated (not over 165°F), at which time it can be fitted over the support member and conforms to vertical, horizontal, or combined vertical and horizontal curves within certain limitations. Lateral bends should have a minimum centerline radius of not less than 2 times the width of the plastic section or 2 1/2 to 3 times the width of the support section, whichever is greater. Mitered corners

should be used if sharper turns are required. Combined vertical and horizontal turns can be formed by twisting the moulding. The material can be joined by thermal welding, and end caps can be shaped using a knife, a file, or abrasives. The use of a cleaning solution for removing grease and foreign material is recommended, after which a solvent is used for polishing or removing abrasive scratches. Normal cleaning requires only soap and water. 701

Construction Details and Finishes STAIRS Wall Handrail Brackets Wall rail bracket of conventional cast design, malleable iron, aluminum or bronze. 3/8" bolt Into wall. Wall rail bracket of conventional oast design, malleable iron, aluminum or bronze, 3/8" stud into wall, tapped into arm of bracket. Wall rail bracket of aluminum with fittings to handrail adjustable to any pitch. 3/8" stud into wall. Wall rail bracket of extruded aluminum, made to set at right angle to wall rail or set vertically. 3/8" bolt into wall. Two-piece wall rail bracket of aluminum. Wall plate bolted Into wall through expansion type anchor. Outer sleeve screwed to rail. Outer sleeve fastened to wall plate, by set screw. Optional fastening Wall rail bracket of formed steel. Filler and anchor bolt through gypsum board on masonry. Bracket fastened to filler by three screws or by 3/8" bolt through center. 702

Construction Details and Finishes STAIRS Steel Stairs [1/2"X1/2"i"OC ' .TYPICAL NOSING FOR CONC FILLED TREAD -NOIZ 8A COMBINATION RISER • SUi-TREAD FACE OF RISER I 1/2 ■ NO.IiaA.NCWCLPOST- t TOP KAIL .J/R'FMTAP »««»€■ •tp *5T & T :V ANGLE welded to stringer 10" J. • L.CHAN NCI. FACE STRING-R.S* face of riser JL PLAN OF NEWEL POST M0W1Z0NTAL RA'L AT TOP LANOIXO ANGLE WELDED TO STRINGER ■ TREA0 WITH NON-SLIP METAL NOSINS PLUS WELD CLOSED EHO CLOSED (NO NO. II OA NEWEL POST- TYPICAL DETAIL OF STEP 1- DESCENDING PAIL J/i F. M. TAP SCREW- ASCENDING PAIL- -3/«"fLA' «tAO -MP SC»t» -Closed end CLOSED END - •SOLID CINDER -ILOCK WALL - WALL STRINOER rCONC PLATFORM -I LiW'xik."*1/," CARRIER ANGLE/ 3"CMANNEL - HEADER-*.7* 10'jtL ID" EQUALI CHANNEL STRINI-f.S* WOT ROLLED SECTION MIN THICRNESS .I'S'I i- closeo Euro DETAIL OF RAILING AND NEWEL POST 1/4 I D PIPE RAIL - IALUSTERS E'xi/2* i'OG , j JT EWEL POST .P XI/Z'CHANNCI

SOTTOM RAIL ,5/IS'SENT PLATE -1/4" F X EXPANSION SOLT -FIN
UPPERMOST FLOOR T -ANCHOR S.OLT , CONC REAM CLOSED
END L ROTTOM RAIL NO It f A COMtINATION RISER » TREAD
TYPICAL SECTION -NEWEL POST -WALL STRIMER TYPICAL
FLOOR -r—* r-J-j. .* ANCHOR ROOT "I CONC SLAR CONC NAM
CLOSED END Typical il*tl Hair cemlruction 703

Construction Details and Finishes STAIRS Steel Stairs PLAH-OF-TRIlSS-
ELSEll-AT-HEWEL.-X-[NTEk- AUDJATE-lAMDlHQ-DETAIL-AT-
NEWLL-T-SIMIiAH- Seal*. 3A-l-0" 704

Construction Details and Finishes STAIRS Steel Stairs Pipe cap
tB.Pipehaad roil^-i bracket £LLWT!OK-OF-I«A:iaT •SECTION-THRU-
TBUSS- -Nora-slip nosirag ^Nosing anchor ^ETAIL-OF-MOH-SLIP-MOS'
•IHQ-rOB-TEIAD- •li'-l-O' — ■ { 705

Construction Details and Finishes STAIRS Steel Stairs t pasf-OAti. 706 5 N
SECTION imil ST&1RS AID/ii"«l'-o"

Construction Details and Finishes STAIRS Steel Slair Details At JOI6 &rtr
rsrf- ■o,«*v. «n- pins - ! <a*-wi «^- pipe- : |5*dUlKIO|, |'4> | / -V- ip-SfKJt-
J&l&e, 4- st PIMIW&1 * |og, |>L^rif9ieH j? \ P&1A1L <rf- HAMPK^L /fr-
SpMfc. *£ AI^/SOvLe.; i fe." =!'-<?" \V?4> <v>lv, &ru pins
&fKfc*fU«»JL«[L.. *fu© 707

Construction Details and Finishes STAIRS Steel Slairs STEEL STAIRS
WITH TERRAZZO TREADS *IO**.OH. PA&mcMrajJ 01*.—^^N.
PR.fcas>iE.n* ftTceL ri&br.%, n> MINIMUM R-feCO/A/ASHPED
THICKN&^a FOR. TfeRRA,t2Q TREAP, RISERS,4- STRmSeRS ©
■5TA.tR. P&TAsIt- fTg.RLR.A'ZgO TR-giAD^ STEEL STAIRS WITH
TERRAZZO TREADS AND RISERS ■*IO GAyGB. FA&R.ICATE'
R.\SSTR.-& TAEA.D5 £(. PLATFOR./A* (D ■=—PL*>5TB1\ ^.1/e"
cuo»ep terr«zq - STRStMOEr*. (AIM.) AFidiUklD TR£AvP«i MOT 4Mi^.
THAN "/tC" VICE. • K* PI-ATE STRMOBR. HOTfc .' RUA.TB.
STRINGIER. PUHGH&O TO ... R.ECOIVE. AHCHOH. OF PIHISHBD
*\AI^feLE . ftTRINSER STRINGER. D&TAIL STAIR P&TA11- WITH
TfeRRA'X'ZO R.I=* .R>5> 4 TREADS 708

Construction Details and Finishes STAIRS Spiral Stairs Construction

Material may be steel, stainless steel, cast iron, or aluminum. Treads are supported in cantilever fashion by the column, each consecutive tread being rotated at a predetermined angle. The platform attaches to the column and is fastened to the floor structure to hold the column secure. The spiral railing is supported by balusters attached to the outer ends of the treads. Tread Designs Fabricators provide several standard types and designs of treads and platforms. These include open riser, closed riser, and cantilever types, with surface of checkered plate, abrasive plate, steel grating, or plain surface to receive wood, resilient flooring, carpet, or other covering. Pan type treads to receive concrete or terrazzo fill are also available. Stair Height Spiral stairs are adaptable to any height, the height being equal to the distance from finished floor to finished floor. Stair Diameter Spiral stairs are available in various diameters from 3'6" to 8'0", normally in 6" increments. A 4'0" diameter is considered minimum for general access purposes; a 5'0" diameter provides a comfortable general purpose stair. Larger diameters are used chiefly for architectural effect. Note that the diameter of the finished well opening should be at least 2" greater than the stair diameter, to provide hand clearance.

Hand of Stairs Left-hand stairs: User ascends in clockwise direction, with handrail at left. Right-hand stairs: User ascends in counterclockwise direction, with handrail at right.

WELL OPENING Quarter column ; , : ' tsja—■ **TYPICAL ELEVATION - RIGHT HAND OPEN RISER STAIR EXIT AT LOWER FLOOR** Minimum clearance between well opening and handrail is 1" Well opening may be either square or circular

EXIT AT LOWER FLOOR |Z ^\~/p j PC// r\ y> ~\^y **PLAN AT TOP FLOOR** Left Hand, 30°, Full Circle Fig. 20 **TABLE 5 Riser Heights for Various Tread Angles** **PLAN AT TOP FLOOR** Right Hand, 22V2°, 'A Circle Tread angle

Min. height of riser* Treads per % circle Treads per full circle 30° 27° 24°/2° 22 22° 8'Vi6" 8" 7B/16" 6"/16" 9 10 11 12 12 13 = 351° 15 = 367°/2° 16 'Minimum height to attain 6'6" clear headroom using a 90° landing, 2" thick, **TABLE 6 Chart for Selection of Number and Height of Risers** Floor to height 7'0" 7'4" 7'8" 8'0" 10 8.4 8.8 9.2 9,6 11 7.6 8.0 8,4 8,7 Number of risers 12 7.0 7,3 7,7 8,0 13 6.8 7.0 7.4 and height of each in inches 14 15 16 6,9 17 18 8'2" 8'4" 8'6" 8'8" 8'10" 9,8 8.9 9.1 9.3 9,4 8,2 8.3 8.5 8,7 7,5 7,7 7.8 8,0 8.2 7,0 7,1 7,3 7.4 7,6 6.7 6.8 6.9 7.1 6.6 9'0" 9'2" 9'4" 9'6" 9'8" 9'10" 9.0 9.2 9.3 8.3 8.5 8.6 8.8 9.0 9.2 7,7 7.9 8.0 8.1 8.3 7.2 7.3 7.5 7.6

7.7 7.9 6.7 6,9 7.0 7,1 7.2 7.4 5og notes below 3'0" RESIDENTIAL 3'6"
 OTHERS Plalloim oilher square special Gliapo lo til well opoiwirj 10'0"
 10'2" 10'4" 10'6" 10'8" 10'10 11 '0" 11 '2" 11 '4" 11' 6" 9.3 8.7 8,9 9,0 9.1
 8.0 8.1 8.3 8,4 8.5 8.6 7,5 7,6 7,7 7.9 8,0 8.1 8,7 8,9 9,0 11'S 12'0" 8.2 8,4
 8,5 8.6 8.7 9.0 6.7 6.8 6.9 7.0 7,2 7,3 7.4 7.5 7,6 6,9 7,0 7.1 7.2 7,8 7,9 8.0
 8.1 8,2 8,5 7.3 7,4 7,6 7,7 7.8 8.0 709

Construction Details and Finishes STAIRS Spiral Stairs Diameter and head
 room Spiral stairs may be made in diameters from 3'6" to 6'0" or greater,
 with 4'0" usually considered the minimum for easy travel. The well hole
 should be at least 3" larger in diameter than the stair, for railing clearance.
 Spiral stairs are usually constructed with 12 or 16 treads to the circle. Head
 room should be calculated on the basis of three-fourths of a circle. On a 12-
 tread circle, 9" is approximately the minimum rise, providing 6'9" head
 room. On a 16-tread circle, 7" rise will provide TO" head room, A rise up to
 12" per tread may be employed, $m^{\pm}y^i^{\wedge} <^{\wedge} <^{\wedge} p^{\wedge}$. CLEARANCE PLAN
 AT PLATrORM PLATFORM CAST METAL SPIRAL STAIR
 CONSTRUCTION: Spiral stairs may be constructed with cast metal treads
 and landings of plain or checkered surface, made with or without abrasive
 material. Each tread has a collar which slips over a stand pipe of steel.
 Risers are not desirable as toe room near the stand pipe is limited, The
 treads are bolted together at the outer edge, either by baits or by using the
 railing posts as bolts. Fabricators may have several types of standard design
 treads and landings which may be satisfactory without preparation of
 special patterns. FRONT EILEIVATION 12 TREADS PtR CIRCLE.-30*
 14 TRCAD5 PE.R CIRCLE.- 22"30' 3 TART PLAN AT 3 TART FRONT
 E.LELVATION "Zzft ~"J 710

Construction Details and Finishes STAIRS Circular Stairs plan A-A. r©
 WALL STRING, • 1- -, I ; 5 i rACE. 5TR.INC. Circular stairs placed
 between walls may be built self-supporting at the inner string and be
 supported by concealed struts or hangers at the outer string, When
 completely exposed a circular stair may be designed to require few supports
 between floors. In constructing a circular stair the overall size of the well
 and the tread length of the stairs may be adjusted to fit the particular
 conditions of the structure. Treads should be a minimum width of 8" at a
 distance 15" out from the Inside railing. The treads may be qf steel, abrasive

cast iron, abrasive non-ferrous metal, cement, tile, linoleum, wood, marble or other material. Landings and platforms may be constructed as part of the stair, and may be supported by beam or cantilever construction. Wall rails and brackets may be constructed with handrail sections matching the railing. Face strings and railings may be similar to those used on straight stairs but should be designed of shapes adaptable to abrupt curved construction. The small radius to which these are constructed offers possibilities of design that should not be overlooked. Combinations of contrasting metal colors can be effectively employed in such installations.

711

Construction Details and Finishes STAIRS Concrete Stairs 712

Construction Details and Finishes STAIRS Concrete Stairs 'M'.D. PIPE HANDRAIL S DETAILS t-STD, MALLEABLE IRON BRACKET SECTION THRU STEP TYPICAL FLOOR SECTION 'A-A' Typical rairtforcad concrete scissors stair 713

Construction Details and Finishes STAIRS Concrete, Steel, and Terrazzo Stairs CAAT AftBAflIVK _S6IH« ,T _CK _U=.«.%, fiTniP AVOOOO <Sf*OUMP« ;,iTMflri)'l'''m'''l°''''''l'''ri''''''''l'''M! |} V -£-- d> &T&JR D-TMl- WITH CARp-RTIHG _ CRWIWf .J*IUt.~—^, 4©-or«s« *t*»c.—> STAIR. D _TA.IL WITH TSRRA^^O TR^AP^ 4 Rl--R- _«tra..J«*n _iJJ«.. HWWW *1- 714 5TEEL RISER. CE.MENT Tg^ADSN ca. tf'ti-tf' 2CV«5 _B»OVIDje-Wl 4 *IA» HU»T«»1-TIMU«K V MOM _D£1« IHIMCM WtJC-nchl _l.* _* auir _■ l.-.* o _» _rain,» Kaou raw .-soit. r:4aj>poicT .of LjSthimo b> -5TEEl-. BSBc. _Ta__-AXZO. _TgJ _AP.-t

Construction Details and Finishes STAIRS Marble ITeads MORTAR SETTING BED STEEL STRINGERS ABRASIVE (NON-SLIP) INSERTS MARBLE RISER ANCHORED TO STRINGER MORTAR SETTING BED BRASIVE (NON-SLIP) INSERTS Hu ■■■'■■■■v. -.■'-.l/v DOWEL mmm ^MARBLE RISER ANCHORED TO STRINGER Kiil^^a \$\$\$W«mtf*&? ■A ,●■■;■ * :j^.:>'V.'j.r.;vV;;,.-.;: MORTAR SETTING BED (1) Fig. 21 Sections (a) and (b) are marble treads and risers supported by steel stringers; section (c), marble treads only; section (d), cubic marble treads supported by concrete or steel stringers. NON-SKID INSERTS

INSTALLED AT SHOP IN 1" x 1" x 1/4" FOR W.I. OR MARBLE
BALUSTERS, EYE-BOLT AT 1 EACH JOINT STAIR BALUSTERS
V PIN key to materials MAG6 LE TILE- CEMENT 1/2" x 1/2" x 1/2" PLASTER
Hi STEEL VM 715 Fig. 22 Interior marble details.

Construction Details and Finishes STAIRS Slate Treads Stair fluted
Fica/Gwgt Back Surtea Shown, Miy A! B Niiunl Clift Fk Plnlh with
Gauotd A k Syrfict Nnunl Cl ft SUil TrMdi mtf flMri w/ Back Surntci
GjHfltdi \$ w/14" mlrt Oround Joina on CoiKTtn Q*. S* m tar S*nd
flubbfd flnlth. Expand edgn HormJ. OmtnlMarurltfMl* Flc j OH High. W
Bottom Of Tru4.5pHnk!i Sicdnfl S d With Partwid Ctmwit. SprinkH With
W ir And Fore* Trtid Down Onto Sttfn) Bid. S t RlnrWhh fipoiry Ctqitnt
Sonant QfoovM Miy Bl Addrt Far Additional StaWMiy. S*kJ Hubtotd 9ft
Trwrdi ind fHnn w/ flack SurfK* gugid. wt w/ It" mln. Qmt*t Jolpa an 5t*1
pan. 3*rnt for Natural Claft f Inlth. Sondaot Qroov f optloml, All ixpowd
Edgti flubbfd Smooth. Slate treads and risers set on metal pan stair /y//.
S an Brick orh r vifinwt S(H, All Suggested detail for slate treads and
risers, exterior and/or interior J'-StA TE. R.I5ER. STEEL- AN&LE-
WITH EXPAW-SOM &OLT AWCHOR 7T-T Bx1^ ft-. 716

Construction Details and Finishes STAIRS Planning Data Any potential
hazards must be eliminated. Stairs should be "easy going," that is, there
must be an appropriate relationship of riser to tread. Treads are of nonslip
material which is also extended onto platforms and landings for a distance
equal to the width of the stair treads. Double handrails, one higher than the
other, are provided on stairs for each line of short or tall pupils. The posts,
which support the center handrails of double stairs, are extended high
enough above the top handrail to prevent pupils from sliding down. PLAN •
VIEW- 717

Construction Details and Finishes STAIRS Barrier-Free Design Data (b)
Elevation of Center Handrail (a) Plan NOTE- X is the 12 In minimum
handrail extension required at each top riser. Y Is the minimum handrail
extension of 12 In plus the width of one tread that is required at each
bottom riser. **H ^ £ £ r 4- (c) Extension at Bottom of Run X Extension at
Top of Run Fig. 23 Stair handrails. I 1"vf/ll 1Vl_ ^ Hi Q Handrail 1MH 32-

38 j J (b) Handrail m^ j 1 „,1V» <<0 OrabBar Fig. 24 Size and spacing of handrails and grab bars. I (d) Handrail 718

Construction Details and Finishes STAIRS Barrier-Free Design Data
11mlnJ llVamax 2SO *B^ £ ras -^ (b) Angled Nosing (a) Flush Riser Fig.
25 Usable tread width and examples of acceptable nosings. Steps and Stairs
Steps and stairs should have nonprotruding nosings so that people with stiff
joints, braces, artificial legs, or other leg or stability problems will not catch
their toes as they climb. Handrails should be oval or round with 1 1/2"/4 cm
hand clearance between the rails and the wall: 1W/4 cm clearance will
provide ease of grip but will prevent the hand or wrist from slipping
between the handrail and the wall if the person loses balance. Handrails
should be positioned on both sides of steps and stairs and should extend
beyond the first and last steps on at least one side and preferably on both to
allow people with long leg braces to pull themselves beyond these points.
To guard against falls and to help children, some codes require another,
lower handrail. Steps, stairs, and handrails should not be made of slippery
material. (c) Rounded Nosing M^M^^ Recommended stairs
W//////////Mk Not acceptable vEZZZZZaN JZZZZZfr Fig. 26
V//////////A Not acceptable y-fg

Construction Details and Finishes STAIRS Ladders 101/2 p. ritEAO v in
ENLARGED DETAIL -£T ENLARGED VIEW'A-A" H i f^' - "i 56°
WOOD twjjwp ncinw A-A m •m !*' Vs" y im^go mm B-B fHOJICttO
l«I« —i|o.D. 56°STEEL J WOI 1 M(H{ J, u SCC1CN A-A SEC1Ow B 68°
STEEL i^+f t ""B -*-*■ 68° ALUMINUM -I HOIl-Dflitl MMttMI »*>
MHemMll HI mCLUSCD OH %!"l(|{>» trt«! <<] ^oU lift ■1-1 H !,°i -li
720

Construction Details and Finishes STAIRS Ladders; Open Steel Stairs
LADDERS Ni ■ «OL1B SF-A.CIHS TS.6E DETeSt/AII-IBD 0
■&HGIHSB.R-<o LA.PPSR. \VIT+t CAST Afef^A'alVe TRftfe.P'a OPEN
STEEL STAIRS f Si r\ ■K "^ -ttUMi ^* n,OUHO BA.F*. i ♦it I IU a +"* -
f=AC& OF- VA.LU -FLOOR rV»T>"x>i>" .4 »«.«:ke.t TO I9B.
06tEFt/MWB.D (D V6R.T ICA.U LADDfeR i-8-H--i-i-nn aP-MtHG
OEIPTH TO »fc ^ RfSTEB^INeu © ST/MR. DETAIL (GR.ATING
TRfe^P) WIDTH QVER.M.U S'IWCt & 7 & 9 k> - its. , "i i CENTER TD

CBHTI5.R SPACING . . . Z.IMGHS.'0,C. -,• | ••3>at■...].-... ■"•4. i ■. »
 '•'. J.." . « ■ -.-1 - . - -?r aTA.NPA"RD -UOL'g 5PACIN& » •FOR. CA«=>T
 A&RAfoNt TESSAO ■— \s^ TB6AD >"\: vi'v "trvvs;"^""^ i/t \^f hols
 apfr,ci ii£-.L, MHEO| © CiftT AE.R1.SIV& TREAD THICKHB.^ in
 wcHea 5/14' ■ya" t/kS y/z- ^6" CViT IROU UP TO 0." WID6 IS" Z4" •bO"
 41' CAft>T 4LU;mhu«\ up to a" wide. ia" as? TIC" 4»" CA-VT
 R1R.0MX& UP TO 6" WIPE. 16* 24* T>Q" ■he," CWbT HICItBI.-
 S.ROHZ6 UP TO t> WIOE. IB" z-t" ■so" ■bd' TWCKNE.'i^
 LI/AITATtOMS -FOR. VARIOUS NVIPTU CA'STIHS'b 721

Construction Details and Finishes STAIRS Steel Pan Cement-Filled Stairs
 "hosihq /"pNSeCTION AT LANDIN 5; P06T 3 WITH POST NOT& -
 TXtTSE EWO WELD 4 ORINO 9MOTH -^AC& CM* WA.LL -
 PLA«T«F1. HOEjIMffl- ■ T-.l^i" «WHT F-IU- 2"XZ* CLIP *.HOI + I | L-
 vV-FUftRING C PLANTER. OH /A.L. ABRASIVE 'SAF£Ty NOSINS w
 STR.N<SE.R. OSTA.ll- jJEXSi 4.TRIHQE.B DETAIL. «UMLM<. FOR.
 *TAJFI"i WITH T£P.I**-ZIQ OR- /AAF^PMJtt. TT*&AS»& 4 FLI ■!£
 (*.% C1.05H 6HD OP <3TRIN(3I|F« -FLOOR. < ifl pv SECTION AT
 COSIC- SLA& fLOOR 722 -CLOSE EMD ol« STRINGER. AVSLD .4
 tmiND %AQOTM A > aeCTION AT LANDING WITHOUT RO«i>T

Construction Details and Finishes STAIRS Barrier-Free Ramps Slopes and
 rise Provide the least practical slope for any ramp or curb ramp subject to
 the following new construction requirements: 1, Maximum running slope
 shall not exceed 1:12 (8.3%) 2. Maximum rise for any run shall not exceed
 2'6" (760 mm) Width Ramps and curb ramps shall have a minimum clear
 width of 3'0" (915 mm) exclusive of edge protection or flared sides. Cross-
 slope and surface Cross-slope of ramp surfaces shall not exceed 1:48 (V4
 in/ft). slope 1:12i. < 1:16 1:16i. < 1:20 maximum rise in I mm 30 30 760
 760 maximum projection ft Im 30 40 9 12 projection ramp slope maximum
 rise & projection new construction elevation 12mln -w wall vertical guard
 rail iMn curb N fNJ V 1 CO railing with extended platform Fig. 27
 Examples of edge protection and handrail extensions. maximum rise
 maximum projection slope I in mm I ft m 1:10.a 1:8 1:12.o 1:10 75 150 0.6
 1.5 maximum rise & projection alterations to existing construction section \
 % mln 9/wmM 36min W////////M 723

Construction Details and Finishes FIREPLACES The function of the fireplace today differs dramatically from its role of years ago. Whereas its original function was primarily to provide heat for warmth and/or cooking, today it serves more as a decorative asset and as the focal point of interior spaces and conversational groupings, providing the esthetic pleasure and comfort of firelight. Of particular interest to the interior designer is the proportion and scale of the fireplace opening, the treatment of wall surfaces surrounding the fireplace, the design of mantel pieces and hearth extensions, and the array of fireplace accessories available. Accordingly, the information contained in this section addresses these considerations. Drawings include elevations, plans, and details of various fireplaces; elevations of a wide selection of prefabricated mantel, types; and a sampling of fireplace accessories including andirons, wrought iron fire sets, and log grates. It should be noted that, aside from their decorative aspects, the fireplace and chimney have important structural implications and require special foundations. Moreover, the fireplace must be designed to carry smoke away safely. With respect to hearth extensions, most building codes require that for fireplaces having an opening of less than 6 ft² (0.56 m²), the hearth must extend a minimum of 16 in (406 mm) beyond the face of the opening and a minimum of 8 in (203 mm) on each side. For fireplaces whose openings exceed 6 ft², the hearth must extend a minimum of 20 in (508 mm) beyond the face of the opening and 12 in (305 mm) on each side. Most building codes also require that woodwork or other combustible materials not be placed within 6 in (153 mm) of a fireplace opening, and that combustible material within 12 in (305 mm) of a fireplace opening not project more than 1/4 in for each 1-in distance from such an opening. Since building codes may vary, it is important that the designer have her or his plans checked for conformance with the applicable local or state codes. Any structural modifications to an existing fireplace and chimney or the design of a new fireplace and chimney should be reviewed by a professional engineer or registered architect. A fireplace that draws properly can be assured by applying proper principles of design. The size of flue should be adequate and should be based upon the size of the fireplace opening. One rule commonly used is to take one-tenth of the area of the fireplace opening to find the minimum area of the flue. For example, if a fireplace had an opening 3 ft wide by 2 ft 6 in high, it would have an area of 1080 in². One-tenth of 1080 in² equals 108 in². The standard-size flue nearest to this

requirement and readily available is a 13- by 13-in flue lining, which has an inside cross-sectional area of 126.56 in². One could also use a 13-in round flue that has a cross-sectional area of 113.0 in². The front of the fireplace should be wider than the back and the upper part of the back should tilt forward to meet the throat in order to throw heat into the room instead of up the chimney. The arch over the top of the fireplace opening should be only 4 in thick, and the throat should project toward the front as much as possible to form the smoke shelf behind it. The area of the throat should be 4 times the area of the flue, with minimum and maximum width of 3 and 4 ft, respectively, so that the narrow throat will cause a quick suction into the flue. The sides of the fireplace above the throat are drawn together to form the flue, which always starts exactly over the center of the width of the fireplace. The smoke shelf is very necessary to stop back drafts. The depth of the fireplace should be one-half the height of the opening, with a maximum of 24 in. The back should rise one-half the height of the opening before sloping forward and should be two-thirds the opening in width. The back, sides, and parts of the hearth that are under the fire must be built of heat-resistant materials. Firebrick laid in fire clay is the best combination. The damper is a large valve that can be adjusted to regulate the draft. Many types of commercial damper units are manufactured. The position of a damper unit is important. The damper is generally set about 8 in above the top of the fireplace opening and is concealed by the brickwork. One advantage of these units is that they are correctly designed and have correctly proportioned throat damper and chamber to provide a form for the masonry and to reduce the risk of failure in the function of the completed fireplace. The hearth consists of two parts, the front or finish hearth and the back hearth under the fire. The front hearth is simply a precaution against flying sparks and, while it must be noncombustible, it need not resist intense prolonged heat. Because the back hearth must withstand intense heat, it is built of heat-resistant materials. In buildings with wood floors, the hearth in front of the fireplace should be supported on masonry. The front hearth should project at least 16 in from the front of the fireplace. At the back part of the hearth it is customary to have an ash dump for dropping the ashes into the ash pit, which is generally located in the basement with a door for cleaning out ashes. 724

Construction Details and Finishes FIREPLACES Components and Terminology 1—y- Flg. 1 Construction details of a typical fireplace. 725

Construction Details and Finishes FIREPLACES -■ Wood Marrbsl -^ Smoke il/ , t Firs - brick- SECTION SECTION SHOWING. ALTERNATE HEARTH PLAN 726 TABLE 1 Recommended Dimensions for Fireplaces and Size of Flue Lining Required (Letters in column heads refer to Fig. 2; c Size of fireolace ODenina Width, 24 28 30 36 42 48 54 60 54 60 66 72 w Height, h 24 24 28-30 28-30 28-32 32 36 36 40 40 40 40 Depth, d 16-18 16-18 16-18 16-18 18-20 18-20 18-20 20-22 20-22 20-22 22-28 II dimensions in inches) Minimum width of back wall, c 14 14 16 22 28 32 36 44 36 42 44 51 Height of vertical back wall, a 14 14 14 14 14 14 14 17 17 17 17 Height of inclined back wall, b 16 16 18 18 18 24 28 28 29 30 30 30 /•.reaa+Br Size of flue lining required <^ Standard rectangular (outside dimensions) 8'> x 13 8 1/2 x 13 8 1/2 x 13 8 1/2 x 13 13 x 13 13 x 13 13 x 18 13 x 18 13 x 18 18 x 18 18 x 18 18 x 18 i i Standard ^SSr^^O^-x, round (inside diameter) 10 10 10 12 12 1 itSsStta-^MIK 15 A^ 15 KjKy<y ^ III, IfifHJMf 15 >c>c><r^vS^^ 15 XXXyvv is ^xs?S<x: 18 x >C/ 18 Air inlet',

Construction Details and Finishes FIREPLACES Design Data SECTION WITH DAMPER ELEVATION Scale Va'-I'-o' SECTION WITHOUT DAMPER MATERIALS Location FRONT HEARTH BACK H EARTH 4 REFLECTING WALLS Maferlaland. Thickness Any Masonry 2"to 4" fire Brick 4" Soapstone 4" Hard-burned Tile V • (/ndudes bid) i (min)— (Rec.min.) LintF°L h«m7 PLAN Stait \-May be ltft void or expanded Ye'-i'-a" to accommodate flue FIREPLACE DIMENSIONS (In Inches) W H D , FLUE (effective Area) T (Arec) T (Width) U fo 64 7A fo Vi W 'A i.M u/it to j4(R«c)hr(wi /l fo ft Mt|8 to?4(Recj formed V6 WH for unllned flue /lo WH for recfanjular limn? '/? WH /or circular lining Va fo% FLUE AREA 3'minimum to 4'2" maximum RECOMMENDED FLUE SIZES (In Inches) FIREPLACE WIDTH w 24 30 fo 34 36 to 44 46 fo 56 56 fo 68 70 to 64 RECTANGULAR FLUES Nominal or Outside Dimension d'A X6tt AV4- XI3 13 XI3 13 X18 16 XI8 20 X24 Inside Dimension IVi X T/i 7 XUft lmxim IP/4 X fc'/4 15HX5V4 17 X21 Effective ^rea 41" 70°" 99" 156"" 195" \278* EQUIVALENT ROUND Inside Diameter a 10 12 15 18 12 Effective Area 50.3" 78.54" 113.0"" 17fe.7" 2544" 380.13" 727

Construction Details and Finishes FIREPLACES Through or Two-Way
SECTION X-X FURNACE FLUE PLAN Fig. 3 Fireplace open on both
sides. TABLE 2 Table of Dimensions and Equipment (In Inches) Width of
Height of Damper Smoke Old flue size New flue size Angle Ash Ashpit
opening, A 28 32 36 40 48 opening, S 24 29 29 29 32 height, E 30 35 35 35
37 chamber F 19 21 21 27 32 6 13 13 13 18 18 H 13 18 18 18 18 6 12 16 16
16 20 H 16 16 20 20 20 (2 req'd),* J A-36 A-40 A-42 A-48 B-54 L 36 40 44
48 56 Tee 35 39 43 47 55 dump 58 58 58 58 58 door 12x8 12 x 8 12x8 12 x
8 12 x 8 "Angle sizes: A-3 x 3 x %; B-3 1/2 x 3 Vb x >/. ". Note Y from Fig. 3:
The damper and the steel T should not be built-in solid at the ends, but
given freedom to expand with heat. 728

Construction Details and Finishes FIREPLACES Corner Design J<, 1—
*~7 A h 'i! / s 1 L I— I — V -1 t T K N Furuace Floe U EARTH "Damper
Joists ^ Ash Pit & C. TIOKJ Plakj A 28 32 2Kb ■4--0 4-ft B 2fe^ 2<&^ 26^
29 29 C 20 20 20 20 24 D 14 14- 14 14 14 T E 20 20 20 20 24 ABLE, op F
29^ 32 a>5 35 43 D lAAtkiSIOMS old Flue 5 I -Z.E-S IKJ G 11^ 11^ 11^
ll# II& OUT 13 13 13 13 13 IKJ H ■* ">4 11^ I5fc 15% OUT 13 13 13 IS
16 AMD E«OIPM.£WT Mew fLUE S IZ.E-S IKJ G out \o\ |2 \0[/A 12 10^.
12 \b\ 16 13^ 16 IU H Out lojf. 12 13 1/2 16 13^ 16 I3& 16 13^ 16 (iN
IMCH6J L 36 40 44 48 56 M 16 16 16 16 20 >) Steel Aki<;lf *A.5fe *A-42
*A-+8 *B-54 * B _60 PUATE LIWTEL K IUI6 II xl6 II "16 11*16 11x16
CoRUta Post N 3+x26^ 34>x2fe^ 3+«2&!£ 3+x29 3**29 * Awgle. Sizes
»Aj^k% Fig. 4 Corner fireplace. *B}^ 2.^*4 Fig. 5 A shallow fireplace with
a copper hood, built as shown, throws out considerable heat after the hood
gets hot. The wall should be of fire-resistant masonry. 729

Construction Details and Finishes FIREPLACES T *- '■'"insxAiL section
;;->Lj Aeale 3'- 1 foof; 730

Construction Details and Finishes FIREPLACES
TO^vw^w^^srowOTjwTOTOrwKS ' p|_AN. ^ Scald & '■/■d Stale Acarin
fireplace si/mmelrical on CcrUa-line SECTION SaUtto'-jki ELEVATION
ScaSe toW ELEVATION .W *w Continuous brick, pattern FricAs \\r.l\\
Slush PLAN Scale i/4~-l'o" fireplace symmetrical on center 731

Construction Details and Finishes FIREPLACES 732

Construction Details and Finishes FIREPLACES Marble Mantels "I I
CAP^L Us I MANTEL J" ,£". si*.* imNltLI PLAN £ * i SECTION D XALt
l'2-i'-0" KEY TO MATERIALS MAERLE: Q 3T ('-JOINTS- » VOOD
MACBLfc f LQQg. -y fLtVATION DtSIGN M VOOD planter. Irrzr: WALL
TILL f; BBONZ.E -sCALE l'2"» I'-O" 733

Construction Details and Finishes FIREPLACES Marble Mantels E L E VA
T I O N •M A R b L E PANELS ANCHORED TO MASONRY WITH
CONCEALED IV I » E ANCHORS -PLASTER OF PA RI5 SPOTS '8
MAR&LE -Z'i SETTING SPACE FACE OF WOOD PANELING .FACE
OF FINISHED WOOD CLOSURE STRIP NAMING STRIPS
ANCHORED IOROL" MASONRY WAIL S^l gas FIRE BRICK F I Nl 5 H
£ D WOO D jfPANE L IV A L L Mi I) kU l 0 II G H M .', S C N t V WALL
m m \?/7P~xy//;////////A i 3 -STEEL SUPPORT ANGLE G'« S'« V 6"
LONG-? RE51; PLAN AT DETAILS 734

Construction Details and Finishes FIREPLACES Marble Mantels 735

Construction Details and Finishes FIREPLACES Wood Mantels V» 10'
Rough Hewn Oah ■ CM , Panel ! -o PinG incj B 4*10 . .-f: B '& *-& Brick-
l' Camber A 3'U" ELEVATION /ECTION- B-B Pa Floor 736 /EC

Construction Details and Finishes FIREPLACES Wood Mantels 737

Construction Details and Finishes FIREPLACES Wood Mantels
tJ16CS.IH.iKIII.Ul/ ijff tk-ICfc, 5k.elf,if Lard wood [ji-isIl, tiud , Live
mouldno rill, aojitsreds to cover end qri.iL. ■PAE-TUL- LLLVATIOM-
Scile. j} ".|-0"- •PALTIAL- E.LLVATIOK- "fohinhle,4", A Vo be flust wiHl.
■Mug.. , •5LCTION- C-C-THEOUflH-MANTtL- 5cile 5'-|-o'- 738

Construction Details and Finishes FIREPLACES Wood Mantels m^pu Fig.
6 These wood mantels are readily available. 739

Construction Details and Finishes FIREPLACES Wood Mantels t ■ 9"- -t- -
t- \S* »-"f «5=i] D + s- + ^2T "■^ ^■'Wa u U I I [<w 740 Fig. 6
(Continued)

Construction Details and Finishes FIREPLACES Wood Mantels u
!^»L^I*JIUlug^gii£aW«W'.MMWMMM»g»JWIM«»M^»j^ 'tAjuwMaw^
*^S IH tZFC 1 ^ D "OFCZ? H

Construction Details and Finishes FIREPLACES 'Fireplace Accessories <#
rx Fire set of wrought metal, stand and four tools. Fire set, polished brass,
stand and four tools. Andirons of wrought metal, smooth or hammered
black finish. Andirons of wrought and cast metal, smooth or hammered
black, or polished brass finish. T^= Fire set, wrought metal, back plate with
jamb hooks, five tools. Fire set, wrought metal stand and three tools. Crane,
pivoted, of wrought metal. SCALE 2»l-0 Crane, pivoted, of wrought metal,
movable hook. Crane, pivoted, of wrought metal, ornamental adjustable
hook. Footman, wrought metal. Trivet, wrought metal. SCALE I" {■ a' Fire
lighter, kerosene torch, can and drip pan, wrought metal. Spit with
removable clamps. Hand turning crank may be replaced by pulley and
operated by weight or spring clock mechanism. 5SCALE r-l'-O" a f a c "— 2
J i Log grate, wrought metal Log grate, wrought metal Fender with wire
Fender with stamped mesh screen. grille. Log grate, wrought metal ©
Fender of Fender of wrought metal. wrought metal Moulding on top may be
replaced by padded seat. Fig. 7 Fireplaces offer opportunities for the use
and display of a variety of metal Items of decorative value. These may be
selected or designed to match other material In the room. Metals used for
wrought and cast fireplace products are usually cast iron, steel In a dark
hammered finish, or polished brass. Combinations of these metals and other
metals may be used very effectively. 742

Construction Details and Finishes LIGHTING Planning Data: Minimum
Shade Heights Although lighting design is a discipline in and of itself, the
interior designer and architect must be knowledgeable about the interface
between lighting elements and the interior architecture. This section,
therefore, focuses primarily on the detailing of this interface. Details from
actual contract drawings, prepared by various interior design and
architectural firms, are provided for the reader's reference. Among the
details are those for valence and cove lighting and for the lighting of stairs,
columns, and skylights. This section also provides some basic planning data
including illuminance values for residences, offices, stores, and industrial
spaces. Bottom of shade at eye level when seated — approx. 40"-42" off

floor. Exceptions: Sewing, Piano Light colored shade should transmit light generously, Fig. 1 Measuring when the lamp is at the side - when sitting, lying down, or playing the piano. When bottom of shade is above eye level, lamp stem should be about 10" behind shoulder — near rear corner of chair Fig. 2 Measuring when the lamp is behind - when sitting. 743

Construction Details and Finishes LIGHTING Planning Data: Minimum Shade Heights Recommended Minimum Shade Dimensions LAMP TYPE
Sr. Floor Swing Type Jr. Floor — Swing Type DiSuser Type Eridge End Table Diffuser Type Sr. Table Wall Lamp Diffuser Type Study Type — pair Make-Up — pair Double Dresser — pair Top Dia. " 10 10 10 14 8 8 14 14 8 4 6 7 8 Depth " 10 10 9 6 8 10 6 13 8 6 7 7 8 Bottom Dia. " 18 16 16 16 13 16 16 16 13 14 10 9-10 12-14 Base Height (measure from table to shade bottom) + Table Height = Seated Eye Height (approx. 40"-42" off floor)
FLOOR LAMPS Measure from floor to bottom of shade. SHADES Measure top and bottom diameters, and depth vertically through center. Minimum 15" to shade bottom. Shade fairly dense, or opaque, in a light but not strong color [30"—»| —x9—9- Exception; Makt-Up White or ivory highly translucent shades •36" »| Standing Fig. 3 Measuring when the lamp is in front - when studying, sewing, or grooming oneself. S»d»d 744

Construction Details and Finishes LIGHTING Floor Lamps 1 I 17"- 1 745

Construction Details and Finishes LIGHTING Table Lamps f i *tf ffi TI T - r- T ll'A" 1 T 12" 1 T T 1 I 13" 1 T T T 13" 1 " 1 1 T 14" i I 30VV T 746

Construction Details and Finishes LIGHTING Desk Lamps .22" 1 J_ i _IW | 17-- T 14" 1 T r—IMlilMJ! I I I 9'1_ | 1 r r 3% T I 15" 1 14'/. " 1 Vh" 14'/i" I 14" 1 i __mum I | 10" 1 ' I 14"—) r -2?" 1 I 27" 1 iy. ■I 3Vi" 15" 1 I W 1 -25V/- 1 ■ 29" 1 | 24 V/ 1 r T r_ T r-r | 261/2" 1 T I 4'l- -26" 1 V 747

Construction Details and Finishes LIGHTING Wall-Mounted Task Lamps -10"- I 27V2,i 1 r-'M 2'/i"l H6'H H10'H 1-fH ■14'i -23i- L-10S-I | 14W: I 14"-Ji^11V<" T 10" 1 H°H -40«- 55» |—10'«"—| -6"H hJ I? 748

Construction Details and Finishes LIGHTING Planning Data: Residential Valance Lighting A \ \ w \L N S> (150 mm) FACEBOARD TILTED 15° TO 20" \SHIELDING Fig. 4 Valance faceboard may be tilted. M100 mm) 2

MIN. (BO mm) SIDE-MOUNTING CHANNEL (BALLAST ENCLOSED) DRAPERY Fig. 5 With side-mounting channels, no extender is necessary. 1 K 3J <- U ' / { (13 mm) STRAP IRON BRACKET SUPPORTS MOUNTED OVER LAMP HOLDERS Fig. 6 intermediate brackets are required to support long faceboards. 2**4' (BO x 100 mm) FLUORESCENT LAMP AND CHANNEL WINDOW ■11"12' (280 - 300 mm) ORAPERIES (19 mm) V •n. I (BO mm) / (127 f«-2'-*7 mm) VAUNCE BOARD 5" Fig. 7 Variation of valance lighting, (if distance between wail and lamp is increased, light will be distributed more evenly, but shielding may be required at the bottom of the faceboard.) 7 »-8" (175-200 mm) (50 mm) Fig. 8 Minimum dimensions for cornice lighting Installation. BALLAST ENCLOSED 2 TUBE CHANNEL SHIELDING MAYBE NECESSARY Fig. 9 Cornice lighting with two tubes may require shielding. 749

Construction Details and Finishes LIGHTING Planning Data: Residential Down Lighting Flg.10 Common types of rJownlights. yU* 2" (50 mm) APERTURE 12" (300 mm) 8' (2.8 m) 9' (2.9 m)' CAVITY OR PLENUM UNOBSTRUCTED FINISHED MAT WHITE SOX DIFFUSING TYPE SHIELDING 45% TRANSMITTANCE SOfo(600 Ix) AVG WALLS 50% FLOOR 25% —fare Fig. 15 Basic relationship for the design of luminous panels. [A light level of 60 fc (600 Ix) is produced by seven rows of three 40-W fluorescent tubes on 18-in (457 mm) centers. Light distribution and surface luminance are approximately uniform.) TABLE 1 Illuminance Values for Residences* ADJUSTABLE SHUTTERS V 1" (26 mm) APERTURE Fig. 11 Pinhole spot, a recessed downllght with adjustable shutters to shape beam pattern. TOZZy prau LOUVER Fig. 12 "Eyeball" semirecessed fully adjustable downlight. \^^\«w^^;\|^ METAL CHANNEL VL VLV FLUORESCENT LAMP' OAIN T CAVITY WHITE - FURRED OOWN STRUCTURE 12" (300 mm) LOUVERS —' ——— 14" (300 mm) Fig. 13 Luminous panel or soffit lighting, used over a kitchen or bathroom counter. //////////A?f)^//////////A r T T-BAR SUPPORT DIFFUSER Fig. 14 Critical dimensions for luminous panel and luminous ceiling lighting. (S should not exceed 154 to 2 times L.) Specific visual tasks Dining Grooming, shaving, makeup Handcraft Ordinary seeing tasks Difficult seeing tasks Very difficult seeing tasks Critical seeing tasks Ironing (hand and machine) Kitchen duties Food preparation and cleaning Serving and other noncritical tasks Laundry Preparation, sorting, inspection Tub area-

soaking, tinting Washer and dryer areas Reading and writing Handwriting, reproductions, and poor copies Books, magazines, newspapers Reading piano or organ scores Advanced (substandard size) Advanced Simple Sewing (hand and machine) Dark fabrics Medium fabrics Light fabrics Occasional-high contrast Study Table games General lighting Conversation, relaxation, entertainment Passage areas, for safety Areas other than kitchen involving visual tasks Kitchen Illuminance Foot- candles 15 50 70 100 150 200 50 150 50 50 50 30 70 30 150 70 30 200 100 50 30 70 30 10 10 30 50 Luxt 150 500 700 1000 1500 2000 500 1500 500 500 500 300 700 300 1500 700 300 2000 1000 500 300 700 300 100 100 300 500 •Minimum on the task at all times. tLux is an SI unit equal to 0.0929 footcandle. 750

Construction Details and Finishes LIGHTING Cove Lighting Details §=T —IL M//^yrr ,*>^tf" » >*u*/» nr ^rrrr^ w" ■ r»H\$ _0"A A.^ »r.»"l \ . i!>if», »,■ » ■ J^ JI • li'""rrn ""l"l"l " " "l"" f"" M I '■"»■" ' "" "" ' *' 5T»«5<sasca(2)r«eur). , piAsosi^ T/re -?A - W fc)'-*' fx«a*-ur I*** PBTA/L Cf L.[gfl-Tr<23VE RECESS 751

Construction Details and Finishes LIGHTING Cove Lighting Details I -...':w ,.s.,^i-«a*».irf<^rfe TT r*iUTeo wf.t>o. pmcia- u; Pl»TU*£ • 4tt IfcHTiWO 4CHtPUUi ftUOCKIN*- TufUC! fAiMTtP -o MA-tH PR.Y.4A.1. - o3 . t- • Mtt^ tTJD* \luUMM —»o«re*JOK> Ax.-pi.fi «ilhJ6 4- 16' /T\ P£T^it - LI6H7 6075 <? p.&c&"T'Q|s- Az&A PAIKITEP Wr W, tEitiNiCi —I fin up CAR,i0Pi TUfte fWTUKE ■ *tt LIOHTiWC 5CWfcPUL& frliOtKINt MUB 'AifJTEP '0 TAihT To MATCH P*Ma.ll * /£ \ PETAit - li6)4; cms *> SLfcVATQH L-o»rr 752

Construction Details and Finishes LIGHTING Cove Lighting Details 'piqu«6 - «i O1- SOVS WMITt MMWW ♦usrtMPfco <WK 60. aiUU& P6TAIL- - UGH" CP^E a TVPS "P1 cl*Tl'Rt. AT l4M.Lt ©PE.TAIL- L.I&HT OO^fc ** 6TA|g.,TVPt 'It.' FI>CTUe.t To tw^» A»oVE, TM '.'Ulnae - id £.t.-iU6 nu>«skic W^Lt *J « r- o n t a --< T<- . = kt *\ si K£ / 0- ttt-f-x.t- <VO«.Att itr ?-'/*' roe ual,, •*vre gj> P£TAIL - LIGHT 6P./& li "P* g)fl.El AT HL&& V = ,'-0 753

Construction Details and Finishes LIGHTING Cove Lighting Details u * "KL /': -."} _v_ —KXI#-flM<! *PA0 —I)-' 'C' t-HHHtft-Mm —W Mtt-W W/PlM. _hhs /f rin> enw £ ttfHf p|X, S>B£ uMHf ■ PIK. pefAUS. «L j»V,

Construction Details and Finishes LIGHTING Cove Lighting Details —
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IWNP. .HcrAfiurl*" fn.*- •««aaU»A|l<?' 755

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Construction Details and Finishes LIGHTING Fluorescent Cove Lighting
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Construction Details and Finishes LIGHTING Cove Lighting lor
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Construction Details and Finishes LIGHTING Miscellaneous Lighting
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4e^ie^3 lew uyui vac^e. Tjeeu^HT Vcv-.wnw -HOZEH LUCIFER, LIGHT
SrflilP |U PLACE. AT LEAAv TWieff 6Ai Piece - locipbs uaHr «rRiP r«o at
12 V. - MOUMT AS «HOWN. 11V /A0.-II LMAP (FLOOD) FHISTING
WALL -^ secricM -type. L •iiALa.:S**"-t3" g>E£Tl£>M - TYPE M Sc*LE
: H*U= -SHE -F«UJ6flU LAHfwCUPeF- eecr^ -Type. c5 i^gftfcl-rEt?
PU*J j> Auc?tg vi^.PC AfgeA 760

Construction Details and Finishes LIGHTING Miscellaneous Lighting
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THJ<?et? •euu^iu *ML^.7>» WITH I UW.IT WMfS %.■£>. si. IB e?e
iTfWMcrec at 12. va-i=6. rpcfr\& op- wow venter fiVJet/s AT UaHTeoK".
M«5uuriU.s» eertt*. same.. ua-J v«5UTAiae Utws. pep TTFT& A -UI-TEL-
A& MI<Sf'£UIMe LENGTHS AS *SSUJP*.C DUTCH METAL S.OLD

LEAP geCTIOsi-TYfEA f^ne^rer? p'usi'g? Lusht ass- lJi^56?Ve)Jee^
<*/HJei- exi-tmuzi couktsi uw& - with i matt i>Mr% sv o-<;. i*'£>.ii. -mi*
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Construction Details and Finishes LIGHTING Miscellaneous Lighting
Details onu & rf pb P «wi-y SiSOARE OK, UiMfiaoftSLfi' # TYPE Fl. «fc-
fl Soft WHITB I* Tftfi K flSHrMJUKJ <J2***T M3UJTEC A* (SaSffl't?
eY ELe£TW£Ai- £9UTf*A£1Bp*. iM J**?* <*<■ nsoce mcth^^. • RELAY
THIS DIM6N TPCiOIUETMAKKP, Optional while plastic or frosted glut
lop W wide metal angle bracken on 24* centers Toggle bolu or other
appropriate fastener* on 16* centen Single- or double>lamp fluorescent
light atrip W hardwood faceboard and end returns, finished to suit outside,
flat white finish Inside Wall bracket cross section «U3£KliJS(it' muertH
htm ti) ia watt HAU3<aOJ l>Mf% <=fgfMrr&£> A.T" 12 VO.-PS
rmniH.'feuaHec' e**e>& pvs^TiiK MoAJTiiJa flATB. -ro ee. ArrA«aHec to
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cAe*Jerr type. j-^iJba uwav- itJiwJctoeaie^r fop. uscxnarJtb. 762

Construction Details and Finishes LIGHTING Miscellaneous Lighting
Details I' iCUPCOKf pan HABDW00P FKAM6 s| !#&YP.|!0. PLYYdSOO
pH»LVi. SHE AU 'L MflULP . . . _rw__ i / *"j * i J—t" UINB <fl»
KXIW. O-fc L* l-tlK(.Uir,-T«Ae«t RJU- WI&WT l<IB^UWC -
<wnNuou<» w«f buxk UU* <SF ■XHT1N& W*Tm<7N ©
PLAM'MCOM/Kl^e-gMTIgY' lW'.|l-0" au*»j <H/vtir upto »mnz »P MS.
CAPACITY N t'-l/lf N«CNTUBB,UXAT» ■pWSRHMSB IN CAB. IM
ZAFUAH: WBSN NICHE' -WJCY iPm^ T^G?VE UWTIM&1 TYPICAL
~5ECfi*JN \4>-"-y 2'z! fTBWL'milP kite anale guEcntx* LAMP l'n*,BY
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PLY***? «««• rtUJiU Jtoaaff fnrruat yi4to-p i*im 5 Vl6HT FimiKH
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Construction Details and Finishes LIGHTING Miscellaneous Lighting
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 DE.TAJL-OWK. UdfiT* fa** •s'-»" jUfHT FIXTUW »y i"»r-o" f\$\
 CWML •■ COfe L.I6KPM6 • SloPCD flML » U*U j^ftt'. Tl y v 4" Lv; i ^^
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 DSJA tU KWOJaFPTSK fc'-M 764 e *+ pr ?r

Construction Details and Finishes LIGHTING Handrail Lighting Details 7"
 J * \ Gypsum board backing Continuous fluorescent light strips between
 posts Lighting cutoff line Support posts 4' to 6'to the Inside per lamp
 lengths V2" tempered glass baluster panels Floor line ', ^— Hardwood 7
 trim trim Continuous cold cathode lamps, exposed or concealed Full-width
 partition at corner Metal stud and gypsum board construction ^+ T" T ^ •
 Extruded aluminum top railing Continuous fluorescent lamps between
 support posts Support posts 4'to 6' to the inside per lamp lengths I.
 Pavement line Lighted wood and glass guardrail 6" maximum Elevation of
 lighted guardrail planter demonstrates the use of combined 3-foot and 4-
 foot fluorescent light strips to achieve overall lengths in 1-foot multiples. To
 minimize dark areas between lamps, use strips without end caps and install
 lamps back to back. Lighted low-partition guardrail <f 2"-hlgh aluminum
 handrail Hardwood trim Gypsum board backing Continuous fluorescent,
 neon, or low-wattage incandescent lamps Metal balusters, 6" on center
 Hardwood vertical supports 4' to 6'on center Metal-lined planter for set-In
 pots or direct planting Metal stud and gypsum board construction Finish
 floor line Lighted guardrail planter section Extruded aluminum light rail
 Sight line for shielding lamps from below Open circulation areas can be
 illuminated with lighted railings, as shown in this section of a lighted
 guardrail planter. 765

Construction Details and Finishes LIGHTING Stair Lighting Photometries
 PHOTOMETRICS V SERIES - EXTRA BRIGHT MAXIMUM
 FOOTCAHDLES LIGHTED WIDTH ■ (AVERAGE FOOTCAHDLES
 OVER THIS WIDTH) ^ HT. ABOVE FLOOR IH. 6 a - 10 12 18 24 30 36

LIGHTED WIDTH 11 14 1ft 21 32 42 S3 63 FOOTCANDLES (1)
 AVERAGE CO 3.6 3.0 2.5 1.7 1.3 1.0 O.B (2) (3 MAXIMUM 13.0 s.e 7.6
 6.5 4.4 3.3 2.6 2.1 NOTES: 1. LOWEST WIDTH TO PCXHT
 FOOTCANDLE LEVEL FALLS TO 10% OF MAXIMUM 2. AVERAGE
 FOOTCANDLES OVER LIGHTED WIDTH 3. FOOTCANDLES
 DIRECTLY BELOW LIGHT FIXTURE ■ CARPET TREAD NOSING
 PVC EXTRUSION LITE FIXTURE Fig. 16 Surface-mounted step light.
 Fig. 17 Recessed step light. 766

Construction Details and Finishes LIGHTING Stair Lighting Details
 ADJUSTABLE SNAP-ON LAMP REFLECTORS RITE ANGLE
 ELECTRODE LAMPS fabricated to-shape and length of architectural
 design continuous line of light — no shadows RITE ANGLE ELECTRODE
 LAMP HOLDERS 94" o.c. max. LUMINOUS COVER (by others) to
 protect lighting from weather and vandalism.
 Clear/translucent/acrylic/lexan as desired with foam gaskets & tamperproof
 screws. Schematic perspective LUMINOUS COVER with foam neoprene
 gaskets (to protect lighting from vandalism and weather) #RRF FEED
 LAMP HOLDER ADJUSTABLE SNAP-ON LAMP REFLECTORS it! I
 |*2 1/2"*— 3K" mln.—j ____ • • • «wJ 7 |f i __J ■ * ah 1 ■ In i ■ fc to
 remove lamp for stairs over 8' long continuous line of light 2 1/2"- recess at
 lampholders only ■» 94" o.c. max. -BVi"- Typical section through riser
 (exterior or interior) -3" mln. - Typical section through riser (Interior)
 Alternate stair lighting designs ■ UPHIGHTING Assymetrical spread
 parapets, dwarf plants TYPICAL DESIGNS/SUGGESTED
 CONSTRUCTIONS, Exact construction as required within parameters of
 illumination desired, lamps and lampholders. SIDE LIGHTING desks,
 counters 767

Construction Details and Finishes LIGHTING Stair Lighting Details ■
 CARPET Carpeted cast-in-place step light CAST IN PLACE!!? . NOSING
 • Y pvcextrusion' "'<^ • ; lite fixture • -CAST IN PLACE TREAD. • ; '.Ota.
 • . ', Exposed cast-in-place step light CAST IN PLACE NOSING PVC
 EXTRUSION LITE FIXTURE RISER TREAD ■ d(S ■ . ■ ■ cr^ 1—
 WOOD TREAD CONCRETE—f TREAD -'R'EXTRUSION, SILICONE
 TO TREAD . Ob ■ /.•;■■■. <>e "•■&■'• C&:\::-/s<^:c6- ;0. do • .•' / " ■ /
 •': P-: v>: [■\$£>&.-::; CONDUIT TO NEXTSTEP I LIGHT FIXTURE

Concrete surface-applied bonded extrusion R' EXTRUSION, SILICONE TO TREAD LEAD- WIRE HOLE FOR LEAD WIRE IN TREAD Wood surface-applied bonded extrusion LIGHT FIXTURE RISER - BRICK (Stone-Gr«nlte-M«rble) NOSING ■PVC EXTRUSION - LITE FIXTURE (Plaesa extrusion end align brick before grouting) -GROUT JOINTS ROUGH SLAB BASE eg" /"V*^ •*<d*.* STOP RACEWAY ATTHIS POINT- PULL WIRES OUT APPROX. 12 INCHES- FILL END OF RACEWAY WITH SILICONE SEALANT. Yi" RACEWAY TO SECONDARY SIDE OF TRANSFORMER Brick steps with recessed light 768

Construction Details and Finishes LIGHTING Stair Lighting Details f mx? \Jc*pcn i» ft& rrp] > I .. i ij— ■ * ! wr&.i * t«36 if ■SPNIHI OMI^fJ ^PE-T^iu of- Tfce/M? <j &toE.e *t SKli»*1 <f *£ PETAIl- of S<TKlMGfEa.»rS]ijK&*14i*4 fi\v&&4\ - gf rvwsLg. ■e.f&r foV; n»: © PE.TA1U <?<= err&rJ J6HT& »' u9Usl6£, * o& 769

Construction Details and Finishes LIGHTING Skylight Lighting Details TYPICAL PLAN ■ 3 SPIRAL LAMPS Iwlthout lumlnoiM •tamtfll) - Lumlnout ElMmrtr (h talMtadt SECTION THRU TYPICAL IKVIITI 3 3) 3) 3 M" mix. Curb Opining— TYPICAL PLAN - U LAMPS jc £Jc Mdx, Curb Optntng TYPICAL PLAN - ITHAIOHT LAMPt IB I] ■MI I] I 3 3 (c lJt 3) 3) 3 — 32" nux. Curb Opening <— TYPICAL PLAN- M LAMPS Fig. 18 Skylight lighting. Skylight serves as fixture - does not interfere with natural lighting, will not cast shadows on luminous element. Spiral, M, U, and straight lamps fabricated to fit curb opening. 770

Construction Details and Finishes LIGHTING Lighted Column Details an. fwisfl LIClHf ftXf.rf^ ^\ ae>Afuasi <* ucw&p cauxwi VA/ 6C*U-: iW'- iuo" 771

Construction Details and Finishes LIGHTING Lighted Column Details © c&JO*., WAJ-tota, . K-Ui-t*.; -non <MU0S&- -Fff FtAsW'feKl IS-'* *U«* «KP*- - (-VkH)»»AH< pMA' 3j.« <t BCW*1 *i»C? lie**!' MMte4,AuT - dpim- S? ft**/*? *i|a*». wKtSif ^fljigp fXnei* feKiarf) - ■osa KM^a»Aht-f-f»H*U ran —g—pp—; , 772

Construction Details and Finishes LIGHTING Lighted Column Details
 Http-teaw._ UkaHTPiyr, 'p' S> PEP- «3s>UJr"W titavfriMUOU^- A,
 F^utep rep pVf'wP, p>iUTEP iXr«.T|M£i cAndtJ PINE. *pff p»SMlt{?
 KEP' Plwe,/ PN>ijtp facet's A ^J4\ P&T^L^ Lir^riTgOLUHKl vg7 ^ALE-I
 iys|t-©'i 773

Construction Details and Finishes LIGHTING Ceiling-Mounted Cold
 Cathode Lighting Details Exposed/sculpture lamp lighting Cold cathode
 lighting, an architectural lighting tool with unusual flexibility. Lamps
 fabricated to the architectural design, continuous line of light- low
 brightness - no glare - high efficiency - long life - approaches a permanent
 light source. Remote transformers - no wiring troughs, ballasts, ballast
 failures, or hum. Only 2 leads for up to 120 feet of lamps. Excellent
 uniform dimming - no premature flickering of individual lamps as with hot
 cathode lighting. ■ RITE ANGLE ELECTRODE LAMPS Continuous line
 of light ?H RITE ANGLE ELECTRODE LAMPHOLDERS 94" o.c. max,
 •Clear nylon lamp ties RITE ANGLE ELECTRODE LAMPS continuous
 line of llgth fabricated to shape of soffit Tnl Tnl Tft. |hi jMi Stem hanger ■
 Lampholder housing RITE ANGLE ELECTRODE LAMPHOLDERS
 RITE ANGLE ELECTRODE LAMPS - continuous line of light • Double
 row warm white on dimmer ■ Clear glass neon on dimmer 95" o.c. max.
 RECESSED LAMPHOLDER n. Wff fc#y± LAMP TIES-for hanging down
 lamp only. O Q RITE ANGLE ELECTRODE LAMPS Continuous line of
 light 774

Principal types of lamps for general lighting purposes Construction Details
 and Finishes LIGHTING Planning Data: Principal Lamp types Category
 Type Maximum lamp efficacy lm/W life hrs Characteristic features Typical
 application areas Incandescent Lamps Normal incandescent lamps and
 reflector lamps 22 1,000 Easy to install, easy to use; many different
 versions; Instant start; low cost price; reflector lamps allow concentrated
 light beams General lighting in the home; decorative lighting; localized
 lighting; accent and decorative lighting (reflector lamps) < Halogen 27
 2,000 Compact; high light output; white light; easy to install; long life
 compared with normal incandescent lamps Accent lighting; floodlighting
 Fluorescent Lamps 31 Tubular 104 20,000 HE Wide choice of light colors;
 high lighting levels possible; economical in use All kinds of commercial

and public buildings; streetlighting; home lighting SL* 61 10,000 Energy-effective; direct replacement for incandescent lamps Most applications where incandescent lamps were used before PL- BO 10,000 Compact; long life; energy-effective To create a pleasant atmosphere in social areas, local lighting; signs, security, orientation lighting and general lighting Gas-Discharge Lamps Self-ballasted 28 12,000/16,000 Long life; good color rendering; easy to Install; better efficacy than incandescent lamps Direct replacement for incandescent lamps; small industrial and public light projects; plant irradiation an— r> i,=i .) High pressure mercury 63 24,000 + High efficacy; long life; reasonable color quality Residential area lighting; sports grounds; factory lighting Metal halide 94 15,000 Very high efficacy combined with excellent color rendering; long life Floodlighting, especially for color TV; industrial lighting; road lighting; plant Irradiation <o High pressure sodium 125 24,000 + Very high efficacy; extremely long life; good color rendering Public lighting; floodlighting; industrial lighting; plant irradiation EL; direct replacement for mercury lamps QUI Low pressure sodium 200 18,000 Extremely high efficacy; very long life; high visual acuity; poor color rendering; monochromatic light Many different application areas; wherever energy/cost-effectiveness is important and color is not critical 775

Construction Details and Finishes LIGHTING Planning Data: Incandescent Bulb Sizes INCANDESCENT BULBS Q C-7 Decor C C-15 0 r~\ A Bulb designation consists of a leiici(s) 10 indicate the shape and a l'figure(s) io indicate the approximate major diameter in eights of an inch. Bulbs are measured through their greatest diameter, in eights of an inch. Thus, a F-15 bulb is a Hume shape, 15/8 of an inch or IV, inches in diameter. CA A-15-19 GT PS ER EAR t 1 KR PAR 38 PAR 46 PAR 56 PAR 64 Lumiiime 776

Construction Detaits and Finishes LIGHTING Planning Data: Incandescent Base Types USES V" 8 Mini Can Screw Mini Cdti Candelabra E-14 Screw European Basa Intermediate Inter B-15 Bayonet European Base B-22 Bayonet European Base I I o Single Contact Bayonet Candelabra SC Bay Double Contact Bayonet Candelabra D S Bay .B. ^ ^J e U TJ Candelabra Prafocus SC PI DC PI Single Pin (T 12 5limiinu> Miniature Bipin Min Bipin IT 6 F Lamp! Single Pin IT 6 Slimline! Single Pin (T 8 Slimhnei Mini Screw MS Terminal Scr. Term Ext Mog Mogul End Prong EndProns Mog E

Pi f=\ Medium Prefocus Mod PI 3 Kon Tact Medium 3C Mt'd ff J A Mogul
 Bipin Mog Bipin IT17F Lii-njil kJJ @ ^ Red RSC Hecqssed Single Contact
 Disc iLiimiimci Recessed Dbt Contact (T 12 F L amp) «Q f? Med Biposi D
 Medium Bipin Med Bipin IT8r Umpl D B Medium Bipin Med Bipin (T 12
 F Lamp LXD RSC Recessed Single Contact Mini Cun Socket 7 £g>*=®
 Ceramic Tubular Medium Side Prong 4 Pin (Cwchiufl Mogul Bipost Mog
 Bip Position Orientad Mogul Pos Of Mi«i Type Q Axial Leud Medium
 SkirtHd Mt;d bkf W^ Medium Skirted Mod Ski \^=f Mogul Prefocus Mog
 PI M ogul Mog Three Contact Mogul 3CMoy 777

Construction Details and Finishes LIGHTING Planning Data: Light Level
 Recommendations OFFICE LIGHT LEVEL RECOMMENDATIONS Type
 of Work Corridors, lobbies Easy tasks (Typed originals, ball-point pen
 handwriting, large print) Medium tasks (Poor copies, medium hard pencil,
 small print) Difficult tasks (Very poor copies, hard pencil writing) Foot
 Candles* 10-15-20 20-30-50 50-75-100 100-150-200 "Choose an
 illuminance value in the mid-range for your type of activity- Then decide
 upon a specific value (Same, lower, or higher) within that range by
 considering the age of the workers and the importance of the work.
 SELECTING THE PROPER FIXTURE • Light Output/Efficiency The
 more light, the fewer fixtures needed in new lighting systems and lower
 operating cost. • Visual Comfort Fixtures should direct light to the task and
 away from the eyes. The fixture's VCP rating, available from the fixture
 manufacturer, should be 70 or above. • Maintainability Check ease of lamp
 replacement, cleanability, and permanence of finishes. • Fit In Application
 Should look right and cover the area to be lighted (consider smaller fixtures
 closer together, such as 2 x 2s instead of 2 x 4s, for lower ceilings, or lower
 light levels or high-panelled work stations). STORE LIGHT LEVEL
 RECOMMENDATIONS Circulation Merchandising Feature Displays High
 Activity Area 3Q 1QQ (Mass Merchandiser) JU luu buu Medium (Family
 Depl. Store) 20 70 300 Low (Boutique. Specialty Stores) 10 30 150
 Shielding Materials Comparison of lighting characteristics for typical 2x4
 troffer luminaries: Clear Lens Polarizer Deep Cell Parabolic Louver
 Diffuser Plastic Louver Panel (45°) White Metal •Louver (45°) -Parabolic
 Louver Panel (45°) Toned Lens Dark Metal Louver 50-70 55-60 45-60 40-
 60 45-55 35-45 40-50 30-60 25-40 55-85 60-70 70-85 40-50 50-70 65-85
 99 70-85 70-90 ; INDUSTRIAL LIGHT LEVEL RECOMMENDATIONS

Footcandles Maintained on the Task GARAGES-SERVICE fj repair □
active traffic areas LOADING PLATFORM MACHINE SHOPS AND
ASSEMBLY AREAS □ rough bench/machine work, simple assembly D
medium bench/machine work, moderately difficult assembly
WAREHOUSES, STORAGE ROOMS 50-1 00fc . 10-20fc ,20fc 20-50fc
50-1 00fc 100-200fo . 200-500fc' . 20-50fc 15fc/30fc .5fc "Higher
illuminance values may be achieved through a combination of
supplementary and general lighting. 778

Architectural Woodwork Standard joinery and casework details 781
Woodwork details 804 Cornices and mouldings 866 Furniture hardware 887

Architeclural Woodwork INTRODUCTION Most residential and commercial projects require the design of a certain amount of architectural woodwork. Such woodwork may be in the form of built-in furniture, cabinets, display cases, reception desks, credenzas, work counters, kitchen cabinets, etc. The extent of detail necessary to intelligently communicate and identify the scope and character of required woodwork is an important consideration in the preparation of contract drawings. It is necessary, therefore, that the designer have a knowledge of basic wood joinery and understand how to apply it in the preparation of construction details. Accordingly, the information in this section can be used as a general guide in the detailing of most woodwork items and addresses four areas of concern. The first deals with basic joinery and typical casework details. This information is fundamental to an understanding of the detailing of woodwork. The typical joints illustrated vary in sophistication and structural integrity and represent the most common methods of joining any two wood members. The casework details are intended to illustrate the construction of routine casework and are divided into three categories: exposed face frame, flush overlay, and reveal overlay. The second area deals with custom woodwork and includes details of woodwork items selected directly from contract drawings contributed by various interior design and architectural firms. This information should prove helpful in providing the reader with a more global perspective of how different firms approach the detailing of some common types of woodwork items and the extent of that detailing. The third area of this section deals with standard cornices and mouldings, and is intended to simply provide the designer with

dimensional and design information relative to the many standard items available on the market. Since many woodwork items involve some moveable elements, the fourth area of this section deals with furniture hardware. 780

Architectural Woodwork STANDARD JOINERY AND CASEWORK
DETAILS Typical Joints Characteristics of Joints Joints may be divided into four general types: butted, shiplapped, tongued-and- grooved, and mitered. Used in their simple basic form, none is satisfactory for cabinet work except the tongued-and-grooved type in certain instances. However, when variously combined or when reinforced with gluing and dowels or splines, satisfactory joints can be developed. Butt joint A simple but weak joint that opens easily and may show end wood when used at angles. Strength and range of use is greatly increased by use of the mortise and tenon and dowels and even more when a straight spline is included. Use of a glued butterfly spline with a butt joint produces an extremely strong joint. These variations are widely used to produce large flush surfaces of solid wood or backing for veneers. Shiplap joint Stronger than a butt joint but subject to opening from shrinkage. Rarely used in a simple form in cabinet work except for door rebates. It is often moulded to conceal shrinkage in quirks or combined as a miter and shoulder for corners. Another variation is the shoulder joint. Tongue-and-groove joint A strong joint, widely used for re-entrant angles. Effect of wood shrinkage is concealed when the joint is beaded or otherwise moulded. In expensive cabinet work glued dovetail and multiple tongue-and-groove are used. Miter joints are weak and difficult to fit if used alone. Joints with miter brads are sufficiently strong for short lengths. Joints made in combination with other forms, as a tongue-and-groove miter, are tight and sturdy. Use of Joints Use of certain types of joints depends to a large degree upon the type of work and skill involved. The following notes indicate use of joints in various categories, but cannot be regarded as an inclusive check list. For panels, shelving, etc., or wherever the end of one piece butts against the face of another; housed joint, with or without cover mould, or some type of tongue-and-groove joint. Omit glue to avoid splitting due to swelling or shrinkage. For joining stiles and rails: mortise and tenon, glued in better work. Dowels may be used or hardwood wedges may be driven and glued into ends of tenons in high grade work. For re-entrant corners: shoulder joints for inexpensive

work. Tongue-and-groove is sturdier. Both should be glued, are often screwed together, and may be glued to a rough frame. For external corners: simple miter and quirk and miter both lack strength. Miter brads are practical only for short lengths. Miter and shoulder glued and face-screwed or nailed is satisfactory (generally "millwork"). Miter and spline is preferable, in high grade work exterior corners are reinforced by gluing to a corner post or short lengths of blocking. Glued joints: when screws, nails, etc., cannot be used, or when fine work is to be veneered, strength of the joint depends on accuracy of milling and total glue surface. Glue surface may be tremendously increased by using multiple or offset tongues and grooves, by forming miter cuts into waves, multiple shoulders, tongues and grooves, etc. Such work is cabinet work. If done by a reliable cabinet maker, a guarantee should be obtained and joint detail and composition of glue left to him or her. Mouldings should be applied in continuous lengths if possible. Use simple miter for necessary joints, cope re-entrant angles unless excessively undercut, miter external corners. TYPICAL JOINTS

Letters indicate class of woodwork in which joint is commonly used. See text. A.- Carpentry B, - Millwork C - Cabinet Work D - furniture.

BUTTED.... Burl" (not recommended) Doweled Butterfly Spline • Bull (Mortise & Tenon Dotted) SHIPLAPPED Shiplap (staggered etc) Shoulder Shoulder & Bead TONGUED & GROOVED. Torque & Groove (Finger & Groove etc) TO T-Bead RAIL/PLP* B^C-D Lapped Dove Tail Tongue & Groove (Offset) Torque & Bead Multiple Tongue & Groove MITERED. Miter (Mouldings & Baseboards) Mitered Shoulder Miter & Spline TtG and Miter

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Architectural Woodwork STANDARD JOINERY AND CASEWORK DETAILS typical Joints TYPICAL JOINTS BUTT TONGUE & GROOVE SPLINE HALF LAP HALF LAP SCARF FINGER DOWELED -C*- MORTISE & TENON, SLOTTED MORTISE & TENON, STUB MORTISE & TENON, BLIND MORTISE & TENON, THROUGH RABBIT LOCK DOVETAIL DOVETAIL DADO -A-, A DOVETAIL. BLIND DADO DADO, BLIND OR STOPPED DADO, BLIND OR STOPPED MITER SPLINED MITER MITER, SHOULDER MITER, LOCK 782

Architectural Woodwork STANDARD JOINERY AND CASEWORK
 DETAILS 'typical Joints Terminology Spline joint Used for gluing plywood in width or length, Since the spline serves to align faces, this joint is also used for items requiring site assembly. Stub tenon Joinery method for assembling stile and rail type frames that are additionally supported, such as web or skeleton case frames. Conventional mortise and tenon Joinery method for assembling square-edged surfaces such as case face frames. Dowe joint Alternative joinery method for serving same function as conventional mortise and tenon. Haunch mortise and tenon joint Joinery method for assembling paneled doors or stile and rail type paneling. French dovetail joint Method for joining drawer sides to fronts when fronts conceal metal extension slides or overlay the case faces. Conventional dovetail joint Traditional method for joining drawer sides to fronts or backs. Usually limited to flush or lipped type drawers. Drawer lock-joint Another joinery method for joining drawer sides to fronts. Usually used for flush type installation but can be adapted to lip or overlay type drawers. Edge banding Method of concealing plys or inner cores of plywood or particleboard when edges are exposed. Thickness or configuration will vary with manufacturers' practices. Through dado Conventional joint used for assembly of case body members - dado usually concealed by application of case face frame. Blind dado Variation of conventional dado with applied edge "stopping" or concealing dado groove, Used when case body edge is exposed. Stop dado Another method of concealing dado exposure. Applicable when veneer edging or solid lumber is used. Exposed end detail Illustrates attachment of finished end of case body to front frame using butt joint. Exposed end detail Illustrates attachment of finished end of case body to front frame using mitered joint. Paneled door details Joinery techniques when paneled effect is desired. Profiles are optional as is the use of flat or raised panels. Solid lumber raised panels may be used when width does not exceed 10 in. Rim raised panels recommended when widths exceed this dimension or when transparent finish is used. Spline Joint French Dovetail Joint Stub Tenon Conventional Mortise and Tenon Joint Drawer Lock-Joint Cr--.-1 Dowel Joint Haunch Mortise and Tenon Joint Through Dado 783

Architectural Woodwork STANDARD JOINERY AND CASEWORK
 DETAILS Typical Joints Blind Dado Stop Dado Exposed End Detail
 Exposed End Detail SOLID RAISED PANEL SOLID OVOLO STICKING

- Paneled Door Detail SOLID O.S. STICKING - Paneled Door Detail FLAT OR RAISED PANEL Typical Frame Parts APPLIED RAISED MOULDING Paneled Door Detail 7B4

Architectural Woodwork STANDARD JOINERY AND CASEWORK DETAILS Cabinetwork A C D B im .U»*VU»ttt |A E DIAGRAMMATIC PLAN (No Scale) TYPICAL CABINET CONSTRUCTION Nail Cover Mould to ground, nor ro Gabincr Grounds are carefully located before erecting GROUNDS I COVER. MOULD Grounds Any rrim Opened L_ (min. clearance) DRAWER CLEARANCE FOR CLOSET CABINET 785

Architectural Woodwork STANDARD JOINERY AND CASEWORK DETAILS Cabinet Work Lapped Coverall' Joint Dovetail concealed by joint The sides of drawers should be dove- tailed to fronts. Usual methods shown. Dovetail i concealed by moulds SECTIONAL PLAN When any type of patented drawer slide is used, consult man'f's catalogue for this dimension. The lapped front- conceals slide Dusl-fknels e. top fax of drawer Rurwers should, be flustt 1. 4-H 2. B3Z Bottomless-For towds,efc- Flu* Lattice Bottom- Pcrmite air circulation for linens, etc "-BACK FRONTS | Bottom rabered to front and sides; secured to front only. Runners 6 Guides preferably fimt- wood; Panels either yeneer or solid SIDES A Guide at side of drawer, fastened to Drawer Runner B G\i\<k (Hardwood) rsMcd into side of Drawer. DRAWERS fir Sliding Doors"this space should be slight- ly greater than depth, of wheel grooves to permit the itmoval of doors Any suitable type track (See mM's catalogue) SLIDING DOORS (Removable.) ^m^m Flush front Panel Doors flt Backing Plate Glass Mirror Removable Mouldings DOOR WITH MIRROR Miter i Shoulder Joint Removable Mouldings, mm DETAILS FOR GLAZED DOORS Miter Join? t Hardwood Spline (invisible Hinges Optional DOORS DETAILS FOR PANELED DOORS (Panels may be either plain or ply-mod) This detail permits use of different woods Detail what Cabind does not extend to Ceiling Backf CORNICE Shelf on cleats fastened to grounds in plaster wall Men my type of patented. Shelf-Adjusters are used, consult marif's catalogue for recommended dimensions Shelf edge may be Hardwood. Adjustable Shelf on vvbod Pegs SHELVES Tension Rod with washer £ nut concealed by wood plucj The Section of a Cabinet above Counter (Upper Section) may be supported as shown. With drawer below With moulding

Door- Drawer With Backboard Backboard with Gove Corner COUNTERS
This detail may be used if Cabinet extends to Finish. Floor BASES Toe
Space with Sanitary Cove Floor Covering may finish, against bottom of
Cabinet top^Any Blocking Rough. Floor www) .Poor Floor Covering butts
against base TOE SPACE 786

Architectural Woodwork STANDARD JOINERY AND CASEWORK
DETAILS Cabinet Work Purpose The following information outlines
methods of assembly and installation of common cabinet work, Solutions of
typical problems are presented without attempting to detail specific
cabinets. Assembly High-grade cabinet and veneered work is assembled as
far as possible at the shop. Joints are glued and blocked, and sometimes
secured with finishing nails or screws. Carpentry and millwork are
generally put together with finishing nails if of soft wood, or with screws if
of hardwood. Hardwood should be drilled to prevent splitting before using
nails or screws, and heads should be countersunk and concealed by cover
moulds, moulding quirks, or putty, plastic wood, or other filler, colored to
match the finish. No nails, screws, or joints should be visible unless they
are intentionally incorporated in design. Shrinkage and warping effects can
be largely eliminated by proper detailing and construction. Wide flat
surfaces (solid or veneered) should be made up of several narrow strips
glued and doweled, splined, or dovetailed together. Cleats may also be
screwed or keyed to backs of wide surfaces. Joints in corners, sheathing,
etc., should be concealed within quirks of moulds (as in moulded tongue-
and-groove) or return faces (shoulder joints). Panels should be rigidly
secured on one side only, and are often left entirely loose. Housed joints,
not glued, permit panels to expand and contract without splitting. Large
moulded surfaces (such as cornices or mantels) should always be shop-
assembled and delivered with scribe-moulds (see "Scribing" below) loosely
tacked to assembled units. Installation All grades of woodwork should be
preservative treated or back painted before erection, preferably before
delivery to the job. Satisfactory priming coats are aluminum paint or white
lead in linseed oil, thinned with turpentine or mineral spirits. Preparation
On frame walls plaster may be limited to one or two coats, may be recessed
between studs, or may be omitted. In the latter case, building paper should
be used between woodwork and studs. On masonry, plaster may consist of
one or two coats or may be omitted. Masonry surfaces, particularly exterior

walls, should be waterproofed or woodwork should be protected by a layer of waterproof paper and should always be furred out. When finish of the interior of cabinets is plaster, either plain or canvas covered, the final coat of plaster is applied after erection of cabinet. Grounds of soft wood for attaching cabinet work must be accurately located, are secured directly to framing members or furring, and must be concealed. Blocking of rough lumber should be erected for supporting raised floors and large or heavy cabinet work, if it can be concealed. Blocking must be accurately placed and secured with nails. Shimming Minor irregularities in blocking, furring, or placement of studs may be corrected by using shims (wedge-shaped pieces of wood, often shingles) to bring completed work to plumb and level lines. Shimming should be concealed. Scribing is the practice of fitting edges of cabinet work accurately to all irregularities of finish plaster, masonry, or other abutting surfaces. Wood mouldings, panel frames, or cabinet returns to be scribed should be provided with a beveled edge. Prefabricated woodwork is generally delivered knocked down for assembly on the job and is erected similarly to custom-made work. Consult manufacturers' data. Note, turn wide- faced mould is obtained from, minimum stock. NAILING TO GROUNDS ' Ophonal | Mouldy Solid or-H Paneled This portion is cut (scribed) an job to conform, to lrrt\$- ula.rU.tis in adjoining plaster or masonry surfaces. SCRIBING AGAINST PLASTER OR MASONRY Drill holes before setting screws or driving nails in hardwood.. Concealed faces of ail mtlwork should be dampmbfedby backpbitiwj. CONCEALING ATTACHMENT OF HARDWOODS Blocking cilticr short" lengths or continuous This joint typical, of millwork Nailing' £ juiind a, mortise and shoulder provides a slrond.reas- onably permanent pint GLUING AND BLOCKING Either Haidwood Dowels or Haidwood Wedges for wedies, make saw-cut La tenon, before assembling. HoLts for dowels may be offset to dnuv joint tight. SECURING MORTISE £ TENON Plastic Wood . Compound or Puffy NAILING IN QUIRKS Mi to stiks or mils; m/otdini panel NAILING PANEL MOULD Cope by hand on job unless excessively undercut: if so, use mU.tr. flat numbers may be bulled or Tonjued-and- Grooved COPING MOULDINGS RE-ENTRANT ANGLE 787

Architectural Woodwork STANDARD JOINERY AND CASEWORK
 DETAILS Plastic-Covered Casework Casework Definitions 7. Exposed

portions All surfaces visible when doors and drawers are closed. Underside of bottoms of cabinets over 4'0" above finished floor. Cabinet tops under 6'0" above finished floor or if over 6'0" and visible from an upper building level or floor. Visible front edges of web frames, ends, divisions, tops, shelves, and hanging stiles. Sloping tops of cabinets that are visible. Visible surfaces in open cabinets or behind glass for premium grade only. Interior faces of hinged doors for premium grade only. Visible portions of bottoms, tops, and ends in front of sliding doors in custom and premium grades only. Semi-exposed portions Shelves. Divisions. Interior face of ends, backs, and bottoms. Drawer sides, subfronts, backs, and bottoms. The underside of bottoms of cabinets between 2'6" and 4'0" above the finished floor. Interior faces of hinged doors, except premium grade. Visible surfaces in open cabinets or behind glass for economy and custom grades and all rooms designated as storage, janitor; closet, or utility. Visible portion of bottoms, tops, and ends in front of sliding doors in economy grade only. Concealed portions Toe space unless otherwise specified. Sleepers. Web frames, stretchers, and solid subtops. Security panels. Underside of bottoms of cabinets less than 2'6" above the finished floor. Flat tops of cabinets 6'0" or more above the finished floor, except if visible from an upper building level. The three nonvisible edges of adjustable shelves. The underside of countertops, knee spaces, and drawer aprons. The faces of cabinet ends of adjoining units that butt together. LEGEND EXPOSED SEMI-EXPOSED CZZ1 CONCEALED H| TOP SEMI-EXPOSED WALL CASE SLIDING GLASS DOORS OPEN CASE CONCEALED LOWER CASE Fig. 1 Inside surfaces of open shelf cabinets and behind glass are considered exposed for premium grade and tops of tall cabinets and upper cabinets 6 ft above the floor that are exposed from upper levels are considered exposed.

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Architectural Woodwork STANDARD JOINERY AND CASEWORK
 DETAILS TVplcal Base Cabinet Details CASEWORK CONSTRUCTION
 DETAILS - BASE CABINET 1\{ } '< © ECONOMY CUSTOM 81
 PREMIUM © ECONOMY \JM ¥ © cu STOM WEB FRAME OPTIONAL
 © ECON OMY © ECON OMY / "T ® CUSTOM & PREMIUM \ t
 ©CUSTOM & PREMIUM © ECONOMY V. ^s^^ ^^--^^c^--5',r r r s >>, \ ^
 1 1 1 1 1 1 X- i\ 1 Y A © CUSTOM & PREMIUM (jj) ECONOMY (1)
 CUSTO (J3) PREMIUM 1 \A ^3 (j?) allgrades y th / ©CUSTOM &

PREMIUM @ PREMIUM m ® fin r ..^L- ECONOMY (j?) CUSTOM & PREMIUM (j?) CUSTOM & PREMIUM A M ^ -»■*- \^Z. t2S r ~tz- r 3 20) ECONOMY (ij) CUSTOM & PREMIUM (5) ECONOMY \\\ (5) ECONOMY Ipf X -A (5) CUSTOM & PREMIUM ynpr (a) CUSTOM& PREMIUM k^S: Fig. 2 Casework construction details - base cabinet.

Architectural Woodwork STANDARD JOINERY AND CASEWORK DETAILS TVplcal Upper Cabinet Details 790 Fig. 3 Casework construction details - upper cabinets at the ceiling.

Architectural Woodwork STANDARD JOINERY AND CASEWORK DETAILS Typical Drawer Details DRAWER DETAILS --* CUSTOM & PREMIUM Qj GRADES © FLUSH OVERLAY CUSTOM & PREMIUM GRADES CUSTOM & PREMIUM GRADES ECONOMY GRADE © FLUSH OVERLAY CUSTOM & PREMIUM GRADES © FLUSH OVERLAY CUSTOM & PREMIUM GRADES © ECONOMY GRADE © CUSTOM & PREMIUM GRADES CUSTOM & PREMIUM GRADES — ^ CUSTOM & PREMIUM (JO) GRADES -vfc. '>SSS>>>^ m yw.^< -A '/// ;<^Vr><, <;>);■ / -f % X ® ALL GRADES (j2) ALL GRADES ® ALL GRADES ®P PREMIUM U5J GRADE DRAWER GUIDE DETAILS ® SIDE RUNNER ALL GRADES © SIDE ALL GRADES ® CDRNER ALL GRADES CENTER ALL GRADES (1) Premium Grade requires top edge Df sides to be shaped or stop shaped. (2) Dado not required. Must be securely attached. Fig. 4 Drawer details, 791

Architectural Woodwork STANDARD JOINERY AND CASEWORK DETAILS Typical Flush Overlay Casework Construction FLUSH OVERLAY TYPE CASEWORK COIMSTRUCTIONIOIM DETAILS M CD © OPTIONAL -> © optional UMIUINAL, CEILING ■m-^- >/ m -f&yw* H^/^ 0 Alignment filler L_ Q NOTE: GRAIN OF DRAWER FRONTS.MAY RUN VERTICALLY OR HORIZONTALLY UNLESS SPECIFIED OTHERWISE. OPTIONAL -» i V OPTIONAL DRAWER CONSTRUCTION ® W1 lass &Mt OPTIONAL (2) Alignment fill! ® Sections shown are typical; however, variations are permitted as provided In the standard established for each grade. Fig. 5 Flush overlay type casework construction details.

Architectural Woodwork STANDARD JOINERY AND CASEWORK DETAILS Typical Flush Overlay Casework Construction SECTION AT DOOR MOTION AT OPENING
ADJUSTABLE AND DO NOT EXCEEDS 1/8" WAC6FOH DHAWCft SLIDES as required (typical) 793

Architectural Woodwork STANDARD JOINERY AND CASEWORK DETAILS typical Flush Overlay Casework Construction 10 / 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100. PARTICLE BOARD - 2% SSxX> H1 ^ 13 OUST PANEL OPTIONAL ^ i ^ 1 ' ! i ' 14 794

Architectural Woodwork STANDARD JOINERY AND CASEWORK DETAILS Solid Core Wood Flush Doors TOP AND BOTTOM RAILS - MINIMUM 1 INCH (25.40mm) WIDTH OF STILES - MINIMUM 1 INCH (25.40mm) CORE OF MAT-FORMED COMPOSITION BOARD COMBINED THICKNESS OF EACH FACE PANEL - MINIMUM 1/12 INCH (2.12mm) FACE VENEER TOP AND BOTTOM RAILS - MINIMUM 1 INCH (25.40mm) CORE OF MAT-FORMED COMPOSITION BOARD WIDTH OF STILES - MINIMUM 1 INCH (25.40mm) COMBINED THICKNESS OF EACH FACE PANEL - MINIMUM 1/12 INCH (2.12mm) FACE VENEER Fig. 6 Mat-formed wood particleboard core (7-ply construction Illustrated). Fig. 7 Mat-formed wood particleboard core (5-ply construction Illustrated). Fig. 8 Mat-formed wood particleboard core (3-ply construction Illustrated). Fig. 9 Glued block core (5-ply construction Illustrated). 795

Architectural Woodwork STANDARD JOINERY AND CASEWORK DETAILS Solid Core and Hollow Core Wood Flush Doors SOLID CORE WOOD FLUSH DOORS TOP AND BOTTOM RAILS - MINIMUM 1

INCH (25.40mm) WIDTH OF STILES ■ MINIMUM 1 INCH (25.40mm)
 WOOD CORE BLOCKS - ANY LENGTH -JOINTS STAGGER ED -
 BLOCKS GLUED TOGETHER COMBINED THICKNESS OF EACH
 FACE PANEL -MINIMUM 1/12 INCH (2.12mm) FACE VENEER Fig. 10
 Framed block glued core (7-ply construction illustrated). TOP AND
 BOTTOM RAILS WIDTH OF STILES CENTRAL WOOD BLOCK
 CORE FRAMED BLOCK CORE ILLUSTRATED CORE LINER
 COMBINED THICKNESS OF EACH FACE PANEL -MINIMUM 1/12
 INCH (2.12mm) -FACE VENEER Fig. 11 Wood block lined core (7-ply
 construction Illustrated). HOLLOW CORE WOOD FLUSH DOORS TOP
 AND BOTTOM RAILS -MINIMUM 2X INCHES (67.2mm) WIDTH OF
 STILES • MINIMUM 1 INCH (25.4mm) WOOD OR WOOD
 DERIVATIVE STRIPS - STRIPS MAY BE STAGGERED OR FULL
 CORE SIZE ■ STRIPS MAY RUN VERTICAL OR HORIZONTAL LOCK
 BLOCKS -REQUIRE -MINIMUM LENGTH 20 INCHES (508mm) -
 MID POINT OF LOCK BLOCK LOCATED AT MID POINT OF STILE
 WIDTH AS SPECIFIED IN 3.1.4 COMBINED THICKNESS OF EACH
 FACE PANEL ■ MINIMUM 1/10 INCH (2.54mm) FACE VENEER Fig.
 12 Ladder core (7-ply construction illustrated). - ■, TOP AND
 BOTTOM RAILS -MINIMUM 2X INCHES (57.2mm) WIDTH OF
 STILES ■ MINIMUM 1 INCH (25.4mm) WOOD OR WOOD
 DERIVATIVE STRIPS ■ PAPER HONEY COMB MATERIAL
 ILLUSTRATED LOCK BLOCKS -REQUIRE - MINIMUM LENGTH
 20 INCHES (508mm) -MIDPOINT OF LOCK BLOCK LOCATED AT
 MIDPOINT OF STILE ■ WIDTH AS SPECIFIED IN 3.1.4 COMBINED
 THICKNESS OF EACH FACE PANEL • MINIMUM 1/10 INCH
 (2.54mm) FACE VENEER Fig. 13 Mesh or cellular core (7-ply
 construction Illustrated). 796

Architectural Woodwork STANDARD JOINERY AND CASEWORK
 DETAILS Cabinet Door and Banding Types a. FLUSH CABINET DOORS.
 All WIC Grades TYPE "T" Veneer/tape banding, 1/16" maximum. TYPE "2"
 Solid banding. TYPE "3" — Medium Density Fiberboard. Banding not
 required for Economy and Custom Grades. Band required for Premium
 Grade. b. LIPPED CABINET DOORS. TYPE "4" Veneer/tape banding,
 1/16" maximum, required. TYPE "5" Solid banding. TYPE "VZ 3Ply Skin TYPE
 "6" — Medium Density Fiberboard. Banding not required for Economy and

Custom Grades. Band required for Premium Grade. >±wk TYPE "4" TYPE "5" TYPE "5" TYPE "6" > <> <r c. STILE AND RAIL CABINET DOORS. All WIC Grades. TYPE "7", S4S Stop. TYPE "8", Solid Stuck. TYPE "9", Moulded Stop. d. The top and bottom edges of sliding doors do not require an edge band. 797

Architectural Woodwork STANDARD JOINERY AND CASEWORK
DETAILS Stile and Rail Joinery Details 0 Vertical Section Fig. 14 Full-height stile and rail raised paneling. Stile and rail wall paneling accented by raised panels creates a beautiful effect of traditional architectural woodwork. Framed within the stiles and rails and accented by the shadow lines, this construction offers limitless opportunities for various effects through the use of different wood species and veneer cuts. Each design creates a unique atmosphere complimented by the finely proportioned paneling. 798

Architectural Woodwork STANDARD JOINERY AND CASEWORK
DETAILS Paneled Wainscot and Doors @ Solid Stile and Rail with Flat Panel Product WW/ © Solid Stile and Rail with Up Moulding and Panel Product with Solid Rim rSWgtAB. © Panel Product Stile and Rail and Panel with Raised Moulding Fin. Floor (5) Vertical Section Fig. 15 Flap paneled wainscot. Flat panels set within the frame of the stile and rail create a rich effect of traditional architectural woodwork. Different results can be produced through the use of veneer selections with transparent finish or painted finishes chosen by the architect or designer. ^^f^KP^ e Veneered Stile and Rails with Loose Mouldings, Flat Panel Product or Glass (5) Veneered Stile and Rails with Loose Moulding and Panel Product with Solid Rim aeaoH ©Veneered Stile and Rails with Solid Profile Moulding and Panel Products with Solid Rim @ Solid Lumber Construction Throughout Fig. 16 Paneled doors. Stile and rail doors designed to accent the adjacent wall paneling whether traditional or contemporary, or used alone, beautify an entryway or area. 799

Architectural Woodwork STANDARD JOINERY AND CASEWORK
DETAILS Raised Paneling (e) Inside Corner (T) Wetting Sillej (5) Jamb VUU Fin. Ceiling ® Top Rail (3) Mullion (4) Bottom Rail (7) Bottom Rail Fig. 17 Full-height contemporary raised paneling. This design,

distinguished by its simplicity, is a contemporary expression of the stile and rail construction. 800

Architectural Woodwork STANDARD JOINERY AND CASEWORK
DETAILS Sash, French, and Panel Doors PROFILES OF STOCK
STICKING FOR SASH & DOORS OGEE SQUARE B & C OGEE '////
/,1-v //</// / % aU- w -'■U/- J///)7Th, W//ti / / / / ut OGEE OGEE SQUARE
SQUARE SASH & FRENCH DOORS B & C OVOLO OVOLO OGEE
SQUARE PANEL DOORS —\jL-L iuA. SQUARE 801

Architectural Woodwork STANDARD JOINERY AND CASEWORK
DETAILS Double Hung Windows Fig. 18 Double hung windows: stock
designs. Standard widths are 1'6", 2'0", 2'6", 3'0", 3'6", 4'0", 4'6", and 5'0".
Standard heights are 2'0", 2'6", 3'0", 3'6", 4'0", 4'6", 5'1", 5'6", and 6'0".
Standard thicknesses are W and W. Stock thickness is W. Standard glazing
is s.s.b. glass -not bedded. n = Fig. 19 Single sash: stock designs. Standard
widths are 2'0", 2'6", 3'0", 3'6", 4'0", 4'6", 5'0", 5'6" and 6'0". Standard
heights are 2'6", 3'0", 3'6", 4'0", 4'6", 5'1", 5'6", 6'0", 6'6", and 7'0".
Standard thicknesses are W and W. Stock thickness is 13/8". Standard
glazing is s.s.b. glass - not bedded. 802

Architectural Woodwork STANDARD JOINERY AND CASEWORK
DETAILS Casement Windows Fig. 20 Casements In pairs: stock designs.
Standard widths are 2'6", 3'0", 3'6", 4'0", 4'6", and 5'0". Standard heights
are 2'0", 2'6", 3'0", 3'6", 4'0", 4'6", 5'1", 5'6", and 6'0". Standard thicknesses
are 1" and W. Stock thickness is 1". Standard glazing is s.s.b. glass - not
bedded. Fig. 21 Windows and sash: typical diamond light cut-ups. 803

Architectural Woodwork WOODWORK DETAILS Wood Paneling
ELEVATION /cale : V - i' - o" • DETAIL-on-LINE-A-A 804

Architectural Woodwork WOODWORK DETAILS Wood Paneling 4 5 ••**
am K5 •\<4- IC|-*<4»- ELEVATION PLAK-A-A' paneled wainscot
.SECTION-CO 1 ^Wood Mold Plasfev .1 <0 -2-5*- _ — _ ?i" -11^- 4"
ELEVATION =BE*: S3E =t&££2 -5-0' FRENCH PANELTnO " V •s } r
ELEVATION &T-B- SECTION-C-C 805

Architectural Woodwork WOODWORK DETAILS Corner Cupboard lin
teilma . r\ ELEVATION S c & I e : Va 806 PLAN 1-0"

Architectural Woodwork WOODWORK DETAILS Corner Cupboard tin
plaster ceiliraa-i> |5wood cornice E ILE V ATI ON S o V4' - V O" SEC-A-
A SECBB" PLAN ¥V - l'-o" 807

Architectural Woodwork WOODWORK DETAILS Bookcases fin plaster
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.LI Mv .,1 ■^•SCALE-ELEVATION- -DOOKbACiC wmct- r- to-1
Showmct- book. sb«lv m<i-»ni-secret Door, to-&■ silver closer- • ■
(©pencil) \ ,/^ / ■ %'■ SCALE PLAN- of ■bOOK^SHELVES- ■a secret-
door- Vertical- Section -elevation- • thpovoh • door^ of pilaster. •SwJe.: ite'
- U-O" ■SECTION OH- LINE -"A-A'-THRV-SHELVING- •Section- on-
line ■'^"■thrlz-cvpeoard- ScsJc- i'&" — j:—o" 808

Architectural Woodwork i aming '^ blocking 2" SJ1CT 16 wM^m4 » rCo)d
ro ■Adjustable shelf Aifppcfia, Ji'dd- jusrmepT. Finish. asdeiji'red..
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place 10 aiinoaro- ^ 5XC1]IOM-THBll'JAMD-D-I\$ | -^upporK standards r
^^^^^^^MI ^ KfMkX -rrammg- Scale {or Scale (or details elevations
SICTIC N-TH2.U- S!LL-"A-A' -Ir -Jhvr- •£>" Opening between rooms-
Fir. line*— I ' <*■ 5U imH-BCSDia^SES-IOTH-Slg^q- P-QDM-
DnD^WAY- WOODWORK DETAILS Bookcases 809

Architectural Woodwork WOODWORK DETAILS Bookcases E i _ 1
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plasjer ■! «—Franking ^ blocking- Jn design of boot cases, keep in rnind
type of books (he She yes are to .accommodate for the home a pi" to
lO'snelf 15 sufficient, for the office, business.etc. having bg books greater
depth IS required. |!" 10"-+ ^mssmsw|«s S£(lj.TI(p^THfiU;HIAD°1-DD- -
Taming ■Aidal corner bead. '■Mffl^mum Veneered backer lO)i-
THEU0MD:£-£1 :=±—If ' lr WMSMSMSSM 777777, The use o(adjustable
jkhelf supports is recommended wmMM2RWM Fnarn'mg - Base.
SECTION-THEODASE: Mr SECTION-BOOKCASE UNIT 810

Architectural Woodwork WOODWORK DETAILS Bookcases FULL JIZE
JECTION-B- F..5..5ECTIONt-A

Architectural Woodwork WOODWORK DETAILS Library Cabinets and
Bookcases i7/e±i. VERTICAL SECTION LIBRARY CABINETS
VERTICAL SECTION-LIBRARY BOOKCASE 812

Architectural Woodwork WOODWORK DETAILS Miscellaneous Details
19'-IO'/2' It vhi w ~-ir S%B" 3i w s_jJHf -t l59/»' r i i] fr..". Zzzi] OI.E
ELEVATION AT BAR ¥ '■"--■»=- >i i' w ELEVATION AT PORTAL
J^^^^S !^_ Ji-Jr =^ J, CONCRETE +- «««« - -.-/ \$UB t i ELEVATION AT
BAR VERTICAL SECTION AT BAR 813

Architectural Woodwork WOODWORK DETAILS Trader's Wall 814 Fig.1

Architectural Woodwork WOODWORK DETAILS Trader's Wall WAJ-U
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vyvEtgricag. MgwiFSgg.^ I6»"Q.C. l»3V V>a!.»il»a-g.«TA.'a.P*.MT
MWT1 W*laHT "SOP" TS5AJC PLAAtC UA.MINA.Tt RMVCICINfi
*HBET AX9 «UH.FI»A.Ma. IfcV DRYOND *ECTION wHixm. j"lai»tic
L_a.MiH*.-rm _A.MIN.WTt C4.MNCT HTKBLIOra- TO »■ V*."
BP1MLCU ^HKLI* vyN*.-Tua>sL. LiHB OF F4ACIA. IPUVOHO
r»L/w»TIC u;wM,r-4*.T& Dooft w pivot HiHae.«a 4 »/»■' wiiaa puuu« 4.
M.W-NMLTIC fiA.T«£H»* ill CXRPET 3! n<k.i°>E.o Floor. l=l_ooya.
>&u/fcft» 2 /,», V-I'-er' Fig. 1 (Continued) 815

Architectural Woodwork WOODWORK DETAILS Storage Cabinets "5
'A- plywood, 1/4" PLYWOOP 1/Z" PLYWOOP ± >1^>6S/^_A I M"
SCALE V8"SCALE ^cX^rcotZcm— ADJUSTABLE SHELVES DOOrtS
NOT SHOWN W PLYWOOD PANEL* ADJUSTABLE SHELVES B ■J/4"
SUPPORTING PARTITIONS POTTED \7- IT I__E 816

Architectural Woodwork WOODWORK DETAILS Wardrobe B- DETAIL
PLAM t 3h«lf 3= ihown. poor not "VD te Door nof ahown ELEVATION
DETArl PLArt D DETAIL 3ECTION 817

Architectural Woodwork WOODWORK DETAILS Bar Sink DTTT E. fo f
I Ar roniL r l A. n ruj- c* u&rM*¥ th- ;| ',^STIW* i. o w.01. n*e* «"cnw of
iw.sc JO M&TCW CWKNKT BCTWBW Ojvt t, oeeatrbM, , . MATCUIfJC
W, C* TRIM t, "T ^^--tBJCWJeSS 0»t DOOH.TXJM. , Q V— STTLES (l,
ft*.*.*) SWUS Uci^ajrceta wen* tof*ey**n op ooore. tium - UWE OP
DOOO.T1UM 3 I D E .E'L t V U 1 O N 818

Architectural Woodwork WOODWORK DETAILS Display Case
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CAS& 819

Architectural Woodwork WOODWORK DETAILS Display Case OK M.t-i
820

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Architectural Woodwork WOODWORK DETAILS Display Case "Sgrfrieer
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UAmiL'UiM^ IHIIIWlldllml,!1 JL PislR*v mefi«N PlaM-^ ...wall-
.JcTy^iNer^l IA HLEVATIOH c?f= . V/ALL - c^SINCT 1 ■•N RETAILS
OF V/ALL-CABIN^T^i -HtfEira-nAl. -7D nsvsWRso »*jee» - Fig. 2 822

Architectural Woodwork WOODWORK DETAILS Display Case X f»n»
www r-^s f*. l ¥f 4, .ivm'A' EfW^I^FPF?5? 15" <I»H»T> sjew 12" «m-B-
uiH. _ u. IW-^if^LS1- ■* ^P^ -jWl (q«M n>£S.o««<iiru:. *Ma«Mtt.L)
SiHish) . prmw •nK/PfMao !i4'*U0lN<f.PL<fcA«««.paMS c-
nOKUAsaawmg pnwce «BS»*«ff»/iS nuc/tr PIHKft). PR»Vlt><r .»«*• N-
450 WDW-IHL004 1-Blto^TCUIS DOOM f «M9CViM£5B T*""*r not
ihnwi -HciNi* eoats. %) .vegna^L.. ■•gesrt^N .- WALL- ^ABINET^i
<SU£S PSOK.T134CK L_ Vi J3" *TO»(f»' /IPBR") ^Z3li TVPKAL STILE
«HlHc5re EMP Fig. 2 (Continued) 823

Architectural Woodwork WOODWORK DETAILS Work Counter
iiMFfMiir eukixi - ® SQTU CL,g VATIOU SfA.CC I I A 8QUTU CI n«e»
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H,*TmM<* |V« I1!'. 'i' kUMILLM 1'- ^iTi.i.m.t\$ tr ***** —►> K hvito
^b#^in>4L-n I file-*-' '••' 4=i fa I.V*^ Fig. 3 Countertop. W1U-,
ELSWiTIOM or tAch COUUTS*. tfWiUBfluJ I im IIM.OLS H"t,o+ ar

»V'l'-«- rmV -tM r ><■»• i; >*a* !•'■ 10' fTliL *HAW<t [I'M* H.fQ 'IU.
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h^fc4. E~ KihT: I?.. V ■4^ ^ ^buiTic um '+. rfi 4 fldl 1AM FMITiA UAM. -
***« •* 4tlUMU l9Ki£fc? Of TUb» "MA.WJ Fig. 4 Back counter. (/>n)
\Wm\9j (\$ << ***.». I'll 824

Architectural Woodwork WOODWORK DETAILS Work Counter ruufi.
Fig. S Word processing counter. LM bsd "=E1 I sc Fig. 6 Cashier counter. -
peoe :sr £ fa-A£,UM. --ss&H^p f5&eri-rt(7ri -z.'-o'.ojly i*or 14*1 A*.
«rtLUPfe iiiifeBfefeY *>*" —^ r ■tiXa.LW*. 6Pla>Uc£ t>iH e -S£ H^=i
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(S>eoMt-ier ANP COTO-IT _f ACE £** SCM6P mJ -^au4>raw&
i&v£ufc\$. Fig. 7 Base and work counter. 'f zV - la 032 Fig. 8 Walkup
counter. ** &au>Y\U \9&(dhq 825

Architectural Woodwork WOODWORK DETAILS Work Counter 4KjO
Hasp jTWO'f !fti<-2.lll&»}.}- •- —•--■ ■ A SH^KW* Vj iiHtf'C. AU i>H
>j/,«|«fcj-«ae TC 6LA6 ASCIIS- JO tfiiOb ?0 eHMIMA'e ----af A 5^ 60Nf
fiM, 6flecgt4A 6 AM 4^6 frH&LF jhlT ^^^ pyrtwr -f Wt^#.-;
£i!ea.pU>H)-- HAW -SOME. .1 i '% VA«6» 1 v-t ^W<n !:Jt?t3 -4 / '=5' 1
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hNTiP 69JNTEP, ilNIT- TAX Ue^ftf -1 nTlfer\TKN CQ^TB^ rW 826

Architectural Woodwork TYP FiLt U<SHT WOODWORK DETAILS
Overfile Cabinet (P-0 OVERFILE FILE EMD CCftiL " 827

Architectural Woodwork WOODWORK DETAILS OverHle Cabinet
rrV*fTH r»LAFti6 Lam,kA**I PibhIM PANTbf 'I- cB — * AfJiMUfc.1
IU«LWtf * M«TAt« tM>TI «3'-<t« '•'•' 2 50UTU S.UeV>WTlOlJ . -IU
V^V^U^~UJ^ "LA 5, -AM,- -UITM.*<-IJ »T JTdl»T— ■- r> ?-fej 1
OTW»M#- ^ ^ .l^^lft -UA*Bw«» *U*h PA.1WT wmTf Sui^tt-oto ci-uiua- -
^^^^^v^ / J ■ AOJUiTikU tlWCU nvwwg fJiLvn aw. sa-mcv tnl .XV- ^ tf N
<IUi.TI- 'tv»T ^S^ - ^ .sr^ (fi> Mot&: a,,»i»Cj T«MAvft ua7u*ia« ej .»!>
ill! Fig. 9 Wall unit with overhead cabinets, ^W 328 Fig, 10 Free-standing
Island unit, PLAftIC 1-4*1. •uArrn bib!, ,; 'j >^b fixl

Architectural Woodwork WOODWORK DETAILS Display Cabinet ® 829

Architectural Woodwork WOODWORK DETAILS Credenza •t * CfiMHX\
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VEfJeep, wo pweL.fw-0 oH V4" cup «rV«. • I" qwsiTt, to" KvtuEc? cast
<S-2 *e*3 ATfflCWtEMT 830 ££>UNTE£. "fi?P

Architectural Woodwork WOODWORK DETAILS Plantar and Storage
Cabinet WD TUMI ^jr^vV.v.p; (&*§ 831

Architectural Woodwork WOODWORK DETAILS Book and Shawl
Cabinet <0> m .gLHVATlc^Hg &F SO«PKiy SKAWL GAglNST-g. &y4-
\\z\\ FT-AH CXF g^g^fgH-AWL <£4glNETS ±o" * I ^- Notched puwtwap
Pws pi>W«sc> tiv/.eiae. 3D \\,, . "pivorrtNdSBc-1 H^W.V.p: easorrs
1YPICAL ^T/LEl PET&IL. Fig. 11 832

Architectural Woodwork 3 eOQ&SUML CA51MET DETAILS
WOODWORK DETAILS Book and Shawl Cabinet Flfl.11 (Continued) 833

Architectural Woodwork WOODWORK DETAILS Storage Shelves
«A,gU4TH('Wl »«M«»UL>) PLiWR tL&VATIOM OP STOfiASC **<
4fMI«l.»* *lh<M 11*1' Pi*** 1* 4LA, *Ah*>f 1-TAHIAhH MAU *\$/
WMk ••••«# M»T<M »M»k> A*»UWI» I k»*t *p «H*tJ> -r«» or vam. ^i
834 f(,-S-tAMftA* >* ir A'V i n ' PAMU. -LIMi 0»T*» J»TT#M llll, 1 l!^
"^ ill' WT»PrV»UA1* 'I ■*il MAX. « :l *v eLBVATIOM OP STOrSAgR
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Architectural Woodwork WOODWORK DETAILS Coat Closets ClM-rt*
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OF UUIT 83S

Architectural Woodwork WOODWORK DETAILS ///yy/y //////////>
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*C*V»* TO WAUI '••s' w ^ « 6 Fig.12 Credenza £ 4 Etc - I'. I'. 'j'
JLUMIdUU 4H«>.f "r fir. r-o- -ins** J^ ALUM/MUM MUDAH.EX £ <S
FUUL titC 836 »vt. l.h jity. !>■-•' ® A \ >uO>^tu ai-av*.Tioki 5r>.ca. u., ig
Fig. 13 File countertop. Ti|. >IU> CAklI-KT ^-9fl»i.U T» WALL (m> e 'i
full s<z.»

Architectural Woodwork WOODWORK DETAILS Base Cabinets r-Ut HO
tu'vATlOIJ on»un.* a»ii.uT)» ruuu (x-riMrwu 3«ni,w«»i CUBIC 2. PILt
PM\«*h HULL UtVf S»V*rfT SLIM I l^hrrricr l£**l»t T* WU.1* axmsv
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re an Al-L imTIHIO*^i TO_uU>Vg uatuiha.. ,A.CQuep-> s »•« r-«- 837

Architectural Woodwork WOODWORK DETAILS a*. 7T 3- I=r d
kLOHTU gL5iV-*.TIOM SrxCSS UnlS, if < 2.Q « » J t >»■ I'-o" v.v. ;fffg
■v«u«»iTt ShMUKlt 1 CLAMfff JflUTJ—J rCblkJIIiJAUIIUTi &W-> <
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c*»"« OTut*,« UOTEl_c2 i 'uu. siit C IV 5 838

Architectural Woodwork WOODWORK DETAILS Pantry Cabinets I ABU-
*M6l£ • *tos>. lW"» «^*3t>, t e-e., \V*<. catch C*£AMIG TILC. Of-,
VIKl-TU &*5t SEVH.ED CUAT FIXED 10 CABINET ^\KJTRY
£AE>INET£> SUP JOINT ON <I«HT Of -XMHSIOM JOINT VERTICAL
SECTION-PANTRY CABINETS 839

Architectural Woodwork WOODWORK DETAILS Altars Pulpit
3'9"Highx3,ffWidexl,8V4"D Table 27"Highx5'0"Longx2'D Center Pulpit
Chair 4'Highx2'2'2"Widexl'10"D Side Pulpit Chair 3'9"High x 2'21/2"Wide
x 1'10"D Communion Chair 3'4"Highx 1W Widex 17>/2"D Flower Stand
2'6" High xl'3" Square Pulpit 4'0" High x3'6" Wide xl'8"D Table
2'9"Highx6'0"Longx2'0"D Flower Stand 2 '0" High xl'2" Square Pulpit
3'9"Highx3'4"Widexl'8"D Table 2'9" High x6'0" Long x2'0"D Flower Stand
2'0" High xl'2" Square Clergy Pew End Pulpit 4'0"Highx3'8"Widexl'9"D
Table 2'9"Highx4'6"Longx2'0"D Flower Stand 2'0" High xl'2" Square
Clergy Pew End 840 Fig. 14

Architectural Woodwork WOODWORK DETAILS Pews W6 PLY BACK
1W * VSOLID SEAT BLOCK 2W9EATPITCH 17"SEATWIDTH 13/16"
FLAKE BOARD 4" POLY-FOAM 1W"K 1" SOLID SEAT BLOCK 2V4-
SEATPITCH 17"SEATWIDTH Fig. 15 Fully upholstered seat. All exposed
surfaces of the seat and back are fully upholstered, ' N46 BODY SAME
EXCEPT WOOD ON BACK OF BACK. Fig. 16 Combination
uphoistered/wood seat. An uphoistered seat with a wooden back (either
solid or veneer laminate). «" 6 PLY SEAT AND BACK 15"SEATWIDTH
Fig. 17 Aii wood seat. Either solid wood or veneered seat and back.

Generally the most expensive option. Wood seats and back can be contoured for increased comfort. Cup holder Card and pencil holder Wood, with bonded foam, upholstered pad Metal, with bonded foam, upholstered pad Fig. 18 Pew accessories. Fig. 19 Screens are adaptable for many uses in the sanctuary. They serve as modesty screens in front of the first row of pews, or they can be adapted for use as communion rails, as choir boxes, or as wainscoting. 841

Architectural Woodwork WOODWORK DETAILS Sanctuary Doors and Miscellaneous Details ELEVATION- SANCTUARY DOORS SECTIONS- SANCTUARY DOORS SECTION AT SANCTUARY COLUMN SECTION AT WAINSCOT PLAN AT INTERSECTION OF CHANCEL RAIL AND COLUMN TREFOIL LAYOUT 842

Architectural Woodwork WOODWORK DETAILS Lectern G> - FIMI5MEI7 OP*X- PACE- pull size section thru b-imah top >'PIAM. WJLLMOST CVKM. STAINeD CAK. CVUHBfcC LO£-K. BCASi PULL-5 tOHOCALEC IH3ET MIHOC TO? * POTfOM !"VEJ&HTel7 OASfc. POR. 3TA&IL.ITV fc\SE., CARX- OAK. .STAIN ®- Ffr-ONT ELEVATION (EAST SIDE '1 _____ |j-^— r_0. 1*' li _5Z L. _t-v *> FROVIPft * INSTALL »JCSS5 ADJUSTING M/MipVMle TO TILT t l- IOLD TOP J"ltOM O' fMOKJ-ONTAL re>»TION)lOl^" 1.1. i LIMB. OP i^MJIMET PC.O'V TOP MNlSHtp, 80MC (3AK. tl?i>Ey CAK. PET k d> 5IPE ELEVATION (SOUTH SIDE) ■SCALE-.' I>_ " © PLAN OF frlMAli TOP SCALE.: lk_ " " I'-O" 843

Architectural Woodwork WOODWORK DETAILS Lectern iJ« <mi. 1152, use ism* PUTt* 473S".W7<* SO. HWU |Wef/8«U1W. ^ ELBWflokI frWf Vi£V e £,UE-VAT10SJ »! [?£ ■ ALL «<T1SR()*U 1» G>6 qLuW 5/+ " HkH1 WuiJ? y/fwufisp i.op -e HaJHia? neifcL. -fa***. ftKM4T, OB. 6!). 2'V.I" 01^*PJW£|- Trii* *IP*. I "«■ } vnneiw *p<^i*, W.WeVT vPtm*v»,& TOW**- I'-lite' PRiW. fHkltJt«lJ (*R, aetMef pcf. p£} (?) MOTt j -JBf Cf R*PR pp» - wsf M.i(jhl lb PMIEL rIOJU>l>J4 jstj^rs pun* cip; "e*wsil>J erm Klft 41*1; 113Z, ilK 0W* fl*b» i.735,-1737. -*3+t oft (ifl. ftf (4) Mr"x3&' LWV* HkiwjKl* r*. rv*r> 0«tw»bJ tf*aucr(fuerK)fc" ttWEt ,1V. fre^-fiPKl 3 \ 844 £lxyftpokl EA^ Vie-V

Architectural Woodwork WOODWORK DETAILS Lectern SECTION
VIEW THRU LECTERN STANDS Kncil drawer PLAN VIEW OF
LECTERN STANDS PLAN SECTION CROSS SECTION a PODIUM 84S

Architectural Woodwork WOODWORK DETAILS Lectern & ZW "/^
30J&-Z. "^ 3&- \& sis^y +—*■ tf»» 17 l&XM fWPLAH ^> is* y HN , >
21-*1 0& , 0<k 0A PIK|'& H,i 1 ^ . nv\7 4\ve.v\zti W* .| ■if ax*r&& T* I ^
V—i «\ i ~<& ,i ..>*. V -tv r r r - r

Architectural Woodwork WOODWORK DETAILS Boartfroom Table
VERTICAL SECTION-BOARDROOM TABLE :'/«V*MAHOGANY
RIBBON' MAHOGANY ACCESS PANEL SECTION 847

Architectural Woodwork WOODWORK DETAILS Bpardroom Table
CROSS SECTION AT BOARDROOM TABLE

}<—End support <H§) l^iM PARTIAL FLAN OF BENCH ■ Poc« of
courtroom S ECTI O N D - D Clip on^le TYPICAL SECTION THRU
REaf? BENCH Architectural Woodwork WOODWORK DETAILS Bench
-Tthick s«of i bock support Com lack device Carpet SEAT AND BACK
SUPPORT OAK CAPfINO-^ QUHK zzvz* "iff RADII -WO. Jt
couNHRrop 7- -2>0FUTFORM 16 OC IH SUBROOR . pawooo !A.
SECTION THRU JUDGES BENCH 849

Architectural Woodwork WOODWORK DETAILS Stair Screen and Seat
BLfiV. •l3(!ZJlBH'ABOVg^EAir tfof 4DX£D •MOUbDgD-BOARDINQ •
SEAT-BACK- BAUUSTGE.- Maun--ftp- ■^8C:THE17-3E£l' ^ f V. 0/
JUBSt* Jft^g -5 3BAT-8KD- MS f.uaunoa: DBTAIL'OMTArB.^r.R.BRN'/:-
SB^T'MAIK'HALL- 850

Architectural Woodwork WOODWORK DETAILS Window Seat 851

Architectural Woodwork WOODWORK DETAILS Wood Railing and Gate
PLAN OF RAI L AND GATE Solid z%'- , - 4—^"plywood ■plyu 1 1 Solid
h-\ r-Carpef Solid 2%.'- -/£'plywood .Cafe -Ji" pi plyi Mardudre as specif
ied- C Court room floor Carper Wood t>c Roil '= \- E L E VATI ON OF RAI
LAND GATE C-Courtroom fl.— L3rPef'^ *— dip anqte "" 'Zja -M m i n.
j» plywood i SECTI O N A - A \p anql S ECTION 0 - B P—Cam-lock

device SECTION C- C Cou r floor Carpel- urtroe Key scram Alum, cam
lock Key sere ai Alum, cam lock 'Vverl, h< If 'A'verfical riol W Tapping
sereuj Tappincjscre ■■■', ' ni i- i Pi a slic anchor » Plastic onchor m: E LE
V.ATI ON ■CAM ..J..Q r.k A^EMBL Y Solid uj o o d Courtroom fl. I Carpet
852

Architectural Woodwork WOODWORK DETAILS Hidden Wood Door
3'-9"tf ORYWALL OPENING DETAIL-HIDDEN DOOR SOUO FILLER
BLOCK- LET INTO TW OF DOOB ANO PANEL HEAD DETAIL 4'2"
ZO'8' TvX S^<^z >^^~<£; l»T/8' DOOR OPENIN6 DETAIL-HIDDEN
PASS THROUGH DOOR DETAIL 853

Architectural Woodwork WOODWORK DETAILS Raised Panel Wood
Doors 4'-S STONE OPBMWe LtAPeP GLASS 'r—fi' m 1)—(1 p=—j*1 U
!* ELEVATION-FRONT ENTRY JAMB DETAIL AT ENTRY fo' LiT1^*
HARDWOOD BDdB V6 HARDWOOD TACB SECTION "A" TVPICAL
DOORS ELEVATION OF SPECIAL POOR 854 JAMB DETAIL AND
DOOR DETAIL

Architectural Woodwork WOODWORK DETAILS Miscellaneous Details
i'-o" Section SECTION SECTION CREDENZA VERTICAL SECTION-
RECEPTION DESK VERTICAL SECTION AT TELLERS COUNTER
855

Architectural Woodwork WOODWORK DETAILS Cornices
TYPICALTRIM AND CORNICE SECTION AT ENTABLATURE <v<* if
PANELING Sf SECTION-TYPICAL CEILING MOULDINGS 856

Architectural Woodwork WOODWORK DETAILS Cornices — : ' 'j~r<- \ I
—T-l '-*■ ,3 EC f/OH 7HZ0 £#A/£ n KA/L Jfa. a* DETAIL °^OOD Comet
MP PAMU/tG 857

Architectural Woodwork WOODWORK DETAILS Cornices
■MATEK.IAU- KNOTTY PINE .STAINED.SHfil.LACEP&HWlia
LECTIO. fitiUH Wood nocvz?- DETAILb-
OF'WOOD'WALL^'CORNICE'LIBRARY' 858

Architectural Woodwork WOODWORK DETAILS WO 00 CONCRETE
m&/((//A DETAIL-FIREPLACE MANTEL DETAIL AT COLUMN
CAPITAL OAK VMECR 2>/z" DETAILS AT COLUMN

Architectural Woodwork WOODWORK DETAILS Detail of Wood-Paneled
Wall and Bookcase rimm flwa caiiriG :b* ■51CTION- . *fl* 'ONE H ft! surf
-A— ALF INCH ELEVATION- mum :K suit fit* KTM 1 ■SECTION "E>'
•% FULL SIZE' MATUUAL-KNQrrr PINl, STAI tJCD AUD WAVED.
PLAN :A i full size 860

Architectural Woodwork WOODWORK DETAILS Cornice and Baluster
Details ■g\5CAbE-B&BVATION- OUTER'STRING- -0TA1R.- MOULD
HAND SAIL M^ / CLOCK ^■H\\ dQfLNICE PAKBb MOUbd <"*4*e#Z
"BALUftTBE. scALi)y-io' 861

Architectural Woodwork WOODWORK DETAILS Display Cases I'-io^a"
■SATIN STAINLESS STEEL FRAME m =s ~cs=S to V4"ADJUSTABLE
SHELF \'-r/zm INTERIOR Or CABINET PAINTED TO MATCH
EXTERIOR SEMI-OUISSB.ACK *:,, LL! , Xr--- ' BALLAST 1" UAI
SHOWCASE SECTION VERTICAL SECTION-DISPLAY CASE
U0UL0I4I VERTICAL SECTION JEWELRY CASE PLAN SECTION
JEWELRY CASE 862

Architectural Woodwork TYPICAL VERTICAL SECTION AT WALL
UNIT WOODWORK DETAILS Display Cases * 1 VERTICAL SECTION
AT DISPLAY CASE ILOCKINO- trOTHEM yj^^r^ Ui (VK I'-tW iWMfr
*n VERTICAL SECTION SHIRT DISPLAY 863

Architectural Woodwork WOODWORK DETAILS Bars SECTION AT
BAR — VENTILATION *UJTS I'****■ 4'OC Z"D\A BRUSHES) BfWSS
SLEEVE LENGTH VARIES CHCKRY TRIM -REMOVAE.LE CHERRY
FASOA STAINED TO MATCH SAMPLE VERT SECTION AT BAR a
OVERHEAD CABINETS SECTION AT BACK BAR 864

Architectural Woodwork WOODWORK DETAILS FINISHING
MATERIALS CKNKIUC TYPE System #1 Lacquers Catalyzed Lacquers
System HI Varnishes Conversion Varnishes System #3 Polyurethane
Catalyzed Polyurethane System #4 I: poxy System #5 Penetrating Oils

System #6 Synthetic Enamels System #7 Vinyl Lacquer Catalyzed Vinyl
 System #8 Fire Retardant Coatings (luminescent) RECOMMENDED
 USAGE For all wood surfaces except medium to heavy acid areas; Interior
 use. For wood surfaces requiring medium acid resistance; Interior use. For
 all wood surfaces; Interior use; Exterior use — spar varnishes. For all wood
 surfaces; Some acid resistance; Interior use. For all wood surfaces; Interior
 use. For all wood surfaces; High acid resistance; Interior use. For all wood
 surfaces; High acid resistance; Interior use. For all wood surfaces; Performs
 "well on Oak, Teak, Walnut, etc Most wood and wood product surfaces;
 Interior use; Most colors available. For all wood products; Interior use;
 Light acid resistance. For all wood products; Interior use; Excellent for
 residential kitchens, etc.; Better acid resistance. For surfaces of wood
 products requiring flame spread protection. (See WIC Technical Bulletin
 No. 423 — Section 19.) Interior use only. UL Rated-UL-723; NFPA-255;
 and ASTM E-84; Tested for flame spread, fuel contributed, and smoke
 developed. CHARACTERISTICS Good coverage; Easy to apply; Sands
 easy; Poor water resistance. Tough wearing surface; Good weather resistance;
 Can be repaired. Good build; Tends to yellow with age; Slow drying. Good
 build and solids; Can be repaired. Tough surface; Excellent wear and
 abrasion resistance; Can be repaired. Tough surface; Excellent wear and
 abrasion resistance; Can be repaired. Very hard surface; Excellent wear and
 abrasion resistance; limited pot life; High water resistance. Easy to apply;
 Makes touch-up easy; Average wear and abrasion qualities; Easy to repair.
 Good coverage; Tough wearing; Can be re-coated or repaired; Easy to
 apply. Tough surface; Good wearing; Resists light chemicals. Tough
 surface; Good wearing; Repairs not easy. Leaching will result if exposed
 directly to high humidity or direct water. Can be coated with compatible
 overcoat system or waterproofing materials. Available for transparent and
 opaque finishes.

Architectural Woodwork CORNICES AND MOULDINGS Deep Sculpt
 and Crown Mouldings Deep Sculpt Mouldings Crown Mouldings
 & Melifete O

Architectural Woodwork CORNICES AND MOULDINGS Crown
 Mouldings i^h-^hULJiiryH 867

Architectural Woodwork CORNICES AND MOULDINGS Crown
Mouldings " ^" n" ^=g ^= ^-r- ^ss- ...,=

Architectural Woodwork CORNICES AND MOULDINGS Door Mm
Mouldings 869

Architectural Woodwork CORNICES AND MOULDINGS Miscellaneous
Mouldings Chair Rail Mouldings Picture/Mirror Hanging Mouldings Panel
and Trim Mouldings

Architectural Woodwork CORNICES AND MOULDINGS Crown
Mouldings Solid Crowns 871

Architectural Woodwork CORNICES AND MOULDINGS Crown
Mouldings 872

Architectural Woodwork CORNICES AND MOULDINGS Miscellaneous
Mouldings AWI 5522 Picture Mouldings AWI 11421 '."» I'A" AWI 11521
IV'x IW" 11/16x13/4 11/16x13/4 Picture Moulds 11/16x13/8 11/16x21/16
11/16x13/8 Picture Frame 7/8x17/8 11/16x15/8 Back Bands 11/8x21/4
1x21/2 Step Moulds 873

Architectural Woodwork CORNICES AND MOULDINGS Miscellaneous
Mouldings 11/8x51/2 & 11/8x71/2 Pilaster 3/16x13/4 Tongue and Groove
Siding 1 j/ lb xz 15/8x1 1/2 1 3/32 x 2 Brick Moulds 21/4x11/2 874
15/8x11/16 Drip Caps 11/16x15/8 11 lb 1 1 Fluted Pilaster 11/8x11/4
17/8x2

Architectural Woodwork 3/8x 1 3/4 Mullion Centers 3/8x15/8 Astragals m r
^5/8x3/4 3/8x1/2 Nosings 11/16x15/8 Panel Mouldings Skew Back
7/16x3/4 9/16x1/2 3/8x3/8 Sash Beads 15/16x2 3/8 "T" Astragal
CORNICES AND MOULDINGS Miscellaneous Mouldings tWtv El lxl
3/4x3/4 1/4x3/4 1/4x5/8 1/4x3/4 1/4x5/8 Shelf Edge I 1/4x3/4 Screen 5/8x1
1/8 Sanitary Cap 1 1/2x11/8 Back Band Parting Strip I Cornerguards 875

Architectural Woodwork CORNICES AND MOULDINGS Base Mouldings
11/16x41/4 11/16x31/2 11/16x21/2 ,,,,,,, ,,,, , , , , ^^^T ^^m HIM— Wttm
11/16x31/2 9/16x31/2 9/16x31/4 7/16x3 1/2x3 11/16x21/2 1/2x2 1/4

9/16x31/2 11/16x31/2 9/16x31/4 1/2x3 7/16x3 9/16x21/4 7/16x21/4
 9/16x51/4 9/16x41/4 11/8x11/8 7/8x7/8 Quarter Round Moulds E^ it at ■1
 Ha HI 3/4x3/4 11/16x11/16 5/8x5/8 HfHI Hill af™ KB1 —a] Ha Bk
 1/2x3/4 7/16x3/4 7/16x11/16 1/2x3/4 1/2x1/2 3/8x3/8 1/4x1/4 Base Shoe &
 Floor Moulds 876

Architectural Woodwork CORNICES AND MOULDINGS Base Mouldings
 AW) 6422 AWI 6S2I W'x \W >/, "x \Vt» AWI 1341 Base Cap Moulds 877

Architectural Woodwork CORNICES AND MOULDINGS Casings
 11/16x2 1/4 11/16x31/2 5/8x21/4 11/16x21/4 11/16x3 1/2 11/16x21/2
 11/16x31/2 11/16x25/8 878

Architectural Woodwork CORNICES AND MOULDINGS Casings 879

Architectural Woodwork CORNICES AND MOULDINGS Headed Casings
 Specially casings

Architectural Woodwork CORNICES AND MOULDINGS Cove
 Mouldings r r r r r r Bed Moulds 881

Architectural Woodwork CORNICES AND MOULDINGS Parel Mouldlrgs
 AWI 12422 %"x IK." AWI 12423 V."x IV." AWI 12424 7,."x ls/,» 882

ArchHsctural Woodwork CORNICES AND MOULDINGS Chair Rail
 Moulding* AWI 10463 %" x 5/,." AWI 10431 Wx 2Vi"

Architectural Woodwork CORNICES AND MOULDINGS Handrails,
 Balusters, and Rounds Handrails 19/16x19/16 Baluster 11/4x11/4 1/2x1/2
 3/4x3/4 11/8x11/8 19/16 Round Moulds Half Rounds 884

Architectural Woodwork CORNICES AND MOULDINGS Clam Shell and
 Slop Mouldings CLAM SHELL MOULDINGS 11/16x21/4 STOP
 MOULDINGS 3/8x11/8 3/8x11/8 1/2x13/8 Sanitary Sash 9/16x21/8
 9/16x21/4 9/16x21/4 9/16x21/4 Lattice 00 MBS ("I ;H ?l 1/2x15/8 1/2x15/8
 1/2x11/8 3/8x3/4 3/8X7/8 3/8x11/16 3/8x15/16 3/8x15/8 3/8x21/4 Clam
 Shell a 3/4x15/8 Sash 3/4x13/8 7/16x15/8 7/16x13/8 7/16x11/8 3/8X3/4
 3/8X7/8 3/8X11/8 3/8X13/8 3/8X15/8 3/8X21/4 Sash Colonial 885

Architectural Woodwork CORNICES AND MOULDINGS Hand-Carved
 Brackets Fluted & 1/2" 10" W x 11" H Left & Right Scroll 8" V W x 11" V a" H
 Bracket 6" 1/4" W x 12" 1/4" H x 5" D. Bracket 8" W x 12" H x 4" D. Bracket
 4" W x 8" 1/4" H x 4" W D. Bracket 4" 1/4" W x 13" H x 9" V i" D. Bracketed W i" W x
 12" H x 7" D Bracket 7" V x" W x 14" H x 7" f e" D

tWJ*>'li>.:";'!ffl-;SVi...l,'yl>.a^ B !0 o o ® i ll.iill .111.1 t 0 # @ i l : \ i i z z x
 3 3 TABLE 1 Butt Hinges No. Type Comments Standard brass butt: A, solid
 drawn type; B, pressed pattern Back flap hinge Strap hinge Lift-off butt 5
 Loose pin hinge; ball-tipped hinge Stopped hinge Piano hinge 8 Clock case
 hinge General usage With wide plates for table leaves and rebated or
 rabbeted fallflaps For narrow sections For doors which have to be removed
 from time to time without disturbing setting Where It is necessary to throw
 door clear of carcass frame with the whole or the hinge knuckle protruding
 Opens through 90° only for box lids, etc. Continuous strip form for
 supporting long lengths; supplied in drilled and countersink or undrilled
 blanks One plate is wider to allow for a projecting door

Architectural Woodwork FURNITURE HARDWARE Hinges

Architectural Woodwork FURNITURE HARDWARE Hinges ^-^^lls-^^
 889

Architectural Woodwork FURNITURE HARDWARE Hinges Left- and
 Right-Hand Hinges Hinges with screw-mounted flanges should be viewed
 as if in mounted condition with the countersunk screw holes facing you, If
 the female flange is uppermost on the left, the hinge is a left-hand hinge and
 vice versa. •mil* fling* "M*1*fl*ng* ■ LalMwnd hlng* Cranked Hinges
 and Their Uses The position of the door relative to the side panel can vary
 considerably, being decided at the design stage in accordance with the final
 effect required. A wide variety of hinge types has developed from va
 riations in door mounting methods, which must be coordinated at the design
 stage. Straight hinge For butting, (lath or front-hung doors. The barrel is
 positioned centrally between the two flanges, 4= Crank B Mounting with
 set-back doors, One flange is cranked by an amount equivalent to the
 thickness of the material, ut= Crank C Similar to crank B but for forward-
 set doors. 4t Crank D Mounting with rebated doors with flanges of non-
 uniform width (reduced female flanges). y= Crank L.1 Mounting with

butting front-hung doors Door opening range 270°. Crank features internal roll. Butt Hinge Designations Butt hinges for cupboards, windows, and doors, and hinges with mortise-type flanges should be viewed with the barrel facing you. If the female flange is positioned on the left of the barrel, it is a left-hand hinge and vice versa. Left-hand flange (with offset flange) Right-hand flange (with offset flange) The same rule applies to hinges with symmetrical flanges. Butt hinge The hinge is made with offset flanges for mortised mounting. The door-mounted flange is secured with screws or pins from the rear or front (giving exposed heads). Double butt hinge Suitable for mounting two doors to a single centre panel. The door opening range is 180° each (centre flange only mortised). -I- 890

Architectural Woodwork FURNITURE HARDWARE Folding Table and Flap Hinges FOLDING TABLE HINGE Functional diagram Self-supporting hinge, for folding and sewing machine tables, flush-mounted r*B 21 -0.B /"O JL «P\ m- [J #]: t B5 L_J Functional diagram (seen from below) Folding table hinge, flush-mounted nsi i. ig Functional diagram Card table hinge Two-way table-leaf hinge, flush-mounted FLAP HINGES -66- 3jp 111 I'I l|E= I' I I 3 ♦. t. *i v*T^ -66- 2 (i—' /, i , t-* -l i A , ~—* 55 Flap closed Flap Lo war ■IwK 6 Flap Flap open Specimen mounting Flap ^ ShaH =*= OapA Flap closed Dimension sin mrr 1. Ovariay *d|iatal)la ■ from 6 to 8 nun ~f j»- Flap ^ n«p J 1 """"T- LL . t , Qapfl Flap open -4 ' Upper tlwlf upearana.. Flap Flap closed Bap open 891

Architectural Woodwork FURNITURE HARDWARE Mitred and Concealed Hinges MITRED HINGE The hinges are suitable for wooden doors and side panels from 16 mm to 22 mm In thickness, chamfered at an angle of 45°. Door closed For the first time a concealed hinge is available for modern furniture Incorporating 45° mitre angles. Door open An all-metal mitred hinge, specially designed to enable door and carcass edges to meet at an angle 0(45°. ® ® ® M >EJJ ® Since both bosses have the same adjustment facilities, the complete hinge can be adjusted In three directions, vertically, laterally and (rom (rant to back. CONCEALED HINGES Specimen installation of a butting, flush-fitting cupboard door. Doors may, however, be set back or forward if preferred, provided the housing recesses are appropriately offset. If doors are set back, care must be taken to ensure

that the opening angle is restricted as little as possible. .-MopMOl Door Door -4— IT Mm* Door closed Door open Specimen installation of a butting, front-hung door, fitting flush with the cupboard side In the conventional manner. Doors may, however, be hung with inset edge if preferred, provided the housing recesses are offset accordingly. It is important in such cases to ensure that centre doors are not mounted with groove gap clearance. .. auopuwi Door 3 J3t Hkigo Door closed Door open Specimen installation of a butting, front-hung, flap-type door. On opening, the flap projects downwards by its own thickness. Thus, if doors or other panels are situated below the flap, a degree of clearance exceeding the flap thickness will be necessary. pn . ^Flop-door Hlngo Flap closed LoworihoM i—p — Hlngo Flap open Specimen installation joining two panels. In this way folding doors can be constructed for furniture or room dividers; ==z:ci- - _r _z- _z:^ qplqp- PlIWl -4- Hingt —*F He' W Typical folding door Door closed Door open ml o A o 80 f-W-H /25\ 31 ®. iWooOMcfaiMoiipto v®/ Dimensions in mm. 892

Architectural Woodwork Ptvot hinge, without slop "----» drawn, with one short flange FURNITURE HARDWARE Pivot Hinges Corner pvtot hinge, without slop WHh outer knuckle, straight Ptvot hinge, without slop Smooth drawn, with one short flange Comer pivot hinge, without stop With outer knuckle, cranked Pivot hinges for writing bureau drop leaf mounting, smooth drawn. Dimensions in mm. 893

Architectural Woodwork FURNITURE HARDWARE Glass Door Hinges For flush-fitted doors ISr-a-H One upper and one lower hinge per mounting For flush-filled doors ^ 1 _ -33— 19 t 16 LM 1 1,5 I tktapaMl _ — aim doer iki< p«wi One left and one right ' hinge per mounting For flush-fitted doors r mln l%|2 Iktopiiwl 1,5 |0 tktopml 0 aim dow One left and one right hinge per mounting For flush-fittlrig doors Mitoiwwl Dim door > Without 2 ttoMtwuitt 30 QlMI door One pull per mounting UnwMwH For front-hung doors 16 10 19 One left and one right hinge per mounting Magnetic Pressure Catches Chom DotlMa Dimensions in mm. 894

Architectural Woodwork FURNITURE HARDWARE Flap Stayi with Brafc Flap ClOMd Flap olosad 1 6 \ _J+] -u Vertical mounting Length (mm) 250 325 450 Internal carcase height (mm) 300 to 400 350 to 450

40010 500 Distance A (mm) 240 308 430 Distance C (mm) Determine by
 Wei mount- *g Horizontal mounting Length (mm) 250 325 450 Internal
 carcass height (mm) 300 to 400 350 to 450 400 to 500 Distance A (mm) 65
 100 150 Distance C (mm) Determine by trial mounting Vertical mounting
 Length (mm) 160 190 220 250 260 Internal carcass height (mm) 25010 350
 300 to 400 350 to 450 400 to 500 450 to 550 Distance A (mm) 127 151 175
 196 222 Distance C (mm) Determine by trial mounting Horizontal
 mounting Length (mm) 160 190 220 250 260 Internal carcass height (mm)
 200 to 300 230 to 330 260 to 360 29010 430 320 to 460 Distance A (mm)
 82 105 127 148 171 Distance C (mm) Determine by trial mounting Length
 (mm) 150 200 250 /Internal carcass height (mm> mln.130 mln.170 min.210
 Distance A (mm) 70 105 140 Distance C (mm) Determine by trial mounting
 Dimensions in mm. 895

Architectural Woodwork FURNITURE HARDWARE Lid Stays Length
 (mm) 250 325 450 Lid height (mm) up to 300 up to 450 over 450 Distance
 A (mm) 170 205 275 Distance C (mm) Determine by trial mounting Lanoth
 (mm] 200 .. 250 300 CtrciM daolh (mm) ato 260 310 , Dlilanc* A (mm 150
 180 230 DUlanoa C (mm) Determine by trial mounting PlapopM UppwiMI
 -X, -4- FUpdOMd "paiMI 896 UwvilhK * Length (mm) 130 250 DIIs.
 tanceA (mm) 150 ZW Distance C (mm) Determine by trial mounting
 Dimensions in mm.

Architectural Woodwork FURNITURE HARDWARE Drawer Runners
 Extension type Soft-Roller systems are capable of varying degrees of
 extension, depending on design. Basically, three types are employed: E —
 Single extension The withdrawal distance offered by single extensions is
 designed to be less than the installation length. Drawers cannot be opened
 clear of the carcass. 3E D V— Full extension The full extension model
 incorporates a pull-out distance as great as, or greater than, the installation
 length. Drawers can be opened completely clear of the carcass. -G= ^ T —
 Telescopic extension 3- :sa Telescopic extensions are fully extending
 systems. Their particular design is such that all the elements travel on a
 central axis resulting in a particularly neat, space-saving, compact assembly.
 Mounting method An indication as to how the rails are secured to the
 drawer or pull-out element. 1 Base mounted Single extensions with friction

bearing mounted nylon rollers Telescopic extensions guided by means of ball cages 897

Architectural Woodwork FURNITURE HARDWARE Magnetic and Spring Catches; Bolts Magnetic catches, screw-mounted Magnetic catches, for metal doors Twin ball catches Rexa-Touch drawer latch Magnetic catches, mortised II 1 / clip-lit Magnetic catches, for installation in series-drilled holes Ball catches, with stop plate or ball headed screw Magnetic push-latches, surface- mounted or mortised Magnetic catches, (heat resistant) Elbow catches, screw-mounted Twin roller catches, screw- mounted Plinth spring catches, screw- mounted and press-fit Magnetic catches, for double doors Roller catches, P" 0^ screw-mounted f^ . ^S^? Spring catches, screw-mounted iy >J 01 CST\^ K) 3 > f / y y Pulls and counterplates for magnetic push latches • i k/W/ lluf *v Furniture bolts, screw-mounted and press-fit Flush bolts, barrel bolts, tower bolt 898

Architectural Woodwork FURNITURE HARDWARE Locks Automatic door bolt Rim locks 1 Mortise locks Sliding door locks pushbutton cylinders Central locking systems with anti-tilt mechanism Cylinder moduli system Lever-type rim locks Central locking, cylinders Inlaid-and Inlaid flap locks Espagnolette locks Glass door locks HCgj ip3*^ Lever locks Locker locks 899

Architectural Woodwork FURNITURE HARDWARE Shelf Supports Shelf supports, pin mounted .., Plug-in, 03mmhole ., Piug-in, 05mmhole ./ y Shelf supports, with added screw fastening Shelf supports, with splng dip .., Plug-In, 0'Ain, hole .. Plug-in.fSemm hole Shelf retainers, In 0 5 mm hole Shell retainers, in 0 5 mm hole Shell supports, screw-in type Shell supports, surface- mounted Mirror mount, lor mirror with backing panel .Plug-in, 04 mm hole Shell supports, plastic Shelf supports, with sleeves (steel) Glass shelf supports, in 5 0 mm holes Shelf support systems, screw-mounted Shelf supports, with sleeves (plastic) Glass shelf supports, In 0 5 mm holes Shelf supports systems, groove-mounted < „■' >' S St "s* ' ^ ./ S" M^ <\v^ vZ \ V (k&? \>¥l M w^" I I 900

Architectural Woodwork FURNITURE HARDWARE WARDROBE RAILS AND SUPPORTS OVA wardrobe rails and supports 018 mm, 0 20 mm, 0 25 mm wardrobe rails and supports Cloakroom fittings Thiefproof

cloakroom fillings Nylon tubular rails and coat hangers Wardrobe rails Tie
 rails and hal racks Shoe racks Universal storage rack Pull-relief plug
 Wardrobe lifts, hanging rails Hal and coal hooks Single and double hooks,
 pew hooks Cloakroom, wall and ceiling hooks, nylon Cloakroom hooks,
 wood <5dJ Wardrobe hanging bars and rails CASTORS AND GLIDES
 Twin-Wheel Castors Audio and Audio-Slop Enclosed Castor* Furniture
 Glides i & Furniture Glides Cylindrical Furniture Legs Furniture Glides
 Bed Box Castor* Ball Castor* Coal Box Castor* Light Duty Swivel Castors
 ^> 901

Architectural Woodwork FURNITURE HARDWARE Furniture Glides
 knock-In fumlturt glld«i Furniture glide With two pins Finish: white plastic
 *!=U [Size 45 X 20 X 5 mm © \$ ^_1 «—45- Dlmantloria In mm Chiirteg
 glide (pin type) Finish: nickel-plated steel i 1 1 i 1 1 1 | Size | 13 mm | 15
 mm | 18 mm | 20 mm | 23 mm | 25 mm | 30~ ChilriM cup* with three li
 Finish: nickel-plated steel ty kM tfiJ ^J [Size | 13mm | 15mm | 18mm |
 20mm | Glide (pin type) With rubber washer Finish: nickel-plated steel;
 black rubber a. 4. J- i. J- I 18 mm | 20 mm [23 mm | 25 mm [30 r Size
 Feltgiid* Finish: felt, In plastic case, grey ± sJs 24 mm | 30 mm Size 20 mm
 Plastic glid* Finish: while plastic | Size | 10mm | 13mm | 16mm | 19mm |
 22mm) Mettic giida (pin type) With rubber washer Finish: white plastic;
 black rubber 1 ± J. JL size 15 mm 20 mm | 25 mm | 30mm~ 902

Specialties Landscaping Signage and graphics Audio-visual systems
 Auditorium seating Security Color theory Window treatments Elevators
 Indoor recreation Accessories 906 931 949 961 977 998 1015 1060 1069
 1075

In most instances, the design process requires a knowledge of, or at the very
 least, an awareness of, certain specialized elements that can contribute
 heavily to the success or failure of a project in terms of aesthetics or
 function, or both. These elements may take the form of manufactured "off-
 the-shelf" products or consist of design theories, standards, and guidelines
 for certain areas of expertise. Accordingly, this section deals with ten such
 elements, ranging from landscaping to accessories. Information can be
 found concerning the height, spacing, and diameter of indoor trees and floor
 plants. Also included are planting standards, details, and maintenance

information. The section dealing with signage and graphics provides information on signage systems, symbols, mounting heights, and locations. Other sections provide data on audio-visual systems, including projection room layouts and details, and auditorium seating arrangements and sightlines. The section dealing with security includes information on door and window hardware, mailbox rooms, lighting, and security systems. Still other sections provide information on color theory and window treatments, including draperies and curtains, shutters and shades, and rods, holdbacks, and ties.

Specialties PLANTSCAPING Design Guidelines DESIGNING WITH PLANTS Any successful design uses plants that are compatible not only in an aesthetic design sense, but also in their growing requirements. No matter how beautiful the design, if neighboring plants are not matched to the correct growing conditions, parts of the design will either deteriorate or require elaborate maintenance. The aesthetic design considerations involve choosing the proper variety of plant textures, heights, and spacing to give the desired effect. The growing considerations involve the proper matching of light intensity soil, and water, as well as proper container size, to the plant environmental requirements. Of all the growing conditions, the most important is the light intensity. It is easy to underestimate the amount of available light, since the human eye can easily see in 20 footcandles of light, while even the plant needing the lowest light requires 50 to 75 footcandles to remain healthy. If the light intensity is to be below 100 footcandles, even these "Sow-light" plants must be slowly acclimatized prior to installation. No matter if the space to be planted is a small office, a large interior garden, or a cafeteria, the first step is to ascertain the actual level of the existing or planned lighting. To allow maximum creativity in the planting design, the light intensity should be considered in the initial planning stages, especially in large areas such as those in shopping malls or corporate interior gardens. Adding the needed lighting fixtures after the initial electrical installation is often expensive or impossible. In smaller-scale situations, such as offices or homes, extra light fixtures should be added or the plants should be chosen according to the available light. If the plants do not have the proper light intensity they will die. The lower the light intensity below the minimum needed by the species, the faster they will do so. Since the light source (incandescent, fluorescent, sun, or other)

is not important, but the light intensity is, accurate intensity measurements are essential. For these measurements we recommend the General Electric Model 213 or 214 light meter or its equivalent. The measurements must be made at the level of the plant foliage; they must be made several times a day on several days typical of the location if sunlight is used; and they must take curtains, tinted glass, and other light-shielding devices into account. Only light hitting the top of the leaves is effective. While underlighting with spotlights can create dramatic effects, it does very little to help the plant. After the light intensity is determined, the plants should be selected from the appropriate light-level group (see Fig. 7), consistent with the design aims. Plants that will be growing near one another should also have similar water requirements (also given in Fig. 7). If plants with different watering requirements must be close, they should be kept in their own growing containers so they can be watered separately. An interior planting designer creates the mood through the interplay of plant texture and plant height, working only with those plants that will live under the predetermined light intensity. Color cannot really be used as a design element, since the average interior light intensity is seldom more than 100 footcandles and brightly colored plants or blooming flowers need up to 1000 footcandles. If flowering plants are used where the lighting conditions are normal, they will generally have to be replaced every few days.

Plant Texture The good designer will provide for design variety through the clever use of plant texture. The term is used here to describe the general structure, shape, and appearance of the plant, regardless of height. It includes the size, shape, edging, and thickness of the plant's leaves, as well as its overall shape and the arrangement and number of leaves on the plant. Five general rules concerning texture should be kept in mind.

1. Juxtapose fragmented foliage (such as that of a palm) with solid foliage (say, that of a dracaena).
2. Avoid too much of the same type of foliage (e.g., large flat leaves) in one area, unless a border or hedge effect is desired.
3. An exception to these previous rules on groupings is the palm. Although all palms have similar foliage, they vary slightly in color and interest, so that different types of palms may be planted together.
4. To create interest, mix small-leaved with large-leaved plants, and narrow-leaved with broad-leaved plants.
5. When using plants as specimens, especially as interior design elements in offices or homes, pick the plant with the background fabric, carpet, or wallpaper in mind. For example, a "busy" foliage plant will fight with a "busy" fabric.

Plant Height Plant height not only determines the scale of the design, but it adds variety to the plant groupings. There are six general rules regarding plant height selection to keep in mind. 1. In the plant grouping, build up with the low plants in front. If the grouping can be seen from all sides, the grouping must be well balanced throughout and built up to the center height. 2. If a plant has canes with no lower foliage, try to place the lower plants in front to conceal the absence of foliage of the taller plants in the rear. 3. Uneven sizes throughout a grouping add more interest than consistent levels of foliage. 4. If a single plant is desired to hide a column or some other object, be sure that the plant height, including its container, is about three-fourths the height of the object to be concealed. 5. Keep the scale of the surroundings in mind when choosing the plant height. A 3-foot plant is fine next to a desk, but a plant of at least 6 feet should be selected if it is to be viewed when entering a room. 6. By convention, interior plant heights are measured from the bottom of the root ball or planter, while exterior plant heights are measured from the top of the root ball. The reason is that interior plants are usually placed in a container or raised planter, and the total available height from floor to ceiling is fixed.

Plant Spacing Under certain conditions, the plants of an interior landscaping design will grow. Therefore, any possible change in the plant size must be considered by the designer. If the lighting intensity is at or below the recommended level, there will be little or no plant growth and the plant size and relationships will change little over time. If the lighting intensity is well above the required level, there will be plant growth, with different plant species growing at different rates. Unlike outdoor plants, indoor tropical plants seldom grow outward; most of their growth occurs upward. The main exceptions are the Ficus family, the schefflera, and the Philodendron Selloum, which will spread somewhat outward. If a full plant design is desired, the required number of plants should be placed close together at the time of installation since future growth will seldom fill in the bare spots. Even if the light intensity is high enough, before the plant can grow significantly its root system must be able to expand. Thus, the best way to ensure that the size relationships of the plants do not change is to keep them in their original growing containers and not to replant them into a growing medium. If they are kept in the original containers, they will become pot-bound and future growth will be automatically limited. Plant material is sold on the basis of height or growing-container size, and one must be

familiar with the particular species to know what the spread will be. For each plant species considered here, Fig. 1 lists the height range for each plant in each standard growing-container size and gives a recommended minimum center-to-center plant spacing. This recommended spacing is based on experience with the plant's branching habits and growth patterns and will give a full plant design. If an open or a less full design is called for, the spacing should be increased. When the plants are to be displayed in individual planters or decorative containers, each plant, still in its growing can, is placed directly into the planter or container, on top of a layer of drainage material of the appropriate depth. However, many standard planters have lips that reduce the interior diameter to less than the overall diameter. This inner diameter should be larger than the growing cans diameter so that the plant can be placed directly into it without being repotted and risking the attendant danger of root damage. To emphasize this requirement, Fig. 2 gives the standard planter diameter needed for each standard size of growing container. The size of the lip changes when a nonstandard type is used. If space is limited, this measurement should be carefully checked. 906

Specialties PLANTSCAPING Plant Height, Spacing, and Diameter
 INDOOR TREES Species Fiddle-leal fig (*Ficus lyrata*) Indian laurel (*Ficus retusaj* Rubber plant {*Ficus elastlca* cv. 'Decora'), tree standard Rubber plant [*Ficus elastlca* cv. 'Decora'), bush type Weeping fig (*Ficus benjamlna*) Norfolk Island pine (*Araucaria heterophylla*) Schefflera (*Brassala actlnophylla*) FLOOR PLANTS Bamboo palm (*Chamaedorea erumpens*) Corn plant [*Dracaena (ragrans* cv. 'Massangeana') Corn plant bush (*Dracaena fragrans* cv. 'Massangeana') Dwarf date palm (*Phoenix Roebelenll*) Height Range 3- 4 ft 4- 6 ft 6-11 ft 5- 7 ft 7- 9 ft 9-12 ft 4-5 ft 5-6 ft 6-7 ft 1)4-2 ft 2 -2)4 ft 3 -4 ft 4 -5 ft 4 -5 ft 3- 4 ft 4- 5 ft 5- 7 ft 6- 8 ft 9-10 ft 9-12 ft 10-12 ft 1)4-2 ft 2 -3 ft 3 -5 ft 4 -5 ft 4 -6 ft 6 -7 ft 3- 4 ft 4- 5 ft 5- 7 ft 7- 8 ft 8- 9 ft 9-12 ft 3-4 ft 4-6 ft 5-7 ft 7-9 ft 1)4-2 ft 3)4-4 ft 4)4-6 ft 5 -7 ft 1 -1)4 ft 1)4-2)4 ft 3 -4 ft 4 -5 ft 5 -7 ft 2-3 ft 3-4 ft 4-5 ft 5-6 ft 5-6 ft Recommended Center- to-Center Plant Spacing 24-36 In 30-42 In 42 In & up 42-54 In 48-60 In 60 In & up 48-60 In 54-66 In 60-72 In 12-18 In 12-24 In 24-36 In 36-48 tn 48-60 tn 24-36 in 30-42 In 36-48 in 48-60 tn 60 In & up 60 In & up 72 In & up 18-30 in 24-36 In 30-42 In 36-48 in 42-54 in 54-66 in 36-48 in 36-48 In 55-66 In 60-72 In 60 in & up 72 In & up 30-42

in 36-48 In 42-54 In 48-60 In 24-30 In 24-36 In 30-42 In 36-48 In 18-24 In 18-30 In 24-36 In 30-42 In 36-48 in 30-42 in 36-48 in 42-54 in 48-60 in 54-66 In Growing Can Diameter 10 In 14 In 17 In 14 in 17 In 22 In 10 In 12 in 14 In 6 In 8 In 10 in 12 In 14 In 10 In 12 in 14 in 17 In 22 In 28 In 36 In 6 In 8 In 10 In 12 In 14 In 17 In 10 In 10 In 14 in 17 In 22 In Metal tubs 10 In 12 In 14 In 17 in 6 In 10 In 12 In 14 In 6 In 8 In 10 In 14 In 17 In 10 In 12 In 14. in 17 In 22 In Fig. 1 Spacing recommendations.

Specialties PLANTSCAPING Plant Height, Spacing, and Diameter Speclee Dwarf dragon tree (*Dracaena marginata*) Dwarf Jchefflera (*Brassala arboricola*) False aralla (*Dlzygothecn elegantlsslma*) Green dracaena {*Dracaena deremensls* cv. 'Janet Craig'} Green pleomele (*Dracaena reflexa*) Kentla palm (*Howea Forsierana*) Lady palm (*Rhapls excelsa*) Mock orange (*Plttosporum Toblra*) Narrow-leaved pleomele (*Dracaena angvslffolla honorall*) Neantha hella palm (*Chamaedorea elegans*) Ponytall (*Beaucarnea recurvata*) Reed palm (*Chamaedorea SelMill*) Self-heading phllodendron (*Phllodendron Selloum*) Southern yew (*Podocarpus macrophyllus* var. Makl) Height Kange 3-4 ft 4-5 ft 5-7 ft 7-9 ft 1 -IK ft 1K-2K ft 3 -4 ft 4 -5 ft 1J4-2 ft 3 -4 ft 5 -7 ft 7 -8 ft 1 -IK ft IS-2)4 ft 3 -4 ft 4 -5 ft 5 -7 ft 1)4-2 ft 3 -4 ft 4 -5 ft 5 -6 ft 3-4 fl 4-5 ft 5-8 ft 3-4 ft 4-5 ft 5-7 ft IK-IK ft 1K-2Kft 2 -3 ft 3-4 ft 5-6 ft 6-7 ft 1 -IK ft 1K-2Kft 2K-3K ft 4 -5 ft IK-2 ft 2 -3 ft 3 -4 ft 4 -5 ft 4-6 ft 6-7 ft 7-9 ft 3 ft 4 ft 5 ft 4-5" 5-6 ft 5-6 ft 6-7 ft

Recommended Center' to-Center Plant Spacing 24-36 In 30-42 In 36-48 In 48 In & up 18-24 In 30-42 In 36-48 In 42-54 In 18-30 In 30-42 In 36-48 In 42-54 In 18-24 In 18-30 In 24-36 In 30-42 In 36-48 In 12-18 In 18-30 In 30-45 In 36-48 In 36-48 In 42-54 In 48-60 In 36-48 In 42-54 In 48-60 In 24-36 in 30-42 In 36-48 In 24-36 In 30-42 In 36-48 In 18-30 In 24-36 In 30-42 In 36-48 In 24-36 In 30-42 In 36-48 In 42-54 In 36-48 In 42-54 In 48-60 In 30-42 In 42-54 In 54-66 In 36-48 In 42-54 In 48-60 In 54-66 In Growing Can Diameter 10 In 12 In 14 In 17 In 6 In 8 In 10 In 14 In 6 In 10 In 14 In 17 In 6 In 8 In 10 In 14 In 17 In 6 In 10 In 14 In 17 In 10 In 12 In 14 In 10 In 12 In 14 In 10 In 12 In 14 In 12 In 14 In 17 In 6 In 8 In 10 In 14 In 10In 12 In 14 In 17 In 12 In 14 In 17 In 10 In 14 In 17 In 10 In 12 In 14 In 17 In Fig. 1 (Continued) 908

Specialties PLANTSCAPING Plant Height, Spacing, and Diameter Sp*c/« Htlght Hmjigt tt*Komm*)nd*d Ctattr- lo-Ctatmr Plant Spicing Growing

Cam Dhmtmtll TABLE OR DESK PLANTS-GROUND COVER Boston fern (*Nephrolepis exaltata* cv. 'Bostoniensis') Common philodendron (*Philodendron scandens oxycardium*) Chinese evergreen (*Aglaonema commutatum* var. *maculatum*) Dumb cans (*Dierfeybachia maculata* cv. 'Rudolph Roehrs') Golden pothos (*Epipremnum aureum* or *Scindapsus aureus*) Grape Ivy (*Cissus rhombifolia*) Jade plant (*Crassula argentea*) Prayer plant (*Maranta leuconeura*) Swedish Ivy (*Plectranthus australis*) Wax plant (*Hoya carnosa*) White flag (*Spathiphyllum* cv. 'Clevelandii') White-striped Dracaena (*Dracaena deremensis* cv. 'Warneckii') Green dracaena (*Dracaena deremensis* cv. 'Janet Craig') Neanthe bella palm (*Chamaedorea elegans*) Self-heading philodendron (*Philodendron Selloum*)

1 ft 1 in - 1X ft 1K-2 ft 1 ft 1K-IX ft 1X-1X ft 1X-1X ft 1X-2 ft 2 - 2X ft 1 ft 2 ft 3 ft 3 - 3X ft 3X-4 ft 1 ft 1 ft 1K-IX ft 1X-1X ft 1 ft 1 - 1X ft 1X-1X ft 1X-1X 1 ft 2 ft 2 - 2X 1 1 2X-3X ft 1 ft 1 ft 1-1X ft 1 ft 1 - 1X ft 1X-1X ft 1X-1X ft 1 ft 1 ft 1 ft 1X-1X ft 2 - 3 ft 2X-3X ft 3 - 4 ft IX-IX ft 2 ft 3 - 4 ft 4 - 5 ft 4 - 7 ft 1 - 1X ft 1X-2X ft 3 - 4 ft 1 - 1X ft 1X-2X ft 2X-3X ft 1 ft 2 ft 3 ft 24-30 In 30-36 In 36-42 In 18-24 In 24-30 In 24-36 In 18-24 In 24-30 In 30-36 In 18-24 In 24-30 In 30-36 In 36-42 In 42-48 In 12-18 In 18-24 In 24-30 In 30-36 In 18-24 In 18-30 In 24-36 In 24-36 In 18-24 In 24-36 In 30-42 In 36-48 In 18-24 In 24-30 In 24-36 In 30-42 In 36-48 In 18-24 In 24-30 In 30-42 In 18-30 In 24-36 In 30-42 In 18-24 In 24-36 In 30-42 In 6 In & 70. 10 In 8 In 10 In 12 in 6 In 8 In 10 In 6 In 8 In 10 In 12 In 14 In 6 In 8 In 10 In 12 In 6 In 8 In 10 In 12 In 8 In 10 In 12 In 14 In 8 In 10 In 12 In 6 In 8 In 10 In 12 In 6 In 8 In 10 In 6 In 8 In 10 In 14 In 6 In 8 In 10 In 12 In 14 In 6 In 8 In 10 In 6 In 8 In 10 In 6 In 8 In 10 In Plant Onmta§

Cwifl/Mr blamnttr 4 In 6 In 8 In 10 In 11 In 12 In 13 In 14 In 17 In 22 In Rt —manM Planter Exurt* DUmtitt 6 In 8 In 10 In 12 In 14 In 14 In 16 In 16 In 18-22 In 24 In Fig. 2 Planter selection. These recommendations are based on the fact that most standard planters have either a 1-Inch lip or no lip at all. Because the growing cans sometimes have ridges or have become deformed, It is always best to allow for a little extra leeway, even for planters with no lip. Some manufacturers, however, put 2-inch lips on their planters, a possibility that should be checked. If the planter is an automatic watering type, the Inside and outside diameters will be quite different, depending on the manufacturer. Fig. 1 (Continued) 909

Specialties PLANTSCAPING Design Guidelines Writing Specifications

The interior landscaping business is very competitive, and a common practice is for the architect or designer to send out the landscaping specifications for bids. Unless the specifications for the job are well-written, however, there are many ways for the contractor to cut corners and still be within the specifications. Consequently, the final installation may not be what the designer had in mind. The lowest bid is not necessarily the best bargain, unless the specifications are very tightly written or unless the architect is dealing with a well-established landscape contractor with a reputation for high-quality work. The following are some suggested guidelines to use in writing specifications. If they are observed, the bids received will accurately reflect the design requirements of the job, 1. Specify the plant heights within a 6-inch bracket. For example, designate 5 to 5½ feet or 5½ to 6 feet. If the specification were simply "5 to 6 feet," the supplier could use all 5-foot plants, which are considerably less expensive than 6-foot plants, 2. For corn plants, dwarf dragon trees, and the like, specify the number of canes and approximate number of foliage heads, as well as the height. The difference in cost between a two-cane and a three-cane corn plant of the same height is not minor. 3. For reed palms, bamboo palms, and the like, specify the number of stems desired, five to six being medium full. 4. For the green dracaena and white-striped dracaena, list the number of main foliage stems desired, They range from one to three stems, 5. For ficus trees, it should be specified whether the bush style or standard tree style is desired. In the bush style, the plant has multiple stems (ranging from two to five in number) branching out from the base of the plants. The standard tree or "lollipop" style has one main 5- to 6-foot stem with a sheared, ball-shaped foliage head. 6. Small plants should be specified as to single plants or combinations or several plants, Examples are dumb cane, Chinese evergreen, and white flag. 7. If ivy trailers are desired, their length should be specified, The trailers take up to eight months to grow, depending on the length, so the designer must plan for these up to a year before installation. Examples are grape ivy, Swedish ivy, golden pothos, common philodendron. and wax plant. 8. Specifications should call for plant cleaning and spraying before installation. 9. Perlite should be specified as the drainage material for both planters and decorative containers. Styrofoam, which is much cheaper, is often used but has little long-term value. 10. The amount of ingredients in large planters (soil mixture, drainage material, soil

separator) should be specified, as should the composition of each of the ingredients. 11. If bark chips, moss cover, or other soil coverings are desired, they should be specified. 12. Special attention should be given to the description of specimen plants, including the number of heads, stems, or canes, and any unusual stem structure that is desired. If canes with character (such as angle and peculiarity of growth), tufts of foliage at various heights, or other unusual features are wanted, they should be specifically mentioned. 13. If the landscape contractor will not maintain the plants after installation, provision should be made for a training program for the maintenance crew. Also, the contractor should provide for two weeks' initial maintenance of the plants and replacement of any that fall below specifications during the period. 14. If the landscape contractor is to maintain the plants after the installation (usually the best all-around solution), such an agreement should be reached before the plants are installed and a maintenance contract should be signed. This contract should include a provision for the replacement of any plant that falls below specifications because of faulty maintenance. This stipulation gives the contractor incentive for professional-quality maintenance. 15. If a large garden is planned and the landscape contractor is given design responsibility for it, the contractor should provide a floor plan of the garden for the designer's approval, before the installation. 16. If the architect or designer provides the landscape contractor with a detailed planting floor plan and the contractor finds it impossible to meet all the specifications (because of unavailability of certain species, etc), the contractor and the designer or architect should agree in writing on any changes.

USE OF INTERIOR PLANTS AND PROCEDURES The general rule of interior planting design is to vary the plant heights, shapes, and textures to give the desired design feeling consistent with the available light level and planting space. The best way to learn to apply this rule to specific situations is to study successful designs. Interior planting designs have usually been found to fall into one of two categories: (1) interior gardens, both large and small, such as those seen in residential and hotel lobbies, corporate headquarters reception areas, and enclosed shopping mall public spaces; and (2) open plan or specimen design, like office landscaping designs and designs that use individual plants as living sculptures. In both categories of design, the main requirements to be considered are the available light intensity, the scale of the design, and the client's wishes and budget. After these basic requirements are determined, however, the design

considerations are somewhat different for the two types of design. Interior Gardens Interior gardens are planting areas, sometimes contained in built-in planters, that have a variety of plants and that convey their design feeling through plant arrangements rather than through individual plant specimens. Small gardens generally contain only a single grouping of plants, act as a single design element, and have uniform lighting and watering requirements throughout. Large gardens have a variety of plant groupings and varying design feelings among the groupings, and they can encompass areas of different lighting and watering requirements. Since any garden conveys its effects through the juxtaposition of different plants, a single dominant plant cannot be considered a garden from the design point of view, even if it is in a built-in planter with ground-cover plants. In designing any built-in planter, enough planter depth must be provided to allow the root ball or the planting can to be covered with soil and to rest on 4 to 8 inches of drainage material. Since soil and gravel are expensive, it is best not to overdesign the planter, by making it larger than necessary, and not to buy too much soil to fill in between the plants. (For example, a depth of 1 1/2 to 2 feet is usually enough for most small gardens.) Figure 1 lists the size of the growing can for different sizes of plants of each species. The depth of the largest growing can, plus the depth of the drainage material, yields the minimum planter depth for the garden. The volume of the planter minus the total volume of all the growing cans indicates the amount of additional soil and drainage material to be provided. If the planter is already in place, its depth may limit the size of the plants that may be used. Since soil must reach to the top of the root ball or can, the only way to utilize too shallow a planter is to put the large plant in the center and to build up from the edge inward. The planter must be wide enough to slant the soil gradually so that the slope is not too great. Small gardens While a garden may be large enough to have only a single design function, that function can be quite varied, provided that the lighting intensity is appropriate. It can serve as a small glen or a space separator, or it can be simply a large decorative planter. The garden can be airy and open or it can be dense and closed. Planter depth of 1 1/2 to 2 feet is usually sufficient. Also, some small gardens can be designed to be changed with the seasons. Often, flowering plants, such as chrysanthemums or azaleas, are used, but the plants must then be replaced every two weeks. If the seasonal or flowering plant changes are desired, the plants should be left in their containers so that they

may be easily moved, Some care should be given to the planter design so that the growing cans are not obvious and do not detract from the arrangement. Creative additions of volcanic rocks, small ponds, or fountains can be quite attractive and set off and enhance the plants. However, with the usually limited space in the small garden, these additions can produce a crowded or overdone appearance. Overcrowding will give a jungle effect that is rarely desired. Just as in other design fields, good proportion and good sense will create a pleasing design that is neither overlooked or overbearing. Large gardens Large gardens are simply larger versions of small gardens, but their very size opens up more design possibilities, since they may be subdivided, into related sections. The shape, height, and texture of the planters may be varied from section to 910

Specialties section. The plants may be chosen to reflect varying design moods and functions. The lighting and watering requirements may differ between sections. In fact, variety is often necessary for good large-garden design, since a large mass of similar plants or plant groupings will create the impression of a monotonous forest or field. Because large indoor gardens usually are in areas of high ceilings, the light level must be very carefully considered. Just the presence of windows or skylights does not guarantee enough light. In addition, if the light sources are distant from the plants, the taller plants may effectively block some of the light from reaching the lower plants and foliage. When large areas are to be planted, there is a tendency to use rocks, pools, gravel, or fountains to cut down the plant costs and simplify the maintenance. Care is essential when using these elements to prevent the plant arrangement from looking bare and sterile. Large gardens are most commonly used in shopping malls. The skilled designer will take this illumination into account, as well as design the garden to enhance the shopper's view of the stores. The designer will always remember that large gardens achieve their effectiveness by both the proper variation of plant groupings and the proper variation of plants within the groupings. Procedures for planting gardens As pointed out earlier, a successful garden needs proper planting, since improper procedures can inflict severe damage. Correct planting involves not only correct technique and design but also correct organization. The techniques of proper drainage, spacing, and handling will ensure that the plants remain healthy once they are installed, Experienced supervision of the installation staff will be

important in this regard, since a large installation of expensive plants is no place for the on-the-job training of the supervisor. Proper planning and organization will ensure that the plants remain healthy between unloading and planting. If the plants are left on an unheated loading dock or stored in an unlighted or unheated room until they are installed, irreversible damage may occur. Drainage Overwatering of plants leads to root rot and is often more harmful than underwatering. To minimize this danger, the planter or container should be installed with proper drainage. The simplest technique is to provide a porous reservoir below the planting soil; any excess water will then drain into it from the root ball and be slowly fed back to the soil as the soil dries out. To prepare the planter or decorative container, the drainage material is poured into the bottom and leveled. The plant growing can is placed on top of the drainage layer and surrounded with more of the drainage material. For the smaller plants (in pots 6 inches or less in diameter), a 1-inch depth of drainage material is usually enough. For the larger plants, a layer of 3 to 4 inches is suggested. For very large gardens, about one-third of the planter depth should be the drainage layer, provided it leaves enough room for the root ball or planting can. The drainage material can be perlite (a readily available synthetic material) alone or mixed with small pebbles or gravel. The perlite is suggested since it is porous enough to feed back the excess water to the soil as the soil dries out. If only gravel or pebbles are used, the excess water will sit and stagnate in the reservoir and will not be fed back to the plants. Even with the proper drainage layer, overwatering is possible if so much excess water is used that it fills up the reservoir. The water level in a small container can be determined by tapping the container at various intervals and listening for the change in sound. In large planted areas, it is wise to provide for "dipstick" readings of the water level. To take such a reading, rigid hollow plastic tubes, with a cloth over their lower ends, are "planted" at intervals along with the plants. The hollow tubes reach from the top of the container to just above the drainage layer and the cloth on the bottom prevents soil or drainage material from entering the tube. A dipstick is lowered into the tube until it touches the cloth. If the stick, upon removal, shows more than $\frac{1}{4}$ inch of water, there is too much water in the bottom of the planter. If gravel is used as part of the drainage material, it should be $\frac{1}{2}$ inch to $\frac{1}{4}$ inch in diameter. Under no circumstances should limestone be used, since it is alkaline and will raise the pH of the water to a level that is too high for most tropical plants. Soil

separator If the plants are removed from their growing cans and replanted in growing soil, it is usually best to use a soil separator between the drainage layer and the planting soil. The separator is a semiporous sheet, often composed of fiberglass wool, which serves to keep the soil from falling into the drainage material. If the separator is not used, soil will clog the drainage material. Fiberglass wool of building material grade should not be used, as it contains chemicals that will damage the plant (Fig. 3).

TOPSOIL,,,SOIL SEPARATOR /•£? -?3?-S O^ScSJ, —DRAINAGE w" ^
 ■- -CSKLXXi- MATERIAL Fig. 3 Soil separation. Planting medium

Because the root systems of tropical plants are much finer than those of outdoor plants, pure topsoil is too heavy and too easily compacted to be used as a planting medium. It will constrict the plant roots and will retain too much water. For the common tropical plants discussed here, we recommend the use of the foliage plant mix developed by Cornell University. Because it is easiest to calculate the quantity of needed soil in terms of the volume of the planter to be filled, the formula given here is PLANTSCAPING Plant Use and Procedures for 1 cubic yard of soil. For conversion purposes, 1 cubic yard equals 21.7 bushels, 765 liters, or 27 cubic feet.

Sphagnum peat moss: V_2 cu yd = 383 lit Vermiculite#2: $1/4$ cu yd = 191 lit Perlite, medium fine: $1/4$ cu yd = 191 lit Ground limestone, dolomitic: 0.85 gal = 13.5 cup = 3.2 lit Superphosphate 20 percent solution: 0.21 gal = 3.4 cup = 0.79 lit 10-10-10 fertilizer: 0.32 gal = 5.1 cup = 1.2 lit Iron sulphate: 0.11 gal = 1.7 cup = 0.41 lit Potassium nitrate: 0.11 gal = 1.7 cup = 0.41 lit While this Cornell foliage plant mix gives the best all-around results, a simpler mix that gives good results in most cases is as follows: $1/3$ by volume sterilized commercial mix of peat moss and vermiculite V_2 by volume sterilized topsoil $1/3$ by volume perlite This mix is particularly effective for container planting. If it is to be used in a larger garden planting, such as a shopping mall garden, more perlite should be added for improved drainage. The peat and topsoil mix is considerably heavier than the Cornell mix and both are heavier wet than dry. If the garden is not situated at grade level, this weight can be an important consideration. Figure 4 gives guidelines to be used in estimating the weight of the planting medium.

Planting organization The basic ingredients for a large planting installation are drainage material, planting medium, soil separator, plant material, material-handling equipment, light, water, and labor Organization of all these ingredients is important since every one must be ready and

available for a successful installation. Arrangements for all these factors should be made ahead of time, and they should be ready and waiting when the plants are delivered. The amount of interior volume in the planters and containers determines the amount of needed drainage material, soil separator, and planting medium. If detailed blueprints are not available, actually measuring the planters is generally a good way to obtain this volume. The relationship between planting medium, drainage material, and soil separator can be determined using the guidelines of the previous subsection. If the plants are to be left in their cans (as generally recommended), the space between the plants is filled with drainage material. If they are removed from their cans, the space between plants is filled with planting medium. In either case, the amount of volume displaced by the plants is simply the sum of the volume contained in the growing cans. Information for each standard size of growing container is given in Fig. 5. The installation should not be started unless all lights and water connections are operating, as the plants will need both light and water during the installation - especially the light. If the plants are delivered dry, they should be watered in their cans unless they are to be planted at once and watered immediately after planting. If the plants are removed from their cans and placed into dry planting medium, they and the planting medium should be thoroughly watered immedi-

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Specialties PLANTSCAPING Plant Use and Procedures ately afterward. Fewer design mistakes will be made if the plants are installed one section at a time, under the direction of a supervisor familiar with the design of the section. If the installation is in an office building, it may be necessary to arrange for a workroom and a freight elevator with access both to the loading dock and the workroom. Depending on the exact arrangements, a crew of four to six workers per supervisor is generally optimum. It is recommended that each section be planted in the following order. First, leftover building material and other debris are removed from the planting areas. Second, drainage material is added to the proper depth and leveled. Third, the plants, either in or out of their growing containers, are placed on top of the drainage material and the soil separator if present, and arranged according to the design. The spaces between the plants are then filled in with drainage material or planting medium, depending on whether the plants are in or out of their growing containers. If planting medium is used,

it should be lightly compacted to prevent its settling later, (If the light intensity is below specifications and periodic replacement of the plants is expected, the plants should be left in their cans.) After the spaces between the large plants have been filled in, the groundcover, if any, is planted. The use of decorative bark or marble chips on top of the soil is not recommended as they easily mix with the soil and are hard to remove if the plants are replaced. After all the spaces have been filled, the plants should be thoroughly watered and the maintenance schedule begun. If dry planting medium is used, it should be watered thoroughly several times during the first week to ensure that it is completely wet, Removing plants from cans or burlap A healthy root system is necessary for the maintenance of a healthy plant. It is the new, very fine, feathery roots that are the most important and also the most easily damaged. This damage is very likely if the soil between the fine roots is dislodged in the course of repotting. Whether the tropical plants are delivered in growing cans or with their roots wrapped in burlap, the root system must be handled with care. The best procedure for removing a plant from its container is to lean the pot on its side, tap on the container sides and bottom, and carefully slide out the plant. In large container-grown plants (in 17-inch or larger cans), the root system may be held very tightly in the can. In this case, a can cutter, which works on either metal or rubber cans, may be the most gentle way of removing the can. Once the can is removed, the root ball of soil and roots should be scored by making 1/8-inch-deep vertical cuts at 3-inch intervals around the root ball from top to bottom. If the can removal and ball scoring are done near the planting site, the exposed root system is subjected to minimum handling. Very large plants and trees are frequently field-grown rather than container-grown. The root balls of such plants will come wrapped in burlap. When planting them, only the upper half of the burlap should be removed. The lower portion will disintegrate in the soil after the plant is installed.

Planting Bed Material
 Cornell foliage mix . Peat/topsoil mix Topsoil (loam) Gravel Sand Note: For conversion to metric system: 1 lb ■ Dry Weight 12-18 lb/cu ft 38-42 lb/cu ft 80-100 lb/cu ft 120-135 lb/cu ft 95-110 lb/cu ft ■ 0.454 kg; 1 cu ft » 0.028 cu m. Wet Weight 25-35 lb/cu ft 70-90 lb/cu ft 100-120 lb/cu ft 120-135 lb/cu ft 120-130 lb/cu ft Fig. 4 Planting material weight. These figures are the normal weight for each of the materials in both the dry and the wet state. The exact weight depends on the degree of compaction of the material. Pot Size 6 In 8 In 10 In 12 In 14 In 17 In 22 In 30 In 32 In 36 In

So" Volume 1 gal 2 gal 3 gal 4 gal 7 gal 10 gal 20 gal 35 gal 65 gal 95 gal
Pat Dlneter x Height 6Xlnx 6 In 8 In x 7 In 10 Inx 9Xln 11 Inx1OXln 1314
In x 12 In 17 In x 16 in 21 In x 17 In 29 In x17' In 32 In x 22 in 36 In x 24
In Note; For conversion to different units, use the following factors. 1 gal -
.00495 cu yd - .0038 cu m - .134,cu ft - 3.79 lit; 1 In - 2.54 cm. Fig, 5 Pot-
size and volume proportions. Rock formations and decorative pools Natural
elements, such as rock formations, decorative pools, water fountains, and
waterfalls, can add an artistic touch and turn an unimaginative large
planting arrangement into a full garden. Unfortunately the overuse of such
design elements is tempting, since they are usually inexpensive compared
with the cost of filling the same area with plants. Provided they are not
overused, they can serve as natural sculpture or as the answer for areas with
too little light to support plants or where conditions limit the variety of
plants that can be used. In rock formations, volcanic rock is the most
commonly used type because it is much lighter than ordinary rock. This
weight factor can be of considerable importance when the weight of the
garden must be limited. This type of rock is also easy to shape with a
hammer and chisel. Although a large decorative pool or fountain must be
custom-designed, there are small fiberglass pools that can be purchased in a
variety of sizes and are available in kidney, free-form, or rectangular
shapes. They are usually no longer than 6 feet, but they are of a standard
16-inch depth, which is deep enough to accommodate any water plants,
recirculating pump, and a filter tray with mat and gravel. Their high-
capacity, low-pressure pumps are usually adequate for small fountains and
waterfalls. If decorative pools are used, some thought might be given to
using water plants in them. These plants are very attractive and can be
easily grown indoors. As with all plants, different species have different
growing and flowering habits. A reputable dealer should be consulted for
information. The use of fish in pools should be carefully studied in light of
the plant maintenance requirements. Fertilizer, plant chemicals, and
limestone runoff from the planting area may enter the circulating water
system and kill the fish. Fish can be an attractive design element, but their
maintenance requirements must be considered along with the maintenance
requirements of the plants. Open Plan and Specimen Design Modern offices
are sometimes sterile places in which to work. The introduction of live
plants into such an environment is one way of making the space seem less
austere and more comfortable without disrupting the integrity of the

original design. For windowless offices, plants provide an attractive natural setting appreciated by the occupants. For offices and other windowed areas, the 912

plants provide a transition which makes indoors and outdoors seem to flow together. In all locations, however, the light intensity must be at the proper level before the plants are introduced. The intensity cannot be taken for granted, since artificial lighting designed for office vision is seldom enough for any but the lowest-light plant species. Even a large window will not provide enough light if it has an overhang or a northern exposure. If the light intensity cannot be directly measured or calculated from detailed ceiling plans, one must assume the worst and use only low-light material. There is sometimes a tendency to use plants to fill in otherwise forgotten spots, such as corners, stairwells, and hallways. Such areas are often poorly lit and no plant will survive there unless additional lighting is installed. In large areas with barely enough light, the usual design problem is how to arrange the limited number of low-light species so that different areas stand out from one another. Design interest can be accomplished by using different types of foliage (for example, fragmented and solid) in the different areas, varying the plant sizes among the areas or using specimen plants selectively. Specimen plants usually have fuller foliage or an unusual stem structure and hence appear to be different from other plants of that species. The true specimen plants are more expensive than ordinary plants of the same species, but can solve many a design problem. However, a plant with fuller foliage than most will also require more light than most to maintain the foliage. If the office has floor-to-ceiling walls, the best design procedure is to select specimen plants that act as living sculptures. Since these plants are used for visual emphasis, the plant height and container size should conform to the scale of the rest of the interior design. The plant texture and container finish should blend with each other and with the wall and floor treatments. The particular plant specimen chosen should have an inherently interesting shape and texture. If the office area is very large or is designed along an "office landscaping" plan with movable partitions, the plants can become an integral part of the design. They can be used with the partitions as space dividers and are excellent for indicating the importance of the space. They also may be effective in relating widely separated areas with one another. They break the monotony of the partitions with both color

and texture. They act as sound absorbers. Also, specimen plants can be used in the office landscaping scheme for visual emphasis. Planting into individual planters Individual decorative containers are used for individual plants or small plant groupings. The plants are left in their growing containers and placed directly into the decorative planter on top of 4 to 6 inches of perlite as the drainage material. The decorative planter or container must be tall enough to accommodate the growing can and the perlite, and wide enough to accommodate the width of the growing can. The space between the growing can and the inner wall of the planter can be filled with additional perlite. (See Fig. 2 for size-selection guidance.) As a decorative finishing, bark chips or sheet moss may be placed on the surface of the soil in the growing can. This decorative cover can be easily removed if the plant is replaced and it does not mix with the soil, as sometimes happens in large gardens. Removing the plant from the growing can and repotting it directly into the planter is not generally recommended. Replacing the plant, if necessary is a messy job unless drainage material and soil separator are added to the bottom of the container. Also, once removed from its growing container, the plant may take up to four weeks to adjust fully to its new environment.

CONTAINERS Decorative Containers: Different Types A plant container should be more than decorative. Its proper selection is the first element of proper maintenance, since the container must provide the plant roots with sufficient growing room and with adequate drainage. All small to medium-size plants are received from the grower in growing containers, usually metal cans or rubber tubs. Large plants are either in large growing containers or their root balls are wrapped in burlap. As a rule, these growing cans provide the proper volume of soil for the size of the plant and have a hole in the bottom for drainage. There is seldom any need to remove the plant from its growing container, especially since rough handling of the root system can shock the plant. Only the smaller plants, such as ivy, can be repotted without much disturbance of the root system. If it is absolutely necessary to repot a larger plant, it should be done carefully as outlined earlier, and it should be always into a larger volume of soil, never into a smaller volume. The decorative container should be chosen so that its inside dimensions are large enough that the plant-growing container can be dropped directly into it. In addition, it

Specialties PLANTSCAPING Plant Containers should be deep enough for the growing container to rest on at least 2 inches of perlite or other drainage material, and leave about 1 inch

between the top of the growing can and the top of the decorative container. Some care must be taken in the choice since the interior dimensions of the decorative container are often not uniformly related to the exterior dimensions. For example, some fiberglass containers have a large lip which limits the size of the growing can that can be dropped directly into them. Also some containers have a large false bottom, which makes the interior depth much less than the outside height. With these simple size-selection rules in mind, the proper decorative container can be selected using Fig. 6 as a guide. This figure lists the decorative pros and cons of the most common types of containers. Excess Water in Container Overwatering of plants is more harmful than underwatering. This problem is most likely to occur when the plants are in individual decorative containers that do not allow the excess water to flow off. To minimize this danger, we have recommended that a plant in a decorative container be double-potted. In the bottom of the decorative container, below the plant growing can, there should be at least 2 inches of perlite or other drainage material to act as a reservoir for excess water. Nevertheless, if the plant is continually overwatered, this reservoir will fill up and lead to root rot because the roots are in a pool of water. If the plant soil is continually wet to the touch, excess water may be the problem. The water level in the container may be determined by tapping the sides of the container. If the water level indicates excess water, the container is tilted on its side, the plant gently pulled from the container, and the excess water drained from the perlite. If the perlite is completely saturated or appears old, it must be discarded and replaced with new drainage material. If the plant has been sitting in a pool of water for some time, the root ball should be allowed to dry before repotting. If a very large container or garden has been overwatered and there is no way to drain out the excess water, not really much can be done short of using a small electric pump. One must simply avoid watering the plant or garden at all until the soil has begun to dry out and feels dry to the touch. 913

Specialties PLANTSCAPING Water Requirements Detlgn Watering Type
 Plant Name Requirement! HIGH-LIGHT PLANTS—150
 FOOTCANDLES AND UP T T T T T T FP FP FP FP FP FP FP DTP DTP
 DTP Fiddle-leaf fig (*Ficus lyrata*) Indian laurel (*Ficus retusa*) Rubber plant
 (*Ficus elastica* cv. 'Decora') Weeping fig (*Ficus benjamina*) Norfolk Island
 pine (*Araucaria heterophylla*) Schefflera (*Brassia actinophylla*) Dwarf date

palm (*Phoenix Roebelenii*) Dwarf schefflera (*Brassia arborescens*) False
 aralia (*Dizygotheca elegantissima*) Lady palm (*Rhaphis excelsa*) Mock
 orange (*Platanus Tobleri*) Ponytail (*Beaucarnea recurvata*) Southern yew
 (*Podocarpus macrophyllus* var. *Maki*) Jade plant (*Crassula argentea*)
 Swedish Ivy (*Plectranthus australis*) Wax plant (*Hoya carnosa*) MED1UM-
 LIGHT PLANTS-100 TO 150 FOOTCANDLES T T T FP FP FP FP FP FP
 FP FP FP FP FP FP Indian laurel (*Ficus retusa*) Schefflera (*Brassia
 actinophylla*) Weeping fig (*Ficus benjamina*) Bamboo palm (*Chamaedorea
 erumpens*) Corn plant (*Dracaena fragrans* cv. 'Massangeana') Dwarf date
 palm (*Phoenix Roebelenii*) Dwarf dragon tree (*Dracaena marginata*) Dwarf
 schefflera (*Brassia arborescens*) Green dracaena (*Dracaena deremensis* cv.
 'Janet Craig') Green pleomele (*Dracaena reflexa*) Kentia palm (*Howea
 Forsteriana*) Narrow-leaved pleomele (*Dracaena angustifolia* *honorata*)
 Neanthes bella palm (*Chamaedorea elegans*) Reed palm (*Chamaedorea
 Seemannii*) Self-heading philodendron (*Philodendron Selloum*) W W W W LF
 W LF W W w w LF LF LF MF MF W W W MF W LF LF W W w w w w
 MF LF Detlgn Watering Type DTP DTP DTP DTP DTP DTP DTP DTP
 DTP DTP Plant Name Requirement* Boston fern (*Nephrolepis exaltata* cv,
 'Bostoniensis') Chinese evergreen (*Aglaonema commutatum* var,
maculatum) Common philodendron (*Philodendron scandens oxycardium*)
 Dumb cane (*Dieffenbachia maculata* cv. 'Rudolph Roehrs') Golden pothos
 (*Epipremnum aureum*) Grape Ivy (*Cissus rhombifolia*) Prayer plant
 (*Maranta leuconeura*) Swedish Ivy (*Plectranthus australis*) White Hag
 (*Spathiphyllum* cv. 'Clevelandii')- White-striped dracaena (*Dracaena
 deremensis* cv. 'Warneckii') LOW-LIGHT PLANTS-50 TO 100
 FOOTCANDLES FP FP FP FP FP FP FP FP FP DTP DTP DTP Corn plant
 (*Dracaena fragrans* cv. 'Massangeana') Dwarf dragon tree (*Dracaena
 marginata*) Green dracaena (*Dracaena deremensis* cv. 'Janet Craig') Green
 pleomele (*Dracaena reflexa*) Kentia palm (*Howea Forsteriana*) Neanthes
 bella palm (*Chamaedorea elegans*) Reed palm (*Chamaedorea Seemannii*) Self-
 heading philodendron (*Philodendron Selloum*) Chinese evergreen
 (*Aglaonema commutatum* var. *maculatum*) Common philodendron
 (*Philodendron scandens oxycardium*) White Flag (*Spathiphyllum* cv.
 'Clevelandii') W LF W LF LF W W w w w w LF w w w w w w LF w w A
 Final Word about Lighting Intensity The preceding lighting-Intensity
 recommendations are based on experience and the assumption that these
 levels will be provided eight hours a day, five days a week, and that the

plants have been fully acclimatized. If light can be provided for more hours each day or more days each week, the plant material will look its best for longer periods. On the other hand, often the energy costs of the longer lighting exposure are more than the costs of plant replacement. However, if the plants are not to be maintained by the landscape contractor with a plant replacement guarantee, provision should be made for giving the plants light exposure seven days a week. Fig. 7 Growing requirements. Design type: T==tree; FP=floor plant; DTP = desk or table plant or ground cover. Watering requirements: W=water weekly; MF=water more frequently, as required; LF=water less frequently, as required. 915

Specialties PLANTSCAPING Plant Containers Automatic Watering Devices In areas where regular maintenance would be difficult, the use of automatic watering devices can be of considerable help. Even when they are used, however, the plant must be checked periodically to see that the device is working properly, that its water reservoir is full, and that no other maintenance problems have developed. Automatic watering devices are either external to the container or are built into the planter. The external devices tend to work well only with small plants, and also, they are likely to detract from the design. For these reasons, the built-in type of device is preferred. The planters with this type come in both cylindrical and rectangular shapes and in several colors. The planter has a hollow space within its double-wall sides which serves to hold a three- to four-week water supply feeding the water to the plant soil by a wick mechanism, sensor, or capillary action. Most types have a float to indicate the amount of water remaining in the reservoir. Since the soil must be in contact with the wick or capillary tubes for the device to work, the plant must be removed from its original growing can and repotted directly in the planter. As the soil never dries out, the plant must be watched for symptoms of over-watering. Because different plants use water at different rates under different humidity and temperature conditions, a timetable should be kept for each container so the maintenance staff will know when to refill each reservoir. The use of automatic watering devices will not eliminate maintenance personnel, but it will reduce the number of workers needed. One person can handle many more plants, devoting more time to cleaning and trimming, since the reservoir has to be refilled only every month or so. Occasionally however; one will find a client who will resist the use of the automatic devices

because he or she likes the assurance of seeing a person with a watering can once a week. The use of the automatic watering devices is expected to increase in the future as more architects and designers become aware of them and convince their clients of their usefulness, and as the manufacturers produce more colors and styles and improve the efficiency of the devices.

Cooltlotr D/p* Fiberglass. Ceramic Metal Baskets, traditional Baskets woven around metal Plastic Hanging planters (Heavy; must be used with a rotating hook which can support the weight and allow for easy plant access.) Pro* Large selection of sizes, shapes, and colors. Light weight, easy to move. Some types have casters. Reasonable prices. Many manufacturers. Large selection of sizes, shapes, colors, and textures. Rich appearance. Can be put on casters. Large selection of sizes and styles. Rich appearance. Polished or brushed finish. Good range of styles and textures. Combines well with all furniture styles. Reasonable prices. Good texture range. Reasonable prices. Do not sag. Need no saucers. Combine well with all furniture styles. Least expensive. Good for table plants. Versatile. Available In ceramic, fiberglass, plastic, and metal. Ceramic In various shapes and textures, metal In various finishes, Plastic and fiberglass are Inexpensive. All are versatile. Caw Easily scratched. Some types have large lips. Expensive. Easily broken In shipping and handling. Expensive. Limited sizes. Tend to sag. Need saucer under plant can to prevent water spillage. Sizes limited. Available mostly In green or white. Sizes largely limited to standard pot sizes. Need saucers underneath, Cheap appearance. Makes plants susceptible to drafts from heating and air conditioning. Difficult to water without spilling on floor. Metal very expensive. All need Inner pot to allow for drainage. Ceramic Is porous and presents condensation problem. Fig. 6 Comparison of container types. 914

(False) Aralia A plant of grace and elegance with narrow, ribbonlike, notched leaves of dark green, usually born on slender, single stems. The aralia is attractive if two or three plants are planted together in one pot. It grows very quickly so prune the stem tips from time to time to prevent the foliage from thinning at the bottom. Temperature The aralia is tolerant of warm temperatures if there is plenty of humidity. Light/sun The plant likes a semisunny to semishady window; an east or west window is ideal. Water/humidity Keep the soil damp but not soggy. The false aralia likes a humid atmosphere. Place your plant on a pebble tray and mist the foliage

daily. Soil The soil should be equal parts loam, sand, and peat moss. Special care You can rejuvenate leggy plants by drastically cutting the stems back to four to six inches from the pot. Do this in the spring and leave the plant in a sheltered location, being sure to fertilize and water frequently. African Violet The African violet, a longtime favorite house- plant, does insist on more care and attention, but its beautiful blossoms make the effort worthwhile. Temperature African violets are more contented and grow best within a temperature range of 65 to 80 degrees. Be careful that your plants are not in an open window or a draft. Light/sun The African violet enjoys a place in an east or west window. Direct sun is too strong, unless filtered through a curtain. Excess sun will cause spotting and loss of color, and too little light causes elongated stems and no blooms. Water/humidity African violets should be watered from the saucer underneath in the morning with lukewarm water. Water when the soil begins to dry out. Do not keep it soggy If the air is dry in your home, place the potted plant in a tray of moistened pebbles. Soil The soil should be porous for good drainage and should contain ample organic matter such as compost or peat moss. Commercial African violet soil mixture is specially prepared for these plants; however, add sand or perlite to ensure adequate drainage. A plastic pot is less likely to cause the lower leaves to rot where they touch the pot. Asparagus Fern - Emerald Feather The bright feathery green of this delightful plant is best displayed in a hanging container. The long branches drape gracefully and are studded with tiny white flowers that ripen into red-orange berries. Temperature Asparagus fern is not fussy about temperatures, but prefers a range of 60 to 68 degrees. Light/sun The bright filtered sun of an east or west window is a good location for this plant. Water/humidity Soak the soil in the pot thoroughly and allow it to become dry to the touch before rewatering. Soil A well-drained potting soil or a mixture of equal parts of loam, peat moss, and sand or perlite. How to start new plants Allow the berries to ripen and when dry sow the seeds they contain. Asparagus fern can usually be grown from seed quite well.

Specialties PLANTSCAPING Typical Plants Avocado The avocado comes easily from seed and is grown for its ornamental foliage. It makes a nice tree for your indoor garden. Allow the plant to reach the desired height and then begin regular pinching to force branching and encourage bushy growth. Temperature Temperatures between 60 and 70 degrees suit the

avocado well. Light/sun Keep your avocado in bright light but protected from direct sun. Avocados are easily sunburned, especially when they are first moved outside. Water/humidity Use tepid water and keep the soil moist. Place the plant on a pebble tray to raise the humidity level around it. This plant likes a fair amount of humidity and benefits from regular misting. Any signs of browning or crispness at the tips and along the edges of the leaves means the plant needs more humidity. Soil Use a mixture consisting of equal parts of sand, loam, and peat moss. AVOCADO *Persea americana* BOSTON FERN *Nephrolepis exaltata* Boston Fern Exaltant is a good adjective for this family of ferns that can fill a corner with rich green foliage. These ferns are excellent for hanging baskets. Initially the ferns may need a lot of attention until the right combination of environmental factors is achieved, but the effort is well worth it. The leaflets grow on a midrib that is covered with fine brown hairs and vary from smooth-edged to feathery and even ruffled. A mature fern can have fronds ranging in length from two to three feet and two to three inches across. Temperature With lots and lots of humidity ferns will do well in house temperatures in the 60 to 70 degree range. Light/sun Ferns need a location with good bright light, but this means filtered sunlight. Avoid direct sunlight Water/humidity It is essential that the roots of the ferns never dry out at any time. Soak the soil regularly. Clay pots and hanging baskets can be soaked in a bucket or the sink for half an hour and then drained. The soil should be checked daily to make sure that it is not drying out. Humidity is the most important ingredient to successful fern growing. Place pots of ferns on a pebble tray. Mist the foliage daily with room temperature water. Soil Ferns need a soil that is loose and easily penetrated by their dense root system. The soil mixture should be rich in peat moss and organic matter with a liberal amount of sand for drainage. A sprinkling of charcoal mixed in the soil helps to keep the soil from becoming sour from the frequent waterings. When potting ferns, place a layer of bits of broken pots or gravel in the bottom of the pot. Ferns do not take kindly to having their roots tampered with, so be careful not to damage them when repotting. CHINESE EVERGREEN *Aglaonema moenkeanus* Chinese Evergreen This beautiful foliage plant has waxy dark green leaves. The leaves grow on a canelike stem and are oblong, tapering to a thin tip. Some of the varieties are variegated with splashes of creamy white or yellow. Under optimal conditions, it will produce a flower spike surrounded by a white spathe. The

flower is similar to a calla lily. The great thing about this plant is that it will adapt to a variety of environments which makes it a good plant for a beginner or a difficult location. Temperature A range of 60 to 70 degrees suits this plant well. Light/sun A shady spot, an artificial light, or any other location will suit this plant. The Chinese evergreen is an excellent plant for a north window. Water/humidity Keep the soil moist but not soggy. To avoid waterlogged soil, allow the surface soil to become dry to the touch before rewatering. The Chinese evergreen can be grown in water. The roots are attractive so a clear glass container shows them off to best advantage. It is important to wash the leaves regularly to keep them dust free. Soil The soil should be equal parts of garden loam, peat moss, and sand. 917

Specialties PLANTSCAPING Typical Plants Dracaenas There are several varieties of dracaenas which vary in foliage color variegation, and size. Here are three that are commonly available. *Dracaena deremensis* "Warneckeii": is a good choice for a location without much light. The gray green foliage is striped with white and gray *Dracaena marginata*: has clusters of narrow deep green leaves edged with red and gray stems strongly marked with leaf scars. This variety will reach a height of five or six feet. *Dracaena sanderiana*: resembles a corn plant in the brightness of the green and the size and shape of the leaves with the difference that the leaves are striped with white. Temperature Moderate household temperatures in the 60 to 70 degree range suit Dumb Cane The cool-looking foliage of this plant is yellow-green, mottled with white. The leaves are pointed ovals that become quite large as the plant matures. The *dieffenbachia* is known as the "mother-in-law" plant or the dumb cane because when a piece of the stem is placed on the tongue it causes temporary numbness and loss of speech. All joking aside, this plant is poisonous. Temperature The *dieffenbachia* prefers warm temperatures and will tolerate hot dry places with added humidity these plants best. It is important to keep plants away from heating vents. Light/sun The *marginata* and *sanderiana* should get only filtered sun or bright light. The *Warneckeii* will fare well in a spot with very little light; it will flourish when more light is available. Water/humidity These plants all like soil that is kept evenly moist but not soggy. Soak the soil in the pot thoroughly and then rewater when the soil surface feels dry to the touch. Humidity is a must. Brown crispy leaf tips and margins mean too little moisture in the air. It is a good

idea to place the dracaenas in pebble trays and mist the foliage daily. Soil Commercial potting soil is adequate but added drainage material such as sand or perlite is advisable. Light/sun This plant does well in an east or west window where it can bask in the sun for a few hours. Water/humidity The soil should be allowed to dry out for a few days before rewatering. The plants indicate a need for water when the leaves show signs of dropping. Regular misting keeps the foliage dust-free and luxuriant. Soil A porous soil of equal parts loam, peat moss, and sand is fine. DUMB CAME

Dieltenbachta ataculata GARDENIA *Gardenia raaicans* *lioraptam* Gardenia The gardenia is a handsome foliage plant with intensely fragrant blooms, but it has an extremely temperamental nature. It is a challenging plant to grow successfully indoors. The most frequently available varieties are *Gardenia radicans floraplana*, a low spreading plant with small double flowers, and *Gardenia florida*, which blooms in summer. Temperature The temperature must be kept above 65 degrees to maintain healthy foliage and flower buds. These plants hate drafts. Loss of flower buds is often due to sudden changes in temperature. Light/sun The gardenia needs lots of light, but avoid strong sun that might burn the leaves. Water/humidity The soil must be kept constantly moist without becoming soggy. Submerge the pot in a bucket of lukewarm water and allow it to soak for half an hour or until the soil is moist on the surface. Do not allow the pot to sit in water as that will cause the roots to rot. Gardenias need very high humidity at all times, Place the pot in a tray of moistened pebbles. Mist the foliage daily with tepid water. Leaf or bud drops indicate the air is too dry. Soil Potting soil should be a mixture of equal parts peat moss, loam and well- decayed manure with sand or perlite added for drainage. 918

Specialties PLANTSCAPING typical Plants OHAPE IVY *Cissus* *titiomblloli* Grape Ivy Grape ivy is a climber or trailer. The olive colored green leaves look a bit like those of holly without the stiffness or the sharp tips. The leaves form attractive groups of three and are accompanied by furry tendrils. Temperature The plant is fairly tolerant of a wide temperature range. Increase the amount of humidity as the temperature goes up. Light/sun Grape ivy will do all right in low light and is often used in low light areas. But it flourishes with bright light or filtered sunlight. Water/humidity Soak the pot and soil thoroughly and then allow the soil to become dry to the touch before rewatering. Mist frequently and wash the

foliage regularly to remove dust and restore the luster of the leaves. Soil A potting soil that is rich in organic matter is the best. Be sure to add plenty of drainage material to the soil mixture. JADE PLANT The jade plant is a tough plant well-suited to the hot dry conditions so prevalent in office and apartment buildings. The rounded leaves are in pairs on the branched treelike stem. A plant that is six to eight years old will produce clusters of lacy-looking star-shaped flowers. Temperature Temperatures ranging from 65 to 75 degrees are fine. Lower and higher temperature will be tolerated. Light/sun The jade plant will require full sunlight with shade at midday if possible. A west or south window would be good locations. If you put the plant outside in the summer, place it in a lightly shaded spot. Water/humidity The soil should remain dry for several days between waterings. The fleshy leaves soak up the soil water and store it for future use. Too much water will cause stem and root rot and certain death. Soil The jade plant will do well in rich garden soil that has coarse sand or fine bits of broken pots added to it for drainage. Each year give the pot a top dressing of humus. A new pot will be necessary only after about three or four years. NORFOLK ISLAND PINE The delightful symmetry of this evergreen makes it a desirable house plant. The branches grow in tiers of six, each tier representing a year's growth. The bright green needles are soft and pleasant to touch. Temperature The ideal temperature is between 50 and 60 degrees. High temperatures are tolerated when sufficient humidity is available. Light/sun The filtered sun of an east or west window is best. Yellowing of the needles might mean too much sun. Water/humidity Provide the plant with a well-drained soil and pot. Water thoroughly and allow the soil surface to become dry before rewatering. Daily misting is necessary for the warmer temperatures of most houses and offices. A pebble tray will help to add more moisture to the air around the plant. Soil Garden loam mixed with equal parts of sand and peat moss makes a suitable potting mixture. Repot the Norfolk Island pine only when it has become potbound (the pot is crammed with roots). This would be about every two or three years. NORFOLK ISLAND PINE *Artocarpus alatus* 919

Specialties PLANTSCAPING Typical Plants *Chamaedorea elegans* Parlor Palm The palm trees are not the easiest plants to grow. However, once you have discovered their basic needs they are a delightful addition to your indoor garden. This palm grows to about four feet tall. It is most attractive

when two or three plants are grouped together in a pot. The long feathery fronds grow out of a single stem. Other varieties to try are *C. seifrizii*, *C. ecumpens*, and *C. costaricana*. Temperature The best growing temperatures for palms range between 60 and 75 degrees. Light/sun Palms are good plants for locations without much light. They do not like direct sun light. Water/humidity During the active growing season, between March and October, the palm needs moist soil but it will not tolerate soggy soil. In the winter months, allow the soil to dry on the surface before rewatering. If the foliage shows signs of browning and drying on the tips, it needs more humidity. Misting regularly is recommended to keep the foliage healthy. Soil The palm needs well-drained soil of equal parts rich garden loam, peat moss, and sand. It will need repotting only every two or three years. It prefers being a bit potbound.

COMMON PHILODENDRON *Philodendron oxycardium* *Philodendron* By nature, the philodendron is a climbing plant, but it also trails. It looks best on a bracket beside the window frame, and for good effect must be kept strongly pinched back so that the plant is full of bushy young growth and does not deteriorate into two or three stringlike stems, temperature Normal house or office temperatures are fine, Light/sun The philodendron is quite hardy and robust and will grow almost anywhere, However, it will fare better in a well-lighted Water/humidity The plant should be kept evenly moist and never allowed to dry out. Be certain water does not remain in the saucer after watering, The foliage should be misted daily and the leaves cleaned of accumulated dust. Soil Potting soil mixed with perlite, vermiculite, or sand and peat moss is recommended.

WINDOWLEAF PHILODENDRON *Philodendron portisum* "Monslara oallciosa" Windowleaf *Philodendron* This philodendron has large heart-shaped leaves that are slashed irregularly. It is an enthusiastic climber and needs a piece of bark or totem for support. The aerial roots can be inserted in the soil or encouraged to attach to the totem, Keep the growing tips pinched back so that the plant doesn't get leggy Temperature The windowleaf prefers temperatures between 65 and 70 degrees. Light/sun Bright light is best for this plant. However, avoid putting the plant in a location where the plant would get direct sun, Water/humidity Soak the plant thoroughly and allow the soil surface to remain dry for a day or two before rewatering, Mist the foliage daily and wash the leaves weekly to remove dust, Soil A soil mixture of equal parts garden loam, peat moss, and sand is fine. 920

Specialties PLANTSCAPING Typical Plants PURPLE PASSION PLANT.

VELVET PLANT *Gynura aurantiaca* Purple Passion Plant-Velvet Plant The strikingly rich royal purple coloring and velvety texture of the foliage and stems attract many growers. The green leaves and stems are covered with tiny purple hairs, The straggly growth habit is best kept in check by frequent pruning. Temperature The purple passion plant likes temperatures in the 65 to 70 degree range. Light/sun the color. Direct or partial sun will promote Rubber Tree Plant This house plant with dark green glossy leaves can grow to be four feet high with a little care and not too much water, Temperature Due to its hardy nature, the plant does well in any normal household temperature. Light/sun The plant will do well in almost any light, but a well-lighted area is best for the rich green foliage characteristic of the rubber tree plant, Water/humidity Water only when the soil is completely dry all through the pot, You should set the entire pot in a bucket when WANDEWNQJEW *Zibtrina pendula* Wandering Jew This is a particularly attractive hanging plant. It is hardy and easy to grow with only one special requirement, which is regular pinching to keep it full and bushy. There are several plants called Wandering Jew, distinguished from each other by their different colorings and markings, The illustration is a *Zebrina pendula*. The leaf is a pointed oval with a deep purple underside, and the upperside is dark green striped with pale silvery-green, *Tradescantia fluminensis* has small oval green leaves marked with white, silver and white, or yellow. Water/humidity It is important that the velvet plant not dry out. Keep the soil evenly moist at all times, A humid atmosphere is important to keep the brilliant color, Mist the foliage frequently and place the pot in a tray of moistened pebbles to raise the humidity. Soil Use potting soil of equal parts garden loam, peat moss, and sand, This plant will also grow in water. watering, so that moisture can penetrate the deepest roots. Clean the leaves every two weeks or so with a damp cloth. Do not artificially shine the leaves as this clogs the plant's pores and does not allow it to breathe! Soil Soil should be a well-drained mixture of equal parts of sand, peat moss, and garden loam, If pot is plastic or rubber, be sure to provide plenty of drainage material in the bottom of the pot. Temperature These plants prefer warm temperatures. Light/sun Bright Indirect sunlight keeps the foliage brilliant, Avoid direct sunlight as they are susceptible to sunburn. Water/humidity Water generously, keeping the soil moist at all times.

During the winter months it will not need quite as much water. Soii This plant grows in a well-drained potting soil, or water, 921

Specialties PLANTSCAPING typical Plants SCHEFFLERA - UMBRELLA TREE Schetllera vonvloaa Schefflera - Umbrella tree If you are looking for a tree for your indoor garden, a schefflera is a good choice, It has handsome deep green leaves that radiate out from a long slender stalk rather like the ribs of an umbrella, Temperature The umbrella tree does well in a room where the temperature ranges from 55 to 75 degrees. Light/sun The schefflera does not like direct sunlight. It grows best in good light from a shaded window. Snake Plant Seen in many homes and offices, this spikey, banded plant will take almost any abuse, Temperature Normal household temperatures are best, but do not allow the plant to become suddenly chilled! Light/sun The snake plant is a good low light plant but needs sun in order to bloom, Water/humidity When watering your schefflera, soak the pot thoroughly and then allow the soil to dry before rewatering, The plant likes a humid atmosphere and responds well to daily misting with warm water. This is essential if the plant is in a room with forced hot air heat. This plant needs a pebble tray, Soil The soil mixture for the umbrella tree should be equal parts of peat moss, garden soil, and sand, The pot should have a layer of gravel or bits of broken pots underneath the soil to ensure good drainage. Water The plant likes the dryness of the home and should never be overwatered, The leaves should be cleaned with clear water every two weeks. Soil Garden loam, peat moss, and sand mixed together provides the best soil for the snake plant, SNAKE PLANT Sansaviera Irllascialo 922 SPIDER PUNT CNoroptiytum eiatvm vlll&luw Spider Plant With its green and white foliage, the spider plant makes one of the best hanging plants. The graceful trailing runners have plantlets and white star-shaped flowers. There are all- green varieties but the more commonly seen one has a green leaf striped with white. Temperature The plant lives best in a warm location. Light/sun This lovely plant does very well hanging in indirect sun or a moderately lighted area. Water/humidity The spider plant should be allowed to dry out before rewatering, Drying leaf tips usually indicates lack of humidity, To tidy up the plant just snip these off, Soil The plant grows contentedly in a rich soil composed of garden loam, sand, and peat moss.

Specialties PLANTSCAPING Typical Plants; Planting Detail Zebra Plant
 The zebra plant is one of the showiest house plants one can grow, Its spike
 of waxy yellow flowers and deep shiny green leaves veined in white makes
 it a striking specimen, Temperature The zebra plant needs warm
 temperatures free from drafts, Light/sun The plant wants bright light but not
 direct sunlight. Water/humidity It is important never to allow the soil to dry
 out, Set the pot in a pebble tray and mist the foliage daily. Soil The zebra
 plant likes loose soil consisting of one part garden loam, one part sand or
 perlite, and two parts peat moss. £0IL LEVEL ■KIUKSEEt £fkIT&iUEK
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Specialties PLANTSCAPING Plant Containers SLATE - SILL. ANCHORS
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Specialties PLANTSCAPING Planters PLANTS. IN POTS PEA GsyWet- AT TOP CAULKIN6 \"£ wooc? '■RJPS> /(• O.C Vi ' ONE OP" TWO JA COFTEC PEAIN4 COHNleCTED, TRAPTHDjANO THEM counbstep to a v*vsre ?/ b Section of window planter in a south wall <\\J^ r*, BLACK FACE 6RICK -#3 COPPER. PIPE ONE TOR EACH ^'-O'OPUNEAR PLANTER. COMtinuOUb ©HINDUS 6HIM<> TKB^TEP WTH «.E050Te BLUE (3LAZEP fACE BRICK 'A copper, pipe with 5CRESU TO Be. TRAPPED ANPiSoWNEC-TEP TO A BUIQSIN^ WASTE UNe Planter for a warm climate FINI= WHITPSANP- PEKFOKATEP PIPESCREWEP INTO PIPS AT THB BOTTOM T3PSCXL. ■ CRUSHED ROCK -=^rr ■;SLATE OK. OOARK,V TILE GALVANIZED STEEL PAN PAINTED WrTH HEAVY COAT OF CHLOEINATEP "" RUBBER PAIMT

■ 1" PIPE TO WALL, PRV WELU Floor planter Planter for a restaurant or store Fig. 9 Detail 1: The plants remain in their own clay pots. The use of pea gravel at top and only 4 In of 3A-'m gravel at bottom permits easy changes of the plants. To take care of watering and drainage, the copper pan is simply sloped to one side and two screened drains are connected, trapped, and joined to a waste line. This takes care of any excess water, as It Is eliminated by gravity drainage. Detail 2: This planter Is for areas where freezing does not occur, and the drainage of excess water can be taken care of by simply extending small pipes directly to the exterior. Detail 3: In this planter the plants remain within the planter and excess water Is carried off by a screened pipe at the bottom. Pea gravel is used as a 1-in topping so that odds and ends dropped Into the planter can easily be removed. Detail 4: A planter in a commercial lobby or entrance Is shown In this detail. The plants are permanently installed and the tall drainage pipe takes care of any top applied water. The white sand at the top Is to bring contrast to the colors of the plants. 928

lo'- O" Specialties PLANTSCAPING Benches =£h ir 4^ zaz i i :~ =£ -
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CONCRETE .SECTION 929

Specialties PLANTSCAPING Benches l'-o'u AL0" -z tcy m ~7F -m X IW
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W/PLUWaED ENC6- ..,... TOW ELEVATION)'-7^4 O) W* ISTfEKNAL
6ALVAHIXED P0t*p SLOPED fECHAED SPACEK W KJBC£6^ m fciJ06
PLUfafafcd tl"4>TrFf) ♦ ICLT' M'IVtlW &T6EL PLATE. ■P/IU-* EYE-
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6CREW rX4")(W STEEL TU&E ^l3'Yi"tl" 'oTBEL PLATE EXPAl^ION
SHIELDS* &CU6 INTO EXISTING ^O^KETE. 5ECTION

SIGNAGE SYSTEM DESIGN CRITERIA Initial consideration should be directed toward determining the basic parameters required in developing the sign system. Each of them merits discussion here.

Performance Requirements Signs usually must be designed to meet specific performance requirements. The good designer will determine how a system is to perform within given space relationships. The sign system may function entirely on its own merit, or it may be supplemented by staff personnel at major decision-making locations, such as the main lobby and reception areas. Sign devices may become decorative amenities to be featured within the environment, or they may be subtle and low-key elements of minor importance. Supergraphics may be considered in certain areas simply as an art form, or as a functional graphic device presented in large scale for emphasis of context. Certainly, a combination of the two is feasible. These are only several performance considerations that should be addressed prior to the development of the signage system. The designer must evaluate the needs of the client, the unique traffic flow requirements and mounting restrictions dictated by the structure, and the basic performance requirements desired of the signing devices to be utilized.

Usage Considerations The general nature of the building complex often defines how signs are to be used. They may be given an appearance of being fixed and an integral part of the architecture by the appropriate selection of materials, colors, and mountings, or they may appear changeable and temporary should need so dictate. Some signage requires constant change to properly relate information to people or people to facility, while most sign devices are considered permanent fixtures within a given space. The designer is responsible for determining how signs are to be used most effectively, and at the same time, for enhancing the environment.

Durability Requirements Prior to the selection of materials for a signing system, durability requirements must be considered. The vast assortment of materials available for signs covers a wide spectrum of durability from soft plastics to metals. The sign copy and background material should be evaluated both individually and jointly when considering durability requirements.

Vandalism Considerations Signs located in controlled spaces are often free from destructive vandalism; however, in many instances vandalism becomes rampant and uncontrolled. There are no materials that may accurately be labeled "vandal-proof." However, some materials are more Helvetica Medium Clarendon Fig. 1 vandal-resistant than others,

Where vandalism is of prime importance, only materials and graphic techniques engineered to resist destruction should be considered. Flexibility to Accommodate Changes and Additions Modern architectural structures are designed to accommodate inner spacial changes to meet tenant needs. Partition systems, prehung door units, room dividers, and modular furniture have ensured ease of change in officescapes. The sign system may also require alterations to preserve continuity. Changes and additions to a sign system should be considered by the designer prior to the selection of materials, graphic techniques, and mounting methods to be used.

Readability Factors Sign readability is determined by the letter style selected, size of copy, interletter spacing, copy position relevant to background, colors, and angle of observance. **Letter style** Letter styles are classified as sanserif and serif. Sanserif letters, such as Helvetica, are more contemporary than serif letters, such as Clarendon (Fig. 1). Each letter style has its own unique personality and flavor. Printers carry alphabets in most letter styles, including lowercase letters as well as uppercase (Fig. 2). Test results indicate that messages starting with an initial uppercase letter and followed by lowercase characters are more recognizable than messages formed with uppercase characters only. Lowercase letters have more "personality" because their shape is varied by ascenders and descenders, resulting in characteristic word forms that are much easier to recognize than all-uppercase word forms. Also, people are more accustomed to reading text in upper- and lowercase than in all uppercase. The proper selection of a particular alphabet should be carefully considered, not only from a legibility point of view, but also from a "personality" standpoint. The letter style should make a concise and meaningful impression in the environment it serves.

Readability Readability is directly related to the size of copy, **Visibility** studies indicate that 1-inch-high Helvetica Medium, for example, is readable from a distance of 40 feet. Using this as a measure for comparison, 1-inch-high Clarendon style would be readable from a somewhat lesser distance, approximately 25 feet. The distance visibility per 1-inch height may be used as a guideline to determine distance readability for larger letters; that is, 2-inch-high Helvetica Medium will be readable at 80 feet, and 3-inch-high at 120 feet. This direct proportion may be helpful for determining copy (text) sizes for signs used in pedestrian situations. However, the direct proportion may not hold true for vehicular traffic applications where many other factors are involved. The designer must

exercise HELVETICA MEDIUM CLARENDON Fig. 2 Specialties

SIGNAGE AND GRAPHICS Signage System Design Criteria caution after selecting the alphabet and copy size to make certain the lettering will fit properly on the sign background. The sign size should be determined using the longest line of copy and the maximum number of copy lines that may be required. Letters and line spacing Interletter spacing and interline spacing of copy greatly affect the overall readability of a sign. Message legibility and ease of recognition are increased when proper visual relationships are established between individual characters, words, and lines of copy. Copy with spacing too tight becomes very difficult to read; copy with too open spacing tends to break the message down into fragments (Fig. 3), Proper spacing depends largely on the distance from which the message is to be read. Messages to be read at close distances should employ tighter spacing than messages that will be read at greater distances, Spacing is also affected by the angle at which the message is to be viewed: Greater angles of observance require wider interletter spacing to prevent the characters of the message from appearing to run together. Copy position The position of copy on the sign background influences the overall readability. Signs on which copy occupies most of the background are not as readable as signs that have sufficient background material surrounding the copy to form a visual barrier separating the message from the environment (Fig. 4 and 5). Emphasis should be placed on selecting an appropriate sign size to best accommodate the sign message. There are nine basic copy placement positions to be considered in determining the important relationship of copy to sign background. They are; upper left, upper centered, upper right, centered left, centered, centered right, lower left, lower centered, and lower right. Traditionally, the most popular placement selections have been the centered and upper left positions. Color Color of copy and sign background greatly affect readability. Strong contrasting colors are more readable than less dramatic color combinations. White copy on a black background offers the greatest contrast and readability. Color also influences the apparent relationship between the copy size and the background. For example, white copy on a black field appears larger than black copy on a white field, although letter height, size, and copy position remain the same in both examples (Fig, 6), Colors in a signage system should also relate harmoniously with the pallet of colors selected for the building and its environment. The designer may choose to select colors that blend with the

environment or vibrant primary colors that accent the sign system and perhaps contrast with the architectural color scheme. Architectural Signage Systems Architectural Signage Systems Architectural Signage Systems Fig. 3

Specialties SIGNAGE AND GRAPHICS Signage System Design Criteria Architectural Signage Systems Architectural Signage Systems Fig. 4 Flg.fi

The viewing angle The angle of observance is influential in the design of a signage system, since it affects interletter spacing and overall readability. Normally, interiorsigns are viewed chiefly from a straight-on position; however, exterior signs are frequently seen from more than one angle, Signs to be read from vehicles moving at varying speeds with different angles of observance may require a compromise in letter spacing to best communicate the message. Multilingual Needs The jet age is a contributing factor in bringing people together from all over the world to visit and transact business. Transportation terminals and public facilities that may be used by visitors unaccustomed to reading English should employ sign systems that bridge any visual communication gap. Multilingual messages in English and the dominant foreign languages used by visitors may be combined and presented on one sign background. However, sign design and graphic formats become very critical to prevent confusion. A more popular solution involves the use of pictorial symbols as word substitutes. Pictographic signs are bold, recognizable images not bound by language barriers. Regulatory Considerations The designer should become aware of regulations governing signs. Federal regulations concerning safety signs are enumerated in Occupational Safety and Health Administration (OSHA) publications. American National Standards Institute (ANSI) publishes standards concerning signage for the physically handicapped. Underwriters' Laboratory (UL) issues standards applicable to illuminated signs. State and local codes contain regulatory information concerning sign sizes, mounting locations and heights, quantities of Fig. 5 signs allowable in various zoning areas, and other restrictions relating to exterior signs. These rules, and those of other regulatory bodies, should be taken under advisement prior to completing a comprehensive signage program. Need for Illumination Many signs are required to relate their messages after dark as well as during natural daylight. The careful designer will determine which signs require artificial illumination and decide on the method of illumination. Signs can

be externally illuminated by readily available stock fixtures produced by many manufacturers, or they can be internally illuminated. Fluorescent lighting is the most common source of internal illumination, although metal arc lamps, incandescent lamps, and neon are frequently employed.

Need for a Graphics Manual Many signage programs are developed for institutions that have a continuing need not only to maintain, but also to augment or change, their signage systems. The preparation of a signage manual containing all the information required to create additional signs or components would benefit the client and ensure continuity in the system as changes and additions are made. The designer should determine this potential need and include the manual with other documents developed for the signage program.

SIGN TYPES CATEGORIZED BY FUNCTION

Signage systems should be logically broken down into various types of signs to be utilized on a particular project. Many categories of sign types may be developed, but one of the most conclusive listings is based on function. The following discussion of signage system components, including sign requirements for specific applications, covers these functions.

Exterior Signs Exterior sign system components are normally viewed from vehicles or by pedestrians who have parked their vehicles and are walking toward their destination.

Primary Identification All architectural projects require some form of identification that is both easily readable and recognizable. A person's first association with a building is the identifying device selected to "label" the structure. The importance of the first impression created by this device should be recognized. A sign that produces an image in keeping with the environment it serves reflects the quality of the people associated with that environment. Major corporations spend large sums of money on corporate identity programs to ensure the visual image presented to the public best reflects corporate philosophy and product desirability. Equal emphasis should be placed upon the image presented by the device employed to identify an architectural structure.

Secondary Identification Many complexes containing more than one basic structure require secondary identification signs to properly identify the various elements within the complex. A systems approach to design will provide continuity in the relationship of primary to secondary identification signs.

Vehicular advance notice A system of road signs suitably located in advance of decisionmaking points will allow vehicular traffic to execute the proper decisions smoothly and safely at the appropriate times. Vehicular

directional Intersections and parking facility entrances are major decisionmaking locations requiring directional devices to guide drivers toward their destination. Traffic regulatory and control Vehicular traffic can be systematically controlled by employing signing devices. Traffic codes are usually clear as to what signs are required, where they are to be located, and the height at which they are to be mounted, Usually, colors, sizes, and shapes are standardized by the traffic authorities. Stop, yield, and speed limit signs are representative of this classification of signs. Instructional Frequently, signs are required to instruct vehicular and pedestrian traffic. These notices must be properly installed in carefully selected locations to be effective. Examples include parking procedures, delivery and service directions, and the like. Informational Signs are required to present information that is both relevant to the location and important to the viewer. This information may pertain to parking rates, hours of operation, and security, or it may relate to items of interest within the environment. Decorative Decorative graphics may be employed to enhance the beauty or decor of a particular area; form, color, and design may be utilized to create interest and to become features of the exterior landscape. 932

interior Signs Interior sign system components should assist visitors to travel from the building entrances throughout the complex until they reach their desired destination. Identification Multiple-occupancy buildings require tenant identification; frequently, buildings with only one tenant will also utilize identification in the main lobby or reception areas to reinforce the corporate signature. Criteria for multiple-tenant signage are very important and should be included in lease documents to provide for visual continuity and architectural harmony. When individuals are allowed to implement their own desires concerning signage, each will attempt to outdo the other, resulting in clutter, confusion, and visual pollution. Signs that are too big, too gaudy, too competitive, and poorly conceived and executed will become commonplace unless controls on tenant identification are established and enforced. Primary directory Information relevant to one's location within a complex should be clearly enumerated on the primary directory, usually located in a very visible area of the main lobby. Alphabetized listings of tenants, departments, and individuals should be concise and should designate the floor and room numbers. Such directories may be flush or recessed wall mounts, horizontal projected wall mounts, or

pedestal or kiosk mounts, and internally illuminated or not, depending upon the ambient lighting conditions. Elevator lobby floor directory High-rise structures require well-positioned signage that not only identifies each individual floor, but also serves as a secondary directory system for that floor. Frequently, the floor identification, directory, and corridor directional signage may be included in one device. When a visitor exits from an elevator on a chosen floor, a sign showing the floor number and also the direction of the office or room number sought is both helpful and reassuring. Pictorial "you are here" indicators Pictorial schematic maps may become an integral part of directory systems, or they may be utilized separately as visual aids in depicting one's intended passage through a complex. Hospitals, sports complexes, and transportation centers, are good examples of structure that may require pictorial maps to supplement word messages. Caution will be exercised by the expert designer to keep the pictorial map simple and correctly oriented in the building according to where the viewer is standing, and to evaluate the need of color coding as part of the visual aid. Too frequently, designers employ a complicated color-coded system that becomes very confusing to the viewer and, in fact, compromises the effectiveness of the system. Primary directional The maze that often results from interior corridor layouts creates many decision-making points for a visitor. Primary directional signs may be ceiling-mounted, wall-mounted, or floor-mounted as kiosk-type units in open areas. Areas with heavy pedestrian traffic should have directional signs located so that people do not obstruct the line of sight to the sign device. Normally, ceiling-suspended or kiosk-type units are the best choice to enhance visibility. Secondary directional Directional signs should be considered in locations where traffic flow and corridor layouts do not demand primary directional devices but do require some guidance for direction control. Corridors within suites of offices and corridors that change direction should be considered as decision-making points that may require a secondary directional signage device. Area identification Specific areas within a complex should be properly identified. These areas may be tenant spaces, divisions, or departments. When occurring along main corridors, they are usually designated by wall-, door-, or transom-mounted devices. Ceiling-suspended signs are a good solution in open office spaces. Room identification Wall- or door-mounted room identification signs are required to "label" the function of a particular room. Work functions are properly

identified within tenant areas, while service and maintenance functions should be suitably designated in most situations. Desk Identification Reception areas may require a sign device located on a desk or counter to identify a particular service or individual rendering assistance to visitors. Such signs may be permanently affixed or removable, and may provide for changeable name inserts. Personnel identification Persons rendering a service to the public, such as nurses, maintenance personnel, and food service personnel, generally are identified by name badges or pins. Regulatory and control signs Signs that authorize or prohibit certain functions are required, frequently by law or code, to inform people using the facility. Examples include signs for the handicapped and signs relevant to no smoking areas, elevator capacities, "no entry" areas, fire control, and "authorized personnel only" areas. These signs are usually mounted on doors or their adjacent walls; they may employ colors which deviate from the standard colors used in the comprehensive signage system to emphasize a dangerous situation or the need for caution. Exits Exit signs are required by codes to designate exits effectively in times of emergency. Supplemental devices are used to give additional information pertaining to a particular exit such as "Emergency Exit Only" and "Alarm Sounds When Door Is Opened." OSHA-approved exit signs are standard items manufactured by many lighting companies, and are generally provided by the electrical contractor. Information exhibit cases Notices, posters, attractions, and promotional pieces should be contained within an appropriately designed case to control the display of this type of information. Standard units featuring vinyl-covered cork panels housed within extruded aluminum frames with lockable doors are available from many directory manufacturers. Specialties SIGNAGE AND GRAPHICS Signage System Design Criteria Decorative features Decorative designs may be reproduced on walls as interior features. Reproduction processes include appliques, painting, and screen printing on location; or mural processes, which are applied much like wall-coverings, may be considered. Doors may also receive supergraphic treatments in which copy may become an integral part of the design. Dedicatory plaques Building dedication plaques should be carefully conceived and implemented, using materials that reflect favorably upon the talents involved in the realization of the project. Historically, these plaques have been bronze or aluminum castings. However, modern technology has provided photographic methods and photochemical

processes which offer the designer a freedom of size, format, letterform, and color not available in the casting operation. Donor recognition Buildings constructed in part by contributions from donors require special recognition for the donors. Hospitals, performing arts centers, and service institutions rely on gifts to assist in financing buildings, additions, and furnishings, and usually stipulate that donors will be remembered and recognized in some prestigious location in the building. The designer is responsible for establishing controls and developing a system that fulfills promises made by those soliciting funds, while allowing flexibility to expand the system as future needs may dictate. Location selection is very important in the overall effectiveness of the donor recognition signage. Mechanical, instrumentation, and control system markings Many industrial and mechanical installations require equipment, control, and pipe markings to meet codes, assist maintenance and service personnel, and ensure safety. Often, these locations are not public spaces, and require an industrial, rather than an architectural, approach to signage. Elevator floor-indicator panels, however, should receive special attention and be considered in a comprehensive signage program. CONCEPTUAL DESIGN OF THE SIGN FACE Emphasis will not be placed on the graphic design of each sign required in a comprehensive signage program. However, the following considerations will help to ensure continuity, correctness, and aesthetic acceptability. Alphabet selection An alphabet must be carefully chosen that best exemplifies the graphic image to be portrayed to the public without compromising legibility and performance requirements. More than one alphabet may be selected should need dictate. However, good design practices should be maintained in choosing the family of alphabets to be employed. Interletter, word, and line spacing Each alphabet has its own "personality" and visual impact; therefore, spacing between characters, words, and lines of copy must be carefully developed to give the best legibility and visual harmony possible (Fig. 8).

Specialties SIGNAGE AND GRAPHICS Signage System Design Criteria
 WaN Mount Ceiling Mount Fig. 7 Architectural Signage Systems Planning
 • Design • Implementation Fig. 9 Fig. 10 Fig. 11 Arrow selection
 Directional arrows should be designed to reflect the "personality" of the letterform selected. Stroke width and size relationships are important considerations (Fig. 9). Copy determination The message for each sign must

be accurately determined and the copy condensed to the fewest words that will still relay the desired message. Wordy signs are frequently misread or not read at all. The message must be concise, clear, and Informative (Fig. 10 and 11).

Copy placement format The placement of copy on a sign face may take one of the nine basic positions or a custom format for special situations (Fig. 12).

Size determination of the sign face After the copy for each sign is in final form, the sign with the greatest amount of copy is selected from each of the sign types utilized and the desired copy height is determined for each type. This height should be based upon the distance from which the sign will be read and the graphic design portrayed. Using this letter height, the message should be laid out with photographic type or transfer lettering to scale, incorporating the copy placement and spacing requirements. The most pleasing shape and size for the message to be contained are then determined, realizing that this particular layout is for the maximum copy required for that particular sign type. A shape and size format should be chosen that works well as a module which can be proportioned and become applicable to the entire family of sign types. While this may be ideal, frequently the proportional system is not applicable. An example of each sign type should be drawn to scale and fully dimensioned to serve as a production guide for signs within that type. (Fig. 13).

Color selections Selection is then made of the copy and background colors that offer good contrast and harmoniously blend with the prominent colors in the environment. It is also wise to consider any corporate colors required by the client.

SIGNAGE SYSTEM DEVELOPMENT CHECKLIST The completed sign schedule, location plans, scaled drawings of typical examples from each sign type, construction or assembly details or both, mounting details, and specifications form the documents required to bid competitively or to negotiate signage projects. Well-prepared documents prevent individual interpretation by vendors and result in comparable competitive bids. The following systematic approach to the design and development of a comprehensive signage program will serve as a guideline to problem solving, employing the concepts contained in this chapter. This checklist may be expanded or condensed to meet individual project parameters. The basic systematic thought process, however, is applicable to all projects.

1. Develop the signage system design criteria based on:
 - a. Performance requirements
 - b. Usage considerations
 - c. Durability requirements
 - d. Vandalism considerations
 - e. Flexibility to accommodate

changes and additions f. Readability factors g. Multilingual needs h. Regulatory considerations i. Need for illumination j. Need for graphics manual for ongoing implementation and system maintenance 2. Study the traffic flow patterns, determine all sign locations, and draw the location symbols on the site and floor plans. 3. Evaluate and select the sign types required from the following list, categorized by function, that meet the design criteria: a. Exterior sign types: Type A- Primary identification Type B-Secondary identification Type C- Vehicular advance notice Type D- Vehicular directional Type E-Traffic regulatory and control Type F- Instructional Type G-Informational Type H- Decorative b. Interior sign types: Type I - Primary identification Type J - Primary directory Type K- Elevator lobby floor directories Type L-Pictorial "You Are Here" indicators Type M - Primary directional Type N-Secondary directional 934

Specialties SIGNAGE AND GRAPHICS Signage System Design Criteria Type O-Area identification Type P- Room identification Type Q- Desk identification Type R - Personnel identification Type S - Regulatory and control Type T - Exit Type U - Information exhibit cases Type W- Dedicatory Type X- Donor recognition Type Y-Mechanical, instrumentation, and control system markings Type Z~Other (to be specified by designer) 4. Select the best signing devices for each sign type designated above from the following lexicon of signage system components that most effectively satisfy the design criteria established: a. Elevated pylons b. Monolithic sign structures c. Panel and post assemblies d. Illuminated sign cabinets e. Directory and informational systems f. Die-cut pressure-sensitive lettering g. Dimensional graphics h. Plaque signage i. Environmental graphics). Other (to be defined by the designer) 5. Conceptually design the sign face for each sign type selected, indicating: a. Alphabet selection b. Interletter, word, and line spacing c. Arrow selection d. Copy determination e. Copy placement format f. Size determination of copy and sign face g. Color selections 6. Complete the location plans by filling in the symbol indicating sign number and type. 7. Prepare scaled drawings of typical examples from each sign type. 8. Prepare the detailed sign schedule. 9. Prepare typical construction and assembly details, mounting details, and engineering drawings for wind loading, foundations, and illumination. 10. Prepare detailed specifications for all materials, techniques, and components required in the system. Riverside Your Family

Hospital ?? Riverside Your Family Hospital @ Emergency Health Center
(\$) Phycinns Visitors Outpatients Receiving M B Fig. 13 FIfl. 12 935

Specialties SIGNAGE AND GRAPHICS Standard Sign Type and Mounting
 Heights STANDARD SIGN SYSTEM tlllllllllllll 12" X 48" 12" X 72"
 18" 36" max jllpy; ||l 9"x9" 12" x 12" Copy 9"x9" 6"x6" Overhead
 Directional Area Room CODE SIGNS fire evacuation map i stair markings
 5" m 111-1/2" 1-1/2" on glass doors and sidelites Elevator Stair Distraction
 strps STANDARD MOUNTING HEIGHTS AND LOCATIONS Copy
 Overhead Dtactbna Door 936

Specialties SIGNAGE AND GRAPHICS Sign TVpe DIRECTIONAL (CEILING-HUNG) Material Choices MDO board, acrylic Finishes Painted, plastic laminate, metal laminate Graphics Vinyl die cuts, silkscreen, dimensional applied letters Standard Mounting Detaii 1. Threaded rod: pendant, flush 2. Scissor clip Sqp_x_ 10"x24" 12"x48" 12"x72" Standard sizes rnnvriinimn iiiiiiiiiiiiiiiimiiiiijiiiiimim scissor clip to tee bar threaded rod to structure above Section Sign Type DIRECTIONAL (WALL-MOUNTED) Material Choices Acrylic, aluminum, acrylic with metal laminate face Finishes Painted acrylic or aluminum, natural aluminum or brass (satin or polished), laminates available in standard laminate finishes Graphics Silkscreen, front surface or reverse Standard Mounting Materials 1. Backpanel: backplate with countersunk screws with shields; magnetic, form, or vinyl tape with adhesive 2. Strips: vinyl tape ' should not exceed 36" high 18" 2" TENANT 18" DIRECTORY Info strip directional strip Standard sizes header strip 1/4" acrylic backplate information strip Section Sign Type AREA DESIGNATION (WALL-MOUNTED) Material Choices Acrylic, aluminum, acrylic with metal laminate face Finishes Painted acrylic or aluminum, natural aluminum or brass (satin or polished), laminates available in standard laminate finishes Graphics Silkscreen, front surface or reverse; vinyl die cuts Standard Mounting Materials Vinyl or magnetic tape, foam tape, silastic adhesive Standard sizes acrylic sign adhesive & tape mount Section 937

Specialties SIGNAGE AND GRAPHICS Sign Type ROOM IDENTIFIER
(WALL-MOUNTED) Material Choices Acrylic, aluminum, acrylic with
metal laminate face Finishes Painted acrylic or aluminum, natural

aluminum or brass (satin or polished), laminates available in standard laminate finishes Graphics Silkscreen, front surface or reverse Standard Mounting Materials Vinyl tape, foam tape, magnetic tape, silastic adhesive 9" Copy 2" 12" Copy Standard sizes acrylic sign adhesive & tape mount Section Sign Type ROOM IDENTIFIER, CHANGEABLE MESSAGE (WALL-MOUNTED) Material Choices Holder, acrylic; insert, vinyl Finishes Painted (surface or subsurface) Graphics Silkscreen or vinyl die cuts Standard Mounting Materials Vinyl tape, foam tape, silastic adhesive 6" Copy Name 3 1" 6" 9" 6" Copy 3 1" 6" 6" L.0 BL 3" 1" Standard sign sizes Standard insert sizes 1/4" 6" 1 i Elevation and section Sign Type FRAMED PLAQUE SIGNS, WALL- MOUNTED (previous plaque types are insertable into standard frame signs) Material Choices Molded acrylic, aluminum, brass Finishes Painted, satin, polished Standard Frame Mounting Materials 1. Frame: screw mount, tape and adhesive 2. Insert: adhesive or magnetic tape, Velcro, magnet 1-5/8" Copy 9" Standard insert sizes insert r.\7.e Elevation and section 938

Specialties SIGNAGE AND GRAPHICS Sign Type COUNTERTOP/FLAG MOUNT Extrusion Material Aluminum Insert Material Acrylic, aluminum, acrylic with metal laminate Graphics See area and room plaques Standard Mounting Details 1, Counter: free-standing with extruded aluminum base 2. Flag mount: countersunk screws and shields counter top Standard sizes wall r"u-nt Elevation and section Sign Type DESKBAR (DESKTOP) Material Choices Aluminum, molded acrylic Finishes Painted, satin, polished Graphics Vinyl die cuts; silkscreen on acrylic plaque, front surface or reverse Standard Mounting Detail Free-standing on desks or countertops \ 2" x 9" 1-5/8" x 9" Standard sizes Note: Changeable face available by using acrylic sign plaque. Sign Type CUT LETTERS: FLUSH, PROJECTED Material Choices Acrylic, acrylic with metal laminate face, brass, aluminum Finishes Polished, painted, brushed, sand blasted Standard Mounting Details Adhesive mount, flush pin mount, standoff mount Standard Letter Sizes Varies from 2" to 18" a Front View Projected Mount Flush Mount (least vandal resistant) Sections 939

Specialties SIGNAGE AND GRAPHICS Medical Symbols MpHiral Nursing Homes Picto'grafics not shown: ivioui^ch Medical Complexes 1,516 Parking 1.413 Health First-Aid Centers 1.372 Playroom 2.531

Warning 1.150 Library or Reading 1.147 Chest Hospital Pharmacy Dental
 Care Wheelchair X-Ray Physiotherapy General Medicine, Female General
 Medicine, Male Coronary Care Hematology Urology Eye Podiatry Mental
 Health Ear, Nose & Throat Oxygen Shower Isolation Nursery Laboratory
 fflHQ 1.508 1 518 1 184 mHH 1.148 1.148 □ Q] 1.448 1 417 1 448 1 123 □
 □ 1 129 1 473 1,137 1 368 1 376 1 411 1 302 i 359 Conference
 Occupational Therapy Rehabilitation Ambulatory Patients I 406 940

Commercial Cocktail Lounge Pub Coffee Shop Liquor Store Mens'
 Furnishings Furniture Cinema Camera Store Gift Shop Florist Dress Shop
 Shoe Store Restaurant Soda Fountain Grocery Store Tobacco Shop
 Bookstore Record Shop Fuel Toy Shop Theater Van Beauty Salon Barber
 Shop Shopping Centers Plcto'graf ics not shown: Stores & Shops 1.218
 Concrete Mixer 1.226 Flatbed Truck Eating Facilities 1.219 Cushman
 Vehicle 1.250 Pickup Truck Community Services 1.222 Dump Truck 1.304
 Basket 1.344 1,362 1.339 1 354 1.361 1,315 1 341 Specialties SIGNAGE
 AND GRAPHICS Commercial Symbols 1.314 Vegetable Produce 1.363
 Newspaper Vendor 1.370 Record Store 1.394 Cooking 1.338 □ BESS 1305
 1455 1336 1.372 B I 290 1 192 941

Specialties SIGNAGE AND GRAPHICS Commercial and Travel Symbols
 « t III ua P o ® 51 0 942

Specialties SIGNAGE AND GRAPHICS Travel Symbols Travel Airport
 Departures Arrivals Car Rentals Bus Subway Train Taxi Monorail Ferry
 Cable Car Automobile Lost & Found Porter Locker Fuel Baggage Claim
 Customs Immigration Money Exchange Motorcycle Moving Sidewalk
 Lodging Ice Cubes Plcto'graf Ics not shown: 1.350 Motel 1.266 Seaplane
 Base 1.253 1.255 1.254 1.202 1.208 1.268 1.278 1.225 1.215 1.201 1.310
 1.319 1.308 1.336 n P3 1.125 1.211 1.145 1.173 1.353

Specialties SIGNAGE AND GRAPHICS Ttavet Symbols W © O gn <P X
 IP f ♦ If •© l^r © «e 8 944

Specialties SIGNAGE AND GRAPHICS Recreation and Sports Symbols
 Recreation and Sports Campers Picnic Area Midway Trailer Train Water
 Swimming Canoeing Sailing Sports Arenas Picto'graf ics not shown: Parks
 1.112 Curling 1.138 Hockey Recreation Facilities 1.115 Dancing 1.183

Tobogganing Amusement Parks 1.140 LaCrosse 1.276 1.471. Wintersports
1.387 Outdoor Recreation Marina, Boating Life Preserver Snowmobiling
Camping Judging Bicycling Women's/Girl's Toilet Fishing I 370 Skiing
Soccer ice Skating Football Hunting, Shooting Golf Baseball Tennis,
Badminton 1.182 945

Specialties SIGNAGE AND GRAPHICS Universal Symbols Universal
Entry Exit Ramp up Ramp down Applicable to any building or facility
Pictographics not shown: 1.110 Children 1.144 Man with boy 1.340 Fragile
1.372 1.410 1.472 1.469 Playroom Church Synagogue Police 1.493 1.516
1.488 1.155 Smoke Parking Keep Dry Janitor 1.404 Emergency Women's
Toilet Men's Toilet Stairs Handicapped No Smoking Telephone Escalator
1.188 3.316 Elevator Down No Parking Drinking fountain 1402B 3.516
Mail Box Check Room Up No Entry 1 326 1402A 946 Shower Waiting
Room Telegraph Office Information International symbols of SMwessihility

Specialties SIGNAGE AND GRAPHICS General Type Styles Alternate
GOTHIC NO. 3 Americana Americano IIAUC Aster Avant Garde GOTHIC
MEOIUM Baker Dan mark 2 Baker Sans MONO REGULAR Baskerville
BOLD HAIC Bodoni Bookman Caledonia BOLD ITALIC Caslon BOLD
Century SCH00LBD0K Century SCH00LB00K B0L0 Cheltenham medium
Columbus COPPERPLATE GOTHIC LIGHT Craw Clarendon BOOK.
Craw Clarendon Craw Modern Delta MEOIUM Eastern Souvenir
MEDIUM Eurostile Eurostile BOLD Eurostile EXIENOE Eurostile
BOIO EXIENOE Firmin Dklol Folio MEOIUM Folio MEOIUM
EXIENOE Fortuna LIGHT Franklin Gothic MEOIUM Futura MEOIUM
Futura DEMIBOLO Garamond BOLD Gerstner Program Gill Sans Horry
f« fx6xX6XXXO woe Helvetica LIGHT Helvetica Helvetica MEOIUM
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Specialties AUDIO-VISUAL SYSTEMS Planning Guideline Summary
 DESIGNING THE SYSTEM The formulation of a communications
 program is based on the functional requirements delineated in the feasibility
 study. The presentation modes to be utilized are a part of such a program.
 They might include slides, films, videotape, and a sound-recording and
 playback system. The detailed design of the facility includes the selection of
 basic equipment, possible modification of that equipment, and provision for
 additional optical elements, as well as the engineering of the electrical
 control circuitry and the design of the electromechanical devices that may
 be needed. The implementation of a proposed A-V system is not merely an
 exercise in mechanical assembly. It is a highly complex process of logistics
 that involves providing specific functional requirements within architectural
 and economic constraints. Careful engineering and balancing of the
 alternatives available will generally achieve optimum results. A large
 number of variables is encountered in every A-V design problem. As an
 example, the dimensions of the presentation room have a significant effect
 on the audience size, the acoustic characteristics, the size of the projected

image, the choice of equipment, and the location and the interrelationship of the components. The A-V consultant who is responsible for the program planning, the design, and the engineering of this complex, multifaceted discipline should be intimately familiar with the problems of fabrication, installation, and operation of such systems. This knowledge will enable the consultant to plan a facility whose execution will not create difficulties and whose construction and operation can be effected without costly changes. However, even when the consultant has experience as an adviser to members of the architectural and engineering professions, the creation of a well-integrated facility is not necessarily assured. His or her work and the completed facilities should be viewed and evaluated.

Optical Aspects It is of critical importance for an A-V system to have the ability to display bright, sharp images to all viewers and to maintain the stability and consistency of those images in a simple and straightforward manner. The picture quality is a function of a number of factors requiring careful attention during all phases of the project. These include: ■ The quality of the original photography or artwork ■ The density contrast, and sharpness of the actual material being projected ■ The output intensity of the projector light source ■ The optical characteristics of each projection unit ■ The optical characteristics of the integrated system ■ The ratio of the projection distance to the image size ■ The centering integrity of the light path from the material being projected to its image on the screen ■ The characteristics of the projection screen or other viewing surface

Projection Engineering

Room size Ideally, the dimensions of the viewing room should be an outgrowth of the estimate of the audience size that was established in the original A-V study. In many cases, however, the A-V design engineer must utilize a predetermined space. Given the characteristics of that space, the designer can determine the ideal audience size for each type of seating arrangement, and also ascertain whether a front or rear projection mode is feasible and what the image size should be. The type of relationship that is desired between the person making the presentation and the audience will determine the seating configuration: theatre, lecture, or conference format. That configuration will in turn dictate the number of viewers that can be comfortably seated for optimum viewing (Fig. 1). As an illustration, a room 20 feet by 32 feet can accommodate about 49 people in a theatre configuration (Fig. 2); in a lecture arrangement, the audience size would be 24 (Fig. 3); a U-shaped table would seat 18 (Fig. 4); and 15 people could fit

comfortably at a conference table (Fig. 5). Circular and multi-use arrangements (Figs. 6 and 7) are additional examples of the relationship of seating configuration and audience size. Other seating configurations have been devised for other types of communication program modes, each with a direct relationship between room size and audience size. The audience size is also affected by the angle of view between each member of the audience and the screen (Fig. 1). Whenever the A-V design engineer has the opportunity of establishing the dimensions of the presentation room, he or she should be aware of the important fact that a longer projection throw for a particular image size results in more even light distribution and sharpness as well as a better angle of view. Consequently a larger audience can be accommodated than would be possible using a system with a short projection distance and a narrower angle of view. This question of projection distance applies to both front and rear projection systems. However, as the throw is normally quite short when a rear projection screen is used, this factor of design in rear projection facilities is an extremely critical planning element. Distortion, sometimes called "keystoning," will result if the viewing surface is not precisely parallel to the plane of the image being projected. Therefore, the light path, which is usually perpendicular to the projected material, must be carefully controlled in relation to the projector and the screen. The size of the audience and the room, as well as the mode of projection, will determine whether the screen will be vertical or at an angle (Fig. 8). Normally, a rear projection screen will permit a vertical viewing surface.

7.5 1 TOTAL DIST. RP ■ REAR PROJECTION FP ■ FRONT PROJECTION P 0 = PROJECTION DISTANCE VD. ■ VIEWING DISTANCE 3.1 IO-PD. = 2:1 RATIO 15'-PD -3:1 RATIO BEST VIEWING AREA 375 MAX VD Fig. 1 The interrelationship of projection distance, image size, and viewing area. 949

Specialties AUDIO-VISUAL SYSTEMS Typical Projection Room Layout and Sightlines 5'-0" REAR PROJECTION SCREEN V 3'-0" uccooou ccouoot COOQCCO Fig. 2 A room 20 feet by 32 feet, seating 49 people In theatre style. 5'-0" REAR PROJECTION SCREEN uuuu Fig. 4 A room 20 feet by 32 feet, seating 18 people at a U-shaped table. REAR PROJECTION SCREEN / uuuuuu uuuuuu \ uuuuuu 20'-0" Fig. 3 A room 20 feet by 32 feet, seating 24 people In lecture style. I S'-O" _J ^ .REAR

PROJECTION SCREEN 950 Fig. 5 A room 20 feet by 32 feet, seating 15 people at a boat-shaped conference table.

Specialties AUDIO-VISUAL SYSTEMS Typical Projection Room Layout and Slight Rear PROJECTION SCREEN 950 Fig. 6 A room 20 feet by 24 feet, seating 13 people at a circular table. With the depth increased to 32 feet, from 7 to 14 observers can also be accommodated. Fig. 7 A modified classroom arrangement including both lecture and separate tables. Fig. 8 Cross section of a theatre-style auditorium with stepped seating showing both front and rear projection. Note that the projected light beam is perpendicular to the screen in both cases. 951

Specialties AUDIO-VISUAL SYSTEMS Planning Guideline Summary
Screen image area The most useful screen is one that is square, as it will permit both vertical and horizontal images, as well as square ones, of course (Fig. 9). A single image format will need one such screen, while a dual format will have a viewing surface that is the width of two images placed side by side (Figs. 9 and 10). It is an easy matter to determine the minimum image size necessary for a room of a given size. For a front projection screen the minimum size is the distance between it and the farthest viewer divided by 6. For a rear projection screen, the division factor is 7.5. As illustration: When the distance between the front projection screen and the last row of viewers, is 45 feet, the minimum image size would be 7.5 feet; with a rear projection screen, the minimum image should be 6 feet. These calculations assume that the original artwork from which the projection materials are made meets the generally accepted basic minimum standards.
Front projection The projector in a front projection system transmits the image in the form of a light beam to an opaque screen where it is reflected back to the viewers, creating the image. As the screen reflects any light falling on its surface, the general light level in the room during a presentation must be extremely low. If the full color and contrast of the projected image is to be retained, the ambient light should be no greater than 0.3 percent of the average screen brightness. Projectors are generally noisy and should be separated from the audience to avoid distractions. If the

space is available, a separate projection booth can be built behind the room's rear wall. Besides insulating the viewers from unwanted sound and light spill, this arrangement provides the opportunity for equipment to remain in place ready for use. There are other possible arrangements when space is constricted (Figs. 11, 12, and 13). Creating an A-V front projection system that is both aesthetically pleasing and functionally efficient requires a high level of technical expertise and design skill. The results of such a combination can be effective yet unobtrusive. Fig. 16 provides an example of a multimedia front projection system that is compatible with the decor of the room and its formalized seating arrangement.

Rear projection The image in a rear projection system is focused on the back of a translucent screen and is visible to the audience on the other side. Since the light passes through the screen rather than being reflected off its front surface, there can be a reasonable light level in the viewing room during the presentation without affecting the quality of the image. It is only in the immediate vicinity of the screen that the room lights need be dimmed. As is the case with all projection systems, for minimum distractions the equipment should be separated from the audience. This can be effected by means of a separate projection booth or by an enclosed cabinet within the viewing room. A separate room usually requires more space, but it may be the best solution for a particular situation. A cabinet within the viewing room permits front access to the projectors, enabling the

**IE MM MOVIE SLIDES
OVERHEADS SLIDES OR OVERHEADS HORIZONTALLY
ORIENTED IMAGE VERTICALLY ORIENTED IMAGE** Fig. 9 A square screen will permit horizontal, vertical, and square **SQUARE IMAGE** to be shown. **SLIDES OR OVERHEADS SLIDES OR OVERHEADS 10'-0" ' 16 MM MOVIE SLIDES OVERHEADS b "in HORIZONTALLY
ORIENTED DUAL IMAGES HORIZONTALLY ORIENTED SINGLE
IMAGE SLIDES OR OVERHEADS SLIDES OR OVERHEADS SLIDES
OR OVERHEADS VERTICALLY ORIENTED DUAL IMAGES
VERTICALLY SINGLE IMAGE SLIDES OR OVERHEADS SLIDES OR
OVERHEADS SLIDES OR OVERHEADS SQUARE OUAL IMAGES
SQUARE SINGLE IMAGE** Fig. 10 A dual-Image format uses a screen that is a double square - giving the same flexibility of image shape for each of the two Images or for a single central image. presenter to load the equipment without assistance. While technical expertise and design skill are needed for the creation of a front projection A-V system, they are even

more important for a system intended for rear projection, as a rear-mode arrangement has more inherent problems to overcome. A rear projection system utilizing the indirect deep method in a separate projection booth requires a considerable amount of space. In addition, if more than one projector is used in such a system, either the projectors must be optically aligned each time a change is made (as there is only one true screen axis), or they are permanently positioned a little off axis, resulting in a slight "keystone" or distortion effect in the projected image (Fig. 17). For good image clarity, the distance between the image source and the screen must be at least twice the picture size. To achieve this clarity within a limited amount of space, the folded light-path method can be used (Figs. 18 and 19). As the name implies, the light path from the projector is "folded" by means of a large mirror, usually placed some distance away. This arrangement has the advantage of reducing the depth required behind the screen while retaining an adequate projection distance. As a further advantage, several projectors can be aligned in optically true positions by the use of a movable mirror with preset position stops. The use of the folded light-path method of projection and a movable mirror can also be engineered in a cabinet that is directly accessible from the presentation room for "hands-on" operation by the person making the presentation. Both single-image and side-by-side dual-image systems can be designed in this manner (Figs. 20 and 21). A great number of variations are possible using the same basic engineering concepts. 952

Specialties AUDIO-VISUAL SYSTEMS Equipment Arrangement These variations can accommodate different functional requirements, spatial limitations, and image-quality parameters. Figures 14, 15, 22, and 23 illustrate some of the possible arrangements. User requirements and job conditions will guide the A-V engineer in the design of a specific system. **The Optical Design Factor** A projection system — of whatever nature — is only as good as the quality of the image on the screen. The clarity, sharpness, resolution, and angle of view that can be expected are a direct result of the thought and care that go into the optical design of the system. The more complex the system becomes, the more critical is the system optics. The need for larger images, sharper images, multiple images, multiple image sources and the existence of physically constraining parameters all add to the conflicting requirements that must be satisfied.

And they must be satisfied if an acceptable image quality is to be achieved.

The Sound System The quality and the functional characteristics of the sound system that is part of an audiovisual facility are as important as the quality and functional characteristics of the optical system. The two aspects of a facility are mutually complementary and the one should not be neglected in relation to the other if the goal of an effective and useful facility is to be attained. The quality of the sound, as perceived by the listener, will be influenced by such factors as:

- The sensitivity of controls
- The quality of the amplifiers
- The quality of the speakers
- The location of the speakers
- The elimination of extraneous sounds
- The overall acoustical characteristics of the space

The design factors that govern the functional characteristics of the sound system might include the following:

- Sound sources: voice, movie soundtrack, videotape, audiotape
- Telecommunication facilities for outside program sources
- Mixing and control requirements
- Quantity and placement of speakers
- Room size and function: conference room, classroom, auditorium
- Provision for flexibility and future expansion

The Remote-Control System Most people who make informational presentations are not audiovisual specialists. Their primary concern is with the material they are presenting and not with the mechanics of how it is to be presented. As a result, any control devices they may be required to operate should be simple and logical. The presenter should be asked to make only a minimum of effort to determine how to manipulate the controls in order to achieve a desired result. The fewer operations necessary to reach a particular goal, the better. For example, in order for a change to be made from one presentation mode to another, it may be necessary to alter the ambient room lighting, reposition a mirror, turn one machine

WALK SPACE MOVIE PROJ. 30" TO 36" EQUIPMENT SPACE -GLASS PORT VIEWING AREA Fig. 11 A typical rear-access equipment arrangement for front projection. [SLIOE] I PROJ. I Fig. 12 A rear-access reduced-depth arrangement Fig. 13 A front-access equipment cabinet for (rant of equipment for front projection. projection. 953

Specialties AUDIO-VISUAL SYSTEMS Equipment Arrangement J 4/ SLIDE /V, PROJECTOR I OVERHEAD PROJECTOR -MOVABLE PARTITION SLIDE PROJECTOR Fig. 14 An arrangement of two enclosed rear projection systems serving a single large room. MOVABLE PARTITION SLIDE PROJECTOR MOVIE PROJECTOR Wm I Fig. 15

The two enclosed and pivoted rear projection systems positioned to serve the two separate rooms that are created when a hidden dividing partition is extended. 954

Specialties off and then another on. If all these things can be accomplished merely by flipping one clearly marked switch, the presenter is freed from mechanical distractions and can concentrate full attention on the message being delivered. The location and spacing of the various switches on the panel, as well as the use of nomenclature unmistakable to a nontechnical person, are important parts of the design of a remote-control system that will aid the presenter in the use of the audiovisual facility. Other considerations that may affect the design of a remote-control system include: The seating configuration The room lighting The number of control points required The use of a lectern incorporating a control module The number and type of functions to be controlled The degree of automation required to meet system objectives

SUMMARY An audiovisual presentation facility is made up of many components and subsystems which are interdependent and must perform as an integrated unit. Regardless of its size or scope, the A-V system must be conceived, designed, and installed to function as a totality - as a single entity that works with optimum efficiency and effectiveness in an unobtrusive manner. In order to achieve this goal - that of developing a logical and workable solution to any particular communication problem - careful and detailed preliminary investigations must be made. These will determine the functional requirements that make up the design program. From this program, the space needs for the equipment and for the audience can be established early enough in the development of the project to avoid undesirable procrustean solutions later. The selection, adaptation, manufacture, assembly, and installation of equipment and components should be carefully coordinated to ensure their functional integrity and performance. Ultimately, a successful audiovisual system is one that serves as a logical and natural extension of the human capabilities of the person using it. It should respond easily and unobtrusively to the communicator's needs, and it should reproduce the material being communicated with the highest possible degree of fidelity.

RECAP Front Projection 1. Viewing distance factor is 6. (For example, if image size is 5 feet the alphanumerics would be clear at a maximum distance of 30 feet to a viewer with a 20/40 vision if characters are Ma inch

on 6- by 9-inch original copy area.) 2. Advantages a. Good angle of view b. Good for checking laboratory quality of all projectuals c. Virtually no apparent falloff to the sides 3. Disadvantages a. High ceilings are required to utilize a square screen to accommodate vertical as well as horizontal images. b. Distraction occurs when the presenter or viewers interrupt the light beam. c. Any ambient light adversely affects image quality. The room must be relatively dark to achieve the desired picture contrast. d. An overhead projector cannot be used most effectively. Rear Projection (Rigid or Flexible Material) 1. Viewing distance factor is 7.5, (For example, if image size is 5 feet the maximum viewing distance would be 37.5 feet.) 2. Advantages a. A 20 percent smaller image than is required by front projection permits minimum standards to be met in low- ceilinged rooms. b. Can be used in higher ambient light conditions. c. No distracting light beam. (Presenter can more comfortably point at details). d. In a brighter room, the presenter easily maintains eye contact. e. An overhead projector can be used, so that neither it nor the presenter blocks the image from the viewers.- 3. Disadvantages a. The inherent grain and directional quality of the rear screen eliminate it as a viewing medium to determine laboratory quality of projectuals. b. The projection system must be designed to overcome apparent illumination falloff at the sides and improve the angle of view. c. Mirrored image is required for proper use. d. More space is required than with front projection. e. Usually costs more. Seating (Plan should permit several arrangements.) 1. A U- or V-table layout provides for best AUDIO-VISUAL SYSTEMS Planning Guideline Summary viewing and viewer/presenter interaction (lowest audience capacity). 2. Conference table (boat-shape or oval) provides good interaction for conferences but not so good as the U- or V-table layout for audiovisual communication. 3. Random seating style (usually with writing tablets) is frequently selected for high- level visitor presentations. It permits larger capacity and creates a more luxurious atmosphere than the two arrangements above. 4. Classroom style (shallow tables parallel to front wall with chairs behind) is the next best method but less conducive to student interaction. a. Stepped, curved seating (lecture hall) provides unobstructed viewing. b. When classroom style is contemplated, study and programmed-learn- ing carrels should be considered. 5. Auditorium style provides the largest capacity seating and is generally used for large group-orientation and overview types of presentation. Rear Projection System Factors 1, The physical center of all

projector lenses must be in perfect alignment with the physical center of the screen to eliminate any "keystone" effect. (For dissolve mode, 2" off center vertically is permitted.) 2. A front-surface mirror should be used to reverse the image so the equipment can be loaded much as it is for front projection; slides in magazines need not be reversed, and special reversed prints are not needed for motion pictures. The use of a mirror can also extend the projection distance appreciably by folding the light path. Remember, the longer the projection distance, the better the viewing angle. Minimum projection distance should be at least 2 times the image size. 3. The screen-image area should be considered to be square to accommodate vertical and horizontal images unless the system is to be used for a special, limited requirement. 4. Apparent light falloff at the sides can be diminished or eliminated by increasing the projection distance and projector illumination. Another minor contributor is slide density. A dense or underexposed slide reduces the amount of light transmission. This condition increases apparent light falloff. 955

Specialties AUDIO-VISUAL SYSTEMS Equipment Arrangement
OVERHEAD ^ REAR PROJECTION BOOTH ^ KH c'_n" /y PROJECTOR
7, 6'-0" VIEWING AREA VIEWING AREA v?7Z>9/m//m;////////; Fig. 16 A
custom-designed recessed front-access equipment cabinet for a multi-image
front projection system. M//////////M | ^J-v\ SLIDE O
PORT \^J> PROJECTOR 7, MOVIE PROJECTOR Fig. 19 An indirect rear
projection arrangement using the folded-light-path method and the
minimum recommended 2 to 1 ratio of projection distance to image size.
This permits a flexible equipment arrangement within tight space
limitations. w//////////% Fig. 17 A deep, indirect-method,
rear projection arrangement using the minimum recommended ratio of 2 to
1 between projection distance and image size, MOVIE PROJECTOR
VIEWING AREA /, I' / REAR PROJECTION BOOTH 956
y//////////7/. Fig. 18 An Indirect rear projection
arrangement using the folded-light- path method, resulting in a ratio of 3.5
to 1 within the same depth. This improves the Image quality and increases
the possible viewing angle as well as allowing rear projection of overhead
transparencies with the overhead projector in the presentation room.

Specialties AUDIO-VISUAL SYSTEMS Equipment Arrangement
 '/////////////////////////■////ja £—OVERHEAD 2 PROJECTOR I Fig.
 20 A front-access rear projection arrangement using the folded-light-path
 method for single- image presentations. w/////////////////////////^ SLIDE
 PROJECTOR J0S WT^f 1. MOVIE projector! SLIDE PROJECTOR (T7\ ^
 Op V^OVERHEAD PROJECTOR Fig. 21 A front-access rear projection
 arrangement using the foided-iight- path method for dual-Image
 presentations. </////////////////////////^^^
 />;/;;/;;/;;/;;/;;/;;/ /OVERHEAD ' PROJECTOR VIEWING AREA Fig. 22 A
 rear-access rear projection arrangement using the folded-light-path method
 for dual-image or single central- image presentations.
 I/////////////////////////, 4, SLIDE V;
 PROJECTOR r DEEP METHOD SLIDE PROJECTOR FOLDED LIGHT
 PATH REAR PROJECTION BOOTH 1 VIEWING AREA Fig. 23 A rear
 projection arrangement for dual-image and single central-Image
 presentations utilizing both deep indirect projection and the folded-light-
 path method. 957

Specialties AUDIO-VISUAL SYSTEMS Typical Projection Room Layout
 5'0"x5'0" REAR PROJECTION SCREEN Front wall supplied with system.
 5'0"x10'0" REAR PROJECTION SCREEN Front wall supplied with
 system. OVERHEAD PROJECTOR W/ TRANSPARENCY CHANGER
 EQUIP. ACCESS- DOOR ■20'- 0" 958 EQUIP. ACCESS DOOR
 OVERHEAO PROJECTOR \W/ TRANSPARENCY CHANCER* ~
 EQUIP. ACCESS DOOR

Specialties AUDIO-VISUAL SYSTEMS Typical Projection Room Layout
 3-00 , 5'0"x10'0" REAR PROJECTION SCREEN -^^CrXCct^ 26'-0" 13'-0'
 SCREENWAW SPEAKER TYPICAL CONFIGURATION 20'-Q" 13'-0"
 TYPICAL CONFIGURATION 959

Specialties AUDIO-VISUAL SYSTEMS Typical Projection Room Layout
 13'-0" -20-0 13'-0" 6'-0" X 6'-0" REAR PROJECTION SCREEN TYPICAL
 CONFIGURATION -26-0- *&& ^ r-OVERHEAO PROJECTOR W/
 TRANSPARENCY CHANGER ■ REVERSING MIRROR WORK TABLE
 DUAL 6'-0" X 6'-0" REAR PROJECTION TYPICAL CONFIGURATION

6'0"x12'0" REAR PROJECTION SCREEN .3 :.*: 01 -U J- rr •Iffl-
•*^ESfv--p'£K(•(AUDIENCE AREA)- 960

AUDITORIUM SEATING Sightlines; Building Codes Sight line studies vary depending on the particular event and seating configuration. The following are some basic design elements. {Note: Remember to review and verify slope, riser heights, tread depths, etc., with pertinent national and local code requirements.) The visibility profile shown in Fig. 1: Angle A: Shifting position to look between heads in row immediately in front of spectator and over all other heads. Angle B: Shifting position to look between heads of two rows immediately in front of spectators and over all other heads. Generally, the variables considered in determining these angles are: ■ 3'8" eye level in the seated position ■ 5" minimum eye clearance ■ Row spacing and row rise Angle A is commonly used in determining floor slope for auditorium, performing arts or theater type seating configurations. When angle A profile is used in conjunction with a staggered seating arrangement (chairs staggered or alternated in arrangement of sizes opposite every other row) it allows unobstructed view of spectators to a determined focal point at screen on stage. The final analysis is to have all sight lines to intersect the desired focal point (usually 5'6" elevation either at screen or 12'0" back from front of stage). Angle B is most commonly used in determining riser or stepped applications for gymnasium, arena, or stadium type seating configurations. When the angle B profile is used (generally associated with an aligned seating arrangement) it allows unobstructed view of spectators to a determined focal point at court line or line of play. The final analysis is to have all the critical sight lines to intersect the focal point or line of play at generally a 3'0" elevation. Legal responsibility lies with the owners and users of equipment in acquiring acceptance with local officials. The following are some basic guidelines. Standard Seating 1. Row spacing shall provide a clear space of not less than 12" (30.5 cm) from the back of one chair to the front of the most forward projection of the chair directly behind it when measured with the self-rising seat in the up position. 2. Rows of chairs shall not exceed 14 chairs between aisles and exceed seven chairs from an aisle to a row end. 3. Aisles serving 60 seats or less shall be a minimum of 30" (76 cm) wide. Aisles serving more than 60 seats shall be at least 3' (91 cm) wide when serving seats on one side and at least 3'6" (107 cm) wide when serving seats on both sides. These minimum

widths, measured at the point furthest from an exit, cross aisles, or foyer shall be increased 1" (2.5 cm) for each 5' (152 cm) in length toward the exit, cross aisle, or foyer. Where egress is possible in either direction, aisles shall be uniform in width. Dead end aisles are not allowed over 20'0" (61.0 m) in length. 4. Cross aisles, foyer or exit widths shall be not less than the sum of the required width of the widest aisle plus 50% of the total required width of the remaining aisles that it serves. Continental Seating 1. Row spacing shall provide a clear space of not less than: 18" (45.7 cm) between rows of 18 chairs or less; 20" (50.8 cm) between rows of 35 chairs or less; 21" (53.3 cm) between rows of 45 chairs or less; 22" (55.9 cm) between rows of 45 chairs or more to a maximum of 100 chairs per row, measured from the back of one chair to the front of the most forward projection of the chair directly behind it with the self-rising seat in the up position. 2. There shall be exits of 66" (168 cm) minimum clear width along each side aisle of the chair rows for each five rows of chairs. 3. Aisles shall not be less than 44" (112 cm) in clear width. Fin.1

Specialties AUDITORIUM SEATING now Length Row length = 1 to £ + (2A) measured at the Chair Size Line Note: End dim. A varies 2" to 3" for end tablet arm applications. EXAMPLE; (3) 20" chairs = 5'-4" row length. DIMENSIONS: Center QTY 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 . SIZE 18" 3'-0" 4'-6" 6'-0" 7'-6" 9'-0" 10'-6" 12'-0" 13'-6" 15'-0" 16'-6" 18'-0" 19'-6" 21'-0" 22'-6" 24'-0" 25'-6" 27'-0" 28'-6" 30'-0" 31'-6" 33'-0" 34'-6" 36'-0" 37'-6" 39'-0" 40'-6" 42'-0" 43'-6" 45'-0" 46'-6" 19" 3'-2" 4'-9" 6'-4" 7'-11" 9'-6" 11'-1" 12'-8" 14'-3" 15'-10" 17'-5" 19'-0" 20'-7" 22'-2" 23'-9" 25'-4" 26'-11" 28'-6" 30'-1" 31'-8" 33'-3" 34'-10" 36'-5" 38'-0" 39'-7" 41'-2" 42'-9" 44'-4" 45'-11" 47'-6" 49'-1" 20" 3'-4" 5'-0" 6'-8" 8'-4" 10'-0" 11'-8" 13'-4" 15'-0" 16'-8" 18'-4" 20'-0" 21'-8" 23'-4" 25'-0" 26'-8" 28'-4" 30'-0" 31'-8" 33'-4" 35'-0" 36'-8" 38'-4" 40'-0" 41'-8" 43'-4" 45'-0" 46'-8" 48'-4" 50'-0" 51'-8" 21" 3'-6" 5'-3" 7'-0" 8'-9" 10'-6" 12'-3" 14'-0" 15'-9" 17'-6" 19'-3" 21'-0" 22'-9" 24'-6" 26'-3" 28'-0" 29'-9" 31'-6" 33'-3" 35'-0" 36'-9" 38'-6" 40'-3" 42'-0" 43'-9" 45'-6" 47'-3" 49'-0" 50'-9" 52'-6" 54'-3" 22" 3'-8" 5'-6" 7'-4" 9'-2" 11'-0" 12'-10" 14'-8" 16'-6" 18'-4" 20'-2" 22'-0" 23'-10" 25'-8" 27'-6" 29'-4" 31'-2" 33'-0" 34'-10" 36'-8" 38'-6" 40'-4" 42'-2" 44'-0" 45'-10" 47'-8" 48'-6" 51'-4" 53'-2" 55'-0" 56'-10" ne to center line (1 to 1) QTY 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 SIZE 18" 48'-0"

49'-6" 51'-0" 52'-6" 54'-0" 55'-6" 57'-0" 58'-6" 60'-0" 61'-6" 63'-0" 64'-6"
66'-0" 67'-6" 69'-0" 70'-6" 72'-0" 73'-6" 75'-0" 76'-6" 78'-0" 79'-6" 81'-0"
82'-6" 84'-0" 85'-6" 87'-0" 88'-6" 90'-0" 19" 50'-8" 52'-3" 53'-10" 55'-5"
57'-0" 58'-7" 60'-2" 61'-9" 63'-4" 64'-11" 66'-6" 68'-1" 69'-8" 71'-3" 72'-10"
74'-5" 76'-0" 77'-7" 79'-2" 80'-9" 82'-4" 83'-11" 85'-6" 87'-1" 88'-8" 90'-3"
91' -10" 93'-5" 95'-0" 20" 53'-4" 55'-0" 56'-8" 58'-4" 60'-0" 61'-8" 63'-4"
65'-0" 66'-8" 68'-4" 70'-0" 71'-8" 73'-4" 75'-0" 76'-8" 78'-4" 80'-0" 81' -8"
83'-4" 85'-0" 86'-8" 88'-4" 90'-0" 91'-8" 93'-4" 95'-0" 96'-8" 98'-4" 100'-0"
21" 56'-0" 57'-9" 59'-6" 61'-3" 63'-0" 64'-9" 66'-6" 68'-3" 70'-0" 71'-9"
73'-6" 75'-3" 77'-0" 78'-9" 80'-6" 82'-3" 84'-0" 85'-9" 87'-6" 89'-3" 91'-0"
92'-9" 94'-6" 96'-3" 98'-0" 99'-9" 101'-6" 103'-3" 105'-0" 22" 58'-8" 60'-6"
62'-4" 64'-2" 66'-0" 67'-10" 69'-8" 71'-6" 73'-4" 75'-2" 77'-0" 78'-10" 80'-8"
82'-6" 84'-4" 86'-2" 88'-0" 89'-10" 91'-8" 93'-6" 95'-4" 97'-2" 99'-0"
100'-10" 102'-8" 104'-6" 106'-4" 108'-2" 110'-0"

Specialties AUDITORIUM SEATING Row Spacing EXAMPLE: 4 rows (3 spaces) at 33" = 8'-3" I— See elevation drawings for envelope dimensions 4 rows (3 spaces) at 33" = 8'-3" ■ See elevation drawings for dimensions *

Notes: 1. Refer to applicable building codes. 2. Spacing varies with tablet arm applications. 3. Row space dimension will be the sum of "clear space" (see building codes) plus "chair envelope" (see chair dimensions) plus any additional space as desired for convenience to permit patron easy access to concessions, restrooms, etc. OVERALL SPACING OF CHAIRS (back to

back) Number of seating rows (spaces +1) 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 ROW SPACE

DIMENSIONS 32" 2'-8" 5'-4" 8'-0" 10'-8" 13'-4" 16'-0" 18'-8" 21'-4" 24'-0"
26'-8" 29'-4" 32'-0" 34'-8" 37'-4" 40'-0" 42'-8" 45'-4" 48'-0" 50'-8" 53'-4"
56'-0" 68'-8" 61'-4" 64'-0" 66'-8" 69'-4" 72'-0" 74'-8" 77'-4" 80'-0" 33" 2'-9"
5'-6" 8'-3" 11'-0" 13'-9" 16'-6" 19'-3" 22'-0" 24'-9" 27'-6" 30'-3" 33'-0"
35'-9" 38'-6" 41'-3" 44'-0" 46'-9" 49'-6" 52'-3" 55'-0" 57'-9" 60'-6" 63'-3"
66'-0" 68'-9" 71'-6" 74'-3" 77'-0" 79'-9" 82'-6" 34" 2'-10" 5'-8" 8'-6" 11'-4"
14'-2" 17'-0" 19'-10" 22'-8" 25'-6" 28'-4" 31'-2" 34'-0" 36'-10" 39'-8" 42'-6"
45'-4" 48'-2" 51'-0" 53'-10" 56'-8" 59'-6" 62'-4" 65'-2" 68'-0" 70'-10" 73'-8"
76'-6" 79'-4" 82'-2" 85'-0" 35" 2'-11" 5'-10" 8'-9" 11'-8" 14'-7" 17'-6" 20'-5"
23'-4" 26'-3" 29'-2" 32'-1" 35'-0" 37'-11" 40'-10" 43'-9" 46'-8" 49'-7"
52'-6" 55'-5" 58'-4" 61' -3" 64'-2" 67'-1" 70'-0" 72'-11" 75'-10" 78'-9" 81'-8"
84'-7" 87'-6" 36" 3'-0" 6'-0" 9'-0" 12'-0" 15'-0" 18'-0" 21'-0" 24'-0" 27'-0"

30'-0" 33'-0" 36'-0" 39'-0" 42'-0" 45'-0" 48'-0" 51'-0" 54'-0" 57'-0" 60'-0"
 63'-0" 66'-0" 69'-0" 72'-0" 75'-0" 78'-0" 81'-0" 84'-0" 87'-0" 90'-0" 37"
 3'-1" 6'-2" 9'-3" 12'-4" 15'-5" 18'-6" 21'-7" 24'-8" 27'-9" 30'-10" 33'-11"
 37'-0" 40'-1" 43'-2" 46'-3" 49'-4" 52'-5" 55'-6" 58'-7" 61'-8" 64'-9" 67'-10"
 70'-11" 74'-0" 77'-1" 80'-2" 83'-3" 86'-4" 89'-5" 92'-6" 38" 3'-2" 6'-4" 9'-6"
 12'-8" 15'-10" 19'-0" 22'-2" 25'-4" 28'-6" 3V-8" 34'-10" 38'-0" 41'-2" 44'-4"
 47'-6" 50'-8" 53'-10" 57'-0" 60'-2" 63'-4" 66'-6" 69'-8" 72'-10" 76'-0" 79'-2"
 82'-4" 85'-6" 88'-8" 91'-10" 95'-0" 39" 3'-3" 6'-6" 9'-9" 13'-0" 16'-3" 19'-6"
 22'-9" 26'-0" 29'-3" 32'-6" 35'-9" 39'-0" 42'-3" 45'-6" 48'-9" 52'-0" 55'-3"
 58'-6" 61'-9" 65'-0" 68'-3" 71'-6" 74'-9" 78'-0" 81'-3" 84'-6" 87'-9" 9V-0"
 94'-3" 97'-6" 40" 3'-4" 6'-8" 10'-0" 13'-4" 16'-8" 20'-0" 23'-4" 26'-8" 30'-0"
 33'-4" 36'-8" 40'-0" 43'-4" 46'-8" 50'-0" 53'-4" 56'-8" 60'-0" 63'-4" 66'-8"
 70'-0" 73'-4" 76'-8" 80'-0" 83'-4" 86'-8" 90'-0" 93'-4" 96'-8" 100'-0" 41"
 3'-5" 6'-10" 10'-3" 13'-8" 17'-1" 20'-6" 23'-11" 27'-4" 30'-9" 34'-2" 37'-7"
 41'-0" 44'-5" 47'-10" 51'-3" 54'-8" 58'-1" 61'-6" 64'-11" 68'-4" 71'-9" 75'-2"
 78'-7" 82'-0" 85'-5" 88'-10" 92'-3" 95'-8" 99'-1" 102'-6" 42" 3'-6" 7'-0"
 10'-6" 14'-0" 17'-6" 21'-0" 24'-6" 28'-0" 31'-6" 35'-0" 38'-6" 42'-0" 45'-6"
 49'-0" 52'-6" 56'-0" 59'-6" 63'-0" 66'-6" 70'-0" 73'-6" 77'-0" 80'-6" 84'-0"
 87'-6" 9V-0" 94'-6" 98'-0" 10V-6" 105'-0" 963

Specialties AUDITORIUM SEATING Chair Dimensions 27W (69.9 cm) -
 —20V4" (52.1 cm) _18J4"(47.0cm) — i—16" (40.6 cm) 22'/a"(57.2cm) —
 —16'/" (41.9 cm)— -15V4"(39.4cm)- 22V4" (57.2 cm)- -17V4"(44.5cm)-
 -£x 15° E S a 2y*\" ? (7.0 cm)1" CI T \"+t-"^ -\l \ if 1 cm) - 61.0 (17" (43.2 f
 4 7 5/16" (18.6 cm) I- ^ 7 5/16" (18.6 cm) H H 5V4" max. (41.0 cm) 964

Specialties AUDITORIUM SEATING Chair Dimensions and Row Seating
 In :3 1 23W I- 21" to 22"- 17%" I-7WH 25" 29Va" 21" to 22' Minimum
 Radius 15'-0" 31 ' /a T 14'A" dJ 29" 25'/2" 1 |. 14"—t—11WH 20" or 21"
 Minimum Radius 15'-0" / 24 W' I- 21" to 22" H 17W i I i4" H—1V--H 21"
 to 22' Minimum Radius 15'-0" 25W lJ*. 24 V2" Minimum Radius 15'-0" /
 965

Specialties AUDITORIUM SEATING General Seating Arrangement
 Seating arrangements in an assembly space will either be identified as
 "multiple- aisle" or "continental." These terms are commonly found in
 design standards manuals, building codes, and similar architectural reference

documents. Each is unique with specific guidelines governing row size, row spacing, and exitways. Basically, a multiple-aisle arrangement (Fig. 2) will have a maximum of 14-16 chairs per row with access to an aisleway at both ends. If an aisle can be reached from one end of a row only, the seat count may then be limited to 7 or 8. It should be noted here that the maximum quantities will always be established by the governing building code. In a continental arrangement (Fig. 3) all seats are located in a central section. Here the maximum quantity of chairs per row can greatly exceed the limits established in a multiple-aisle arrangement. In order to compensate for the greater length of rows allowed, building codes will require wider row spacing, wider aisles and strategically located exit doors. Although more space would appear to be called for, a continental seating plan is often not any less efficient than a multiple-aisle arrangement. In fact, carefully planned, a continental arrangement can frequently accommodate more seating within the same space. For early planning an average 7.5 sq. ft. per person may be used. This will include both the seating area and space necessary for aisleways.

Fig. 2 Multiple-aisle arrangement.

Fig. 3 Continental arrangement,

Specialties AUDITORIUM SEATING Layout Information Design

Considerations

1. Layout per applicable building and life safety codes, regulations, and ordinances.
2. Allow sufficient distance between aisles for desired quantity and size of chairs plus end space.
3. Space rows to allow for proper seat to back clear space.
4. Determine radius or straight rows and locate by the chair size line.
5. Allow 1" minimum clearance from either side or rear of chair to any adjacent side wall, end walls, etc.
6. Provide adequate sightlines for either sloping or stepped (riser) floor configurations.
7. Seating area should be free of obstructions.
8. To allow for sufficient aisle illumination: Aisle lights are generally located in the end panel standards at least every other row. Locate aisle light junction box 6" from the standard.
9. Provide adequate floor or riser materials for sound anchorage.

TYPICAL PUN OF SEATING AND TERMINOLOGY

FRONT OF AUDITORIUM

STANDEE RAIL

REAR OF AUDITORIUM

EXIT

967

Specialties AUDITORIUM SEATING Row Seating Seat Widths Seating comfort is initially established by individual chair widths Available sizes range from 18" to 24" however, all may not be produced by a single manufacturer. The most commonly used chair widths are 20", 21", and 22" It should be noted that these dimensions are nominal, being measured from center to center of the support legs. If seating comfort is a high priority, thought must be given to a particular width and the space taken up by chair arms to determine an actual size Usually, smaller sizes of 18 and 19" have limited application due to the minimum clear width provided Typically, all manufacturers size their chairs along an imaginary line which may be referred to as a datum line, "Chair radius line" or a similar name For accurate planning in an assembly area, this line must be identified so as not to over or underestimate the potential of a row of chairs Row Spacing Row spacing or "back to back" spacing of seats is also very important in developing a comfortable assembly area A minimum dimension occasionally used is 2'-8" (32") This spacing provides marginal clearance between a seated person's knees and the back of the chair in the next forward row At the same time, however, it will require that a seated person stand to permit the passage of another individual As the row spacing is increased to 3'-0" (36"), seating comfort is dramatically improved and passage along a row of seated persons is accomplished with less disruption

RadtutUne \. \-- Floor Design Seating comfort will also be affected by the design of the assembly space floor. Flat or less steeply sloped floors will usually allow a person to extend their knees and legs even under minimum row spacing dimensions. Here an individual can take advantage of the open area under a seat and the free space created by the pitched back of a chair As the floor slope is increased, this "free" space diminishes. The extreme condition exists where a large elevation change between rows is combined with a minimum row spacing An example would be a 12" high riser and a 32" wide row spacing. At this point, it becomes necessary to consider increasing the back to back dimension to provide more leg room The 'free' space under a chair is also lost when a row of seats is located directly behind a low wall In this case a recommended minimum clearance would be 11' measured from seat edge in the lowered position to face of wall. The back to back dimension of a row of seats abutting a rear wall should also be carefully studied Normally, the pitched back of a chair will overlap a riser face, automatically reducing the width of that row unless succeeding rows

are similarly positioned Where a rear wall exists The recommended procedure is to increase the dimension of the last row sufficiently to accommodate any overlap plus a minimal space between the wall and the edge of the chair back

Specialties AUDITORIUM SEATING Visibility Visibility in an assembly space is a function of seat location. As stated earlier, building codes, comfort guidelines, floor design and the overall form of an assembly space will play a part in seating arrangements. This information combined with a basic understanding of sightline analysis and related planning guidelines can result in achieving an acceptable, if not optimum, level of viewing for spectators. Perhaps film projection requires the most critical sightline analysis, since poor seat location will result in distorted images. For this activity the seating parameters are established by the screen or image size. An angle of 30° up to 45° measured perpendicular to the far and near edges of the screen can establish a side to side seating limit, while the screen or image height may determine the maximum distance. The minimum dimension or closest recommended seat will also be set by the screen height. (It should be noted that these figures are approximate and apply principally to flat screen projection.) 969

Specialties AUDITORIUM SEATING End Stage; tt Arena Basic Theater Form End Stage Quantity of Seats Seating Area Space per Seat Row Spacing Most Distant Seat Stage Elevation Floor Design 55 450 Sq. Ft. 8.23 Sq. Ft 2'-9" 22'-0" None Flat/One Risers" Basic Theater Form % Arena Quantity of Seats Seating Area Space per Seat Row Spacing Most Distant Seat Stage Elevation Floor Design 56 622 Sq. Ft. 11.1 Sq. Ft, 3'-3" 32'-0" None Risers 4" 970

Specialties AUDITORIUM SEATING End Stage Basic Theater Form Quantity of Seats Seating Area Space per Seat Row Spacing Most Distant Seat Stage Elevation Floor Design End Stage 80 700 Sq. Ft. 8.75 Sq. Ft. 3'-6" 25'-0" 3'-6" Risers 6" Basic Theater Form Quantity of Seats Seating Area Space per Seat Row Spacing Most Distant Seat Stage Elevation Floor Design End Stage 92 955 Sq. Ft. 10.4 Sq. Ft 3'-0" 37'-0" 12" Flat V Projection Screen Stage Jt 971

Specialties AUDITORIUM SEATING End Stage Basic Theater Form
 Quantity of Seats Seating Area Space per Seat Row Spacing Most Distant
 Seat Stage Elevation Floor Design End Stage 99 953 Sq. Ft. 9.62 Sq. Ft.
 3'-5" 32'-0" 12" Risers 12" Basic Theater Form Quantity of Seats Seating
 Area Space per Seat Row Spacing Most Distant Seat Stage Elevation Floor
 Design End Stage 105 903 Sq. R. 8.6 Sq. Ft. 3'-0" 45'-0" 5" Risers 12" 972

Specialties AUDITORIUM SEATING $\frac{1}{2}$ Arena; End Stages Basic Theater
 Form Quantity of Seats Seating Area Space per Seat Row Spacing Most
 Distant Seat Stage Elevation Floor Design 'A Arena 108 1200 Sq. Ft. 11.1
 Sq. Ft. 3'-0" 45'-0" Varies Risers 12" Basic Theater Form Quantity of Seats
 Seating Area Space per Seat Row Spacing Most Distant Seat Stage
 Elevation Floor Design End Stage 120 1088 Sq. Ft. 9.1 Sq. Ft. 3'-3" 41'-0"
 1'-6" Iscidual Slope 2.4°, 4.7°, 71° 973

Specialties AUDITORIUM SEATING End Stages; Wide Fan Basic Theater
 Form Quantity of Seats Seating Area Space per Seat Row Spacing Most
 Distant Seat Stage Elevation Floor Design End Stage 180 1218 Sq. Ft. 6.8
 Sq. Ft. 2'-8" 50'-0" 1'-6" Iscidual Slope -1° to 10° Basic Theater Form
 Quantity of Seats Seating Area Space per Seat Row Spacing Most Distant
 Seat Stage Elevation Floor Design Wide Fan 253 1790 Sq. Ft. 7.1 Sq. Ft.
 3'-0" 45'-0" None Risers—Varying Height 10' to 16" 974

Specialties AUDITORIUM SEATING End Stage Basic Theater Form End
 Stage Quantity of Seats Seating Area Space per Seat Row Spacing Most
 Distant Seat Stage Elevation Floor Design 430 2586 Sq. Ft. 6.0 Sq. Ft.
 2'-8" 58'-0" 2'-6" Sloped 7.1 ° Basic Theater Form Quantity of Seats
 Seating Area Space per Seat Row Spacing Most Distant Seat End Stage 224
 1660 Sq. Ft. 7.4 Sq. Ft. 2'-9" 60'-0" Stage Elevation None 975

Specialties AUDITORIUM SEATING Wide Fan Basic Theater Form
 Quantity of Seats Seating Area Space per Seat Row Spacing Most Distant
 Seat Stage Elevation Wide Fan 208 1557 Sq. Ft. 7.5 Sq. Ft. 2'-8" 48'-0" 5"
 Floor Design Risers 14 Basic Theater Form Wide Fan Quantity of Seats
 Seating Area Space per Seat Row Spacing Most Distant Seat Stage
 Elevation Floor Design 207 1428 Sq. Ft. 6.9 Sq. Ft. 3'-4" 40'-0" 2'-3"
 Sloped 6.6° 976

Specialties SECURITY General Control Guidelines CONTROL OF

GROUND Fencing can be a very effective means of limiting access to secondary exits and to vulnerable ground-level dwellings. Fencing functions as a control by requiring entry through a single, limited, highly visible area. The fencing surrounding most single-family homes does not have locked gates. It is intended primarily to protect children, pets, and gardens, and to define the area immediately around the home as the private outdoor space of that household. Any intrusion into the area within the fence is therefore noticeable. As a security measure, such fencing, used symbolically, is of minimal value against premeditated crime, but it does make criminal intent visible and so is an important deterrent. A

conventional use of fencing in multi-family complexes is to limit access to backyards and windows of a housing cluster. On conventional city blocks, backyards of row housing are accessible only through one of the houses.

However, in many superblock designs, such backyards are left open to public access. In this situation, addition of a limited amount of fencing can protect a large group of homes (see Fig. 1). This approach can also

subdivide the superblock and so create small, natural clusters. The Lobby Improving visibility is the most important ingredient in providing a naturally secure lobby. It is crucial that a tenant entering a building be able to see what is going on in the lobby from the outside. Hidden nooks and blind curves provide perfect hiding places. Where such features cannot be removed structurally, the use of mirrors, windows, and improved lighting may ease the situation. Ideally, a person walking down a path to enter a building should be able to see anyone standing in the lobby and elevator waiting area. In fact, it is often advantageous if the arriving person can see into the elevator from across the lobby. **CONTROL OF INTERIOR**

PUBLIC SPACES OF MULTIFAMILY DWELLINGS The most vulnerable locations in multifamily buildings are the interior public spaces; lobbies, elevators, stairwells, and corridors. These are areas open to the public but without the attending surveillance given a public street by passersby and police. The crimes that occur in these interior public spaces are the most fearful types of crimes, involving acts of personal confrontation such as robbery, assault, and rape. Limiting access to these spaces through the use of a doorman or intercom/door lock system can be of substantial benefit. Lobby visibility discourages a number of different kinds of crime. Crimes of personal confrontation may be deterred primarily because the potential

victim can readily perceive and avoid a suspicious person in the lobby. The potential criminal must also fear the possibility that another tenant or the police may be viewing the crime in the well-lit open area. Fig. 1 Use of fencing to define and secure large semiprivate areas. Mailbox crime — generally the theft of checks — can be deterred when mailboxes are located in a highly protected area of the lobby. This protection can consist of placing the mailboxes behind an intercom or in a locked mailroom. It is essential that the mailboxes be visible from as many different viewpoints as possible. Improved visibility in this context can be a significant deterrent to crime. Some managers designate an area of the lobby as a legitimate resting place, where chairs and other lounging items are provided. Lounging may aid security, particularly if the building includes a high proportion of elderly. The best locations for such seating are areas with high visibility. Often tenant patrols use this space as a station and provide still another dimension of security. A bulletin board is an inexpensive device that can improve lobby security by providing a diversion. If, for example, a tenant enters the lobby and sees someone she doesn't recognize waiting for an elevator, she may need a reasonable excuse for not taking the same elevator. The bulletin board provides the tenant with a natural excuse to pause and survey the situation. The area around the main entry to a multifamily building should be clearly distinguished from the public walkway which leads to it. A person entering through the main door should feel distinctly that he is entering a space controlled by the residents of the building. The main entry should be well lit and clearly visible from outside. Entry doors should be constructed of a transparent material covering as large an area as possible. In vandalism-prone areas, the main entry doors should be made of unbreakable glass or other similar, very sturdy transparent material. Because of the need for good visibility, replacing glass panels with metal or other material should be avoided. For window walls and doors where the incidence of vandalism is extreme, glass panels less than 2 feet from the ground and higher than 7 feet from the ground may be replaced by solid materials. Fire Doors and Fire Stairs Secondary exit doors are the weakest link in security of buildings. An ideal secondary exit door would be one that allows exit but not entrance. Unfortunately, there is no acceptable emergency exit system that allows egress only. In the design of any security system there is a continuing clash between the need for security against crime and the need for safety in case of fire, Fire doors are frequently used

for entry and exit by criminals. Installation of panic hardware and the absence of exterior hardware sometimes prevent criminal use, These measures will not suffice, however, where tenants do not cooperate in avoiding use of secondary exits and ensuring they are kept closed, To a large extent, the design and location of secondary fire exits determine tenant attitudes about the exits. For example, a building's main entry may face the street, but the parking lot may be to the rear of the building. If the secondary exit is also at the rear and close to this destination, the temptation to use the fire door as an entry or exit will be difficult to resist, Similarly security is decreased in buildings where the main entries face the interior of the project while the fire doors face the surrounding streets with their parking and shopping facilities, Where the fire exit does not represent any shortcut or improved convenience to the tenant, it is far more likely to remain closed. A securely designed building is one in which the fire door exits to an area that is less convenient or desirable than the area outside the main door. In cases of persistent breaks in security of secondary exits, it is possible to modify the building plan at the ground level and open a new doorway in a better location. However, this improvement is costly and can only be done where architecturally possible. Another architectural modification to improve security involves making a fire exit into a legitimate secondary entry and developing a security system that protects both the main and secondary entries. If a fire door exits to a 977

Specialties SECURITY General Control Guidelines; Doors and Hardware parking area, for example, this modification may be more successful than efforts to prevent tenants from using that exit. If the main entry is equipped with an intercom system, the secondary entry should be similarly equipped and made easily surveillable through the use of lighting and windows. Other mechanisms can be used to limit access to and prevent circulation through the emergency exit system. A fire exit passageway, for example, can be modified by installing a second door inside the building a short distance from the existing exterior door. Both doors should be equipped with hardware so that they can be opened only from the inside. The point of this system is that it is unlikely that both doors will be p rapped or jam med open at the same time. A tenant entering an open exterior fire door which leads only to the locked second door will have to exit and use another door. A few experiences of this kind will convince most tenants that it is probably

more convenient to go directly through the main entrance. This double-door system generally does not conflict with fire codes. An extension of this concept is to have the fire door on each floor above ground level openable from the corridor only. Thus, once someone has gone into a stairwell he can only exit at the ground level. This system may be somewhat inconvenient to tenants accustomed to moving easily between floors, but it does create roadblocks for anyone attempting to enter the building from the ground-level exit door. The improvements outlined above are generally applicable to all dwellings. In buildings which have such security personnel, additional measures are possible. A doorman or security guard can only be effective if he controls all access to the building, including access through fire doors. In a well-designed building, the doorman can see the fire doors from his position at the main entry. Where this is not possible, an inexpensive and effective solution is to install panic hardware with an alarm, and make sure the doorman can hear and respond to the alarm. Where the doorman or guard has access to closed-circuit TV, this may be used to monitor the fire doors. If the doorman can also be given a device for controlling the secondary door, it becomes very difficult for a criminal to use the fire entry.

Elevators There are virtually no structural modifications that can improve security within elevators. The only possible improvements are use of mirrors, communication devices, emergency buttons, or an electronic surveillance system. Security modifications to other areas of a building improve security within the elevator. If the elevator waiting area and the elevator cab are a visible extension of the lobby, the residents are afforded some protection. Similarly, if the fire door and fire stairs are secure, there is less chance of a criminal entering the elevator on an upper floor. In this sense, the safety of the elevator is dependent upon the general security of the building.

SECURING THE DWELLING Illegal entry into dwelling units is traditionally prevented by use of hardware. However, there are building design features which in themselves limit access, improve surveillance, and promote neighbor recognition.

Windows Ground-level windows are generally most vulnerable to illegal entry and breakage. (All windows whose lower ledges are less than 7 feet off the ground should be considered ground level.) There are three ways to discourage criminal entry through ground-floor windows: design ground-floor areas which need few windows; house activities on the ground floor which hold no interest to the burglar; and assign the grounds immediately adjacent to the building for the

use of the neighboring resident and fence off the grounds for his protection. Elaborate architectural details - protruding ledges, for example - often increase the vulnerability of lower windows. Fences, garbage containers, and parked cars, when located near windows, are used as stepping stones to an otherwise inaccessible window. Care should be taken to prevent this type of situation. Most windows above the ground floor are relatively inaccessible, with very important exceptions. Fire escapes make windows accessible. Little can be done to modify fire escapes, except in terms of hardware, because of fire safety and fire codes. One solution is to ensure that the ladder from the lowest fire escape is at least 12 feet above the ground. The ground area under the fire escape should be highly visible. Another point of entry to the fire escape is the roof, which can be secured with panic hardware and possibly patrolled. The roof also provides possible entry to windows or balconies on the top floor. Therefore, security of the roof is quite essential, particularly to top-floor residents. Other accessible windows are those located diagonally across from a stairwell window. The criminal can open a stairwell window and cross from the stairwell into the units. It is not advisable to board up stairwell windows, as they provide the security of visibility to the stairwell and may have a fire safety function. Accessible windows are also those located above or near door canopies. Criminals can reach the canopy by climbing onto it from the ground or from a stair or hall window. Doors Security of doors, beyond the hardware aspect, depends upon surveillance and neighbor recognition. An experienced burglar needs just a few seconds to enter a locked apartment door equipped with minimal hardware. Within this interval, the crucial factors are: Will the intruder be seen or heard by tenants, will the viewer perceive that the potential criminal is in fact an intruder, and will the viewer respond by calling authorities or in some way challenge the criminal? Physical design can directly influence the opportunity for surveillance of doors. Corridors that are open to view, either single loaded or with windows, are more easily surveillable by residents and police. Thus the opportunity for the criminal to attempt entry undetected is reduced. In most single-family homes (detached or row) where the entrance door is on the street, the only means of improving surveillance is to avoid placing trees and shrubs where they hide the doors and windows, and to locate lighting to improve visibility around these openings. In multiple-family dwellings, the apartment doors, located on interior corridors, are generally difficult to keep

undersurveillance. Any windows, mirrors, or lighting that allow someone inside an apartment or outside the building to view the hallway and doors can be helpful.

HARDWARE This section describes hardware devices that secure the individual residential dwelling and the multifamily dwelling. Much of this material is intended to prevent burglary. However, some of the measures, particularly those directed at multifamily dwellings, will also deter forcible entry, robbery, and vandalism.

THE RESIDENTIAL DWELLING

Door Materials The major security tests of door material are its ability to withstand efforts to force entry by brute strength and its ability to retain securely the locking devices attached. Materials most commonly used for doors are wood, aluminum, steel, and glass, often in combination with hardboard, fiberboard, asbestos, and plastic. The two most common door designs are panel and flush. Panel doors consist of vertical and horizontal members framing rectangular areas in which opaque panels, panes of glass, or louvers are located. Flush doors consist of flat panels running the full height and width of the door. (See Fig. 2.) Solid-steel flush doors, although most secure, are rarely used except in very high-security areas such as banks and prisons. Steel-clad doors, which are flush doors constructed of 24-gauge sheetmetal facing bonded to a nonresinous, kiln-dried wood interior, provide an optimum weight-strength situation for ordinary residential use. Hollow steel doors (1½-inch flush type) are satisfactory in multiple-dwelling buildings. Aluminum doors can provide sufficient protection but may be comparatively expensive. While less strong than steel-clad doors, wood doors can be secure. All exterior wooden doors should be of solid-core construction with a minimum thickness of 1½ inches. Although flush doors provide better security, if panel doors are desired for aesthetic reasons, the panels should have a minimum 1½-inch thickness (see Fig. 3). Both hollow-core wood doors and thin-wood panel doors are unacceptable where security is a factor. 978

Specialties

SECURITY Doors and Hardware

Door Frames The sides and top of a doorway are provided with a door frame which holds the door in position. The side members of the door frame are called jambs; the top member is called the head (see Fig. 4). The strike is the portion of the jamb which is cut out or drilled out to allow installation of a metal plate, which accepts the latch or bolt from the door lock (see Fig. 5). Wooden frames provide an unacceptable level of security unless they are at least 2 inches

thick. Metal-covered wood frames provide an optimum cost-security investment when used in combination with metal-covered wood doors. If a hollow steel frame is used, the residual air space behind the frame should be filled with a crush-resistant material such as cement grout, especially in the area of the strike (see Fig. 6). This will prevent an intruder from wedging a crowbar between the door and frame and crushing the frame to free the lock. For doors swinging in, rabbeted jambs should be used. These are jambs containing a metal extension that protrudes beyond the edges of the closed door, thus preventing tampering in the area of the strike (see Fig. 7). For doors without rabbeted jambs, an L-shaped piece of angle-iron at least 2 feet long, mounted in the area of the strike, gives extra protection (see Fig. 8). The iron acts as a lip which protects the strike from attack. Fig. 2 Door types. 1K" MINIMUM Fig. 5 Door strike. CEMENT FILLER BEHIND STRIKE TO PREVENT FORCING OF THE FRAME Fig. 6 Hollow metal door frame. Fig. 3 Panel door. Fig. 4 Doorframe. Fig. 7 Rabbeted jamb. 979

Specialties SECURITY Doors and Hardware For doors opening out, a flat metal plate, called an escutcheon plate, can be mounted to the face of the door in the area of the lock. This plate, which extends beyond the edge of the door and fits flush with the jamb when the door is closed, will protect the lock from attack in the area of the strike (see Fig. 9). All plates located on the outsides of doors should be attached with tamper-resistant connectors such as round-headed carriage bolts or one-way screws. Door Hinges and Closers Spring hinges close the door automatically by using spring force. A spring hinge prevents a criminal from slipping in behind a resident who has neglected to close the door immediately upon entering. Also, spring hinges prevent the resident from leaving the door open when he exits. Door closers (see Fig. 10) serve the same purpose. These are for more heavy duty and are commonly used in lobbies and commercial facilities. Hinges should be mounted on the inside of the door so that burglars cannot remove the door from the hinges to enter. If hinges must be placed on the outside, they should have nonremovable pins. Pins can be made nonremovable by peening the straight end or by drilling and tapping a machine screw into the middle portion of each pin from the inside of the open hinge (see Fig. 11). Doors with outside hinge pins can also be protected by screwing two screws halfway into the jamb edge of the door. One screw is placed near each hinge, and a receiving hole is drilled into the jamb for each screw. These

protruding screws hold the door when it is closed, even if the hinge pins are removed. T^h Fig. 10 Door closer. Fig. 8 Protective angle-iron for doors opening in. 0 0 S> 0 * 0| 0 0 ® Fig. 11 Nonremovable hinge pin. Escutcheon plate for doors opening out. 980

Specialties Doors and Hardware Door Locks Locks must withstand or seriously delay not only a simple forced entry but also sophisticated criminal attack. Locks may also guard against window entry-door exit crimes. Parts of the a lock are defined as follows: Cylinder: A cylinder is that part of the lock into which the key is inserted. If the proper key is used, the cylinder will allow the key to turn, thus moving a bolt or latch. Deadbolt: A deadbolt (or bolt lock) is a heavy metal bar which moves horizontally into the strike of the door jamb, thus locking the two together. It is called a deadbolt because it cannot be pushed back unless the knob is turned by the correct key. Latch: A latch (or spring lock) is the part of the lock that keeps the door in a closed position by extending into the strike automatically when the door, is closed. The latch is most often operated by the doorknob. Most latches can be pushed back by external pressure without having to turn the doorknob. Deadlatch: In a deadlatch, the latch is positively held in the projected position by an automatic mechanism which is depressed against the strike plate (see Fig. 12). Strike: The strike is the portion of the jamb where a metal plate has been placed to receive the deadbolt and/or the latch (see Fig. 5). Stopworks: Stopworks consist of two buttons located under the latch, Pressing the top button in allows the doorknob to turn freely and operate the latch, from both inside and out. Pressing the lower button in allows the inside doorknob to operate the latch, but "freezes" the outside doorknob. Throw: The throw of a lock is the length (in inches) that the deadbolt extends beyond the face of the lock. Primary locks Primary locks operate in conjunction with the latch. There are two major types: mortise locks and cylindrical or bore-in tubular locks (commonly called key-in-the- knob locks). Mortise locks (see Fig. 13) are more common than key-in-the-knob locks and will provide good security. All mortise locks with latches should contain a deadbolt with at least a 1-inch throw constructed of case- hardened steel, brass or zinc alloy, or bronze. Federal FF-H 106a heavy-duty series 86 mortise locks or 185 latch and 190K modified deadbolts are recommended. The deadbolt and latch should be key-operated from the exterior and operated from the inside by a

device not requiring a key. Mortise locks with latches used in residences should not contain an automatic spring latch with stopworks. Although stop-works prevent the outside knob from being turned, they leave the premises open to easy entry because they do not prevent the latch from being pushed back. An intruder need only insert a credit card into the strike area, push back the spring latch, and open the door (called "loiding" or "shimming" the lock). In locks without stopworks, the deadbolt (which cannot be loided) must be thrown by the key of the resident. Eliminating the stopworks prevents the resident from relying on the stopwork and latch mechanism alone. Key-in-the-knob locks (see Fig. 14) are less secure than mortise locks. Although inexpensive due to easy installation, key-in-the-knob locks can be easily gripped by a tool and twisted until they break. A key-in-the-knob lock can include a deadbolt, at a comparable to slightly higher price than a mortise lock. Secondary locks A secondary lock (rim lock) operates independently of the latch. "Secondary" is perhaps a poor name, since this type of lock is essential for good security. Secondary locks are usually mounted above the primary lock at shoulder level. They are operated by a key from the outside, and by a turnbolt from the inside. Both mortise and secondary locks may require keys to open them from inside and outside - useful where access to premises may be gained through a small opening other than the door (window transom), since this will prevent the thief from using the door to remove large objects or to escape. There are three major types of secondary locks: spring bolt, horizontal deadbolt, and vertical deadbolt. The spring bolt lock operates much the same as the primary door latch. Because the bolt must be spring loaded and bevelled to allow automatic latching, the bolt can be easily opened. A button (slide stop) may be set to deadlock the bolt. However, the button must be set from the inside and can only be used when another means of egress is available. The spring bolt is not recommended as a secondary lock (see Fig. 15). Horizontal bolt rim locks operate much the same as deadbolts on primary locks. While horizontal deadbolts afford much better protection than spring bolts, they still can be easily overcome. By inserting a crowbar between the door and the jamb, the intruder can pry them apart to release the bolt from the strike. For this reason, the longer the throw of the deadbolt, the greater protection it affords. However, throws of over 1 1/2 inches may have excessive cantilever. The recommended minimum

throw is 1 inch (see Fig. 16). Vertical bolt deadlocks should be used as secondary locks wherever possible. These utilize two deadbolts that fit vertically into eyeholes or sockets attached to the jamb. This creates a firm bond between the door and the jamb. The vertical bolt deadlock made by Segal is highly recommended, both for its pressed-steel construction and for its ability to hold up under heavy use (see Fig. 17). For additional security, a pick-resistant cylinder should be installed in a good vertical deadbolt body. This combination provides excellent security. 981

Specialties SECURITY Doors and Hardware Fig. 14. Key-In-knob lock. Fig. 15 Spring bolt. Fig. 16 Horizontal bolt. STEEL BAR FLOOR RECEPTACLE Fig. 18 Buttress door lock. Fig. 19 "Magic Eye" lock with thumb turn. Fig. 20 Buttress door lock with deadbolt. Fig. 17 Vertical bolt. 982

Specialties SECURITY Doors and Hardware The locks discussed so far rely on the rigidity of an existing door frame to resist attacks on the lock. Since older buildings may contain weak door frames, a buttress-type door lock is advisable. Locks of this type include a bar set against a plate on the door and into a receptacle in the floor, thus forming a triangular buttress (see Fig. 18). Most of these locks can be operated only by a key from the outside. The Magic Eye Company buttress lock can be operated from the outside by a key and from the inside by a turnbolt to prevent accidental locking (see Fig. 19). One model contains a heavy-duty deadbolt as well as the buttress bar, and affords still further protection (see Fig. 20). The double-bar lock may also be used to increase the strength of a door, by means of two steel bars that extend up to 2½ inches into each side of the jamb (see Fig. 21). The cylinder is protected on the outside by an escutcheon plate to prevent forcible removal. A pick-resistant cylinder can be installed for added protection. The Fox Police Lock and the Fichet Locking Bar are examples of high-quality double-bar locks. Cylinders Regardless of the type of lock purchased, the cylinder is critical in providing protection. It must withstand efforts by sophisticated criminals such as lock pick experts. The cylinder is the part of the lock into which the key is inserted. The most common type of cylinder is the pin tumbler which operates as follows; As the key is inserted, spring-loaded pins are raised to the proper position to allow the barrel and the key to turn; the turning

causes the bolt or latch (or both) to move. If the wrong key is used, the pins will line up incorrectly and prevent the barrel from turning (see Fig. 22). Recently, cylinders have become available which utilize special keyways and keys to make the cylinder pick proof or pick resistant (see Fig. 23). Medeco, Illinois Duo, Sargent, Keso, Eagle Three Star, Mela, Fitchet, and Miracle Magnetic are highly pick resistant. Such cylinders provide improved security, but may require registered keys that can be duplicated only at the factory upon receipt of a signed request. A compromise is the use of a key type whose blank is not available normally, but for which spare blanks are kept for replacements. < Of all cylinders on the market, Medeco has proven most difficult to overcome. Medeco utilizes twisting tumblers operated by a key with angular or criss-cross cuts. Only if the proper key is inserted will the pins twist the exact amount needed to allow the barrel to turn. If special keyway cylinders are deemed unnecessarily secure or costly (Medeco cylinders cost about two times the next adequate), the cylinder used should be of solid- bar-stock bronze and machined for a tight fit. The cylinders of a master-key system of locks are constructed so that individual keys fit only one lock, but a single master key can open all locks in the system. Use of a master- key system makes maintenance and other authorized access simpler, but the dangers of improper use of a lost or stolen master key far outweigh the benefits. From a security standpoint, a cylinder should have at least six pins. This often results in the cylinder being longer than the

Specialties SECURITY Doors and Hardware thickness of the door. In mortise locks (which are recessed into doors), a six-pin cylinder often extends slightly beyond the surface of the door thus making it susceptible to forcible removal by use of a gripping tool. To prevent use of such a tool, protruding cylinders should be protected by one of the following: Spinner ring: A hardened steel ring that forms a collar around the cylinder and which spins freely around the cylinder when gripped (see Fig. 24). Bevelled-ring cylinder guard: A case- hardened steel ring that prevents the cylinder from being gripped by a tool because of its bevelled shape (see Fig. 25). Scotsman makes a flat, very secure, cylinderguard ring. Escutcheon plate: A metal plate mounted to the door, which covers all of the cylinder except the core (the part where the key is inserted), thus protecting the cylinder from attack. The escutcheon plate should be constructed of

malleable cast iron and attached to the door with one-way screws. Machine bolts should not be used to mount escutcheon plates on mortise locks, as the increased pressure can have an adverse effect on the mechanism (see Fig. 26). Sliding Doors Sliding doors opening onto a ground-level patio or accessible balcony (on the first floor or top floor, or adjacent to other balconies) should be constructed so the movable section of the door slides on the inside of the fixed portion. Sliding doors should be break resistant (plate glass) and equipped with a vertical-bolt Segal lock (see Fig. 27), which uses a hook-type bolt to grip door and frame together, or a Loxem Sli-door lock that hoods at top and bottom (see Fig. 28). Fig. 24 Spinner ring. Fig. 25 Bevelled ring. IN 984 Fig. 23 Key ways. Fig. 26 Escutcheon plate covering cylinder mortise lock.

Specialties SECURITY Doors and Hardware Fig. 27 Segal lock. Fig. 28 Loxem Sli-door lock. Doors with Large Glass Panels Exterior doors containing panes of glass are not recommended for security. French doors that open out should have hinges with nonremovable pins. The vertical stile incorporating the lock should withstand a concentrated horizontal load of 300 pounds. The doors should contain a mortise-type lock that is key operated from the inside and outside. The lock should contain a pin-tumbler cylinder with at least six pins (a pick-resistant cylinder can be used for extra protection). Even when fitted with key-operated locks inside and outside, doors with large panes of glass are a security problem. Use of break-resistant glass substitutes is one modification. Bars or metal grilles, while providing good security, may be aesthetically unacceptable. Alarms may also be used on these vulnerable doors. Double Doors On double doors, the active leaf should be equipped with a mortise-type lock. The inactive leaf should be equipped with flush bolts with at least a 3/4-inch throw at head and foot (see Fig. 24). Private Garage Doors Many rolling overhead doors operated by electric motors offer adequate security because the motors are controlled by a key switch inside the garage or by a low-power radio transmitter. Manually operated doors should be provided with slide bolts on the bottom bar (see Fig. 30). Chair-operated doors should be provided with a cast-iron keeper and pin for securing the hardened- steel chain. Door Interviewers Interviewers are devices installed on an opaque door to allow residents to see and hear who is outside the door without opening it. Fig. 29 Flush bolt on double door. Fig. 30 Slide bolt on garage door. 985

Specialties SECURITY Windows and Hardware glass has a thin, hardening coating and, while no stronger than plate glass, will not cut someone who breaks it. Several companies have developed unbreakable, transparent polycarbonate materials which look like glass but are very difficult to break. GE's Lexan, for example, is guaranteed unbreakable. It costs two to three times as much as glass and has low resistance to scratching. An improved material, Lexan MR-4000, is slightly more expensive but is much less easily scratched. These polycarbonate materials have not yet been extensively used for private dwellings. Another type of durable "glass" is fabricated much like the safety glass used in automobiles: two layers of high-quality glass are bonded together with a layer of tough vinyl between. This is sold by one company as Secur-lite. While Secur-lite can eventually be broken, the noise and trouble required to do so are considerable deterrents. Oversized glazed areas should be avoided. Anything beyond standard size (6 feet by 8 feet for glass, for example) is expensive and may be difficult to obtain. The only reliable devices are those with a key-operated locking mechanism. Yale and Ideal Security manufacture a window lock which is a modification of the pin-type lock. It can be locked in either of two positions, one of which allows the window to be open slightly at the bottom for ventilation {see Fig. 38). Fox makes a window lock combining a pin-type lock and a hasp and padlock. Although somewhat unsightly, it provides excellent protection. Ideal Security manufactures a modification of the crescent sash lock which requires a key to operate. All of these devices provide adequate security for normal residential use. A set of keys should be convenient to the window for use in emergencies but far enough away so that a burglar cannot reach them.

■■■■■ M ii i .36 Pin latch. Fig. 34 Crescent sash lock. Fig. 35 Thumb screw lock Fig. 37 Slide bolt. 987

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Specialties SECURITY Windows and Hardware Window Bars, Grilles, and Gates Where tighter security is desired, metal bars, grilles, and gates have proven most reliable. If a wire mesh grille is used, the metal should be at least one-eighth of an inch in diameter and the openings should not exceed 2 inches (see Figs. 39 and 40). The grille should be attached to the window frame with machine or roundhead bolts which cannot be removed from the outside. If bars are used, they should be placed not more than 5 inches apart. The bars should have a diameter of at least three-quarters of an inch and be set at least 3 inches into the masonry Sliding gates afford excellent protection and can be pushed aside or opened for emergency exit. The gates should be set in tracks on the top and bottom to prevent them from being pulled or pried away from the window (see Fig. 41). Protect-A-Guard gates are highly recommended for residential and commercial use. All of these devices should be installed inside the window for maximum security. Skylights The best protection for skylights is installation of metal bars, grilles, or mesh. Bars should be made of steel not less than three quarters of

an inch in diameter and should be placed not more than 5 inches apart (see Fig. 42). If mesh is used, it should be at least one-eighth of an inch thick and the spaces should not be greater than 2 inches. Mesh should be secured firmly by machine or roundhead bolts that cannot be removed from the outside. If metal is undesirable, a securely fastened hasp and padlock will discourage entry and exit through the roof, if the glass is not removed. Both hook-in-eye and sliding-bolt devices are unacceptable security measures for skylights. Fig. 39 Mesh window grille. Fig. 40 Wire mesh dimensions. Fig. 38 Keyed window lock. Fig. 41 Window guard.

Specialties SECURITY Doors and Hardware; Elevators MULTIFAMILY DWELLINGS Lobby Doors and Walls All lobby entrance doors should provide maximum visibility of the lobby. This often requires large glass areas in the lobby doors. Where there is a high degree of vandalism and crime, use of Lexan is recommended. In all cases, oversized glass sheet should be avoided. Glazed areas should be divided so that sheets larger than 6 by 8 feet are not needed. The doorframe should be constructed of rugged, heavy-duty metal. The vertical jamb incorporating the lock should withstand a concentrated load of 500 pounds and be a minimum of 5 inches thick so that it can receive heavy-duty mortise lock sets. The main outer lobby door should have a key-operated lock with a pin-tumbler cylinder containing at least six pins. The key for this lock should not open any other door (such as an apartment door) as this makes the lobby- door cylinder susceptible to picking. An antifriction latch (see Fig. 43) and a sturdy door closer should be used in conjunction with the lock. Lobby doors, especially if locked or equipped with intercoms, should open out for fire safety and to reduce vandalism (tenants who have misplaced their keys can kick an in-swinging door hard enough to break the locking mechanism). Secondary Exits In multifamily dwellings, exit doors leading to fire stairwells on each landing should have self-locking deadlatches to allow free egress while prohibiting entry. The stairside surface of the door should be free of hardware to prevent access to one floor from another via the stairwell. Hardware should limit access to the roof or ground-floor exits via the stairwell. Panic hardware, if required, should be in the form of vertical-bolt latches on the top and bottom of the door. This hardware makes the door more sturdy and makes entry from the outside difficult (see Fig. 44). Doors leading into the buildings from garage areas should have self-looking dead-

latches with a minimum throw of one-half inch that allow free egress but require a key for entry into the building. The door should be protected in the area of the strike. All exit doors should be equipped with a self-closing apparatus that can be adjusted to the desired tension. Since fire doors are required by law to be operable from the inside, they are often a means of escape. Exit alarms (see Fig. 45) bring immediate attention to fire doors that are opened when there is no apparent fire. A panic bar or other device simultaneously opens the door and sounds a local alarm. However, effectiveness of the alarm as a security measure depends upon the speed and consistence of response to the signal. Exit alarms on fire exits leading to roofs keep burglars from using the roof for escape or for access to top-floor apartments. However, the alarm may prove more a nuisance than a good security measure if teenage vandalism is prevalent. Teenagers often set off the alarm to harass the local official, who must respond to the signal and reset the alarm. Fig. 42 Skylight protection. Fig. 43 Antifriction latch bolt.

Elevators In most middle-income multifamily dwellings, vandalism of elevators is relatively rare. However, in many high-crime areas and low-income housing developments, this vandalism is reaching a critical level. In New York City Housing Authority projects, vandalism to elevators and elevator equipment is responsible for almost 60 percent of elevator outages. Parts of the elevator most commonly vandalized are the hall buttons, indicator lights, hatch door glass, hatch door interlock, and buttons located inside the cab, especially the emergency and light switches. Hall buttons are most commonly vandalized because of their accessibility. Impatient tenants push the buttons excessively and often kick or smash them in frustration. To prevent damage to the button and the electrical contacts inside, a stainless steel mushroom-type button should be used (see Fig. 46). The shape of the button prevents the contacts from being damaged by the buttons being pushed too heavily against them. Another stainless steel button has been developed on the same principle, except that the stopper is inside the mechanism so that the button has the more familiar stunted-cone appearance. Use of indicator lights for the lobby, the cab, and the other floors should be decided by the management. In some projects, indicator lights are so vandalized that it is easier to eliminate them. In other developments, indicator lights dampen user impatience and the result is less wear and tear on the buttons. If indicator lights are used, they should be protected by a heavy-duty plastic shield. There are two types of elevator

doors: swing and slide. This nomenclature refers to the doors on each floor; the cab door is always a slide door. Slide doors, which are automatic, are becoming increasingly popular despite higher initial cost, because they increase protection against vandalism. Swing doors are inconvenient and more subject to vandalism (short-circuiting of door interlocks, jamming of closing mechanisms, and joyriding on top of cabs).

Specialties SECURITY Elevators In many older elevators (especially the swing-door type), the hatch and cab door contain small glass windows which allow people to see inside before entering and allow passengers to see what floor they're passing. In high-crime areas, this glass has proven more dangerous than helpful. Vandals smash the glass readily, even if wire glass is used. The opening left when the glass is broken presents a very dangerous situation. Hatch door glass should be eliminated by welding or bolting a piece of metal over the opening. Where this is prohibited by a strict building code, a variance is often granted in a high-crime area. A less desirable modification is to install a heavy steel grille over the opening and replace the glass with Lexan. Interlocks are more commonly vandalized on swing-door elevators. Causes of damage are excessive pulling on the elevator door while the cab is at another floor and short circuiting due to water or urine damage. The latter problem can be solved by installing interlocks with hydrophilic (non-water-absorbing) contacts. When damaged, this type of interlock requires replacement of only the contact plates rather than the entire mechanism. Damage caused by excessive pulling may be alleviated by signs cautioning tenants against such pulling. Closing mechanisms (keepers) can be made to fit more securely when the bolt is in place to prevent too much play in the door. The emergency stop button presents a problem because it is often misused. The button may be activated to stop the elevator between floors to commit crimes such as mugging, rape, and drug abuse. Because every elevator has several automatic safety mechanisms that prevent it from falling freely down the shaft, the stop button is primarily a psychological comfort to passengers. Wherever possible, the stop button should be eliminated. The building code requirement for stop buttons is being challenged in New York and several other cities. If code change is unlikely, a variance should be applied for where elevator crime is common. A constant-pressure alarm switch is also somewhat better than the conventional toggle switch, **EMERGENCY EXIT**

ONLY 6 a" ALARM WILL SOUND ' -J • -J Fig. 45 Exit alarm. Fig. 44
Vertical bolt on exit door. Fig. 46 Mushroom button. 990

Specialties SECURITY Elevators Secer Light and Kendall are among the manufacturers of elevator dome lights that are highly vandal resistant (see Fig. 47). They are constructed of durable steel and contain a shatterproof plastic plate to protect the bulb. Where use of these lights is economically prohibitive, Lexan or an equivalent should be used to protect the light bulb. Aside from vandalism, joyriding on top of elevator cabs is becoming prevalent in high- crime areas. Injury occurs most often when children are struck by the counterweight when the cab and counterweight pass each other. In other cases, children are crushed between cabs, struck by dividing beams, or squashed under a cab in the pit. There are numerous means of access to elevator roofs and shafts: door interlocks are jammed by using simple household tools; emergency stop switches are abused; and roof escape hatch doors are forced. Once on top, children often abuse passengers inside the cabs and interfere with normal elevator operation. It is difficult to prevent crime by modifying elevator equipment. Restricted access to the building through the use of a buzzer-reply system, tenant patrol groups, or doormen is more likely to be effective. Closed-circuit television and audio-intercom systems mounted on elevators are other possible crime control devices. A common device used to increase visibility in an elevator is a convex mirror placed in the upper back corner of the elevator. This allows a person to see if anyone is waiting inside the elevator before he walks into a possible assault situation (see Fig. 48). An elevator modification that may deter crime is the up-discharge, down-collect system. When controlled in this way, an elevator will only stop for a person who has selected "up" (discharge) at the ground-floor level. Passengers on the upperfloor can only enter the elevator on its way down (collect). The advantage is that a person entering the elevator on the first floor can be assured that the elevator will not stop at another floor to allow a suspicious person to enter. Such a system may be inconvenient for residents-a person wishing to go from the fifth to the seventh floor would have to travel down to the ground floor and then up again. The system is far from foolproof, as criminals can operate in other ways; but the modification is inexpensive and may deter crime in buildings without security personnel. Fig. 47 Unbreakable light fixture. Fig. 48 Elevator mirror. 991

Specialties SECURITY Garages, Secondary Entries, and Mailbox Rooms access to the building (intercom or doorman), mailboxes should be located inside the protected area. Mailboxes may be located in a locked room. Such a room must contain a large window to make it visible from the lobby, and be lighted 24 hours a day to reduce its potential as a location for muggings and other crimes. The door to a mailbox room should have sturdy self-locking hardware. Where back-loading mailboxes (generally secure) are used, a separate mail-loading room is often provided (see Fig. 49). The better mailboxes are constructed of 16-gauge metal, The doors are tightly fitted and without holes to prevent prying them open and to prevent matches from being dropped in. The metal may be corrugated for additional strength. Cylinder locks with at least five pins should be used. Door size should be kept to a minimum to further limit the possibility of prying doors open (see Fig. 50), American and Gorth manufacture such mailboxes. Fig. 49 Mailroom and loading room. Garage Doors and Secondary Entries Doors to interior garages provide a means of entry that circumvents many security 'precautions. If access to the building is to be limited, entry through the garage door must be carefully controlled. The most practical solution is to have a locked door which tenants can open but which automatically closes behind them, usually within 15 seconds. A large number of manufacturers provide such self-closing doors. The major variation is the means for opening the garage door. Radio-controlled devices, requiring each auto to have a transponder, are expensive and far from foolproof. If a device is stolen from one car, all the devices should be replaced (an expensive procedure). A convenient and less elaborate system has a key-operated switch mounted on the driver's side of the garage, allowing the driver to use a key without leaving his car. Despite these controls, the garage door should be monitored by tenants, security personnel, or electronic equipment if a building is to retain a high level of security. A door leading directly from a parking area to the building interior must be treated the same as a main entry, Such a door will be used continually, and requires equivalent security measures. The secondary lock recommended for storage rooms containing valuables is the Fox double-bar lock. Mailboxes and Mailbox Rooms Mailboxes are a major target for criminals within multifamily dwellings, particularly in low-income communities, The mail includes welfare, social security, and veterans' checks as well as others, These checks are particularly vulnerable because they arrive on set

days of the month, The bank of mailboxes should be located in the most secure and easily surveyed space available. Some brands of mailboxes do provide security, but any mailbox can be opened in the 10 minutes required to force open the door. If there is any control of 992

LIGHTING Good lighting in a residential development permits adequate visibility and surveillance. Generally, the higher the lighting level, the better the security. An appropriate level of lighting should be provided in each area; the light should be without excessive glare and generate no heavy shadows; and lighting should be resistant to vandalism and easy to maintain. Fluorescent lamps are tubular glass lights that require special current-control devices called ballasts. Operating costs of fluorescent lamps are significantly lower than for incandescent bulbs: fluorescent tubes typically produce 3 to 4 times as much light per watt and operate 7 to 10 times longer than incandescent bulbs (due in part to lower operating temperatures). Interior Lighting Lobbies, elevators, stairwells, and corridors must be well lit. Interior lighting normally requires only conventional incandescent bulbs, but low-glare or "frosted" incandescent or fluorescent luminaires are preferable. Low wattages of 25 to 200 watts generally suffice. It is usually desirable to install low-wattage fixtures at close intervals to minimize shadows and glare. The most common problem of interior lighting is vandalism. Naked bulbs provide maximum illumination at minimal installation cost, but they are so often and so easily broken that maintenance costs are very high, and crime is encouraged by lack of lights. Recessed lighting suffers less from accidental breakage and vandalism. Transparent bulb protectors allow nearly total passage of light, but since the bulb can be seen, a vandal will likely try to break it. Translucent bulb covers are therefore preferable, even though some of the light is blocked by the cover. Secer and Kendall have developed fixtures that are vandal resistant. They are made of plastic and come in a variety of shapes and sizes. Exterior Lighting All heavily used spaces such as paths, entries, and parking areas should be lit by 5- to 10-foot candles. Higher fixture locations have a variety of advantages. As a general rule, the useful ground coverage of an elevated light fixture is roughly twice the height of the fixture. Thus, a 150-watt incandescent lamp mounted 8 feet above the ground can provide adequate light for 16 feet along a walk. Higher luminaires are safer from vandalism. However, lighting fixtures mounted higher than the second floor

may create a feeling of being in a "compound." A variety of specialized, high-intensity light sources can illuminate large outdoor areas such as recreation facilities and parking lots. Mercury-vapor and sodium-vapor lamps are available in sizes up to 1500 watts; the eerie bluish light of early mercury-vapor lamps may be avoided by selecting one of the newer "color-corrected" models. Once again, the point is to provide an appropriate level of light without creating glare or shadows. Lamp and fixture breakage can be controlled in part by installing fixtures of tough, break-resistant plastic. The spherical, white glass fixtures so common today are less vulnerable, though not as tough as the more expensive plastic models. A final comment on lighting is specifically relevant to a building or residential development inhabited primarily by the elderly. The pupil in the human eye gradually decreases in size due to advancing age. As a result, about twice as much actual brightness is required to create the same degree of brightness on the retina of a 60-year-old as on the retina of a 20-year-old (the ratio reaches 3 by age 75). Therefore, lighting levels in residences for the elderly should be well in excess of conventional standards and much higher than what seems adequate to a (younger) management staff.

Fig. 50
Mailboxes.

Specialties **SECURITY** Electronic Systems Electronic security equipment includes alarms designed to detect unauthorized entrance; closed-circuit television systems, apartment-to-lobby intercom locks, and various audio equipment. While the initial cost of many of these systems is high, each could reasonably be installed in moderate-income residential complexes and could prevent future need for more costly measures.

ALARMS An alarm performs two functions: it detects the presence of an intruder, and it reports the intrusion. The quality of an alarm mechanism is measured by its ability to perform these two functions. A wide range of devices detect intrusion of a criminal into a building. These fall roughly into two categories: contact devices and motion-detection devices.

Contact Devices Contact devices are mechanical switches that detect movement or perhaps the breakage of glass. A common type consists of a contact on the door (or window) and a contact on the frame. When the door is closed, the two contacts form part of an electrical circuit. When the door is opened, the contact is broken, the circuit is opened, and the alarm circuit is activated (see Fig. 51). A similar device, called a string-pull alarm, employs a slight

variation in that the opening of the door pulls a string, which closes a switch that trips the alarm. Many contact devices are purely mechanical (as just described), while others include magnetic and mercury switches.

Usefulness of a contact depends upon its sensitivity (how much the device can be jarred without being activated) and its reliability. Most situations call for a device sufficiently sensitive that a skilled burglar cannot enter without setting off the alarm, but not so delicate that an innocent jostling will disturb it. Foil strips are a related mechanism used primarily to detect breakage of glass in windows and doors. A delicate strip of metal foil is glued or taped to the glass. The foil strip acts as one long, continuous electrical circuit. If the glass is broken, the foil is broken, which interrupts the circuit and activates the alarm. Foil can be circumvented if it is possible to break the glass or release a lock without breaking the foil. Primarily because of their unattractiveness, foil strips are seldom installed in residences. Contact devices can be made part of a lock mechanism (see Fig. 52). This type of alarm is set off whenever an attempt is made to force or pick the lock. Contact devices themselves are very inexpensive; a simple magnetic contact pair costs about \$2. But each contact device can protect only one opening; therefore, even a single-family house requires several devices to protect all points of entry. In addition, it is often expensive to install the alarms and connect them to an alarm-reporting device. Contacts may be hidden so criminals cannot locate and dismantle them easily. Hiding an alarm system lessens its value as a deterrent, but increases the criminal's chances of being apprehended while com-

Specialties SECURITY Electronic Systems mitting a crime. Since deterrence is the primary goal of residential security efforts, it is quite common to advertise the existence of an alarm without revealing the location of the mechanisms. This advertising is sometimes done where no alarm system exists. Considering the minimal expense involved in such a ruse, it may be worth the cost, but even very unsophisticated criminals can pick out such fake systems. Heat-sensitive devices are sometimes combined with contact switches to provide an inexpensive fire-security alarm system.

Motion-Detection Devices These devices detect the motion of an intruder as he moves about the protected space. This detection can be accomplished in a variety of ways. Seismographic devices are turned on by vibrations or weight upon the floor (these devices have been perfected so they are not

triggered by a passing truck). Photoelectric cells ("seeing-eye" mechanisms) use a beam of light to detect any motion across a protected span. Ultrasonic devices send inaudible sound waves through a room (see Fig. 53).

Movement by an intruder changes the pattern of reflected sound waves and thus triggers an alarm. Increased sensitivity improves the effectiveness of each of these systems, but also raises their costs. Motion detectors are far more expensive than contact devices, but one motion device can protect an entire area, regardless of the number of points of entry. Installation costs are often minimal, as the detection device need not be connected to any part of the structure. Motion detectors are most useful in spaces not used during scheduled periods of time, such as in commercial establishments which are totally empty at night and in homes left empty during vacation. More

expensive motion-detection devices can protect limited areas, such as a single door or window. Alarm Reporting Systems The term "alarm-reporting system" describes the mechanism that receives the message of an intrusion and reacts. Essentially, there are only two kinds of alarm-reporting systems: Intrusion is reported either by a loud alarm on the premises (called a local alarm) or via wires to a security force which is prepared to react when notified (called a central alarm or silent alarm). A local alarm has a bell or buzzer connected to the intrusion device which produces a loud audio signal on the premises when the alarm is activated.

This is the simplest type of alarm and can be installed readily. The deterrent effect is dependent upon the burglar's being intimidated and driven off immediately by the noise. Noise of the local alarm can also stop a crime in progress and aid in apprehension if someone responds to the alarm. Local alarms are often operated by batteries (see Fig. 54). Instead of an alarm being sounded, lights in the building can be turned on by an alarm system, or lights and alarm can both be activated. This local system also protects people sleeping in a house by alerting them that a break-in is being attempted. Generally, keys are required to shut off local alarms. Fig. 54

Local alarm. A central alarm-reporting system sounds an alarm at a remote point usually connected to the detection device by wires (telephone lines are used in many cases). The remote point is sometimes the residence of the owner of a protected business establishment and sometimes the local police station; but generally, it is the headquarters of a private protective agency. These agencies have guards stationed at this headquarters who will respond to the alarm signal. Usefulness of the alarm system is dependent upon the

speed and reliability of the response. A local alarm signal is often activated at the same time as a central alarm, thus simultaneously frightening the criminal and alerting the authorities. If only a central alarm-reporting system is activated, the criminal is not warned that an alarm has been sent. This system (called a "silent" alarm) increases the possibility of apprehension while eliminating the possibility of driving the intruder off with noise. A variation on this central-alarm arrangement is to utilize regular city police to respond to the central alarm. In high-income, low-density high-burglary-risk communities, the city police allow alarms to be hooked up to the police headquarters, where the dispatcher serves as monitor. Another arrangement is for the detection device to trigger a tape-recorded message that is automatically telephoned to the police, telling them the location of a burglary in progress. The single major problem of all alarm systems is the possibility of false alarms. They can be caused by defects in the intrusion-detection device or the reporting system. False alarms diminish the credibility of the entire system. If neighbors experience repeated false alarms, if security guards are called out unnecessarily, or if police are accidentally telephoned a tape-recorded message, response by all of these persons slows dramatically and will eventually cease. Thus, the intrusion device must be designed so that it is not accidentally activated by noncriminal occurrences. Related to the false alarm issue is the question of how the alarm is turned off. The most common method is for the alarm to operate after a 20-second delay; that is, the alarm will not sound for 20 seconds after a contact is broken or motion detected, allowing the resident a brief period in which to switch off the entire system. The switch can be simply a button located in a hidden place. A key-operated switch is more secure, but the possibility of false alarms increases because residents often forget or cannot locate their keys. However, the turnoff mechanism should not be so simple or accessible that the criminal can activate it. Selecting Alarm Systems The security alarm business is large and complex. It is therefore impossible to specify manufacturers or even types of alarm systems for general use. The quality of installation and the maintenance program that backs up the system are crucial elements that should outweigh initial price in the selection of equipment. The best advice is to deal with firms that have a verifiable history of quality installation, a reliable guarantee/warranty record, and an established repair and maintenance program. The concept of a consistent "level of security" avoids excessive

expenditures for one piece of equipment while other means of entry are unprotected. Equipment characteristics should fit specific installation situations. It is often difficult to install contact switches in older houses because window frames often have warped or buckled. String- pull devices have to be set from the inside and therefore cannot be used for a normal exit door. Selection of alarm equipment should be based on specific system characteristics desired: Is deterrence of crime or apprehension of criminals the primary goal? Should the system be visible to deter attempted burglary or should it be hidden to increase the likelihood of apprehending a burglar? CLOSED-CIRCUIT TELEVISION When used in residential settings, closed- circuit television (CCTV) is intended to provide "electronic windows"; that is, a visual surveillance where physical design has obviated unaided surveillance. The purpose is to create an environment in which residents know that normal restraints of surveillance by citizens and their authorized agents exist, albeit aided by electronics. While initially costly, CCTV often reduces security personnel requirements or obviates the need for expensive redesign of existing structures. 995

Specialties SECURITY Electronic Systems Electronically aided surveillance is not equal to personal surveillance. A corrective response to a detected crime is obviously a step further away if the viewer sees the crime on a TV receiver rather than on the spot. The deterrent of having a policeman or other person on hand is lost. There is also the possibility of equipment malfunction. But CCTV has a quality of its own: being watched while unable to ascertain who, if anyone, is doing the watching is somehow unnerving, and definitely is a deterrent. A remotely controlled surveillance camera can be fitted with an automatic panning device so that the camera swings from side to side continuously, even when no one is monitoring the system. CCTV System Requirements In general, a CCTV system should perform at approximately the same level as commercial broadcast receivers. Specific equipment and the quality of image needed are determined by characteristics of the area under surveillance, schedules of operation, makeup of the monitoring staff and their expected responses to emergencies, and use of special equipment. American and foreign manufacturers have TV cameras suitable for security work. All equipment should meet the standards of the Electronic Industries Association for CCTV. Service and maintenance are generally more difficult and expensive

than installation; therefore, the capability and reputation of a local supplier is crucial. City police or traffic departments often have had experience with manufacturers, suppliers, and maintenance operations. To encourage reliance on the system by users, and to prevent criminals from taking advantage of a lapse, the CCTV system should break down as infrequently as possible and be repaired quickly in the event of a breakdown. Picture resolution depends primarily on camera quality and lighting levels: Higher lighting levels permit the use of less sensitive, less expensive cameras. The entire system should operate unattended. This requires electronically stable equipment, meaning, for example, that no one should be required to constantly adjust the lens of the camera. It is difficult to project costs of CCTV systems because of the variety of system sizes and configurations and the range of equipment costs. Camera prices start as low as \$200, but more sophisticated models, such as those sensitive to very low light levels, cost up to \$10,000 each. Complicated accessories including zoom lenses, remote pan (side-to-side movement) and tilt (up-and-down movement) mechanisms, and low-light equipment can increase installation and maintenance costs tremendously. The cost of monitoring equipment can be as low as the cost of a conventional television receiver, but more specialized and sensitive equipment is far more expensive. Camera Locations Locations of a CCTV camera and the light level at that point are key cost-effectiveness factors. A camera's location defines the area to be observed by the camera, and the nature of the location greatly influences the camera's vulnerability to theft and vandalism. CAMERA HOUSING - CAMERA J LENS &-; "7 MIRROR PROTECTIVE WINDOW Fig. 55 Recessed camera. Available lighting dictates the type of camera needed to produce a final image of adequate quality. Of course, supplemental lighting may be provided at additional cost. The camera must be able to view an area that is significant in terms of crime control. Wide-angle or other special lenses should be avoided by choosing a different camera location. Most importantly, the camera itself must be protected from theft and vandalism. This means that the body and lens of the camera should be in an inaccessible place. A mirror is often used to reflect the image into the lens, so that the expensive lens will not be broken by pointed instruments, thrown objects, or bullets (see Fig. 55). All interior cameras should be placed inside sturdy housings which are installed with tamper-proof connectors. Cameras must be accessible for maintenance and repair, however. A number of

locations meet all of these requirements. An elevator in a high-rise building is often protected by CCTV. The camera is generally mounted on the outside of the elevator cab wall so that the image passes via a mirror in a corner of the elevator to the protected lens. In case of camera failure, the elevator must be stopped so that the camera maintenance man can step onto the top of the cab and reach over the side to repair the units. This is not overly inconvenient for repairmen, but it does make access to the camera more difficult for a potential thief. Building lobbies are another common location of interior cameras. Lobby cameras are commonly hung from the ceiling or recessed into the ceiling. The elevated locations require that the repairmen use a ladder. Use of a ladder, however, would make a thief very conspicuous. Outdoor locations usually depend upon inaccessibility to protect equipment from theft and vandalism. Cameras are located atop steel poles or on poles extending from roofs or walls. An alternative is to place the camera in a wall or window of an accessible apartment. Lighting for CCTV Systems Lighting plays a key role in the cost and effectiveness of a CCTV system. For camera locations inside buildings, it is almost always less expensive to raise the light level than to use low-light-level equipment. The required lighting level is only slightly higher than normal for building interiors, can be achieved without glare, and has an intrinsic value as a crime deterrent. Exterior lighting can be very expensive. Cameras used outdoors are almost always more flexible and sensitive, being capable of adapting to full sun, cloudiness, and dusk. But as indicated earlier, camera costs rise dramatically for low-light-level equipment. While increasing of lighting levels is also expensive, well-designed extra lighting again has an intrinsic value as a crime deterrent. Monitoring of CCTV Systems Effectiveness of CCTV depends on the nature and quality of monitoring. Many people may be used as monitors: city police, project security personnel, members of organized tenant patrols, tenants acting as individuals, and various combinations of these groups. The choice depends principally upon availability of personnel and their monitoring costs. City police will monitor CCTV systems only if they believe it is the most efficient use of manpower. Thus an area being surveyed must suffer large numbers of crimes to warrant hiring a policeman or civilian whose function is simply sitting, watching, and adjusting. Crime reduction or criminal apprehension through CCTV monitoring would have to be substantial to justify continued use of such manpower. Police use of CCTV systems is

generally limited to shopping districts and city-center areas. Police normally monitor large systems that include several cameras (each equipped with pan, tilt, and zoom capability) and a monitoring console, so that the viewer can watch activity in several places at once and adjust his equipment to concentrate on a particular place, incident, or individual. Commercial and industrial facilities often hire private security personnel to monitor CCTV systems. Guards are used less frequently in residential complexes. The major advantage of use of guards is that a single guard can control several entrances to a building or complex of buildings. Usually the guard can see all entrance doors, the lobby, and the elevator interiors on the monitor screens. He can be given audio contact with the lobby area. With the use of an intercom

Specialties system, he can also control garage and front door entrances. He can also be given the ability to stop the elevator in midflight. Thus the security guard can see and hear every person entering the premises; he can prevent them from entering; and he can even exert some control after they enter. It is also possible to staff a monitoring panel with members of tenant patrols. Use of volunteer personnel eliminates payment of guard salaries. Because they are personally acquainted with the project residents, tenant monitors can easily pick out strangers and perhaps distinguish a minor argument among friends from an impending fight. But, there are serious drawbacks in using tenant monitors. It is difficult to guarantee the performance of unpaid people. The novelty of working with TV monitors will wear off quickly, and declining interest increases the likelihood of patrol members simply not showing up. Additionally, tenant patrol members are not equipped or empowered to take much action. The tenant monitoring the CCTV has no real authority over police or security personnel. Finally, there is the problem of tenant patrol members using their position to harass or intimidate other tenants. An alternative is in-apartment tenant monitoring. Tenants of a building or housing project can monitor CCTV on their home TV screens. By connecting CCTV equipment to a master antenna within a building, tenants can have the option of tuning into unused TV channels to monitor lobby, elevator, playground, or parking lot activity. Tenants may watch CCTV when they are expecting someone to arrive, or when a child is playing within viewing range of a camera in a playground area. Older people may watch for less specific reasons. Obviously, this does

not assure continuous monitoring, but if one or more of 200 tenants is watching, it would be risky for intruders to take chances. An in-apartment tenant monitoring system requires that a cable TV or master antenna system be in operation in the building. CCTV is clearly most suited to large, high-rise dwellings. Picture quality of the CCTV systems should be comparable to that of commercial broadcasting to promote tenant usage. While some picture disintegration may be acceptable in a conventionally monitored CCTV system, there should be no distortion in a system designed for in-apartment monitoring. It is desirable (and generally not expensive) to install a microphone system so that sound accompanies the TV picture, which makes the system more interesting and enjoyable. It is possible to organize a voluntary in-apartment monitoring program to improve coverage. A tenant organization could arrange for persons to watch CCTV in their homes during specified hours. Such a scheduled system would promote better coverage and facilitate participation because there would be no requirement that residents leave their apartments. Also, CCTV monitors should be placed where responsible individuals, such as management staff and patrolling guards, are at work or pass by continually.

INTERCOM SYSTEMS Most urban multifamily dwellings are equipped with buzzer-reply systems to limit access to the building to tenants and to people who have been interviewed by tenants on an intercom system. An atypical buzzer-reply intercom system in an apartment building functions as follows: A panel located outside the lobby entrance door lists the names and apartment numbers of all tenants in the building. Next to each tenant's name is a call button that when pressed rings a bell or buzzer within that tenant's apartment. The tenant responds to the call by walking to a panel mounted on the wall of his apartment and speaking via an intercom system to the person outside the door. When identification is satisfactorily established, the tenant pushes a button on the panel which momentarily allows the entrance door to be opened without a key. Because the costs involved in installing wiring for such a system in an existing building are very high, buzzer-reply systems should be installed in all new buildings during the construction phase. A modified version of the traditional buzzer-reply system has recently come into use. Local telephone companies install and service front-door intercom systems that use existing telephone wires instead of a separately wired system. The panel mounted **SECURITY Electronic Systems** outside the lobby door differs from a conventional panel in that it

is supplied with a telephone receiver and the list of residents has a three-digit number next to each name. A person wishing to enter the building dials the appropriate three-digit number which makes the phone of the tenant buzz (not ring). The tenant then speaks with the person over the phone. If recognition is established, the tenant dials "4" to open the front door. If a tenant is speaking on the phone when the buzzer sounds, he can depress the receiver once, speak to the person in the lobby, buzz him in by dialing "4," and then depress the receiver again to return to his initial telephone conversation. For tenants without telephones, a special unit that can be used only for the intercom can be installed. Fees for installation and service are billed by the phone company and added to the tenants' monthly rent.

ELEVATOR AUDIO SYSTEMS Use of audio systems in elevators is rapidly increasing. An elevator audio system is an uncomplicated sound-transmission installation consisting of a microphone and speaker located in the elevator cab and connected to similar devices near the elevator doors on each floor. The system allows someone inside the elevator to speak to anyone standing in the elevator waiting area, and vice versa. In office buildings or high-income residential buildings, an additional connection is made so that a doorman, guard, or maintenance man can respond to persons inside the elevator. In low-income housing, the equipment in the cab is simply connected to the elevator on each floor. Some systems are designed to remain on at all times, but most require the person in the cab to push a button before he can talk to the outside location. A continuous voice relay system reassures the elevator rider that he can communicate with the outside if any trouble arises, whereas the need to push a button limits the usefulness of a noncontinuous audio system in crime situations. Any elevator audio device is useful when breakdowns occur and someone is trapped inside the cab. 997

Specialties

COLOR THEORY Primary colors The longest extended slices on the color wheel (Fig. 1) show the three primary colors - red, yellow, and blue. They're called primary because all the other colors come from combinations of these three colors. Secondary colors Mix any two primary colors and you get the secondary colors: Orange (red and yellow) Violet (red and blue) Green (blue and yellow) Tertiary colors All of the other six colors on the wheel are called tertiary, or intermediate, colors. They are a mixture of the primary colors plus an adjacent secondary color. Thus:

Yellow orange (yellow and orange) Yellow green (yellow and green) Blue green (blue and green) Blue violet (blue and violet) Red violet (red and violet) Red orange (red and orange) Color has three dimensions: the hue, distinguishing one color from another - such as red, green, blue, etc.; the value, denoting lightness or darkness; and the tone or intensity, which is the brightness or dullness. These hues, values, and intensities can appear to change when different ones are used together. Two or more light values combined afford little contrast; nor will darker values in combination provide much interest. But, when a light value is used with a dark, the light appears lighter while the dark appears darker, White is the lightest of all colors, and values range from it through varying gradations of gray to black. Colors that are nearer white in value are called tints and colors that are closer to black in value are called shades. Intensities or tones also have similar effects. A brightly upholstered chair will appear brighter and will stand out when used with a carpet of dull color, as it will produce a spot of interest. In contrast, a few dull-colored pieces of furniture will sink into the background if the room contains brighter-colored rugs, draperies, and other furnishings. Contrasting or opposite hues will emphasize one another. Red with green will make the red look redder and the green appear more orange, while the red-purple will take on a bluish tone. There are many ways of combining colors for interest. Related color schemes such as reds, purples, and blues together can produce very pleasing effects. Contrasting hues, such as blues with oranges, can also be combined to give more vibrant results. Some people enjoy excitement. Warm colors such as yellow, orange, and red are exciting because they are associated with things like sunshine, fire, heat, and even blood. Warm colors tend to "advance," and a predominantly warm-colored wall will seem to come forward. They are especially effective in rooms that are on the east or north side of a house, because light entering from those directions seems to be a cool light. The warm colors and cool light complement each other and make the room seem cozier and warmer. Cool colors are those associated with water, verdure, and the sky - blues, greens, and violets. These tend to "recede," and under most conditions, light, cool-colored walls will create an illusion of greater space. They are good choices for rooms on the south and west side of the house, since these areas receive a lot of sunlight all year around. There is a cooling effect in the warm-light areas, another complementary association. Black, white, gray, and brown - and the tones of the latter two, known as griegie and beige

- are not considered to be colors so much as neutrals. In practice, they are the "no-color" colors, which are used with other colors to modify them or to contrast with them. But they are far from being negative. As you work with color, you will find that all colors are influenced by the company they keep. This is particularly true of the tints, shades, and so-called neutral colors. A juxtaposition of two muted colors, such as a gray and a tan, will bring out latent greens, lavenders, and pinks you did not see before. Colors also have visual weights. Dark and bright appear heavy, while light or dull seem to weigh less. Remember that a dominant color is the one that "controls" a room, while the others are accents.

Basic Color-Scheme Planning
Successful decorating often depends on how well the total effect is anticipated. Here are four types of schemes that professional decorators have in mind when they start to plan a job. They are no guarantee of perfect results, but they do make an unwieldy subject easier to handle.

Monochromatic This scheme is built around one color, using it somewhere in its full intensity and then varying it with a number of shades, and tints of the same color. For example, in a monochromatic scheme of yellow, the range could be from dark shades of gold, through clear yellow, to light, pale-yellow tints. A monochromatic color scheme can be restful, create a feeling of spaciousness, and provides a good background for art objects, collections or similar decorations. But generally, when employing a monochromatic color scheme, the interest of the room comes through by using a variety of textures and patterns.

Analogous or related Because it's the easiest color scheme to work with, an analogous scheme is the one that enjoys the greatest popularity at the present time. It is based on two or three colors, such as yellow, yellow orange, and red orange, that lie close to each other on the color wheel, with "relief" provided by tints and shades of the same that have been tinged with adjacent greens or vermillion. The analogous color scheme is restful and refreshing also, and the colors are more interesting because of their variations in intensity and value. It is the kind of color scheme that is easily changed; a slight shift of emphasis here and there is all that is necessary to completely change the character of the room.

Complementary or contrasting This scheme, which is rapidly coming into favor, uses colors that are opposite each other on the color wheel - blue and orange, red and green, yellow and violet. One color is usually a primary color and the other a secondary color. Using such contrasting colors will give a lively and vibrant room, but it is a color scheme that must be used

with caution. One color should always dominate, with the others being primarily dramatic accents. The "shock" impact of a complementary color scheme can be softened by selecting unexpected shades and tints of the two colors. That is, a vivid color and its complement can be quieted, if you prefer, by graying them, or reducing their values. Employing a pair of opposites in this manner means that there will be both cool and warm colors in a room, which makes a mutually complementary association. A complementary color scheme tends to make a room seem smaller. Accented This is a combination of adjacent, related, or analogous colors - call them what you will - accented by a bold touch of color from the opposite side of the wheel. An example would be a scheme ranging through a number of strong, soft, and grayed yellows, spiked with purple or violet. There are also other color schemes, such as triad and split-complementary that you can adapt from the color wheel, but the four suggested above are the easiest to visualize and to carry out. In whatever basic scheme you use do not forget the neutrals: black, white, the grays, and browns - to which you might add metallic gold and silver. Since they will appear, of themselves, in the wood and metal of your furnishings, they must at all times be considered for the part they play in the total effect. If you wish, the neutrals can constitute a fifth, and very sophisticated, color scheme of their own! But usually they must be more or less just "accepted," and played up or played down by the colors you combine them with. Incidentally, some black and white is an asset to almost any color scheme, but too many and indefinite neutrals, used with stronger colors, tend to compromise a color scheme and make it look confused or merely drab. It is best to think of any neutral as a distinctive note of color, whether it is the fieldstone of a fireplace or a hardwood floor.

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Specialties COLOR THEORY WHITE * BLACK Fig. 1 The basic wheel with a three-dimensional projection of the attributes of color- hue, value, and intensity- as shown in their relation to one another. The circular band represents the hues: G, green; B, blue; P, purple; R, red; and Y, yellow. The upright center axis is the scale of value. Paths leading from the center indicate color intensity. 999

THEORY Color Schemes and Combinations PRIMARY- WARM {^
COLORS | *gg and tones which they dominate. COOL COLORS' Blue

Green [Violet and, tones NEUTRAL J White 1 atones I ^hthev COLORS 1
 Black *mi***t*h dominate. { J grey or frown. EXAMPLES OF
 ANALOGOUS (RELATED) COLOR SCHEMES DIRECTIONS FOR
 USING COLOR COMBINATION CHARTS COLOR WHEEL NUMBER
 1 Illuslrarei color comblnolloni In which the ihre* prlmorlei or* used
 together, or the three secondary colon, or three terllory celeri, The three
 point of each triangle link lthe colon used In this TRIAD COLOR
 SCHEME. The rule for succeii li lo uie only on* of I hem In o ilrong, bright
 ion*, In small ar*ai, with lthe olher two In roftened (or grayed] ronoI.
 COLOR WHEEL NUMBER 2 lluiifrotes lthe color pairs which or* effective
 loaether through contrast. ThU Is tolled lthe COMPLEMENTARY COLOR
 SCHEME. As the arrowi Indicate, lh* pain or* exactly opposite •oeh olher
 on the wheel, On* should b« used In a brlghl Ion* for imaller areas, the
 olher lit grayed lanes ond larger atelai, COLOR WHEEL NUMBER 3
 lluilro.es lthe ui« of o color wllh the two which ore rtextl la lti oppoill* on
 the wheel, one on each ilde. This SPLIT COMPLEMENTARY COLOR
 SCHEME follawi lthe rule for comptemenlortes, and may Include lthe dlrecl
 contrast color olio, If deilred. For example, yellow may be used with blue-
 vlolel, and red*vloUi, with or without lthe lruue vlolel ihode which com*!
 between them. To avoid confuilon, not oil of lthe lrlanglei or* Indlcoled.
 Cul a piece of pope* the size of lthe lrlongle ond loy lIt wllh lh* top polnl ot
 ony color you cheat*. TW olhor polnli will reit on the correct colon.
 COLOR WHEEL NUMBER 4 lluilrotei how on ANALOGOUS COLOR
 SCHEME U developed by uilng colon which are relaled because lthey or*
 side by side M the wheel. Any group con be used, oil around lthe wheel, oi
 Indlcoled by lthe dolled lines. Cor on accent color you can use a contrail
 color oppoillie ony one of your group. For lmlonce, In lh* yellow-orange lo
 red group, complementary blue could be used for occenl (ihown an color
 wheel 2). alack, white, fl»oy ond olher definitely neutral tones can be used
 wllh any combination of colors. BASIC PRINCIPLES FOR WORKING
 OUT A COLOR SCHEME 1. DOMINANT OR CONTROLLING COLOR
 Dteldo on your dominant or controlling color, which may damlnalo by
 caytr. Ing o lorgo area or by ilronglth af color In o tmailtr orta. Docldo
 wholhor your ioundollan ar background color ll lo bo lho damlnonl or o
 locondary color. Hon lo g» a largo omounl of qulol background calar, o
 imall omaunl of bold, ltrong color. All largo faundollon oroai ihould bo In
 light ar grayod form, 5. GRAYING Cltar colon oro gayor, mart chtrful, but

grayed lanol oro mora rtllful, lh.li harmonlti rtior. tubllo. Mining gray wllh brlghl colon bring, thorn Into rtlallon with olhor colon In lh. room. Ai . . . rod ond yollow In brlghl lon.i loom la cloih. Mlxod wllh gray, thty bocamo roi« and ton and ga vary wall tog.lh.r. Um trill principle olio In buying malorlali. Avoid loo much graying. ll glvti muddy lonti, dlrly grayi, flat groom. A llrtlo gray goal far. 3. RELIEF AND CONTRAST COLORS Dtelldo on rtlltf and conlrait colon ond bring th*m Into all porli of Iho room compoilllon. R.flltmb.r Iho ordtr In lh. omaunl of ipaco ollowod «och ont~foundallon, thin nlllof, lh.n conlroil. All colon—including background colon—ihould b. koytd lo tht damlnonl color. Soflin itrong conlroil colon wllh whllo, Conlroil ll lon In llghlor llnil. Soflon dorkor conlralli wllh gray. 4. ACCENT COLORS Um purt brlghl liUmm colon only In occ.itorlii, tic. Dliiribult lhtm to lhty will nol bt ipolly. Tht tmolltr lht ano lht brlghl.r lh. color may bt. Th. largtr lht orta lht lafltr lht lont ihould bt. Don'l um largt omounli of purt bright color. 3. KEYING Thli ll another mtont af crtollng harmony, A kty color It tht ont about which lht calar Khtmo It built—tho dominant, or controlling color. All olhor colon In lht room muil bt "ktytd" wllh ll—hormonlztd. Two colon In which ony port of a third color ll prtntl will bt linked togthtr. Examplti To kty rid and ytllo to ooch olhtr, mix thtm both wllh a Unit of lht third primary huc~blui. Yloltt ond grttm will rtiult, and thtit oro hormonloui to um with your ltrong tonti. Rtmimbtr thli principle In buying al wlll al mixing colon. A lovtly print or ort ob|tcl will havt lhtM lon.i ktyod for you, ond you cot um lhtm for yaur own competition. Tht lofo rult li to avoid loo mony colon ond loo ilrong lon.i txcopl In acctnli, tic, Moil colon will "go logtlhor" If you lofltn thtm.

Specialties COLOR THEORY Terminology and Combinations WHAT DECORATORS MEAN WHEN THEY USE THESE COLOR TERMS (HUE) Eoch l«ellon In the color wh.*l U calUd a hu*. To chang* a hut, onolhor color (not black, whltt or purt gray) mutt b* added to It. Every hut hai a different wave length (ram *v*ry othir hui. Mlx*d with In camp l*m*nt •qually It product! gray. PRIMARY COLORS) Alio calltd "normal," alw "lundamintat." frlmarlil on tht thrtt plgmtnl colorl which cannot bt praducod by any nilxtur* af olhtr pig. m*nti. Thtit art rod Mk« that o(a geranium fiow*r, yollow Ilk* that o(rip* ltmom, blut Ilka tht d*«p door hut of a tunny touthtrn iky. SECONDARY COLORSt 5tcondarltl art tht thrtt colon which art productd by mixing two of tht thrtt prlrnarlti In

tqual amounti. Rtd -f. ytlaw := arangt) rod -f- hlut = purplt tor viol*!})
 ytlow 4- blu* rs gr*tn. TERTIARY COLORS. Ttrllarlti art tht colon
 praductd by mixing a primary with a Itcandary, tht txact ihadt dtp«ndlng
 upon tht proportion. Rtd -f. orang* productl lhadtl luch ai ruit*t, burnt
 orangt, coral, tic. R*d -f purplt—mul- btrry, amthylt, orchid, ttc, Blut -4-
 purplt—htllolropt, ptrlwlnxlt, lavtn- dtr, ttc. Blut -f gr**n—lurquolt*,
 aquamarln*, bolllt grtn, tic. Ytlow -f orang*—malx*, prlmroM, flam*,
 tic. Ytlow 4* sr*«n—ladt, Nil*, «llv*, charlrtuM, tic. MUtural of
 complmtntarlti not Includtd btcauw th*M pro. due* thadei at gray—a
 neutral. Som* authortl*! conildor, alio, th* ihad*l produced by mixing two
 i*condarl*i ai ttrtlarlti, iuch ai ilatt, citron, buff, lagt, ttc. COMPLEX
 COLORSi All colorl which art madt vp of mora com'pllcattd mlxturti than
 thoit producing itcondary and ttrtlary colon art calltd compUx. NEUTRAL
 COLORSi Black and whit* or* conlld*r*d n*ulral. Alio all thai* Hull and
 lhad*l In which tan*l of gray or brown predominate. TINTSi Th* light lon.i
 mulling whin whit* ll mlxid wllh a color. Much whit* makll a color cold.
 SHADES* Th* dark ton« r*cu(tlng when black U mixed with a color.
 Much block d*ad*m th* color. TONEi Each hu* hot many ton*!. By Ion*
 —or lonal valu*—w* m*on th* relative •Irangth of th* hu* ai It approach*!
 black or whit* at th* oppoilt* *ndi of th* valu* teal*. Mlx*d with while, a
 color It "pel*" In Ion*) mlx*d wllh black. It U "dark" In ton*. Th* upper
 ond low*r *xlr*m*l of any color would b* whit* (or v*ry pal* gray), and
 block. CHROMAr Thlt l*rm ll ui*d Interchangeably wllh valu*, tonal
 valu*, ond ln< tonilty. Th* chroma of a color luch oi ytlow W "light") th*
 chroma af a color luch ai Navy blu* ll "dark." Wh*n a color fadil, II Iomi
 chroma. LUMINOSITY) Thli l*rm ll uiid lo d*icrlb* a quollly of warm
 dear colon In light- r*fl*clng lon*i and finish*)*, luch ai light
 goldon.yollow. Cltor whltt li olia lumlnoul, Llltratly "lumlnoui" ar* only
 metall In gold, llv*r, platinum, ar cl*ar plaltlu. SUGGESTED COLOR
 GROUPS COLOR TO USE WITH IT ,td 0«*n, gray, blu* (for accent)
 SCA'LET Light blu*, icru (or Navy and rM**| CRIMSON pMrl groy,
 mauv* OARNET Sapphire blu*, mauv*. p*arl grar CARDINAL , Marin*
 blu*, turquolM, gray WINE Hack, aid blut, bilgi «°SE Fleih, light blut.
 grim CI" «°SE Blu. In varloul lhadtl CEDAR ROSE Blut, crtam "NK •
 Gr**n, orchid, blu* for accint ORANGE Vlolt, light blu.. Indigo for accent
 BURNT ORANGE Electric blu., light brown SALMON , Turquallo,
 lavender HENNA , Peacock green, royal blue, g)ay 'EACH Run, blue, Ian

MAIZE Powder blue, pink YELLOW Violet, blue, groan PRIMROSE ,
 Lownder, duity role, toll green SOFT YELLOW Brown, French blu*
 GOLD Soft groy.grtan, d**p rid DARK GREEN Brown, bilg* (or lag*
 grim and gilil MYRTLE H.llatrop., y,||aw SOFT GREEN Rouwaad, deep
 violet TARRAGON Hellalrap., pearl gray CHINESE JADE Ran, Ivory
 NILE ,. ..Cornflower, orang* LIGHT OREEN Rale, dork green, mauve
 BLUE Yellow, land, orange for occinl COPENHAGEN BLUE , Burgundy,
 gray FLEMISH BLUE Ollv.-gr.en, cardinal LIGHT BLUE Orchid,
 champogn* DEEP PURPLE Orang*. gray VIOLET Grain, llghl and dork
 ihadoi LAVENDER Gr**n, mauv*, gray HELIOTROPE Llghl blu*, cream
 HYDRANGEA Old ran, prlmrou yillow MAUVE Emerald gr**n, dark r*d,
 brawn BROWN Orang*, Ion, cardinal lor accint °*>Y Vlol.t, crlmion,
 lavender, WALLS MEDIUM To LIGHT COLOR AREAS AND SAMPLES
 PROPORTIONATE SAMPLE SIZES 1. WALLS WxI*" 2. FLOOR 18" x
 18" J. DRAPERIES lo"xlo" 4. CEILING U"xM» 3. COUCH, ETC 12" xU"
 o. WOODWORK 10"x14" 7. LARGE CHAIR 8" x 8" 8. LARGE CHAIR
 o" x 8" 9. SMALL CHAIR o" x 7" 10. ACCESSORIES «"x7" 11.
 ACCENTS 4"xo" 12. TRIMMINGS S" x 6" FLOORS DARK TO
 MEDIUM VARIED ai UI*d h.r* meant cholc* of light,' dark, or medium
 tan.i, clear ar grayed colorl. Drapery varied CEILING LIGHT LARGE
 COUCH 5E.D SOFA VARIED POINTS TO REMEMBER |N MATCHING
 SAMPLES FOR COLOR 1. Uf* largor tamploi If ponlbl*, *ip*c|ally In
 patterned rnatorlall, but k**p approximate proportion! of chart. 5lx*1 ar*
 d*1*rmInld ac. carding to araa and Inloroit. Celling and floor areai, far
 example, ar* *qual—but floor In* tereit li gr*at*r, h*nc* th. larg.r lampl*.
 If MV.ral ll*mi ar* th. lam* color add lh.m to mak. on. lampl*. 2. Mak.
 allowance far texture. Soft rough lurface In paint, paper, or fabric mak.t
 colon appear dork.r. Hard glatiy lurfacet appear lighter, 3. Mak* allowance
 for dlilance. Colon look brighter when lhey are cloitf farther away th.y ••em
 lafl.r, grayed by almaiphore. Calari which match exactly I ft. away may
 Mem quite dlffer.nl at IS ft. Thli li Important In a large hlgh-colllngod
 room. WOODWORK VARIED VARIED 4. Mak. allowanc. lor proximity.
 Wh*n lid* by lido) Complimentary colorl brighten *och alherj related
 colon, when both llshl ar both dark, deaden each other) neutral colon
 brighten clear colon, but pure itrong prlmory calan deaden n.utrall luch 01
 grayi, brawns, •tc,) light and dark lon.l brlght.n »ach ath.r, •specially while
 for dark colon ond black for light lonoi) one calar may M*m to chano*

another! hu. ai wh.n a itrang cliar color glv.i a tinge af III complementary to a n.utral —rod, far .xampl*. may glv* a gr**nlih calt lo gray unl*n a lllll. red hai b.«n mixed wllh th. gray, 5. Mak. allowanc. for proportion, Th. larg.r th. orea th* darker the calar wjll appear. ChooM a wall color llghlly llght.r Ihon you really Want It. Dan't d.cld. .xocl ihad. of a palnl.d wall until alt olh.r mat.rlali hav. been choien. It ll .oil.r lo malch paint lo fabric and fiap*t than th* olh.r way around. LARGE, CHAIR s. LARGE CHAIR BRIGHT SMALL CHAIR 9 ACCESS' 0RIE5 IO BRIGHT ACCENTS rciMMINGS II 12 COLOR COMBINATION CHARTS S = SECONDARY T = TERTIARY 1001

Specialties COLOR THEORY Munsell System of Color One of the best-known and widely respected systems of color standardization used in the United States today is that developed by Albert H. Munsell. He became greatly interested in the practical application of color and was disturbed by the fact that the popular names for colors did not describe them adequately for professional purposes. They are named after flowers or plants, such as violet, indigo, old rose, primrose; after fruits, such as peach, pomegranate, grape, avocado, plum; after places such as french blue, naples yellow, or Prussian blue; or after persons, such as Davy's gray or Hooker's green. Essentially the system consists of an orderly arrangement of colors in the shape of a three-dimensional color solid. The system is based on a color circle of ten major hues made up of five principal hues, red, yellow, green, blue, and purple, and five intermediate hues, yellow-red, green-yellow, blue-green, purple-blue, and red-purple. Each hue is indicated by a symbol as follows: Red: R Yellow: Y Green: G Blue: B Purple: P Yellow-red: YR Green-yellow: GY Blue-green: BG Purple-blue: PB Red-purple: RP Each of the above major hues has been given a value of 5 in the inner scale around the hue circle (see Fig. 2, hue symbols), i.e., 5 R, 5 YR, 5 Y, 5 GY 5 G, 5 BG, 5 B, 5 PB, 5 R and 5 RP. Between each of the major hues are values of 2.5, 10, and 7.5 for rough indication of hue. The outer scale of the hue circuit is divided into 100 segments to provide greater accuracy for indicating hue where needed. In the Munsell color tree each hue (H) is allotted ten segments of the hue circle, making 100 hues, and these hues form the horizontal center, or equator, of the color solid. The center segment of each color is considered the true color, and the remaining segments in each hue section vary according to their proximity to adjoining colors; for

example, as red gets closer to yellow it contains more yellow, and this is indicated by the numerical designation. The value (V) notation denotes the lightness or darkness of a hue, which is determined by a neutral core at the center of the hue circle. The core contains ten gradations from a supposedly perfect white (one having 100 percent reflectance) at the top to 0, a perfect black (having 0 percent reflectance) at the bottom. The chroma (C) notation indicates the saturation of the hue, or the strength of the color. The chroma scale extends outward from the central core or axis, and the increments vary from 0 at a neutral gray to as high as 16, according to the amount of saturation produced by a given hue at a given value level. Since colors vary in chroma, or saturation, some colors extend farther from the neutral axis than others, and the solid is therefore not symmetrical, Pure red, with a chroma of 14, for instance, extends farther than blue-green, with a chroma of only 6 (see Fig. 1). A Munsell notation indicating hue, value, and chroma (H V/C) might be given as follows: Vermilion: 5R 5/14 Rose: 5R5/4 With this information it is possible to describe exactly any given hue and to locate its place in the color solid. Furthermore, as Munsell stated, one can "select one familiar color, and study what others will combine with it to please the eye," by the use of three typical paths: one vertical, with rapid change of value; another lateral, with rapid change of hue; and a third, inward, through the neutral center, to seek out the opposite color field. All other paths are combined by two or three of these typical directions in the color solid. 55 ° Fig. 2 Munsell hue symbols and their relation to one another. 1002

THE RED FAMILY REPRESENTATIVE MEMBERS OF THE RED FAMILY . . . FLESH . . . DUSTY-PINK . . . SHELL-PINK . . . CORRECT USES ... In light tints, chormlno bockground for fairly light, lorge rooms ... In bright tones, ROSE . . . DUSTY ROSE . . . OLD ROSE . . . CARDINAL OR CRIMSON . . . RASPBERRY . . . RED- highly decorative when used for comporoHvely smoll oreas, as in accessories, occents, etc. . . . In darker BURGUNDY OR WINE RED . . . MAROON . . . ETC shodes, rich and warm for draperies, carpets, upholstery, especiolly in large rooms with heavy furniture. CHARACTERISTICS . . . Warm, advancing . . . Cheerful, hospitable, active ... In strong tones, stlmulat- CAUTION ... Do not use too much red—especially In clear, bright tones, and En light rooms . . . Safer to ing, hold, viral, dramatic, exciting. soften a little by "graying," except when

used for accent ... Do not use the Red Family for background WHAT THEY CAN DO . . . Make objects seem closer, larger . . . Make room seem smaller by bringing back- of a room that is small or crowded ... Do not use bright red for large objects, unless you want to call ground closer . . . Focus attention on wall or object . . . Bring life, brightness, warmth, to drab, dark attention to them ... Do not forget when using the Red Family to include cool colors In your color or too-cool! rooms. schemes ... It takes two cool colors, or a large area of one cool color, to balance red. SUGGESTED COLOR SCHEMES IN WHICH MEMBERS OF RED FAMILY PLAY A DOMINANT ROLE

DOMINANT COLOR SHELL-PINK MELON PINK (like Pompeiian terra cotta) VENETIAN PINK DUSTY PINK BOIS DE ROSE DAWN ROSE LIGHT ROSE ROSE (1) ROSE (2) ASHES OF ROSES DUSTY ROSE (1) DUSTY ROSE (2) OLD ROSE (1) OLD ROSE (2) OLD ROSE (3) RED (1) MAJOR WALL COLOR Shell-pink Melon Pink Venetian Pink Dusty Pink Woodwork, cocoa brown Sail de Rose Dawn Rose Light Rose Warm Gray Oyster White Ashes of Roses Deep dusty Rose Pickled wood Old Rose Paper, Old Rose, cream and gray Paper, Blue and: rose pattern Soft grayed Green MAJOR FLOOR COLOR Light Blue Brown Patterned rug, soft tones, green, mauve, red Deep Brown Sage Green Dark Green Deep Blue-green Rose Taupe Rose Red Cafe au lait Brown Pinkish Beige Warm Gray Mulberry Ivory Patterned rug with red background

DRAPERIES AND UPHOLSTERY Off-white background, powder-blue and shell-pink In pattern Light tan background, melon-pink with green olive in design Draperies, yellow and white stripes Upholstery, green and dull pink stripes Draperies, pale mauve (dusty pink curtains) Chair and bed covers, cocoa and pink stripes with cocoa brown trimming Draperies, pearl gray, trimmed with bois de rose Upholstery, pale gray with sulphur yellow Neutral background, dawn rose and leaf-green in pattern Draperies, dusty rose background, blue-green and foliage tones in design Upholstery, blue, blue-green and green, plain, pattern and stripes Draperies, rose Upholstery, rose and cream stripes Draperies, oyster white Upholstery, oyster white with reds, blues and soft green White Lighter dusty rose, white and brown Draperies, dusty rose Upholstery, foam green and natural Blue and pale yellow, with touch of old rose Old rose and gray Draperies, Ivory background, rose, blue and green in pattern Upholstery, old rose, and drapery fabric Draperies, pale gray background, red In pattern Upholstery, plain green, red and green stripes

ACCENT COLORS 1 ration Red Green

Olive Red x White Green and Sulphur yellow Wet Leaf-green Blue Blue or Green Red Strong Blue Silver Pale Green Jade Green Blue and Silver Green Blue and Orange REMARKS Charming for bedroom. In slightly darker tones of same colors, could be adapted to living rooms. Feminine feeling Good for living or dining room with mahogany or walnut furniture Good for living room with 18th Century furniture Interesting for modern bedroom with furniture in cocoa tones Living room color scheme with feminine feeling Attractive for bedroom Suitable for fairly large room with average tight Appropriate for almost any living or bedroom Charming for bedroom or woman's living room Adaptable to bedroom or living room Good color scheme for living room or dining room Adaptable to living room, traditional or modern This color scheme is charming and delicate, suitable for bedrooms and dressing rooms With woodwork painted worm gray, this scheme would be attractive for bedroom or dressing room Attractive for formal living room Suitable living room or dining room

RED (2) RED (3) RED (4) RED (5) SOFT RED U) SOFT RED (2} CRIMSON (1} CRIMSON (2) CRIMSON (3) CHINESE RED LACQUER RED (1) LACQUER RED {2} ITALIAN RED CRANBERRY RED WINE RED {1» WINE RED C2> WINE RED (3) AMERICAN BEAUTY RED DAHLIA OLD RED BURGUNDY (1) BURGUNDY (2) BURGUNDY (3) BURGUNDY (4) Paper, granite flecked with red Paper, red toile design; woodwork, white Old white Woodwork, dark green Paper, light background, red with green and brown in pattern Neutral White Bone white Soft Gray Slate Gray Grayed soft Green Gray Paneling, Red Lacquer, silver trim Yellow Paper with light ground and cranberry pattern Paneled fireplace end painted blue Green Soft grayed Blue Paper in yellow, pale gray and white Linen color Gray Soft light shade Old Red Pale dear yellow Burgundy Warm Gray with pinkish cast Setge with pinkish cast Patterned rug, red predominating Hooked rugs Floors, painted blue Red rug Dark blue Soft Red Rug, blue, red and white pattern Crimson Crimson Soft Crimson Chinese Red Brown Black Floor, stained dark Patterned rug in old reds and dark blues, red predominating Deep soft red Wine Red Deep Gray Dark Wine Red Fawn Red Dahlia Deep Old Red Burgundy Darker Burgundy Burgundy Burgundy Chintz, red predominating In pattern Draperies, textured beige Upholstery, textured beige trimmed with red Red and white brocade Red Draperies, blue-green background, deep red in design

Upholstery, blue-green Draperies, linen color with red in pattern
 Upholstery, chair seats, blue White background, floral design in red and soft
 green Draperies, white, gray and crimson Upholstery, white, trimmed with
 crimson Draperies, bluish white, trimmed with soft- crimson Upholstery,
 bluish white Draperies, grayed soft green Upholstery, deep beige Gray and
 lacquer red Silver gray, with red and black in design Draperies, Italian red
 damask. Upholstery, red damask, yellow, glazed chintz with white ground
 and red in design Draperies, rose Upholstery, blue, And chintz-like
 wallpaper pattern Draperies, wine red Upholstery, grayed green and off
 white Draperies, wine red Upholstery, wine red, grayed white and blue
 Draperies, Ruby taffeta Upholstery, glazed maroon chintz with yellow
 flowers Chair seats, bright yellow leather Draperies, fawn and American
 beauty Upholstery, American beauty, fawn, blue and gray Draperies, light
 neutral background with dahlia and melon green in pattern Upholstery,
 melon green Draperies, old red, beige and white stripes Upholstery, light
 tan Draperies, white, valance and trimming Upholstery, burgundy and white
 Burgundy and natural Draperies, primrose yellow Upholstery, pale grayed
 blue and white Draperies, beige background, shell pink and burgundy In
 pattern Upholstery, shell pink Green and Pewter Blue (chairs painted blue)
 Green Blue Yellow Silver White Black Gold Light Beige Green Silver Gold
 Green Crystal Silver Gold Blue Larkspur Blue Copper Gold Chartreuse
 White Grayed While Good for Early American room, with ptiw or maple
 Colonial type living room or library Living room fn period feeling with
 walnut furniture Early American dining room, with pine and maple Nice for
 living room 18th Century dining room Mahogany furniture of very nice
 design would be lovely with white walls A smart sophisticated color scheme
 Adaptable to modern styles Good for living room or dining room Colorful
 living room with some red lacquer furniture Modern library Living room
 English In feeling; also good for Federal American Cheerful living room
 color scheme Attractive for living room with cool North or East light For
 fairly formal living room, average size Specially appropriate for Victorian
 dining room Dramatic color scheme for fairly large living room Adaptable
 to various types of living rooms Warm, colorful plan for a room inclined to
 be cold Rich color scheme for formal room For large room, living room or
 library Good for any room not too small or too sunny Sam* as above

THE ORANGE FAMILY REPRESENTATIVE MEMBERS OF THE
 ORANGE FAMILY . . . IVORY . . . PEACH . . . CORAL . . . BEIGE five
 for "toning" up a room In too dull or quiet color, or warming a cold room. .
 . . RUST . . - TERRA COTTA . . . WARM BROWN. CORRECT USES . . .
 Best En off shades, except for accent ... In softened tones, excellent
 background color CHARACTERISTICS . . . Always warm, advancing . . .
 Cheerful, welcoming, gay, vibrant, glowing ... In for dark or cold rooms . . .
 Wonderful accent color En proper combinations. itrong lone., akin to red ...
 In softer Jon«, o good mixer. CAUTION ... Do not use too much of the
 clear color ... In large areas It has the disturbing quality of WHAT THEY
 CAN DO ... In slightly less degree. Orange repeats the activities of the Red
 Family . . . Effec- red ... Do not use for background of small room unless
 you want it to be very "cozy." SUGGESTED COLOR SCHEMES IN
 WHICH MEMBERS OF ORANGE FAMILY PLAY A DOMINANT ROLE
 DOMINANT COLOR IVORY PEACH (1) PEACH (2) PEACH (3)
 PEACH (4) PEACH (5) PEACH (6) APRICOT CORAL (1) coral m
 WARM BEIGE (1) WARM BEIGE (2) WARM BEIGE 13) WARM BEIGE
 (4) WARM BEIGE 15} HENNA (1) HENNA (2) TERRA COTTA
 COPPER (1) COPPER (2) BURNT ORANGE WARM BROWN CD
 WARM BROWN (2) WARM BROWN (3) WARM BROWN (4) MAJOR
 WML COLOR Ivory Woodwork, Ivory Peach Peach Peach Paper, shades of
 yellow, through peach to brown Fiorol pattern paper, white ground, peach
 and green Yellowish Pink Apricot Paper, silver ground, coral- rose design;
 Woodwork, coral Coral Woodwork, soft Blue Warm Beige Pink and Beige
 wallpaper Warm Beige Worm Beige Beige with pink cast Gross cloth, fan
 and gold Mint Green Woodwork, Cream Pink Terra Cotfa Pine paneled
 Rough plotter with oak paneling Neutral Woodwork, walnut Tobacco
 Brown Tan Yellow Brown Pale Yellow MAJOR FLOOR COLOR Floor,
 painted ivory Rug* Jade Green Peach Old Green Worm Brown Brown Rust
 Eggplant Rose ond Cream Aquamarine Gray Blue Warm Beige Warm Beige
 Warm Beige Dork Toupe Light worm beige Henno ond block Henna
 Eggplant Floor, Pine Rugs (hooked] In tones of orange, yellow, green Ook
 plonk floor Patterned rug, tones of brown, green, copper Rug, greenish
 background, with burnt orange and henna Worm Beige Warm Brown
 Orange Brown Orange Brown DRAPERIES AND UPHOLSTERY Ivory
 background, rose and blue green in pattern Apple green Draperies ond
 upholstery, peach, with old blue Draperies, brown and coral stripes

Upholstery, brown background, with coral, beige and tan Draperies, brown
 Upholstery, light and dark peach Draperies, peach Upholstery (choir seats),
 green Draperies, peach background, blue in design Upholstery, peach and
 blue Draperies, old rose Bed and furniture covering, apricot, trimmed with
 black Aquamarine Draperies, off-white background rose and green in
 design. Coral valance; Bed and furniture covering, copper rose and same
 chintz used for hangings Brown and Copper Draperies, old white and beige
 Upholstery, dusty pink and pale olive green Draperies, burgundy
 background, -white and beige in design; Upholstery, burgundy and natural
 Draperies, russet (beige glass curtains) Upholstery, brown with tan
 cushions Brown, beige and white stripes or checks Henno, green and gray
 Draperies, henno with valance of bedspread material Bedspread, henno,
 light and dark green and tan stripes Upholstery, same combination in
 patterned material Draperies, pinkish yellow, trimmed with terra cotta
 Upholstery, bats de rose Draperies, copper toned background with orange,
 yellow and green in pattern Draperies, copper colored, coarsely woven
 material Upholstery, neutral green, trimmed with brown and copper Burnt
 orange and henna Chintz in clear yellow, beige and worm brown in design
 Draperies, copper and topaz; Upholstery, worm browns Burnt orange and
 apple green Shades of warm browns and orange ACCENT COLORS Rose
 and Jade Green Bittergreen Old Blue Copper Yellow Green Coral and
 yellow Orchid Aquamarine and Silver Off-white Bright Green Terra Cotta
 White Tan Sky Blue Gold White Pale Green Blue and copper Green Blue
 White Orange, French Blue Greenish Blue Silver REMARKS Charming for
 bedroom or lady's sitting room with ivory painted furniture Attractive for
 bedroom, modern or traditional For a room that needs warming up, with
 cool touches Modern color scheme appropriate for living room or library
 Good modern living room color scheme Very good for late Colonial dining
 room with Duncan Phyfe style furniture Good color scheme to lighten dark
 bedroom Charming for young girl's room Charming for woman's bedroom
 Attractive for bedroom with furniture painted blue Modern living room or
 library Restful living room color scheme Good for living room, library, or
 man's bedroom Restful, chromatic color scheme Appropriate for informal
 living room or boy's room Living room or man's bedroom Man's bedroom
 Charming for dining room in Directoire feeling with furniture painted
 yellow and gold Appropriate for living room in Early American feeling with
 Early American style furniture Good color scheme for large, formal,

English-style living room, with furniture In natural oak and walnut Good
 for dining room with walnut furniture Modern living room or library
 Appropriate for bay's room Restful, cheerful color scheme for library
 Attractive for modern living room or dining room

THE YELLOW FAMILY "f^rou? MEMBS °F TM«oEUOW FAMILY "■
 CREAM ' * * BUFF * " STMW " CANARY of yft!,ow Mmtm Itt ■*»*■
 qua,, "y « « *o"" «,or- rHA»i*rT»ieri/-c * w " " * " CORRECT USES . . .
 Excellent background for all average rooms ... In light tints, bait wall-
 background for <-HAJW.,ItRJ5TIC5 - • • Warm, somewhat advancing . . .
 The sunlight color—gay, happy, bright, cheerful poorly lighted roomi ... In
 clear, bright tone*, safe accent color almost everywhere. ... In light ton**,
 luminous, radiant. CAUTION . . . Do not use yellow without totting under
 artificial Hghf, and providing lamp shades to offset WHAT THEY CAN DO
 ... Diffuse and increase light by reflection, making dark rooms seem lighter
 and brighter color changes ... Don't us* in wide expanses in a very sunny
 room ... Don't um bright tones without ... In pale tints, yellow lights up a
 smalt room without makino it seem smaller because reflective radiance
 restful combination color. SUGGESTED COLOR SCHEMES IN WHICH
 MEMBERS OF YEUOW FAMILY* PLAY A DOMINANT ROLE
 DOMINANT COLOR CREAM BUFF PALE YELLOW (1) PALE
 YELLOW (2) BRIGHT LEMON (1) PALE LEMON 12) JONQUIL
 YELLOW (1) JONQUIL YELLOW (2) YEUOW (1) YELLOW (2)
 YELLOW (3) YEUOW (4) GRAYED YELLOW SOFT YEUOW EMPIRE
 YELLOW CITRON YELLOW LEMON YELLOW SULPHUR YEUOW
 MAJOR WALL COLOR Cream Woodwork, cream Buff Pale Yellow Pale
 Yellow Bright Lemon Pale Lemon Yellow Woodwork, white Jonquil Yellow
 Jonquil Yellow Yellow Yellow Morbleixed yellow paper Woodwork, deep
 green Paper in Yellow and Ivory stripes, divided by narrow plum lines
 Grayed Yellow Soft Yellow Slate Gray Citron Yellow Lemon Yellow
 Sulphur Yellow MAJOR FLOOR COLOR Potterned rug, mulberry, green
 and cream Buff Soft Beige Pale Yellow Beige Tobacco Brown Gray Soft
 Blue-Green Brawn Brown Deep Green Yellow Tan Brown Deep Brown
 Lime Green Citron Yellow Tele de Negre Olive Green DRAPERIES AND
 UPHOLSTERY Draperies, green, cream trimming Upholstery, green and
 green yellow Copenhagen blue and burgundy Draperies, light yellow
 background, soft reds, greens and blue In pattern Upholstery, soft blue,

chintz of draperies Turquoise Draperies, beige background, yellow green and lavender Seat covers, wet leaf green Draperies, white with yellow trimming Upholstery, emerald green; some pieces white, yellow, dull orange Draperies, white, trimmed with Chinese 'red Upholstery, warm gray and white Draperies, white and yellow Upholstery, soft blue, green and white Blue and apple green Dutch blue and white Draperies, green, yellow trimming Chair seat upholstery, yellow Draperies, gray background, yellow, plum and rose in pattern Upholstery, blue and light tan Draperies, brown and beige stripes Upholstery, yellow, beige, moss green Draperies, yellow Upholstery, cinnamon brown Draperies, Empire yellow Upholstery, Strong clear yellow Coral Brown and henna Shades of green and sulphur ACCENT COLORS Mulberry Orange, Tete de Negre Green and Red Coral Wet Leaf Green Orange Chinese Red White Black Bright Red Blue Old Gold White Chartreuse Silver Silver Orange Coral REMARKS Appropriate for bedroom, especially in Directoire* feeling with cream and gold furniture Glowing color scheme for living room or inon'i bedroom Good color scheme for medium-sized dark room Colorful for bedroom or small sitting room This color scheme will brighten up a dark dining room Suitable for living room with north light Charming for living room, modern or traditional Good for any room without too much light Good for room with cold light Attractive for Informal living room Charming for dining room, especially in Directoire feeling Good combination for dark maple woodwork Pleasant for living room or man's bedroom Attractive and restful for library or living room Suitable for living room or dining room Modern or traditional living room Distinctive for modern living room Colorful modern living room

DEEP YELLOW MUTED GOLD GOLD (1) GOLD (2) TAN (1) TAN (2) TAN (3) TAN (4) BROWN (1) BROWN (2) BROWN (3) BROWN (4) BROWN (5) SABLE BROWN (1) SABLE BROWN (2) CHESTNUT BROWN TAWNY BROWN GOLDEN BROWN NUT BROWN TOBACCO BROWN CHOCOLATE BROWN DARK BROWN (1) DARK BROWN (2) BROWNS Deep Yellow Coromel Yellow, flat finish Woodwork, Olive Green Gray and pale Yellow paper Neutral Tan Tan linen color Tan Brownish Tan Brown paneled Pine paneled Warm Beige with brown carpet Pine paneled Chalk White Sable Brown Sable Brown Fawn Tawny Brown Pine Knotty Pine Nut Brown Pine Tobacco Brown Chocolate

Brawn Deep Beige Light Chortreuse Light Brown Red Bleached wood,
 rubbed with gold and waxed Old gold carpet Harvest Gold and Gray Dark
 Ton light and dark Tan Brown Taupe Floor, Ook Multi-colored scatter rug*
 Deep Brawn Pine Brown Hooked Rugs Brown Off-white Deep warm Beige
 Chestnut Brown Light beige and taupe Golden Brown Navajo rug Moss
 Green Beige Eggshell Dark Brawn Dork Brown Deep Brown Draperies,
 gold damask Upholstery, pivot, wine red, gold with red, blue, lavender in
 partem Draperies, soft caramel taffeta, trimmed with brawn Upholstery,
 brown and yellow Draperies, old gold, trimmed with green Upholstery,
 alive green and paprika Draperies, oyster white, trimmed with multi-color
 fringe Upholstery, rust-shot silk Draperies, burnt orange Upholstery, brawn
 and bumf orange Draperies, fan and rose stripes Upholstery, linen color
 with rose, tan, gray and blue in pattern Dark green and vermilion Draperies,
 brown with red, yellow and blue pattern Upholstery, old red and yellow
 chintz Beige background with brilliant gold, scarlet and orange tones of fall
 foliage Old chintz in blue ond brawn Draperies, brown, beige and dusty
 pink Upholstery, brown and off-white Yellow, copper and blue chintz
 Turquoise Draperies, off-white with turquoise Upholstery, shell pink and
 off-white Draperies, bright yellow Upholstery, plain chartreuse with white
 pattern Champagne background, beaver, turquoise and apricot in pattern
 and trimming Blue on light ground Draperies, colorful hunting print
 Upholstery, red leather Draperies, dark linen Upholstery, green, brown and
 white Clear yellow chintz with beige and dark brown in design Draperies,
 white Upholstery, chartreuse, brown and eggshell Draperies, pale, dear blue
 Upholstery, cinnamon brown Light tan and brown Draperies, off-whit*
 Upholstery, Wedgwood green and off-white Gold Ebony Light Green Brass
 Rich Chocolate Rose Brown Blue Blue Silver and Pewter White Green
 Peach Turquoise Earth Brown Turquoise Yellow Blue and Green Yellow
 White Chartreuse Pale Clear Blue White Gold Suitable for period room
 with Queen Anne, Sheraton, and other Georgian style furoilure Unusual
 modern living room scheme where there is plenty of light Charming for
 modern living roam with blond wood Traditional or modern dining room
 Good with natural wood rones Attractive for living room or library with
 walnut woodwork and furniture Rich, warm color scheme for living room
 or library, especially with walnut furniture and paintings Suitable for
 bedroom with oak furniture in Early English feeling Attractive for living
 room with plenty of tight Dining room in French Provincial style

Appropriate for modern living room Suitable Early American living room with maple furniture Charming bedroom color scheme Distinctive modern living room Same as above Appropriate for living or dining room Good for dining room in French period feeling Library or Den Restful living room color scheme Suitable modern dining room with much sunlight Interesting modern color scheme for living room or dining room Good for dining room or living room Good for modern living room Suitable for living or dining room

THE GREEN FAMILY REPRESENTATIVE MEMBERS OF THE GREEN FAMILY . . . NILE . . . LETTUCE . . . PEA . . . GRASS . . . Hon to room. ' " " UVE ' " * BOTTIE * • • ETC- * ■ CORRECT USES . . . , One of best background color* far average rooms, especially where reitfulneu ii Impor-CHARACTERISTICS . . . Cool, receding—except when mixed with a warm color . . . Most restful color . . • tont . . . Great corrective value for rooms too small or too warm . . . Suitable in proper tones for Friendly with all other colors, refreshing, versatile . . . Endless variety of tones and combinations, background In any port of room—floor, walls, ceiling. WHAT XHEY CAN DO ... In light, toft lints, makes rooms seem larger because the wall seems farther CAUTION ... Do not use in quantity in cold, dork or overlarge rooms—choose warm, advancing colors foil away » . . Mokes objects seem farther away, therefore smaller . . . Brings atmosphere of rest and relaxa- backgrounds, keeping green for smaller oreas. SUGGESTED COLOR SCHEMES IN WHICH MEMBERS OF GREEN FAMILY PLAY A DOMINANT ROLE DOMINANT COLOR PALE GREEN CD PALE GREEN (2) LIGHT GREEN CI) LIGHT GREEN (2) LIGHT GREEN (3> APPLE GREEN (1) APPLE GREEN (2) APPLE GREEN (3) SOFT GREEN U> SOFT GREEN (2) SOFT GREEN (3) FOAM GREEN IVY GREEN DEEP LIME MAJOR WALL COLOR Pale Green Pole Green White Pickled Pine Off-white Apple Green Apple Green Pale Apple Green Soft Green Soft Grayed Green Pale Soft Grayed Green Slate Gray Clear Beige Deep Lime MAJOR FLOOR COLOR Dark Green Plum Light Green Light Green Soft Light Green Plum Yellow Green Floor, brown walnut Rug, blue and tan Deeper Green Deeper Green Ivy Green Foam Green Ivy Green Deep Lime DRAPERIES AND UPHOLSTERY Draperies, off-white background, with pale blues, greens, and mouve in pattern Upholstery, darker blue Draperies, natural linen color, with flowered

plum and green in design Upholstery, plum, gold, green Draperies, white with dark green pattern Upholstery, dark green and white Draperies, off-white and light green Upholstery, light green She! I-pink, given and off*white Draperies, apple green Upholstery, gold, yellow and ivory Gray, blue, and touches of light yellow Draperies, royol blue background, with rase and green leaves in pal tern Upholstery, some drapery chintz, also rose, antique salmon, apple green ond cream stripes Draperies, plum background with beige and green in pattern Upholstery, plum, beige and green Draperies, corn yellow Upholstery, grayed green and off-white Draperies, soft grayed green Upholstery, golden yellow and white Draperies, lemon yellow and while Upholstery, lemon yellow and gray Draperies, beige with light and dark green floral design Upholstery, same chintz and some clear beige White, green and melon pink ACCENT COLORS Mauve and Violet Gold Yellow Brown Crystal Gold Light Yellow Black, gold, while and ruby Orange Pine Green Lacquer Red Gold Mack White REMARKS

Appropriate for average living room and bedroom Especially good for traditional living room Pleasant color scheme for modern room Attractive for living room ar library Charming and cool for small living room or sitting room Good for small living room This combination makes cool room Early American living room with maple or cherry furniture Restful Color Scheme Adaptable to living room, dining room or bedroom Excellent to add feeling of space to small room Good modern color scheme Appropriate for living room Dramatic modern scheme, especially good with blond wood

NILE GREEN MINT GREEN JADE GREEN SAGE BOTTLE GREEN CHARTREUSE GREEN CHARTREUSE GREEN TURQUOISE BLUE GREEN (11 BLUE GREEN {2} BLUE GREEN (3) WET LEAF GREEN DEEP GREEN (1) DEEP GREEN (2) DEEP GREEN (3) DEEP GREEN (4) DEEP GREEN (5) GEORGIAN GREEN DARK GREEN (1) DARK GREEN (2} DARK GREEN (3) GREEN OLIVE Nile Green Pure White Woodwork, while Pale Jade Slate Gray Pale Apricot Chartreuse Green Gray Light Turquoise Deep Cream Blue Green Dull Blue Green Deep Lime Deep Green Gray and White paper. Block pattern Green, lighter than carpet Greenish Gray Deep Soft Green Deep Georgian Green Ivory Green Warm Gray Dark Green Green Olive Green Painted Mint Green, spatter- dashed with turquoise and yellow Dark Blue Soft Deep Sage Bottle Green Shades af Tete de Negre Chartreuse Patterned rug. Green with Mots Jtose Blue

Green Blue Green Bug, light field, red violet and green leaves In pattern
 Bronze Gray 'Deep Green Deep Soft Green Deep Green Light Brown Deep
 Green Dark Green Dark Green Tan ReaSCoral Glazed chintz with green
 background and while In design, red lining and trimming Turquoise, yellow
 and mint green Draperies, blue Upholstery, blue and jade green Gray and
 blue with green touches Green, apricot and topaz Draperies, chartreuse
 green Upholstery, shades of heliotrope Chartreuse and bright blue
 Draperies, turquoise, green and rose stripes Upholstery, turquoise Apple
 green, greenish blue, touch of burnt orange Draperies, light grayish tan
 background, turquoise, rose and green in pattern Upholstery, same print and
 some soft rose Draperies, white Upholstery, red violet While background,
 wet leaf green, dawn rose and bright lemon in pattern Draperies, sky blue
 chintz with rose and green pattern Upholstery, emerald green and gray
 Draperies, yellow Upholstery, yellow flowers and pale green leaves on gray
 background Draperies, off-white Upholstery, off-white and Wedgwood
 green Draperies, apple green Upholstery, grayed greens and white Golden
 yellow arid white Draperies, soft golden yellow Upholstery, golden yellow
 and deep green Draperies, white with dark green pattern Upholstery, off-
 red, off-white and dark green Draperies, dark green or gray background
 Upholstery, light green Draperies, chintz In blue-green and soft red
 Upholstery, same chintz, also some soft red lime green, red-coral, antique
 white Crystal Whit* Silver Silver Topaz Silver Silver Rose Burnt Orange
 Green Dark Green Iose Gold and Hose White and Green Yellow Salmon
 White Gold Black Yellow Copper Coral and While Cool, airy bedroom
 Adaptable for informal living room, dining room or bedroom Attractive for
 modern living room Excellent color to make small sunny room seem larger
 and cooler Good for living room or dining room Modern bedroom. Good
 with furniture painted chartreuse Adaptable to any modern room Attractive
 for bedroom with mahogany furniture Versatile color scheme for average
 room Very restful for living room or bedroom Dining room in period feeling
 with walnut furniture Sophisticated modern color scheme Suitable for living
 room or man's bedroom Modern living room or dining room Very cool and
 fresh Good combination to make small room seem larger Modern or
 traditional setting Very cool and restful for period living room or library
 Subtle color combination. Good for living room or dining room Cool and
 restful for living room Charming for sunny Hying room Daring modern
 color scheme. Good with traditional or mod- era furniture in light finish

THE BLUE FAMILY REPRESENTATIVE MEMBERS OF THE BLUE
 FAMILY . . . PALE . . . BABY . . . 5KV . . . POWDER . . . medlar because
 they Mem more dttant ... In dark tones, make lighter contrast color more
 luminous. A . . . MIDNIGHT . . ETC. CORRECT USES . . , In light tones,
 excellent background for wall, dark, warm rooms . . . Good combining
 CHARACTERISTICS . . , Coldest, most receding, unlets mixed with warm
 color ... Serene, quiet, "spa- color, especially In soft tones - - . Effective
 background for many other colors. clous" . . . Much-loved hue . . . Too
 much of it In dull fanes may be depressing. CAUTION . . . Do not use In
 quantity In cold or dark or over-tore, rooms ... Do not use too much In dull
 WHAT THEY CAN DO . . . Make room seem larger, cooler, more airy and
 spacious . . . Make objects look shades ... Do not use without some warm
 bright accent color. SUGGESTED COLOR SCHEMES IN WHICH
 MEMBERS OF BLUE FAMILY PLAY A DOMINANT ROLE
 DOMINANT COLOR PALE BLUE U) SKY BLUE POWDER BLUE (1)
 POWDER BLUE (2) LARKSPUR BLUE (1) LARKSPUR BLUE (2)
 BLUE (1) BLUE (2) BLUE (3) HYDRANGEA BLUE (1) HYDRANGEA
 BLUE (2) COPENHAGEN BLUE PENCIL BLUE MAJOR WALL
 COLOR Pale Blue Garden Sky Blue Powder Blue Powder Blue Larkspur
 Blue Larkspur Blue Pate Blue, deep Rose and Ivory paper Faded Blue
 {middle value} Striped wallpaper in tones of light and medium blue and
 White Pale Hydrangea Blue Hydrangea Blue Copenhagen Blue Lemon
 Yellow MAJOR FLOOR COLOR Dark Blue Champagne T>elft Blue
 Powder Blue Pale Gray Blue Blue Floor, dark Brown Carpet, Gray and
 Yellow Dark Blue Eggplant Deeper Blue Burgundy Pencil Blue
 DRAPERIES AND UPHOLSTERY Tan draperies and upholstery
 Champagne, sky blue and orchid Draperies, canary yellow Upholstery,
 yellow and powder blue White draperies and upholstery Draperies, wine,
 trimmed with white Upholstery, deep blue and white Neutral background,
 blue and pink in pattern Blue, gold and rose with touches of black—in
 stripes or plain Draperies, old yellow Upholstery, yellow, old yellow, and
 touch of Venetian red, also some blue Draperies, blue with white in pattern
 and trimming Upholstery, lemon yellow Draperies, peach background with
 white, copper, gold and hydrangea blue in design Upholstery, same chintz,
 also old blue Draperies, 'salmon pink Chatr seats, black and gold Gray with
 blue and burgundy Blue background with yellow In pattern and Mm
 ACCENT COLORS Silver Orchid Off-white Peach Deep Blue and Wine

Red Dohlia Block Blue Dark Blue Old Blue Gold Rose and Silver Silver
 REMARKS Modern color scheme for bedroom with furniture In lemon
 color Modern color scheme, good with light natural finish woods Cool and
 fresh for bedroom Dainty feminine bedroom Good modern living room
 combination Any period room with enough light Attractive for traditional
 living room Bedroom with Directoire feeling and walnut furniture Good for
 small, low-ceilinged but tight room Good for room with strong light,
 especially with 18th Century furniture Dining room with Directoire feeling
 Attractive for traditional living room Setting for dining room with modern
 furniture

MEDIUM BLUE til MEDIUM BLUE (2) MEDIUM BLUE (3) DUSTY
 BLUE SOFT DULL BLUE OLD BLUE GRAYED BLUE TURQUOISE
 BLUE TURQUOISE BLUE GREEN BLUE ROYAL BLUE (1) ROYAL
 BLUE [2) ROYAL BLUE (3) DEEP BLUE (1) DEEP BLUE 12) DEEP
 BLUE (3) DEEP BLUE (4) DEEP BLUE (5) GARDEN POOL BLUE
 DARK BLUE Blu* Blue Cream Dutty Blue Ivoty Old Blue Gray Blue Pale
 Turquoise Blue Grayed Turquoise Green Blue Wolls, Silver Woodwork,
 Royal Bios Old White Dull White Pate Amethyst Deep Blue Woodwork,
 [vory Deep Blue Pole Yellow Creamy White Ivory Cream and beige- paper
 Mulberry ground Mole Blue Dark Burgundy Soft Dull Blue Deeper Blue
 Deeper grayed Blue Turquoise Grayed Turquoise Plum rug; flowered
 pattern Floor., painted Gray Rug, Blue Royal Blue Rug, Deep Blue ground
 with honey-yellow in pattern Deep Blue Dork Blue rug with Tan and Rose
 In pattern Deep Blue Natural Blue Garden Pool Blue Dark Blue rvg with
 row and tan In pattern Draperies, cherry red Upholstery, chintz In blu*, rose
 and mauve, some pieces in cherry red and gold Lavender, gray, and some
 rose French Blue Draperies, gray, trimmed with soft blue Upholstery, soft
 blue ond gray Ivory background with blue, rose and green in design, some
 pieces in old rose Faded Pink, or chintz-with blue, green and pink Yellow,
 white, and gold Draperies, golden yellow Upholstery, golden yellow and
 white Wine and ivory Draperies, peach Upholstery, green-blue with some
 plum Draperies, bio*, trimmed with silver Upholstery [chain], silver and
 blue leather Clear yellow Draperies, royal blue Choir Soars, royol blue
 Morocco Deep blue, trimmed with gold Draperies, dull ivory Upholstery
 (chairs), ivory or rosy red leather Draperies, deep saa blue Upholstery,
 Canary yellow Draperies, deep tea blue (geld gauze glass curtains)

Upholstery, deep blue Draperies, deep blue Upholstery, yellow with dash of white Draperies, garden pool blue Chair Seats, red leather Draperies, linen color with blues, greens and rosy reds In pattern Chair Seats, dark blue Gold Rose Jade Green Crystal Green Green Red White Polished Brass Peach Silver and Black Silver Silver Amethyst and Gold Red White White Gold, Rose-Pink and Blue Whit* and Silver Blue and Silver Good for living room, especially In 18th Century French feeling Adaptable for lady's bedroom In lighter blues, also to living room In darker shades of duller blues Good for south living room or bedroom Restful for living room or dining room Bedroom or informal living room Charming for living room or bedroom Dining room In Directoire feeling with mahogany Any room not too large or too dark Dignified but friendly living room Simple living room In French Provincial feeling Distinctive modern dining room Modern living or dining room Attractive for modern dining room, especially with lemon wood furniture Cool and charming for living room not too dark Colorful for dining room with plenty of light Modern living room or dining room Good when blue is dominant color In a dark room Good for bedroom, furniture painted blue with flower decorations, and some smaller oyster white pieces Dramatic modern living room with furniture In rich mahogany or walnut tones Good for dining room where tight fit is needed

THE VIOLET (OR PURPLE) FAMILY REPRESENTATIVE MEMBERS OF THE VIOLET FAMILY . . . ORCHID . . . LAVENDER . . . MAUVE ...
 ... Create restful, quiet atmosphere when used in soft tones. . . . PURPLE, CORRECT USES ... In light, soft tints, excellent wall and ceiling background for an overage room . * . CHARACTERISTICS ... Cool when mixed with blue, warm when mixed with red ... In pure form, cold Strong shades good for accent ... In deep, soft tones, attractive for carpets, upholstery, draperies, and formal . . . In purple tone., rich and dignified but not friendly ... May be depressing. CAUTION ... Do not use blue tones of violet in cold, dark, over-large rooms ... Be careful when using WHAT THEY CAN DO . . . Add to impression of room size and coolness, especially when mixed with blue strong shades for dominant color ... Do not use without some warm contrast. SUGGESTED COLOR SCHEMES IN WHICH MEMBERS OF VIOLET FAMILY PLAY A DOMINANT ROLE
 DOMINANT COLOR ORCHID (1) ORCHID (2) LAVENDER (1) LAVENDER (2) LAVENDER (3) HELIOTROPE (1) MAUVE (1)

MAUVE (2) HELIOTROPE (2) VIOLET MULBERRY U) MULBERRY
 12) MULBERRY (3) PURPLE MAJOR WALL COLOR Orchid Paneted
 paper in Orchid and Pale Yellow Lavender Lavender, Blue and White paper
 Pale Lavender Gray Gray Paneled wall painted Pale Mauve Pearl Gray
 Dove Gray Dusty Mulberry Dutty Mulberry Scenic paper Woodwork,
 walnut paper In Gray and toft Purple stripes MAJOR FLOOR COLOR Blue
 Mulberry Lavender with mouve border Plum Rose Beige Mauve Deep
 Violet Heliotrope Black and White Mqhogany Ebony Deep Mulberry Gray
 DRAPERIES AND UPHOLSTERY Champagne, orchid and blue Green,
 yellow and orchid chintz Gray, light blue and touches of fade green
 Lavender Pink, lavender and white Heliotrope draperies and upholstery
 Light blue, Nile green, some rose Mauve and yellow Draperies, heliotrope
 trimmed with silver Upholstery, Tarragon green, gray and heliotrope
 Draperies, vioTet Upholstery, coral and old gold Draperies, clear blue
 chintz with mulberry and brown in partem Upholstery, clear blue Draperies,
 creamy peach chintz with gray, old rose and ebony design Upholstery,
 creamy peach Mulberry Purple, with.blue, green and gray ACCENT
 COLORS Black and Silver Green Jade Green Rose Lavender Violet and
 Silver Rose Crystal and Septa Green Silver and Black White Black Orange
 Burnt Orange REMARKS Attractive .for living room Cool, airy bedroom
 Good for sunny room Feminine bedroom Bedroom with warm light Lovely
 color scheme for woman with gray hair Good for sunny bedroom Especially
 attractive for Louis XVt style bedroom with walnut furniture Cool and
 restful living room Attractive with silver gray painted woodwork and violet
 lines, also white ond gold furniture Very good for traditional living room or
 dining room Good for sunny living room ar dining room Traditional dining
 room Half or living room

Specialties COLOR THEORY Reflective Values; Safety Color Guides
 TABLE 1 Reflective Values Color Approx. percent of reflection White, dull
 or flat White, gloss Light tints Cream or eggshell Ivory Pale pink and pale
 yellow Light green, light blue, light orchid Soft pink, light peach Light
 beige, pale grey Medium tones Apricot Pink Tan, yellow gold Light greys
 Medium turquoise Medium light blue Yellow green Old gold, pumpkin
 Rose Deep tones Cocoa brown, mauve Medium green, medium blue
 Unsuitable dark colors Dark brown, dark grey Olive green Dark blue, blue
 green Forest green Natural wood tones Birch and beech Light maple Light

oak Dark oak, cherry Redwood Black walnut, mahogany 75-85 85-90 79 75
 75-80 70-75 69 70 56-62 64 55 35-50 44 42 45 34 29 24 21 20 10-15 12 5-
 10 7 35-50 25-35 25-35 10-15 10-15 5-15 Safety Color Guides Physical
 hazards: Red: Fire protection equipment and apparatus; danger; stop
 Orange; Dangerous parts of moving machinery Yellow: Physical hazards
 that might cause stumbling, falling, etc. Green: Safety - first-aid dispensary
 or kits, stretchers, safety deluge showers, etc. Blue: Caution against
 movement or use of equipment being worked on such as elevators,
 scaffolding, etc. Black and White: Traffic direction; sanitation Equipment in
 industrial plants: Red: Fire protection systems and equipment Orange:
 Dangerous materials, nonflammable, such as acids, alkalis, toxic materials,
 gases, oxygen Yellow: Dangerous materials, flammable, such as fuel oil,
 gasoline, kerosene, alcohol, propane, butane, acetylene, hydrogen, and
 solvent Green: Safe materials, such as drinking water, service water, brine
 Blue: Protective materials Violet: Valuable materials Black: Electrical
 conduit Recommended ceiling values should be in the range of 60-90%.
 Floor reflection values should be in the range of 15-35%. Overall reflection
 values of a room should be in the 35-60% range.

Specialties COLOR THEORY Color Effect TABLE 2 Color Effects of
 White Fluorescent Lamps Lamp appearance; effect on neutral surfaces
 Effect on "atmosphere" Colors strengthened Colors grayed Remarks Cool*
 While White Neutral to moderately cool Orange, yellow, blue Red Blends
 with natural daylight Deluxe' Cool White While Neutral to moderately cool
 All nearly equal None appreciably Best overall color rendition; simulates
 natural daylight Warmt White Yellowish white Warm Orange, yellow Red,
 green, blue Blends with incandescent light Deluxe! Warm While Yellowish
 white Warm Red, orange, yellow, green Blue Excellent color ren- dition;
 simu laics Incandescent IlKhl Daylight Bluish white Very cool Green, blue
 Red, orange Usually replaceable with CW White Pale yellowish white
 Moderalely warm Orange, yellow Red, green, blue Usually replaceable
 with CW or WW Sofl White- Natural Pinkish while Warm, pinkish Roil,
 orange Green, blue Usually replaceable Willi CWX or WWX Color Effects
 of Mercury and Filament lamps Lamp appearance; effect on neutral surfaces
 Effect on "atmosphere" Colors slrenghened Colors grayed Remarks
 Mercury Greenish blue while Very cool, greenish Yellow, green, blue Red,
 orange Poor overall color rendering White Mercury Greenish white

Moderately cool, greenish Yellow, green, blue Red, orange Color- Improved Mercury Yellowish while Warm, yellowish Yellow, green Blue Color rendering often acceptable, but mil equal to any white fluorescein Deluxe White Mercury While Moderately nml Orange, yellow, blue Green Color rendering good: compares favorably with CWX fluorescent rilamentl Yellowish while Warm Red, orange, yellow Blue fAii'llonl color rendering
 * Greater preference at higher levels, t Greater preference at lower levels.
 1014

Windows are available in many types, each having advantages. The principal types are double-hung, casement, stationary, awning, and horizontal sliding. They may be made of wood or metal. Heat loss through metal frames and sash is much greater than through similar wood units. Glass blocks are sometimes used for admitting light in places where transparency or ventilation is not required. Insulated glass, used both for stationary and moveable sash, consists of two or more sheets of spaced glass with hermetically- sealed edges. This type has more resistance to heat loss than a single thickness and is often used without a storm sash. Wood sash and door and window frames should be made from a clear grade of all-heartwood stock of a decay resistant wood species or from wood which is given a preservative treatment. Species commonly used include ponderosa and other pines, the cedars, cypress, redwood, and the spruces. Double-Hung Windows The double-hung window is perhaps the most familiar window type. It consists of an upper and lower sash that slide vertically in separate grooves in the side jambs or In full- width metal weatherstripping. This type of window provides a maximum face opening for ventilation of one-half the total window area. Each sash is provided with springs, balances, or compression weatherstripping to hold it in place in any location. Compression weatherstripping, for example, prevents air infiltration, provides tension, and acts as a counterbalance; several types allow the sash to be removed for easy painting or repair. Ibe jambs (sides and top of the frames) are made of nominal 1-inch lumber; the width provides for use with dry-wall or plastered interior finish. Sills are made from nominal 2-inch lumber and sloped at about 3 in 12 for good drainage. Sash are normally 1% inches thick and wood combination storm and screen windows are usually V/s inches thick. Sash may be divided into a number of lights by small wood members called muntins, A ranch-type house may

provide the best appearance with top and bottom sash divided into two horizontal lights. A colonial or Cape Code house usually has each sash divided into six or eight lights. Some manufacturers provided preassembled dividers which snap in place over a single light, dividing it into six or eight lights. This simplifies painting and other maintenance. Assembled frames are placed in the rough opening over strips of building paper put around the perimeter to minimize air infiltration. The frame is plumbed and nailed to side studs and header through the casings or the blind at the sides. Where nails are exposed, such as on the casing, use the corrosion-resistant type. Hardware for double-hung windows includes the sash lifts that are fastened to the bottom rail, although they are sometimes eliminated by providing a fingergroove in the rail. Other hardware consists of sash locks or fasteners located at the meeting rail. They not only lock the window, but draw the sash together to provide a "windtight" fit. Double-hung windows can be arranged in a number of ways — as a single unit, doubled (or mullion) type, or in groups of three or more. One or two double-hung windows on each side of a large stationary insulated window are often used to effect a window wall. Such large openings must be framed with headers large enough to carry roof loads.

Casement Windows

Casement windows consist of side-hinged sash, usually designed to swing outward.

Specialties

WINDOW TREATMENTS

Window Typos

because this type can be made more weather-tight than the inswinging style. Screens are located inside these outswinging windows and winter protection is obtained with a storm sash or by using insulated glass in the sash. One advantage of the casement window over the double-hung type is that the entire window area can be opened for ventilation. Weatherstripping is also provided for this type of window, and units are usually received from the factory entirely assembled with hardware in place. Closing hardware consists of a rotary operator and sash lock. As in the double-hung units, casement sash can be used in a number of ways—as a pair or in combinations of two or more pairs. Style variations are achieved by divided lights. Snap-in muntins provided a small, multiple-pane appearance for traditional styling. Metal sash are sometimes used but, because of low insulating value, should be installed carefully to prevent condensation and frosting on the interior surfaces during cold weather. A full storm-window unit is sometimes necessary to eliminate this problem in cold climates.

Stationary Windows

Stationary windows used alone or in combination with double-hung or casement windows usually

consist of a wood sash with a large single light of insulated glass. They are designed to provide light, as well as for attractive appearance, and are fastened permanently into the frame. Because of their size (sometimes 6 to 8 feet wide), 1½-inch-thick sash is used to provide strength. The thickness is usually required because of the thickness of the insulating glass, 1015

Specialties WINDOW TREATMENTS Window Types OPERATION

VENTILATION REMARKS Double-hung window: upper and lower sections slide vertically to open; spring balance; lock at meeting rail
 Projected window: with fixed upper section of glass, vent (hopper type) at bottom opening In; crank-operated
 Awning window: four horizontal sections project out; crank-operated 1016
 Combination window: upper section is of glass block, supported on angle and channel girts attached to columns; lower section is half fixed glass, and half hopper (crank-operated)
 Casement window: with fixed glass section at bottom, two swing-out sections at top; crank-operated
 Projected window: with sections opening out at top, in at bottom; crank-operated
 Sliding window with lower fixed section Substantial airflow, but not directed well; drafty without a shield
 Substantial airflow, but not directed well; drafts are difficult to avoid
 Adequate airflow In most climates; well directed, not drafty Very good airflow, both In quantity and quality (not drafty) Large quantities of airflow are easily controlled, with fairly good draft control Substantial airflow, but hard to control, drafty Adequate, well directed air flow for most climates
 No parts project even when open; sometimes difficult to open if schoolroom has usual shelving at sill level; usually a glass deflector is installed to prevent drafts Easily operated; shades can be drawn without obstruction when window is open (but they will billow in breeze); these windows must be placed carefully—there is danger of children running into them outdoors if open; rarely used In schools Easily operated, can be used with shades or blinds closed over most of its area; view is unobstructed, even when window is closed Easily operated; does provide some ventilation even when partially shaded; view through this type is almost unhindered with few obstructions at eye level Can be opened quite wide even during rainstorms; is easily operated; shades can be drawn without obstruction; framing does obstruct outdoor view somewhat whether window is open or closed No parts project either inward or outward when open, but window is sometimes difficult to slide with the usual schoolroom shelf at sill level

Some types of glass block refract light to ceiling, providing good light distribution across classroom and eliminating need for shades or blinds; however, designer must take care to use properly; this type does not always meet brightness tests for good schoolroom lighting

Specialties WINDOW TREATMENTS Window types TYPE

DESCRIPTION SIDE HINGED CASEMENT materials • wood, steel, aluminum use • common in residences and apartments operation • rotary crank or lever operators hold the vent open to desired position, up to 180°, but usually 90° note • available also as a single vent • generally allow exterior of glazing to be cleaned from inside when outswinging. • provide 100 percent opening in the ventilation area • will be subject to wind pressures when opened. CASEMENT-HOPPER materials • wood, aluminum use • especially appropriate for high-rise, life safety Installations operation • sophisticated hardware note • no protection from rain when open • available to limited extent as "tilt and turn" type which acts as a bottom hung window in normal use, but which can be converted by use of secondary hinges into a side-hung, inswinging type, allowing for easy cleaning. CASEMENT-COMBINATION materials • wood, aluminum, steel in varied quality grades use • commonly known as the "classroom window" operation • combination of in-swinging hopper and out-swinging casement vents offer flexibility for ventilation control BOTTOM HINGED HOPPER materials use note wood, aluminum and steel where vent will not interfere with interior conditions lower cost utility quality is commonly used for residential basements no protection from rain when open 1017

Specialties WINDOW TREATMENTS Window types TYPE

DESCRIPTION BOTTOM HINGED, continued HOPPER-SPECIAL materials • steel, stainless steel use • in commercial and industrial buildings where appearance is not of major importance and resistance to forced entry is- • to prevent forcible exit; sometimes called "guard" windows operation • combined with fixed lights or with projecting vents above, which offer high and low openings that are best for natural-air circulation (due to principles of stratification) • separate vent frames usually swing in as hoppers • jalousie-like vents also available • frames often reinforced with steel rods notes • vents limited in size • muntins usually separate openings of 8 inches square materials • steel use •

in housing for mental patients, to provide protection against exit while minimizing appearance of restraint note • vents have a maximum clear opening of about 6 inches TOP HINGED AWNING / \ / s materials use operation wood, steel, aluminum multiple assemblies are used mostly In steel for industrial buildings separate units are commonly combined with fixed lights, or with hoppers for maximum stratification ventilation (These are available also in wood and aluminum.) are out-swinging projected windows that create a "canopy" against rain penetration when in multiple, vertical stacks, the mechanical operation will allow for the bottom vent to open before the other vents, which will then open in unison PROJECTED OPTIONAL HOPPER materials use operation JALOUSIE materials use operation note steel or aluminum medium quality grade is called "intermediate" and is commonly used in commercial, institutional and industrial type buildings architectural windows are frequently used for schools, hospitals, office buildings, etc. similar to awning windows but with optional fixed glass lights and/or hoppers. wood, steel, aluminum primarily for sunrooms, porches, and the like where protection from the weather is desired with maximum fresh air multiple vents combine unobstructed vision with controlled ventilation the louvers are fully adjustable and can be set in any position can be fitted with storm sash on the inside to provide more weather tightness screens, interchangeable with storm sash, are furnished various types of glass, including obscure and colored, often are used for privacy or decoration 1018

TYPES Specialties WINDOW TREATMENTS Window Types DESCRIPTION GLIDING DOUBLE-HUNG materials • wood, aluminum, steel In different designs and weights to meet various service requirements for all types of buildings use • with combination of fixed windows for maximum window openings • use in buildings other than residential and light commercial has been declining operation • top and bottom openings optimize natural stratification ventilation note • also available In single-hung (only one sash operating) and triple hung (three operable sash) SLIDING —> <— materials • wood, aluminum (with various coatings and claddings) use • mostly in residential buildings operation • provide only one half of opening for ventilation; • sash height to width ratio should not exceed 1 to 2 for good operation note • sash usually removable for cleaning and may be very large DUAL-VENT fr= w i[L_..___. ;CI-. u ax i. T

materials use operation note aluminum mostly In hospitals essentially two sets of double-hung sash—air circulates through the bottom outer sash and then through the top inner sash provides ventilation while protecting from rain and drafts ■ check cost PIVOT VERTICAL PIVOT \ / \ / \ materials use operation note wood, aluminum and steel mostly in air conditioned buildings consists of large vent, usually pivoted in the center of the head and sill of the main frame, which rotates 180° or 360° around its vertical axis for cleaning not primarily designed for ventilation, although may be held open up to 4" with special hardware (unless unlocked by maintenance personnel) 1019

Specialties WINDOW TREATMENTS Window Types TYPES
 DESCRIPTION PIVOT, continued HORIZONTAL PIVOT operation • similar to vertically pivoted but rotates around a horizontal axis
 INDUSTRIAL PIVOT -X1 materials • steel, aluminum use • often used horizontally and vertically to form entire walls • lower cost for use In Industrial and utilitarian buildings note • mechanical operators are available
 SPECIAL CONTINUOUS S • " ^ s / < ■ " y / / -v. \ \ \ \ " V. N " s N materials
 * steel use • for top lighting and ventilation In monitor and sawtooth roof construction operation • hinged at the top to the structural-steel framing members of the building and swing outward at the bottom • two-floor lengths are connected end to end on the job note • mechanical operators may be either manual or motor- powered AUSTRAL REVERSIBLE materials • wood and steel use • schools, hospitals and other institutional buildings • upper and lower sash counterbalanced on arms pivoted to frame • upper and lower sash operate simultaneously note • difficult to screen, shade or curtain materials • wood and steel use • residential and Industrial buildings operation • similar to double-hung In appearance, but may be tilted for better control of ventilation, or reversed for cleaning note • not universally available CUSTOM TYPES VARIOUS CONFIGURATIONS materials • aluminum, steel, stainless steel use • special types for windows in houses of worship, mausoleums, and memorial buildings operation . various arrangements available 1020

Specialties WINDOW TREATMENTS Window Types 1. Round Top over casements 2. Full Round with operating center 3. Round Top with authentic divided lites 4. Separated Round Top 5. Eyebrow with Gothic divided lites

6. Round Top over double-hung 7. Simulated Round Top 8. Round Top with quarter panes 9. Transom Round Top. 10. 3 point Round Top 11. Quarter Rounds 12. Round Top 13. 12 foot Round Top 14. Inverted over picture window with decorative divided lites corners 15. 12 foot wide Round Top with operating center 16. Ovals 17. Spider Web 18. Gothic head 19. Rounded casements 20. Rounded double hung with Gothic lite pattern. 1021

Specialties WINDOW TREATMENTS Window Types Two-light window, the all-purpose window Two-light window, three lights over one light Two-light window, divided twelve lights Two-light window, divided sixteen lights L Two-light window, six lights over one light Two-light window, four horizontal lights high ZED ID ED HD n ' ,T"r" B 1 1 Two-light window, six lights over nine Two-light window, eight lights over twelve Typical styles of windows 1022

Specialties WINDOW TREATMENTS Window types: Projected and Stationary ■CASING HEAD JAMB HINGED SASH SCREEN OR STORM SASH ■\$* yVj- SILL SCREEN OR STORM SASH Fig. 1 Projected window. Cross sections: A, head jamb; B, horizontal mullion; C, sill. HEADER DRIP CAP CASING HEAD JAMB STORM SASH SASH I: -GLASS MUNTIN sj|" PUTTY B SASH RAIL 'N— STORM SASH V 1* c Fig. 2 Stationary window. Cross sections: A, head jamb; B, muntin; C, sill. 1023

Specialties WINDOW TREATMENTS Window Types: Double-Hung and Casement LINTELS SASH BALANCE VRIP CAP CASING — BLIND STOP -SASH SCREEN OR STORM SASH UPPER SASH MEETING RAILS LOWER SASH INCREASE WHEN SASH WEIGHTS USEO CASING PARTING STRIP SASH LOWER SASH SCREEN OR STORM SASH Fig. 3 Double-hung windows. Cross sections: A, head jamb; B meeting rails; C, side jamb;/), sill. 1024 Fig. 4 Outswinging casement sash. Cross sections: 4, head jamb; B, meeting styles; C, side jambs;/), sill.

Specialties WINDOW TREATMENTS Window types: Double-Hung and Casement HEADER ■/DING DRIP CAP CASING METAL CORNER BEAD CALKING METAL WINDOW FRAME PLASTER TC.STUOS CASING METAL CORNER BEAD ■CALKING ■METAL WINDOW

FRAME METAL WINDOW FRAME Fig. 5 Double-hung metal windows. Cross sections: d, head; fl, side jamb; C, sill. Fig. 6 Solid-section steel outswinging casement sash. Cross sections: A, head jamb; fl, side jamb; C, sill. 102S

Specialties WINDOW TREATMENTS Window Types Wood Trim The casing around the window frames on the interior of the house should be the same pattern as that used around the interior door frames. Other trim which is used for a double-hung window frame includes the sash stops, stool, and apron (Fig. 7A). Another method of using trim around windows has the entire opening enclosed with casing (Fig. 1B). The stool is then a filler member between the bottom sash rail and the bottom casing. The stool is the horizontal trim member that laps the window sill and extends beyond the casing at the sides, with each end notched against the plastered wall. The apron serves as a finish member below the stool. The window stool is the first piece of window trim to be installed and is notched and fitted against the edge of the jamb and the plaster line, with the outside edge being flush against the bottom rail of the window sash (Fig. 7A). The stool is blind-nailed at the ends so that the casing and the stop will cover the nailheads. Predrilling is usually necessary to prevent splitting. The stool should also be nailed at midpoint to the sill and to the apron with finishing nails. Face-nailing to the sill is sometimes substituted or supplemented with toenailing of the outer edge to the sill (Fig. 7A). The casing is applied and nailed as described for doorframes, except that the inner edge is flush with the inner face of the jambs so that the stop will cover the joint between the jamb and casing. The window stops are then nailed to the jambs so that the window sash slides smoothly. Channel-type weather stripping often includes full-width metal sub-jambs into which the upper and lower sash slide, replacing the parting strip. Stops are located against these instead of the sash to provide a small amount of pressure. The apron is cut to a length equal to the outer width of the casing line (Fig. 1A). It is nailed to the window sill and to the 2- by 4-inch framing sill below. ■ When casing is used to finish the bottom of the window frame as well as the sides and top, the narrow stool butts against the side window jamb. Casing is then mitered at the bottom corners (Fig. 7B) and nailed as previously described.

OUTSIDE CASING — JAMB- PARTING STRIP , -PLASTER CASING
NAIL STOOL TO SILL STOOL APRON TOENAIL JAMB CASING

NARROW STOOL Fig. 7 installation of window trim: A, with stool and apron; fl, enclosed with casing. 1026

Specialties WINDOW TREATMENTS Curtains/Draperies CURTAINS

Curtains are soft window coverings that generally are shirred (gathered onto a rod) or have headings attached to solid-wood rods, round or oval metal rods, or cafe rods rather than cord-operated traverse rods. Curtains may be either stationary fabric panels or slid open and closed by hand. They are flexible in that they can be short or long, layered or tiered, or used alone or in combination with other soft, or with hard treatments. Curtain is traditionally a term for informal treatments, such as cafe curtains. However, curtains also may be quite formal, as are shirred and elegant tied-back fabric treatments. Even though curtains are generally thought to be shirred treatments, other headings might be included in this category. Indeed, there is a crossover of terminology between draperies and curtains. Generally draperies are installed on cord-operated traverse rods, although they may be stationary pleated panels. Curtains may be installed on traverse rods (as in a pleated cafe curtain, for example), and headings such as the pencil pleat; drawstring pencil pleat; shirred, spaced pencil pleat; alternate pencil pleat; ruffled shirring tape heading; and smocked heading may be called either curtain or drapery treatments. DRAPERIES Draperies are made with pleats. They are hung with drapery hoods onto carriers of conventional, architectural, or decorative traverse rods or into the rings of wood rods or cafe curtain rods; or they may thread onto spring-system traverse rods. Generally draperies are either hung straight to the floor or tied back. Thus they operate, or "draw," by opening and closing with a cord or a wand or by hand. The exception is tied-back draperies, which sometimes are let down at night. However, tied-back draperies are trained to tie back at an angle and therefore should not be handled to any extent. Draperies draw in a pair and meet in the center (center-meet) or draw one way from left to right or from right to left. One-way draw draperies require one-way traverse rods. Draperies that hang at a doorway rather than at a window are called portieres. They may be pleated in any fashion or shirred. They may be placed on a traverse rod, but historically (and they were used extensively in the Victorian era), they were tied-back stationary panels made of a heavy fabric that were let down when privacy or insulation was needed. Draperies can be made of any fabric. The selection will depend on the style, use, and

needs. Sheer fabrics do best as diffusers of glare and as providers of daytime privacy. Medium- to heavyweight fabrics are excellent choices for overdraperies and plain tieback draperies. Lining fabrics are the right weight for privacy liners or underdraperies. If a drapery is given a ruffled edge or a banding, that trim should be a lightweight, semi-crisp, flexible fabric, not a heavy, stiff fabric or a sheer, slippery fabric. VS***- WU*A. M»<. **ttY\ - +GU tr it' <2* 6K0VU \ACW\t-* # ^UteiAIN F^K^-T

Specialties WINDOW TREATMENTS Curtains/Draperies TABLE 1
 Draperies Period style Early English Tudor Jacobean Charles II Anglo-Dutch William & Mary Queen Anne Early Georgian Chippendale Fabric Crewel, embroideries, hand-blocked linen, silk and worsted damask, velvet, brocade Crewel, embroideries, hand-blocked linen, silk and worsted damask, velvet, brocade, India print Crewel, embroideries, hand-blocked linen, silk and worsted damask, velvet, brocade, Indian print Colors Full-bodied crimson, green, and yellow Full-bodied crimson, green, and yellow Full-bodied crimson, green, and yellow Design Large bold patterns: tree branch, fruits, flowers, oak leaf, animals, heraldic designs Large bold patterns: tree branch, fruits, flowers, oak leaf, animals, heraldic designs Jacobean motifs, classic medallions and garlands Upholstery fabrics Tapestry, leather, needlework, velvet, brocade Tapestry, leather, needlework, velvet, brocade Tapestry, leather, needlepoint, velvet, brocade Late Georgian Adam Hepplewhite Sheraton Empire Federal Brocade, damask, chintz, taffeta, satin, toile de jouy Delicate subdued hues of rose, yellow, mauve, green, and gray Classic designs, small in scale: garlands, urns, floral, animals, etc. Damask, brocade, velour, satin, petit point, leather in libraries Louis XIV Louis XV Louis XVI Spanish renaissance Early colonial Early American Modern French provincial Victorian Silk, satin, damask, taffeta, muslin, brocade, toile de jouy Velvet, damask, crewel, India print, printed and embroidered linen Crewel, embroideries, hand-blocked linen, silk and worsted damask, velvet, brocade Toile de jouy, damask, chintz, organdy, cretonne Textured and novelty weaves, all fabrics Chintz, cretonne, hand-blocked linen, velvet Velvet, brocade, damask Delicate powder blue, oyster white, pearl, rose, pale greens, mauve, yellow Rich vigorous colors, red, green, and gold Full-bodied crimson, green, and yellow All colors, but more subdued than in early period All colors, bright to pastel Subdued colors, pastel shades Turkey red, other rich colors Stripes

sprinkled with ribbons, flowers, medallions, lyres, and other classic motifs
 Bold patterns in classic and heraldic designs; also arabesques Large bold
 patterns: tree branch, fruits, flowers, oak leaf, animals, heraldic designs
 Scenic, birds, animals, floral Solid colors, modern designs, stripes Screen
 prints, block prints Solid colors, formal patterns Petit point, satin, moire,
 velour, chintz, damask, brocade, tapestry Leather, tapestry, velvet, linen,
 brocatelle Tapestry, leather, needlepoint, velvet, brocade Haircloth, mohair,
 linen, chintz, velours All fabrics, novelty weaves, plastics Solid colors,
 textured weaves, tapestry Haircloth, needlework

Specialties WINDOW TREATMENTS Curtains/Draperies of Colonial and
 Federal Periods 18th-century colonial: tieback damask drapery with balled
 fringe 18th-century colonial: staggered tieback with plain edge asymmetric
 panels 18th-century colonial: swagged valance over bishop sleeve draperies
 over holdback Federal: tieback panels with fringed raised valance of
 contrasting color Federal: waterfall over holdback with draped valance
 Federal: asymmetric tieback with fringed valance Federal: heavy valance
 over straight draperies 1029

Specialties WINDOW TREATMENTS Curtains/Draperies of Georgian and
 Directoire Periods iiliakiiiiillliul jllj | ■■ ■■■■■■ mmmmmmmmm iUb. \ 1
 1 Georgian: tieback drapery with Austrian valance with fringes Georgian:
 tieback heavy woven drapery with fabric-covered heading - sheer curtains
 behind Georgian: tieback drapery with tapered French pleat heading Late
 Georgian: curved fabric cartridge valance over holdback draperies Late
 Georgian: gilt wood cornice over fixed lambrequins and sheer curtains Late
 Georgian: gilt metal cornice over fixed tieback draperies and sheer curtains
 Directoire: tieback draperies with contrasting edging on decorative brass
 rod Directoire: fringed overdrapery valance on fringed sleeved tiebacks and
 fringed drapery Directoire: painted stepped wood cornice over swag with
 twin cascades and tieback 1030

Specialties WINDOW TREATMENTS Curtains/Draperies of Victorian
 Period Mid-19th-century Victorian: central swag with twin cascades over
 heavy draperies with braided tieback over sheer undercurtain Late-19th-
 century Victorian: looped festoon over decorative brass rod Late-19th-

century Victorian: neo-Greek-style cornice with fringed valance over tieback fringed fabric with lace undercurtains 1031

Specialties WINDOW TREATMENTS Curtains/Draperies of 1940s Fringed finger festoon over brackets over straight line draperies with sheer undercurtains Fringed fabric valance with cascades over bishop sleeve draperies 1032

Specialties WINDOW TREATMENTS Curtains/Draperies Cafe curtains Ruffled tieback curtains Fringed segmented valance with ruffled trim - tieback draperies over cafe curtains Fabric-wrapped-pole draped valance Cafe curtains with gathered valance and ball fringe trim 1033

Specialties WINDOW TREATMENTS Curtains/Draperies Tieback draperies with ruffles on center arch rod - pleated shade beneath High tieback draperies with ruffled multirow valance-woven shade beneath Full-length straight draperies with ruffled valance - decorative bows Shirred, ruffled balloon valance over ruffled, shirred heading on narrow rod , <umL*%'rtmttin-f~f4~'MT*irtn'*n7tf &nimnmw!avnmwum!!Wf<j Shirred heading on brass pole - fixed draperies with rosette tieback Shirred heading on brass pole-fixed panel Shirred balance with shirred tieback draperies over blinds Penta balloon valance over ribbon tieback draperies Shirred valance with horizontal accent banding - tieback draperies with matching edge banding 1034

Specialties WINDOW TREATMENTS Curtains/Draperies Bishop sleeve fringed-tip valance with ribbon tieback draperies Triple-row fringed heading with shirred tieback draperies Fabric-covered straight cornice over paired tieback draperies and scalloped curtain on brass rod Pinch pleated draperies with horizontal tiebacks over standard roller 1035

Specialties WINDOW TREATMENTS Curtains/Draperies Cafe curtains with scalloped edges on brass rods Single pleated draperies over paired double hung windows with tab headed cafe" curtains on rod Fabric-covered cornice board valance - ribbon tiebacks on drapery Ruffled valance heading over brass rod with straight draperies over scalloped cafe curtain 1036

Bow tieback curtains with fringed trim and gathered valance Bow tieback gathered curtains on brass rod Specialties WINDOW TREATMENTS Curtains/Draperies J™ (!■«■ Wf.MV." IW.'U'WI'WiWiIT. •■ Simple traditional swag with cascade draperies Rosette tieback priscilla curtains with continental heading Scalloped cafe curtains on brass rod with pleated valance 1037

Specialties WINDOW TREATMENTS Curtains/Draperies Ill 1 "- I =Ex Overlapping swag on rod with ball-fringed cascades Swag with bow and asymmetric cascades Symmetrical swags and cascades with center rosette Symmetrical draped swag over brass rod Asymmetrical swag drapery over brass rod with sheer curtain Symmetrical draped swag on rod with ties at end over sheer or solid curtain 1038

Specialties WINDOW TREATMENTS Curtains/Draperies Triple-tall cascades with dual swags and rosette holds Swag over rod with rosette holds Swag, draped valance, cascading ends 1039

Specialties WINDOW TREATMENTS Curtains/Draperies Asymmetrical double-rad-supported swags and draperies - contemporary Symmetrical long cascades and swags with center swag and bishop sleeve draperies Overlapped double swag thrown over door-high holdbacks, with draperies billowed at floor 1040

Specialties WINDOW TREATMENTS Curtains/Draperies Flared cornice box with geometric trimmed scallops over reverse swags and asymmetric floor-tip cascades Trimmed cornice box over asymmetric tieback drapery Symmetrical box-pleated draperies on brass rod with double-tiered rosette holdbacks 1041

Specialties WINDOW TREATMENTS Curtains/Draperies Pleated shade Mlniblnd - custom fitted Wooden shutters Rounded valance will) drapery Double swags with balloon drapery Double swags with cascades and drapery Shirred curtain with sunburst Pleated shade with shirred curtain French pleats with shaped top under curtain 1042

Specialties WINDOW TREATMENTS Headings Semiformal box pleits, evenly spaced Cartridge pleat, evenly spaced Triple pleat heading, evenly

spaced Shirred heading on a narrow rod Shirred heading gathered on a wide rod Ruflied shirred heading on an extra-wide rod Pencil heading on a wide rod Shirred heading gathered on an extra-wide rod Cluster heading, spaced evenly Ruflied shirred heading on a narrow rod Standard heading on a narrow rod Spaced pencil pleats heading on a wide rod 1043

Specialties WINDOW TREATMENTS Headings Tripie shirred heading with ruffle gathered on 3 narrow rods Decorative heading on a wide rod Triple shirred heading gathered on 3 narrow rods Grouped French pleats Scalloped heading with rings and bows Tab heading spaced evenly on rod Scalloped heading with rings Double butterfly pleat heading Single butterfly pleat heading French pleats with scalloped heading 1044

Specialties WINDOW TREATMENTS Tiebacks and Holdbacks Straight Tapered Banded Tapered with welting Oversized welting Shirred with welting Shirred with braided trim Braided Rosette Fringed with welting Scalloped Frilled Twin ruffles Straight with scallops Twin welting with box pleating Rosette with cascade Ruffles Bow/ribbon Bow with welting and tassel Tapered with welting, rosette, and tassels 1045

Specialties WINDOW TREATMENTS Tiebacks and Holdbacks Leaf motif Decorative holdback Scalloped with welting 1 i Layered with | Decorative with tassel Decorative bow and Square holdback with ribbon insert Decorative knob with 2 Inserts Decorative knob holdback with tassel Stylized rosette Decorative knob with inserts Decorative knob holdback with two Decorative bow holdback Standard knob with cascade Decorative knob with circular Insert and cascade 1046

Specialties WINDOW TREATMENTS Mounts, Brackets, and Couplers Flush mount - closed top Outside mount - open top Outside mount - closed side Flush mount - open top Outside mount - closed top Inside mount bracket BRACKETS Flush extra- Base-mounted Curved bracket Support bracket Curved support Double rod projection bracket extra-projection bracket bracket bracket COUPLERS Extra-projection Base-mounted base-mounted coupler coupler 1047

Specialties WINDOW TREATMENTS Finials, flings, and Hooks FINALS "Qst^^g RINGS Plain ring Round eyelet ring Oval eyelet ring Square eyelet

ring Round clip-on ring Oval clip-on ring Rounded-end clip-on ring
HOOKS Metal and plastic hooks for standard tapes Metal hooks for
decorative tapes 1048

s Swinging arm separated rod Polyvinyl chloride (PVC) with end caps
curtain rod Wide telescoping curtain rod PVC pipe with elbows for
projection Specialties WINDOW TREATMENTS Rods Flat curtain rod
Double flat curtain rod Cafe curtain rod Fluted wood rod Tension rod with
adjustable screw Sash rods c/r EPs^{±1} Separated curtain rod Extra wide
telescoping projection rod 1049

Specialties WINDOW TREATMENTS Traverse Rods Conventional single
hung traverse rod - A, projecting end brackets; B, end housing; C,
telescoping rod; D, center support; E, master carriers; F, carriers; G, end
bracket; H, cord; I, tension pulley Double traverse rod One-way traverse
rod with two center supports Double traverse rod with valence

Specialties WINDOW TREATMENTS Guidelines for Fabric Panel Widths
and Rod Lengths TABLE 2 Fabric Panel Widths and Pleating Guidelines
Desired pleated panel coverage 16" 20" 24" 28" 32" 36" 40" 44" 48" 52"
56" 60" 64" 68" 72" 76" 80" 84" 88" 92" 96" 100" 104" 108" 112" 116"
120" 124" 128" Flat fabric without hems 43" 51" 59" 67" 75" 83" 91" 99"
107" 115" 123" 131" 139" 147" 155" 163" 171" 179" 187" 195" 203" 211"
219" 227" 235" 243" 251" 259" 267" Hemmed flat fabric 39" 47" 55" 63"
71" 79" 87" 95" 103" 111" 119" 127" 135" 143" 151" 159" 167" 175" 183"
191" 199" 207" 215" 223" 231" 239" 247" 254" 263" Number of 4" flat
spaces between pleats 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23 24 25 26 27 28 29 30 31 32 Number of pleats 5 6 7 8 9 10 11 12 13 14
15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 Width of fabric
in each pleat 3VST 3 1/4" 3 1/2" 3 3/4" 3W 3V4" 3W 3 1/2" 3W 3 3/4" 3 1/2" 3W 3 1/2"
3 1/2" 3 1/2" 3W 3 1/2" 3W 3 1/2" 3 1/2" 3 1/2" 3 1/2" 3W 3 1/2" 3 3/4" 3 1/2" 3 1/2" 3W 3 1/2"
3 1/2" TABLE 3 Rod Lengths Needed for Various Widths of Windows and
Stackback Spaces If the glass is 38" 44" 50" 56" 62" 68" 75" 81" 87" 94"
100" 106" 112" 119" 125" 131" 137" 144" 150" 156" 162" 169" 175" 181"
187" The stackback* should be 26" 28" 30" 32" 34" 36" 37" 39" 41" 42"
44" 46" 48" 49" 51" 53" 55" 56" 58" 60" 62" 63" 65" 67" 69" Your rod
length and drapery coverage should be (add for overlaps and returns) 64"

72" 80" 88" 96" 104" 112" 120" 128" 136" 144" 152" 160" 168" 176" 184" 192" 200" 208" 216" 224" 232" 240" 248" 256" *For one-way draws, deduct 7" from stackback, Note: Figures are based on average pleating and medium-weight fabric, For extra bulky fabrics, add to stackback to compensate for the additional space they require. 1051

Specialties WINDOW TREATMENTS Shades mmmam Overlapping trimmed valance over scalloped shade Banded valance over Roman shade Shirred cornice box over tri-part balloon shade Twin bow cloud shade with gathered heading Inset flush Roman shade with horizontal folds Tri-part Austrian shade with ruffled trim Inset flush bottom pull Roman shade with horizontal folds 1052

c ~tjjp cp -^ Pleated shade Crenellated edge tsxm : _ i i' f ■ _ ^L _ ! - ■ T : r jii t: Shirred shade Rolldown shade , , , -rt -, • - .-. T- 1 * n T i Rolldown shade Pulldown shade Specialties WINDOW TREATMENTS Shades 10S3

Specialties WINDOW TREATMENTS Shades Corner installation Fig. 8 Woven wood blinds. These blinds have horizontal or vertical reeds - long slats of wood from Y* to 1 inch width - that are held together by decorative vertical yarns. They range In designs from those that are made mostly from exposed wood to those that are mainly yarns of several colors creating various interesting effects. Woven wood blinds can be used with many window treatments Including draperies and cafe curtains, and such shade types as Roman-fold, spring-rolls, cord and pulley, and duo-fold. Top treatments include canopies, valances, and arches, while scallops, fringes, and trims are suitable for the bottom. Because woven wood blinds add color and texture to a window, they are particularly adaptable to the "natural" look in decorating.

Specialties WINDOW TREATMENTS Shutters T ■"■' — 1- u u u fl u I 1 ' ' ■ , 1 ' 1 ' ' ' ' ' ' ' ■ ' ' ' n , , K in _M __, ■ ^ ^ ' " " ' Louvers: V/t", W, 3V2", 414"; thickness: W\ width of stile: 2"; width: 8" to 36" In W increments Louvers: VA", W, 3V4", W\ thickness: W; width of stile: 2"; width: 8" to 36" in Vt" Increments. Louvers: 2V4". Fan top (nonadjustable louvers); louvers: 214"; thickness: W. 11111111 Louvers: ZYz"; thickness: VA". Louvers: W\ thickness: W. Tnrnr 3PQO Ttrnn 3DDD 3DOD DPDD 3DDD anna UDDD]DDD 3QDD]QDD aDDD anDn anon annn anna .DDDd

anna PPDD]QDQ nana aPDD DDDD PDDD DDDU aaao onoa DODD
 DDDD DDOD □ □DD ndaa DDDD DDaa anna DPDD DDDD DDDD
 QPDQ DDDD noon nnnoc □ a one DPDDC □ DDDC aanDC DDDPC
 QDDOD PDDDD DPDDD oanaa opapp naaaa □ □□ad DDDdD □ □□□a □
 □pan npoDq □ □□□ PPPPD PDPDD npnpp Louver grid. Standard: 4' x 8"
 unframed; standard framing: W wide x ZW thick. Traditional/Dixie panel;
 louvers: W; thickness: VA". Fig. 9 inside shutters can be used next to
 windows in place of curtains. Some are put under curtains or draperies;
 others are used caffè style either above or beneath cafe curtains. Shutters
 may be made from wood or metal. Natural wood tones are often used to
 enhance the beauty of the shutters. The inside section may be made from
 any of the following materials: fabric mesh, cane, grill cloth, or screening.
 1055

Specialties WINDOW TREATMENTS Shutters STANDARD DOUBLE
 HUNG WINDOW CROSS SECTION OF LOUVER 1-1/4" louver 1/4" to
 1/8 taper . i^Heidrift 11/16" Q— 1 ot mulliple dowi Stile 13/16"itl-1/2" /
 Prefilled light blind Hinging Slop showing mounting posllion on window
 stop behind shutiers. InstiUitlon mdice on window slop MUfT f" 7/16- or
 greiter (see distince "A" cross section Below) / Heivy duty (0 050] offset
 bras* limsh hinge is idfustible up ind down, let! ind rtght illowtng tor
 peHect alignment of shutiers on iny window 1-1/4" louvers spiced Ton I"
 cenlers, Louver projection: 5/16" from beck ol penel Lower louver llghi
 blind 1 of multiple dowels Gotlom rail 11/16" H CROSS SECTION
 IMPORTANT: Distance "A" on window stop MUST be 7/16" or greiler for
 solid, squi/e in- stillilion ot hinging slop. Measure to point X II window
 stop tS NOT removed lock rail -> 2-panel single hung lock rail " :t 2-panel
 single hung lock__ rail ~ 3E Bi-fold single hung Bi-fold double hung 1056

Specialties WINDOW TREATMENTS Shutters and Venetian Blinds
 AUTHENTIC DOUBLE FLUTED WOODEN FASCIA- HIDES ALL
 OPERATIONAL HARDWARE, FRONTY1EW: TUtor Hirdwin with long
 lift wormgiar, Tiltor Cordl Authentic 3" Slsti- (select Alukin yellow eeditj
 gr«c«fuMy rounded it ends. Authentic WO Odin Tilled, Molded Wooden
 Bottom Rill mitchei Till Rill Qrackii imliUs either inside casemeni on
 window ttop moulding (see position e below} or on Ihi outside ol window
 element (Me poiitiom b, c, or d) POSITION i below shows bucket In• tilled

inside of the window e*sement (IBM), Distinct "a" on slop moulding must
 meisu'o 3/8" or greater for solid square li|, POSITION i »nd c shows
 bracket installation on the lice of casement (OBM). POSITION d shows
 bucket insui- lilion on the will beyond or ibove the cisement. In this
 position bracket will allow $\sqrt{2}$ ~ projection in order thai blind will cleir
 casement. CROSS SECTION OF TTPICAL DOUBLE HUNG WINDOW
 Bracket projection allows 1/2* cWeriiKc Irom behind Mind, STANDARD
 DOUBLE HUNG WINDOW CROSS SECTION OF LOUVER -M/2"*J/|"
 Flit Louver with radius edges Preddlld light blind Hanging Stop showing
 mourning position on window nop behind shutters- Installation surface an
 window slop MUST be 9/16" Of greater (set distance "A" cross section
 below) Heavy duly I.06S) ollset brass finish hinge Is adjustable up and
 down, lefl and light si lowing (or perfect alignment ol stumers on sny
 window. CROSS SECTION IMPORTANT; Distance "A" on window slop
 MUST be 9/16* or greater for solid, square installation of hinging slop. Uhe
 ltrtit J spice oedend ottom rails *e shown nailer In right \ ot multiple do
 wets . Bottom tell WW " ' i t / * tif m at Mu**urt to point X If window slop
 tS NOT removed. Measure to point Y If window if nn l* i»m***<* 1057

Specialties WINDOW TREATMENTS Guidelines DECORATIVE RODS
 Measuring for Most Windows Outside mount Decorative rods should be
 mounted on the wall. Measure width of glass; if total glass exposure is
 desired, add for stackback (see Table 4). Rods should be hung so that
 drapery headings (pleated tops of panels) are at least 4" above the glass, so
 they can't be seen from the outside. Figuring Stackback Stackback is the
 amount of wall space needed if open panels are to clear the glass
 completely. This dimension, added to the window opening, gives you the
 proper rod length. Begin by measuring the window opening, then consult
 Table 4. Find your opening measurement and read across for the right rod
 length. DRAPERY COVERAGE- fflf -DRAPERY COVERAGE - STACK-
 I BACK I M I I M 3(> VW- TABLE 4 Stackback: Average Pleating and
 Medium Weight Fabric Window opening 24" 30" 36" 42" 48" 54" 60" 66"
 72" 78" 84" 90" 96" 102" 108" 114" 120" Stackback* 21" 23" 25" 26" 29"
 30" 31" 32" 34" 36" 37" 38" 39" 42" 44" 45" 48" Rod length 45" 53" 61"
 68" 77" 84" 91" 98" 106" 114" 121" 128" 135" 144" 152" 159" 168"
 *Deduct 7" for one-way draw. Measuring for Special Windows Sliding
 doors Measure as for outside mounted rod. Convert rod from two to oneway

draw. Corner and bay windows Decorative rods may be used at these windows. However, it is best to consult your dealer or designer about the measuring. Layered Treatments Decorative traverse rods are often used for overtreatments. If the undertreatment is inside mounted or is an outside mounted mini-blind, pleated shade, Romanette woven wood or a cafe curtain, set the brackets for maximum clearance. Drapery returns will be AW. If you are using an undercurtain, you will want a utility curtain rod. It comes with its own bracket-supports. If you are using underdraperies, double brackets are available. They hold both rods and automatically align the headings. Over- drapery returns will be 6" to 7". If using a decorative rod over an outside mounted vertical blind or a woven wood shade other than Romanette, special brackets are available. Overdrapery returns will be 6" to 7".

PLEATED SHADES Measuring for Most Windows

OUTSIDE MOUNT

Inside mount Measure width at top, center and bottom. Use narrowest measurement. Shades will be made narrower to slip inside easily. Measure length from inside top of opening to sill. A W deep recess is needed for flush mounting. 2 1/4" for Duette in %. Outside mount Measure width of opening. Add at least VA" on each side for overlap. Measure from top of frame to sill or IV2" below opening if there is no sill. (If brackets are to go above window frame, add an extra IV2" for bracket bases.) Ceiling mount Measure desired width and length of blind. Overlap window openings by at least W on each side.

Measuring for Special Windows

Multiple shades

At very wide or sectioned windows and at sliding doors, it's wise to use two shades hung from one headrail. Make a drawing of the window; include measurements of glass, woodwork and overall size. Your dealer will do the rest.

Corner windows

Inside-mounted shades need no special measuring instructions. If outside or ceiling-mounted shades are used, they can be overlapped. Make a drawing of the windows; include measurements of glass, woodwork and overall size. Your dealer will do the rest.

Other special windows

Bays and other unusual windows can frequently be fitted for pleated shades. Make a drawing of the window; include measurements of glass, woodwork and overall size. Or ask your dealer to do the measuring for you.

Layered Treatments

Pleated shades are most often used with an undertreatment. If inside mounted, no extra projection is needed for the overtreatment. If outside mounted, the overtreatment must have a clearance of 2 1/4" to clear the headrail. A cornice used over pleated shades should have a 4 1/2" return. ; 1 ■ 1

TABLES Stack

Chart Shade length 24" 36" 48" 60" 72" 84" 96" STACK UP Stack W 1%
 2" Vtt1 2%" 2%" 2'>" Measurements are from top of headrail to bottom of
 bottom rail. CONVENTIONAL TRAVERSE RODS Measuring for Most
 Windows tie =a Outside or ceiling mount Conventional rods may be
 mounted on the wall or ceiling. Measure the width of glass; if total glass
 exposure is desired, add for stackback (see Table 4). Rods should be hung
 so that drapery headings (pleated tops of panels) are at least 4" above the
 glass, so they can't be seen from the outside. Figuring Stackback Stackback
 is the amount of wall space needed if open panels are to clear the glass
 completely. This dimension, added to the window opening, gives you the
 proper rod length. Begin by measuring the window opening, then consult
 Table 4. Find your opening measurement and read across the right rod
 length. -DRAPERY COVERAGE VW Measuring for Special Windows
 Sliding doors Measure as for outside mounted rod. Use a one-way draw
 rod. 1058

Specialties WINOOW TREATMENTS Guidelines Corner and bay windows
 Measure each window as if it were set flat into the wall. At corners, run one
 rod into the corner and butt the other into it. At bays, butt all rods. Use
 either one or two-way draw rods for corners. For bays, use three two-way
 rods or two one- ways with a two-way in the center. F HAA CORNER
 WINOOW /V\Ai *AAA Layered Treatments Conventional traverse rods
 come in sets specifically for layered treatments, If you are using an
 undercurtain, you will want a traverse and plain rod set; both rods are on
 one set of brackets. Overdrapery returns will be AW to 5W. If you are using
 underdraperies, use a double traverse rod set, Again, one set of brackets
 holds both rods, Overdrapery returns will be 5W to 6V2". If the
 undertreatment is inside mounted or is an outside mounted mini-blind,
 pleated shade, Romanette woven wood or a cafe curtain, use a single rod
 and set the brackets for maximum clearance. Drapery returns will be AW, If
 the undertreatment is an outside- mounted vertical blind or a woven wood
 shade other than Romanette, special extender plates for brackets and
 supports are available. Overdrapery returns will be 5>/2"to 6V2",
 Remember, whenever you change the clearance of the brackets, you also
 change the drapery return. VERTICAL BLINDS Measuring for Most
 Windows INSIDE MOUNT Ceiling mount Measure desired width and
 length of verticals; deduct at least %" for floor clearance. Clrmosa 2000

Ask your designer, decorator, or store to measure for you. Measuring for Special Windows

TABLE 7 One-Way Draw Stackback Sliding doors Use a one-way draw. Measure width from trim to trim. Add to this measurement desired extra width for overlap beyond door. If total glass exposure is desired, also add for stackback (see Tables 6 and 7). Measure from a point 2V_s" above door trim to floor; deduct 6" for clearance. Layered Treatments When layered, verticals are most often used as an undertreatment. If inside mounted, no extra clearance is needed for the overtreatment. If outside mounted, the overtreatment must have a clearance of 6" to clear the open vanes. A cornice used over verticals should have a 6" return. Figuring Stackback Stackback is the amount of wall space needed if open verticals are to clear the glass completely. This dimension, added to the window opening, gives you the proper track length. Begin by measuring the window opening, then consult Table 6 or 7 for the type of treatment you desire - one- or two-way draw. Find your opening measurement and read across for the right track. (Note: Stackback figure for two-way draw is total stack; one-half of this is on each side of the window,) If your window opening is somewhere in between the measurements in the tables, go to the next smallest opening. Add the stackback listed there to your opening dimension.

TABLE 6 Two-Way Draw Stackback butSIDE mount Window opening

Stackback Track Inside mount Measure width at top, center and bottom. Use narrowest measurement. Verticals will be made slightly narrower to slip inside easily. Measure length from inside top of opening to sill. A minimum 3V_a" recess is required for track; AW if open vanes are to be flush with front of opening. Outside mount Measure width of opening. Add for stackback (see Tables 6 and 7). Measure from a point 2 1/2" above top of frame to sill or floor; deduct 6" for clearance. Minimum projection of front of vane from wall is 5"; maximum is 6W". Minimum clearance of back of vane from wall is 1"; maximum is 2 1/2".

24" 30" 36" 42" 48" 54" 60" 66" 72" 78" 84" 90" 96" 102" 108" 114" 120" 9" 10" 11" 12" 14" 15" 16" 17" 19" 20" 21" 22" 23" 25" 26" 27" 29" 33" 40" 47" 54" 62" 69" 76" 83" 91" 98" 105" 112" 119" 127" 134" 141" 149" Window opening 24" 30" 36" 42" 48" 54" 60" 66" 72" 78" 84" 90" 96" 102" 108" 114" 120" MINI BLINDS Stackback 7" 8" 9" 11" 12" 13" 14" 15" 17" 18" 19" 20" 21" 22" 23" 25" 26" Track 31" 38" 45" 53" 60" 67" 74" 81" 89" 96" 103" 110" 117" 124" 131" 139" 146" Measuring for Most Windows Inside mount Measure width at top, center and bottom, Use narrowest measurement. Blinds will be made

slightly narrower to slip inside easily. Measure length from inside top of open to sill, A 1½" deep recess is needed for flush mounting of Mono-Rail minis; however, a difference of W is not objectionable. Outside mount Measure height and width of area to be covered. It is recommended that blinds overlap window opening by at least W on each side. Measure from top of frame to sill or 4" below opening if there is no sill. —. I— T~S %

INSIDE MOUNT OUTSIDE MOUNT Ceiling mount Measure desired width and length of blind. Overlap window openings by at least 4" on each side.

TABLE 8 Stack Chart Blind length 36" 48" 60" 72" 84" 96" 108"

Mini stack 3" 3½" 4" AVa" 4¾" 5½" 5V4" Micro stack 3V4" 3½" AVa" W 5½" 5½" 61/*" Measurements are from top of headrail to bottom of bottomrail. 1059

Specialties ELEVATORS Elevator Types and Planning 1060 ELEVATOR TYPES Hydraulic: For low-rise buildings - speeds up to 200 ft/min. Ideal where design limitations preclude overhead supports and machine rooms. Economical to install and maintain; no penthouse or load-bearing walls required. Geared traction: For low- to medium- rise buildings - speeds up to 400 ft/min. Recommended for all types of buildings where higher speeds are not essential, Gearless traction: Recommended for high-rise applications requiring the ultimate in service - speeds of 500 ft/min and up. ELEVATOR PLANNING Starting Point Recommendations for All Types of Buildings When preparing schematics for a particular type of building, select the quantity capacity and speed from one of Tables 1 to 4 and keep in mind that generally speaking: ■ Passenger elevators should be wide and shallow with center-opening or single slide doors. ■ Service elevators should be narrow and deep with two-speed doors. ■ Combination passenger/service elevators should be almost square with either center-opening or two-speed center-opening doors. ■ Freight elevator size and shape should be determined by the dimensions of goods to be carried and by the loading/unloading methods used. Doors should be of the vertical bi-parting type. The data contained in Tables 1 to 4 are based upon the following criteria. Office buildings 100 square feet per person; an interval of 30 sec; net rentable area = 80% of gross area; 5 minute carrying capacity of 12% of building population; typical floor heights were estimated at 12' and the main floor at 18'. When the building exceeds 250,000 square feet total, it is suggested that consideration be given to the use of separate freight elevators which are

not included in Table 1. Hospitals 5 minute vehicular demand = .04 x the number of beds; interval of 35-50 seconds for vehicular traffic; visitor and staff population = 3 x number of beds; 5 minute carrying capacity equal to 12% of building population. Hotels Registration during conventions = 1.5 x the total number of rooms; maximum 1 hour peak is 1.15 x total registrations; 5 minute carrying capacity = 10% of total 1 hour peak load. Apartments Population est. @ 2 persons per bedroom; 5 minute carrying capacity of 7%; maximum waiting interval of 60-90 seconds; average of 9'0" floor height. Further; Table 4 applies only for average or middle income apartments. For applications beyond the scope of the table such as local-express arrangements, luxury apartment buildings and other considerations, please consult your local elevator company representative. Note: If a restaurant or general assembly area is located in your building (on any but the main floor) and is not served by a separate elevator, the information contained in Tables 1 to 4 may not apply.

TABLE 1 Office Buildings - Passenger Elevators Only

Number of Floors	Including Main	5 floors	6 to 8 floors	9 to 12 floors	13 to 15 floors	16 to 19 floors	20 to 22 floors
GROSS SQUARE FOOTAGE/FLOOR	5000 Sq. Ft.	2 Elev.	25000	25000	25000	25000	25000
2 Elev.	25000	2500#	@ 250 fpm	3 Elev.	25000	350 fpm	4 Elev.
25000	25000	2500#	@ 350 fpm	4 Elev.	25000	400-500 fpm	4 Elev.
25000	25000	2500#	@ 600 fpm	5 Elev.	25000	700 fpm	5 Elev.
25000	25000	2500#	@ 700 fpm	5 Elev.	25000	800 fpm	8000 Sq. Ft.
2 Elev.	2500#	@ 250 fpm	3 Elev.	25000	350 fpm	4 Elev.	25000
400-500 fpm	4 Elev.	2500#	@ 600 fpm	6 Elev.	25000	700 fpm	6 Elev.
30000	@ 800 fpm	12000 Sq. Ft.	3 Elev.	2500#	@ 250 fpm	3 Elev.	2500#
@ 400 fpm	4 Elev.	3000#	@ 500 fpm	5 Elev.	3000#	@ 700 fpm	* * 16000 Sq. Ft.
3 Elev.	25000	@ 300 fpm	4 Elev.	25000	@ 400 fpm	5 Elev.	35000
@ 600 fpm	6 Elev.	3500#	@ 700 fpm	* * *	Because of the complexities such as local and express arrangements and service requirements inherent in taller and larger buildings, we suggest you contact your Armor representative when the limitations of this chart are exceeded.		

TABLE 2 Hospitals

Number of Floors	Upto 6 Floors	7 to 9 floors	10 to 12 floors
BEDS/FLOOR	Up to 20 Beds	3 Elev.	4500#
3 Elev.	4500#	@ 200 fpm	6 Elev.
3-3000	1	@ 500 fpm	3-4500#
@ 500 fpm	3-4500#	@ 500 fpm	7 Elev.
3-3000	*	@ 500 fpm	4-45000
@ 500 fpm	21 to 30 Beds	IT 4 Elev.	300 fpm
300 fpm	7 Elev.	@ 500 fpm	@ 500 fpm
4-3000	C 4-4500	I Elev.	f @ 500 fpm
@ 500 fpm	@ 500 fpm	31 to 40 Beds	5 Elev.
2-3000	@ 350 fpm	3-4500	1 @ 350 fpm
8 Elev.	4-30000	@ 500 fpm	4-4500#
@ 500 fpm	9 Elev.	4-3000	* @ 500 fpm
5-4500#	@ 500 fpm	41 to 50 Beds	2-301
3-4	5 Elev.	350 fpm	350 fpm
THEVT	4-3000	* @ 500 fpm	5-45000

500 fpm 10 Elev. 4-30000 @ 500 fpm 6-4500) @ 500 fpm NOTE: The number of elevators listed above will most likely be reduced if automatic cart and/or container handling equipment for food, laundry, surgical instrument and central supply distribution is used. TABLE 3 Hotel Buildings - Passenger Elevators Only Number of Floors 4 to 5 floors 6 to 9 floors 10 to 12 floors 13 to 15 floors 16 to 20 floors UNITS/FLOOR Up to 14 Units 2 Elev. 3000#@ 150 fpm 2 Elev. 30000 ©300 fpm 3 Elev. 30000 ©300 fpm 4 Elev. 3000# @ 350 fpm 4 Elev. 30000 ©350 fpm 15 to 20 Units 2 Elev. 3000# ©175 fpm 2 Elev. 30000® 350 fpm 3 Elev. 30000 © 350 fpm 4 Elev. 30000® 350 fpm 4 Elev. 30000® 400 fpm 21 to 28 Units 2 Elev. 30000 ©200 fpm 2 Elev. 30000 @ 400 fpm 4 Elev. 30000 © 350 fpm 4 Elev. 30000® 350 fpm 4 Elev. 30000 ©400 fpm 4-1 Service 35000® 400 fpm 29 to 32 Units 2 Elev. 30000® 200 fpm 3 Elev. 3000#@ 300 fpm 4 Elev. 30000 ©350 fpm 4 Elev. 30000® 350 fpm 4 Elev. 30000® 400 fpm 4-2 Service 35000 @ 400 fpm NOTE: Because of the complexities such as local and express and service requ taller and larger hotels, we suggest that you contact you r Armor representatives of this chart are exceeded. TABLE 4 Apartment Buildings - Passenger Elevators Only irements inherent in when the limitations Number of Floors Including Main 6 floors 12 floors 18 floors 25 floors 32 floors 40 floors BEDROOMS/FLOOR 10 Bedrooms 1 Elev. 20000 ©150 fpm 2 Elev. 20000 ©200 fpm 3 Elev. 20000 ©350 fpm 3 Elev. 25000 ©400 fpm 3 Elev. 25000 ©500 fpm 4 Elev. 25000 ©700 fpm 16 Bedrooms 1 Elev. 20000® 150 fpm 2 Elev. 20000 ©200 fpm 3 Elev. 20000 ©350 fpm 3 Elev. 25000® 400 fpm 4 Elev. 25000® 500 fpm • 22 Bedrooms 2 Elev. 20000® 150 fpm 2 Elev. 25000 ©250 fpm 3 Elev. 25000 ©350 fpm 4 Elev. 25000 ©400 fpm * • 30 Bedrooms 2 Elev. 20000® 150 fpm 3 Elev. 25000 © 250 fpm 4 Elev. 25000 @ 350 fpm * • *

Specialties ELEVATORS Low- and Mld-Rlse Elevators HYDRAULIC ELEVATORS Ideal for use in buildings up to six floors. Supported and raised by a powerful hydraulic plunger, the Oildraulic is renowned for smooth performance, quiet operation, and accurate floor leveling. And since it's supported ■from below, no vertical load is placed on the building. That means hoistways can be of lighter construction and no penthouse is needed. The machine room can be located nearly anywhere to let you maintain a flat roof line and save money on construction. HOISTWAY PLAN RIGHT HAND DOOR SHOWN LEFT HAND AVAILABLE RECOMMENDED

SIZES AND CAPACITIES TYPE APART- AVERAGE LARGE
 BUILDING MENT OFFICE/HOTEL OFFICE/STORE MODEL
 FLEETWOOD 21 -H PLAN1 MAROUIS-25 PLAN 2 MAROUIS-25
 SEVILLE-30 SEVILLE-3S CAPACITY [IN POUNDS] DIMENSIONS A1
 B< C Dz E F G 2100 & 5'-8" 4'-3" — 7'-4" 5'-9" 15'-0" 3'-0". 2500& 6'-8"
 4'-3" 3'-6" 8'-4" 5'-9" 17'-0" 3'-6" 2500 h 6'-8" 4'-3V2" 3'-6" 8'-4" 6'-89(J"
 17'-0" 3'-6" 3000 & 6'-8" 4'-9" 3'-6" 8'-4" 6'-3" 17'-0" 3'-6" 3500 b. 6'-8"
 5'-5" 3'-6" 8'-4" 6'-11" 17'-0" 3'-6" ' Inside dimensions. 2 Single car
 dimensions. &These models meet minimum size lor handicapped use.
 Hoistway dimensions are based on no provisions for seismic conditions and
 8'-0" O.A, nominal cab height. Standard speeds available: 75, 100, 125 150
 FPM. POWER UNIT (MACHINE) LOCATION: The most desirable
 machine room location is on the lowest floor, adjacent to the elevator
 hoistway. It may, however, be located remote from hoist- way if necessary.
 Typical size for one-car installation: 7'-10" x 5'-6" x 8'-0" high; for two
 cars: 10'-9" x 6'-6" x 8'-0" high. Enclosure to meet local code requirements
 must be provided. A sound- isolated machine room is recommended for
 quietest operation. Adequate heating and ventilation of machine spaces
 must be provided. 1061

Specialties ELEVATORS High-Rise Elevators TRACTION ELEVATORS
 For use in buildings over six floors. They can serve up to 27 landings and
 can be used in office buildings, apartment complexes, dormitories, hotels,
 and other structures. These elevators can travel up to 1000 ft/min and are
 ideal for high-rise buildings of all kinds, HOISTWAY PLAN -1 HOIST
 BEAM BY OWNER - TEMPERATURE RANGE: 50°F MIN. 90°F MAX.
 TRAFLOMATIC 100°F MAX. COMPUTAMATIC ■rsry ■! I | \ZT TR
 4"SLAB BY OWNER my mm m Smcrfx mcogo UDDER TO PIT BY
 OWNER — "'I PEE-"' iB'-d'COMPUTAMAnC' 7'-6,TTIAFLOMAnC
 ROUGH SILL RIGHT HAND DOOR SHOWN: LEFT HAND
 AVAILABLE RECOMMENDED SIZES AND CAPACITIES TYPE
 SMALL OFFICE/ AVERAGE LARGE BUILDING APARTMENT
 OFFICE/HOTEL OFFICE/STORE MODEL SPF21-H6. SPF25& SPF30k
 SPF35& CAPACITY (IN POUNDS) DIMENSIONS A' B' C D* E F G
 2100 5'-8" 4'-3" — 7'-4" 6'-8" 15'-0" 3'-0" 2500 6'-8" 4'-3" 3'-6" 8'-4" 6'-8"
 17'-0" 3'-6" 3000 6'-8" 4'-9" 3'-6" 8'-4" 7'-2" 17'-0" 3'-6" 3500 6'-8" 5'-5"
 3'-6" 8'-4" 7'-10" 17'-0" 3'-6" 'Inside dimensions 2 Single car dimensions

&These models meet minimum size for handicapped use. Hoistway dimensions are based on 1" out of plumb, no provisions for seismic conditions, and no occupied space below hoistway. If these conditions cannot be met, then consideration must be given for additional required space. MINIMUM PIT, OVERHEAD, MACHINE ROOM DIMENSIONS CAPACITY (IN LBS.) 2100 2500 3000 3500 DIMENSIONS L O P< L O P4 L O P4 L O P4 SPEED (FEET PER MINUTE) 200 16'-0" 15'-4" 5'-0" 16'-0" 15'-4" 5'-0" 16'-0" 15'-4" 5'-0" 350 16'-0" 15'-4" 5'-0" 16'-0" 15'-4" 5'-0" 16'-0" 16'-0" 5'-0" 16'-0" 16'-0" 5'-0" 450 — — — 16'-0" 16'-4" 6'-6" 16'-0" 16'-4" 6'-6" 16'-0" 17'-6" 6'-6" Overhead "O" based on 8'-0" O.A. nominal cab height. 3 Machine beams designed per ANSI/ASME code A17.1 and does not include floor weight and loads on floor. ' 6'-0" min. "P" travel above 250'-0" SPF21 SPF25 1 travel above 225'-0"SPF30 V travel above 200'-0" SPF35 J for speeds up to 350 F/M 1062

Specialties ELEVATORS High-Rise Elevators HOISTWAY PLAN ROUGH SILL RIGHT HAND DOOR SHOWN LEFT HAND AVAILABLE RECOMMENDED SIZES AND CAPACITIES TYPE SMALL AVERAGE LARGE BUILDING OFFICE/APARTMENT OFFICE/HOTEL OFFICE/STORE DIMENSIONS A' B1 C D2 E3 F G CA 2000 b 6'-0" 3'-7" 3'-0" 7'-8" 6'-0" 15'-8" - PACITY (IN POUNDS) 2100k 5'-8" 4'-3" — 7'-4" 6'-8" 15'-0" 3'-0" 2500 b 6'-8" 4'-3" 3'-6" 8'-4" 6'-8" 17'-0" 3'-6" DS1 3000k 6'-8" 4'-9" 3'-6" 8'-4" "a 7'-2" 17'-0". 3'-6" 3500 b 6'-8" 5'-5" 3'-6" 8'-4" "a 7'-10" 17'-0". 3'-6" 4000 b 7'-8" 5'-5" 4'-0" 9'-4" "a 7'-10" 19'-0". 4'-0" 1 Inside dimensions 2 Single car dimensions 3 Add 3" when speed = 500 FPM or above AAdd2"when speed = 1000FPM • Add4"when speed = 1000FPM Hoistway dimensions are based on 1" out of plumb, no provisions for seismic conditions, and no occupied space below hoistway. If these conditions cannot be met then consideration must be given for additional required space. MINIMUM PIT, OVERHEAD, MACHINE ROOM DIMENSIONS CAPACITY (IN LBS.) 2000 2100 2500 3000 3500 4000 DIMENSIONS L O P5 0 0 P5 L O P5 L O F* L O Ps SPEED (FEET PER MINUTE) 200 16'-0- 15'-4" 5'-0- 16'-0" 15'-4" 5'-0" 16'-0" 15'-4" S'-O- 16'-0- 15'-4" 5'-0- 16'-0- 15'-4" 5'-0" 350 16-0- 15-4" 5'-0" 16'-0" 15'-4" 5'-0' 16'-0" 15'-4- 5'-0- 16'-0- 16'-0- 5'-0- 16'-0" 16'-Q- 5'-0- 450 16'-0" 16-4- 6'-6" 16'-0" 16-4- 6'-6- 16'-0- 16'-4- 6'-6" 16'-0" 16'-4-

6'-6" 16'-0" 17'-6" 6'-6" 500 16'-6" 18'-6" io'-r 17'-0" 18'-6" io'-r 17'-0" 18'-6" io'-r 700 16'-6" 20'-6" 11'-5" 17'-0" 20'-6" ir-5- 17'-0" 20'-6" 11'-5" 800 18'-0" 20'-0" ir-6- 18'-0" 20'-6" 1V-6" 1000 18'-0" 20'-0" 11'-6" 1 B'-O- 20'-6" ii'-e- NOTE; These dimensions are for general application to custom designed elevators. Overhead "0" based on 8'-0" O.A. nominal cab height. ' Machine beams designed per ANSI/ASME code A17.1 and does not include floor weight and loads on floor. ■ HOIST BEAM BY OWNER TEMPERATURE RANGE; 50°F MIN, 90°F MAX. TRAFLOMATIC 100°F MAX. COMPUTAMATIC =P ■ II! ^ 18'-6" COMPUTAMATIC* 7'-6" TRAFLOMATIC 1063

Specialties {LEVATORS Passenger Elevator Door/Entrance Details
 CENTER-OPENING DOORS These permit quickest entry and exit, thus speeding elevator service, and provide an attractive balanced appearance both in the hallway and inside the elevator car. They should always be used in high-speed applications. DETAIL 2 (TYP.) DETAIL 2 (TYP.) NOTE: H=opening in 2" increments only [use next higher even dimension for odd size door). C=actual frame opening TWO-SPEED SLIDING DOORS Doors of this type provide the widest possible opening width for small cars but do not afford the entry and exit speed of center-opening doors. The two doors move in the same direction, one sliding behind the other. REFER TO 'D" DIM. ON APPROPRIATE HOISTWAY PLAN. DETAIL 2 (TYP.). FRAME OPENING flight hand entrance shown. Left hand available where required. DETAIL 3 (TYP.) NOTE: H=opening in 2" increments only (use next higher even dimension for odd size door). C=actual frame opening SINGLE SLIDING DOORS This is the most economical type of elevator door, and also the slowest. The single door moves either to right or left from one side of the elevator car, the opening being limited by the width of the door and car. REFER TO ■ D" DIM. ON APPROPRIATE HOISTWAY PLAN VERTICAL SECTION (TYPICAL) DRYWALL CONSTRUCTION 3 1/2" MINIMUM DETAIL 2 (TYP.) Right hand entrance shown. Left hand available where required DETAIL 3 (TYP.) NOTE: H=opening in 2" Increments only (use next higher even dimension for odd size door). C=actual frame opening DETAIL 2 i \\\\\ 2" j 4"x4"x3/8-ANGLE- o .•> v DETAIL 4-SILL SUPPORTS 4" t -i:«-^.;« ,,,.!>• K\\l K ROUGH SILL LINE—) ROUGH SILL LINE - RECOMMENDED ALTERNATE CENTER OPENING AND SINGLE SLIDING DOORS 5" x 5" X 3/8"

ANGLE - 2" V • t. 5-1/2' * . < 5-1/2" v-n i' ' * ' A v ■ ■ *: \ K\l i\ ROUGH
 SILL LINE—] ROUGH SILL LINE- RECOMMENDED ALTERNATE
 TWO-SPEED DOORS F 1064

Specialties ELEVATORS Barrier-Free Considerations Floor plan of elevator cars The floor area of elevator cars shall provide space for wheelchair users to enter the car, maneuver within reach of controls, and exit from the car, Acceptable door opening and inside dimensions shall be as shown in Fig, 1. The clearance between the car platform sill and the edge of any hoistway landing shall be no greater than 3/8 in (9.5 mm). Illumination levels The level of illumination at the car controls, platform, and car threshold and landing sill shall be at least 5 footcandles (53.8 lux). Car controls Elevator control panels shall have the following features: 1. Buttons. All control buttons shall be at least 1/2 in (12.7 mm) in their smallest dimension. They may be raised or flush. 2, Tactile and visual control indicators. All control buttons shall be designated by raised standard alphabet characters for letters, arabic characters for numerals, or standard symbols as shown in Fig. 3(a), and as required in ANSI A17.1-1978 and A17.1a-1979. The call button for the main entry floor shall be designated by a raised star at the left of the floor designation [see Fig. 3(a)], All raised designations for control buttons shall be placed immediately to the left of the button to which they apply, Applied plates, permanently attached, are an acceptable means to provide raised control designations, Floor buttons shall be provided with visual indicators to show when each call is registered. The visual indicators shall be extinguished when each call is answered. 3. Height, All floor buttons shall be no higher than 48 in (1220 mm), unless there is a substantial increase in cost, in which case the maximum mounting height may be increased to 54 in (1370 mm), above the floor. Emergency controls, including the emergency alarm and emergency stop, shall be grouped at the bottom of the panel and shall have their centerlines no less than 35 in (890 mm) above the floor [see Fig. 3(a) and (b)]. 4. Location. Controls shall be located on a front wall if cars have center opening doors, and at the side wall or at the front wall next to the door if cars have side opening doors [see Fig. 3(c) and (d)]. Car position indicators In elevator cars, a visual car position indicator shall be provided above the car control panel or over the door to show the position of the elevator in the hoistway As the car passes or stops at a floor served by the elevators, the corresponding numerals shall

illuminate, and an audible signal shall sound. Numerals shall be a minimum of 1/2 in (13 mm) high. The audible signal shall be no less than 20 decibels with a frequency no higher than 1500 Hz. An automatic verbal announcement of the floor number at which a car stops or which a car passes may be substituted for the audible signal. Emergency communications If provided, emergency two-way communication systems between the elevator and a point outside the hoistway shall comply with ANSI A17.1-1978 and A17.1a-1979. The highest operable part of a two-way communication system shall be a maximum of 48 in (1220 mm) from the floor of the car. It shall be identified by a raised or recessed symbol and located adjacent to the device. If the system uses a handset, then the length of the cord from the panel to the handset shall be at least 29 in (735 mm). The emergency intercommunication system shall not require voice communication. ■ 80min- 2030 36 mtn 915 (a) I 68min 1 ' 1730 T ^ yyV 55 c 1 5 <<*• 2 if) r1 1370 -4 i— ,,— 1 36min 9 15 J MOTE: The automatic door reopening device is activated if an object passes through either line A or line B. Line A and line B represent the vertical locations of the door reopening device not requiring contact Fig. 1 Minimum dimensions of elevator cars. Fig. 2 Hoistway and elevator entrances.

Specialties ELEVATORS Barrier-Free Considerations * u H s c d 0 e In or oor osad o« _ pan ■ncy_ m 3/4 19 control diamet 70 80 50 k 60 -•10 N 20 SO BO \ -• o mo -i o ©o i _ * _*,—i er amargancy_ slop octagon symbol •lull ba tactual, but tha X I* not (a) Panel Detail (b) Control Height J^mmm7 .. . f I L rLJ Alternate Locations of Panel with Center Opening Door (d) Alternate Locations of Panel with Side Opening Door Fig. 3 Car controls.

Specialties ELEVATORS Typical Section of Escalator © MP FLOOR ESCALATOR. N2 1^2 1.; 9 -a" zs- a !a' TO EPfcE OF UPPB. SUI'OuRI. 7MO« BETWEEN W0P.WN4 POINTS. *"iK.'n*K:uwe twrewp Pwre olm« ioiw AUJK. WW*. . ft «e s*s6 to <♦ rMimtfp to *flsxtt,*»«i.' 4 J T-ICf CTION A-A scXUr rv-i'-o" © -MWJ S3WSO- 1067

Specialties ELEVATORS Wheelchair Lifts controls—"dead man" type - platform In lower position doors with Interlock Fig. 4 Wheelchair lilt. In certain installations where ramps may be Impossible due to space

limitations, small mechanical wheelchair lifts can be installed to overcome level changes. Manufactured lifts have a lift range from two to several feet and are either electro-mechanical, hydraulic, or pneumatically operated. Lifts can be semi-enclosed and equipped with entrance interlocks for safety, and either key-operated for limited use or button-type. "Dead-man" controls are recommended for safety. I—32'Vii"—« ACCESS WIDTH IV-h * .(SEE CHART A) 39Vn" , OPTIONAL: UPPER GATE W/KEY CALL-SEND CONTROL (WALL MNT'D OR GATE MNT'D) R.H. OR L.H. GATE SWINGS 42V4" UPPER LANDING FRONT ELEVATION "L" (SEE CHART A) MACHINE HS'G LOWER LANDING 34V,, " 48" — 1 NOTE: ALL DIMENSIONS ARE NOMINAL - *| 15" 4 SIDE ELEVATION 2V PLATFORM ' I HGT HEIGHT "H" 67V, 91V. ', 15 V, 139V. 163', "V 48 72 96 1Z0 144 42V<' ^" Alectrical junction outlet 50" _i 1068

Specialties INDOOR RECREATION Basketball ££L- COURT SURFACE FAN SHAPED BACKBOARD RECTANGULAR BACKBOARD Fig. 1 NCAA basketball. The color of the lane space marks and neutral zone marks shall contrast with the color of the bounding lines. The mldcourt marks shall be the same color as the bounding lines. All lines shall be 2 In wide (neutral zone excluded). All dimensions are to Inside edge of lines except as noted. Backboard shall be of any rigid weather-resistant material. The front surface shall be flat and painted white unless It Is transparent. If the backboard Is transparent, It shall be marked with a 3-In wide white line around the border and an 18 x 24-In target area bounded with a 2-In wide white line. [High school recommended court Is 84 x 50 ft with a 10-ft unobstructed space on all sides (3 ft minimum). Collegiate recommended court Is 94 x 50 ft with a 10-ft unobstructed space on all sides (3 ft minimum).] MM 185'.3M") I6'-7"I 13M«2'.7K") 13M I42*.7 V) 5.8QM(15'.X") 1,80MI5MV'I RECTANGULAR BACKBOARD /— I.SOMfS'-IV) / RADIUS EDGE OF PAVEMENT - UNOBSTRUCTED SPACI COURT LAYOUT Fig. 2 AAU basketball court. All dimensions are to Inside edge of lines except as noted. All lines to be .05 m (2") wide. Backboard shall be of any rigid weather-resistant material. The front shall be flat and painted white unless It Is transparent. If the backboard Is transparent, It shall be marked with a .05-m-wlde white line around the border and a .45 x .59-m target area bounded with a .05-m-wlde white line.

1069

Specialties INDOOR RECREATION One-, Three-, and Four-Wall Handball
 <L I 6'-0" i 6'-0" ; 6'-0" , 6'-0" T J _f 20'-0" 24'-0" — 1V4" LINE,
 PAINTED WHITE, RED OR YELLOW , COURT SURFACE 9 b CO
 ELEVATION ,— 114" MESH XII GAUGE CHAIN LINK FENCE i ! ' M- \\
 —cwcurtiWAU.trHioH y—IMORTUMI . SrOfUNI IMWCtMAmtK-^ r—I
 j— IOHQUW W-fl" I i i J i y^toatDFPAVIMtHTOR'tHCf *• r<- ! COURT
 LAYOUT Fig. 3 One-wall handball. Playing court Is 20'0"wlde by 34T long
 plus a required 11'0" minimum width of surfaced area to the rear and a
 recommended 8'6" minimum width on Bach side. Courts In battery are to
 be a minimum of 6'0" between courts. Court markings: IVHn-wide lines
 painted white, red, or yellow. V -FRONT WALL -LINE OF FOOTINGS
 !"•«" \: £L SERVICE LINE SERVING ZONE -SHORT LINE -VERTICAL
 LINE EXTENDING 3" UP WALL ON BOTH SIDES TO DEFINE
 RECEIVING ZONE. SIDE WALL 20'-0" ff-O' JV _x: LONG LINE EDGE
 OF PAVEMENT 10 2T-8' (SINGLE COURTI Fig. 4 Handball court layout-
 four-wall. All court markings to be 1Vz fn wide and painted white, red, or
 yellow. 1070 FRONT WALL LINE OF FOOTINGS SERVICE LINE
 SERVING ZONE SHORT LINE VERTICAL LINE EXTENOING3"UP
 WALL ON BOTH SIDES TO DEFINE RECEIVING ZONE. SIDE WALL
 20'-0" 10' -f BACKWALL- - ACCESS OOR WITH VISION PANEL.
 UNDER CUT 1" FOR DRAINAGE. Fig. 5 Handball court layout—three-
 wall. All court markings to be 1Vz In wide and painted white, red, or
 yellow. Playing court Is 20'0" wide by 40'0" long plus a minimum 10'0" to
 the rear of the three-wall court. Overhead clearance required Is 20'0"
 minimum.

Specialties INDOOR RECREATION Gymnasium »UTIMAH LOCATION
 OF NtCHAMCU- P _'''-' tDUIFMCNI ROOM WHIM »|7C CONDITIOKI
 |T PONOTMRMITLOQATINIIN* ■ AUMCHT il (tilt A*«taUP«DI ■
 SECTION A-A Fig. 6 Gymnasium plan and section. •r- -METAL FAN
 SHAPED IAIKETIALL KACKSTOPS —. irOLOINO TYPE) ^
 GYMNASIUM .M ,n,V-.»% 'i:n(t» :l"f::i WOMEN/ sV131TIN0 EXEftCISE
 RM 1071

Specialties INDOOR RECREATION Tennis and Paddle Tennis COURT
 LAYOUT ISOMETRIC SHOWING NET Fig. 7 Tennis court. All
 measurements for court markings are to the outside of lines except for those

Involving the center service line which is equally divided between the right and left service courts. All court markings to be 2 in wide. 120 ft k ft h CM to *? ; _ fw C4 b 03 . . _ _ BASE LINE —T 3'-0"—| 12'-0" " " * f? ■ ^ CENTER MARK SERVICE LINE u!i NET 3 ' 1 CENTER SERVICE 1 LINE (SINGLES) j UJ a w wJ _ . 1 27'-0" UBLE) O o z s 5 > 4'-6 36--0" | -3--0" ' 12--0" FENCE ENCLOSURE OR EDGE OF PAVEMENT 60'-0" ISOMETRIC SHOWING NET Fig. 8 Paddle tennis court. All measurements for court markings are to the outside of lines except for those involving the center service line, which is equally divided between 1072 r'9'1' and 'e" senioe court' AH court markings to be 1# in wide. 10'-0' 40'-0" BASELINE r-6- 1 MIN. SERVICE LINE RIGHT SERVICE COURT NET LEFT SERVICE COURT SERVICE LINE 8ASE LINE 10'-0" LEFT SERVICE COURT RIGHT SERVICE COURT 10'-0" 20'-0" r-6" \ _ POST 10'-0" MIN. FENCE ENCLOSURE OR -EDGE OF PAVEMENT COURT LAYOUT

Specialties INDOOR RECREATION Badminton and Volleyball COURT LAYOUT ISOMETRIC SHOWING NET Fig. 9 Badminton court. All measurements for court markings are to the outside of lines except for those involving the center service line which is equally divided between right and left service courts. All court markings to be VA" wide and preferably white or in color. Minimum distance between sides of parallel courts to be 5'0". S'-O 'MIN, r*51 MTMH 9 5 z 5 9 b 9 m 9 to 9 to 9 m 9 N 20-*' (DOUBLES) 17'*(SINGLES) r W V4" RIGHT HALF COURT N SHORT SE LEFT HALF COURT ET ~ w uf _ J U Z RVICE LINE !2 tu z LEFT HALF £ RIGHT HALF > COURT g COURT ^ in z Z Q u • w m LONG o SERVICE LINE [DOUBLES) « BACK BOUNDARY BOUNDARY LINE 1 UJ □ B'-0"MIN, X*' 1 OS § F POST LINE f j EDGE OF PAVEMENT -> 2" VERTICAL TAPE MARKER ISOMETRIC SHOWING NET Fig. 10 Volleyball court, All measurements for court markings are to the outside of lines except for the centerllne. All court markings to be 2 in wide except as noted. 9 8 PE b b 9 8 9 8 i 10'-0" 9 b 0- O b 30'-0" 10'-0" SERVICE AREA 1 i ,— 2" SPIKING LINE j— 4" CENTER LINE y— 2" SPIKING LINE END LINE FrtGEOF PAVEMENT , 50'-0" SIDELINE 10'-0" o } 3'-0" MIN CO T EDGE OF PAVEMENT I.. * COURT LAYOUT 1073

Specialties INDOOR RECREATION Shuffleboard ■ "fa o J 9 to k BLACK DISCS HERE HEAD YELLOW OR RED DISCS HERE— 6'-6" „ BASE LINE EXTENDED 2'-0" OR TO NEXT COURT 4'-0" 3 o r-6' N N 3'-0" p^ ^ 09 » 3-.0", "^^.^ S-^ 3'-0" 3'-0" 12'-0" 39'-0" 52'-0" 3'-0" y-o" 00 00 3'-0" K K 3'-0" u. O O r-6" FOOT 6'-6" "-' *■ COURT LAYOUT COURT MARKING DETAIL Fig. 11 Shuffleboard court, All dimensions are to centers of lines and to edge of court. Maximum line width VA In, minimum % In. Playing court Is 6 0" x 52'0" plus a recommended minimum of Z'O" on each side or 4'0" between courts in battery. 1074

Specialties ACCESSORIES Letter Boxes f il l « r t> iDCT-^ C3 !»■ o -t- 0 □ FKWT <»!•/, '-i-"1 C3 o »ttlTI o it"Ai o CZ] o ■■;•■■■■ ■'<■ i C=3 0 C3 o vIPl I ReCtettet -mumofiM*- Surface Mounting Nesting -1*»»" C3 □ X 5 a □ C3 C3 W; Jt £ 'MHTWHOra J=—11—J- * Sofnl • BewrtMdl J, ^ NO, OF OPENINGS SINGLE 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 36 39 40 41 42 43 44 45 DOUBLE 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 66 70 72 74 76 76 60 62 64 66 88 90 FOR 3-4 ft 5 WIDE NESTS SUGGESTED ARRANGEMENT 3 4 5 3-3 3-4 4-4 4-5 5-5 4-3-4 4-4-4 4-5-4 5-4-5 5-5-5 4-4-4-4 5-4-4-4 5-4-4-5 5-4-5-5 5-5-5-5 4-4-5-4-4 5-4-4-4-5 5-5-3-5-5 5-5-4-5-5 5-5-5-5-5 5.4^4-4-5 5.4.5.4.5-4 5-5-4-4-5-5 5-5-4-5-5-5 5-5-5-5-5-5 5.4.4.5.4-4-5 5-5-4-4-4-5-5 5-5-4-54-5-S 5-5-5-4-5-5-5 5-5-5-5-5-5 5.5.4.4.4.5.5 5-5-5-4-4-4-5-5 5-5-5-4-4-5-5-5 5-5-5-5-4-5-5-5 5-5-5-5-5-5-5-5 4-4-5-5-5-5-5-4-4 5-5-5-4-4-4-5-5-5 5-5-5-5-3-5-5-5-5 5-5-5-5-4-5-5-5-5 5-5-5-5-5-5-5-5-B-5 ROUGH OPENING 17% 23'/. 28"/, 36'/: 42 47 Vi 52'%. 58'/, 66</> 71% 77'/, 82% 88'/, 98 101'/. 106% 112s'/. 117% 125"/, 131% 136'6 142 147'/, 155% 160%, 166 V.. 171'/. 177% 185'/. 190Vi 195"/, 201% 206-y,, 214% 220%. 225% 231'/. 236 Vj 244'/. 219% 255'A 260% 266%, All dimentlont In InclMt, CABINET 16% 22'i 27%, 35'j 41 46% 51'V,, 57% 65 W 70% 76%, 81% 87'/, 95 100'/, 105% 11 iy,, 116% 124"/, 130% 135% 141 M6' /, 154% 159"/, 165 V. 170"/, 176% 184'/. 169% 194"/, 200% 205"/, 213% 219V,, 224% 230'/. 235'i 243'/. 216% 254 V4 259% 26Sy,, NET OVERALL WIDTH 18% 24 W 29"/, 37% 43 46% 53"/, 59% 67 W 72% 7By,, 83% 89'/. 97 102'/. 107% 113%, 118% 126"/, 132% 137'! 143 148'/. 156% 161%, 167'. 172"/, 178% 186%, 191% 196"/, 202% 207"/, 215% 221%, 226% 232'/. 237% 245'/. 250% 256V, 261% 267%, 1075

Specialties ACCESSORIES Mail Collection Boxes .ik.- .■'; kU.s. mail:
 tmwim' urn.. ,d 19 1/4" H X 14 3/8" W x 6" D *.*■\$&** SiEr nv'iTlilniiiiH
 ■ r«frm,'. ,* -SHffEBItHr <i-H 17 1/8" H x 14 1/16" W x 16 1/8" D WW**
 MS* r»"»il" ituuworewg •H;> LETTEKAHO TEOWWUS I PO LOCK
 U.S. MAIL LETTERBOX ■fflff umwBt ;*t— -12"- OUHIOd toon • i I i
 iitWMttiftaiitoM j MWH *»u. -»*«ftM»- 30" H x 18" W x 12" D 36" H x
 20" W x 12" D 1076 LOTEMAND unERiu* HEIWIUWI.B tCHEHHC
 CMDMSERT U.S. MAIL „ _ LETTERBOX .MMMBI -BUHHIHOOIt —

Specialties -ii"- umM ano LITTIN MAH. itio im mnm.ts LETTERBOX
 U.S. MAIL ^ UTTM ^JBjiiSw vuvomum ■»" U»0HW r- t^—r -» 1^
 «MUMIW. -MMBfIMlr ■AM wwwwtu Contemporary 30" H x 18" W x
 10" D Letters lot 6 1/2" H x 14" W U.S. M i , MAIL f 39" AIL i ":"zn
 MAIL /< SCHEDULE CARD INSERT j J I 1' <r— 0 y Plate Front 26"H x
 39"W ACCESSORIES Mall Drops Classic Bronze 36" H x 19" W x 10" D
 ,7/.iENGBAVED <* LETTERS I BOOK DEPOSITORY ,2'V t20" H'i. ■1
 Book Depository 10 1/2" H x 20" W SCHEDULE CARD INSERT
 UNITED STATES MAIL LETTER MAIL TIED IN BUNDLES SO" r *r-
 Plate Front 26" H x 56" W 1077

Specialties ACCESSORIES Mail Chutes, Parcel Lockers, and Key Cabinets
 T -mronueotun rtadOFi'oon A • -XAUWF0ttlT (with cojmm «ct«i) ->(.
 I4W" 14V*" «toiw ouut pahsli n: "n fti U.S. MAIL ELEVATION (Typical)
 TOP FLOOR Wide Chute IXTKUKDMJNL «AS£S£CIK»I „——.— . f n
 "I" . _.... n EXIAUOfO ALUM. MOLDING j . GLASS bpfL- — r- —fM —
 * _ — — •tl '* 1 i?4 kWj FINISH WALL 1 i 1 1 pL «m GLASS ffi 1 ! I" 1 ir
 BOUOH H.0OB WEWNO Wide Chute ROUOH FLOOD OMNI* I" I li"
 HOUGH HALl ^m////////////////////////////////////W I" 111" BOUOH FLOOD OKWM
 Twin fU,5,POSIAI ••A'sizeiox T " I ibw"ifc==*--== H h ^ IUF- ll II L==.*
 = = 1 H ^-=-.=1 PLA CAPTIVE LOCK (W-1 MS',*" *&ri; PLB PLC Wart
 CM Horizontal Parcel Lockers * > m •16V." NEST- Key Storage Cabinet
 The safest way to keep keys stored in one location. Available in four
 different sizes, 128 keys. Holds 32 K«y» 91/4" H x 13 M" W Holds 64
 Keys lGs/ie" H x 13g/«"W Holds 96 Key* ' 23V4"H x 13«/s"W Holds 128
 Key* 30V4"H X 13 5/B*"W u central secure holding from 32 to ft i^J Key
 Keeper For use In a mall room (rear loading) Installation. The arrow lock Is

accessed by the postman to retrieve the mail room door key, and stored safely when not In use. * -<v- -IV - >!V

Specialties ACCESSORIES Telephone Baaths , 7« " r — 2 1 5" «- 1 27 3/8" W/BA8E CUTOFF FOfl MTQ, AQAINST WALL Cluster Cluster Free-standing Free-standing CENTERS Of PANELS ,«,, U—18"- TTK SJ ..-1/2" DIVIDER PANEL GROUND LEVEL - Wall-mounted WITH URtCTOftt Will *-13Trt 4«S1" ■*- nv* IT* H I UNIT) Wall-mounted Free-standing Wall-mounted 1079

Specialties ACCESSORIES types of Grilles Cast metal grille of unit design, cast in one piece. Grille size is governed by unit sizes plus width of border. Cast metal grille of Renaissance design, ferrous or non-ferrous metal. Cast metal grille of unit design, units cast separately and built into frame. Grille size is governed by unit sizes plus width of border. Cast metal grille panels of various sizes cast in units and fitted into cast or wrought metal frame. ®~Q fiRILLIL SIZF UNIT ,| \F "sir h Cast metal grille of unit design. Units cast separately and built into grille spaces. Grille size must conform to unit sizes plus widths of bars and borders. Cast metal grille for ventilator opening; may be fitted with metal screen and may be formed to curved wall or cove. Cast metal grille for ventilator opening of special architectural form. Fig, 1 Cast metal grilles may be designed and built In various combinations, They may be made In small units cast separately or as one complete piece. 1080

Specialties ACCESSORIES types of Grilles nnannnonnnnnnaQ
□□□□□□□□□□□□a □□□□□□□□□□□□ao nnnnnnnnnnnnnann □ £!
□□□□□□□□□□□□ □□□□□□□□□□□□a nnDnnnaannnnnnn nannnnnnnnnnnnn
nnnnnnnnnnnnnnnn nnanDnnnnnnannn □□□□□□□□□□□□a annnnnannnnnnnn
nnannnnnnnnnnnn nnnnnnnnnnnnnnnn nnnnnnnnnnnnnnnn gogogogogogogog
88888888 88888888 88888888 88888888 rS«2ll§l MS TOE Mrs 0P21&I
ElPgIfSI EMS Stamped metal grille, square holes. Holes may be from V*" to)!4" or larger, In Va " steps, with bars or frets from '/'»" to W or greater.
Q-GQQC PQQOT nTTTT Stamped metal grille, square holes diagonally arranged. nnannnnDnnac: nnnnnnnnannc an an an □□ an □□ □□ an oooco
Stamped metal grille, Grecian or Union Jack design, made in a number of unit sizes and variations. ogo ogo O nan OXOddd v ana OaOddd v ana

Annn <XOr-ir-irn ogo ogo nnaac nnaac annac nnaac nnaac nnaac
 Stamped metal grille, shell or fish scale design, made in several unit sizes
 and variations. Stamped metal grille, square holes arranged to form plain
 border with diagonal center. Stamped metal grille, square holes arranged to
 form diagonal border, with plain center. Foundation or Bulkhead Grille /~\
 v. - /->. ^ va y!w//aaaa^^^ Screwdriver Operated Catch YA yJ77y"////A&S
 ssssss Lock and Dead Bolt SCALE ll"" I'-o" AND i" = l" PROVIPE STOP
 A5 REQUIRED \ VA y>///AA/77\^ssss^ S Thumb Latch SPECIFYt Slut of
 holit or unit itampingi Thlckntu Mital and flnith Approximate width of
 marflnt Method af fattening Hlng*t and lockt where required Invisible
 hand hoi* doort. Foundation or Bulkhead Grille Fig. 2 Stamped metai
 griiies are produced in a great variety of designs, metais, thicknesses, and
 stzes, Percentage of free area of stamped griiies may vary from - about 25%
 to over 70%, with a great many designs in the 55% to 65% range. Margin
 widths can be made to accord with requirements of particular instaiiations,
 consideration being given to duct openings and overaii dimensions. Metal
 may be steel, painted or otherwise finished, bronze, aiuminum, monei
 metai, or other nonferrous metais, in thicknesses from 16 gauge to V*".
 1081

Specialties ACCESSORIES Types of Grilles ~T— — I i I .1 r- I I 3 ~1
 Perforated Metal Grilles Si MIS □ tfottof GRItLE 3 PANtLS 77F>'. Grille
 of perforated sheets set into frame of channel, angle, and mouldings.
 Perforated Metal Grilles V4//f/i Flat Band Frame Reversed Angle Frame J'
 XWKM Moulding and Angle Frame Moulding and angle frame to swing in
 fixed frame. Tee Bar Frame SPECIFY: Pattern of perforation* Mttal Finish
 Oaug* or Ihlcknoi* Give from* dttallt Fig. 3 Perforated metal grilles may
 be obtained In several designs and are produced of 19 gauge steel in sheeis
 of standard sizes. Sheets may be cut to any size and placed In frames of
 metal or other material. Perforated metal grilles are used for vent openings,
 panels, covers, shelves, partitions, cabinets, metal furniture, boxes,
 machinery guards, enclosures and many other purposes. They are also
 available In many other patterns In any ferrous or nonferrous metal that can
 be perforated, and in thicknesses from about 24 gauge In the smaller
 perforations to '<" or 3A" In the larger perforations. 1082

Specialties ACCESSORIES types of Grilles Wrought metal grille of heavy close construction, for radiator or ventilator openings. 1 ^7 1 *,» " O, "■ , - ■» C> ' - " " ■ __iaJ ... "f' ...,,: " ." i Wrought metal grille for glass door. Wrought metal grille for railing or ornamental construction. Wrought metal grille of light construction, with silhouette work, leaves, flowers, and husks, for arch decoration. Mr 'i ,i.... r|—; 1 ft U nw " T j; T IK Wrought metal grille of sheet and flat bars, for ventilating opening. Wrought metal grille for counter. Wrought metal grille for large window or opening. Fig. 4 Design of wrought metal grilles Includes the use of other metal forms, such as sheets, extruded mouldings, castings, and stampings. Thus, in addition to plain bar sections and forged Items, use is made of the unlimited number of extruded mouldings In non-ferrous metals; stamped leaves, rosettes, and ornaments of many kinds; cast Iron, bronze, nickel silver, and aluminum Items of every character; and rolled or drawn sections of many shapes, 1083

Specialties ACCESSORIES Types of Grilles rm Grille of round or square vertical bars welded to horizontal bars set In masonry. 4pPPp Grille, with ornament between bars, set in masonry opening. H I II 11 4==» *=== Grille with anchors set in masonry and grille bolted or riveted to anchors. Grille of welded construction set on face of wall. u -r" j. _ m l i i ii. ^ . " TMr" i I*- i Grille with short alternate vertical bars, angle clips fastened to opening jambs. j^tesi mm Grille of welded construction set on face of wall. H TT 1L AL XL ^ tft W A B D 1084 Methods of forming vertical bars. Fig. S Window grilles are of plain construction when used for protection only, and when used for ornamental effect, may be designed with many unique and interesting ideas. Window grilles maybe set In the masonry openings or on the face of the wall, with either plain or ornamental brackets or supports. They may also be attached to window frames, or may be arranged to swing, with hinges and locks. Material sizes In window grilles may vary according to the degree of protection required, and in proportion to (he size of (he grille.

Specialties ACCESSORIES Grille Fastening Methods , GRILLE SIZE j) /["/////VA X/X JJJj 7h DUCT Set in Rabbet GRILLE Size _Z_ xm&'* 1ZL DUCT Reversed Angle Frame on Back GRILLE SIZE.... U 1> /\«%m 17T '\ CONTINUOUS ANGLE. Band and Angle Frame GRILLE 5,1 ZE // xm&M m -1 Angle Frame in Marble GRILLE SIZE Hinged to Woodwork

GRILLE SIZE Secured by Moulding GRILLE SUE J ^T^KHtZZ Z777A- if HL ^ANGLE CLIP DUCT Angle Frame Exposed GRILLE SIZE 7 SP^ V/////A ~JZL / Angle and Plaster Ground GRILLE. SIZE Angle Frame and Moulding L GRILLE SIZE / Y/////A JZL Hinged to Angle Frame SCALE j . I GRILLE SIZE Angle Frame on Back GRILLE SIZE Reversed Angle Frame Exposed GRILLE SIZE 3 Y////////A JZL W Lugs for Marble GRILLE SJZJL 777A JZL K3 Band Frame and Moulding , GRILLE SIZE Round Rod Hinge Fig. 6 Methods of fastening grilles. Cast and stamped metal grilles may be fastened by screws or hinges and locks to walls of wood, plaster, marble, or other material In a variety of ways, depending upon the type of the grille, the type of framing to be used around the grille, and the appearance or effect desired. These methods show a number of ways In which cast and stamped grilles may be fastened. In selecting the method desired, consideration should be given to whether or not the grille will require frequent removal. The size and weight of the grille will have a bearing upon the size of frames, screws, and hinges. 1085

Specialties ACCESSORIES Grille Fastening Methods WELD =1 1< s Two members crossed, one member cut and welded to the other. Two members crossed, each member halved. Pin may be countersunk or peined. Two members crossed, one member split permitting other to pass through. U Two members crossed, one member laid upon the other, with roundhead or countersunk rivet. Two members crossed, one member punched permitting other to pass through. □ Two members crossed, each member crimped. Two members crossed, each member flattened, with round head rivet. Two members linked. Formed members welded to sides of straight member. M s Formed members banded to sides of straight member. -L-V-1- Formed members terminated and banded to sides of straight member. One member cut and welded to side of other. One member set into One member set punched hole In other against other member member and welded, and attached by flat head tapped screw. ^ Corner members Welded. -q ^ Corner members fitted, with two flat head tapped screws. 6 -J-y-J- Formed member at- Two parallel members tached to, the other attached by spool member by band In and pin. slot. One member attached to side of other member by angle clips bolted or riveted. Formed member with knife edge, welded to side of straight member. Formed member with blunt end welded to side of straight member. =3^ RS3 E^J Corner members bolted through angle clips.

Corner members with welded gusset plates. B Methods of splitting members and forming into scrolls, ends, ornaments, or structural elements. SCALE 1/2" = 1'-0" Fig. 7 Wrought metal grilles are fabricated by the use of a great many different methods of crossing and joining members. Some of the more widely employed of these methods are shown here. 1086

VINYL WALL PROTECTION GUARDS Specialties ACCESSORIES
 Protection Guards .078-VINYL .062" EXTRUDED ALUMINUM
 RETAINER MOLDED END TERMINAL RC-40 %• ANCHOR BOLT !
 WmXSQLE FAST-1, TOGGLE RUBBER VWLLSmCER 1' L-STEEL
 BACK UP CHANNa (BY OTHERS) .081" EXTRUDED ALUMINUM
 RETAINER 'A-ANCHOR BOLT W/TOGSLEMSr- "TOGGLE
 STEaBACKUP CHANNa(BY OTHERS) .063 EXTRUDED ALUMINUM
 RETAINER CONTINUOUS TOP 1 BOTTOM VINYL BUMPER ,125"
 VINYL T-31 MOLDED END CAP X" BOLT WITH EXPANSION
 ANCHOR ,07T EXTRUDED ALUMINUM RETAINER W ANCHOR
 BOLT W/7DGGLEfi»Sr- * TOGGLE RUBBER WUJLSRACER
 MOLDED ENO TERMINAL J #TS7S STEEL BACK UP CHANNEL (BY
 OTHERS) ,078" VINYL .090-EXTRUDED ALUMINUM RETAINER W
 ANCHOR BOLT .W/TOGOLEWSr- «, TOGGLE IUBBER WALL
 SPACER ■MOLDED END TERMINAL J RC-10 StEELBACK UP
 CHANNEL (BY OTHERS) ALUMINUM WALL PROTECTION
 GUARDS f"v," :U VINYL BUMPER %■ ANCHOR BOLT
 VWDGaLEfitSr- 1 TOGGLE BOLT COVER ■VINYL BUMPER
 EXTRUDED ALUMINUM END CAP 'STEE. BACK UP CHANNEL (BY
 OTHER) .125- EXTRUDED ALUMINUM RAIL ,062- EXTRUDED
 ALUMINUM RAIL AIR CUSHIONS mam* VINYL BUMPER
 ALUMINUM RETAINER EXTRUDED ALUMINUM END CAP STEEL
 BACK UP CHANNEL (BY OTHERS) EXTRUDED ALUMINUM END
 CAP VINYL BUMPER 'AIR CUSHIONS VS" ANCHOR BOLT
 WTOGGLE msr- 4 TOGGLE ANCHOR BOLT COVER ■VINYL
 BUMPER ■-STEEL BACK UP CHANNEL (BY OTHERS) .125"
 EXTRUDED ALUMINUM RAIL FLUSH-MOUNTED CORNER
 PROTECTION GUARDS .063 EXTRUDED ALUMINUM RETAINER
 18" 00 STAGGERED .110 VINYL (C RADIUS) ,063'ALUMINUM
 RETAINER IOxWSMS Too STAGGERED ALUMINUM STRUCTURAL
 BASE #10X1V."F.H,SMS 1. "7-BASE LOCK .063' ALUMINUM

RETAINER (FULL HEIGHT) y-'IOxI'/.-SMS I FORMED ALUMINUM I
5_ B** (BASE HEIGHT) ST 10B7

ACCESSORIES Vinyl Wall Protection Guards W/PLASTIC SHIELD 1 roc
STAGGERED MO x 1'// SMS 18-00 STAGGERED -.063" ALUMINUM
RETAMER .063-ALUMINUM RETAINER ,—'10xM"SMS \$g 18* 00
STAGGERED 110- VINYL FOAMCUSHION — CLOSURE TOP 4
BOTTOM .063 EXTRUDED ALUMINUM RETAINER H"QYP V*LL
BOARD .110 VINYL GUARD •10x1%' SMS iroc STAGGERED
EXTRUDED ALUMINUM STRUCTURAL BASE COVE BASE
MOLDED CAPS_ TOP&BOTTOM .063 EXTRUDED ALUMINUM
RETAINER Ceiling height-flush mount •10X1WSMS 18" oc
STAGGERED COVE BASE Wainscot installation - surface mount 1088

Specialties ACCESSORIES Handrails VINYL HANDRAILS •rmo moused
return/corner .M? EXTRUDED ALUMINUM RETAINER RC-15
RETURWSORNER .MO-EXTRUDED ALUMINUM RETAINER
^•ANCHOR BOLT W/TOGCLEfiW- « TOGGLE STEEL BACK UP
CHANNEL (BY OTHERS) .078-VINYL .ROW MOLDED
RETURN/CORNER ..OTTVINYL. .061 ■EXTRUDED -ALUMINUM
RETAINER W ANCHOR BOLT rw/roasi£«sr- •VTOGGLE -VINYL
BUMPER L. BRACKET WB-SO R&3|HM0l«D R£TURH«MNER
CONTIMUOU3 RETAINER ■tWUMAOfETCAR .»TEE». RACK UT
CHANNEL IBYCPTHEIUI VUMX BRACKET WB-SO .RC-<0
MOLDED -RETURWCOHNER SXr EXTRUDED -ALUMINUM
RETAINER .078'VINYL t+SMSer , '••ANCHOR BOLT :(-
\\rWTOGOLEWSr- \\l 4 TOGGLE RC10 'RETURN/CORNER
ALUMINUM HANDRAILS -AIR CUSHION -VINYL BUMPER
.125'EXTRUDED "ALUMINUM RAIL -R-2145 RETURN "WALL
BRACKET WB-21 1089

■■tfZ- 4'-o' TC7 titat w it iBMnaH •4MKLK**Am»r (V|t*AO |-ri.} J -
FIU.^L-,1 T»»« MT«hMM(» AT«|»tl TYPE ST ■ CI.AM94* — U»J|T»
SNMMt -cLAhitM&a. «L«T#M ft I unit <i»i*«; ?:- ' - cmj.fcft*A*,» -r
3AWt. ftK TVPE & m AT J« /jt A MTtftUiHt* "■^■●""H *«m ra/we ^
wM.rt.. TVP£ J^T •' — Vr*M«u . , , k'v|T»i»ITKI sham*****

Specialties ACCESSORIES Column Covers FIXED COVER UPPER
 ALIGNMENT CLIP, FACTORY ASSEMBLED TO REMOVABLE
 COLUMN COVER LOWER ALIGNMENT CLIP, FACTORY
 ASSEMBLED TO FIXED COLUMN COVER NAP-FORM* ASSEMBLY
 COLUMN ASSEMBLY PERSPECTIVE MOUNTING In" HOLE
 BRACKET W.'xVi" ALIGNMENT SNAP-FORM* ADJUSTMENT CUPS
 ASSEMBLY GROOVE *•" THICK JOINT DETAIL PLAN FIXED
 COVER FIRE PROOF EXISTING COLUMN AS PER LOCAL CODE
 REMOVABLE COVER COLUMN COVER 7" DIAMETER FIXED
 COVER REMOVABLE COVER COLUMN COVER 10" DIAMETER
 1091

Specialties ACCESSORIES Column Cavers ADDITIONAL BRACING
 ABOVE I COLUMN COVERS BY OTHERS 1£ TOP SLIDER -
 BRACKETS SOFTFORMS® METAL STUDS SCREW FASTEN
 COVERS TO METAL STUDS REINFORCED JOINT COMPOUND
 EMBEDDED REINFORCED TAPE CURVED TROWEL SKIM COAT
 SANDPAPER PLUMB STUDS WITH LEVEL TEMPLATE FOR STUD
 ALIGNMENT BOTTOM BRACKETS SPECIFY OUTSIDE DIAMETER
 — MINIMUM 14" EXISTING STEEL COLUMN WITH FIREPROOF
 INSULATION ROLLED ALUMINUM 1/8" THICK SOFTFORMS® FIN
 R_TAPE & ISPACKLE #6 FLAT HEAD FULL THREADED SELF
 TAPPING SCREWS SOFTFORMS® 16 GAUGE MTL STUD SERIES
 100K TYPICAL JOINT DETAIL NO SCALE SECTION -UPPER
 COLUMN COVER -POP-RIVET UPPER COLUMN COVER FIN TO
 JOINER PLATE 6" O.C. PITTCO WILL PROVIDE POP-RIVET GUN,
 DRILL BIT & ALUMINUM POP-RIVETS SOFTFORMS9 COLUMN
 COVERS SECTION 1092

Specialties ACCESSORIES Column Coven CONDITION 1 — COLUMN
 PASSES THROUGH CEILING; REQUIRES INSTALLATION PRIOR TO
 FINISHING CEILING. TAPE& STF-100 SPACKLE TRIM RING PASSES
 THROUGH GPDW CEILING WITHOUT REVEAL PASSES THROUGH
 GPDW CEILING WITH REVEAL mm^mrm^ r ±& ANGLE SPLIT-
 RING TRIM, SCREW TO COLUMN COVER JL~ isr CEIUNG
 •BRACKET. EXTRUDED ALUMINUM POST -yt" PASSES THROUGH
 LAY-IN TILE CEILING WITHOUT REVEAL CONDITION 2 —

COLUMN INSTALLED BETWEEN FIXED CEILING AND FLOOR
^ftft^M^fflW SGS8pwBShwB3ws^SwB < f • * t • > • , > * ' ri&fianMM*?
fiHnfti rfsai3 <b inr -At FLUSH TO CEILING FIBERBOARD OR
PARTICLE BOARD REVEAL (BLACK) MINIMUM REVEAL EQUALS
TIP-UP CLEARANCE G REVEAL BOARD AT FLOOR H FLUSH TO
FLOOR m SCRIBE RING AT FLOOR 1093

Specialties ACCESSORIES Column Covers 2Vz" METAL STUDS FIRE
RATED GPDW 2 LAYERS SO-9-075 SO-9-100 TAPE& SPACKLE PACK
COMPLETELY WITH FIRE RATED THERMAFIBER 0-9-150 2/2 _J_
METAL "HL STUDS FIRE RATED GPDW 2 LAYERS TAPE& SPACKLE
0-9-250 TAPE& SPACKLE FIRE RATE GPDW 2 LAYERS j^Tf 6" BENT
MTL STUD 2Vi" METAL STUDS FIRE RATED GPDW 2 LAYERS 9-300
METAL-HI STUDS 1094

Specialties ACCESSORIES Column Covert EXISTING STEEL COLUMN
WITH FIREPROOF INSULATION r REFER PAGE SS 36.0 -TAPE ft _
SPACKLE -\V*"X 1V4" MTL ANGLE TWO OFFSET -2V>" MTL
STUDS 1095

Specialties ACCESSORIES Column Coven 1096

3-ARM HIGH-TRAFFIC MANUAL AND ELECTRIC TURNSTILES 3.8?
1.18 p1.19 2» DIA. SHOWN CONDUIT /WIRE HOLES-2TYP. "tf - 24.63
• Plan A r IT ii 4 If Ijj rXi V \> i' ii ii A' "~1 ^ FINISHED FLOOR Side
elevation Front elevation Plan DETAIL "B" CUP ANCHOR BOLT
FINISHED FLOOR ■•CONCRETE HALF SCALE ANCHOR -SEE
DETAIL "B" Vertical section • •]< K^M^i ;\ ■ Square t 'Tl |, .(Socket(or ■\l-
\ I - trf Yokes &■ i'l ii .'Turnstiles' ^T>(1.75" O.D. Anchor details 55° E3 A.
7 90° 34" Side elevation Front elevation

ACCESSORIES Turnstiles 4-ARM MANUAL AND ELECTRIC
TURNSTILES | g<" 1 ~C_ 23V." qi 22V," , J -sty- Basic customer security
Enhanced security «=a right hand on left hand clockwise counterclock
Rotation guide 3-ARM MANUAL AND ELECTRIC TURNSTILES Post-
mounted Tandem (post-mounted) Wall-mounted Portable Yoke Turnstile
Maximum security HIGH-SECURITY TURNSTILES —31H"—4>—30
y," 86" W TYP. Manual 54 '/>' 1098 Plan ol electric type configuration

15- 21'." =F=F EMERGENCY EXIT Q~=" 17*" 24", 30", 36" OPENING
 37 W r 27", 33", or 39" CL Emergency - quick release T I7V 1 38" Q~ i 4
 24", 30", 36", 46" OR 60" OPENING KSJssr £j-JP\\] ^ 27", 33", 39", 5
 1"OR63"CL ' MODEL BA-1 Break-away fingerlatch features an adjustable
 tension setting. SAME mounting as Gate Finger latch CL ' Finger. Double
 rail - (lip sleeve latch 27", 33", 39" or 51"CL 10" 24", 30", 36", 48"
 OPENING cn \- u= 3D I 27". 33", 39" or 51"CL ' Double rail - aisle closure
 with (lip sleeve latch 45" or 48" 30", 36" or 48" OPENING 1 33", 39" or
 51" CL Triple rail - linger latch I SLEEVE BEARING OUTLET rb in T
 SPECIFY LENGTH OF OPENING WHEN ORDERING 12'." 12%" ED
 18W" Single rail - cart security with flip sleeve latch 10%" 44.VS" j 10W
 17 V 33HSTE3E2 13% (3=3 36" or 4B" OPENING A6V," DCE3 Triple rail
 - alarm system Single section welded rail- electrically controlled gate latch,
 wheelchair access Specialties ACCESSORIES Tlirnstlles =SV7 JJ{^ 30",
 36" cr48" OPENING -l-!-r C=% 48" 33", 39" or 51"CL Double rail -39'//-
 34" 38" (. 38.62" -I 15" H Single section welded rail - sell-closing or sell-
 opening 1099

Specialties ACCESSORIES Water Coolers WALL _ FACE ■ 5' MM,
 ALTERNATE JSfc "" SKXROUGH-N MTS. FRAME WIDTH
 /^VtWASTC TRAP OUTLET ,ote, —| w/s'.jsgn iw*wmi. I— MflJISI FOR
 DIFFERENT FOUNTAIN INSTALLATION HEIGHT, DIMENSIONS
 MARKED BY f*> MUST K ADJUSTED ACCORDINSIY. 14133 i.ffffir
 T3&P 1 a. 1.203ul^ 4= I 0 J7M - V NPT SUPPLY (5 ■t 1.04lul kl.035>>
 »39J*" 1,997k) STAWaflo HdtMT 1057-1 * THIS IS A DUAL PURPOSE
 DRINKING WATER FACILITY; THE DIMENSIONS NOTED 1*1 ARE
 RECOMMENDED HEIGHTS ONLY, WHEN INSTALLING THIS UNIT,
 LOCAL, STATE OfI FEDERAL CODES SHOULD BE ADHERED TO
 AND DIMENSIONS ADJUSTED ACCORDINGLY, ■ ■mi. iyfci j)>jtj//js?
 i}?Sj/} VJ/////;j/;///r// »34H" I.B70u) STaMJAH0 HE'tHT 1,38lu> WALL
 FACE- *3«V »29'V 1.74 9 u] WASTE I 1,670u) 7? /; ; ; ? > / > ; s / / ; / ; s /
 ' / -nrr 1100

Specialties ACCESSORIES Water Coolant .044 m .279- CLUJiMCf) 1101

Specialties ACCESSORIES Accessible Drinking Fountains equipment
 permitted in shaded area (a) Spout Height and Knee Clearance (b) Clear

Floor Space (c) Free-Standing Fountain or Cooler £ E 3 30min 760 not to exceed fountain depth (d) Built-in Fountain or Cooler Drinking Fountains and Water Coolers 1102

General Reference Data Space planning Human factors Floor and wall covering Fabric Electrical Columns, capitals, and entablatures Nails, screws, and bolts Mathematical data and formulas 1106 1110 1122 1130 1132 1135 1136 1139

This section provides a variety of time-saving reference material in the form of tables, charts, formulas, and planning guidelines. Included are area requirements for the preliminary space planning of various building types and human factors data related to anthropometrics, space, and acoustics. Also included are a number of tables for determining carpet and wall covering yardage quantities. In addition, a series of tables dealing with electrical data provides typical amperage ratings for office and electronic equipment and for residential appliances. Still other tables and charts contain mathematical data relative to functions of numbers, metric system conversions, and areas of plane figures.

General Reference Data SPACE PLANNING Area Requirements By Use
The first portion of Table 1 shows some of the planning guidelines for several types of office use. Of course, usable areas per employee vary greatly depending on the type of work performed and types of support space and common areas required, such as file rooms, data processing, conference rooms, and so forth. Rules of Thumb Office use: 125 to 150 net sq. ft, area per person. Retail space: 30 net sq. ft. per person on ground floor; 50 net sq. ft. per person on upper floors. Classrooms: 20 net sq. ft. per pupil. TABLE 1 Space Planning By Building Type Building/Use Type
Office buildings, all types Work station, minimum clerical Work station, clerical with VDT Work station, with visitor space Work station, supervisor Manager, private office Law firm Law firm library Law firm conference Insurance company, branch Insurance company, branch Total, Includes common areas and circulation Energy company Conference and dining rooms Restaurants Dining areas (Includes dining room Banquet Cafeteria, college Cafeteria, commercial Counter service Table service, hotel or restaurant Table service, minimum Kitchens Type < 200 Cafeterias 7.5-5.0

Hotels 18.0-4.0 Restaurants 7.0-4.0 Serving and service areas Cafeterias
 Restaurants Add to totals space for food stor Sq. Ft. per Unit 100-250 net 40
 person 55 person 65 person 100 person 150-225 person 450 attorney 25-30
 attorney 25-30 attorney 100 average workstation 155-165 employee 255
 employee 15 person m but not waiting, coat room, etc.) 10-15 seat 12-15
 seat 16-18 seat 18-20 seat 15-18 seat 11-14 seat Meals per Hour 200-400
 400-800 5.0-4.0 4.0-3.5 7.5-3.0 6.0-3.0 5.0-3.6 5.0-3.6 6 person 5 person
 age, administration, waiting. Area Basis usable usable usable usable usable
 usable total usable usable usable usable total usable total usable net net net
 net net net net 800-1300 3.5-3.0 4.0-3.0 5.0-3.0 net net 1106

General Reference Data TABLE 1 Space Planning By Building Type
 (Continued) Building/Use Type Night clubs Bars Hotel 1.5 persons per
 room without extensive conferencing facilities Large stores Public library
 Stack space Reading rooms Staff space Overall Museums, exhibition areas
 Theater and assembly areas Seating area, fixed seats Seating, movable
 seating Theaters, fixed seating (Does not include stage, etc.)
 Stage/backstage Performing arts theater Lobbies Lobbies Elementary The
 following figures are listed. Small classrooms Library Art room Secondary
 The following figures are listed. Cafeteria Small classrooms Large
 classrooms Art classrooms Home economics Laboratory classrooms Library
 Music rooms Physical education Shops/vocational rooms small
 Shops/vocational rooms wood, metal, etc. University Classrooms, small
 Classrooms, large lobby, Sq. 25 18 550-600 Retail 30-50 Cultural 0.08 20-
 35 100 50 15 7.5 15 8-12 100% 3 30% Educationa R. per Unit person
 person room person bound vols. user staff person person person seat seat
 seat seating area person seating area SPACE PLANNING Area
 Requirements By Use Area Basis net net gross net net net net net net net
 net net based on the number of students In the particular space 20-30 40 40
 student student student net net net based on the number of students In the
 particular space 12-15 20-25 15 50-60 50-60 55-70 40 30-35 125 50 120-
 140 20 12-15 student student student student student student student student
 student student student student student 1/3 of total net net net net net 20%
 of total net net net net net net . 1107

General Reference Data SPACE PLANNING Area Requirements By Use
 TABLE 1 Space Planning By Building Type (Continued) Bulldng/Usa

General Reference Data SPACE PLANNING Library Area Requirements
LIBRARY PLANNING Libraries represent a unique building type in that a majority of space is devoted to housing books and not people. The number of volumes to be housed becomes the primary planning parameter, rather than numbers of people. For a detailed layout of book stacks, you can use the figures given in Table 3. For preliminary planning, the following general guidelines are useful, Rules of Thumb Public library: 12—18V2 volumes per sq.ft, Law library: 5-7 volumes per sq. ft. To stack space, add a

"configuration loss" of from 6% to 20%, to account for inefficiencies in stack layout. Minimum aisle between open stacks: 3 ft. Oin. Staff spaces: 100 net sq. ft. per person. Reading room seating: 15-35 sq. ft. per person plus 6% configuration loss. Net/gross multiplier: 1.25. Maximum of 15,000-20,000 sq. ft. per floor. Example A 100,000 volume public library is planned. How much space should be devoted to open stacks? Plan about 15 volumes per sq. ft. (100,000 ÷ 15 = 6667 sq. ft.). Add a configuration loss of 10%, to give a total area of 6667 + 667, or 7333 sq. ft. of stack space.

TABLE 3 Library Shelving - Volumes per Linear Foot of Shelf Based on Subject (Standard stack section 3 ft wide x 7½ ft high with 7 shelves)

Subject	Art (excluding oversize)	Circulating	nonfiction	Economics	Fiction	General literature	History	Law	Medical	Periodicals, bound	Public documents	Technical and scientific	Average for overall estimating
Volumes per foot of shelf	7	8	8	8	7	7	4	5	5	5	6		
Volumes per single face section	147	168	168	168	147	147	84	105	105	105	126	125	

These figures should be reduced by at least 10% to avoid overcrowding and to allow for expansion.

1109

General Reference Data HUMAN FACTORS Anthropometrics

APPROPRIATENESS It is essential, due to the many variables involved, that the data selected be appropriate to the user of the space or furniture to be designed. It becomes necessary, therefore, for the intended user population to be properly defined in terms of such factors as age, sex, occupation, and ethnicity. If the user is an individual, or constitutes a very small group, it may, in certain situations, be feasible to develop your own primary anthropometric data by actually having individual body measurements taken. Surely, if one is prepared to take the time to be fitted for a dress or a suit, one should be willing to spend the time to be fitted for an interior environment or components of that environment, particularly since, in most cases, the latter will reflect a far greater financial investment. The measurements, in the event individual data are generated, should, however, be taken with proper instruments by a trained observer. In situations where specific body dimensions or other data for a particular user population are unavailable, and both time and funds prevent undertaking sophisticated studies, an engineering anthropometrist can be consulted to discuss the statistical methods of obtaining the necessary information.

"AVERAGE MAN" FALLACY As suggested previously, a very serious

error in the application of data is to assume that the 50th percentile dimensions represent the measurements of an "average man" and to create a design to accommodate 50th percentile data. The fallacy in such an assumption is that by prior definition 50 percent of the group may suffer. There simply is no "average man." Depending on the nature of the design problem, the design should usually be conceived to accommodate the 5th or the 95th percentile, so that the greatest portion of the population is served. Dr H.T.E. Hertzberg, one of the country's most distinguished research physical anthropologists, in discussing the so-called average man, indicated, "there is really no such thing as an 'average' man or woman. There are men who are average in weight, or in stature, or in sitting height, but the men who are average in two dimensions constitute only about 7 percent of the population; those in three, only about 3 percent; those in four, less than 2 percent. There are no men average in as few as 10 dimensions. Therefore, the concept of the 'average' man is fundamentally incorrect, because no such creature exists. Work places to be efficient should be designed according to the measured range of body size." REACH, CLEARANCE, AND ADJUSTABILITY The selection of appropriate anthropometric data is based on the nature of the particular design problem under consideration. If the design requires the user to reach from a seated or standing position, the 5th percentile data should be utilized. Such data for arm reach indicates that 5 percent of the population would have an arm reach of short (or shorter) dimension, while 95 percent of the population, the overwhelming majority, would have longer arm reaches. If the design in a reach situation can accommodate the user with the shortest arm reach, obviously it will function for the users with longer reaches as well; it is equally obvious that the opposite is not true, as shown in Fig. Ma). In designs where clearance is the primary consideration, the larger or 95th percentile data should be used. The logic is simple. If the design will allow adequate clearance for the users with the largest body size, it would also allow clearance for those users with smaller body size. Here, too, it can be seen from Fig, 1 [b] that the opposite is not true. In other situations it may be desirable to provide the design with a built-in adjustment capability. Certain chair types, adjustable shelves, etc., are examples of such. The range of adjustment should be based on the anthropometrics of the user, the nature of the task, and the physical or mechanical limitations involved. The range should allow the design to accommodate at least 90 percent of the user population involved, or more.

It should be noted that all the foregoing examples were used primarily to illustrate the basic logic underlying the selection of the body dimensions involved and the particular percentiles to be accommodated. Wherever possible, however, it is naturally more desirable - Fig. 1 (a) People of smaller body dimensions and, correspondingly, the lower-range percentile data should be used to establish dimensions where reach is the determining factor. (b) Larger-size people and, correspondingly, the high percentile range data should be used in establishing clearance dimensions. 1110 BUTTOCK-POPLITEAL v LENOTH <= BUTTOCK-KNEE =& v LENOTH v 1^=> BUTTOCK-TOE LENOTH =^i <faa BUTTOCK-LEO LENOTH oo^ Fig. 2 Body measurements of most use to the designer of interior spaces.

General Reference Data HUMAN FACTORS Anthropometrics Adult Male and Female Miscellaneous Structural Body Dimensions In Inches and Centimeters by Age and Selected Percentiles MEN WOMEN MEN WOMEN In cm 36.2 91.9 32.0 81.3 30.6 78.2 26.8 68.1 B In cm 47.3 120.1 43.6 110.7 41.3 104.9 38.6 98.0 in cm 68.8 174.2 84.1 182.8 80.8 164.4 58.3 143.0 In cm 20.7 52.8 17.0 43.2 17.4 44.2 14.9 37.6 In cm 27.3 69.3 24.6 62.5 23.7 60.2 21.2 53.8 In cm 37.0 94.0 37.0 94.0 32.0 81.3 27.0 66.6 in cm 33.9 66.1 31.7 80.5 30.0 76.2 28.1 71.4 r~%> si Ujg a! _£2. SHOULDER BREADTH j MIDSHOULDER s& height srrrma BUTTOCK-TOE LEHQTH Adult Male and Female Functional Body Dimensions In Inches and Centimeters by Age, Sex, and Selected Percentiles REFT WOMEN —raw WOMEN in cm 38.3 97.3 36.3 92.2 32.4 62.3 29.9 75.9 In cm 48.1 117.1 49.0 124.5 39.4 100.1 34.0 88.4 in cm 51.8 131.1 49.1 124.7 59.0 149.9 55.2 140.2 in cm 35.0 88.9 31.7 80.5 29.7 75.4 26.8 67.6 In cm 39.0 86.4 38.0 96.5 29.0 73.7 27.0 68.8 in cm 88.5 224.8 84.0 213.4 76.8 195.1 72.9 185.2 1111

General Reference Data HUMAN FACTORS Anthropometrics able to accommodate the greatest percentage of the user population. In this regard, there is no substitute for common sense. If a shelf can just as easily be placed an inch or two lower, without significantly impacting on other design or cost factors, thereby accommodating 98 or 99 percent of the user population, obviously that is the correct design decision. The clearances shown in Fig. 3 are intended to introduce general guidelines for barrier-free

design. While we have utilized the wheelchair as our design subject, it does not represent the largest number of disabled. However, it is usually the most demanding for which to design. To provide practical limits for this design, we have chosen to plot the range of reach for the short female to the tall male. The overlapping areas of ability for the handicapped and the non-handicapped demonstrate the field of good design practice common to both. When planning for accessibility, it is important to consider the attitude at which the wheelchair approaches the object desired. Reach limits differ for frontal and side reach. Because of this, range of reach is plotted for each. The elevation targets represent the maximum height at which controls requiring manual dexterity should be located. Wheelchairs vary in size. They are fitted to their users in much the same manner as clothing is. A range of sizes is given, with the dimensions for the "typical" collapsible, manual chair indicated. Electrically powered wheelchairs require more space. Further, the wheelchair must be considered in its "occupied" state, as the user imparts additional space requirements with arms and feet as well as basic maneuvering space. We consider the basic space requirement for an occupied wheelchair to function to be 3 ft wide by 4 ft deep. This same space will accommodate most people who use canes, crutches, and walkers. Blind people using the cane technique for perceiving obstacles can also be accommodated in this space. For a person in a wheelchair to make a complete turn, an area of approximately 5 ft by 5 ft is required. As the elevations of surrounding surfaces change, so do the space requirements. The length of time that one is confronted by close quarters also affects the required clearance. An opening through a wall may be 2 ft 8 in clear as it represents only a short time involvement. As travel distance and traffic increase, passage width must also. The complexity of space also affects minimum clearance. To make a simple 90° turn, adjoining passages 3 ft wide are required (and 3 ft 6 in preferred if a normal walking is to be maintained). A 180° turn around a fixed partition requires more space. As clearances relate to general circulation requirements, space needs again increase with traffic speed and volume. Narrow corridors (4 ft) should be restricted to basically short, one-directional traffic patterns. Generally, maintain at least 5 ft clearances or more as determined by code. range of reach scale: 3/8" = 1'-0" range of reach scale: 3/8" = 1'-0" 3'-0" 19"-31"

*«"typ. wheelchair scale; 3/8" = 1'-0" Fig. 3 cane technique scale: 3/8" = 1'-0"

General Reference Data HUMAN FACTORS Anthropometrics clearance scale: 3/8"= 1'-0" openings scale: 3/8"= 1'-0" ^ 360° turn scale: 3/8"=1'-0" 3'-0"clf. 90° turn scale: 1/4"= V-O" 180° turn scale: 1/4"=V-0" ■© one-way scale: 3/8"= T-0" two-way \J scale: 3/8"= V-O" Fig. 3 (Continued) 1113

General Reference Data HUMAN FACTORS Wheelchair Dimensions Handgrip Back Upright Tire Handrim //fX\ . Wheel l(v\ Spokes Brake ^ Axle ' Tipping Lever \Y ^J^^ Wheelbase \y Caster Wheel Fig. 4 eve level Back Upholstery Overall length: 40" to 50" NOTE: Footrests may extend further for very large people. Fig. 5 Dimensions of adult-sized wheelchairs. Width: open, 24" to 25"; closed, 10" 1114

General Reference Data HUMAN FACTORS Floor Space for Wheelchairs (a) Clear Floor Space (b) Forward Approach (c) Parallel Approach NOTE: x\$ 24 in (610 mm), NOTE: x\$ 15 In (380 mm), <<«) Clear Floor Space in Alcoves NOTE: If x > 24 In (610 mm), then an additional maneuvering clearance of 6 In (150 mm) shall be provided as shown, n f\ '—r~"* ,jF J .— r*~ri 48 1220 12 r 305 1 NOTE; If x > 15 In (380 mm), then an additional maneuvering clearance of 12 In (305 mm) shall be provided as shown, (e) Additional Maneuvering Clearances for Alcoves Fig. 6 Minimum clear floor space for wheelchairs. 1115

Full Slidt s ^> Push Side 18 mln, 24 pmfami 433 o NOTE x= 12 In (305 mm) If door has both a closer and latch. (■) Front Approach*! — Swinging Doors Full Side 0 NOTE: x =■ 36 In (915 mm) minimum If y = 60 In (1525 mm);x» 42 In (1065 mm) minimum If y = 54 In (1370 mm), Push Side 54mln >5f <? NOTE; y • 48 in (1220 mm) minimum If door has both a latch and closer, Hinge Side Approaches — Swinging Doors Pull Side ,» X 24 mln X 24mln Push Side (? \$, E >5f NOTE: y = 54 In (1370 mm) minimum If door has closer. NOTE y = 48 In (1220 mm) minimum If door has closer, (c) Latch Side Approaches — Swinging Doors NOTE: All doors in alcoves shall comply with the clearances for front approaches. L, .ftjjjrtn tog o asr (? Front Approach - Sliding Doors avid Folding Doors <«) Slide Side Approach — Sliding Doors and Folding Doors E X 124 mln TelO P (0 Latch Side Approach — Sliding Doors and Folding Doors NOTE: All doors In alcoves shall comply with the clearances for front approaches. Fig. 7 Maneuvering clearances at doors.

General Reference Data HUMAN FACTORS Wheelchair Clearances at Doors
Two hinged doors in series. 1117

General Reference Data HUMAN FACTORS Telephone Mounting Heights;
Control Reach VM/J/JMMM optional post and L base I 30 min 760 *3»
48 min 1220 10 max 255 Elevation (a) Side Reach Possible Plan
Y/////?)/////A Elevation 'Height to highest operable parts which are
essential to basic operation of telephone. Fig. 9 Mounting heights and
clearances for telephones. 00 o CM CM r- tv 30 760 20 max 1 510 more
than 24 610 6 mtn 150 Plans (b) Forward Reach Required required X
preferred required * preferred 9 § to vending I 3 V/S/////////M. ^ 8 1 2
<* fh -EL. vending a a a a fcj V/////////S. 1118 Forward Reach
Possible Fig. 10 Control reach limitations. S/de Reach Possible

General Reference Data HUMAN FACTORS Space TYPES OF SPACE
Besides needing enough space in order to move about and perform various
tasks, people react to space in a variety of ways. Several researchers have
defined the space surrounding the individual in terms of the limits within
which people categorically respond (see Figs. 11 and 12). Intimate space is
that area in which a person tends not to allow anyone to intrude unless
intimate relationships are expected. Personal space is that area within which
a person allows only selected friends or fellow workers with whom personal
discussion is mandatory. Social space is that area within which the
individual expects to make purely social contacts on a temporary basis.
And, finally, public space is that area within which the individual does not
expect to have direct contact with others. Obviously, the more intimate the
spatial relationship becomes, the more people resist intrusion by others.
Personal space factors are important in establishing the privacy
requirements for architectural design. Distance Relationships Among
People (Hall) Intimate Personal Social Public <C£pt c£p>f cggp^ <qup>*
l&-4ff. H/ 4- \Z ft. I Fig. 11 PUBL\C Fig. 12 1119

General Reference Data HUMAN FACTORS Space TYPICAL
SUBJECTIVE RESPONSES TO SELECTED SPATIAL FEATURES
Although few research data have been generated with regard to how people
respond to specific spatial factors (at least in terms of being able to

prescribe precise, quantitative guidelines), it is important for the designer to reflect on potentially negative reactions that often result when a given space is not made compatible with what the user expects in terms of the size, shape, organization, color, and illumination of a particular space. The considerations listed in Table 4 are suggested as a checklist for the designer.

TABLE 4	Space Characteristic	Probable Response	Size (generally volume)	Shape (generally proportion)	Color and Illumination	Windows	Space organization	Furnishings
1120	If the space is too small for the number of people, furnishings, equipment, or other objects that occupy it, people will consider it to be crowded. Although they may accept a crowded condition on a temporary basis, they will object to living or working in such a space for extended periods of time. If the space is too large for the people, furnishings, equipment, or other objects that occupy it, people will consider it "unfriendly," inconvenient, and/or overly demanding in terms of communicating, travel distance, maintenance, etc. Although they may accept the "barnlike" atmosphere for temporary periods, they will object to living or working in such a space for extended periods of time. If the space is out of proportion (too narrow, wide, long, high, etc.) for the intended use, people will consider it awkward and often distracting or oppressive. Although they may accept proportional distortion on a short-term basis (i.e., as they pass through briefly), they will object to living or working in such a space for extended periods of time. If the space contains such distortions as all curved surfaces, acute wall junctures, and too many projections or surface changes, people will consider it confusing and difficult to maneuver in and/or furnish. Although they may accept such distortions (or even consider them interesting) on a temporary or one-time basis, they will object to living or working in such a space for extended periods of time, it should also be noted that blind people depend on the constant proportions of right-angle corners to aid them in negotiating a space; such individuals are easily confused by curved surfaces, walls that are not at right angles, and periodic projections that imply they may have reached a turning point. When a ceiling is extremely high relative to the lateral dimension of a space, people feel as though they are working in a pit and that the walls are closing in on them. When a ceiling is extremely low and the space in front of the observer is very long, people feel as though the room is "endless" or as if they will hit their heads unless they duck. If a space is dark (unless this is required for a particular operation, such as a motion picture presentation),							

people tend to become lethargic and less active, or they may feel anxious. As a rule, the less bright a room is, the less cheerful it seems. A small space will seem even smaller. If a space is too bright, people will feel overly exposed, or they will complain of glare or thermal discomfort (even though actual glare in terms of accepted light levels or inappropriate thermal conditions for comfort are not present). If there are too many different colors, too large expanses of very saturated color, or too many and too "busy" patterns of color within a space, most people become irritated after more than a brief exposure to the space. If there is too little color, no visual pattern, or no other decorative "break" in the visual environment, people will find the space monotonous, boring, and eventually irritating to the point of wanting to escape. Although isolated points of highly reflective surface provide interest, all-metallic and highly reflective surface treatments create both subjective and directly objective interference for most people who have to work in the space. Generally, most people do not like to live and work in a space that is devoid of windows. First and foremost, people seem to need visual contact with the outside world. Too many windows, on the other hand, can cause the following possible negative reactions: too much glare, too much exposure (fishbowl effect), lack of protection from outside elements, true anxiety (caused by floor-to-ceiling glass at high elevations). The internal components within a space and the traffic corridors and entrance and exit locations will seem either well organized or badly organized. The furnishings, partitions, decorative objects, etc., will appear as being either organized or disorganized, depending on the observer's ability to comprehend what things are and where they are with respect to his or her vantage point. Key behavioral response issues are: apparent capability to find one's way to specific locations, apparent ease for interacting and communicating with others with whom the individual must associate, apparent privacy provisions necessary to perform individual tasks. Although these are sometimes conflicting needs, the people who use a space will perform on the basis of how well each of these factors has been executed for them, not for the designer or the boss. The organization of internal space components obviously interacts with all the other space characteristics; i.e., the individual perceives and reacts to the combined effects of size, shape, color and illumination, windows, and organization simultaneously. A significant behavioral response will be an individual's interpretation of whether sufficient options

are available for local modification of his or her own portion of the space. Even though people may never require a modification, they react to their own space in terms of permanently established restrictions that eventually elicit the feeling that the space is too small, the wrong shape, too dark, or isolated from the rest of the world, for example. As a general rule, people are sensitive to Improperly proportioned furniture, i.e., furniture that is too large, too small, or the wrong shape for the space in which it is placed. Although the designer normally tries to select furnishings that are properly proportioned for the space he or she has created, this may ultimately restrict the efficiency of the Individual (e.g., a desk or storage cabinet may be too small). Thus, although the general visual proportions of furniture in relation to space must be taken into account to avoid negative observational responses, shortchanging the Individual in terms of specific furniture and use requirements soon stimulates an even stronger negative response.

TABLE 5 Comparison of Sound Pressure Levels and Loudness Sensations
 Sound Pressure Level (decibels-A scale) Source Sensation 130 120 110 100
 90 80 70 60 50 40 30 20 10 Jet Aircraft at 100' Bass Drum at 3' Auto Horn
 at 3' Physical Pain Thunder, Artillery, Nearby Riveter Elevated Train
 Discotheque Deafening Loud Street Noise Noisy Factory Truck Unmuffled
 Police Whistle Very Loud Cocktail Party Noisy Office Average Street Noise
 Average Radio Average Factory Loud Noisy Home Inside General Office
 Face to Face Conversation Quiet Radio Moderate Quiet Home Private
 Office Empty Auditorium Quiet Conversation Faint Rustle of Leaves
 Whisper Soundproof Room Very Faint General Reference Data HUMAN
 FACTORS Acoustics Decibel Scale for Sounds Physical damage 12n_
 (jeFtaKecfn Painful (siren) Deafening (power mower) Ver\ Loud (cocktail
 party) Loud (traffic) Moderate (conversation) Faint (rustling leaves') Fig. 13
 100- 80- 60- 40- 20- 0 J Threshold of Audibility TABLE 6 Speech-
 Interference Levels that Barely Permit Reliable Conversation Distance
 between Speech-interference level, dB TABLE 7 Speech Interference
 Levels (SIL) and Noise Criteria (NC) Recommended for Rooms talker and
 listener, ft 0.5 1.0 2.0 3.0 4.0 5.0 6.0 12.0 Normal 71 65 59 55 53 51 49 43
 Raised 77 71 65 61 59 57 55 49 Very loud 83 77 71 67 65 63 61 55
 Shouting 89 83 77 73 71 69 67 61 Type of room Secretarial offices, typing
 Coliseum for sports only (amplification) Small private office Conference
 room for 20 Movie theater Conference room for 50 Theaters for drama, 500

seats (no amplification) Homes, sleeping areas Assembly halls (no amplification) Schoolrooms Concert halls (no amplification) Maximum permissible level (measured in vacant rooms SIL 60 55 45 35 35 30 30 30 30 30 25 NC 50-55 50 30-35 30 30 20-30 20-25 20-25 25 15-20 1121

Reference Data FLOOR AND WALL COVERING Length and Width of Carpet Roll Converted to Area TABLE 8 Length and Width of Carpet Roll Converted to Area Length, 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 ft ft' 18 27 36 45 54 63 72 81 90 99 108 117 126 135 144 153 162 171 180 189 198 207 216 225 234 243 252 261 270 279 288 297 306 315 324 333 342 351 360 369 378 387 396 405 414 423 432 441 9 ft yd' 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 m' 1.67 2.51 3.34 4.18 5.02 5.85 6.69 7.52 8.36 9.20 10.03 10.87 11.71 12.54 13.38 14.21 15.05 15.89 16.72 17.55 18.39 19.23 20.07 20.90 21.74 22.57 23.41 24.25 25.08 25.92 26.76 27.59 28.43 29.26 30.10 30.94 31.77 32.61 33.44 34.28 35.12 35.95 36.79 37.63 38.46 39.30 40.13 40.97 Width of carpet roll ft' 24 36 48 60 72 84 96 108 120 132 144 156 168 180 192 204 216 228 240 252 264 276 288 300 312 324 336 348 360 372 384 396 408 420 432 444 456 468 480 492 504 516 528 540 552 564 576 588 12ft yd' 2.67 4 5.33 6.67 8 9.33 10.67 12 13.33 14.67 16 17.33 18.67 20 21.33 22.67 24 25.33 26.67 28 29.33 30.67 32 33.33 34.67 36 37.33 38.67 40 41.33 42.67 44 45.33 46.67 48 49.33 50.67 52 53.33 54.67 56 57.33 58.67 60 61.33 62.67 64 65.33 15 ft m' ft' yd' 2.23 30 3.33 3.34 45 5 4.46 60 6.67 5.57 75 8.33 6.69 90 10 7.80 105 11.67 8.92 120 13.33 10.03 135 15 11.15 150 16.67 12.26 165 18.33 13.38 180 20 14.47 195 21.67 15.61 210 23.33 16.72 225 25 17.84 240 26.67 18.95 255 28.33 20.07 270 30 21.18 285 31.67 22.30 300 33.33 23.41 315 35 24.53 330 36.67 25.64 345 38.33 26.76 360 40 27.87 375 41.67 28.99 390 43.33 30.10 405 45 31.21 420 46.67 32.33 435 48.33 33.44 450 50 34.56 465 51.67 35.68 480 53.33 36.79 495 55 37.90 510 56.67 39.02 525 58.33 40.13 540 60 41.25 555 61.67 42.36 570 63.33 43.48 585 65 44.59 600 66.67 45.71 615 68.33 46.82 630 70 47.94 645 71.67 49.05 660 73.33 50.17 675 75 51.28 690 76.67 52.40 705 78.33 53.51 720 80 54.63 735 81.67 m' 2.79 4.18 5.57 6.97 8.36 9.76 11.15 12.54 13.94 15.33 16.72 18.12 19.51 20.90 22.30 23.69 25.08 26.48 27.87 29.26 30.66 32.05 33.44 34.84 36.23 37.63 39.02 40.41 41.81 43.20 44.59 45.99

47.38 48.77 50.17 51.56 52.95 54.35 55.74 57.13 58.53 59.92 61.32 62.71
64.10 65.50 66.89 68.28 Length. 51 52 53 54 55 56 57 58 59 60 61 62 63
64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87
88 89 90 91 92 93 94 95 96 97 98 ft ft' 459 468 477 486 495 504 513 522
531 540 549 558 567 576 585 594 603 612 621 630 639 648 657 666 675
684 693 702 711 720 729 738 747 756 765 774 783 792 801 810 819 828
837 846 855 864 873 882 9 ft yd' m' 51 42.64 52 43.48 53 44.31 54 45.15
55 45.99 56 46.82 57 47.66 58 48.49 59 49.33 60 50.17 61 51 62 51.84 63
52.68 64 53.51 65 54.35 66 55.18 67 56.02 68 56.86 69 57.69 70 58.53 71
59.36 72 60.20 73 61.04 74 61.87 75 62.71 76 63.55 77 64.30 78 65.22 79
66.05 80 66.89 81 67.73 82 68.56 83 69.40 84 70.23 85 71.07 86 71.91 87
72.74 88 73.58 89 74.41 90 75.25 91 76.09 92 76.92 93 77.76 94 78.6 95
79.43 96 80.27 97 81.10 98 81.94 Width of carpet rol ft' 612 624 636 648
660 672 684 696 708 720 732 744 756 768 780 792 804 816 828 840 852
864 876 888 900 912 924 936 948 960 972 984 996 1008 1020 1032 1044
1056 1068 1080 1092 1104 1116 1128 1140 1152 1164 1176 12 ft yd' 68
69.33 70.67 72 73.33 74.67 76 77.33 78.67 80 81.33 82.67 84 85.33 86.67
88 89.33 90.67 92 93.33 94.67 96 97.33 98.67 100 101.33 102.67 104
105.33 106.67 108 109.33 110.67 112 113.33 114.67 116 117.33 118.67 120
121.33 122.67 124 125.33 126.67 128 129.33 130.67 m' 56.86 57.97 59.09
60.20 61.32 62.43 63.55 64.66 65.77 66.89 68 69.12 70.23 71.35 72.46
73.58 74.69 75.81 76.92 78.04 79.15 80.27 81.38 82.50 83.61 84.73 85.84
86.96 88.07 89.19 90.30 91.42 92.53 93.65 94.76 95.88 96.99 98.11 99.22
100.33 101.45 102.56 103.68 104.79 105.91 107.02 108.14 109.25 I ft' 765
780 795 810 825 840 855 870 885 900 915 930 945 960 975 990 1005 1020
1035 1050 1065 1080 1095 1110 1125 1140 1155 1170 1185 1200 1215
1230 1245 1260 1275 1290 1305 1320 1335 1350 1365 1380 1395 1410
1425 1440 1455 1470 15ft yd' 85 86.67 88.33 90 91.67 93.33 95 96.67
98.33 100 101.67 103.33 105 106.67 108.33 110 111.67 113.33 115 116.67
118.33 120 121.67 123.33 125 126.67 128.33 130 131.67 133.33 135
136.67 138.33 140 141.67 143.33 145 146.67 148.33 150 151.67 153.33
155 156.67 158.33 160 161.67 163.33 m' 71.07 72.46 73.86 75.25 76.64
78.04 79.43 80.83 82.22 83.61 85.01 86.40 87.79 89.19 90.58 91.97 93.37
94.76 96.15 97.55 98.94 100.33 101.73 103.12 104.52 105.91 107.30
108.70 110.09 111.48 112.88 114.27 115.66 117.06 118.45 119.84 121.24
122.63 124.03 125.42 126.81 128.21 129.60 130.99 132.39 133.78 135.17
136.57 50 450 50 41.81 600 66.67 55.74 750 83.33 69.68 99 891 99 82.78

1188 132 110.37 1485 165 137.96 100 900 100 83.61 1200 133.33 111.48
1500 166.67 139.36

General Reference Data TABLE 9 Paperhanging Walls and Ceilings Size of room, ft 4x8 4 x 10 4x 12 6x 10 6 x 12 8x 12 8 x 14 10 x 14 10 x 16 12 x 16 12 x 18 14 x 18 14 x 22 15 x 16 15 x 18 15 x20 15 x 23 16 X 18 16 x20 16 x 22 16x24 16 x 26 17 x 22 17 x 25 17 x 28 17 x32 17 X35 18 x22 18X25 18 x 28 20 x 26 20 x28 30x34 Height of ceiling 8 ft | 9 ft] 10 ft Single rolls for walls 6 7 8 8 9 10 11 12 13 14 15 16 18 15 16 17 19 17 18 19 20 21 19 21 22 24 26 20 21 23 23 24 27 7 8 9 9 10 11 12 14 15 16 17 18 20 17 18 20 21 19 20 21 22 23 22 23 25 27 29 22 24 26 28 8 9 10 10 11 13 14 15 16 17 19 20 22 19 20 22 23 21 22 23 25 26 24 26 28 30 32 25 27 28 28 27 30 30 33 Yards of border 9 11 12 12 13 15 16 18 19 20 22 23 26 23 24 25 28 25 26 28 29 31 28 31 32 35 37 29 31 33 33 34 39 Rolls of ceiling 2 2 2 2 3 4 4 5 6 7 8 8 10 8 9 10 11 10 10 11 12 13 12 13 15 17 18 12 14 16 17 18 21 Allowance for waste is included in all figures. Deduct one roll for every 36 sq ft of openings. Deduct one roll for every 2 doors. Deduct for windows as area of each opening. One roll of wallpaper equals 36 sq ft (24 ft by 18 in.). FLOOR AND WALL COVERING Covering Capacity of Wallpaper, Paint, and Tile TABLE 10 Covering Capacity Material Surface or use Coverage per gallon, sq ft Exterior Painting Priming paint Flat house paint Oil paint Stain Wood Metal Over primer Repainting 1 coat Masonry Concrete Stucco (smooth) Stucco (rough) Wood shingle siding, first coat Wood shingle siding, second coat 450 500 500 400 300 250 200 150 150 200 Interior Painting Priming paint Metal primer Undercoat (enamel) Flat Semigloss enamel Satin-gloss enamel Gloss enamel Floor enamel Aluminum paint Spar varnish Clear gloss varnish Lacquer Interior stain Wood Metal Over primer Finish coat Finish coat Finish coat Finish coat Floors Aluminum, first coat Aluminum, second coat Finishing woodwork Finishing woodwork Over stain Woodwork, first coat Woodwork, second coat Woodwork, third coat 500 600 400 500 450 450 400 500 600 700 600 600 450 500 600 700 Miscellaneous Barn red oil paint Rust inhibitor (zinc paint) Furniture sealer and stain Repaint barn Metal Unpainted furniture 450 650 600 TABLE 11 Flooring Tile (Net covering capacity per 100 sq ft) Tile size, in. 6 x 6 X 9 x 12 X 12 X 18 X 18 X 18 X 36 X 6 12 9 12 24 18 24 36 36 No. per of pieces 100 sq ft 400 200 178 100 50 45 34 23 11 No. of pieces per sq ft 4.00 2.00 1.78 1.00 0.50 0.45 0.33 0.23 0.11 1123

General Reference Data FLOOR AND WALL COVERING Wall Areas of
Rooms TABLE 12 Wall Area of Rooms (8-ft ceiling), ft²

Feet	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
3	96	112	128	144	160	176	192	208	224	240	256	272	288	304	320	336	352	368	384	400	416	432	448
4	112	128	144	160	176	192	208	224	240	256	272	288	304	320	336	352	368	384	400	416	432	448	464
5	128	144	160	176	192	208	224	240	256	272	288	304	320	336	352	368	384	400	416	432	448	464	480
6	144	160	176	192	208	224	240	256	272	288	304	320	336	352	368	384	400	416	432	448	464	480	496
7	160	176	192	208	224	240	256	272	288	304	320	336	352	368	384	400	416	432	448	464	480	496	512
8	176	192	208	224	240	256	272	288	304	320	336	352	368	384	400	416	432	448	464	480	496	512	528
9	192	208	224	240	256	272	288	304	320	336	352	368	384	400	416	432	448	464	480	496	512	528	544
10	208	224	240	256	272	288	304	320	336	352	368	384	400	416	432	448	464	480	496	512	528	544	560
11	224	240	256	272	288	304	320	336	352	368	384	400	416	432	448	464	480	496	512	528	544	560	576
12	240	256	272	288	304	320	336	352	368	384	400	416	432	448	464	480	496	512	528	544	560	576	592
13	256	272	288	304	320	336	352	368	384	400	416	432	448	464	480	496	512	528	544	560	576	592	608
14	272	288	304	320	336	352	368	384	400	416	432	448	464	480	496	512	528	544	560	576	592	608	624
15	288	304	320	336	352	368	384	400	416	432	448	464	480	496	512	528	544	560	576	592	608	624	640
16	304	320	336	352	368	384	400	416	432	448	464	480	496	512	528	544	560	576	592	608	624	640	656
17	320	336	352	368	384	400	416	432	448	464	480	496	512	528	544	560	576	592	608	624	640	656	672
18	336	352	368	384	400	416	432	448	464	480	496	512	528	544	560	576	592	608	624	640	656	672	688
19	352	368	384	400	416	432	448	464	480	496	512	528	544	560	576	592	608	624	640	656	672	688	704
20	368	384	400	416	432	448	464	480	496	512	528	544	560	576	592	608	624	640	656	672	688	704	720

TABLE 13 Wall Area of Rooms (9-ft Ceiling), IP

Feet	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
3	108	126	144	162	180	198	216	234	252	270	288	306	324	342	360	378	396	414	432	450	468	486	504
4	126	144	162	180	198	216	234	252	270	288	306	324	342	360	378	396	414	432	450	468	486	504	522
5	144	162	180	198	216	234	252	270	288	306	324	342	360	378	396	414	432	450	468	486	504	522	540
6	162	180	198	216	234	252	270	288	306	324	342	360	378	396	414	432	450	468	486	504	522	540	558
7	180	198	216	234	252	270	288	306	324	342	360	378	396	414	432	450	468	486	504	522	540	558	576
8	198	216	234	252	270	288	306	324	342	360	378	396	414	432	450	468	486	504	522	540	558	576	594
9	216	234	252	270	288	306	324	342	360	378	396	414	432	450	468	486	504	522	540	558	576	594	612
10	234	252	270	288	306	324	342	360	378	396	414	432	450	468	486	504	522	540	558	576	594	612	630
11	252	270	288	306	324	342	360	378	396	414	432	450	468	486	504	522	540	558	576	594	612	630	648
12	270	288	306	324	342	360	378	396	414	432	450	468	486	504	522	540	558	576	594	612	630	648	666
13	288	306	324	342	360	378	396	414	432	450	468	486	504	522	540	558	576	594	612	630	648	666	684
14	306	324	342	360	378	396	414	432	450	468	486	504	522	540	558	576	594	612	630	648	666	684	702
15	324	342	360	378	396	414	432	450	468	486	504	522	540	558	576	594	612	630	648	666	684	702	720
16	342	360	378	396	414	432	450	468	486	504	522	540	558	576	594	612	630	648	666	684	702	720	738
17	360	378	396	414	432	450	468	486	504	522	540	558	576	594	612	630	648	666	684	702	720	738	756
18	378	396	414	432	450	468	486	504	522	540	558	576	594	612	630	648	666	684	702	720	738	756	774
19	396	414	432	450	468	486	504	522	540	558	576	594	612	630	648	666	684	702	720	738	756	774	792
20	414	432	450	468	486	504	522	540	558	576	594	612	630	648	666	684	702	720	738	756	774	792	810
21	432	450	468	486	504	522	540	558	576	594	612	630	648	666	684	702	720	738	756	774	792	810	828
22	450	468	486	504	522	540	558	576	594	612	630	648	666	684	702	720	738	756	774	792	810	828	846
23	468	486	504	522	540	558	576	594	612	630	648	666	684	702	720	738	756	774	792	810	828	846	864
24	486	504	522	540	558	576	594	612	630	648	666	684	702	720	738	756	774	792	810	828	846	864	882
25	504	522	540	558	576	594	612	630	648	666	684	702	720	738	756	774	792	810	828	846	864	882	900

540 558 576 8 198 216 234 252 270 288 306 324 342 360 378 396 414 432
 450 468 486 504 522 540 558 576 594 9 216 234 252 270 288 306 324 342
 360 378 396 414 432 450 468 486 504 522 540 558 576 594 612 10 234
 252 270 288 306 324 342 360 378 396 414 432 450 468 486 504 522 540
 558 576 594 612 630 11 252 270 288 306 324 342 360 378 396 414 432
 450 468 486 504 522 540 558 576 594 612 630 648 Feet 3 4 5 6 7 8 9 10 11
 12 13 14 15 16 17 18 19 20 21 22 23 24 25 Feet 12 270 288 306 324 342
 360 378 396 414 432 450 468 486 504 522 540 558 576 594 612 630 648
 666 13 288 306 324 342 360 378 396 414 432 450 468 486 504 522 540
 558 576 594 612 630 648 666 684 14 306 324 342 360 378 396 414 432
 450 468 486 504 522 540 558 576 594 612 630 648 666 684 702 15 324
 342 360 378 396 414 432 450 468 486 504 522 540 558 576 594 612 630
 648 666 684 702 720 16 342 360 378 396 414 432 450 468 486 504 522
 540 558 576 594 612 630 648 666 684 702 720 738 17 360 378 396 414
 432 450 468 486 504 522 540 558 576 594 612 630 648 666 684 702 720
 738 756 18 378 396 414 432 450 468 486 504 522 540 558 576 594 612
 630 648 666 684 702 720 738 756 774 19 396 414 432 450 468 486 504
 522 540 558 576 594 612 630 648 666 684 702 720 738 756 774 792 20
 414 432 450 468 486 504 522 540 558 576 594 612 630 648 666 684 702
 720 738 756 774 792 810 1124

General Reference Data FLOOR AND WALL COVERING Wall Areas of
 Rooms TABLE 14 Wall Area of Rooms (10-11 Ceiling), It2 Feet 3 4 5 6 7 8
 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 Feet 3 120 140 160 180
 200 220 240 260 280 300 320 340 360 380 400 420 440 460 480 500 520
 540 560 4 140 160 180 200 220 240 260 280 300 320 340 360 380 400 420
 440 460 480 500 520 540 560 580 5 160 180 200 220 240 260 280 300 320
 340 360 380 400 420 440 460 480 500 520 540 560 580 600 6 180 200 220
 240 260 280 300 320 340 360 380 400 420 440 460 480 500 520 540 560
 580 600 620 7 200 220 240 260 280 300 320 340 360 380 400 420 440 460
 480 500 520 540 560 580 600 620 640 8 220 240 260 280 300 320 340 360
 380 400 420 440 460 480 500 520 540 560 580 600 620 640 660 9 240 260
 280 300 320 340 360 380 400 420 440 460 480 500 520 540 560 580 600
 620 640 660 680 10 260 280 300 320 340 360 380 400 420 440 460 480
 500 520 540 560 580 600 620 640 660 680 700 11 280 300 320 340 360
 380 400 420 440 460 480 500 520 540 560 580 600 620 640 660 680 700
 720 Feet 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Feet 12 300 320 340 360 380 400 420 440 460 480 500 520 540 560 580
600 620 640 660 680 700 720 740 13 320 340 360 380 400 420 440 460
480 500 520 540 560 580 600 620 640 660 680 700 720 740 760 14 340
360 380 400 420 440 460 480 500 520 540 560 580 600 620 640 660 680
700 720 740 760 780 15 360 380 400 420 440 460 480 500 520 540 560
580 600 620 640 660 680 700 720 740 760 780 800 16 380 400 420 440
460 480 500 520 540 560 580 600 620 640 660 680 700 720 740 760 780
800 820 17 400 420 440 460 480 500 520 540 560 580 600 620 640 660
680 700 720 740 760 780 800 820 840 18 420 440 460 480 500 520 540
560 580 600 620 640 660 680 700 720 740 760 780 800 820 640 860 19
440 460 480 500 520 540 560 580 600 620 640 660 680 700 720 740 760
780 800 820 840 860 880 20 460 480 500 520 540 560 580 600 620 640
660 680 700 720 740 760 780 800 820 840 860 880 900

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General Reference Data FLOOR AND WALL COVERING ESTIMATING
 THE AMOUNT OF HARDWOOD STRIP FLOORING REQUIRED An
 allowance for side-matching, plus 5% for end-matching and normal waste
 are incorporated in these percentages. Take the Square Footage and ADD
 the percentage below opposite the size strip flooring to be used. When using
 3/4x1-1/2" Strip ADD 55% 3/4x2" 42-1/2% 3/4x2-1/4" 38-1/3% 3/4x3-1/4"
 29% 3/8x1-1/2" 3/8x2" 1/2x2-1/2" 1/2x2" 38-1/3% 30% 38-1/3% 30%
 Above percentages are for laying flooring straight across the room.
 Additional flooring should be estimated for diagonal applications and bay
 windows or other projections. CONVERTING SQUARE' FEET OF
 FLOOR SPACE FLOOR SPACE Square Feet 5 10 20 30 40 50 60 70 80 90
 100 200 300 400 500 600 700 800 900 1000 To Board Feet of Strip
 Flooring Requi BOARD FEET REQUIRED 3/4x2V 7 14 28 42 55 69 83
 97 111 125 138 277 415 553 692 830 968 1107 1245 1383 3/4x1%" 8 16 31
 47 62 78 93 109 124 140 155 310 465 620 775 930 1085 1240 1395 1550
 (5% Cutt: 3/4x3V 6 13 26 39 52 65 77 90 103 116 129 258 387 516 645
 774 903 1032 1161 1290 Lng Waste 1/2x2" 7 13 26 39 52 65 78 91 104 117
 130 260 390 520 650 780 910 1040 1170 1300 red. Included) 3/8x1%" 7 14
 28 42 55 69 83 97 111 125 138 277 415 553 692 830 968 1107 1245 1383
 1127

General Reference Data FLOOR AND WALL COVERING Carpet
 Construction VELVET (a) Velvet weave (c) Axminster weave WILTON
 BACKING YARNS IWARPI (b) Wilton weave Fig. 14 CROSSWISE
 BINDING (WEFII wMW^MZM^^Mmm DOUBLE JUTE BACKING
 TUFTS NEEDLED THROUGH WOVEN BACK BACKING
 COMPOUND LOCKS TUFTS (d) Tufted process Carpets are
 manufactured in three different ways: woven, knitted, or tufted. Woven
 carpet The surface pile and backing of woven carpet are interwoven at the
 same time, creating a single fabric. Due to the interweaving, which locks all
 of the yarns together in the single woven fabric, the pile yarns cannot be
 pulled out. Some carpet weaves presently available are velvet, wilton, and
 axminster. Velvet is best suited for solid- color carpet; however, tweeds,

stripes, and salt-and-pepper effects can be produced on velvet looms. The usual velvet is a solid-color carpet with smooth surface and even pile. Sometimes the pile is cut to produce a plushlike surface [see Fig. 14(a)]. It may also be had in loop pile, or twist. Wilton weave comes in almost unlimited numbers of textures and sculptured effects, as well as patterns. The pile is sometimes cut, sometimes left uncut; a combination of cut and uncut may also be obtained. In multicolor wiltons, one color may be seen on the surface pile, while other colors are hidden in the body of the carpet. Embossed and sculptured effects are also made by the wilton looms, and cut and uncut pile can be combined with cut pile for the top level, with loops at other levels. Another variation is to have some pile yarn straight and others twisted [see Fig. 14(b)]. In the axminster weave, which is similar in appearance to handweaving, we find a complete flexibility in the use of color. In this method each tuft is inserted separately and while solid-color carpets can be made by this method, it is nearly always used for multicolored pattern carpet such as orientals, or modern and geometric designs [see Fig. 14(c)]. Tufted carpet In the tufted process, which was only recently perfected, the tufts are attached to a previously made backing, as compared with the methods described above in which the backing and pile are integral. The tufts are held in place by a heavy coating of latex applied to the backing, which is usually cotton, jute, or kraft cord. By the use of this method, a wide variety of textures is possible. For example, the tufted pile can be made in several levels; it can be cut or uncut; and carved or striated effects can be obtained. The pile can be looped or plush. Tufted carpets are made in multicolor patterns with an increasing number of textural effects and refinements [see Fig. 14(01. 1128

General Reference Data FLOOR AND WALL COVERING Location of Door Hardware PANIC, BOLT %TR)KEI PIVOTS T-T } MOUMTEP CLOS.IMC, J HH DOORS THE LOCK IS i LOCATED IN THE METAL I BASE PLKT6 CLOStMC; MECHANISM IW CCWR. 5W PIVOTS PANIC BOLT STRIKE- CLOSING MECH»WI*>M IN FLOOR. ■\ 'r- o l I- 9 UJ 111 w I r- "o "O P flush eou tn-WKsea- FLU5H BOLT CPERATING, MKUANKM HINCiB TOP DOOR tVX-PER IM COOR. OK. 5URB*CE- MOUKTEC? EXPOSED TYPE OP \J CLOSER 4 PUSH PLATE .POOR PULL :<? DOOfKMOB MAIL. 1 ■e l2 ■KICK PLATE MOP PLATE- T il CENTER. HIVJGiE FOR HEAVY POORS HIMCE

BOTTOM PLUSH BOLT STWKF FIMI5H HARDWARE LOCATION FOR. ALL TYPES *} DOORS 1129

General Reference Data FABRIC Characteristics and Suitability HOME-
DECORATING FABRIC CHART In th. chart b.low or* lltod the fabrlei
uiually classified ai primarily decorating certain types af draperies, couch
caven, etc.; sheer cattani iuch at cheesecloth, materials. In addition ta then,
practically all dren material! moy be used, and dimity, plain and dotted
Swiss, lace, lawn, argandy, voile, all used far glass curare often woven In
extra widths far this purpose. Among these arei light weight tains, bed
coverings, etc.i pile fabrics such at corduroy, panne velet, velaur, vel-
caltpns iuch as cambric, challls, chambray, gingham, muslin, percale,
poplin, vet, velveteen, all excellent far upholstery, draperies, .tc,j silk
fabrics such ai seersucker, sltkaMne all uied for Informal draperies, bed
coverings, dressing tables, faille, moire {watered silk), pongee and
shantung, satin, taffeta, all used for etc.) stiff fabrics such as buckram and
crinoline far Interlining curtain lops, valances, draperies, bed coverings,
slipcovers, sometimes upholstery) sheer lllkl such « etc.r caftans such as
Canton flannel used far Interllnlngs, and sateen far linings* chiffon far glass
curtains, lamp shades, etc* heavy utility cottons such at crash, denim, drill,
gabardine, pique, all suited ta FABRIC ARMURE ARTIFICIAL LEATHER
BATIK BOBBINET BROCADE BROCATEUE BROCHE BURLAP
CALICO CANDLEWICK CANVAS CASEMENT CLOTH
CELLOPHANE CHENILLE CHEVRON CLOTH CHINTZ CRETONNE
CREWEL EMBROIDERY DAMASK DRUID'S CLOTH DUCK »
DESCRIPTION Ribbed silk, cotton, rayon {sometimes wool), fabric with
smalt design on the surface. Avatlable under many trade names. A woven
catton fabric, coated with nitrocellulose preparation and stamped surface ta
simulate different kinds of leather. Jovaneie proceis af coloring fabrics by
blackening aut various parts af the pattern with wax before dyeing, Net with
hexagonal openings. Originally handmade with a bobbin. Rich, colorful
fabric with embroidery effects an taffeta, twill, satin or damask weave
background. Gold or stiver metal threads sometimes Introduced In the
figures. Brocade Is also the name deilgnatlng a certain type af Jacquard
weave. A heavy fabric with general characteristics of damask, but figures
mare ratted and vel* vety In quollty giving emboued effect. Brocade with
small floral pot tern. Coarse, plain-weave fabric made af jute or hemp.

Comes In variety of colors and widths. Inexpensive. Light-weight cotton fabric In plain weave. May be printed, plain or patterned, In deep colors. Designs usually small. (Also for drapes.) Cotton yarn used for hand tufting on mullin sheeting. Yarn may be white or In color and design simple or elaborate, A coarse, firm cotton or linen material, rough finish, plain weave. May be bleached, unbleached, starched, dyed, or printed, Light, plain, and usually neutral In color. Made In cotton, linen, mohair, silk, wool or rayon. Sometimes carries In small figures. Glossy, transparent synthetic product woven on warp threads of cotton, Often woven In with other materials and used for many novelty effect.. Various types of fabrics woven with chenille yarn of silk, wool, mercerised cotton, or rayon, Fabric with broken twill weave forming chevron pattern. A firm plain weave cotton fabric usually printed In gay pattern, but may be had In plain color, May be semi-glazed or fully glazed, Some chintz has special finish so that It will retain gloss after washing. Glazed chintz is more resistant to dirt, while Its shiny surface and stiff texture add to Its charm. There are many grades, of chintz, and many have lustrous special finish, Cotton or linen fabric named for French town of Creton, with plain, rep or damask weave background printed In large designs. Does not muss easily and can be washed often. A type of wool embroidery worked on unbleached cotton or linen ground In large floral, bird, or tree designs. The name originated with the beautiful silk woven In Damascus during the 12th Century, Damasks are now made of linen, cotton, wool, or any of the synthetic fibers, or combinations of the two. In taffeta weave on satin ground, this fabric In flat woven pattern is usually reversible. Damask is also the name given to a kind of Jacquard weave. A fabric of loosely twisted cotton yarn, or cotton mixed with jute, In basket weave. Something like monk's cloth but not as rough in texture. Heavy plain weave cotton fabric. A material made by matting together, under heat or pressure, woolen fiber., mohair, cow hair, or mixed fibers. SUITABLE FOR Draperies, Medium-weight upholstery. Medium-weight upholstery. Panels. Other decorative uses, Curtains and panels. Other decorative effects. Glass curtains, Dressing table skirts, etc. Draperies. Medium-weight upholstery. Especially good for rooms of Queen Anne, Chippendale, Hepplewhite or Sheraton furnishings of 18th Century. Draperies on very large sized windows, Heavy-weight upholstery. Lined or unlined draperies. Medium-weight upholstery. Drapery or upholstery purposes. Curtains, draperies, bedspreads, comfortable*,

Excellent with Early American or French Provincial furnishings. Bedspreads, draperies, and other decorative purposes* Awnings, couch covers, etc.} also used for stiff Interlining as at top of draperies. Fine for draw-curtain*; also used for sheer textyrei. Draperies In modern Interiors. Trimmings, etc. Draperies. Yarn used for tufting, fringes, etc. Draperies, etc. May be formal or Informal In pattern. Suitable to any room according to pattern, quality and treatment. Used for draperies, upholstery, slip covers, lamp shades, etc. Draperies, upholstery, slip covers, bed covers, etc. Often more formal than chintz. Draperies and upholstery. Used extensively during Jacobean period, in silk or cotton It is used for draperies and upholstery. Appropriate for Queen Anne, Chippendale, Hepplewhite or Sheraton furnishings of 18th Century. Drapery. Couch cover. Outdoor cushions, etc. Upholstery and couch cover. Rugs.

General Reference Data FABRIC Characteristics and Suitability HOME-DECORATING FABRIC CHART continued FABRIC FILET NET FORTUNE PRINTS FRIAR'S CLOTH FRISSE GAUZE HAIR CLOTH HOMESPUN INDIA PRINTS JASPE LAME LAMPAS MARQUISETTE METALLASSE MOHAIR MONK'S CLOTH MOQUETTE NINON PLUSH REP SAIL CLOTH SUPPER SATIN SCRIM STRIPE TERRY CLOTH THEATRICAL GAUZE TOILES DE JOUY TWEED WEAVES VELOUR DESCRIPTION Cotton or linen net with square mesh. Hand netted filet has a knot at each corner of square mesh. Fabrics produced in Venice by a secret printing process which gives cotton cloth the effect of antique brocades and damasks. Comes in beautiful color combinations. like druid's cloth but with finer basket weave. Uncut Pile fabric of wool, mohair, cotton or linen. Patterns may be printed or produced by using yarns of different colors, or by cutting some of the loops to give sculptured effect. Very durable. Thin, sheer transparent fabric of plain weave, sometimes printed. May be all silk, or cotton, linen, wool, mohair, synthetic fibers, or combinations. A fabric with warp of cotton, worsted, or linen, and horsehair weft, woven plain, striped or patterned. May now be obtained in colors and variety of woven designs. Coarse hand-woven woolen, cotton or linen fabrics. Also trade name given to imitations made on power looms. Printed cotton cloth with clear colors and characteristic designs of India or Persia. Handprinted with many colors on white or natural background. Fabrics having warp threads of different colors giving material streaked or mottled effect, resembling

Jasper. A fabric with silk and metal thread* In plain weave or with a woven pattern. A fabric similar to damask In appearance and brocatelle In weave. Generally all silk with multi-colored pattern on plain background, often classic in design. Sheer cloth in gauze weave of cotton, silk, rayon, often with woven figure. It comes in wide range of colors, and may be dyed or printed. Fabric with brocaded pattern In raised, padded or blistered effect. Various types of fabrics made from the fleece of the Angora goat. Most durable of all textiles. Now woven In combination with cotton, linen, silk or wool Into many types of plain, twill or pile fabrics. Heavy cotton fabric of coarse basket weave. Pile fabric resembling fur, woven on Jacquard loom with small set pattern In different colors. A semi-transparent fabric of silk or rayon. High pile fabric resembling fur, made of silk, wool, cotton or any synthetic fiber. Pile may be cut or uncut. Plain weave fabric of heavy rib made of silk, cotton or Wool, or synthetic fibers. Un* patterned and reversible. Stout, firm, plain weave cotton material similar to canvas In construction but lighter. Has a stiff, hard texture and Is printed In gay, bright colors. Sleek, smooth very heavy satin In rayon or silk may be slightly stiff because of thickness. Fabric of coarse two-ply yarns In plain, open weave. Often mercerized. Term used to designate fabric with uneven streaked effect. This process gives two-toned appearance |o taffeta|o, satin, etc. Light cotton fabric similar to bath towel. Woven with uncut loops. May be dyed or printed, In designs of one or two colors. Rich texture and reversible. Loosely woven, transparent plain weave fabric of cotton or linen. Obtainable In brilliant as well as soft colors. Inexpensive. Printed cotton material with repeat designs showing landscapes, or historical scenes. Reproductions of famous printed fabric woven at Jouy, near Paris, France, Designs and figure groups usually In colors on white or cream background. Term applied to a large group of woolen goods made from worsted yarns, woven in plain, twill, or herringbone twill weaves In homespun type. Really a French word for velvet. Through common usage, a short*pile velvet. SUITABLE FOR Curtains, tablecloths, scarves, etc. Draperies. Wall hangings, screens, etc. Same as druid's cloth. Upholstery, Glass curtains. Upholstery. Used extensively In England and America during middle of 19th Century. Curtains and upholstery In informal rooms. Bedspreads In cotton. Draperies. Wall hangings. Bed coverings, etc. Draperies and other decorative effects. Drapery. Panels. Used as damask is used. Excellent for glass curtains. Fluffy, dainty, tailored spreads. Draperies.

Very durable and widely used for upholstery. Drapery material. Used for upholstery In mohair, wool, or heavy cotton. Glass curtains. Upholstery. Draperies. Upholstery. Draperies. Slip covers. Bedspreads, etc. Drapery and upholstery, bed coverings* etc. Suitable in formal and period rooms for draperies. Curtains, bedspreads, «lc According to fabric. Droperies. Draw-curtains. Glass curtains. Droperies, wall hangings, upholstery, bed covering I. Excellent for French, English and American period rooms of lata 16th Century and early 1^oh Century; also French Provincial. Draperies ond uphalsfary. Vary good for modern or masculine roams. Same os velvet. 1

General Reference Data ELECTRICAL Office and Electronic Equipment
 TABLE 16 Typical Amperage Ratings Equipment Amperage Equipment
 Amperage Electronic Equipment' Electronic Equipment Video Display
 Terminals (Detached Keyboards) Memory Storage Devices (Desk Top)
 Normal maximum Burroughs/MT983TP110 Digital/VT 278 Decmate
 Form-Phase Systems/8115-2 Harris/8680A Hewlett-Packard/2382A
 IBM/3101 IBM/3278 or 3276 IBM/3279 IBM/5251 Modem IBM/6580
 ITTCourier/2790-2A Perkin/Elmer/1251 Prime/PT-45 Raytheon/PTS100
 Sperry-Univac/UTS20(313) Texas Instruments/940/200 Texas
 Instruments/DS990 Wang/5503 Xerox/8000 Series Video Display Terminals
 (Integrated Keyboard) Normal maximum Hewlett-Packard/HP-9845A
 LearSiegler/ADM31 NCR/7900-01 Perkin-Elmer/550B Tektronics/4112
 Wang/5536 Printers (Stand Alone) Digital Equipment/LA-120AA
 Wang/5503 Xerox/8000 Print Server Printers (DeskTop) Normal maximum
 Centronics/7030 Digital Equipment/Decwriter IV Form-Phase
 Systems/8125 Hewlett-Packard/7221 B Hewlett-Packard/H P7240A
 IBM/3287 IBM/5256 Epson/FX185 NEC/3510 ITT Courier/8700 Lear
 Siegler/300 Series Perkin-Elmer/650 Raytheon/PTS 1200 3472 Sperry-
 Univac/0786 Tektronix/4612 Texas Instrumerits/Omni-800 810R0
 Wang/5531-2 w/floor mount Wang/5577 (DW-20 Series) 2.50 .08 1.25 .48
 1.00 .75 1.20 1.33 2.50 2.10 4.80 1.50 1.30 1.00 .80 .50 2.00 2.20 3.00 .50
 3.00 4.50 .50 .60 .80 3.00 2.00 3.00 4.00 11.00 3.50 1.50 1.30 1.60 1.00
 2.50 2.36 4.70 .70 2.50 1.60 1.60 1.50 1.50 4.00 3.30 5.00 1.70 1.20
 Wang/5503 Disk Xerox/8000 NS Digital Equipment/RX02 Hewlett-
 Packard/8290 ZM Hewlett-Packard/9895A Sperry-Univac/8406 Texas
 Instruments/WD-500 Bell/Western Electric/Dataphone 300/1200 Office
 Equipment* General Typewriter Transcriber Microfiche Manuscript holder

Calculator A.C. adapter Electric eraser Pencil sharpener Fan Space heater
 (1,000 watts) Space heater (1,250 watts) Space heater (1,500 watts) Coffee
 pots Copy machine Clock CRT (average) Printer (average) Lighting
 Adjustable task light 2" task light (20 watt) 2" task light (30 watt) 2" task
 light (40 watt) 2" task light (35 watt, energy-saving ballast) Indirect
 ambient light (30 watt/2 lamp) (30 watt/3 lamp) (40 watt/2 lamp) (40 watt/3
 lamp) (35 watt/2 lamp) (35 watt/3 lamp with energy-saving ballast) HID
 light (400 watt) (250 watt) (175 watt) PLP light (20 watt) (40 watt) 4.00
 12.00 3.00 1.00 1.60 1.50 3.00 .08 1.50 .15 .85 .75 .25 .05 .25 .25 1.00 8.50
 10.50 12.50 10.00 15.00 .03 1.50 3.50 .80 .27 .40 .48 .38 .68 1.08 .80 1.28
 .63 1.01 4.00 2.50 1.80 .65 .80 Wore: These figures are for quick reference
 only. For specific information consult the manufacturer. •Some appliances -
 such as large copiers, coffee makers, or space heaters - require most of the
 current available on a 20-amp circuit. It is recommended that such devices
 be supplied with their own receptacle, directly from the building. This
 leaves the capacity of Series 9000 circuits available for the more dynamic
 requirements of the office occupants. 1132

General Reference Data ELECTRICAL Locations of Receptacles In
 Dwelling Units Receptacle required if any wall width is 2ft or more T Wall
 receptacles Walls with sliding glass doors or floor-to-ceiling windows must
 be counted as wall space requiring receptacles Floor receptacles must be
 close to wall to count as required spaced receptacles Fig. 15 From any point
 along wall, at floor line, a receptacle must be not more than 6 ft away. Fig.
 16 Location of the receptacle as shown will permit the plugging in of a lamp
 or appliance located 6 ft on either side of the receptacle. Receptacle
 required at each counter space wider than 12 in. f\ (Wall receptacles
 /Counter space (Combination switch and receptacle) • -i——-^——-' ^ , i,fr
 COUNTER SPACES In kitchen and dining rooms such as shown by arrows
 (above) must be supplied with receptacles if they are over 12 in. wide
 Appliances are frequently used even on narrow counter widths; this
 requirement is designed to remove the dangerous practice of stretching
 cords across sinks, behind ranges, etc., to feed such appliances. Inaccessible
 receptacles. This receptacle is rendered inaccessible by refrigerator
 therefore another receptacle must be installed to serve counter top V -v-
 LOCATION of receptacle will vary, depending upon available wall space,
 Arrow! how lateral possible, A receptacle In a medicine cabinet or In

the bathroom lighting -fixture does not satisfy this rule. Fig. 18 Receptacle required adjacent to wash basin in residence. Counter top Refrigerator RECEPTACLE LOCATED behind an appliance, making the receptacle Inaccessible, does not count as one of the required "counter-top" receptacles, (Neither does it count as one of the appliance-circuit receptacles required to be located every 12 ft.) Fig. 17 Countertop receptacles are needed and must be accessible. 1133

General Reference Data ELECTRICAL Appliance Loads and House Circuits TABLE 17 Residential Appliance, Load, and Circuit Chart*

Appliance Typical wattage Voltage Amps Wires and Size fuse needed load size or breaker { Type of circuit and comments Range Countertop range Oven built-in Dishwasher Waste disposal Broiler Fryer Coffee maker Refrigerator Toaster Frypan Roaster 12,000 6,000 5,000 1,200 500 1,500 1,300 800 400 1,100 1,200 1,500 115/230 115/230 115/230 115 115 115 115 115 115 52 26 22 10 5 13 11 7 4 10 10 13 3#6 3#10 3#10 2#12w/grd { 2#12w/grd 2#12 2#12 2#12 2#12 2#12 2#12 2#12 50A 30A 30A 20A 20A 20A 20A 20A 20A 20A 20A 20A Separate circuit—grounded Separate circuit—grounded Separate circuit—grounded These two can be connected on one circuit; must be grounded. Two or more 20-amp circuits needed for these appliances depending on number used at once. A 115/230 V - "splitwired" circuit provides capacity of two ordinary circuits at any outlet, Ask your wiring inspector about this. Clothes dryer Washer Hand iron Hotplate ironer Workshop Portable heater Television Portable lights—(up to) Lighting, general (each) Air conditioner (window unit) Air conditioner (central unit) Water system Heating plant Electric heaters (built-in) Water heater 5,000 to 9,000 500 1,000 1,500 1,650 1,500 300 300 100 1,500 3,400 500 600 750 to 4,500 1,500 to 4,500 115/230 115 115 115 115 115 115 115 115 115 or 230 115/230 115 115 230 230 25 3#10 to ■ 3#6 9 2#12w/grd 9 2#12w/grd 13 2#12w/grd 15 2#12w/grd - 2#12 13 2#12 3 2#14 3 2#14 11S2#14 13 2#12w/grd or 7 20 3#10 5 2#12w/grd fl 2#12w/grd 7 2#12w/grd to 20 2#10w/grd 30A to 45A 20A 20A 20A 20A 20A 20A 15A 15A ISA 20A - 20A 20A 20A to 30A Separate circuit—grounded. Grounded—advise fused outlet for motor protection. -, A 20-A circuit, will carry >■ only one of these in addition) to washer. Separate circuit grounded. >■ Use on general-use circuits. (Not over 9 per circuit, including convenience outlets.) Requires separate circuit; 230 volt operation

preferred. Check manufacturer's recommendations; should be grounded. Separate circuit—grounded. Provide motor protection (230V. for 14 hp. or larger). Separate circuit—grounded, Provide motor protection. Wiring should be planned with heating. Provide separate circuits for heating, Separate circuit—grounded. { Wire sizes are for copper wire. For aluminum, use next larger size. | Fuses advised in place of ordinary fuses up to 30 amp hrs they do not blow on harmless short-time overloads and cannot be replaced by a larger size. \$ W/grd means with groundwire. This is usually a bare wire run inside the same cable but can be installed separately. Portable equipment is grounded through the third prong on the plug. Permanent equipment is grounded by direct connection of the third wire to the frame of the appliance. • Courtesy Agricultural Extension Service, South Dakota State University. TABLE 18 Typical Office Amperage Loads

Appliance	100-150W	150-200W	200-250W	250-300W	300-350W	350-400W	400-450W	450-500W	500-550W	550-600W	600-650W	650-700W	700-750W	750-800W	800-850W	850-900W	900-950W	950-1000W
CAD station*	.25	.33	.40	.50	.60	.75	.85	1.00	1.10	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25
Calculator	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Coffee pot*	.75	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	3.75	4.00	4.25	4.50	4.75	5.00
Clock	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Radio	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Stereo	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Tape recorder	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Laser printer*	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	3.75	4.00	4.25	4.50	4.75	5.00	5.25
Desktop copier*	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	3.75	4.00	4.25	4.50	4.75	5.00	5.25
Electric eraser	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Fan	.75	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	3.75	4.00	4.25	4.50	4.75	5.00
Freestanding copier*	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	3.75	4.00	4.25	4.50	4.75	5.00	5.25
Pencil sharpener	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Task light (4')	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Adding machine	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Letter opener	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Dictaphone	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Telecopier	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Word processor	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Postage meter	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Tape dispenser	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Personal computer	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Desktop printer	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
CRT	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Space heater*	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	3.75	4.00	4.25	4.50	4.75	5.00	5.25
Typewriter	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Microfiche reader	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
Transcriber	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
A.C. adapter	.05	.07	.10	.15	.20	.25	.30	.35	.40	.45	.50	.55	.60	.65	.70	.75	.80	.85
100-W lamp	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	3.75	4.00	4.25	4.50	4.75	5.00	5.25

*Some appliances such as coffee pots, copiers, printers, and heaters consume most of the amperage available on a circuit. It is recommended that these devices be connected directly to the building power supply, leaving flexibility for other circuit planning. TABLE 19 Common House Circuits

Circuit Type	Ampere rating of fuse or circuit breaker	Veils Load capacity
Uscofcircull Types of outlets	12 15 115-18	Installed huhUIH!, outlets for lamps and low Wallace appliances all over the house
Individual .ipplUiice	Individual power 115-120 1300-210(1	Automatic washer, relrlgcralor. freezer Those for attached ilghl fixtures; ordinary convenience uut- lels Appliance minimum #12
return mended # ()	20 23 115-120 2300-2400 2875-3(XH)	Portable appliances itt Ml the », dining mam, wwkhhti, laundry Ordinary or grounded convenience outlets 30-fW 220 240 66M-14.100 Hai^c, water heater, do I lie b dryer Ordinary or grounded con- veniencce outlets ____
Heeceptai-lefor llItIR Will)	3	heuvy prongs or appliance* maybe attached

General Reference Data COLUMNS, CAPITALS, AND ENTABLATURES Proportions doric ionic corinthian 1135

General Reference Data NAILS, SCREWS, AND BOLTS Nails 2d 3d 4d
6d 8d 9d 10d 12d 16d 20d 30d 40d 50d 60d Common nails Q—iiiMni^: 8d
CE 4d Finishing nails V- v__ W qQpp vf v__ ^> Not. Int. T 3/16" 1-1/2
7/32" 2 1/4" 2-1/2 5/16" 3 3/8" 4 7/16" 6 | 8 1/2" J 9/16" 10 5/8" 12 11/16"
14 3/4" 16 13/16" 18 7/8" 20 15/16" 22 1" Z4 1-1/8" 1T m ? t | y |
STANDARD TACK SIZES ESTABLISHED LENGTHS OF TACKS
MEASURED UNDER THE HEAD 1136

3> Barbed: for fastening shingles or other flexible materials Boat spike:
long spike used In timber construction d Brad: finishing nail less than 1
Inch long 3> :r=~ Box: for nailing thin dry wood close to edge " 11111 - -
n= Cement-coated box: coated with resin that increases holding power C
11111 Casing: similar to finishing, but with dulled point to penetrate thin
trim without splitting 1111111 f> Common: for rough and heavy
construction Concrete: hardened steel nail with diamond point Fluted
concrete: hardened steel nail with fluting to increase holding power (t
Escutcheon pin: small nail used to attach escutcheons _£^ J Fence: iarge-
head nail for holding weathered boards Q I I I I S Finishing: slender nail
that can be set below surface Cut finishing: finishing nail used in historic
restoration General Reference Data NAILS, SCREWS, AND BOLTS Nails
P MM 11 Blunt flooring: nail with blunt tip to prevent splitting hardwood
flooring)mr _c> Built-up roofing: for attaching roofing felt tr Cut flooring:
for historic restoration It Drive-screw flooring: nail with screw-drive to
increase holding power m> Annular drywall: for attaching drywall to
framing; rings Increase holding power HMDDHH ^^ Gutter spike: for
attaching gutter to fascia Hinge: for attaching large hinges such as for barn
doors (Picccccccccco Annular hinge: hinge with rings to Increase holding
power Lath: small nail for installing wood lath I> Offset head: for use with
power nailer D11111: Parquet flooring: thin nail to prevent splitting parquet
:sn Pole barn: large spike for attaching framing to poles :e> Roofing: for
attaching asphalt roofing to underiayment (f No-leak roofing: nail with
rubber gasket to seal metal roofing > Scaffold: nail with double head to
make nail easy to pull for temporary fastening lELTJ n> Shingle: f""
attaching cedar shingles Cut shingle: used in historic restoration hum

Siding: used to install beveled wood siding 3> Screw-thread siding: nail with a screw thread to increase holding power Cement-coated sinker: used to install underlayment Slating: used to attach roofing slates iii[]
[fnrrrniEffln> Ring shank underlayment: nail with rings to increase holding power ill > Spike: common nail 4-1/2 inches or longer 1137

General Reference Data NAILS, SCREWS, AND BOLTS Screws and Bolts
SCREWS isxmG> Flat head wood screw: for fastening wood to wood 35xxxm> Oval head wood screw: decorative :orams» Round head wood screw: used with washer RWWUtt* Sheet metal screw: for thin metal Oven head machine screw: older design Hi Oval head machine screw: attractive
SCREW AND BOLT HEADS Slotted Tamper-proof WASHERS 1138
Internal- tooth Phillips Internal torx G Tamper-proof External- tooth >- Lag bolt: for heavy loads in wood m Carriage bolt: bolt will not turn 1 R Hex head bolt: for heavy loads i:s y- Square head bolt: replaced by hex head II! Round head bolt: older design Stove bolt: finishes flush Combination Phillips/slotted Cfutch Tamper-proof hexagon Finish Internal-external-tooth 8 Fillister head cap screw: small, strong-.headed a Fillister head machine screw: small, strong-headed (\ Step bolt: bolt won't turn Plow bolt: for steel to steel Tamper-proof torx

General Reference Data MATHEMATICAL DATA AND FORMULAS
Units of Measurement LINEAR MEASURE Measures of Length Nautical Units 12 inches 3 feet 6M yards = 16M feet 40 polea = 220 yarda 8 furlongs = 1760 yards! = 6280 feet / 3 milea 4 inches 0 inches = 1 foot = 1 yard « 1 rod, pole or perch = 1 furlong = 1 mile » 1 league = 1 hand « 1 span 6080.20 feet 6 feet 120 fathoms 1 nautioaJ mile per hr. > 1 nautical mile ' 1 fathom = 1 oable length ' 1 knot Surveyor's or Gunter's Measure 7,92 inohea 100 links - 66 ft. = 4 rods SO ohains 33>3 inohes - 1 link = 1 ohain « 1 mile = 1 vara (Texas) Length Equivalents Centimeters 1 2.540 30.48 91.44 100 2012 100000 160935 Inches 0.3937 1 12 30 39.37 792 39370 63360 Feet 0.03281 0.08333 1 3 3.281 66 3281 S280 Yards 0.01094 0.02778 0.3333 1 1.0936 22 1093.0 1760 Meters 0.01 0.0254 0.3048 0.9144 1 20.12 1000 1009 Chains 0.0,4971 0.001263 0.01515 0.04545 0.04971 1 49.71 80 Kilometers 10-5 0.0,254 0.0,3048 0.0,9144 0.001 0.02012 1 1.609 Miles 0.066214 0.04578 0.0,1894 0.0,5682 0.0,6214 0.0125 0.6214 1
Subscripts after any figure, Oj, 9 4, etc. mean that that figure is to be

repeated the indicated number of times. MEASURES OF AREA 144 square inches = 1 square foot 9 square feet = 1 square yard 30K square yards ~ 1 square rod, pole or perch 160 square rods * 10 square chains = 43,560 sq. ft. f " 1 ft 0 re » 5645 sq. varas (Texas) 040 acres = 1 square mile = 1 "section" of U. S, Govt, surveyed land Square Meters 1 0.0,6452 0.09290 0.8361 25.29 404.7 1012 4047 2589998 Square Inches 1550 1 144 1296 39204 627264 1568160 6272640 Square Feet 10.76 0.006944 1 9 272.25 4356 10890 43560 27878400 Area Equivalents Square Yards 1.196 0.0,7716 0.1111 1 30.25 484 1210 4840 3097600 Square Rods 0.0395 0.0,2551 0.003673 0.03306 1 16 40 160 102400 Square Chains 0.002471 0.051594 0.0,2296 0.002066 0.0625 1 2.5 10 6400 Rods 0.0,9884 0.0,6377 0.0,9184 0.0,8264 0.02500 0.4 1 4 2560 Acres 0.0,2471 0.0,1594 0.0,2296 0.0002066 0.00625 0.1 0.25 1 640 Square Miles or Sections 0.0,3861 0.0,2491 0.0,3587 0.0,3228 0.0,9766 0.0001562 0.0,3906 0.001562 1 (1 hectare « 100 aros = 10,000 centiares or square meters)

Subscript after any figure Oi, 9i, etc., mean that that figure is to be repeated the indicated number of times. VOLUMETRIC MEASURE MEASURES OF WEIGHT Measures of Volume 1728 cubic inches = 1 cubic foot 27 cubic feet = 1 cubic yard 1 cord of wood » 128 cu. ft. 1 perch of masonry **16H to 25 cu. ft. Dry Measure 2 pints = 1 quart 8 quarts = 1 peck 4 pecks « 1 bushel 1 std. bbl, for fruits and vegetables = 7056 cu. in. or 105 dry quarts, a truck measure Liquid or Fluid Measure 4 gills = 1 pint 2 pints - 1 quart 4 quarts = 1 gallon 7.4805 gallons <=> 1 cubic foot (There is no standard liquid barrel; by- trade custom, 1 bbl. of petroleum oil, unrefined » 42 gal.) Board Measure 144 cu. in. - volume of board 1 ft. sq. and 1 in, thick. No. of board feet in a log *■ (J«d—4)1*Z>, wherof « diam.of log(usually taken in- side the bark at small end), in., and X> = length of log, ft, The 4 in. deduced are an allowance for slob. This rule is variously known as the Doyle, Conn. River, St. Croix, Thurber, Moore and Beeman, and the Scribner rule. Weights (The grain is the same in all systems) Avoirdupois Weight 10 drams = 437.5 grains = 1 ounce 16 ounces = 7000 grains = 1 pound 100 pounds = 1 cental 2000 pounds = 1 short ton 2240 pounds «=> 1 long ton 1 std. lime bbl., small - ISO lb. net 1 std. lime bbl., large «■ 280 lb. net Also (in Great Britain): 14 pounds 2 stone » 28 lb. 4 quarters » 112 lb, Troy Weight 24 grains 20 hundredweight - 1 atone = 1 quarter = 1 hundredweight (cwt.) *» 1 long ton w 1 pennyweight (dwt.) 20 pennyweights * 480 grains" 1 ounce 12 ounces = 5760

grains = 1 pound 1 Assay Ton » 29,167 milligrams, or as many milligrams
 as there are troy ounces in 1 ton of 2000 lb, avoirdupois. Consequently, the
 number of milligrams of precious metal yielded by an assay ton of ore gives
 directly the number of troy ounces that would be obtained from a ton of
 2000 lb. avoirdupois Apothecaries' Weight 20 grains = 1 scruple 3 3
 scruples, * = 60 grains = * 1 dram 3 8 drams = 1 ounce 3 12 ounces « 5760
 grains = 1 pound Cubic inches 1 1728 46656 1.805 57.75 67.20 231 268.8
 2150 61.02 Cubic feet 0.0,5787 1 27 0.001044 0.03342 0.03889 0,1337
 0.1556 1.244 0.03531 Volume and Capacity Equivalents Cubic yards
 0.0,2143 0.03704 1 0.0.3868 0.001238 0.001440 0.004951 0.005761
 0.04609 0.001308 U.S. Apothecary liquid ounces 0.5541 957.5 25853 1 32
 37.24 128 148.9 1192 33.81 U. S. quarts Liquid 0.01732 29.92 807.9
 0.03125 1 1.164 4 4.655 37.24 1.057 Dry 0.01488 25.71 694.3 0.02686
 0.8594 1 3.437 4 32 0,9081 U.S. gallons Liquid 0.0,4329 7.481 202.0
 0.007813 0.25 0.2909 I 1.164 9.309 0.2642 Dry 0.0,3720 6.429 173.6
 0.0067H 0.2148 0.25 0.8594 1 8 0.2270 Bushels U.S. 0.0,4650 0.8036
 21.70 0.0,8392 0.02686 0.03125 0.1074 0.125 I 0.02838 Liters (0 0.01639
 28.32 764.6 0.02957 0.9464 1.101 3.785 4.405 35.24 1 Kilograms 1
 0.0,6480 0.03110 0.02835 0.3732 0.4536 907.2 1016 1000 Grains 15432 1
 480, 437.5 5760 7000 140, 15680, 15432356 Mass Equivalents Ounces
 Troy and apoth. 32.15 0.0,2083 1 0.9115 12 14.58 29167 32667 32151
 Avoirdupois 35.27 0.0,2286 1.09714 1 13.17 16 320, 35840 35274 Pounds
 Troy and apoth, 2.6792 0.0,1736 0.08333 0.07595 1 1.215 2431 2722 2679
 Avoirdupois 2.205 0.0,1429 0.06857 0.0625 0.8229 1 2000 2240 2205 Tons
 Short 0.0,1102 0.0,7143 0.0,3429 0,0,3125 0.0,4114 0,0005 1 1.12 1.102
 Long 0.0,9842 0.0,6378 0.0,3061 0.0,2790 0.0,3673 0.0,4464 0.8929 I
 0.9842 Metric 0.001 0.0,6480 0.0,3110 0.0,2835 0.0,3732 0,0,4536 0.9072
 1,016 1 Subscripts after any figure, Oa, 9<, etc., mean that that figure is to
 be repeated the indicated number of times, Subscripts after any figure, Oa,
 9<, etc., mean that that figure is to be repeated the Indicated number of
 times.

General Reference Data MATHEMATICAL DATA AND FORMULAS

Metric Conversion Tables METRIC WEIGHT Avoirdupois 1 milligram
 (mg) = 0.0154 gr. 1 centigram (cg) = 10 mg = 0.1543 gr. 1 decigram (dg)
 = 10 cg = 1.5432 gr. 1 gram (g) = 10 dg = 15.4323 gr. 1 dekagram (dag) = 10g
 = 0.3527 oz. 1 hectogram (hg) = 10 dag = 3.5274 oz. 1 kilogram (kg) = 10

hg = 2.20461 lb. 1 quintal (q) = 100 kg = 220.46 lb. 1 metric ton (M.T.) = 10 q or 1,000 kg = 2,204.62 lb. AVOIRDUPOIS WEIGHT Metric 1 grain (gr.) = 0.0648 g 1 dram (dr.) = 27.34375 gr. = 1.7718 g 1 ounce (oz.) = 16 dr. = 28.3495 g 1 pound (lb.) = 16 oz. = 453.5924 g or 0.4536 kg 1 hundredweight (cwt.) = 100 lb. = 45.3592 kg 1 short ton (s.t.) = 2,000 lb. = 907.18 kg or 0.9072 M.T. WOOD MEASUREMENTS Customary Metric 1 board foot (bd. ft.) = 144 cu. in. (1 ft. x 1 ft. x 1 in.) = .00236 nv' 1 cord foot (cd. ft.) = 16 cu. ft. (4 ft. x 4 ft. x 1 ft.) = .4528 nv' 1 cord (cd.) = 8 cd. ft. (4 ft. x 4 ft. x 8 ft.) = 3.625 nv' Metric Customary 1 stere 1 ma 1.3079 cu. yd. or 0.2759 cord LENGTH AND DISTANCE Customary Metric 1 inch (in.) = 2.54 cm 1 foot (ft.) = 12 in. = 30.48 cm 1 yard (yd.) = 3 ft. = 0.9144 m 1 rod (rd.) = 16.5 ft. = 5.0292 m 1 furlong (fur.) = 40 rd. or 1/8 mi. = 201.168 m 1 statute mile (mi.) = 5,280 ft. = 1.6093 km 1 league = 3 mi. = 4.8280 km Metric Customary 1 millimeter (mm) = 0.03937 in. 1 centimeter (cm) = 10 mm = 0.3937 in. 1 decimeter (dm) = 10 cm = 3.937 in. 1 meter (m) = 10 dm = 39.37 in. 1 dekameter (dam) = 10 m = 39.37 m 1 hectometer (hm) = 10 dam = 328.0833 ft. 1 kilometer (km) = 10 hm = 0.62137 mi. SURFACE OR AREA Customary Metric 1 square inch (sq. in.) = 6.4516 cm² 1 square foot (sq. ft.) = 144 sq. in. = 0.0929 m² 1 square yard (sq. yd.) = 9 sq. ft. = 0.8361 m² 1 square rod (sq. rd.) = 30' / 16 sq. yd. = 25.293 m² 1 acre (A.) = 160 sq. rd. = 0.4047 hm² 1 square mile (sq. mi.) = 640 A. = 258.998 ha or 2.5899 km² Metric Customary 1 square millimeter (mm²) = 0.002 sq. in. 1 square centimeter (cm²) = 100 mm² = 0.1549 sq. in. 1 square decimeter (dm²) = 100 cm² = 15.499 sq. in. 1 square meter (m²) = 100 dm² = 1.549 sq. in. 1 square dekameter (dam²) = 100 m² = 119.6 sq. yd. 1 square hectometer (hm²) = 100 dam² = 2.4710 A. 1 square kilometer (km²) = 100 hm² = 247.104 A. or 0.3861 sq. mi. METRIC LAND MEASUREMENTS Metric Customary 1 centiare (ca) = 1.549 sq. in. 1 are (a) = 100 ca = 119.6 sq. yd. 1 hectare (ha) = 100 a = 2.4710 A. 1 square kilometer (km²) = 100 ha or 0.3861 sq. mi. VOLUME MEASUREMENTS Customary Metric 1 cubic inch (cu. in.) = 16.387 cm³ 1 cubic foot (cu. ft.) = 1,728 cu. in. = 0.0283 m³ 1 cubic yard (cu. yd.) = 27 cu. ft. = 0.7646 nv' Metric Customary 1 cubic millimeter (mm³) = 0.00006 cu. in. 1 cubic centimeter (cm³) = 1,000 mm³ = 0.0610 cu. in. 1 cubic decimeter (dm³) = 1,000 cm³ = 0.0353 cu. ft. 1 cubic meter (m³) = 1,000 dm³ = 1.3079 cu. yd. 1 cubic dekameter (dam³) = 1,000 nv' = 1,307.9 cu. yd. 1 cubic hectometer (hm³) = 1,000 dam³ = 1,307,900 cu. yd. METRIC CAPACITY MEASUREMENTS Metric Customary 1

milliliter (ml) = 0.0610 cu. in. 1 centiliter (cl) = 10 ml = 0.6102 cu. in. 1
 deciliter (dl) = 10 cl = 6.1025 cu. in. 1 liter (l) = 10 dl = 61.025 cu. in. or
 1.057 liquid qt. or 0.908 dry 1 dekaliter (dal) = 10 l = 610.25 cu. in. 1
 hectoliter (hl) = 10 dal = 6,102.50 cu. in. 1 kiloliter (kl) = 10 hl = 35.315 cu.
 ft. or 264.178 gal. or 28.38 bu. HOUSEHOLD CAPACITY

MEASUREMENTS Customary Metric 1 teaspoon = 1/6 fl. oz. = 4.9 ml 1
 tablespoon = 3 teaspoons = 1/2 fl. oz. = 14.8 ml 1 cup = 16 tablespoons = 8 fl.
 oz. = 236.6 ml 1 pint = 2 cups = 16 fl. oz. = 473.2 ml 1 quart = 2 pints = 32
 fl. oz. = 946.4 ml 1 gallon = 4 quarts = 128 fl. oz. = 3.7851 L

LIQUID CAPACITY MEASUREMENTS Customary Metric 1 quart (qt.) =
 7.219 cu. in. = 0.11831 l 1 pint (pt.) = 4 gi. = 28.875 cu. in. = 0.4732 l 1 quart
 (qt.) = 2 pt. = 57.75 cu. in. = 0.9463 l 1 gallon (gal.) = 4 qt. = 231 cu. in. =
 3.7853 l 1 barrel (liquids) (bbl.) = 31.5 gal. = 4.2 cu. ft. = 119.24 l 1 barrel
 (petroleum) (bbl.) = 42 gal. = 5.61 cu. ft. = 158.98 l Imperial Customary
 Metric 1 imperial quart = 1.2009 U.S. qt. = 69.355 cu. in. = 1.13649 l 1
 imperial gallon = 1.2009 U.S. gal. = 277.420 cu. in. = 4.54596 l DRY
 CAPACITY MEASUREMENTS Customary Metric 1 pint (pt.) = 33.600
 cu. in. = 550.60 cm³ 1 quart (qt.) = 2 pt. = 67.20 cu. in. = 1,101.21 cm³
 1 peck (pk.) = 8 qt. = 537.61 cu. in. = 8,809.85 cm³ 1 bushel (bu.) = 4 pk.
 = 2,150.42 cu. in. = 0.035239 m³ 1 barrel (bbl.) = 4 cu. ft. = 0.115627 m³
 Imperial Customary Metric 1 imperial dry quart = 1.032 U.S. qt. = 0.9354
 cu. in. = 1,136.5 cm³ 1 imperial bushel = 1.032 U.S. bu. = 1.284 cu. ft. =
 0.03636 m³ 1140

General Reference Data MATHEMATICAL DATA AND FORMULAS

Fraction, Decimal, and Metric Conversion Tables Fractions of Inch or Foot
 & Hi Hi % % . % % X % % % *e % % % M .0052 .0104 .015625 .0208

.0260 .03125 .0365 .0417 .046875 .0521 .0573 .0625 .0677 .0729 .078125
 .0833 .0885 .09375 .0990 .1042 .109375 .1146 .1198 .1250 .1302 .1354
 .140625 .1458 .1510 .15625 .1615 .1667 .171875 .1771 .1823 .1875 .1927
 .1979 .203125 .2083 .2135 .21875 .2240 .2292 .234375 .2396 .2448 .2500

Lin 10 millimeters = 10 centimeters = 10 decimeters » 10 meters = 10

decameters ■ 10 h Bctometers ■ ■ifi a #3| lg *6 H *6 Hi Kb H % H 'Ke >A
 f*6 H % 1 1*6 Wi 1*6 1H 1*6 1M 1*6 i« 1*6 1H 1% 1% 1% 1H 1% 2 2*6
 2H 2*6 2X 2% 2H 2*6 2*6 2% 2*6 2H' 2% a»« 3 DECIMAL OF AN

INCH Fractions of Inch or Foot % % % *6 % % % H u4i % % *6 % % % «
 .2552 .2604 .265625 .2708 .2760 .28125 .2865 .2917 .296875 .3021 .3073

.3125 .3177 .3229 .328125 .3333 .3385 .34375 .3490 .3542 .359375 .3646
.3698 .3750 .3802 .3854 .390625 .3958 .4010 .40625 .4115 .4167 .421875
.4271 .4323 .4375 .4427 .4479 .453125 .4583 .4635 .46875 .4740 .4792
.484375 .4896 .4948 .5000 » ear = 1 centimeter = 1 decimeter = 1 METER
(m) = 1 decameter - 1 hectometer = 1 kilom eter I 10 milli 10 cent 10 deci
10 liter lOdeca lOhect lis &3 3*6 3X 3)4 3*6 VA 3% 1% 3l*6 3H 3'*6 Wi
3'Ke 4 m m 4*6 4)4 m *K 47*6 VA 4*6 m 4l*6 4K 4% m 4'% 5 5*6 5Va
5*6 5)4 5*\$ 5\$ 5% 5}i 5% 5H 5'*6 534 5% 5^ 5% 6 AND OF A FOOT
Fractions of Inch or Foot % % % *6 % % % % *%. % % '*6 % % % M
AETRIC MEA liquid and J [iters = 1 cent liters = 1 deci iters = 1 LIT i = 1
deca iters = 1 hect liters = 1 kilol .5052 .5104 .515625 .5208 .5260 .53125
.5365 .5417 .546875 .5521 .5573 .5625 .5677 .5729 .578125 .5833 .5885
.59375 .5990 .6042 .609375 .6146 .6198 .6250 .6302 .6354 .640625 .6458
.6510 .65625 .6615 .6667 .671875 .6771 .6823 .6875 .6927 .6979 .703125
.7083 .7135 .71875 .7240 .7292 .734375 .7396 .7448 .7500 SURES)ry liter
iter ER(1) liter oliter ter Is? "5 |3£ As 6*6 6H 6*6 6)4 6*6 &Va 6% VA 6%
6H 6% a% 6% «H e1^ 7 7*6 7J^ 7% 7)4 7*6 7Vs 7*. 7H 7% 7H 7% 7«
7J*6 7% 7»*6 8 8*6 8H 8% 8J4 854 8*6 8J* 8*6 8% 8'*6 8H 8»J4 8K 8%
9 Fractions of Inch or Foot <«« % % 13*e % % % « % % % 16*6 % % % 1
.7552 .7604 .765625 .7708 .7760 .78125 .7865 .7917 .796875 .8021 .8073
.8125 .8177 .8229 .828125 .8333 .8385 .84375 .8490 .8542 .859375 .8646
.8698 .8750 .8802 .8854 .890625 .8958 .9010 .90625 .9115 .9167 .921875
.9271 .9323 .9375 .9427 .9479 .953125 .9583 .9635 .96875 .9740 .9792
.984375 .9896 .9948 1.0000 it 11* &£ 9*6 9Mi 9Ke 9)4 W4 9Vs 9% m
9K6 95i 9lKe W 9^6 9Vs 9'Ke 10 10K6 10H 10Ke 10M 10% 10}* 10%
10Jf 10% 10^ 10l*6 lO1^ 10% 10l*6 11 11*6 11)1 11*6 llji 11*6 11*6
11*6 H« H*6 HM 11% 11% 12 Weights 10 milligrams = 1 centigram 10
centigrams = 1 decigram 10 decigrams = 1 GRAM (g) 10 grams = 1
decagram 10 decagrams = 1 hectogram 10 hect j grams = 1 kilogr am
METRIC CONVERSION FACTORS METRIC TO AMERICAN
Millimeters ■*■ 25.4 = inches Centimeters X 0.3937 = inches Meters X
39.27 = inches Millimeters X 0.003281 = feet Centimeters X 0.03281 = feet
Meters X 3.281 = feet Meters X 1.094 = yards Kilometers X 0.621 = miles
Kilometers X 3280.7 = feet Square millimeters + 645.1 = square inches
Square centimeters + 6.451 = square inches Square meters X 10.764 =
square feet Square kilometers X 247.1 <= acres Hectares X 2.471 = acres
Cubic centimeters + 16.383 = cubic inches Cubic meters X 35.315 = cubic

feet Cubic meters X 1.308 = cubic yards Cubic meters X 264.2 = gallons
 Liters X 61.022 = cubic inches Liters X 0.2642 = gallons Liters $\div$ 28.316 =
 cubic feet Hectoliters X 3.531 = cubic feet Hectoliters X 2.84 = bushels
 Hectoliters X 0.131 = cubic yards Hectoliters X 26.42 » gallons Kilograms
 X 2.2046 = pounds Kilograms $\div$ 1102.3 = tons AMERICAN TO METRIC
 Inches X 25.4 = millimeters Inches X 2.54 = centimeters Inches X 0.0254 =
 meters Feet X 304.8 = millimeters Feet X 30.48 = centimeters Feet X
 0.3048 = meters Yards X 0.9143 = meters Miles X 1.6093 = kilometers Feet
 $\div$ 3280.7 » kilometers Square inches X 645.1 = square millimeters Square
 inches X 6.451 = square centimeters Square feet $\div$ 10.764 = square meters
 Acres $\div$ 247.1 = square kilometers AprpR -s- 0. 471 — hpctarps JllilCO .
 $\div$ 1.6093 X $\div$ 1.6093 = Cubic inches X 16.383 = cubic centimeters
 Cubic feet $\div$ 35.315 = cubic meters Cubic yards $\div$ 1.308 = cubic meters
 Gallons (231 cu. in.) $\div$ 264.2 = cubic meters Cubic inches $\div$ 61.022 = liters
 Gallons X 3.78 = liters Cubic feet X 28.316 = liters Cubic feet $\div$ 3.531 =
 hectoliters Bushels $\div$ 2.84 = hectoliters Cubic yards $\div$ 0.131 = hectoliters
 Gallons $\div$ 26.42 = hectoliters Pounds $\div$ 2.2046 = kilograms Tons X
 1102.3 = kilograms 1141

General Reference Data MATHEMATICAL DATA AND FORMULAS

Areas of Plane Figures -s-*iT T X .1 I—»-l 1 H —h- Square Diagonal = $d =$
 $t\sqrt{2}$. Area = $\frac{1}{2}d^2$. Example, $t = 6$, $d = 8$. Area = $\frac{1}{2}(6^2 + 8^2) = 50$. Ans. $d =$
 $6\sqrt{2} = 8.484$ Ans. Rectangle and Parallelogram Area = ab or $tVrf$ —
 6' l'. Example. $a = 6$; $b = 8$; $S = 48$. Area = $8 \times 6 = 48$ Ans Trapezoid Area = $Hh(a + b)$
 Example, $a = 6$; $b = 8$; $H = 3$. Area = $3(6 + 8) = 42$. Ans. Trapezium Area =
 $\frac{1}{2}H(a + b)$. Example, $a = 6$; $b = 8$; $H = 3$. Area = $\frac{1}{2}(6 + 8) \times 3 = 21$. Ans. Triangles Both formulas apply to
 both figures Area = $\frac{1}{2}bh$. Example, $b = 8$; $h = 5$, Area = $\frac{1}{2}(8 \times 5) = 20$. Ans. Area = $\frac{1}{2}S(S - a)(S - b)(S - c)$ when $S = \frac{1}{2}(a + b + c)$. Example. $a = 8$; $b = 3$;
 $c = 4$. $S = 7.5$. Area = $\sqrt{7.5(7.5 - 8)(7.5 - 3)(7.5 - 4)} = 8.1$. Ans.
 Regular Polygons S sides - 1.780477 5' ■■ 8 3.68871 r> ■ 8.5981S0 S' -
 8.46410 r' 7 " - S.6SS87S 5' - 3,87101 r> Area 8 " - 4.888487 5' » S.S1S68
 r' 0 " " 6.18187a 5' - 8,87478 r> 10 " «■ 7.694850 S' - 8.84980 r> 11 " »
 9.S6S67S S1 - 8.88993 r1 18 " » 11.196300 S> = 8.81431) r' n » number of
 sides; r = short radius; S = length of side; R « long radius. $n \times 180^\circ$ n p. .
 360° Area = $\frac{1}{2}S^2 \cot \frac{\pi}{n}$ = $\frac{1}{2}R^2 \sin \frac{2\pi}{n}$ — 4 n x n 180' = nr Ian — Circle
 8.1416; A — area; d «■ diameter; p ference or periphery; r V = wd -

S.1410d. $j > j = irr = 6.88S2r. 8.1416 6.8888 A = 2i - = 0.78S4d > /f =$
 $* \cdot (\blacksquare * \ll S.1416r' \text{ circum- radius. } iVrA = 8.54 - ^{2.4} 4.4 ; tt " * (j / - = 1.188V^3$
 $" 4 / ^ = 0. \ll 64V^4 A = -f - /f p * p * i7 = 18.S7 8 = 4 \text{ Circular Ring Area} = r(R'$
 $- r') - 3.1410(R' - r') \text{ Area} = 0.78 \ll 4(Z) \gg - d >) - 0.78M(D - d)(D + d) \text{ Area} =$
 difference in areas between the inner and outer circles. Example, $R = 4; r =$
 $2. \text{ Area} = 8.1416(4' - \gg) = 37.6098. \text{ Ans. Area Quadrant } rr > \text{ --- } = 0.78 \ll 4r'$
 Example, $r = 8. c \text{ Area} = .78/54 X S \gg 0.3987c'. \blacksquare \text{ chord. } 7.0680. \text{ Ans.}$
 Segment $b \blacksquare$ length of arc, S — angle in degrees $c = \text{chord} = V4(8*r \text{ --- } *)$
 $\text{Area} - J - \pounds [ir - e(r - *)) \cdot 1 \ll (r - *) T 860 8 \text{ When } 9 \text{ is greater than } 180^\circ \text{ then}$
 $j X \text{ difference } rrV \text{ between } r \text{ and } * \text{ is added to the fraction -ttt } 800 \text{ Example,}$
 $r - 8; \ll - 180^\circ; * - 1.f \text{ Area. } 8.1416 X 3' X ^{-g-196VU} \gg S.S878. \text{ Ans. Sector}$
 $rr \gg; \text{ Area--j--. so0,, } 8 = \text{ angle in degrees; } b = \text{ length of arc. Example, } r = 8; \gg$
 $= 180^\circ 180 \text{ Area} = 3.1416 X 3' X ^ = 9.4848. \text{ Ans. Spandrel Area} = 0.41$
 $J0ra \gg 0.107.1c2 \text{ Example, } r \gg 3 \text{ Area} = 0.8146 X 3! - 1.0314. \text{ Ans Parabola}$
 $/ = \text{ length of curved line } \ll = \text{ periphery --- } \gg ' = AtVcti+c) + 8.0S80 X$
 $\log tVc+VT+c)] 'eh' i \text{ which } < 4AV (?) \text{ Area} = - ih \text{ Exmple, } \gg = 3; /i = 4$
 $\text{Area} = jX3X4 = 8. \text{ Ans, } 3 \text{ Ellipse Area } = toA = 3.1416o6 3 T \text{ Circum.} = 2t V$
 $" y" \text{ --- (close approximation) Example, } a = 3; 6 = 4. \text{ Area} = 3.1416 x 3 x 4 -$
 $37.6992. \text{ Ans. Circum.} = 2 x 3.1416 V (3)' + (4' - 6.2832 x 3.5355 = 22.21$
 Ans. FORM TBIANSLE TRAPEZIUM TRAPEZOID
 PARALLELOGRAM RES. POLV60N CIRCLE SECTOR OF A CIRCLE
 SEGMENT OF A CIRCLE CIRCLE SQUARE ELLIPSE PARAOOLA ▲ I
 m ■r: $M 0 \cdot ^ oA" 0 a \cdot A:$ METHOD OF FINDING AR E AS Base $x /;$
 perpendicular height $V 5 (5 - a) (s-b) (s-c), s = ' /:$ sum of the three bides $n, b,$
 c Sum of area of the two tfianglos $' /:$ sum of parallol sides x perpendicular
 height. Base x peifjendicular height. $/.$ suni of udes x ms'de radius $rTr' =$
 $O.705^O x \text{ diam"}. = 007953 x \text{ circomk-ii.-net; } \sim; = 0003/266 rA", - ore x /,$
 tadiui $f y ttA a j \setminus \text{Cumuli!- -vde } * n^MB \text{ Side} = \text{diameter } x 0flti623 \text{ Long}$
 diameter x shoft dimeter $x 0 73iao Bcisc x -/,$ perpendicular height 1142

General Reference Data MATHEMATICAL DATA AND FORMULAS

Functions of Numbers TABLE 20 Functions of Numbers NO. 1 2 3 4 5 6 7
 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
 33 34 35 36 37 38 39 40 41 42 43 44 45 SQUARE 1 4 9 16 25 36 49 64 81
 100 121 144 169 196 225 256 289 324 361 400 441 484 529 576 625 676
 729 784 841 900 961 1024 1089 1156 1225 1296 1369 1444 1521 1600
 1681 1764 1849 1936 2025 CUBE 1 8 27 64 125 216 343 512 729 1000

1331 1728 2197 2744 3375 4096 4913 5832 6859 8000 9261 10648 12167
 13824 15625 17576 19683 21952 24389 27000 29791 32768 35937 39304
 42875 46656 50653 54872 59319 64000 68921 74088 79507 85184 91125
 SQUARE ROOT 1.0000 1.4142 1.7321 2.0000 2.2361 2.4495 2.6458
 2.8284 3.0000 3.1623 3.3166 3.4641 3.6056 3.7417 3.8730 4.0000 4.1231
 4.2426 4.3589 4.4721 4.5826 4.6904 4.7958 4.8990 5.0000 5.0990 5.1962
 5.2915 5.3852 5.4772 5.5678 5.6569 5.7446 5.8310 5.9161 6.0000 6.0828
 6.1644 6.2450 6.3246 6.4031 6.4807 6.5574 6.6332 6.7082 CUBE ROOT
 1.0000 1.2599 1.4422 1.5874 1.7100 1.8171 1.9129 2.0000 2.0801 2.1544
 2.2240 2.2894 2.3513 2.4101 2.4662 2.5198 2.5713 2.6207 2.6684 2.7144
 2.7589 2.8020 2.8439 2.8845 2.9240 2.9625 3.0000 3.0366 3.0723 3.1072
 3.1414 3.1748 3.2075 3.2396 3.2711 3.3019 3.3322 3.3620 3.3912 3.4200
 3.4482 3.4760 3.5034 3.5303 3.5569 LOGARITHM 0.00000 0.30103
 0.47712 0.60206 0.69897 0.77815 0.84510 0.90309 0.95424 1.00000
 1.04139 1.07918 1.11394 1.14613 1.17609 1.20412 1.23045 1.25527
 1.27875 1.30103 1.32222 1.34242 1.36173 1.38021 1.39794 1.41497
 1.43136 1.44716 1.46240 1.47712 1.49136 1.50515 1.51851 1.53148
 1.54407 1.55630 1.56820 1.57978 1.59106 1.60206 1.61278 1.62325
 1.63347 1.64345 1.65321 1000 X RECIPROCAL 1000.000 500.000
 333.333 250.000 200.000 166,667 142,857 125.000 111.111 100.000
 90.9091 83.3333 76.9231 71.4286 66.6667 62.5000 58.8235 55.5556
 52.6316 50.0000 47.6190 45.4545 43.4783 41.6667 40.0000 38.4615
 37.0370 35.7143 34.4828 33.3333 32.2581 31.2500 30.3030 29.4118
 28.5714 27.7778 27.0270 26.3158 25.6410 25.0000 24.3902 23.8095
 23.2558 22.7273 22.2222 NO, - DIAMETER CIRCUM, 3.142 6.283 9.425
 12.566 15.708 18.850 21.991 25.133 28.274 31.416 34.558 37.699 40.841
 43.982 47,124 50.265 53.407 56.549 59.690 62.832 65.973 69.115 72.257
 75.398 78.540 81.681 84.823 87.965 91.106 94.248 97.389 100.531
 103.673 106.814 109.956 113.097 116.239 119.381 122.522 125.66 128.81
 131.95 135.09 138.23 141.37 AREA 0.7854 3.1416 7.0686 12.5664
 19,6350 28,2743 38.4845 50.2655 63.6173 78.5398 95.0332 113.097
 132.732 153.938 176.715 201.062 226.980 254.469 283.529 314.159
 346.361 380.133 415.476 452.389 490.874 530.929 572.555 615.752
 660.520 706.858 754.768 804.248- 855.299 907.920 962.113 1017.88
 1075.21 1134.11 1194.59 1256.64 1320.25 1385.44 1452.20 1520.53
 1590.43 1143

General Reference Data MATHEMATICAL DATA AND FORMULAS

Functions of Numbers TABLE 20 Functions of Numbers (Continued) NO.

46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69
70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 SQUARE
2116 2209 2304 2401 2500 2601 2704 2809 2916 3025 3136 3249 3364
3481 3600 3721 3844 3969 4096 4225 4356 4489 4624 4761 4900 5041
5184 5329 5476 5C25 5776 5929 6084 6241 6400 6561 6724 6889 7056
7225 7396 7569 7744 7921 8100 CUBE 9733C 103823 110592 117649
125000 132651 140C08 148877 157464 1C6375 175616 185193 195112
205379 216000 226981 238328 250047 262144 274625 287496 300763
314432 328509 343000 357911 373248 389017 405224 421875 438976
456533 474552 493039 512000 531441 551368 571787 592704 614125
636056 658503 681472 704969 729000 SQUARE ROOT 6.7823 6.8557
6.9282 7.0000 7.0711 7.1414 7.2111 7.2801 7.3485 7.4162 7.4833 7.5498
7.6158 7.6811 7.7460 7.8102 7.8740 7.9373 8.0000 8.0623 8.1240 8.1854
8.2462 8.3066 8.3666 8.4261 8.4853 8,5440 8.6023 8.6603 8.7178 8.7750
8.8318 8.8882 8.9443 9.0000 9.0554 9.1104 9.1652 9.2195 9.2736 9.3274
9.3808 9.4340 9.4868 CUBE ROOT 3.5830 3.6088 3.6342 3.6593 3.6840
3.7084 3.7325 3.7563 3.7798 3.8030 3.8259 3.8485 3.8709 3.8930 3.9149
3.9365 3.9579 3.9791 4.0000 4.0207 4.0412 4.0615 4.0817 4.1016 4.1213
4.1408 4.1602 4.1793 4.1983 4.2172 4.2358 4.2543 4.2727 4,2908 4.3089
4.3267 4.3445 4.3621 4.3795 4.3968 4.4140 4.4310 4.4480 4.4647 4.4814
LOGARITHM 1.66276 1.67210 1.68124 1.69020 1.69897 1.70757 1.71600
1.72428 1.73239 1.74036 1.74819 1,75587 1.76343 1.77085 1.77815
1.78533 1.79239 1.79934 1.80618 1.81291 1.81954 1.82607 1.83251
1.83885 1.84510 1.85126 1.85733 1.86332 1.86923 1,87506 1.88081
1.88649 1.89209 1.89763 1.90309 1.90849 1.91381 1.91908 1.92428
1.92942 1.93450 1.93952 1.94448 1,94939 1,95424 1000 X RECIPROCAL
21.7391 21.2766 20.8333 20.4082 20.0000 19.6078 19.2308 18.8679
18.5185 18.1818 17.8571 17.5439 17.2414 16.9492 16.6667 16.3934
16.1290 15.8730 15.6250 15.3846 15.1515 14.9254 14.7059 14.4928
14.2857 14.0845 13.8889 13.6986 13.5135 13.3333 13.1579 12.9870
12.8205 12.6582 12.5000 12.3457 12.1951 12.0482 11.9048 11.7647
11.6279 11.4943 11.3636 11.2360 11.1111 NO. =. DIAMETER CIRCUM.
144.51 147.65 150.80 153.94 157.08 160.22 163.36 166.50 169.65 172.79
175.93 179.07 182.21 185.35 188.50 191.64 194.78 197.92 201.06 204.20
207.35 210.49 213.63 216.77 219.91 223.05 226.19 229.34 232.48 235.62

238,76 241.90 245.04 248.19 251.33 254.47 257.61 260.75 263.89 267.04
 270.18 273.32 276.46 279.60 282.74 AREA 1661.90 1734.94 1809.56
 1885,74 1963.50 2042.82 2123.72 2206.18 2290.22 2375.83 2463.01
 2551.76 2642.08 2733.97 2827.43 2922.47 3019.07 3117.25 3216.99
 3318.31 3421.19 3525.65 3631.68 3739.28 3848.45 3959.19 4071.50
 4185.39 4300,84 4417.86 4536.46 4656.63 4778.36 4901.67 5026.55
 5153.00 5281.02 5410.61 5541.77 5674.50 5808.80 5944.68 6082.12
 6221.14 6361.73 1144

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